Materials for Engineers

Lecture 1

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Reference:

1. NPTEL lecture on History and evolution of materials by Prof. Bishak Bhattacharya, IIT Kanpur
2. Materials science and engineering by William Callister, Wiley Publications

Course Structure

Learning Objectives	**To provide overview of microstructure and properties of various engineering materials** • **To explore relations between performance of engineering products and microstructure, properties of materials that are used to construct them.**
Learning Outcomes	After the completion of the course, students will be able: • To explain the microstructure and properties of materials like steels, polymers, ceramics, and composites. • To understand the correlation of microstructure-properties-performance of materials so as to select suitable materials for engineering products.
Contents of the course	• Classification and evolution of engineering materials, crystal structure, defects, crystallographic planes, directions, slip, deformation mechanical behaviour, strengthening mechanisms, microstructure and properties of metal alloys (12) • Properties and processing of polymers, ceramics and composite materials, microstructure-property relationships (9) • Electrical, electronic and magnetic properties of materials, microstructure-property relationships (6) • Introduction to Nano, Bio, Smart and Functional materials. (3) • Introduction to selection of materials, Product based case studies on microstructure-property-performance of materials in the design of automobile; aircraft structures; e-vehicles; energy storage; electronic, optical and magnetic devices; and biomedical devices. (12)

- ***3 credit course***
- ***3 lectures / week***
- ***No laboratory***

- ***Common for all branches***

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Study Materials and Evaluation Scheme

- Class lectures – PowerPoint presentations & lecture videos
- Class notes – students are encouraged to take class notes during the lectures

Essential Reading	**1. William D. Callister Jr., David G. Rethwisch, "Materials Science and Engineering: An Introduction", 10th Edition, Wiley, 2018.** **2. Michael Ashby, Hugh Shercliff, David Cebon, "Materials – Engineering, Science, Processing and Design", 4th Edition, Butterworth-Heinemann, 2018.**
Supplementary Reading	1. V Raghavan, "Materials Science and Engineering: A First Course, 5th Ed, 2007, PHI India. 2. Donald R. Askeland K Balani, "The Science and Engineering of Materials," 7th Edition, Cengage Learning, 2016. 3. Michael Ashby, "Materials Selection in Mechanical Design", 5th Edition, Butterwoth-Heinemann, 2016.

Evaluation Scheme:

Mid-Sem – **40%** + End-Sem – **60%**

History and Evolution of Materials

Materials : "Drivers" of our Society

- Rooted in our culture and have influence over virtually every segment of our daily lives –
  - Transportation
  - Housing
  - Clothing
  - Communication
  - Recreation, etc.
- Early civilizations have been designated by materials development:
  - Stone Age
  - Bronze Age
  - Iron Age

P. K Kelkar library, IIT Kanpur

Fighter Craft

Image: Ontario sea plane association

Home

Antenna

Image: www.goes-r.gov

World of materials

Stone Age

- The term "**Stone Age**" was coined in the late 19th century by the Danish scholar **Christian J. Thomsen.**
- Roughly extended between **15000-2000 BC**.
- Characterized by creation and use of **stone tools**.
- Wood, bones and other materials were also used as tools but have shorter life.

Image: Canadian Anglo-Boer War Museum, Canada

Image: Nubian Museum, Egypt

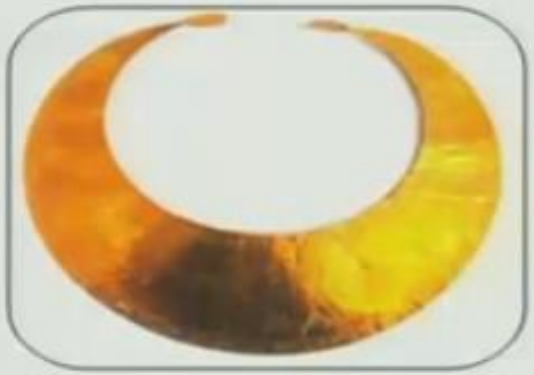
Image : Wesleyan University, USA

Image: Government Museum, Chennai, India

Stone Age Tools

Bronze Age

- Roughly extend between 3500-500 BC.
- Beginning of **metal working**.
- Copper was mixed with tin, to create a new alloy - **BRONZE**, which was stronger than the other two metals individually.
- Used for tools, weapons, armor, decoration, etc.



Neck collar of Gold
Image: http://www.britishmuseum.org/

Daggers (kind of knife) and Swords
Image : The Archaeology Gallery at West Stow, England

Axe Head
Image: Kenilworth Abbey Barn Museum, UK

Spartan Armor
Image: Metropolitan Museum of Art, New York

Bronze Age – India

- Begins around 3000 BC.
- Development of Indus valley Civilisation/ Harappa Culture, first ever urban civilisation.

Indus valley civilization
Image: NCERT

- The famous **Dancing Girl** is a **bronze statuette** (10.5 cm high) casted using the lost wax method and dating around 2500 B.C., from the **Mohenjo-daro site**, Sindh (now in Pakistan) of the Indus Valley Civilization.
- Found by **Ernest Mackay** in 1926.
- Although it is in standing position, it was named *"Dancing Girl"* with an assumption of her profession – 25 bangles in left and 4 bangles in her right hand.

Dancing Girl
(**Image:** National Museum, New Delhi, India)

Ceramic Pot, Indus valley
Image: www.antiques.com

Iron Age

- It is the **last stage** of the archaeological sequence known as the three-age system (Stone Age, Bronze Age, & Iron Age).
- The Iron Age began about 3000 years ago and **continues till today**. Use of iron and steel has changed drastically the human development.
- Witnessed **industrial revolution**.
- Improved modes of transportation –Automobiles, Railways and aero planes.

Transportation modes

IRON PILLAR OF DELHI

- Iron age in South-Asia begins around 1200 B.C
- IRON PILLAR OF DELHI (around 1600 year old) called as "a testament to the skill of ancient Indian blacksmiths".
- Solid shaft of **wrought iron** (high **phosphorous** content) about 7m tall, 0.4 m diameter weighing over 6,000 kg.
- High resistance to corrosion results from an even layer of crystalline iron hydrogen phosphate hydrate, which serves to protect it from the effects of the local Delhi climate.

Iron Pillar, Qutab Minar Complex, New Delhi, India

In-depth study, book and papers published by Late **Prof. R. Balasubramaniam**, Department of Materials Science and Engineering, IIT Kanpur, http://www.iitk.ac.in/infocell/Archive/dimov1/iron_pillar.html

World's First Cast Iron Bridge

- Built over the **River Severn** in Shropshire, **England**.
- **First arch bridge in the world** to be made of **Cast Iron**.
- Opened in the year 1781.
- Longest span = 31m

The Iron Bridge, England

The Steel Story

Steel is an **alloy of iron and carbon** (less than 2% by weight) added with **other alloying elements** such as sulphur, phosphorous, manganese, chromium, etc. to achieve variety of properties.

13th Century BC : Early blacksmiths discovered that iron became harder and stronger when left in charcoal furnaces.

3rd Century BC : "**Wootz Steel** (Ancient India)"
The craftsmen of southern India used crucibles to smelt wrought iron with charcoal to produce 'Wootz' steel – still admired today.

Roman Era : The Romans learned how to temper work-hardened steel to reduce its brittleness.

3rd century AD : Chinese created steel by melting together wrought iron with cast iron.

Modern era in Steel making

1855 : Henry **Bessemer**, steel mass production began with Bessemer convertor – air blast injected into molten pig iron (carbon 3.5-4.5%) to reduce its carbon content – suitable for phosphorous free ores.

1860 : **Gilchrist-Thomas process** - improvement to the Bessemer process – used limestone to remove phosphorous

1865 : **Siemens-Martin/Open-hearth process** - complemented the Bessemer process - melting iron bar or steel scrap with pig iron until carbon content reduced by dilution and oxidation

1953 : **Basic oxygen steelmaking** – used oxygen blast instead of air.

Latest : **Electric arc furnace** - highly efficient recycler of steel scrap

Steel can be recycled endlessly with no detrimental effects on its properties. All steel created as long ago as 150 years can be recycled and used in new products and applications.

Steel lifeline – Rail track

- World has total rail network of about 16 lakh km length.
- **India** has **4th longest** railway network (65,000 km) after USA(2.5L), China(1.2L) and Russia(86k).
- **Rail steel** contains carbon(0.6-0.8%), and other alloying elements such as Mn, Si, S, P, Mo, Cr, V to produce hard, wear & corrosion resistant rails.

Diamond Crossing, Nagpur*

*Place where India crosses from East to West and North to South.

International rail track (Atari, India - Lahore, Pakistan)

Pamban Rail Bridge, Rameshwaram, Tamil Nadu

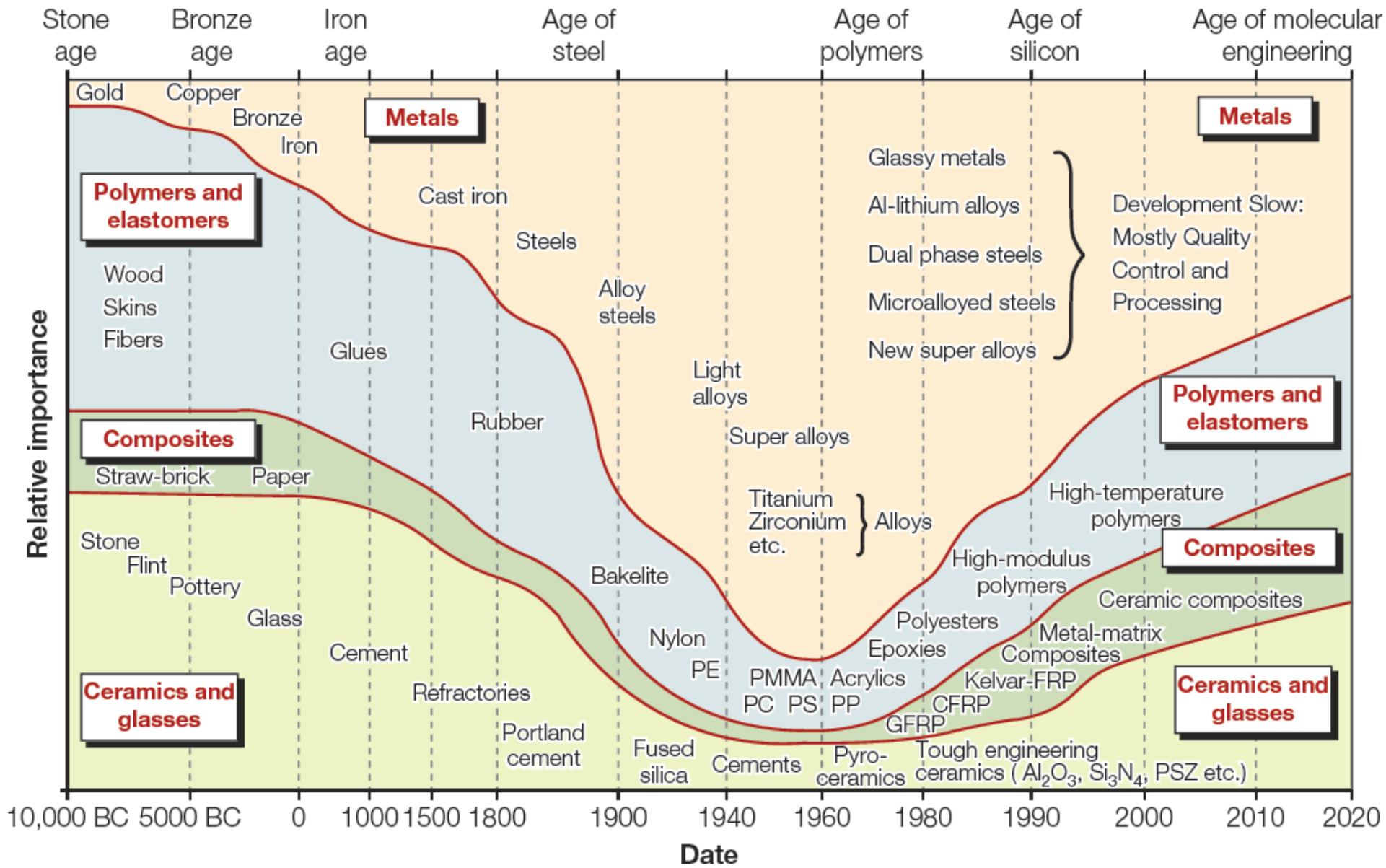
Dudhsagar falls track, Goa

Howrah Bridge

- Howrah Bridge is a **suspension type Balanced Cantilever bridge** over the Hooghly River in Kolkata, West Bengal, India.
- **World's 6th longest cantilever bridge (Longest span = 457 m).**
  1st - Quebec Bridge (1917), Canada (longest span - 549m span).
- Official name: **Rabindra Setu**, named after Gurudev "**Rabindranath Tagore**" (first Asian Nobel laureate).
- Commissioned in 1943.
- Total length - 705m
  Height - 82 m
- **Material – Steel, supplier TATA Steel**
- Consumed about 26,500 tons of steel, no bolts and nuts used - **only Rivets.**

Howrah Bridge, Kolkata

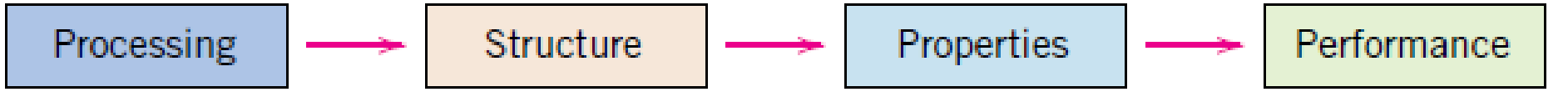
Summary of Evolution of Materials

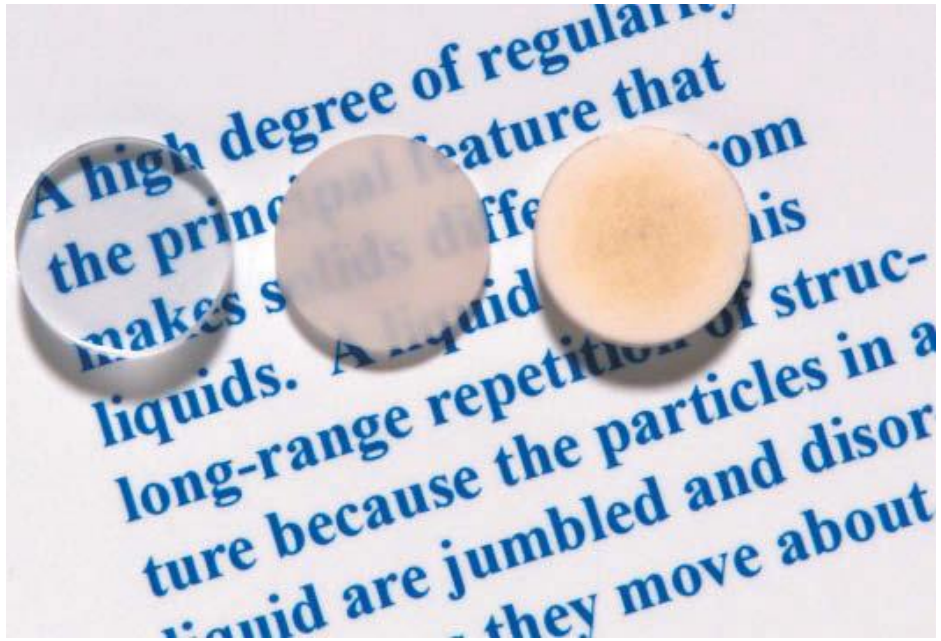


Materials Science and Engineering

Materials Science – Relations between **structure** and **properties** / synthesize new materials

Materials Engineering - Structure–property relations, designing or engineering the structure of a material to produce a predetermined set of properties / develop new products, processing of materials into products





Three thin disk specimens of aluminum oxide

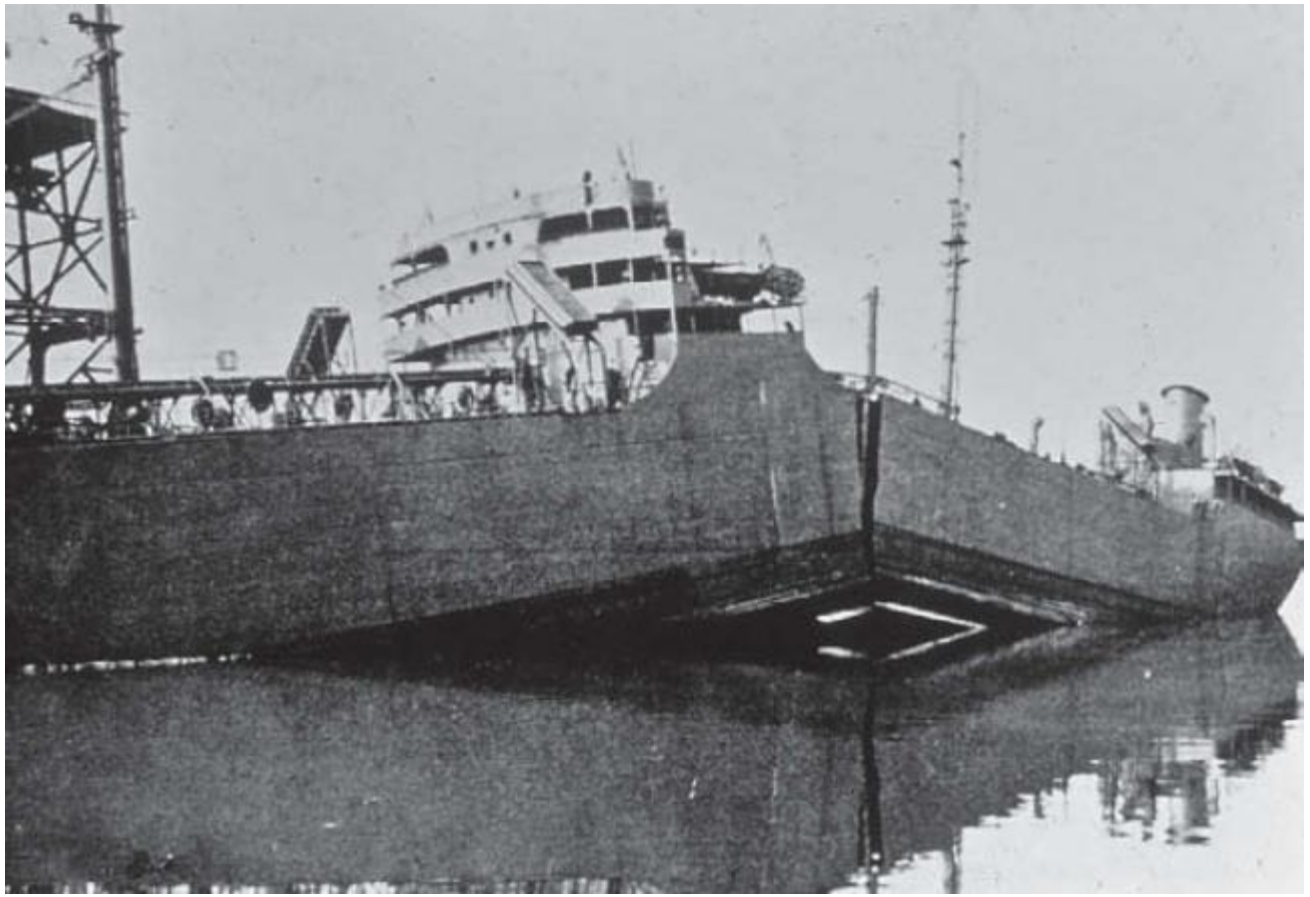
The differences in optical properties are a consequence of differences in structure of these materials, which have resulted from the way the materials were processed.

WHY STUDY MATERIALS SCIENCE AND ENGINEERING?

- Materials scientists and engineers are specialists who are totally involved in the investigation and design of materials for different applications.
- Problem involves selecting the right material from the thousands available.
- It may be necessary to **trade one characteristic for another** – Example - material having a high strength has only a limited ductility
- **Deterioration of material properties** that may occur during service operation – Example – reduction in mechanical strength may result from exposure to elevated temperatures or corrosive environments.
- What will the **finished product cost?** A material may be found that has the ideal set of properties but is prohibitively expensive.
- Understanding of processing-structure-property relation is important

Liberty Ship Failures

Case study involving the role of materials engineers

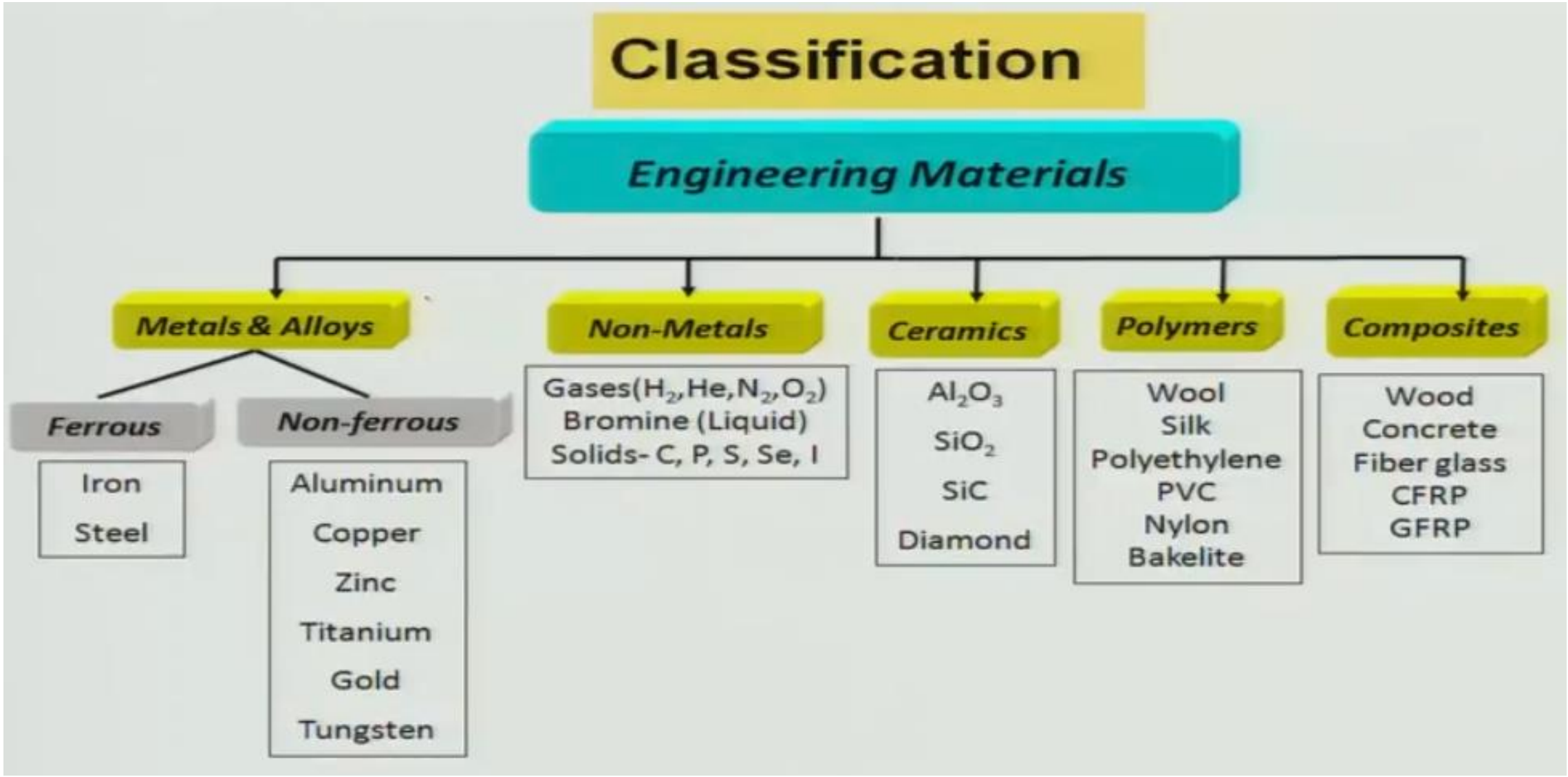


Failure of steel because of ductile to brittle transformation at low temperatures

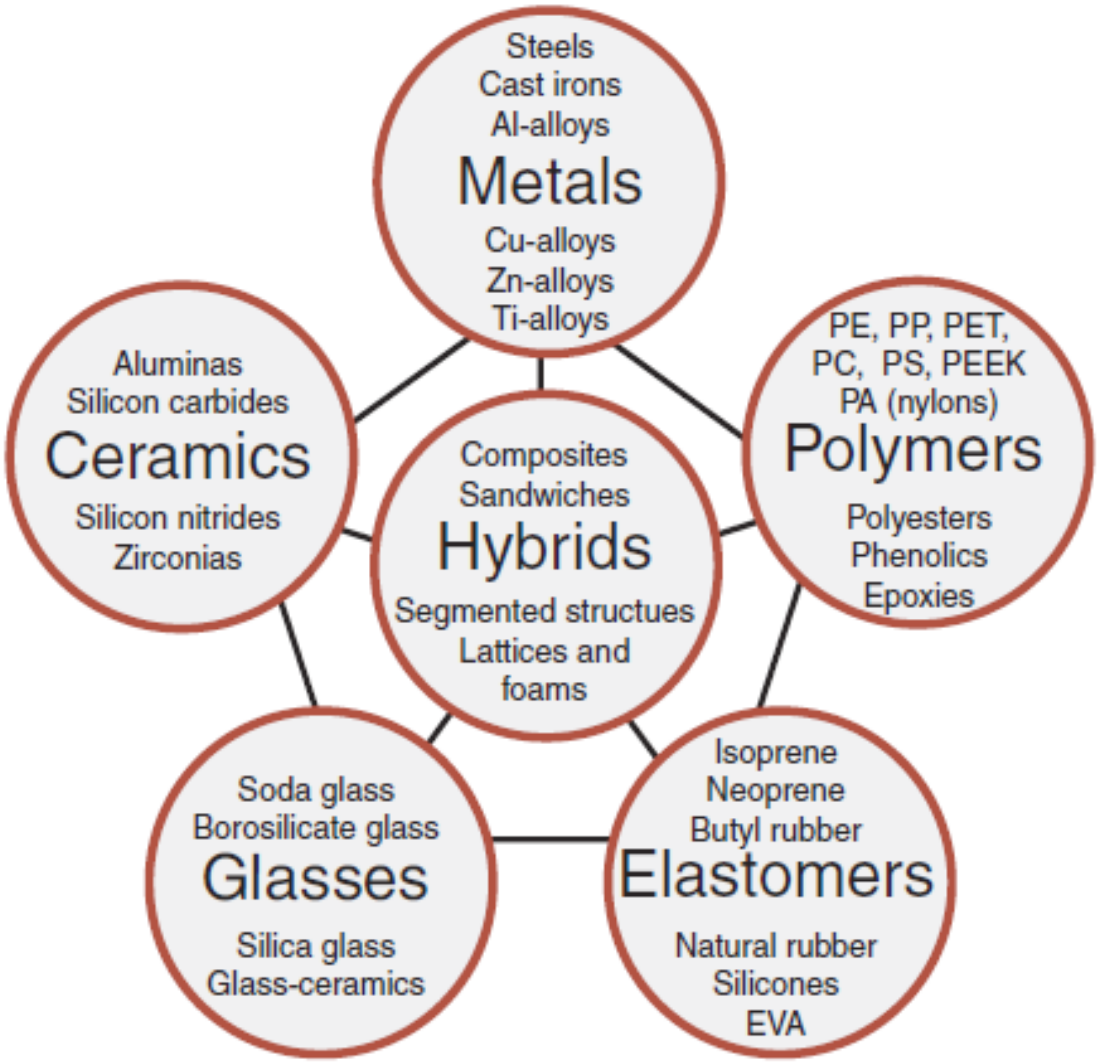
Remedial measures:

- Lowering the ductile-to-brittle temperature
- Rounding off corners
- Installing crack-arresting devices such as riveted straps and strong weld seams
- Improving welding practices and establishing welding codes

Classification of Materials



Menu of Engineering Materials



General Properties of Metals

Steel, aluminum, magnesium, zinc, cast iron, copper, lead, etc

- High electrical conductivity
- High heat conductivity
- Ductile
- Easily deformed
- High thermal shock resistance
- Suitable for structural and load bearing applications
- Alloys utilized for development of high performance metals

General Properties of Ceramics

Bricks, glass, refractory and abrasive ceramics

- Low electrical conductivity
- Low heat conductivity
- Brittle
- High wear resistance in high temperature applications
- Corrosion resistance
- Generally used as insulators and load bearing structures
- Advanced ceramics are utilized in integrated circuits and optical applications due to the improvements on optical and electrical properties
- Biocompatible materials

General Properties of Polymers

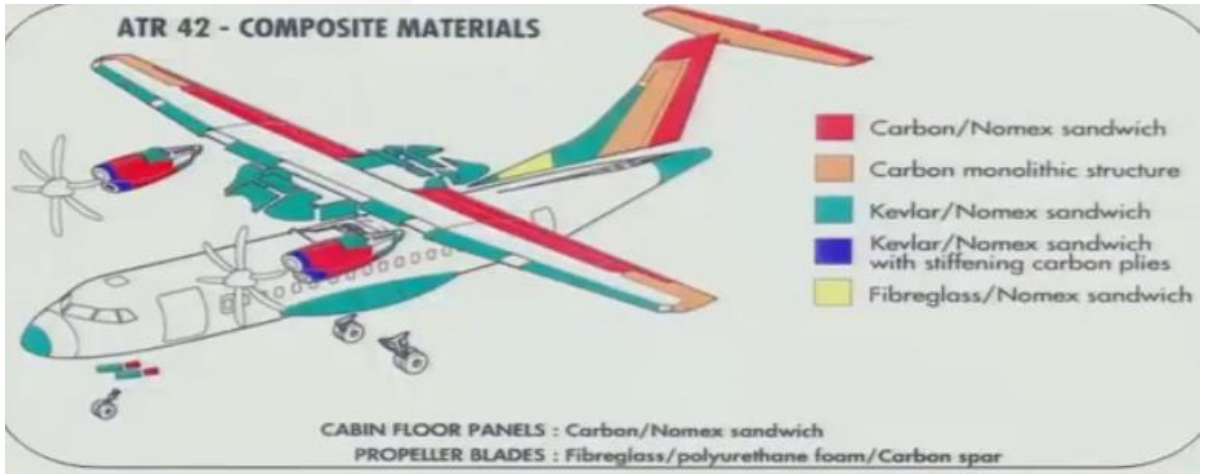
Elastomers, plastics, and adhesives

- Produced by polymerization of organic molecules into large molecular structures
- Low electrical conductivity
- Low thermal resistance
- Low strength
- Light materials

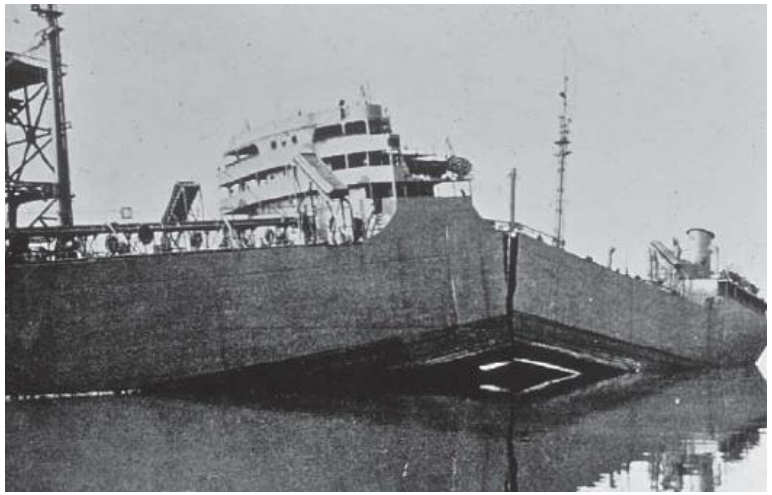
General Properties of Composites

Comprised of two or more materials

- Carbon fiber reinforced polymer, polymer fiber reinforced HA
- Light
- Strong
- High fracture toughness
- High thermal shock resistance



Evolution of Materials Engineering



Advanced Materials

Materials for high technology – a device or product that operates or functions using relatively intricate and sophisticated principles, including electronic equipment (camcorders, CD/DVD players), computers, fiber-optic systems, spacecraft, aircraft, and military rocketry.

Semiconductors – Integrated Circuits

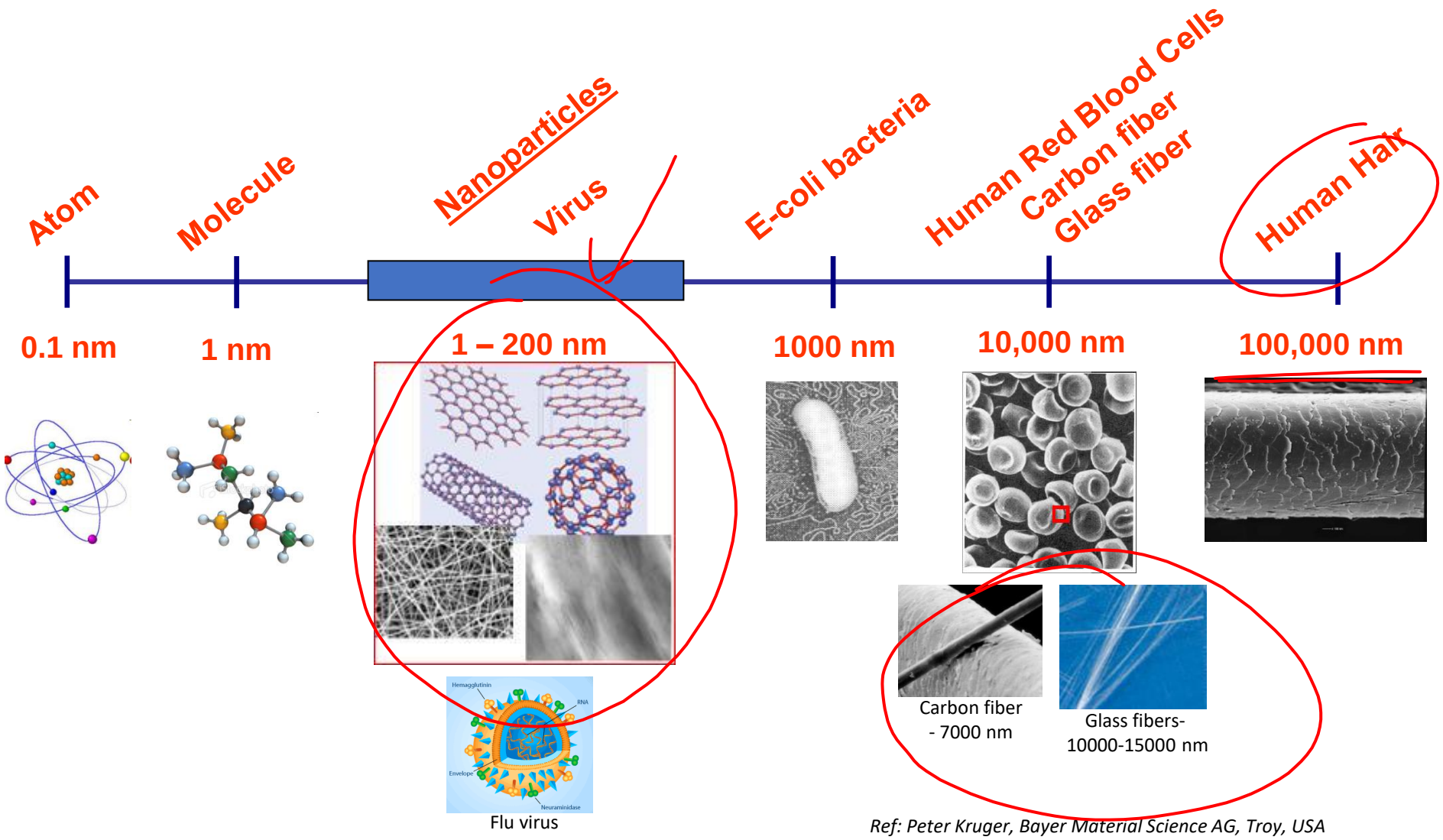
Biomaterials – Replace deceased or damaged parts, compatible with body tissues

Smart materials – Sensors, actuators, energy harvesting, shape memory

Nanomaterials – Mechanical, electronic, biomedical, sporting, energy, industrial applications

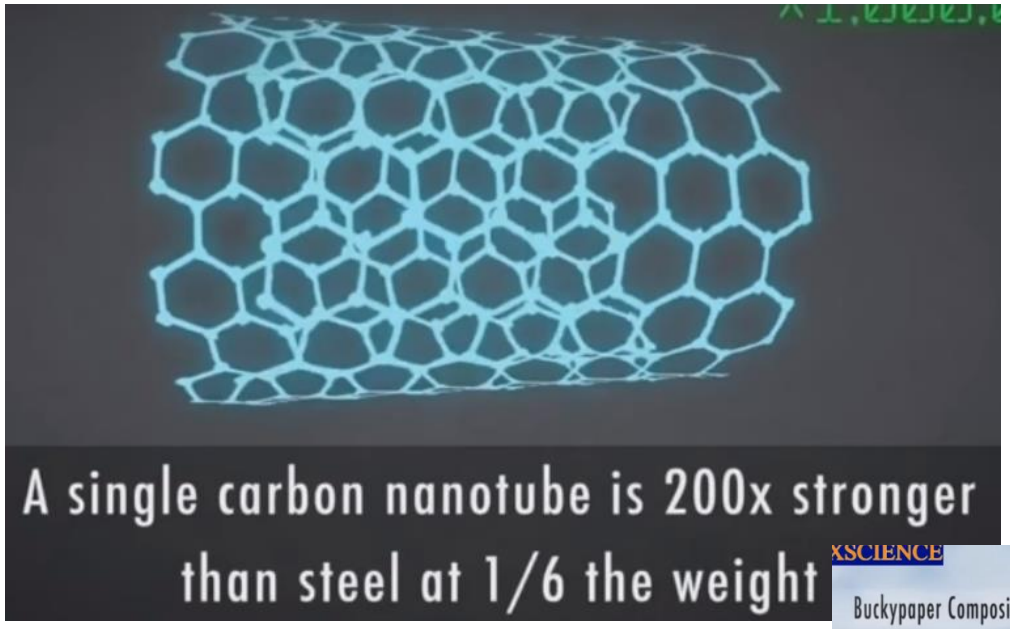
- Nanocarbons—Fullerenes, carbon nanotubes, and graphene
- Particles of carbon black as reinforcement for automobile tires
- Nanocomposites
- Magnetic nanosize grains that are used for hard disk drives
- Magnetic particles that store data on magnetic tapes
- CNTs in automotive, sporting goods

Nanomaterials – Size Comparison

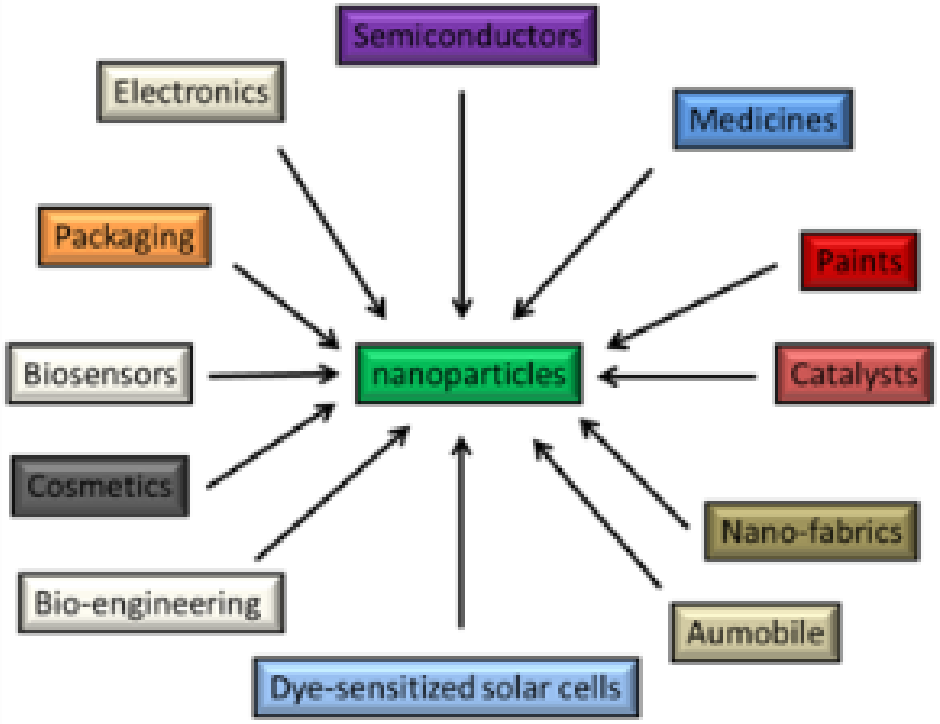


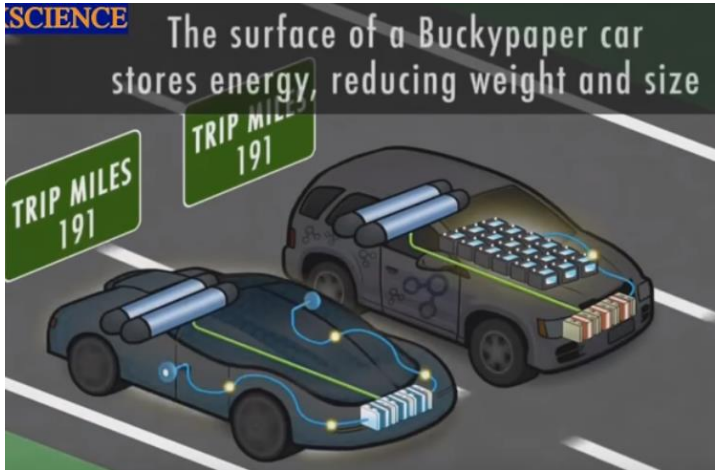
Ref: Peter Kruger, Bayer Material Science AG, Troy, USA

Nanomaterials – Applications



- ✓ Electronics
- ✓ Optical
- ✓ Thermal properties





Modern Materials Needs

Automobiles, aircraft, trains – Weight reduction, fuel efficiency, high temperature operation

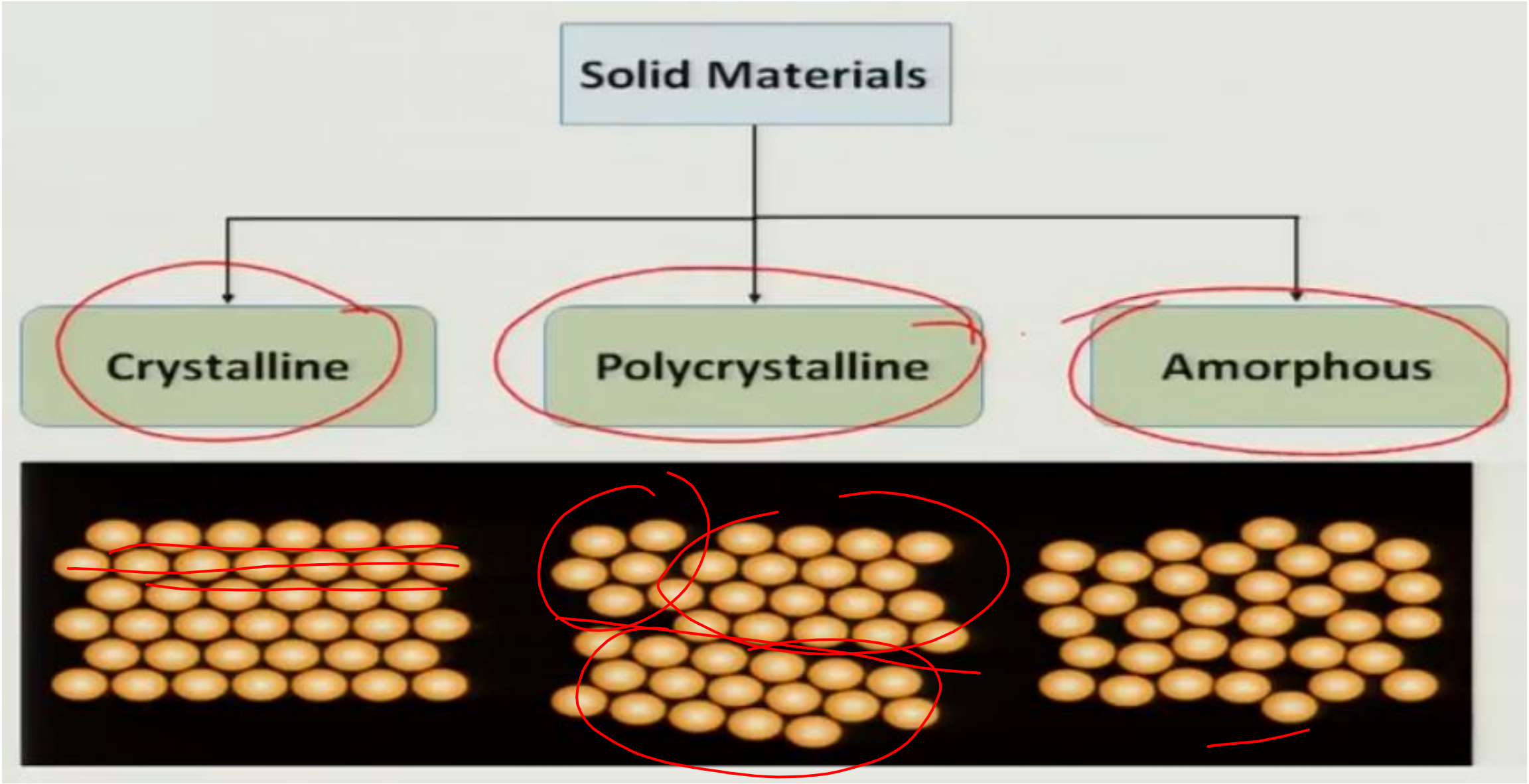
Energy – Economic resources – solar cells, fuel cells, energy harvesting

Pollution control – safe manufacturing and processing, environmental degradation, clean water

Renewable energy resources – additional reserves, new recycling technologies

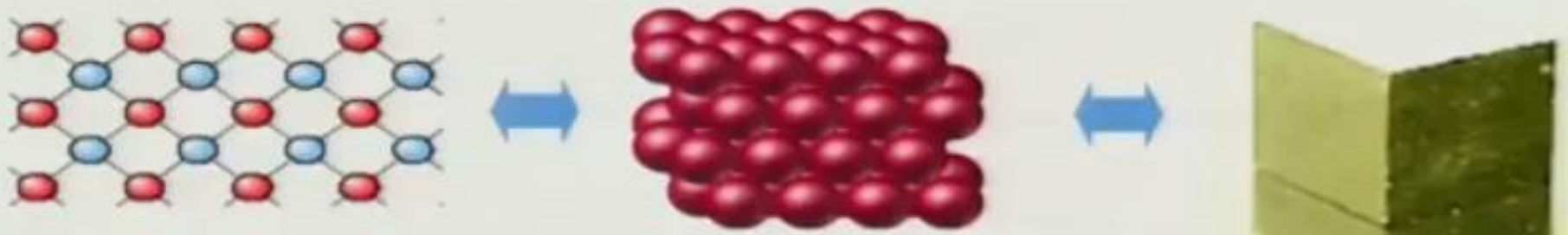
Importance of Crystallography

- **Crystallography** is the study of **atomic and molecular structure**.
- The study of crystal geometry helps to understand how the **atoms** in a material are **arranged** in order to interpret the **relationship** between **atomic structure** and **properties** of these materials namely-
  - ✓ Mechanical
  - ✓ Electrical
  - ✓ Magnetic
  - ✓ Optical
  - ✓ Metallurgical
- Common Technique: *X-ray* diffraction


Solid Materials
Crystalline
Polycrystalline
Amorphous

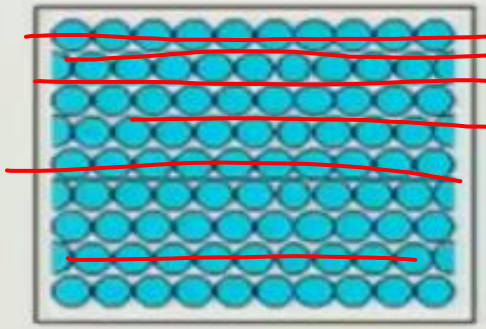
Crystalline Solid

- A **long range** structure exists in which **atoms** are **arranged** into unit cells and the **unit cells repeat** in a **regular orderly pattern**, forming a lattice (3-D).
- Directional properties - **Anisotropic** nature.
- **Sharp melting point**
- Example – Gold, Silver, Aluminum, etc.

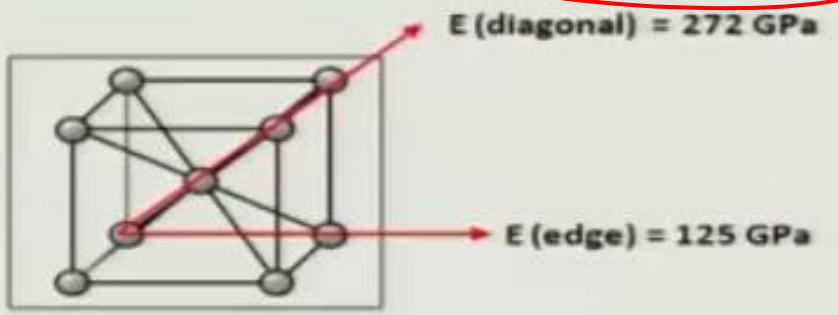


Crystalline solid

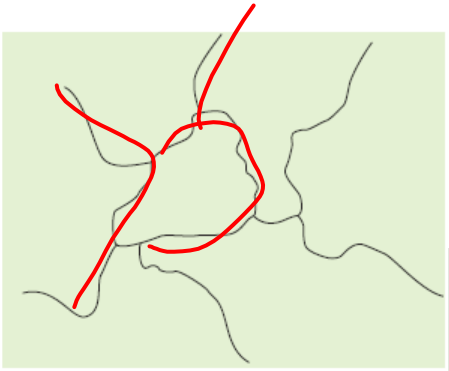
Single Crystal



- Only one grain – no grain boundaries.
- Formed by uniform cooling of liquid material.
- Have **a high degree of order** throughout the ***entire volume of the material***
- Useful for applications where grain boundaries are harmful.
- For example, high temperature deformation or creep resistance (as **creep** takes place by **grain boundary sliding**) - single crystal turbine blades - Ni-based super alloy gas turbine blades (withstand creep at high temperature due to lack of grain boundaries).
- Other examples – Diamond, Semi-conductors.
- BCC* Iron

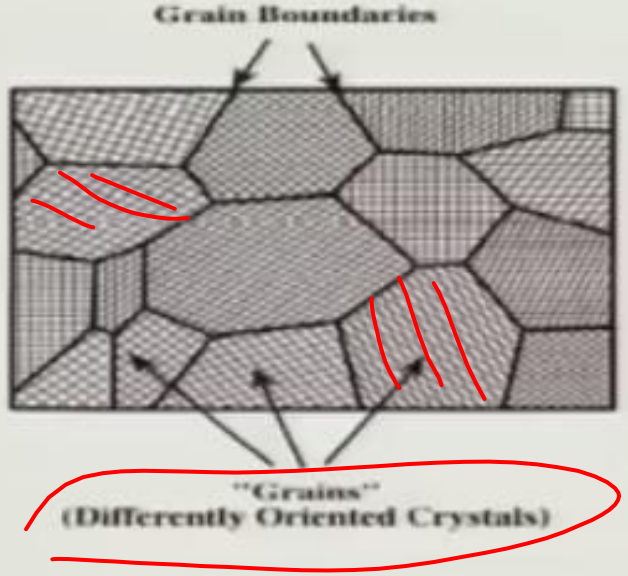


* BCC will be discussed in further slides



Polycrystalline Solid

- Made up of an **aggregate** of many small **single crystals** (also called crystallites or **grains**).
- **Across grain boundary**, the **orientation** of crystal **changes**.
- The grains are usually 100 nm - 100 microns in diameter.
- Polycrystalline material have a high degree of order over many atomic or molecular dimension.
- Polycrystals with grains that are <10 nm in diameter are called Nano-crystalline.
- Most engineering materials are polycrystalline.
- **Properties may or may not vary** with **direction**.
- Example – Common steel is polycrystalline.
  - If **grains** are "**textured** (due to rolling)", then **anisotropic**
  - If **grains** are **random**, then **isotropic** (E ≈ 210 GPa)

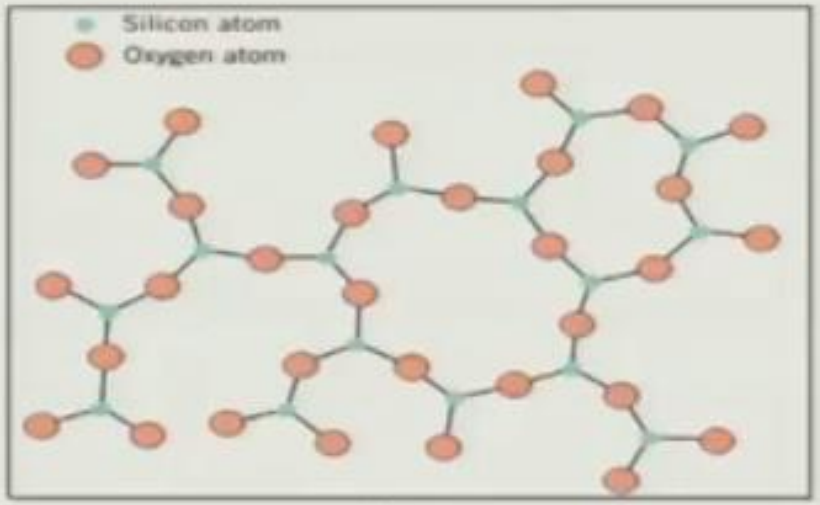


	Modulus of Elasticity (GPa)		
Metal	***[100]***	***[110]***	***[111]***
Aluminum	63.7	72.6	76.1
Copper	66.7	130.3	191.1
Iron	125.0	210.5	272.7
Tungsten	384.6	384.6	384.6

Amorphous Solid

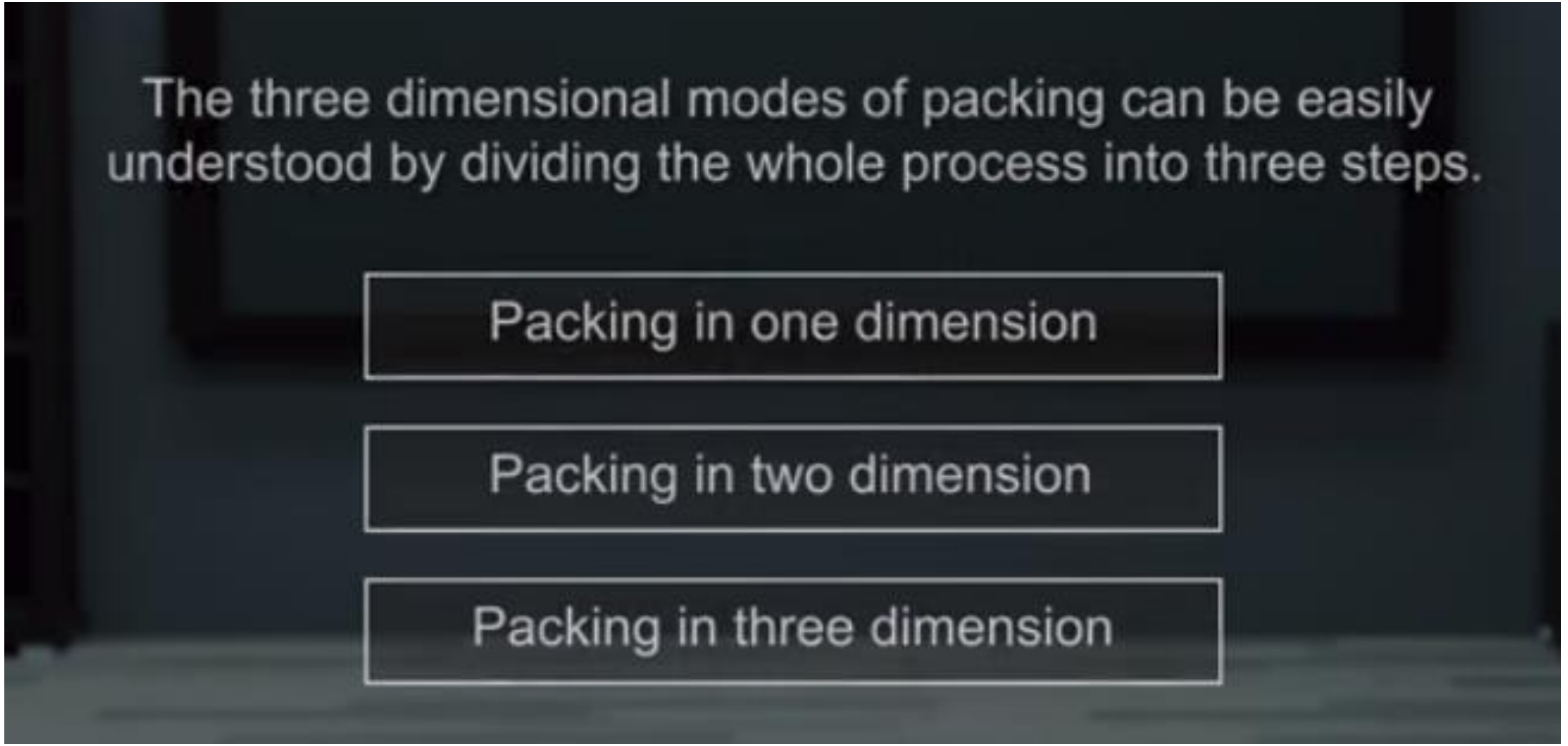
- The atoms or molecules are arranged in an **irregular manner.**
- No periodic packing.
- **No directional properties** - Isotropic substances.
- Occurs for complex structures, rapid cooling.

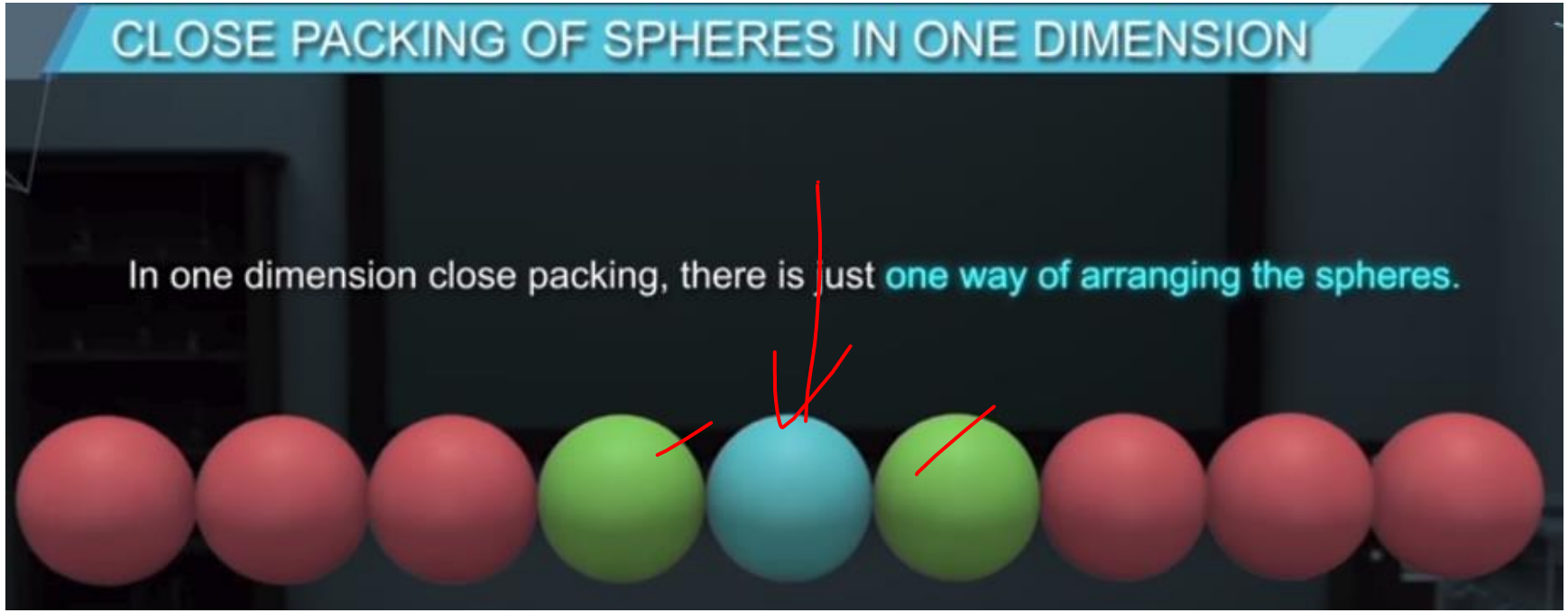
Example : Glass, Plastic, rubber, etc.

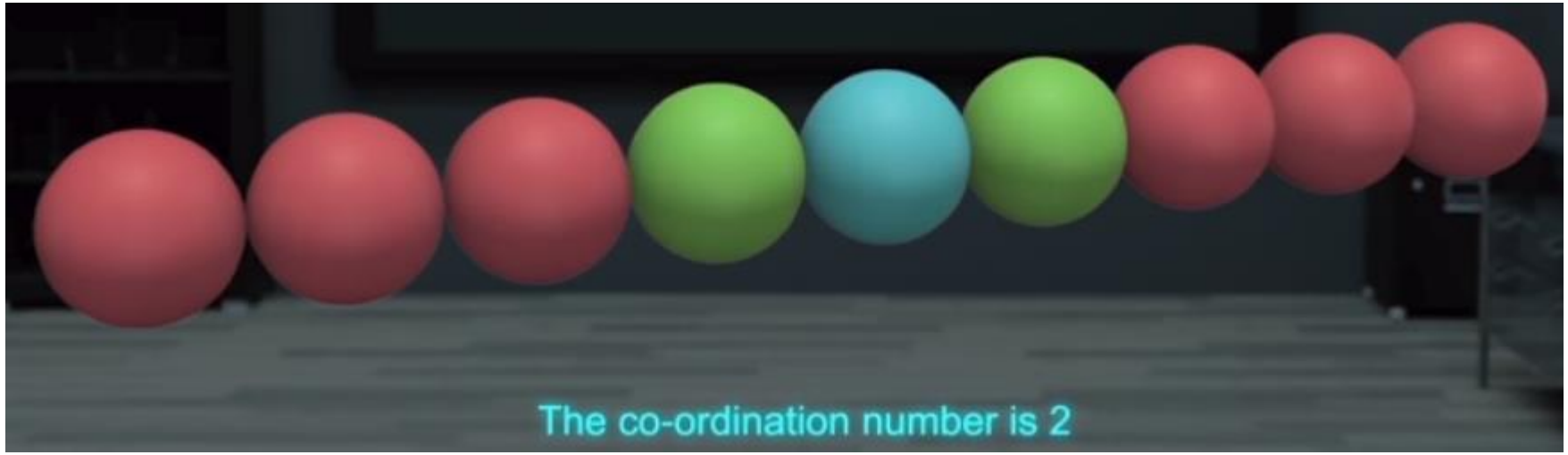


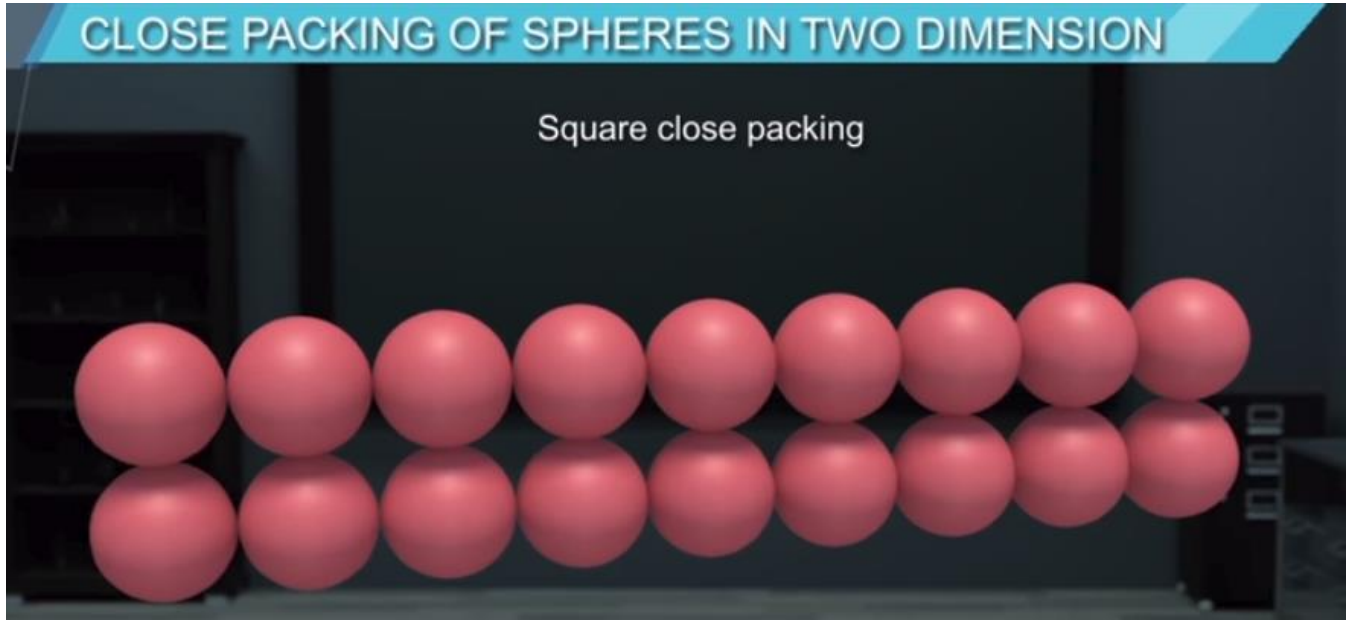
Silica

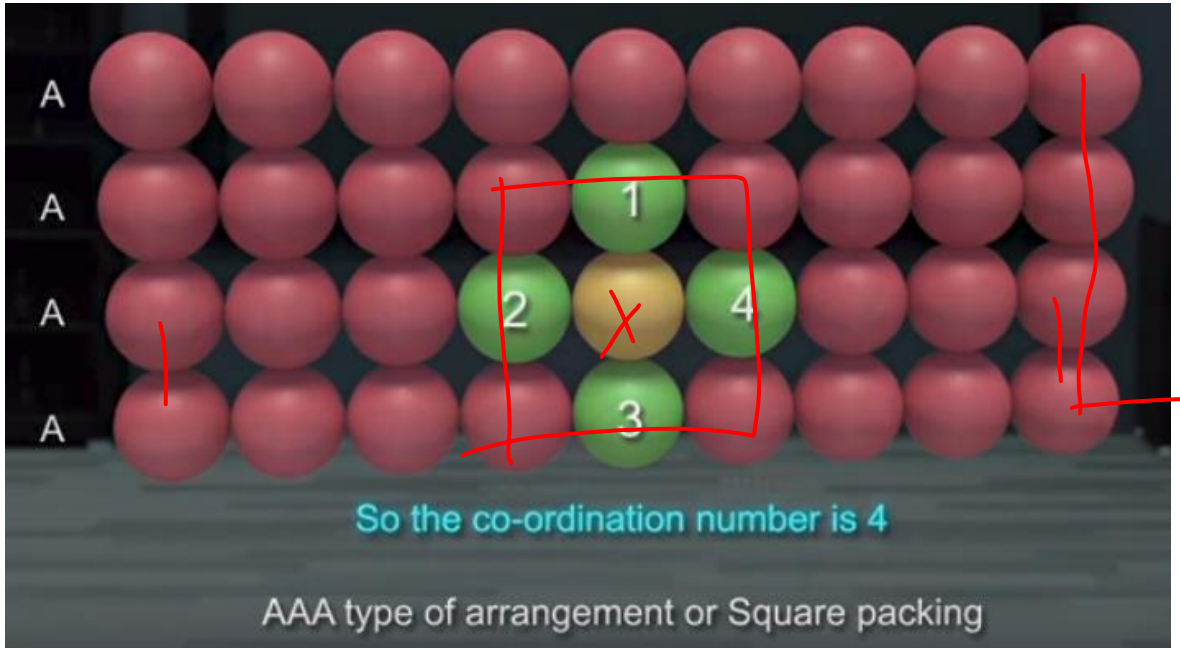
3D packing of atoms



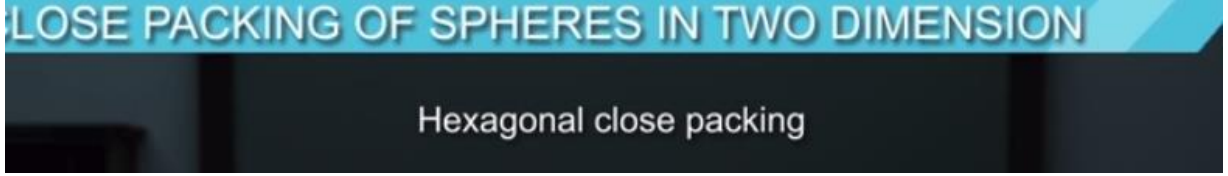

CLOSE PACKING OF SPHERES IN ONE DIMENSION
In one dimension close packing, there is just one way of arranging the spheres.

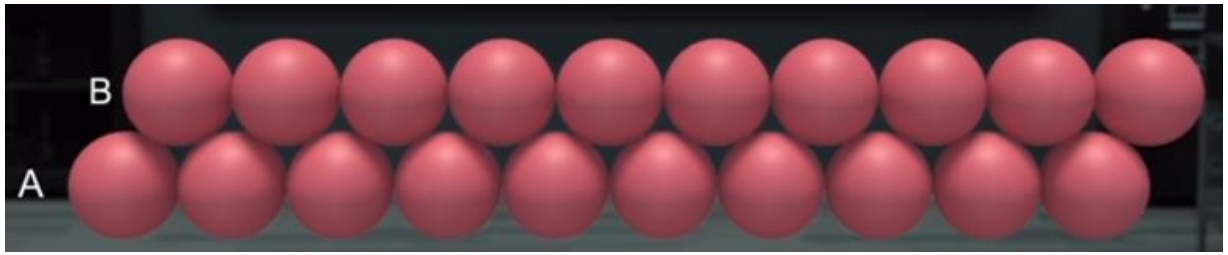

The co-ordination number is 2

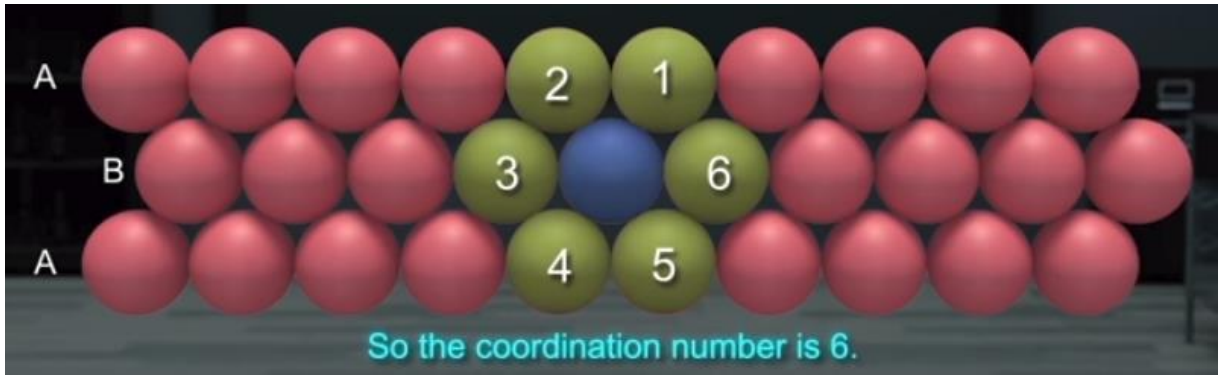

CLOSE PACKING OF SPHERES IN TWO DIMENSION
Square close packing

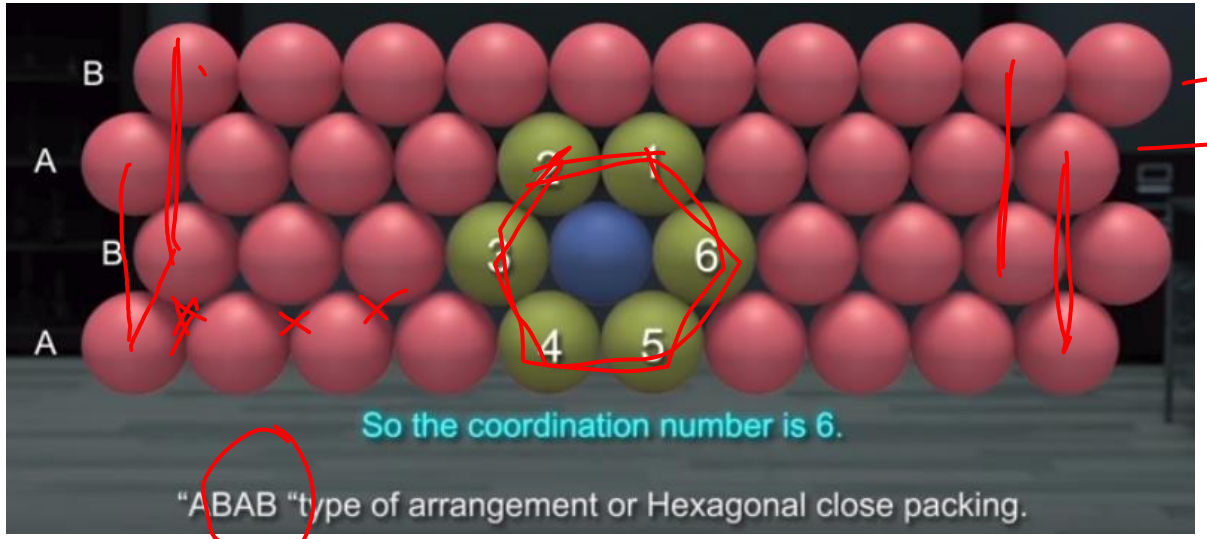

A
A
A
A
1
2
X
4
3
So the co-ordination number is 4
AAA type of arrangement or Square packing

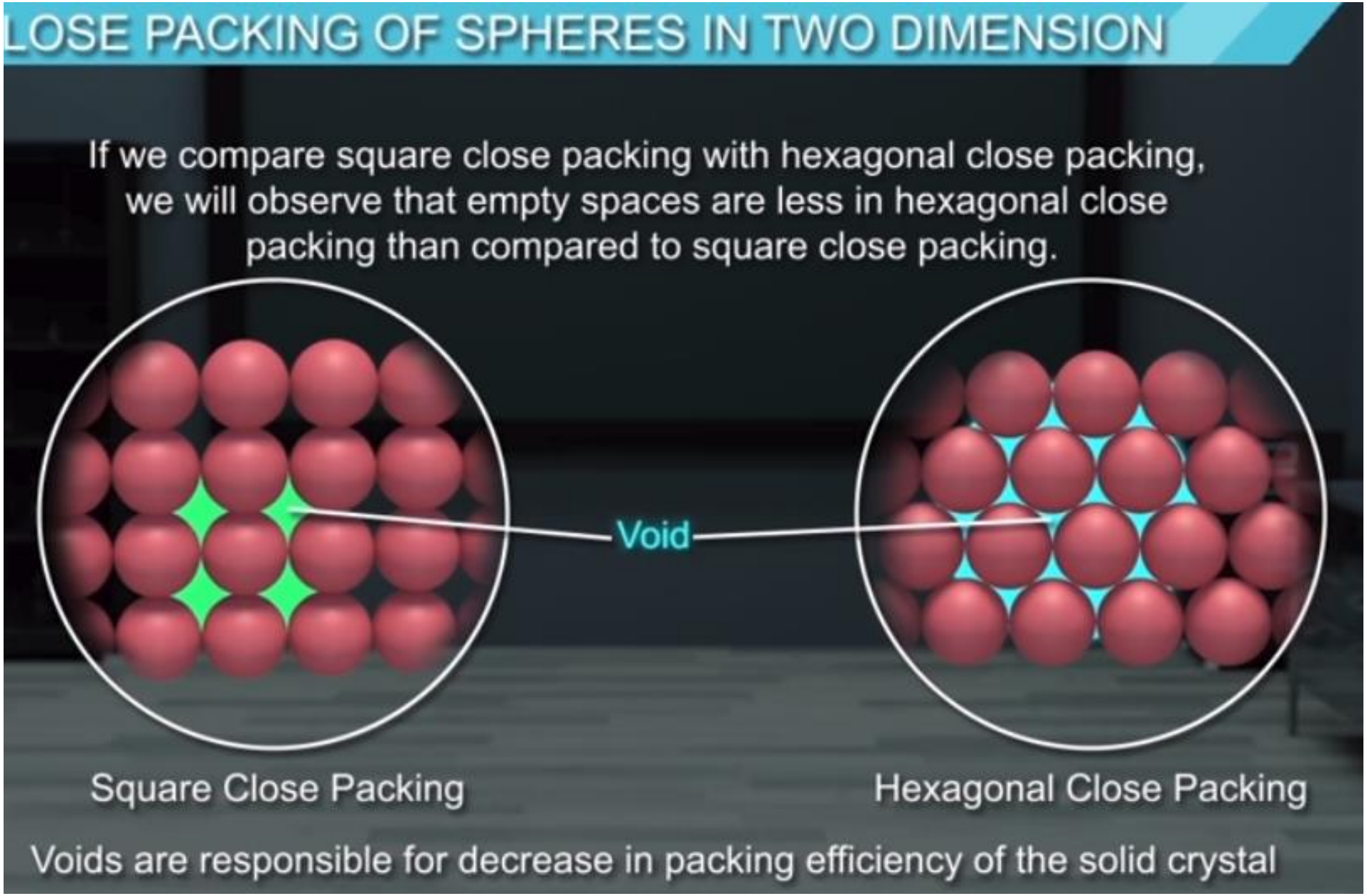
h
A
A


LOSE PACKING OF SPHERES IN TWO DIMENSION
Hexagonal close packing


B
A

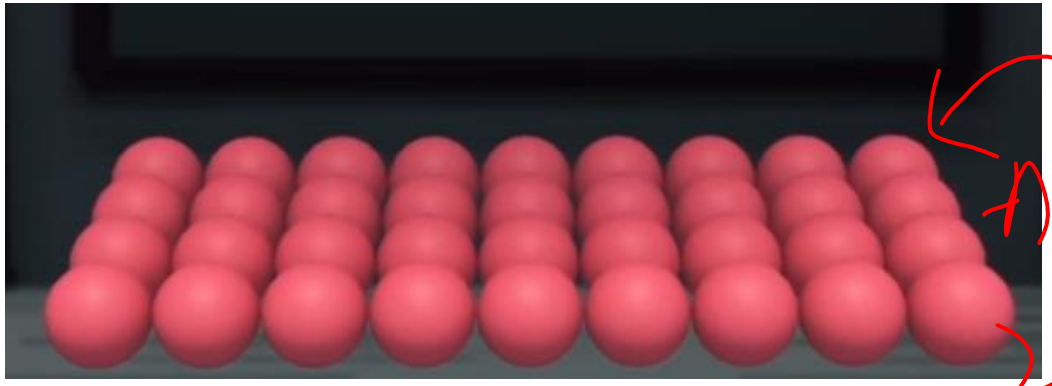

A
B
A
2
1
3
6
4
5
So the coordination number is 6.

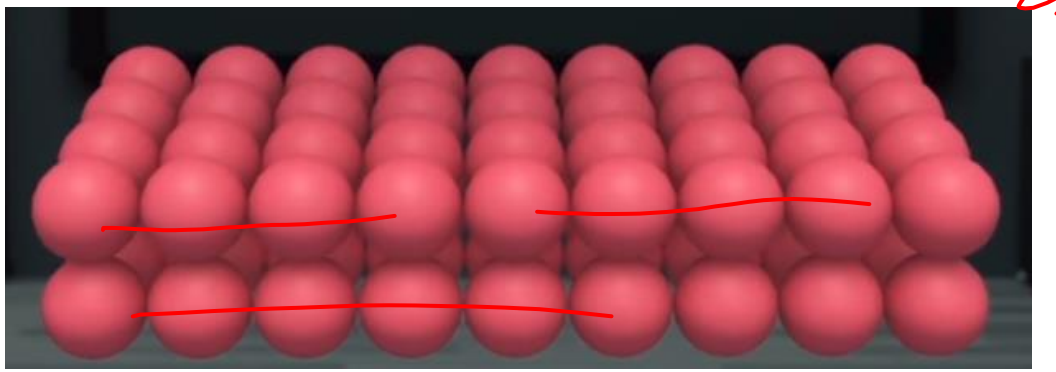

B
A
B
A
2
1
3
6
4
5
So the coordination number is 6.
"ABAB "type of arrangement or Hexagonal close packing.

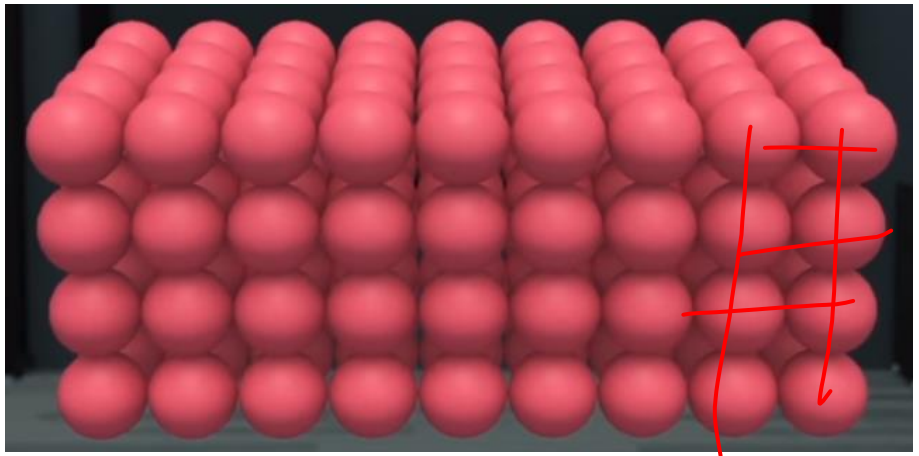

LOSE PACKING OF SPHERES IN TWO DIMENSION
If we compare square close packing with hexagonal close packing, we will observe that empty spaces are less in hexagonal close packing than compared to square close packing.
Void
Square Close Packing
Hexagonal Close Packing
Voids are responsible for decrease in packing efficiency of the solid crystal

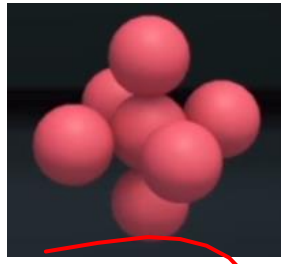
THREE DIMENSIONAL CLOSE PACKING

Two dimensional square close packed



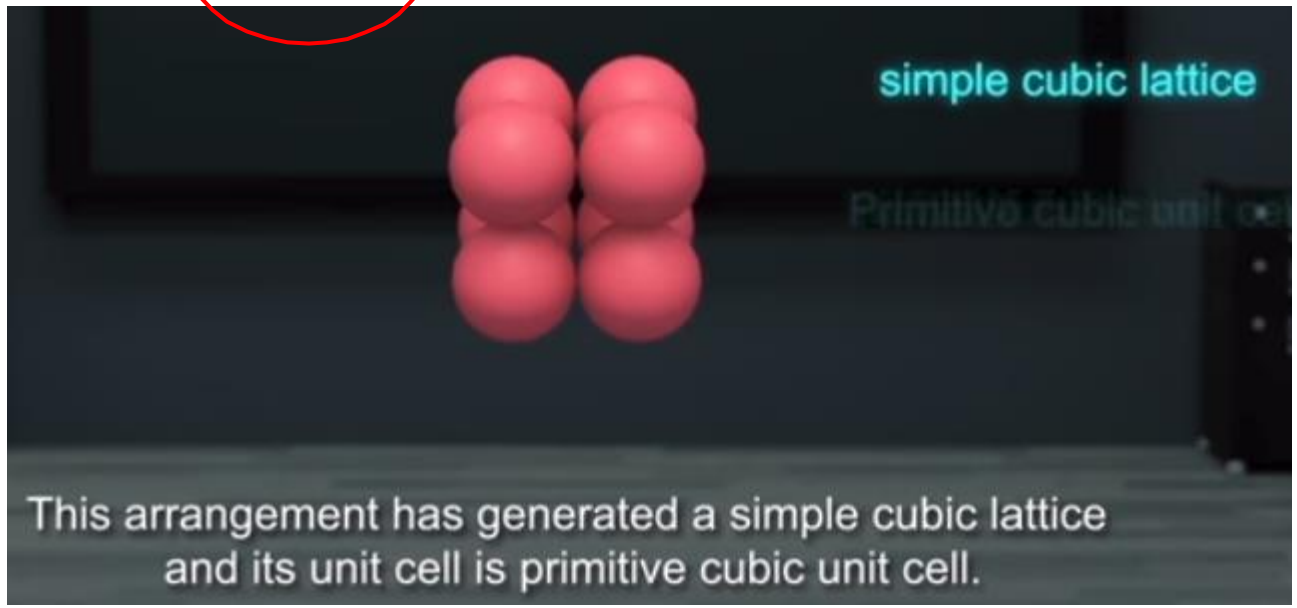


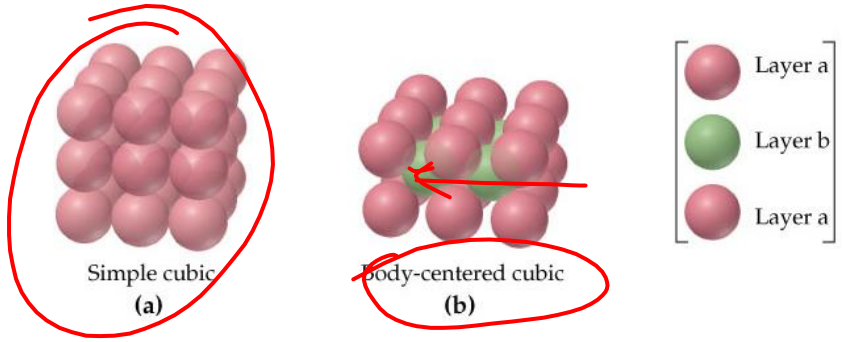


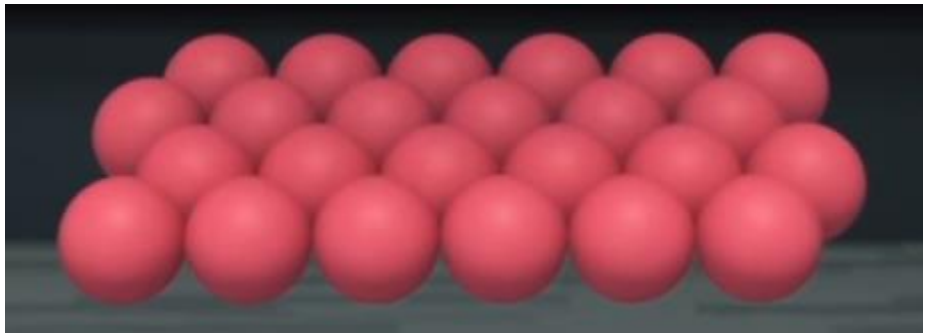


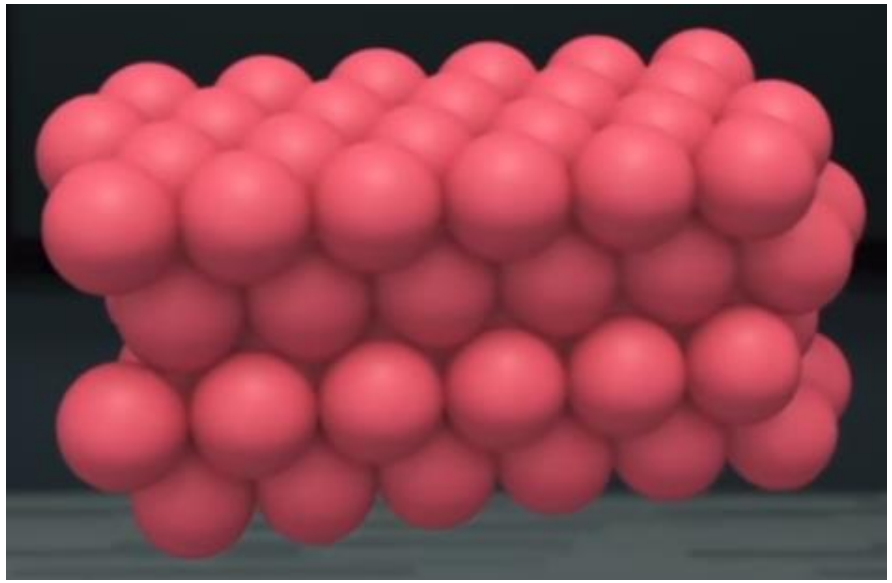
C.No = 6

(has AAA type of arrangement)

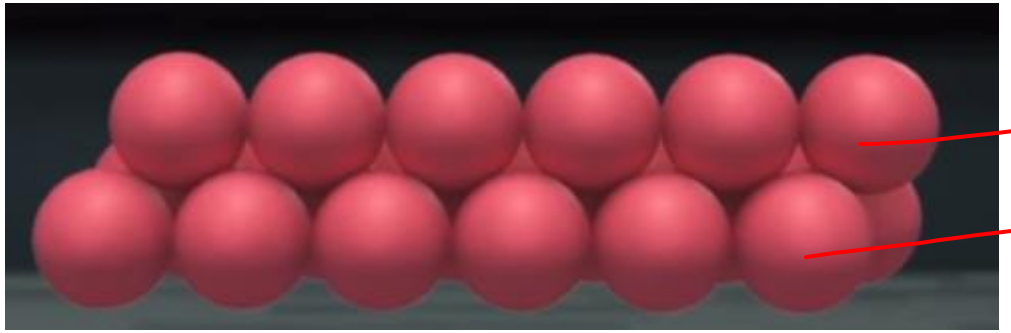


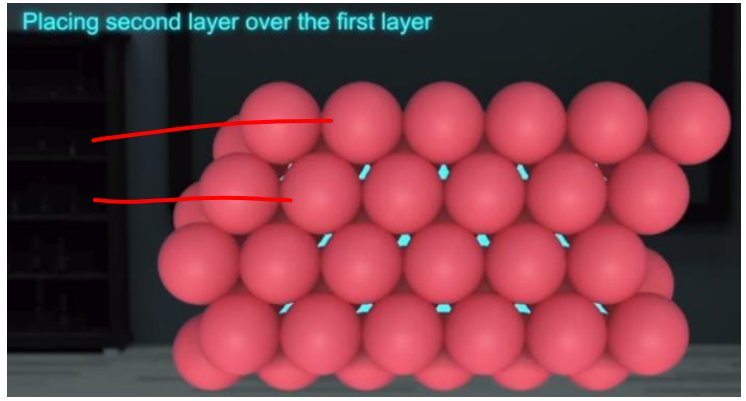




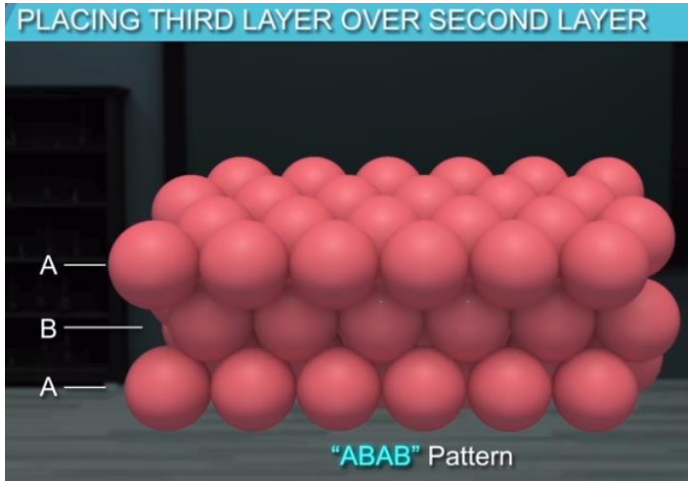


Placing second layer over first layer



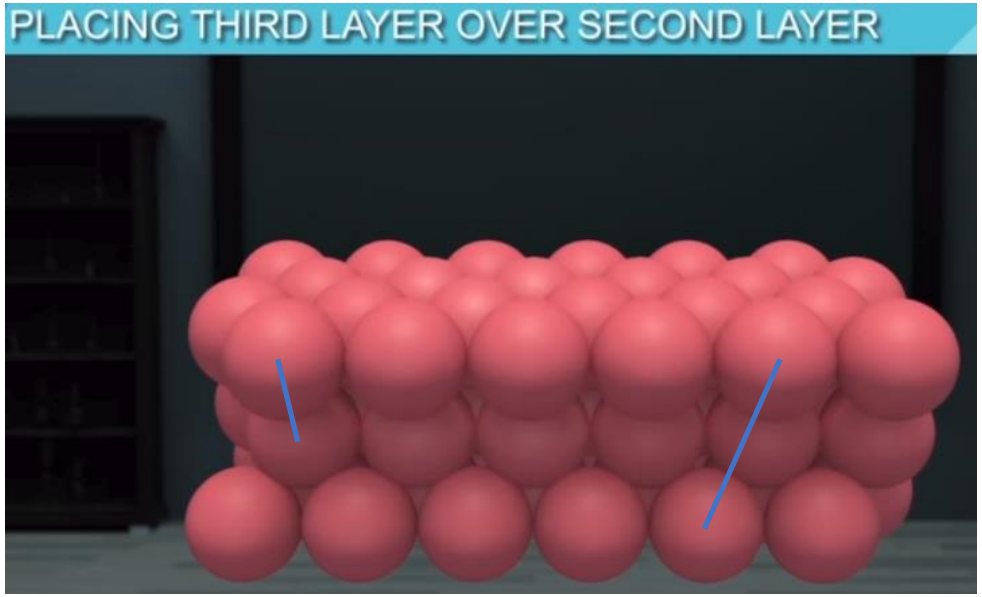


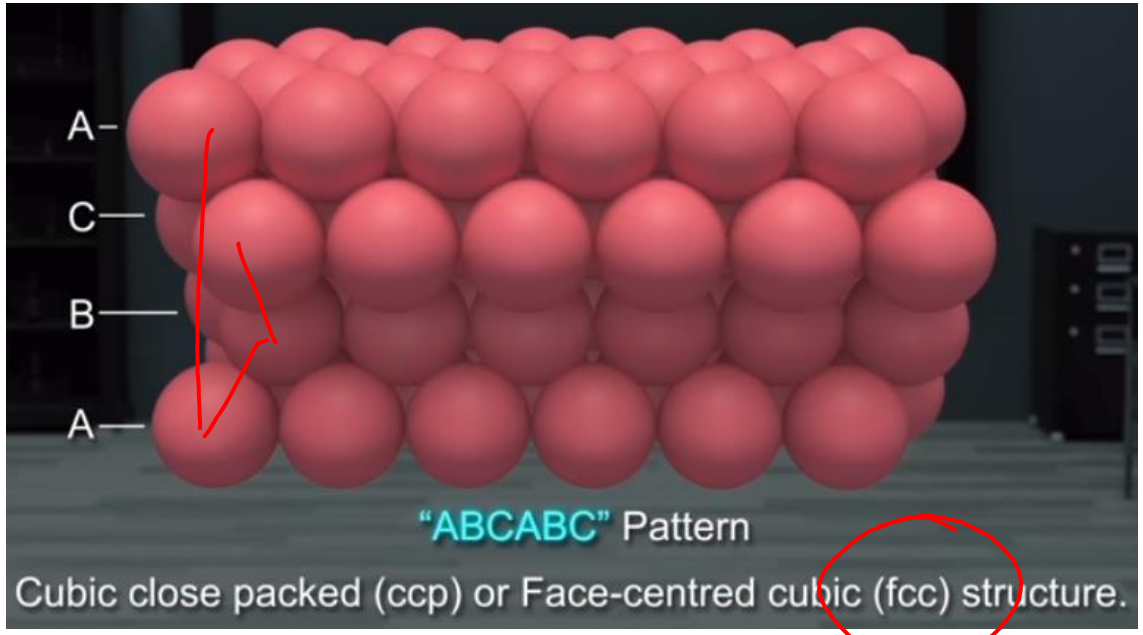
Top view

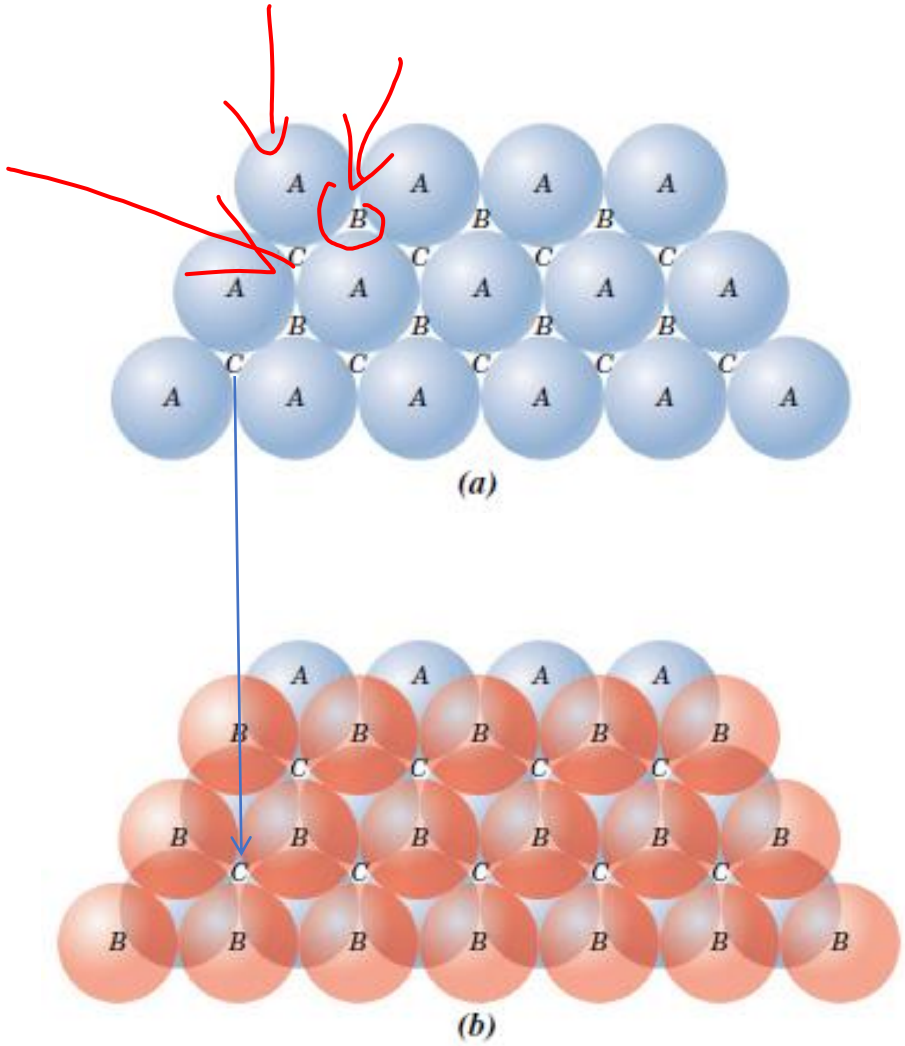


(Front view)

HCP structure


PLACING THIRD LAYER OVER SECOND LAYER

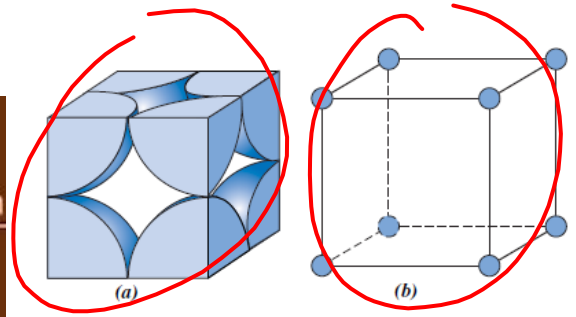

A
C
B
A
"ABCABC" Pattern
Cubic close packed (ccp) or Face-centred cubic (fcc) structure.


(a)
(b)

Simple Cubic Structure

$$APF = \frac{\text{Number of Atoms per Unit Cell} \times \text{Volume Occupied by One Atom}}{\text{Total Volume of the Unit Cell}}$$

Co-ordination Number = 6



$$\frac{\text{Number Of Corner Atoms}}{\text{Unit Cell}} = \frac{1}{8} \times 8$$

$$APF = \frac{\frac{1}{8} \times 8 \times \text{Volume Occupied by One Atom}}{\text{Total Volume of the Unit Cell}}$$

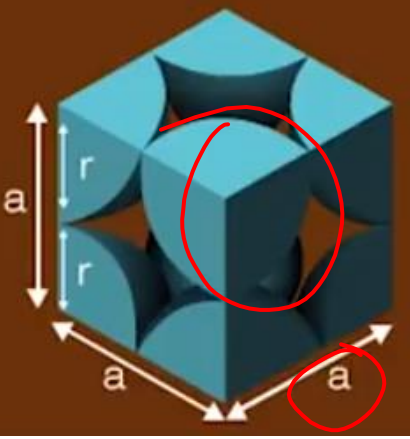
$$APF = \frac{1 \times \frac{4}{3} \times \pi \times \left(\frac{a}{2}\right)^3}{a^3}$$

$$APF = \frac{1 \times \frac{4}{3} \pi \frac{a^3}{2}}{a^3} = \frac{1 \frac{4}{3} \pi \frac{a^3}{8} 2}{a^3} = \frac{\pi}{6}$$

$$APF = 0.52$$

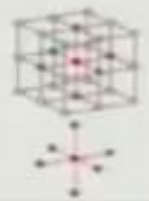
$$APF = \frac{1 \times \frac{4}{3} \times \pi \times \left(\frac{a}{2}\right)^3}{\text{Total Volume of the Unit Cell}}$$

$$r = \frac{a}{2}$$



Simple Cubic Structure (SCC)

- ✓ The unit cell consists of **eight corner atoms** and all these corner atoms contribute only one effective atom for the lattice.
- ✓ **Coordination No.** = the number of atoms immediately surrounding a central atom in a crystal = **6**
- ✓ Rare due to poor packing (only Polonium, Po).



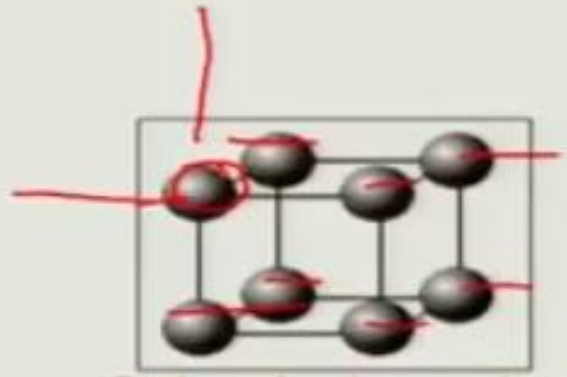
Number of atoms per unit cell, n = (1/8) x 8 = 1

$$\text{Atomic packing factor} = \frac{\textit{Volume of atoms in a unit cell}}{\textit{Total unit cell volume}}$$

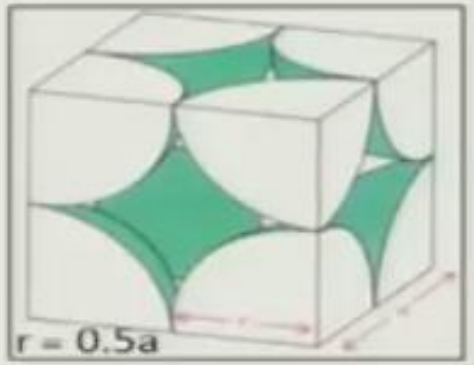
$$= \frac{(\textit{atoms/unit cell}) \; x \; (\textit{volume/atom})}{a^3}$$

$$= \frac{1 \; x \; \frac{4}{3}\pi(0.5a)^3}{a^3} = 0.52$$

- The atoms in SCC occupy 52% of the space and the rest (48%) is void/interstitial space.



Reduced-sphere unit cell



Hard –sphere unit cell

Body-Centered Cubic Structure (BCC)

- ✓ BCC structure has 8 corner atoms and 1 body center atom.
- ✓ The center atom is not shared by any of the unit cells.
- ✓ Coordination No. = 8
- ✓ **Example :** Chromium(Cr) , Iron(Fe) , Molybdenum(Mo) , Potassium(K) , Sodium(Na) , Tantalum(Ta) , Vanadium(V)

Number of atoms per unit cell, n = (1/8) x 8 + 1 = 2

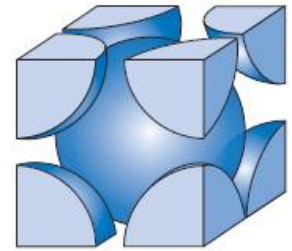
$$\text{Atomic packing factor} = \frac{\textit{Volume of atoms in a unit cell}}{\textit{Total unit cell volume}}$$

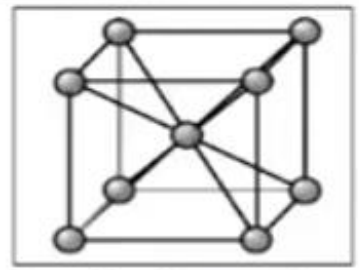
$$= \frac{(\textit{atoms/unit cell}) \times (\textit{volume/atom})}{a^3}$$

$$= \frac{2 \times \frac{4}{3}\pi\left(\frac{\sqrt{3}\,a}{4}\right)^3}{a^3} = \mathbf{0.68}$$

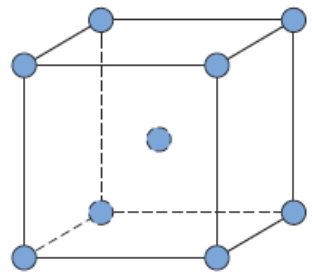
- The atoms in BCC occupy 68% of the space and the rest (32%) is void/interstitial space.
- BCC is tightly packed than simple cubic structure (52%).

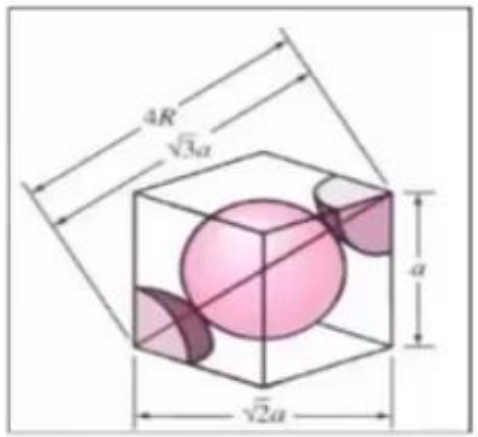
Hard –sphere unit cell





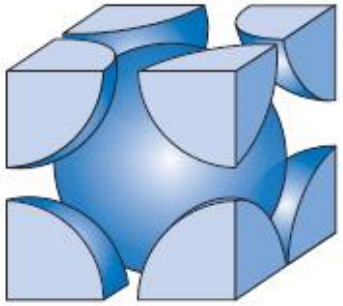
Reduced-sphere unit cell

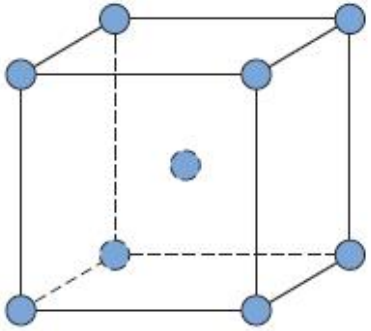


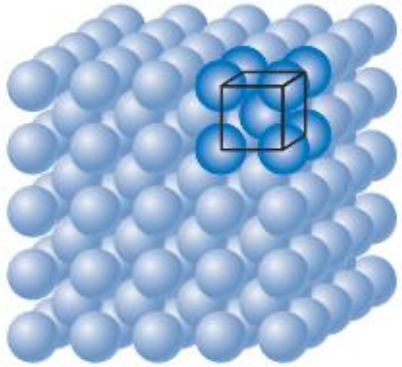


$4R = \sqrt{3}\,a$

Body Centered Cubic Structure



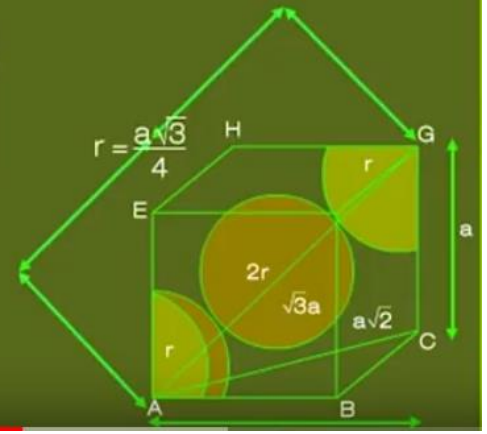




$$APF = \frac{\text{Total Number of Atoms per Unit Cell x Volume Occupied by One Atom}}{\text{Total Volume of the Unit Cell}}$$

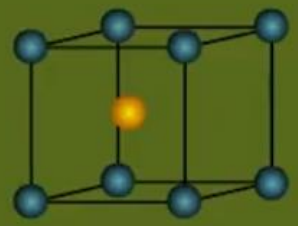
$$APF = \frac{2 \times \frac{4}{3} \quad \pi \quad \left(\frac{a\sqrt{3}}{4}\right)^3}{\text{Total Volume of the Unit Cell}} = \frac{\pi a^3 \frac{\sqrt{3}}{8}}{a^3}$$

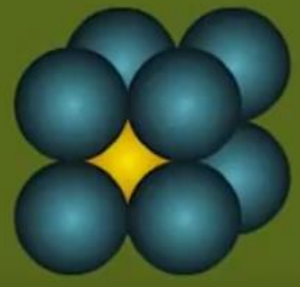
Total Volume of the Unit Cell



$$APF = \frac{\text{Total Number of Atoms per Unit Cell x Volume Occupied by One Atom}}{\text{Total Volume of the Unit Cell}}$$

$$APF = \frac{\text{(1 + Body Centered) Atoms x Volume Occupied by One Atom}}{\text{Total Volume of the Unit Cell}}$$

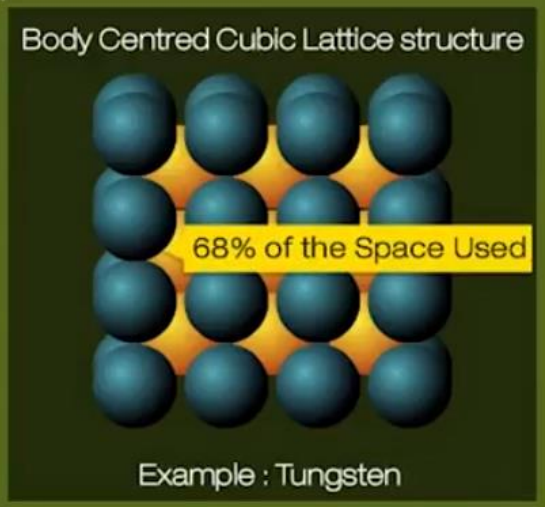




$$APF = \frac{\text{Number of Atoms per Unit Cell x Volume Occupied by One Atom}}{\text{Total Volume of the Unit Cell}}$$

$$APF = \frac{2 \times \frac{4}{3} \quad \pi \quad \left(\frac{a\sqrt{3}}{4}\right)^3}{\text{Total Volume of the Unit Cell}} = \pi \frac{\sqrt{3}}{8}$$

$\therefore$ APF for BCC = 0.68



Simple Cubic
Body Centred Cubic
52%
68%
<
The APF for a BCC is greater than SC

Face-Centered Cubic Structure (FCC)

- ✓ FCC structure has 8 corner atoms and each **face has one center** atom.
- ✓ Coordination No. = 12
- ✓ **Example :** Aluminum(Al), Copper(Cu), Gold(Au), Lead(Pb), Nickel(Ni), Platinum(Pt) , Silver(Ag)

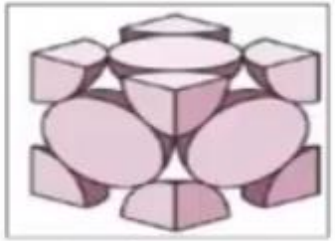
Number of atoms per unit cell, n = (1/8) x 8 + 6 x (1/2) = 4

Atomic packing factor = $\frac{Volume\ of\ atoms\ in\ a\ unit\ cell}{Total\ unit\ cell\ volume}$

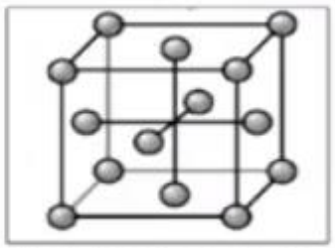
$$= \frac{(atoms/unit\ cell)\ x\ (volume/atom)}{a^3}$$

$$= \frac{4 \times \frac{4}{3}\pi\left(\frac{\sqrt{2}\,a}{4}\right)^3}{a^3} = \frac{\pi}{3\sqrt{2}} = \mathbf{0.74}$$

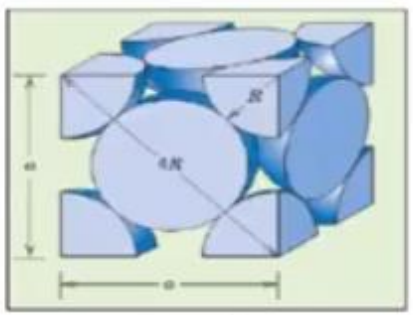
- The atoms in BCC occupy 74% of the space and the rest (26%) is void/interstitial space.
- FCC is tightly packed than BCC (68%).



Hard –sphere unit cell

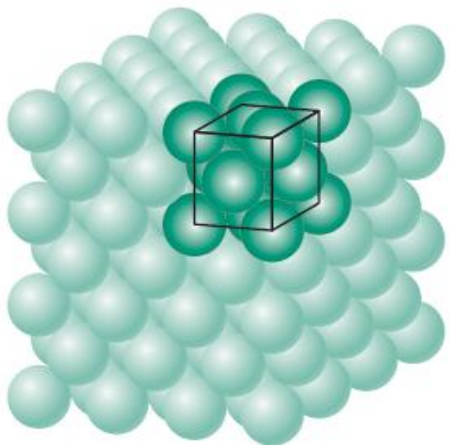


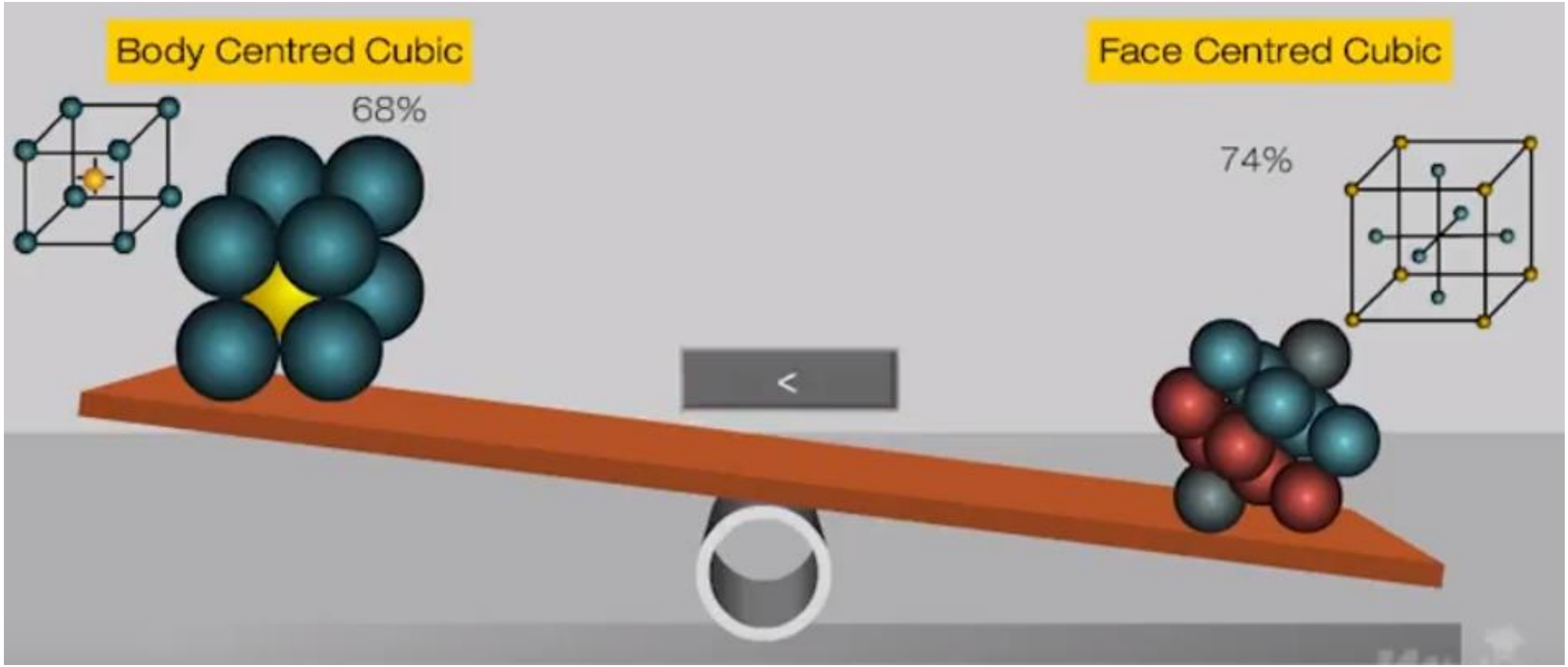
Reduced-sphere unit cell



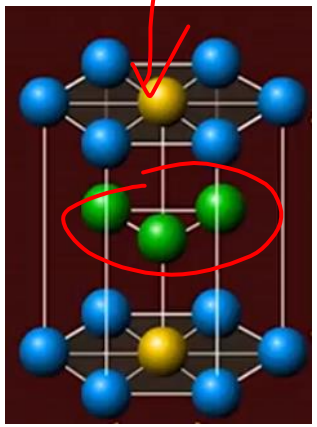
$4R = \sqrt{2}\,a$

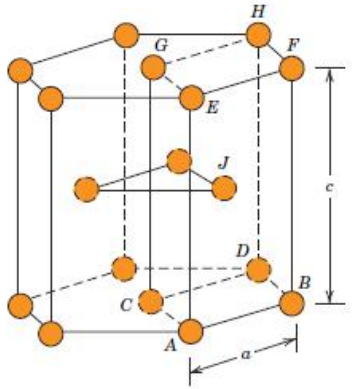
FCC

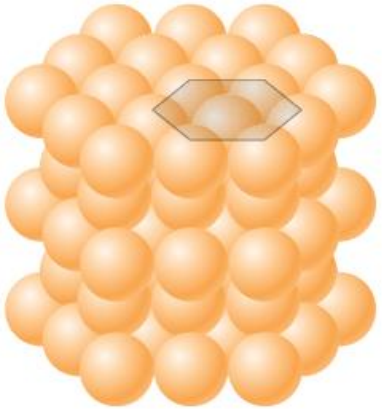



Body Centred Cubic
68%
Face Centred Cubic
74%
<

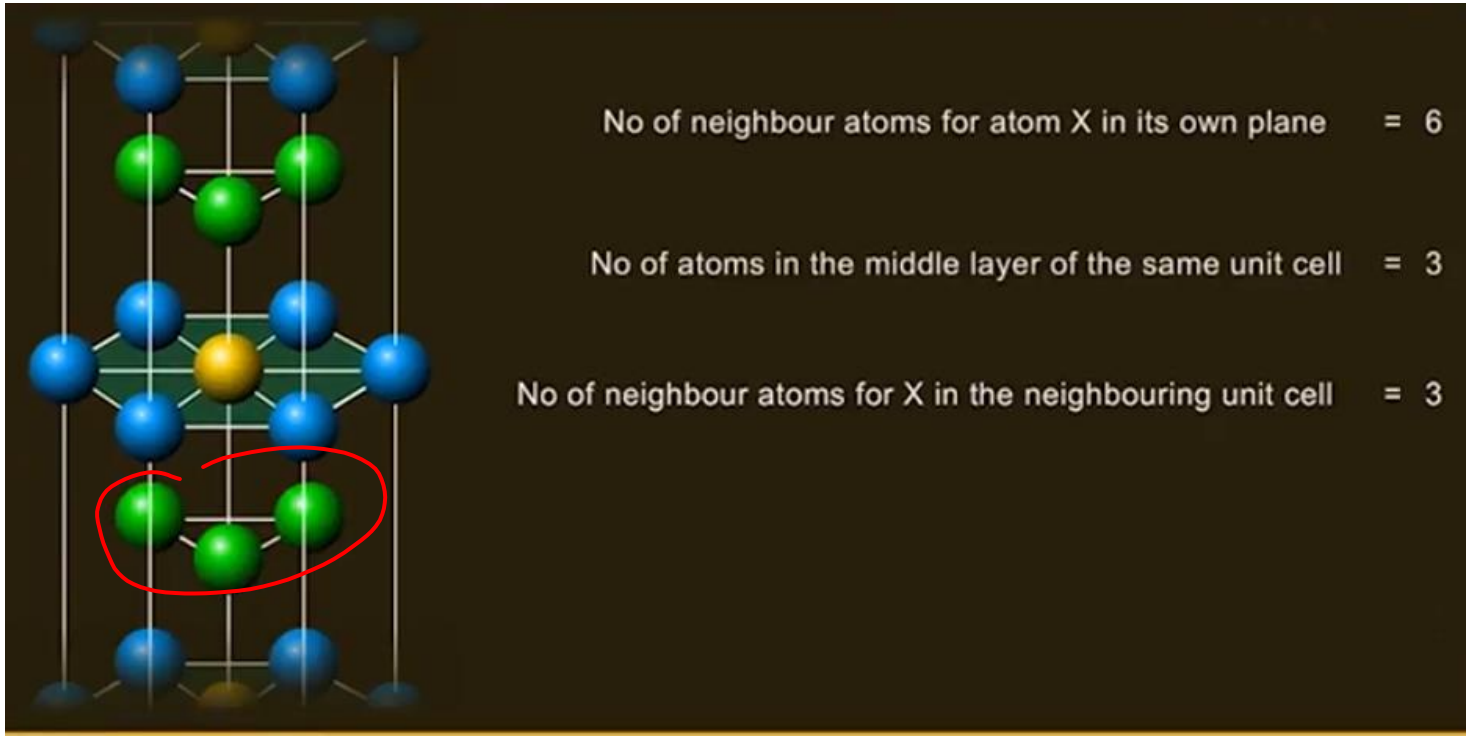
Hexagonal Closed Pack Structure (HCP)

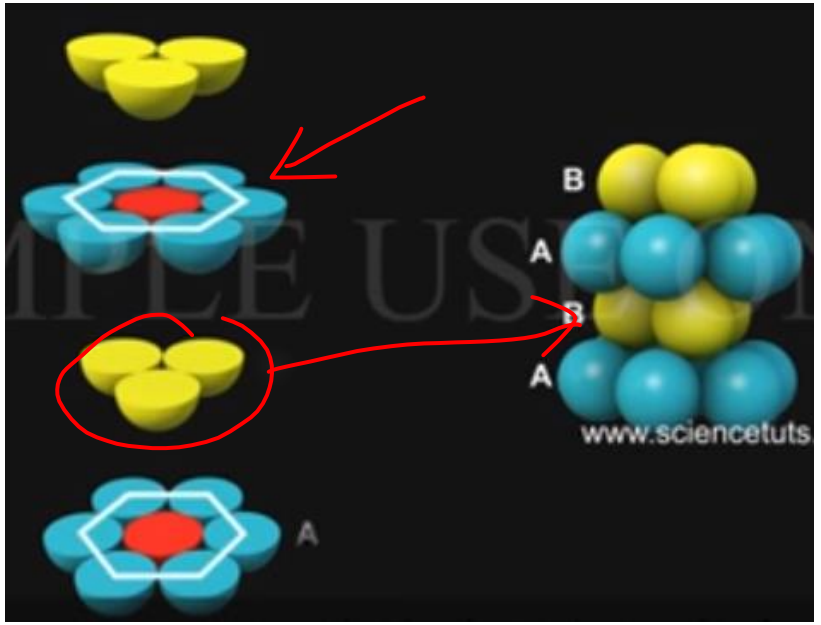


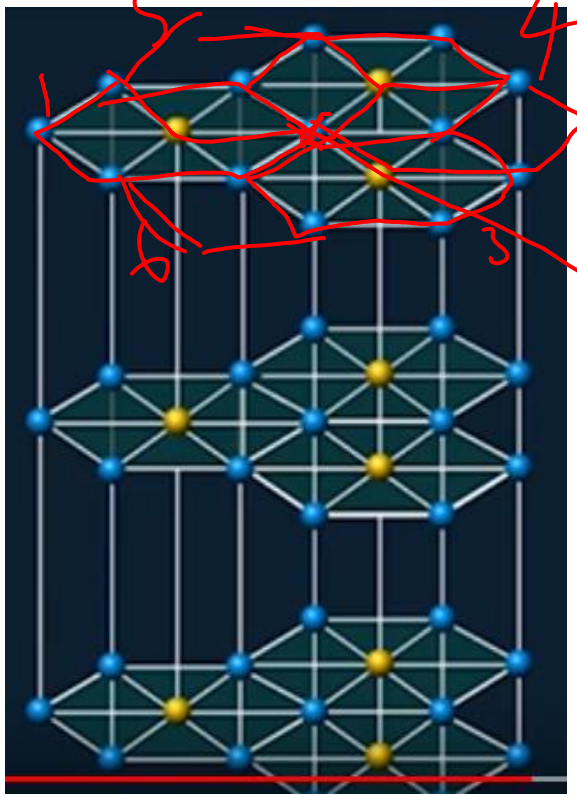




Coordination number: 6 + 3 + 3 = 12

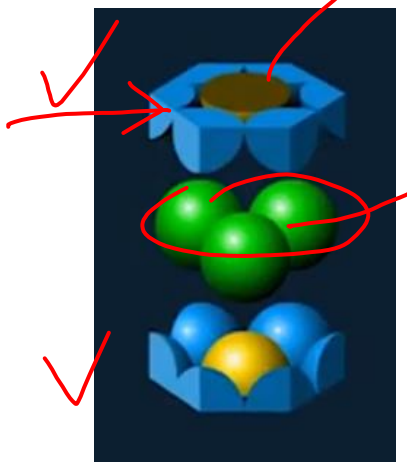






Hexagonal Closed Pack Structure (HCP)

- *Each corner atom contributes only one-sixth to a unit cell*
- *Totally there are 12 corner atoms in a unit cell*



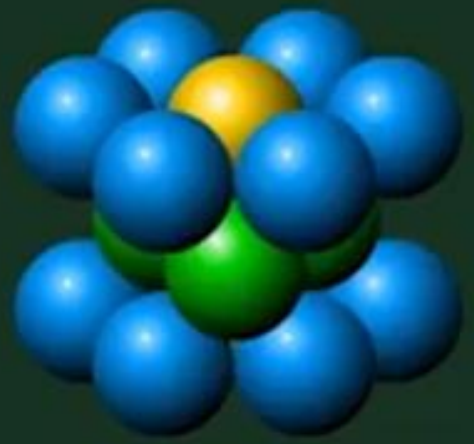
Total Number of Corner Atoms per Unit Cell $= \frac{1}{6} \times 12 = 2$

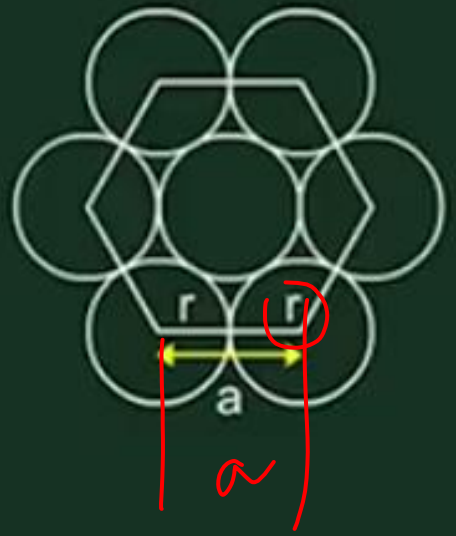
Total Number of Base Atoms per Unit Cell $= \frac{1}{2} \times 2 = 1$

Total Number of Middle Atoms per Unit Cell $= 3$

Total number of atoms per unit cell $= 6$

Hexagonal Closed Pack Structure (HCP)



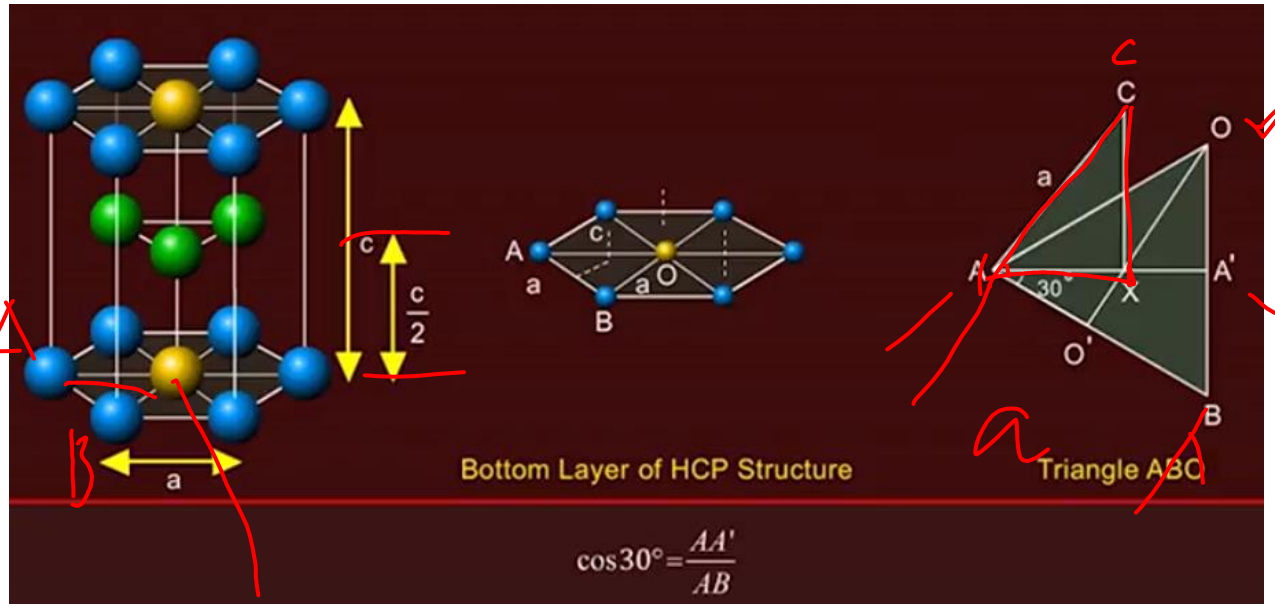


$a = 2r$ (or) Atomic Radius $r = \frac{a}{2}$

Each corner atom touches each other atom in the HCP structure.

Therefore, the distance between any two corner atoms is the atomic radius.

Hexagonal Closed Pack Structure (HCP)



$$\therefore AA' = AB\cos 30° = \frac{a\sqrt{3}}{2}$$

But, $AX = \frac{2}{3}AA' = \frac{2}{3}a\frac{\sqrt{3}}{2} = \frac{a}{\sqrt{3}}$

In the triangle AXC, $AC^2 = AX^2 + CX^2$

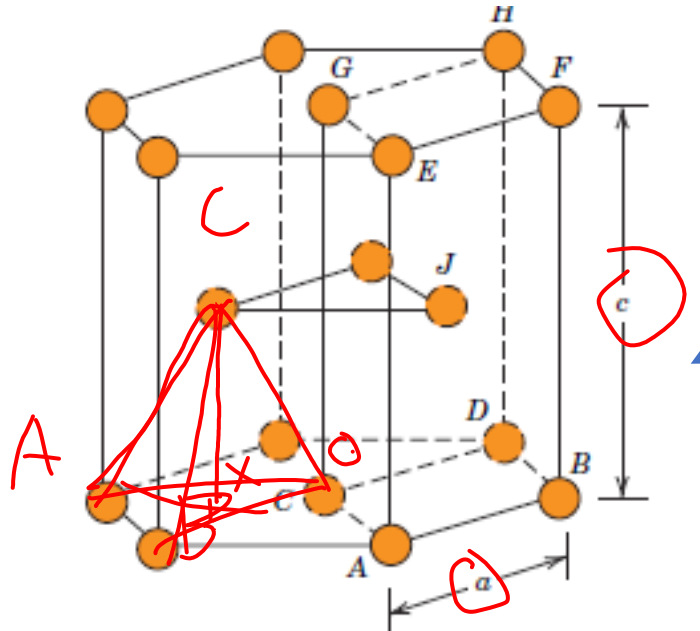
$$\therefore CX^2 = AC^2 - AX^2 = a^2 - \frac{a^2}{3} = \frac{2}{3}a^2$$

But, $CX = \frac{c}{2}$

$$\therefore \left(\frac{c}{2}\right)^2 = \frac{2}{3}a^2$$

$$(\text{or}) \quad \frac{c}{a} = \sqrt{\frac{8}{3}} = 1.633$$

Refer this.



Hexagonal Closed Pack Structure (HCP)

To find the packing fraction of HCP structure:

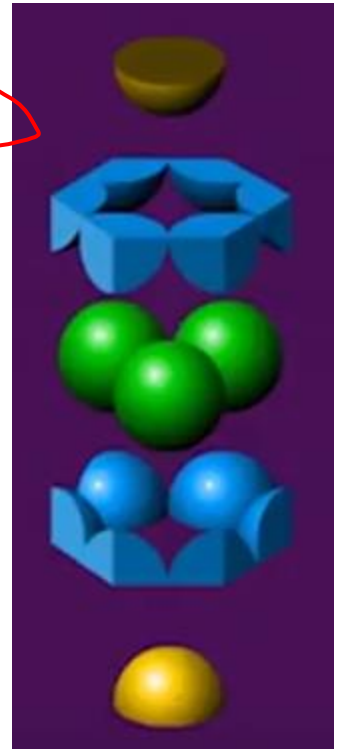
Area of the base = 6 x area of the triangle ABO.

$$= 6 \times \frac{1}{2} \times AB \times OO'$$

$$= 3 \times a \times \frac{\sqrt{3}}{2} a = \frac{3\sqrt{3}a^2}{2}$$

Hence the volume of the unit cell of the HCP structure = Area of the base $\times$C.

$$= \frac{3\sqrt{3}a^2 c}{2}$$



The total contribution from the top layer to the unit cell is $\frac{1}{6} \times 6 + \frac{1}{2} = \frac{3}{2}$ atoms.

Similarly the bottom layer also contributes the same $\frac{3}{2}$ atoms.

The central layer contributes 3 atoms fully because all the central layer atoms are well inside the unit cell.

$\therefore$ Total no. of atoms per unit cell $= \frac{3}{2} + \frac{3}{2} + 3 = 6$

Hexagonal Closed Pack Structure (HCP)

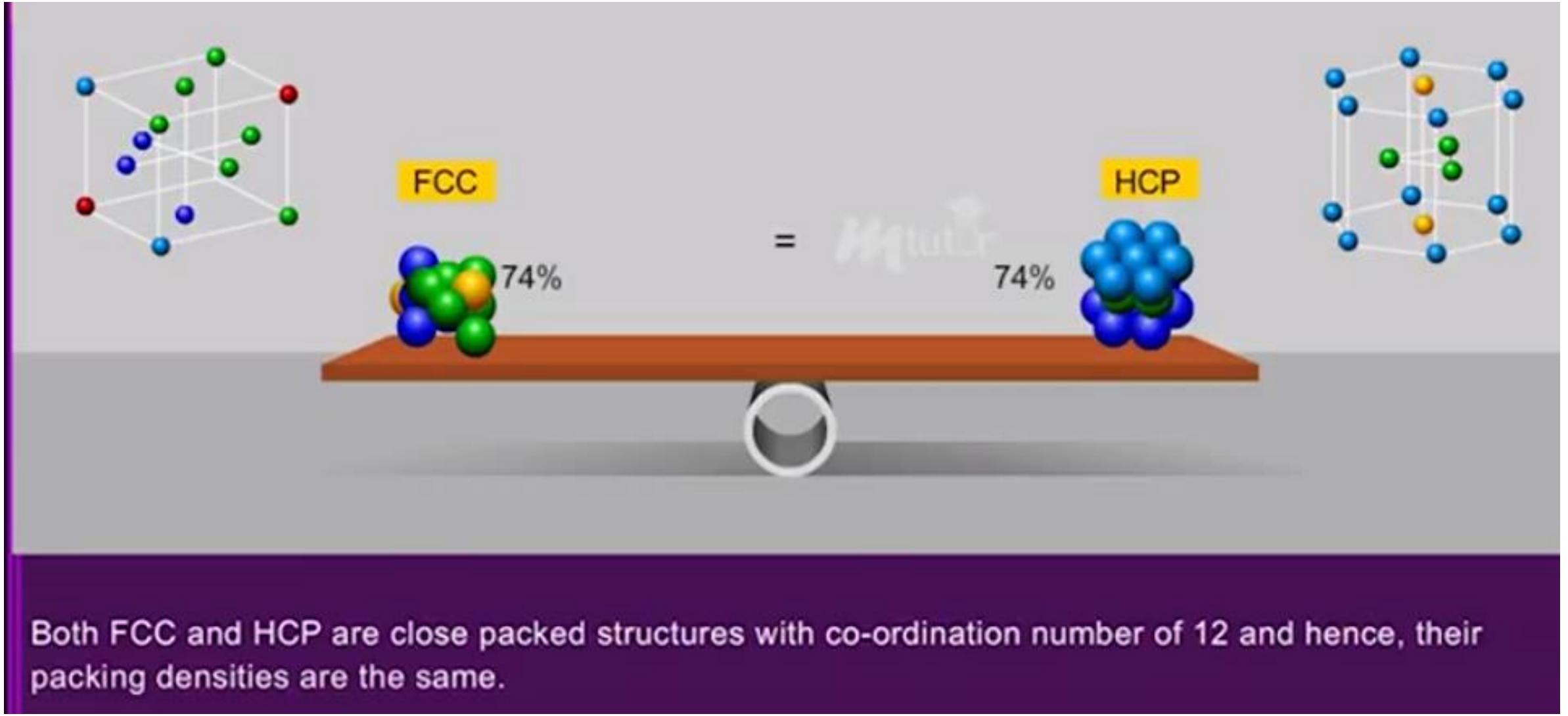
Packing fraction $= \frac{v}{V}$

$$= \frac{6\times(4/3)\pi r^3}{\frac{3\sqrt{3}a^2c}{2}}$$

Substituting $2r = a$ & $c = \sqrt{\frac{8}{3}}\,a$, we get

Packing fraction $= \frac{\pi}{3\sqrt{2}} = 0.74$

Thus the density of packing is 74% and hence it is a close packed structure.



Both FCC and HCP are close packed structures with co-ordination number of 12 and hence, their packing densities are the same.

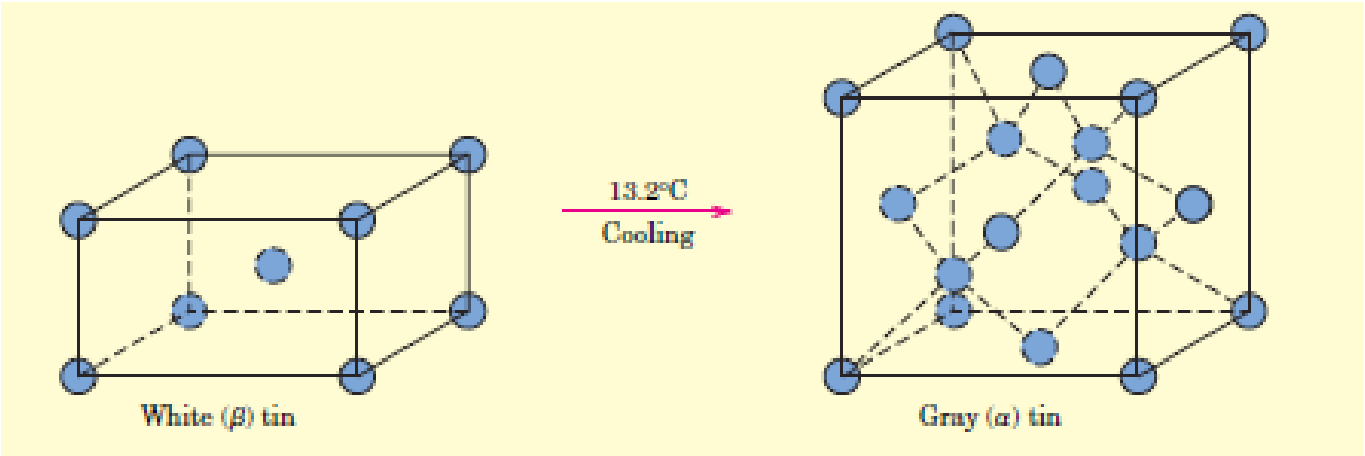
Properties of some crystal structures					
Sl. No.	Properties	Simple Cube S.C.	Body Centred Cube B.C.C.	Face Centred Cube F.C.C.	Hexagonal Close Packed H.C.P
1.	Unit cell volume	a^3	a^3	a^3	$\frac{3\sqrt{3}a^2c}{2}$
2.	No. of atoms per unit cell	1	2	4	6
3.	Co-ordination number	6	8	12	12
4.	Nearest neighbour distance (2r)	a	$\frac{\sqrt{3}a}{2}$	$\frac{a}{\sqrt{2}}$	a
5.	Density of packing	0.52	0.68	0.74	0.74
6.	Examples	Polonium	Iron, Barium, Chromium	Aluminium, Copper, Lead, Gold	Zinc, Magnesium

POLYMORPHISM AND ALLOTROPY

- Some metals, as well as nonmetals, may have more than one crystal structure, a phenomenon known as **polymorphism**

- When found in elemental solids, the condition is often termed **allotropy.**
  - ➢ Pure iron has a BCC crystal structure at room temperature, which changes to FCC iron at 912°C

- Most often a modification of the density and other physical properties accompanies a polymorphic transformation.

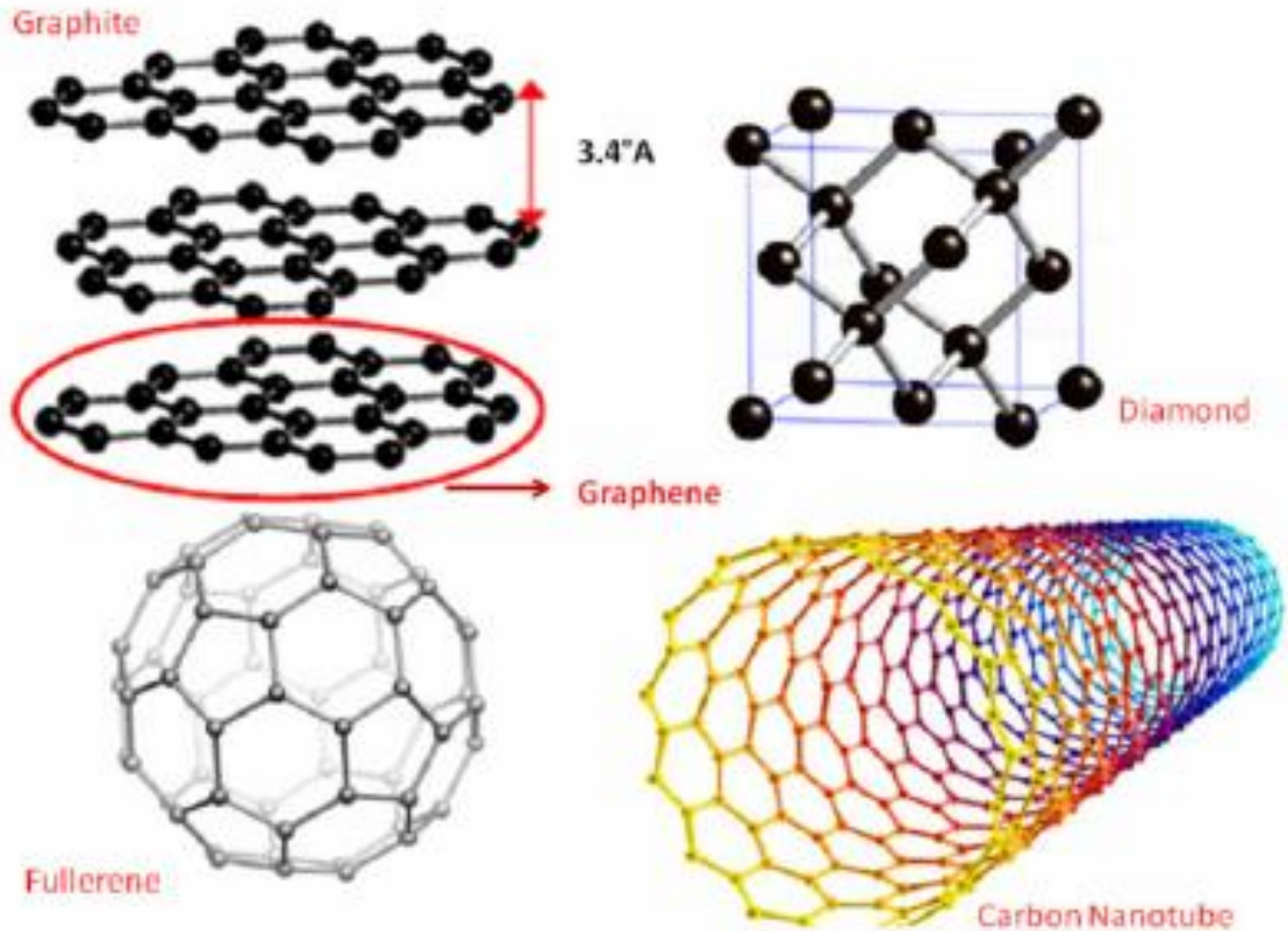


Body centered tetragonal

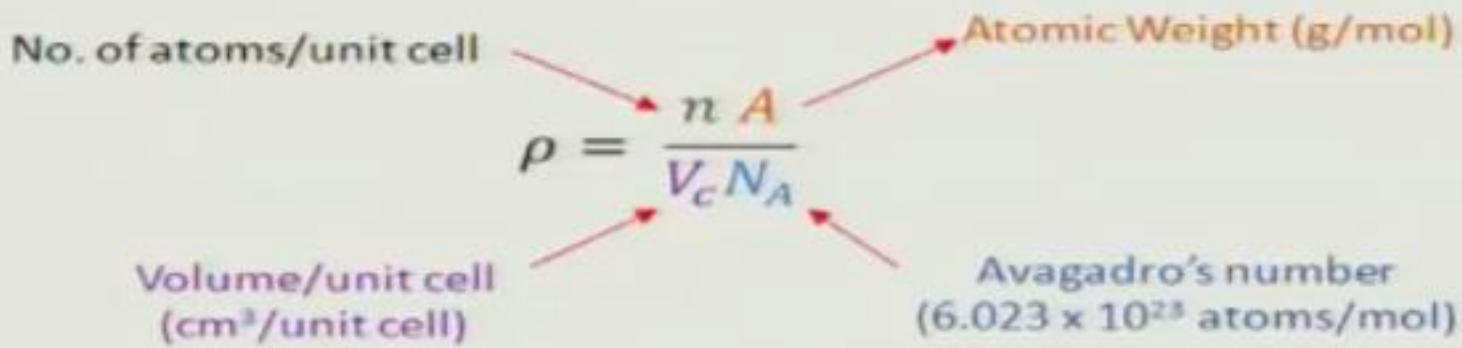
Diamond cubic crystal structure

(Accompanied by increase in volume)

Allotropes of Carbon



Theoretical Density



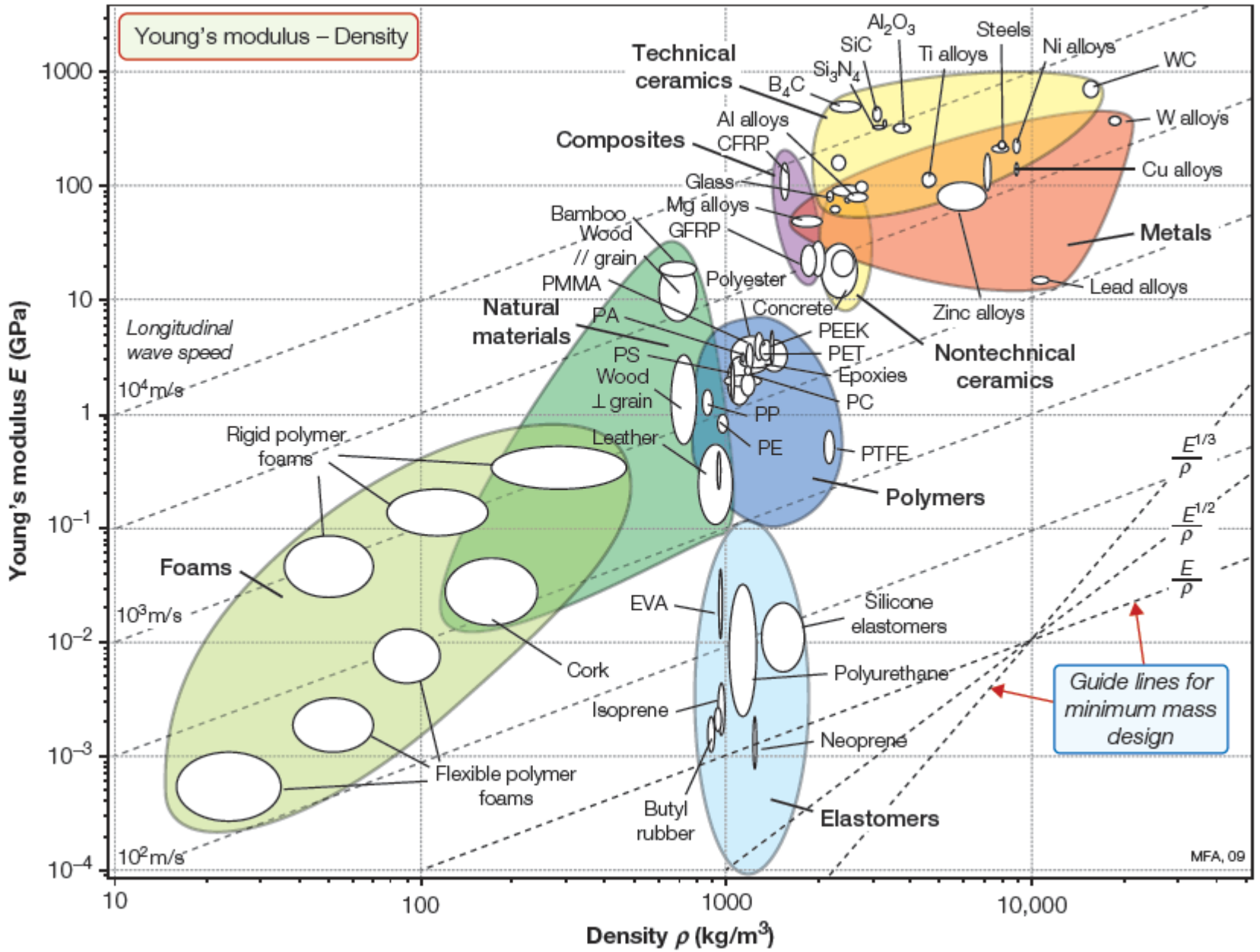
Example : Copper

- Crystal structure = FCC (4 atoms/unit cell)
- Atomic weight = 63.55 g/mol (1 amu = 1 g/mol)
- Atomic radius, R = 0.128 nm (1nm = 10^{-7} cm)
- $V_C = a^3$; For FCC : $a = 4R/\sqrt{2}$; $V_C = 4.75 \times 10^{-23}$ cm^3
- Result : Theoretical $\rho_{Cu} = 8.89$ g/cm^3

  Actual $\rho_{Cu} = 8.94$ g/cm^3

Material Property Charts

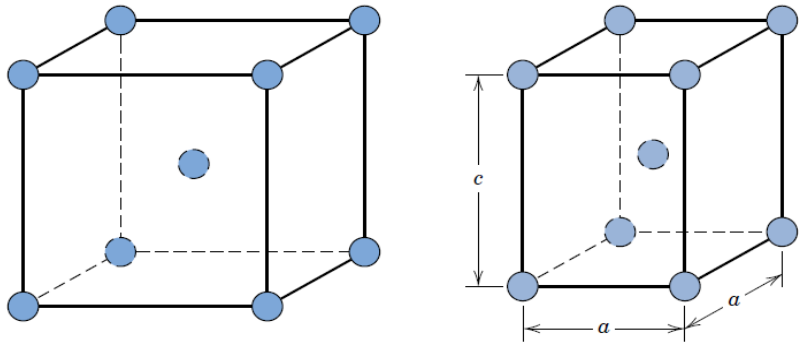
Young's Modulus vs Density



Processing/Structure/Properties/Performance?

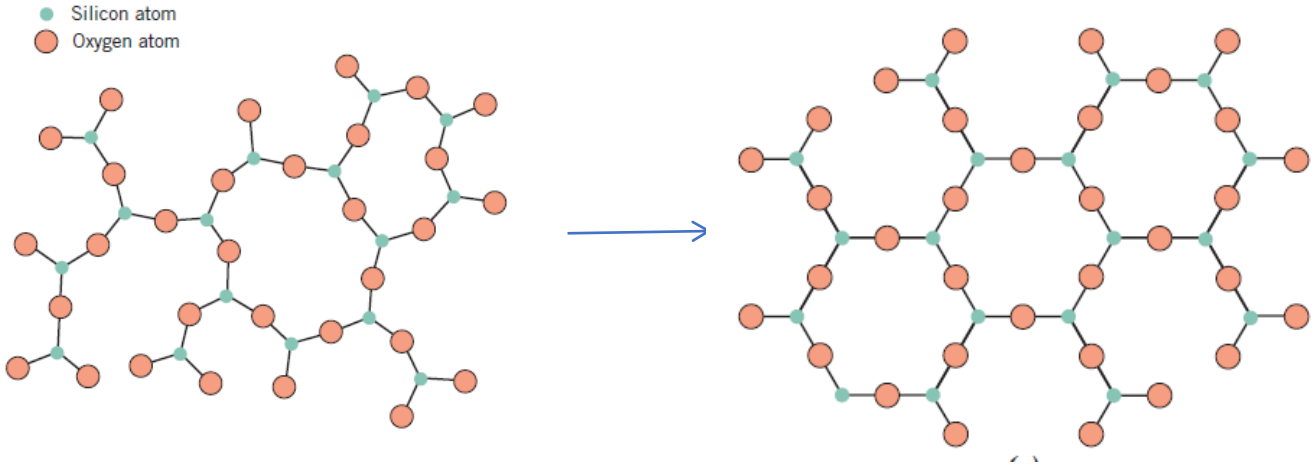
Iron–Carbon Alloys (Steels) (Structure)

Crystal structure for Fe (BCC) → **Crystal structure for martensite** (BCT)



Different properties and behaviour (High strength, brittle)

Non cyrstalline silica → Crystalline silica



Different properties and behaviour (hard, high melting point)

Significance of Crystallographic directions and planes

- ✓ **Deformation** under loading (eg. slip) occurs on certain crystalline **planes** and in certain crystallographic **directions**.
- ✓ Helps to predict **modes** of **material failure**.
- ✓ **Other properties** of materials (electrical conductivity, thermal conductivity, elastic modulus, magnetic property, piezoelectric property etc.) can vary in a crystal with **orientation**.

Example

Turbine blade

Efficient strong Ni based single crystal turbine blades withstand high temperature due to absence of grain boundaries because blades are grown on close packed direction [1 1 0] FCC Ni.

Silicon ingots and wafers in various sizes

Image : http://www.chipsetc.com/

Computer chips come from Si wafers sliced from single crystal Si grown along [1 0 0] direction.

Miller Indices

- Miller Indices are the **designation of the planes and direction** in the unit cell.
- The number of indices will match with the dimension of the lattice or the crystal. Example: In 3-D, unit cell has 3 indices namely **(h k l)**.
- **(h, k, l)** represents a point – but "no comma" is used while representing indices.

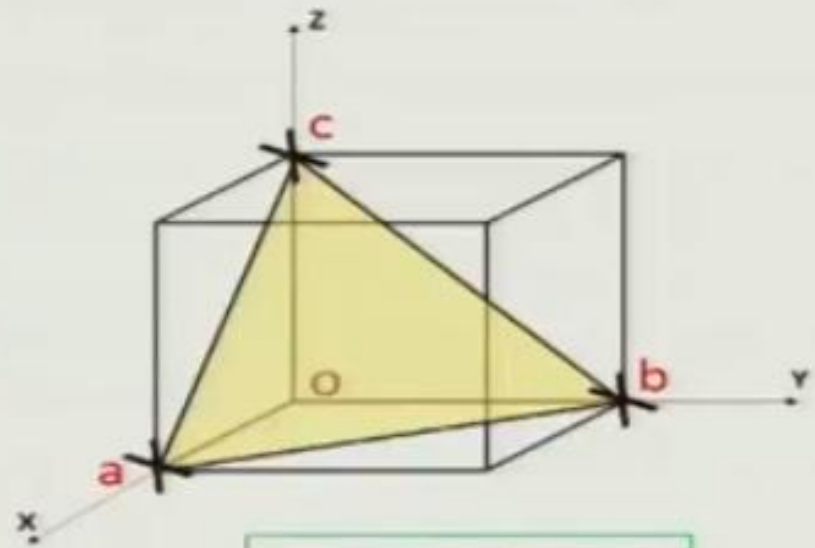
Meaning of the different brackets

Name	Symbol	Meaning
Half-moon Bracket	**( )**	Individual Plane - (1 0 0)
Curly Bracket	**{ }**	Family of Planes {1 0 0} = (1 0 0), (0 1 0), (0 0 1)
Square bracket	**[ ]**	Individual direction - [1 0 0]
Carrot Bracket	**< >**	Family of Directions <1 0 0> = [1 0 0], [0 1 0], [0 0 1]

How to find Miller Indices of a given plane?

Steps involved

- **Choose origin** O in unit cell such that
  - ✓ Plane does not touch the origin.
  - ✓ Plane is closest to origin.
- **Mark intercepts** from chosen origin O along X-axis as '**a**', from O along Y-axis as '**b**' and from O along Z-axis as '**c**'.
- **Take reciprocal** of the intercepts, i.e.,1/a, 1/b, 1/c.
- **Modify the reciprocal**
  - ✓ Reduce multiples.
  - ✓ Eliminate fraction
  - ✓ Put bar above the negative integer in indices (if any)
  - ✓ Place round bracket around indices (no commas)



Answer: (1 1 1)

How to find Miller Indices of a given plane?

Steps involved

- **Choose origin** O in unit cell such that
  - ✓ Plane does not touch the origin.
  - ✓ Plane is closest to origin.
- **Mark intercepts** from chosen origin O along X-axis as '**a**', from O along Y-axis as '**b**' and from O along Z-axis as '**c**'.
- **Take reciprocal** of the intercepts, i.e.,1/a, 1/b, 1/c.
- **Modify the reciprocal**
  - ✓ Reduce multiples.
  - ✓ Eliminate fraction
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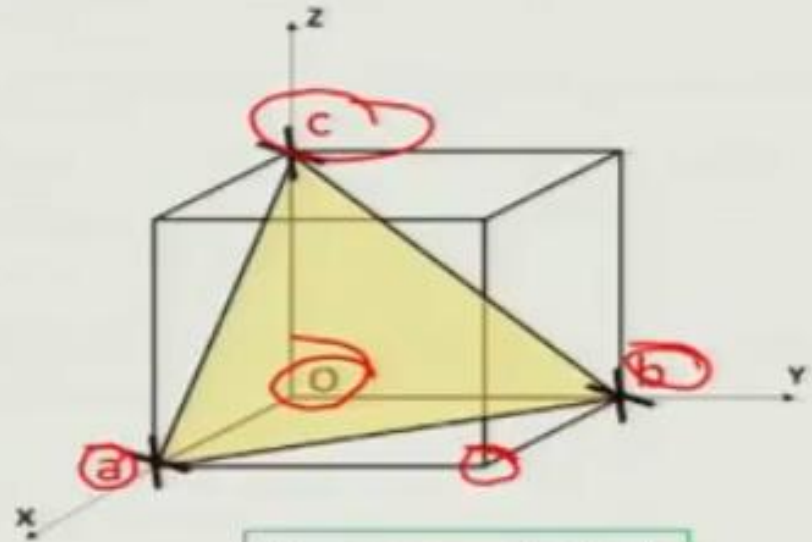
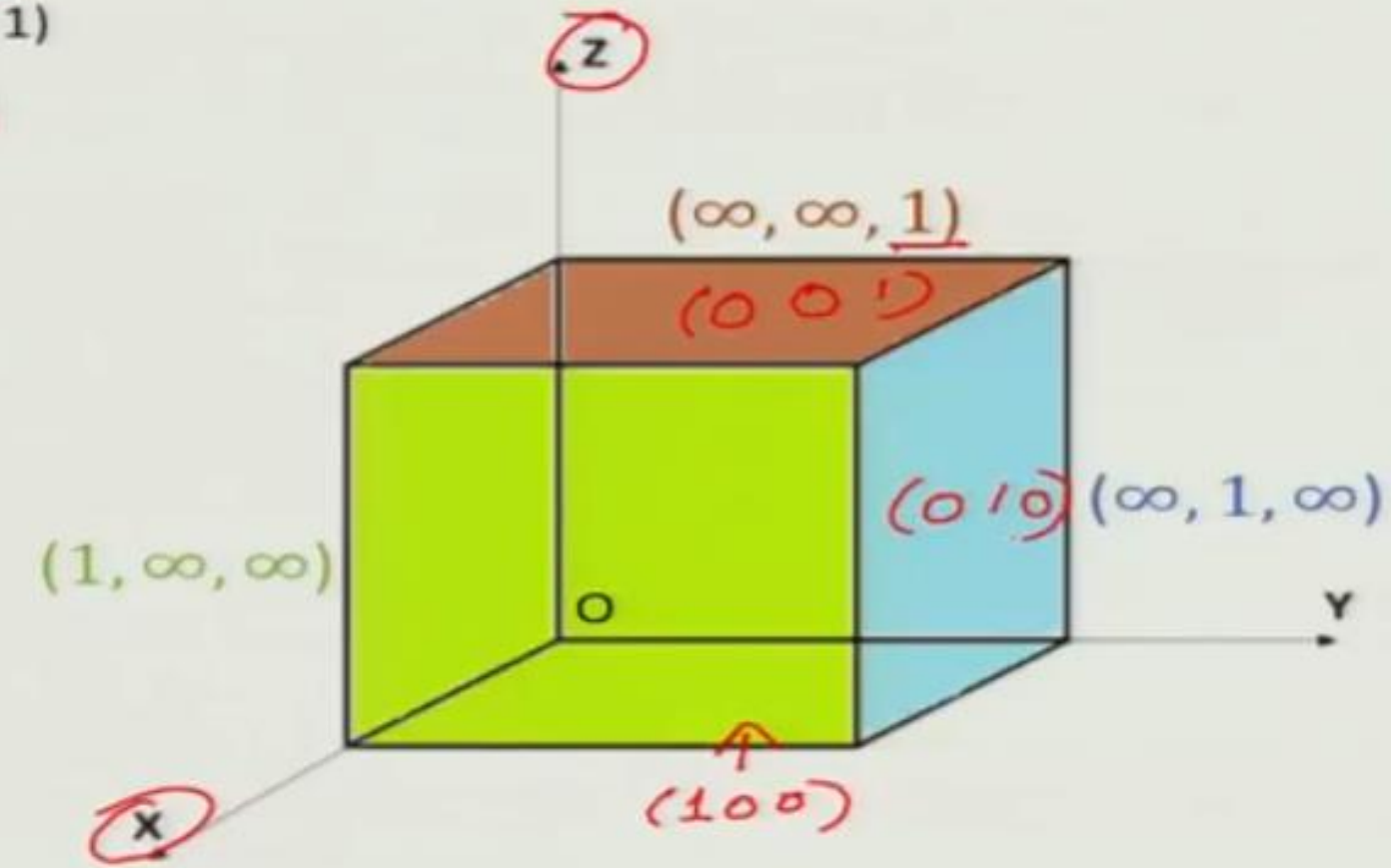


Answer: (1 1 1)

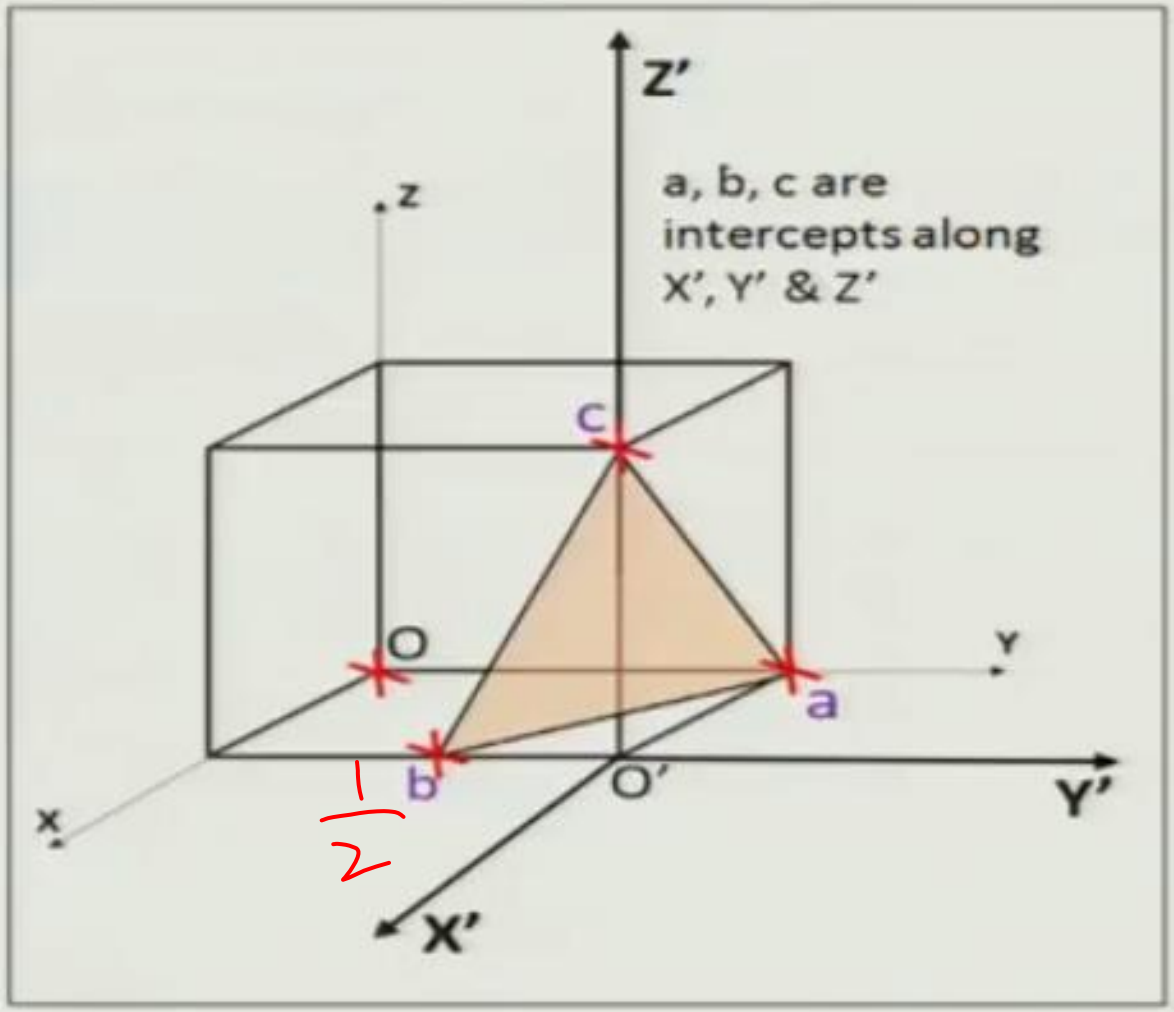
- Green Face = (1/1, 1/∞, 1/∞) = (1 0 0)
- Brown Face = (1/∞, 1/∞, 1/1) = (0 0 1)
- Blue Face = (1/∞, 1/1, 1/∞) = (0 1 0)



Determine the indices for the given plane.

- Plane cannot touch origin
- Choose the closest origin

1.Choose new Origin (O')	1	1	0
Unit cell intercepts	a	b	c
2. Intercept from origin (O')	-1	$\frac{-1}{2}$	1
3. Reciprocal value	-1	-2	1
4. Miller Indices		$(\bar{1}\,\bar{2}\,1)$	



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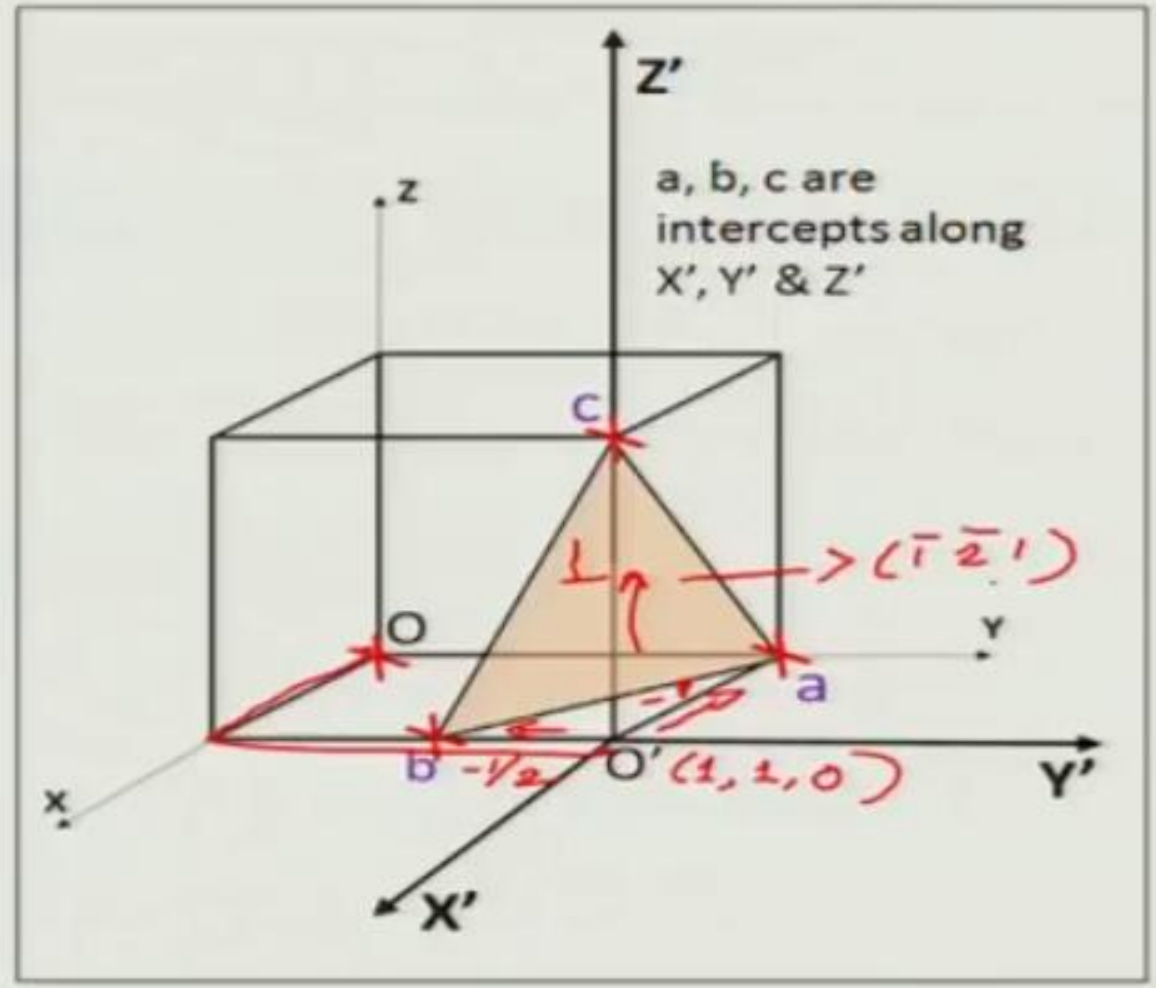
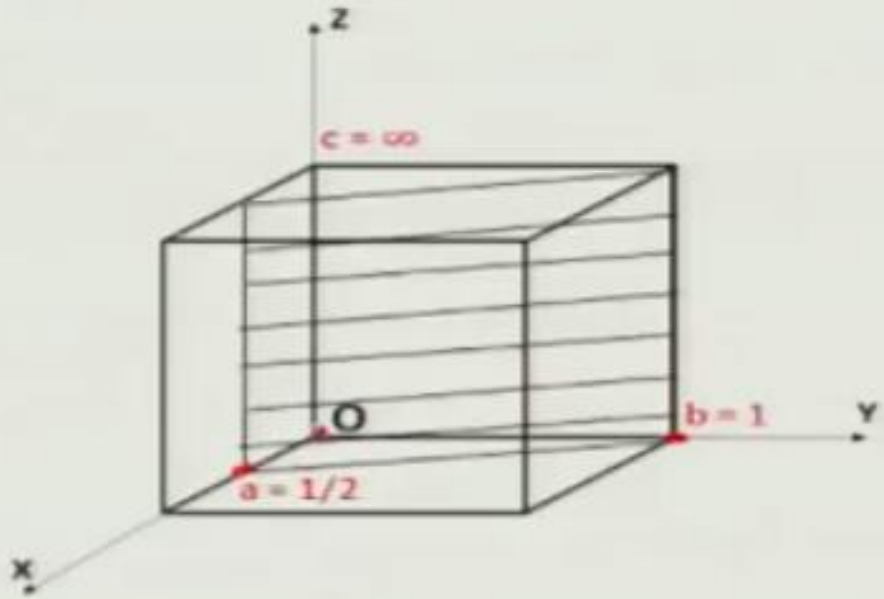


Determine the indices for the given plane.

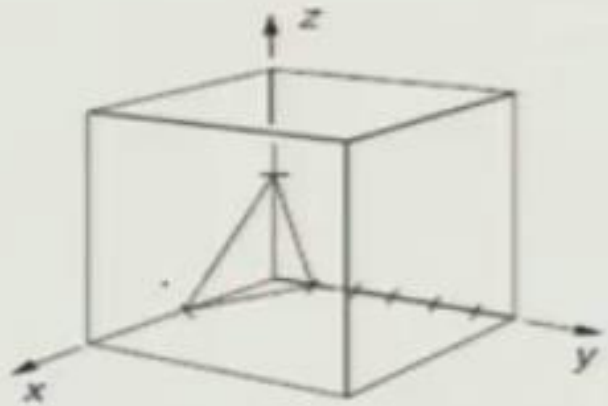
- Plane cannot touch origin
- Choose the closest origin

This plane cuts two of the reference axes(X & Y) at intercepts a & b

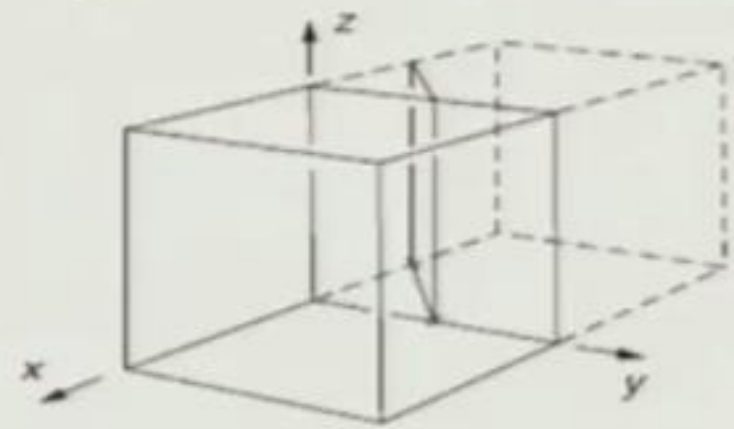
1.Choose Origin (O)	0	0	0
Unit cell intercepts	a	b	c
2. Intercept from origin (O)	$\frac{1}{2}$	1	∞
3. Reciprocal value	2	1	0
4. Miller Indices		(2 1 0)	



Miller indices – more examples (given plane)



	x	y	z
Intercepts	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{2}$
Reciprocals	2	6	2
Lowest integers to give same ratio	1	3	1
Quote	(131)		

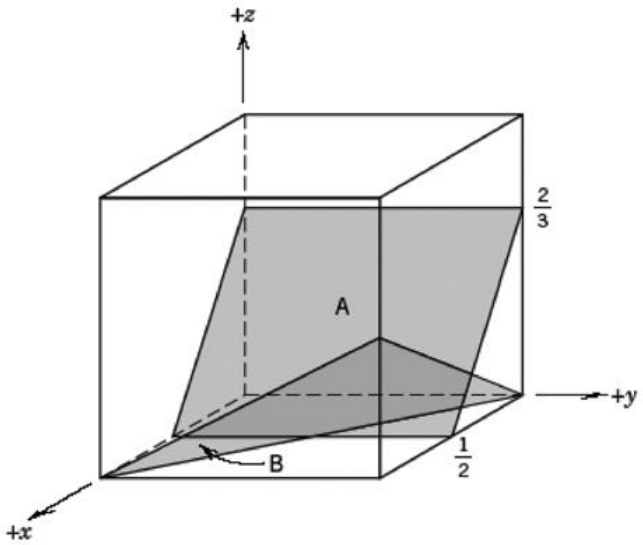


x	y	z
$\frac{1}{2}$	$\frac{1}{2}$	∞
−2	2	0

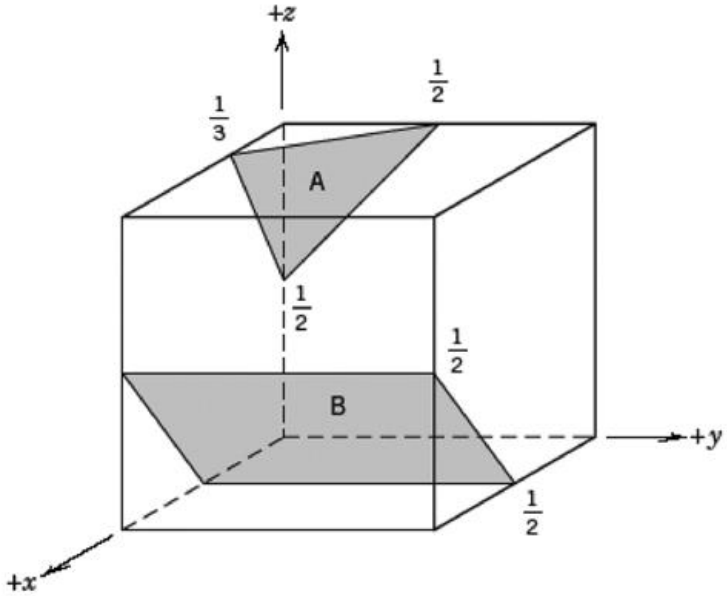
$(\bar{1}10)$

Find the Miller Indices?

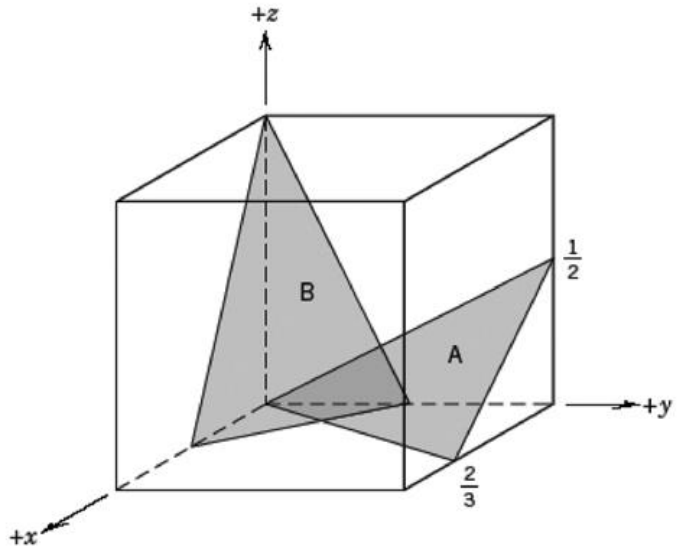
Problem 1



Problem 2



Problem 3



Materials for Engineers

Lecture 6, 7

Dr S. Gowthaman
Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

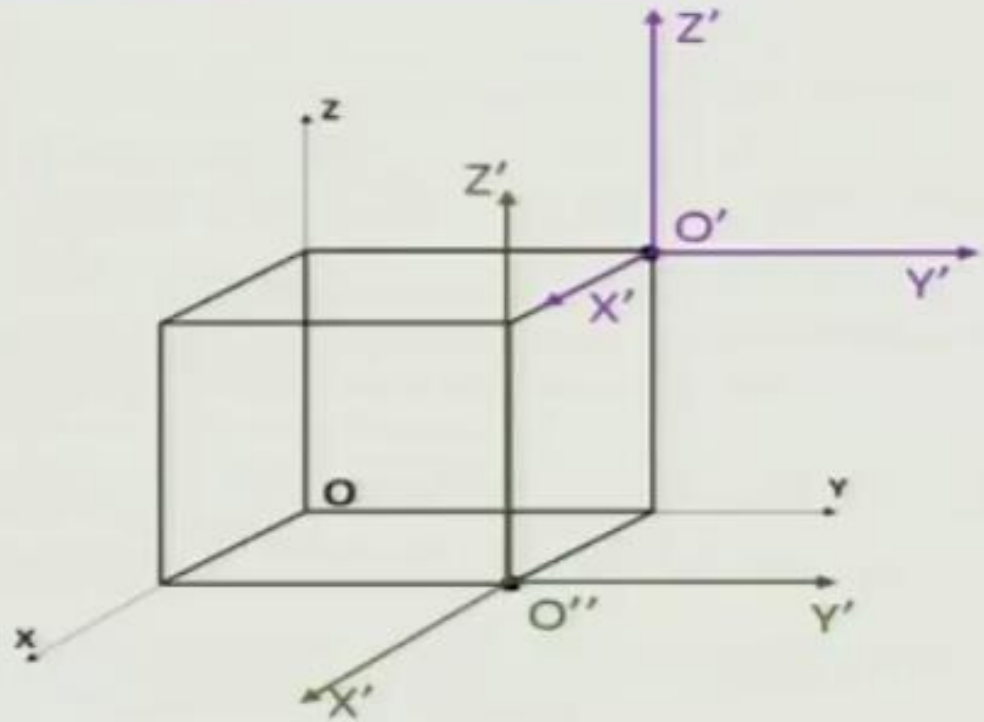
Reference:
1. NPTEL lecture on History and evolution of materials by Prof. Bishak Bhattacharya, IIT Kanpur
2. Materials science and engineering by William Callister, Wiley Publications

Selecting the Origin – Indices with bar

- If there is a bar over the number (x, y or z), the new origin O′ should have a '1' in the same place as number with a bar.
- If there is not a bar over the number, the new origin O′ should have zero '0' in the same place as the non-bar number.

Number with bar, then replace $\bar{N} = 1$
If no bar over the number, then N = 0

Indices	New Origin
(0 1 2)	X = 0, Y = 0, Z = 0 (No change)
$(0\ \bar{1}\ \bar{1})$	X′= 0, Y′= 1, Z′= 1
$(\bar{1}\ \bar{2}\ 2)$	X″= 1, Y″= 1, Z″= 0

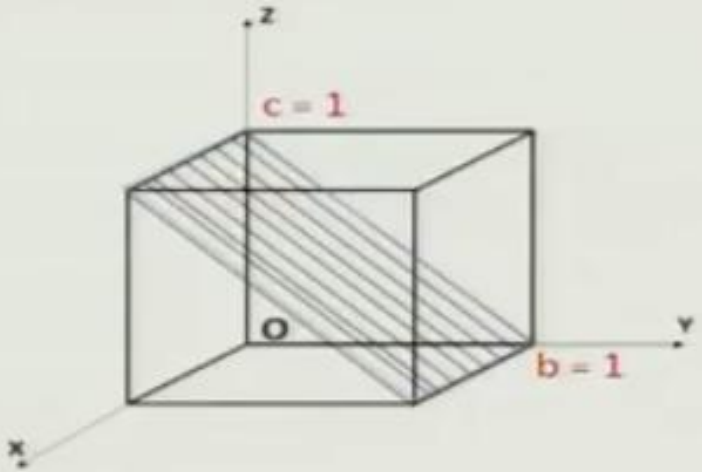


Given Indices (h k l) > Draw the Plane

- Select origin O at (0,0,0) OR if there are negative indices then origin at O' in X', Y', Z' frame.
- Take reciprocal of indices to get 1/h, 1/k, 1/l : which will be the intercept.
- Mark intercept along x, y, z axes at a, b, c.
- Draw plane by connecting intercepts (a, b, c).

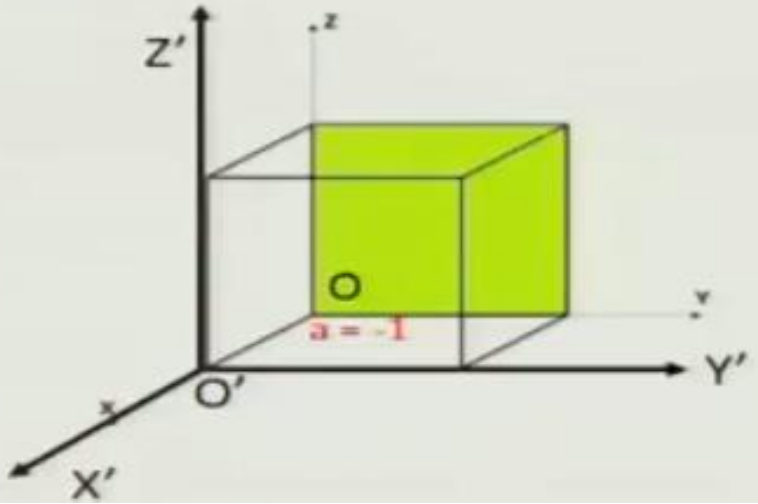
Given Indices (0 1 1) > Draw the Plane

1.Choose Origin (O)	0	0	0
2. Reciprocal value of indices (0 1 1)	∞ (parallel to x-axis)	1	1
Unit cell intercept	a	b	c
3. Mark & connect intercepts	Draw Plane		



Given Indices ($\bar{1}$ 0 0) > Draw the Plane

1.Choose Origin (O')	1	0	0
2. Reciprocal value of indices ($\bar{1}$ 0 0)	-1	∞ (parallel to Y-axis)	∞ (parallel to Z-axis)
Unit cell intercept	a	b	c
3. Mark & connect intercepts	Draw Plane		



Draw Planes?

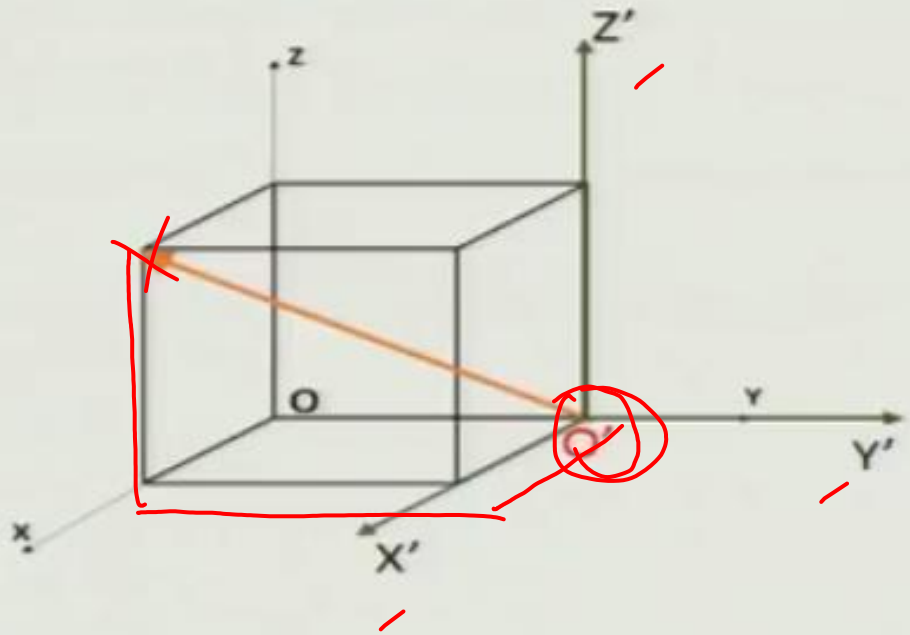
$(1\,1\,\bar{2})$ $(\bar{1}2\bar{3})$, $(1\,\bar{3}\,1)$, $(0\,\bar{1}\,\bar{3})$

How to find Miller Indices of a given direction ?

Given direction > determine **indices** [? ? ?].

- Create **new origin** O' at **tail**, if necessary.
- Label O' and new axes X', Y' & Z'.
- Find coordinates from tail at O' to the head of direction.
- Reduce multiples, eliminate fractions, use 'bar' for negative.
- Place square bracket "[ ]" around integer.

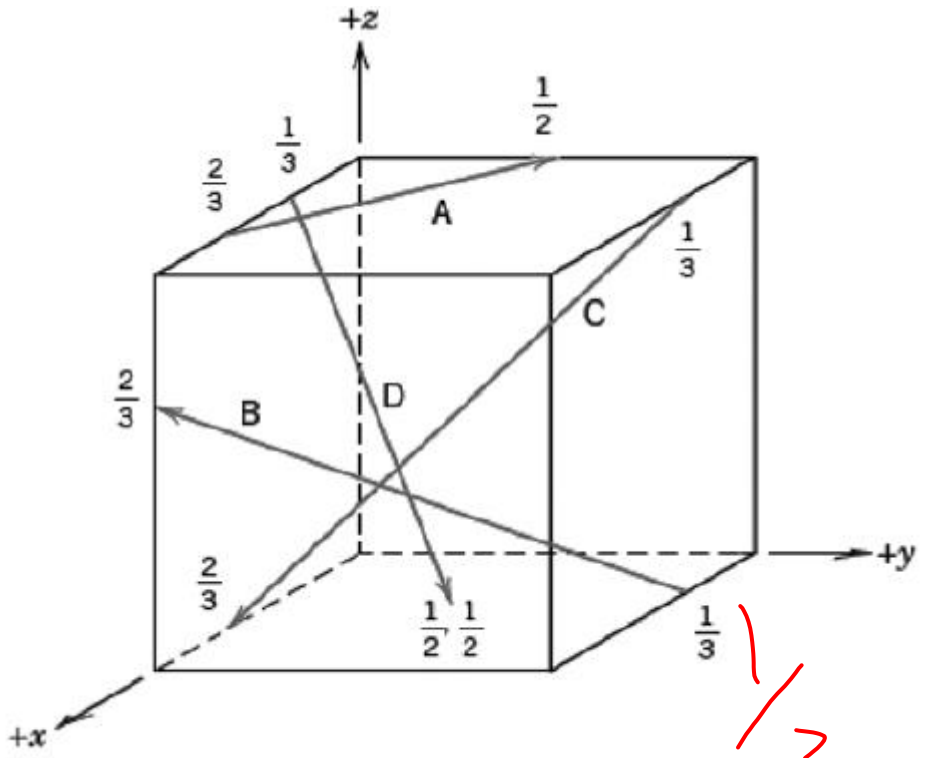
Ans : [1 $\bar{1}$ 1]



1 −1 1

[1 $\bar{1}$ 1]

Find the Miller Indices?



2/3

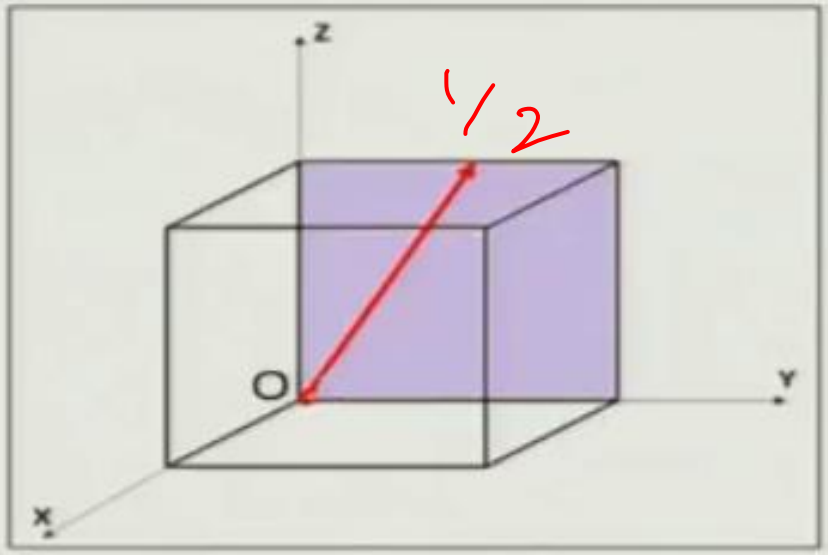
1/3

Drawing directions from Indices

Example

Given indices **[0 1 2]** > draw direction.

- **Select** origin O at (0,0,0) OR choose new O' if there are negative indices.
- **Reduce** indices to keep direction within unit cell (in this case divide by 2).
- **Locate** from origin at (0,0,0) to a position (0, ½ ,1) in the unit cell.
- **Draw** direction from tail (0,0,0) to arrow tip at (0, ½ ,1)

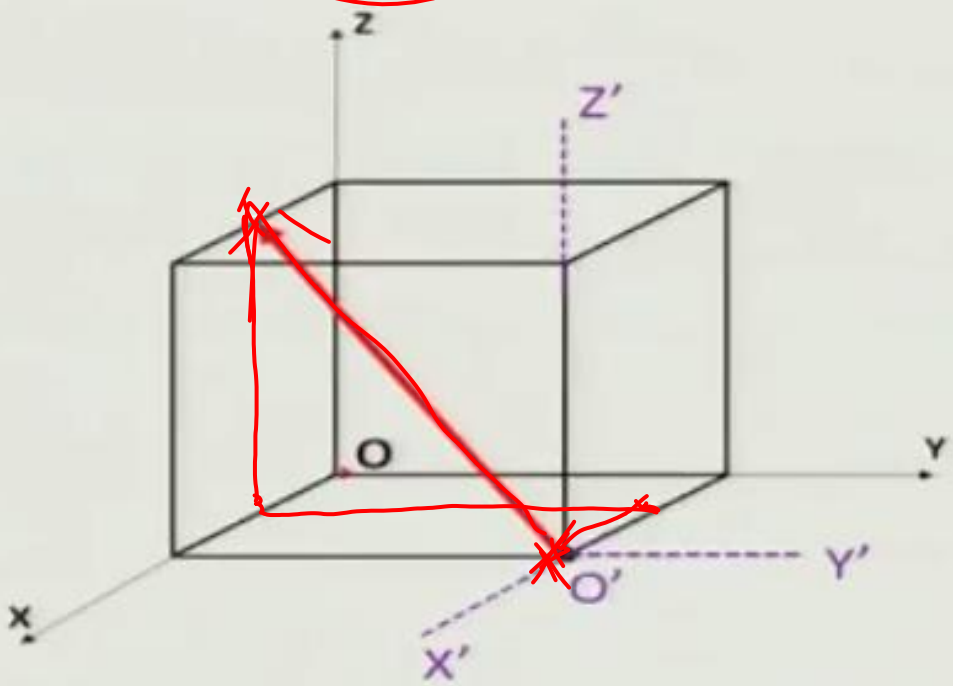


Draw the **specified direction** in the unit cell with **indices given** as [$\bar{1}$ $\bar{2}$ 2].

- Select new origin O' as (1,1,0)
- Reduce [$\bar{1}$ $\bar{2}$ 2] to -1/2, -1, 1 with in the unit cell
- Locate the coordinates.
- Draw direction from origin to the coordinates.

(1, 1, 0)

$$\frac{-1}{2} = 1 \quad 1$$



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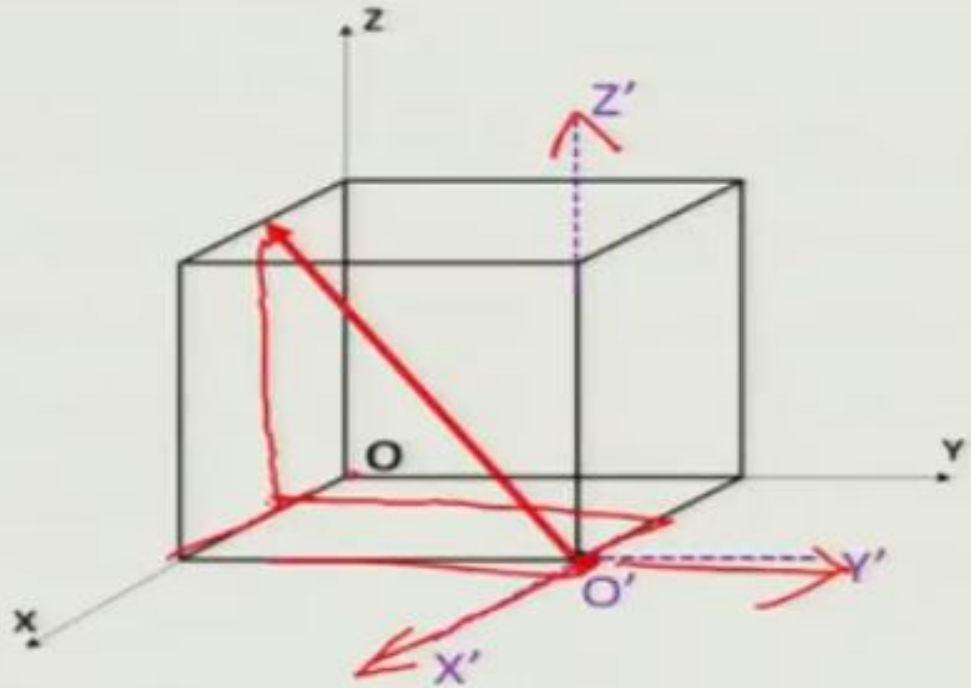
- Reduce [$\bar{1}$ $\bar{2}$ 2] to -1/2, -1, 1 with in the unit cell
- Locate the coordinates.
- Draw direction from origin to the coordinates.

[-1 -2 2]

$-\frac{1}{2}$ -1 1



Draw Directions?

$[\bar{1}10]$, $[\bar{1}\bar{2}1]$, $[0\bar{1}2]$, $[1\bar{3}3]$, $[\bar{1}22]$,

Planar Density

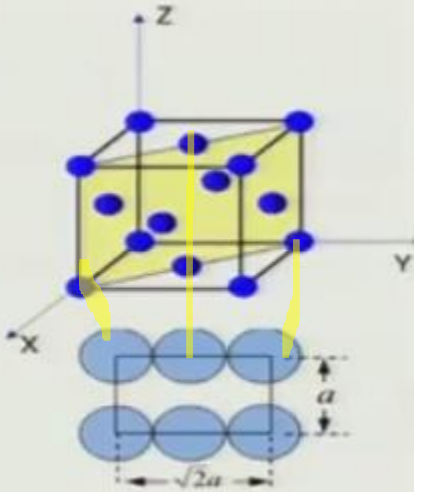
Planar density

Planar density , PD = $\frac{\text{No. of effective atoms per unit area}}{\text{Area of plane}}$

Planar density of (110) plane in the FCC crystal

No. of effective atoms per unit area = $\frac{1}{4}$ x 4 corner atoms + $\frac{1}{2}$ x 2 side atoms = 2

$$PD_{(110)} = \frac{2}{a\sqrt{2}a} = \frac{\sqrt{2}}{a^2}$$



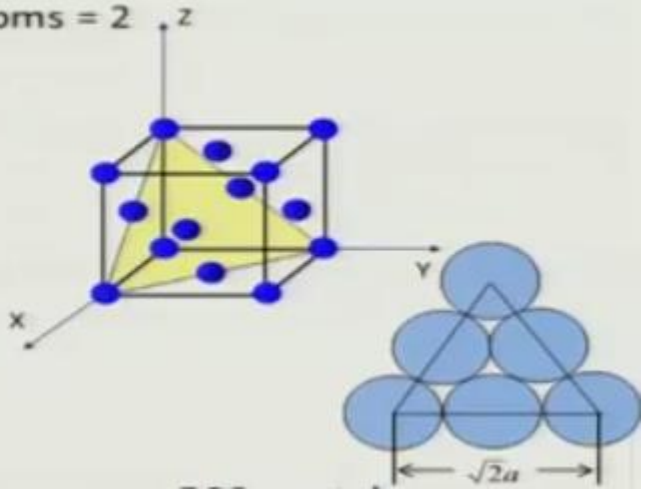
FCC crystal

Planar density of (111) plane in the FCC crystal

No. of effective atoms per unit area = $\frac{1}{6}$ x 3 corner atoms + $\frac{1}{2}$ x 3 side atoms = 2

$$PD_{(111)} = \frac{2}{\frac{1}{2}(base \times height)} = \frac{4}{\sqrt{3}a^2}$$

This is higher than (110) and any other plane. Therefore, **(111) plane** is the **most densely packed plane** in the **FCC** crystal



Linear Density

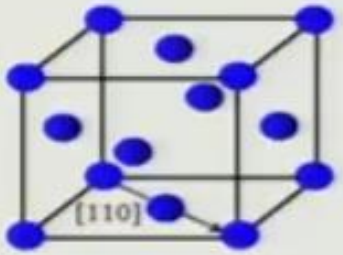
Linear density

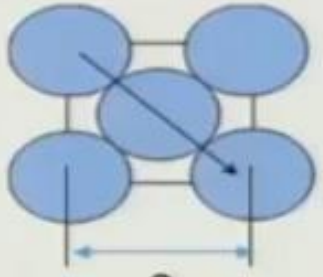
$$\text{Linear density, LD} = \frac{\text{No. of effective atoms on a line}}{\text{Length of direction vector}}$$

Linear density in [110] direction in the FCC crystal

$$LD_{[110]} = \frac{\text{No. of effective atoms on a line}}{\text{Length of direction vector}} = \frac{\frac{1}{2}+1+\frac{1}{2}}{\sqrt{2}\,a} = \frac{2}{\sqrt{2}\,a}$$

This is the most densely packed direction in the FCC lattice.





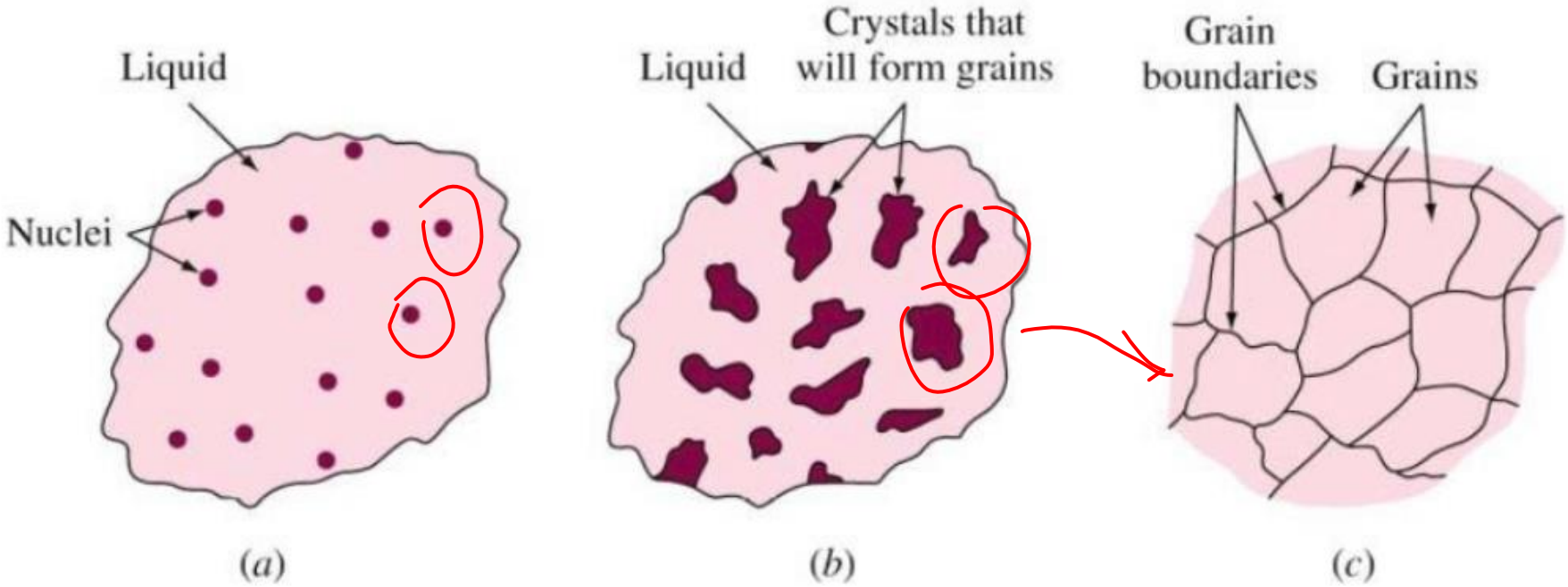
FCC crystal

Similarly for BCC crystal
Highest atomic density plane: (110)
Highest atomic density direction: [111]

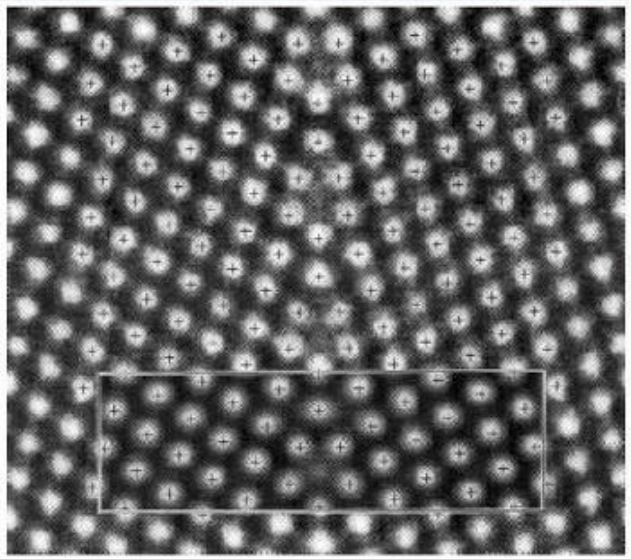
- ✓ Deformation in metals depends on linear and planar density.
- ✓ **Slip occurs** on most **densely packed** crystallographic **planes** and along **directions** having the **greatest atomic packing** due to **lower shear stress/energy requirement**.

Solidification of Metals

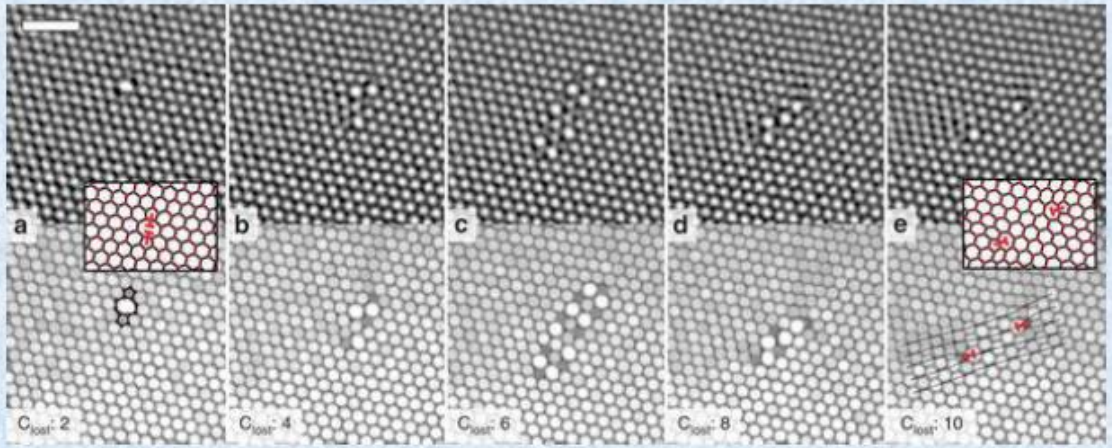
- Nucleation – The physical process by which a new phase is produced in a material. In the case of solidification, this refers to the formation of tiny stable solid particles in the liquid.
- Growth - The physical process by which a new phase increases in size. In the case of solidification, this refers to the formation of a stable solid particle as the liquid freezes.



REAL CRYSTALS



- **Real crystals** are never perfect: there are **always defects**!
- **Controlling** the defects is one of **the main** goal of materials science and engineering!



Imperfections in Solids

Crystal defects are imperfections in the regular geometrical arrangement of the atoms in a crystalline solid

- The properties of materials are profoundly influenced by the presence of imperfections.
- It is important to have knowledge about the types of imperfections that exist and the roles they play in affecting the behavior of materials.

Defects in crystalline structures

- Defects in the crystalline structure can have an tremendous effect on a materials behavior.
- We can modify and improve many of the physical, electrical, magnetic, and optical properties of crystalline materials by controlling the imperfections in their lattice structure.

Applications of Defects / Imperfections in Solids

Importance of defects depends upon the material, type of defect, and properties, which are being considered

Dislocation – Strength/toughness of materials

Vacancies – Electrical and thermal conductivity of materials

Impurities – Mechanical/Electronic properties of materials

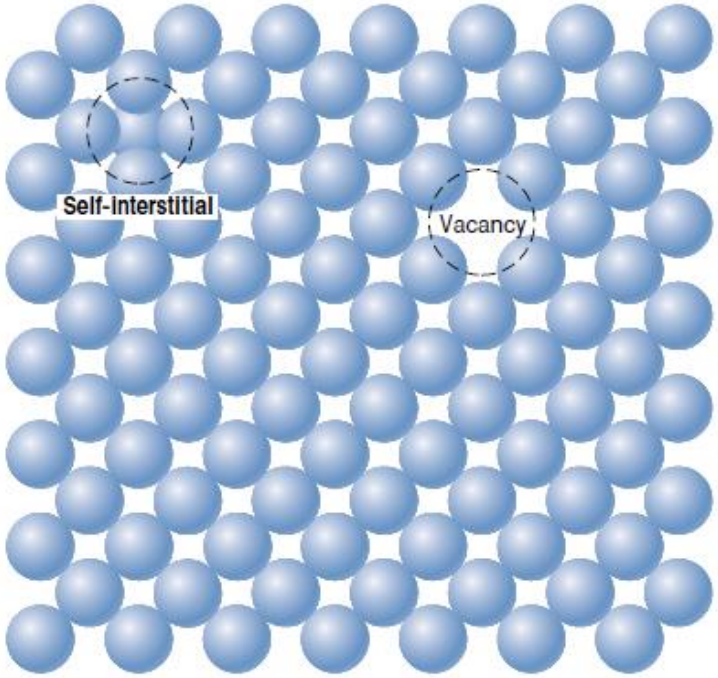
Grain boundaries – Strength of materials

Classification of Defects

Based on shape and size of defects :

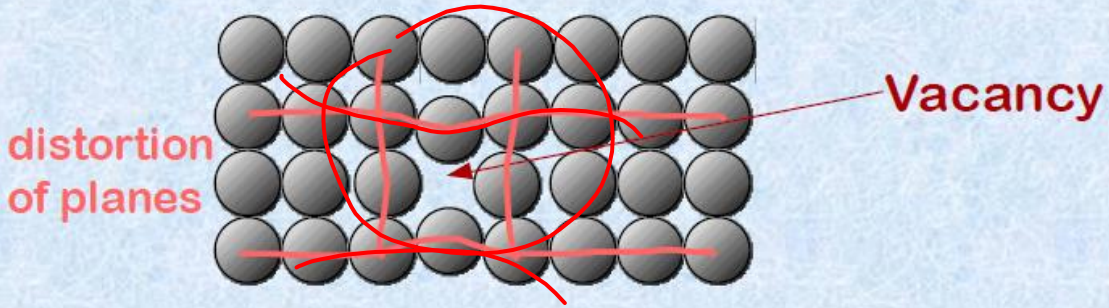
- **Point defects** (zero dimensional)
  - ✓ Vacancy defect
  - ✓ Interstitial defect
  - ✓ Substitutional defect
  - ✓ Frenkel defect
  - ✓ Schottky defect
- **Line defects / Dislocations** (One dimensional)
  - ✓ Edge dislocation
  - ✓ Screw dislocation
  - ✓ Mixed dislocation
- **Plane defects** (Two dimensional)
  - ✓ Grain boundaries
  - ✓ Twin boundaries
- **Volume defects** (Three dimensional)
  - ✓ Pores
  - ✓ Cracks, etc.

Extended defects

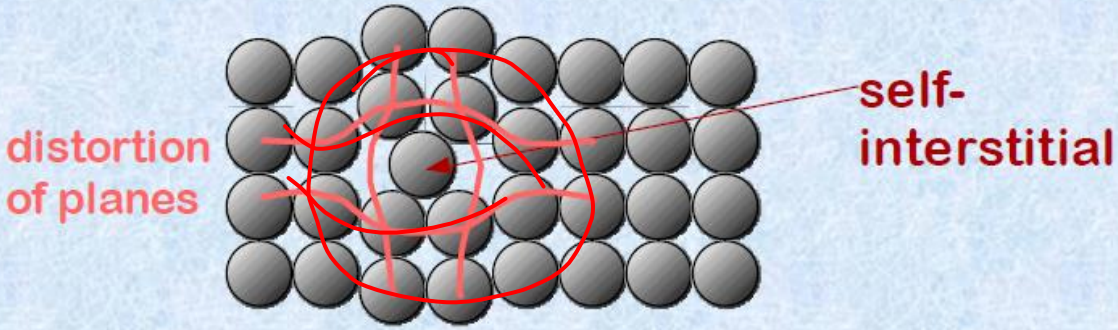


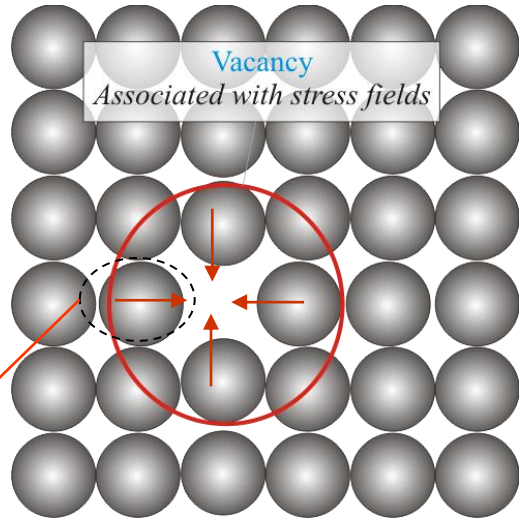
Point Defects (I)

Vacancies: vacant atomic sites in a structure.



Self-Interstitials: "extra" atoms positioned between atomic sites

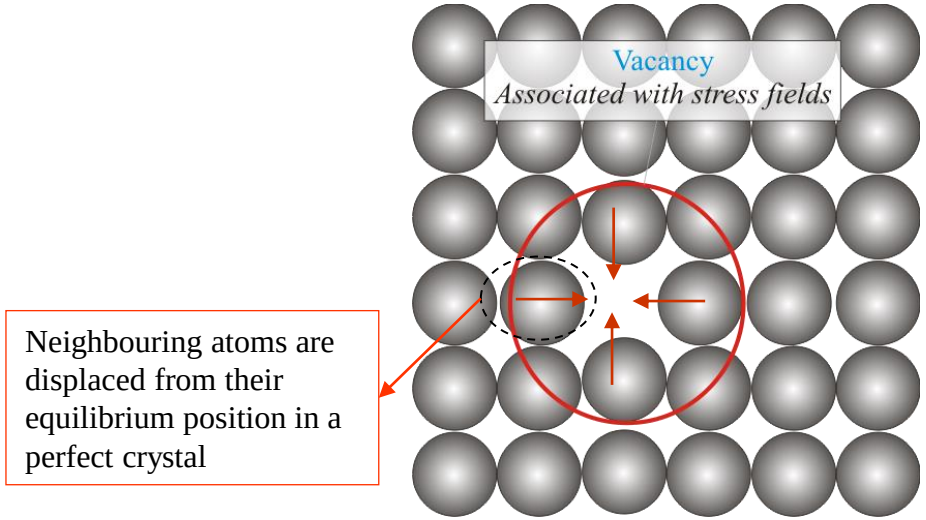




Neighbouring atoms are displaced from their equilibrium position in a perfect crystal

Vacancy

- Missing atom from an *atomic site* is called a vacancy.
- Atoms around the vacancy displaced from their equilibrium positions.
- This gives rise to a stress field in the vicinity of the vacancy.
- Vacancies play an important role in diffusion of atoms and in many other processes/effects in materials science
- Non-equilibrium concentration of vacancies can be generated by:
  - quenching from a higher temperature
  - bombardment with high energy particles
  - plastic deformation.



Number of Vacancies

The number of vacancies (N_D) for a given quantity of material depends on and increases with temperature according to

$$\frac{N_D}{N} = \exp\left(\frac{-Q_D}{kT}\right)$$

where:
N - is number of sites
N_D - is number of defects
Q_D – is activation energy
T – is a temperature
k – Boltzmann's constant
1.38×10^{-23} J/atom K
8.62×10^{-5} eV/atom-K
1 eV =$1.602 \cdot 10^{-19}$ J

For most metals, the fraction of vacancies N_D/N just below the melting temperature is on the order of 10^{-4}—that is, one lattice site out of 10,000 will be empty.

$$N = \rho \times (N_A/A) \times V$$

where: ρ - density; A – atomic mass
V – volume and N_A – Avogadro number
(6.02×10^{23} atom/mol)

Equilibrium Concentration of Vacancies

- Find the equilibrium number of vacancies in $1m^3$ of Cu at 1000 °C.

$\rho_{Cu} = 8.4\ g/cm^3$ $\quad A_{Cu} = 63.5 g/mol$

$Q_D = 0.9 eV/atom$ $\quad N_A = 6.02 \times 10^{23}$ atoms/mole

---

For $1\ m^3$, $N = \rho \cdot N_A / A_{cu} \cdot 1m^3 = 8 \cdot 10^{28}$ sites

$$\frac{N_D}{N} = \exp\left(\frac{-Q_D}{kT}\right) = 2.7 \cdot 10^{-4}$$

Q_D: 0.9eV/atom

T: 1273K

k: 8.62×10^{-5} eV/atom-K

- Answer:

$N_D = 2.7 \cdot 10^{-4} \cdot 8.0 \times 10^{28}$ sites $= 2.2 \times 10^{25}$ vacancies

At what temperature does the first vacancy become stable in a Cu crystal?

Materials for Engineers

Lecture 8-10

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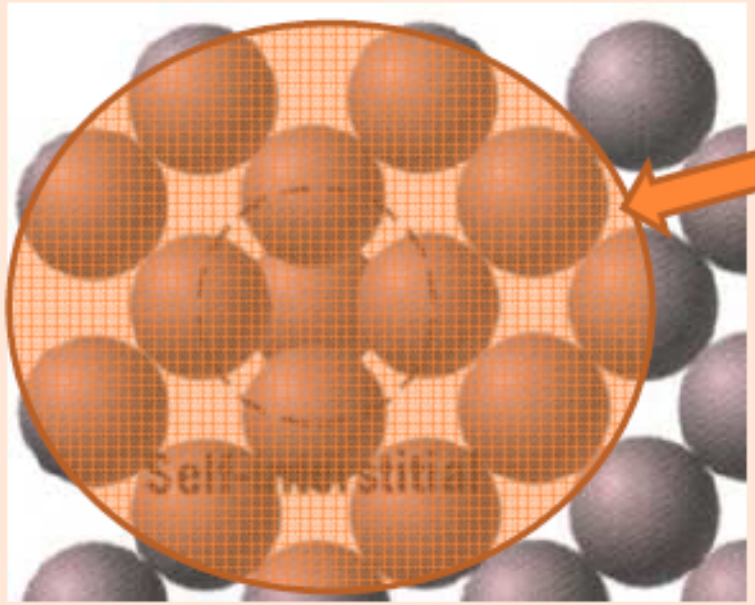
Chennai – 600127

Reference:
1. Materials science and engineering by William Callister, Wiley Publications

SELF-INTERSTITIAL DEFECT



Large distortions in the surrounding lattice because the atom is substantially larger than the interstitial position in which it is situated

The formation of this defect is not highly probable, and it exists in very small concentrations, which are significantly lower than for vacancies.

Alloys

- An **alloy** is a combination, either in **solution** or **compound**, of two or more elements, at least one of which is a metal.
- An alloy with two components is called a binary alloy; one with three is a ternary alloy; one with four is a quaternary alloy.
- The result of alloying is a metallic substance with properties different from those of its components.

Example: **Steel** is a metallic alloy whose major constituent is iron. One classical definition is that steels are iron-carbon alloys with up to 2.1% carbon. With the increased carbon, steel is harder and has a much higher tensile strength than iron, but is also more brittle.

Solid Solution, Solvent, Solute

The addition of impurity/foreign atoms to a metal results in the formation of a **solid solution** and/or a **new second phase**

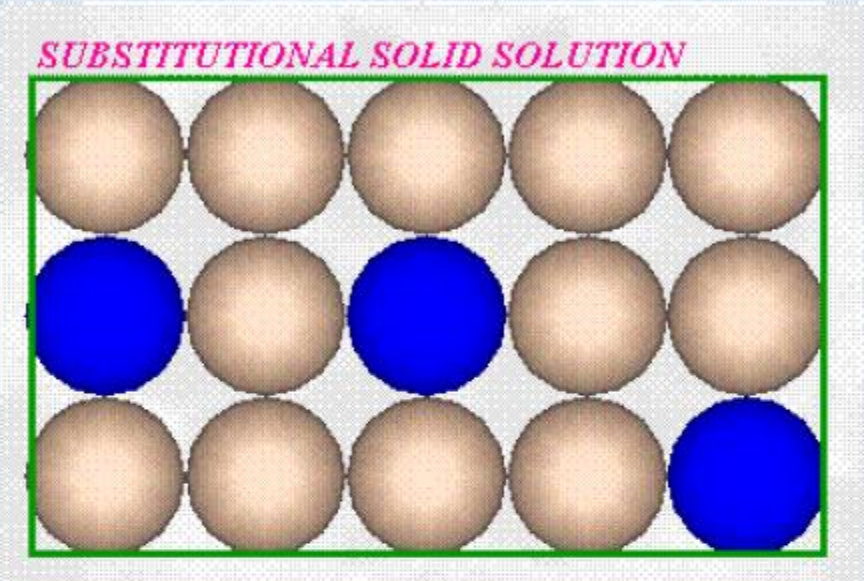
With regard to alloys, **solute** and **solvent** are terms that are commonly employed. *Solvent* is the element or compound that is present in the greatest amount; on occasion, solvent atoms are also called *host atoms*. *Solute* is used to denote an element or compound present in a minor concentration.

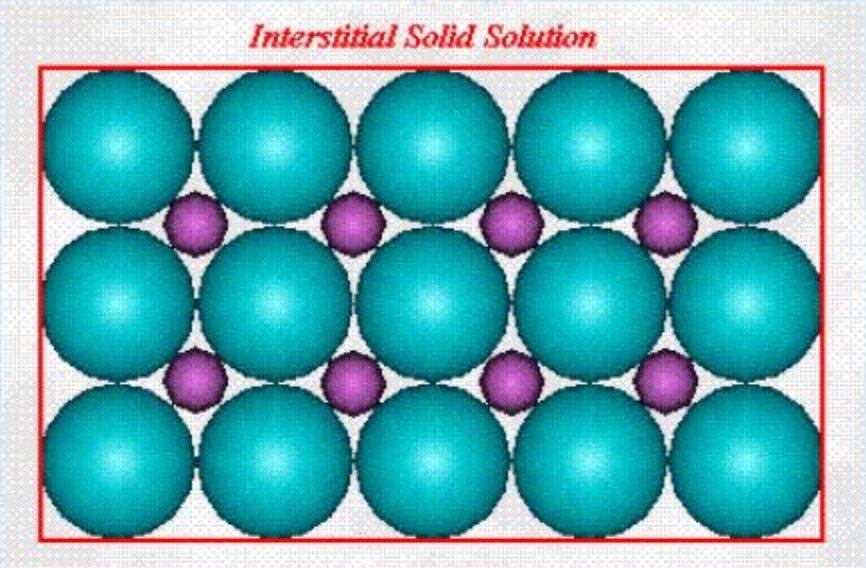
A solid solution forms when, as the solute atoms are added to the host material, the crystal structure is maintained and no new structures are formed. The impurity atoms are randomly and uniformly dispersed within the solid.

(Perhaps it is useful to draw an analogy with a liquid solution. If two liquids that are soluble in each other (such as water and alcohol) are combined, a liquid solution is produced as the molecules intermix, and its composition is homogeneous throughout.)

A solid mixture containing a minor component uniformly distributed within the crystal lattice of the major component.

Types of Solid Solutions





The name – **substitutional-** of this solid solution tells you exactly what happens as atoms of the **parent metal** are replaced or substituted by atoms of the **alloying metal**.

In **interstitial** solid solutions the atoms of the parent or **solvent metal** are bigger than the atoms of the alloying or **solute metal**. In this case, the smaller atoms fit into interstices i.e. spaces between the larger atoms.

In both cases the overall atomic structure is virtually unchanged.

Impurity/Alloying Element/Dopant

Point Defects (II) ➡ Impurities in solids

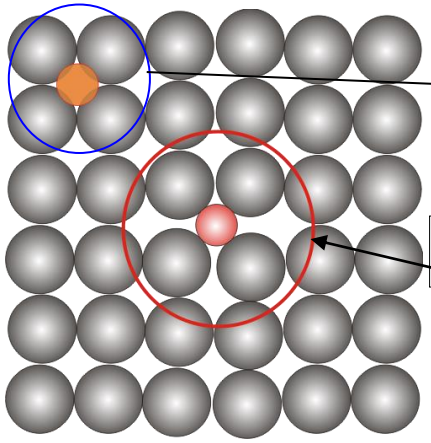
- A 'foreign' element added (called as impurity/alloying element/dopant based on the context) can go to an interstitial site (between atoms) or may substitute for an atom of the host.

Impurity

Or alloying element

Interstitial

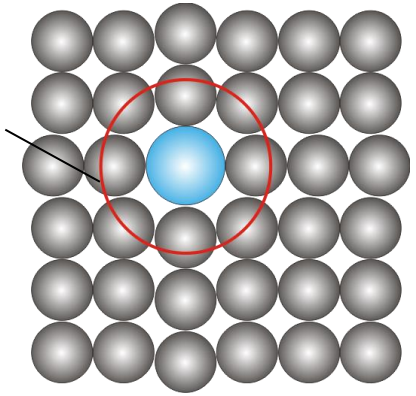
Substitutional

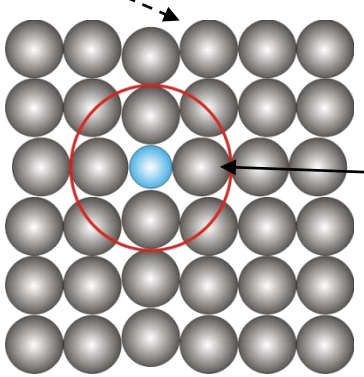


Overlaid to illustrate the relative size of atom and void (usually the insterstitial atom is bigger than the void)

Tensile / Compressive

Compressive stress fields



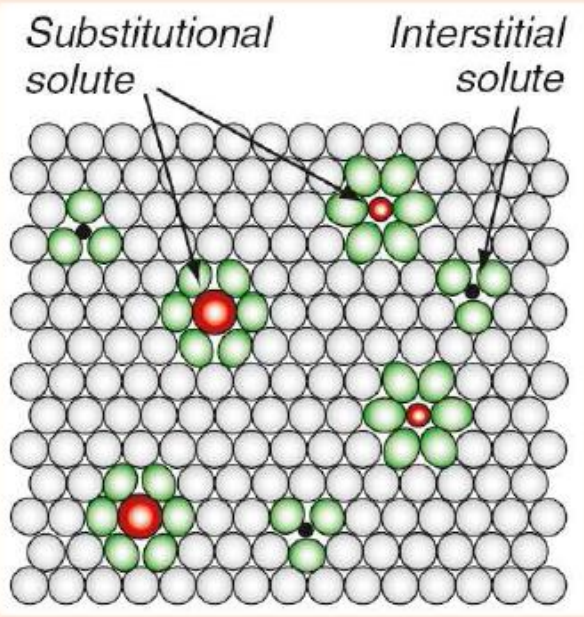


Tensile Stress Fields

Substitutional solid solution – dissolved atoms replace those of the host

Interstitial solid solution – dissolved atoms squeeze into spaces or "interstices" between the host atoms

Dissolved atoms rarely have the same size as the host material, so the surrounding *lattice is distorted*



Range of Solubility in Substitutional Solid Solutions

Substitutional Solid Solutions can

- exist over a *limited composition range* – **limited solubility;**
- exist over the whole composition range – **full solubility**.

HUME ROTHERY RULES: basic conditions for appreciable solubility

1. Atomic Sizes: should not differ by more than 14%;
2. Crystal structures: same for "pure" metals;
3. Electronegativity: The atoms should have similar electronegativity, or compounds will form;
4. Valences:

Impurities in Solids - Examples

❑ Substitutional Impurity/Element
- Foreign atom replacing the parent atom in the crystal
- E.g. **Cu** sitting in the lattice site of FCC-**Ni**

- **Cu-Ni Alloys**

Rule 1: $r_{Cu} = 0.128$ nm and $r_{Ni} = 0.125$ nm.

$\Delta R\% = 2.3\%$ **favorable** √

Rule 2: Ni and Cu have the FCC crystal structure. **favorable** √

Rule 3: $E_{Cu} = 1.90$ and $E_{Ni} = 1.80$. Thus, $\Delta E\% = -5.2\%$ **favorable** √

Rule 4: Valency of Ni and Cu are both +2. **favorable** √

Expect Ni and Cu forms S.S. over wide composition range.

Example II

Is solid-solution favorable, or not?

- **Cu-Ag Alloys**

  Rule 1: r_{Cu} = 0.128 nm and r_{Ag} = 0.144 nm.

  $\Delta R\%$ = 9.4% **favorable √**

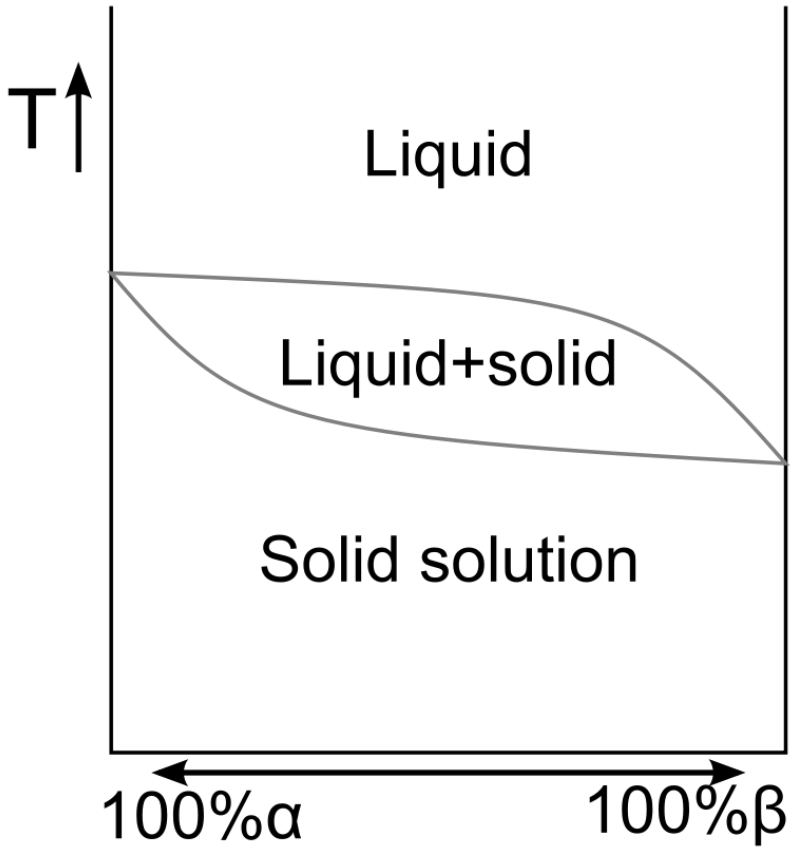
  Rule 2: Ag and Cu have the FCC crystal structure. **favorable √**

  Rule 3: E_{Cu} = 1.90 and E_{Ni} = 1.80. Thus, $\Delta E\%$ = -5.2% **favorable √**

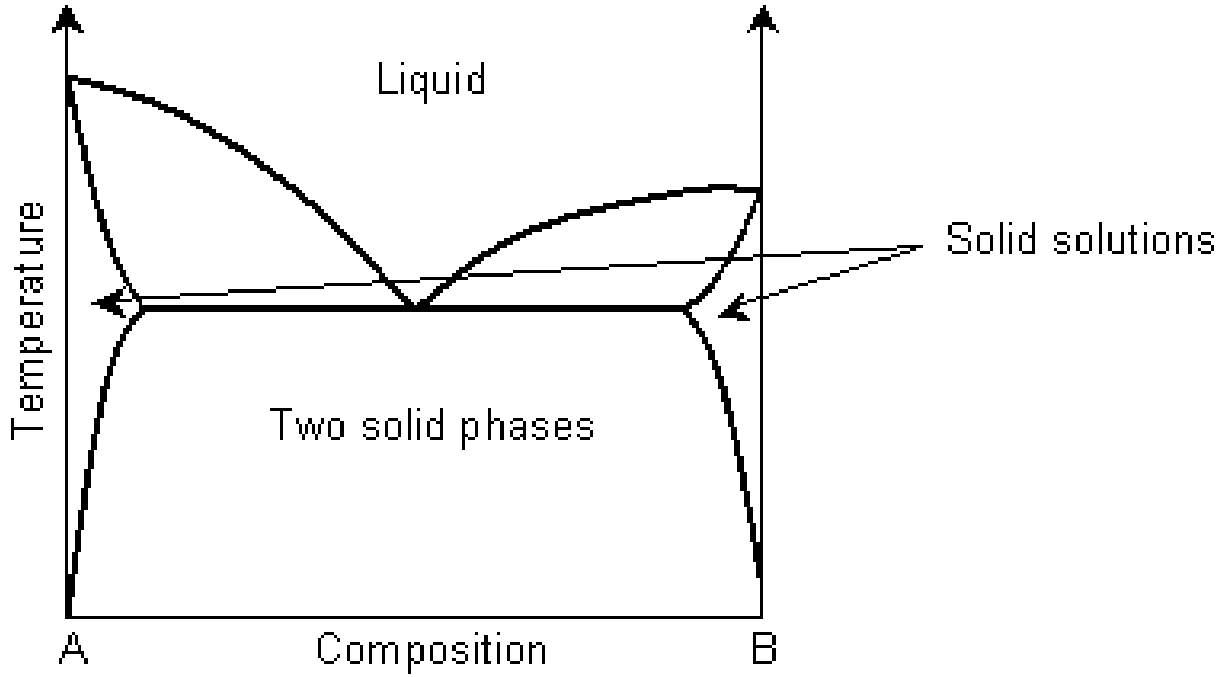
  Rule 4: Valency of Cu is +2 and Ag is +1. **NOT favorable**

  Expect Ag and Cu have <u>limited solubility</u>.

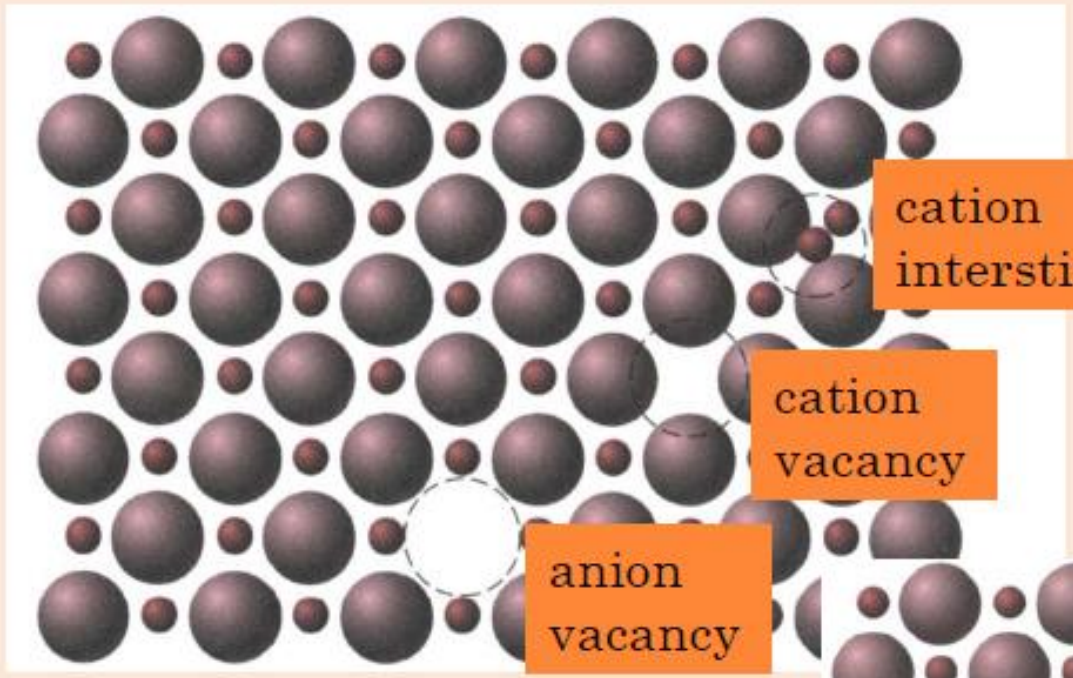
Complete Solubility



Limited Solubility



POINT DEFECTS IN CERAMICS

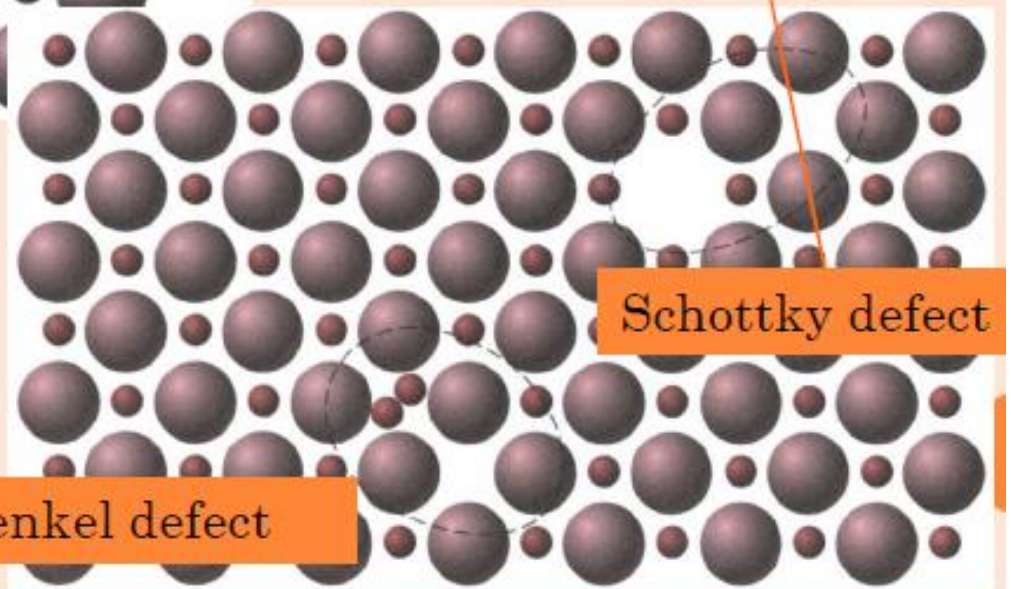


cation interstitial

cation vacancy

anion vacancy

a cation vacancy–anion vacancy pair

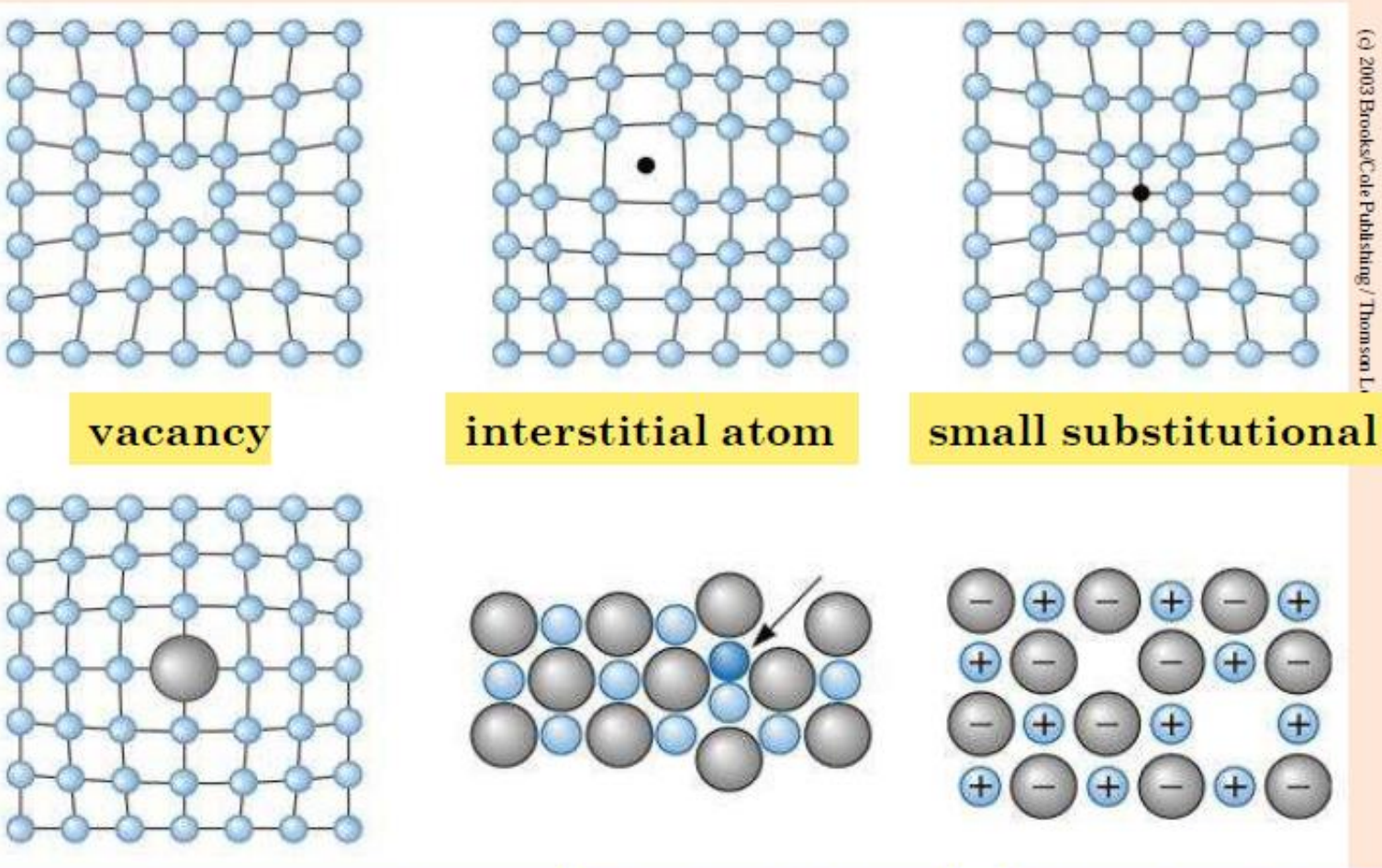


Schottky defect

a cation–vacancy and a cation– interstitial pair.

Frenkel defect

SUMMARY OF POINT DEFECTS



vacancy

interstitial atom

small substitutional atom

large substitutional atom

Frenkel defect

Schottky defect

a cation–vacancy and a cation– interstitial pair.

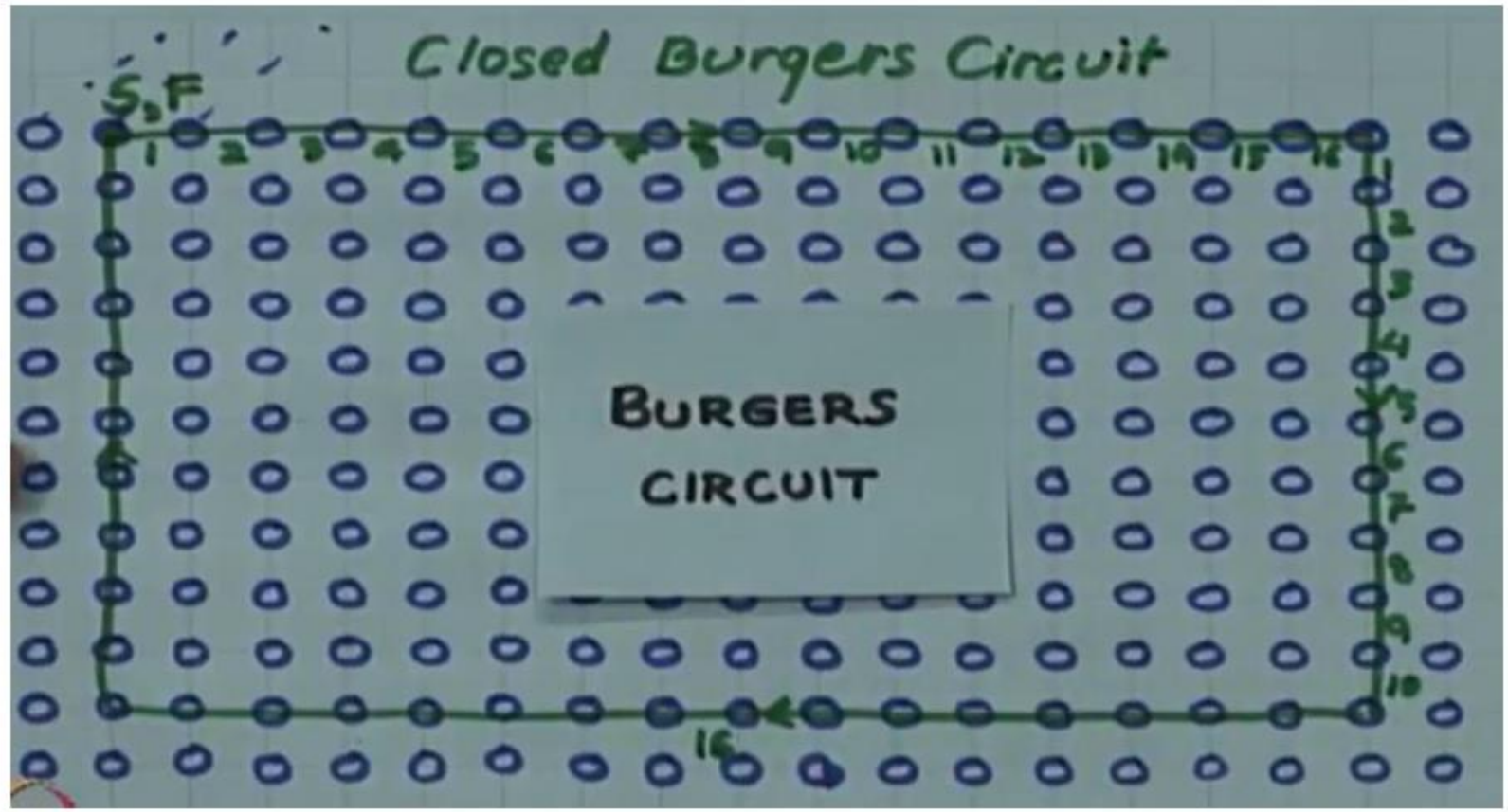
a cation vacancy– anion vacancy pair

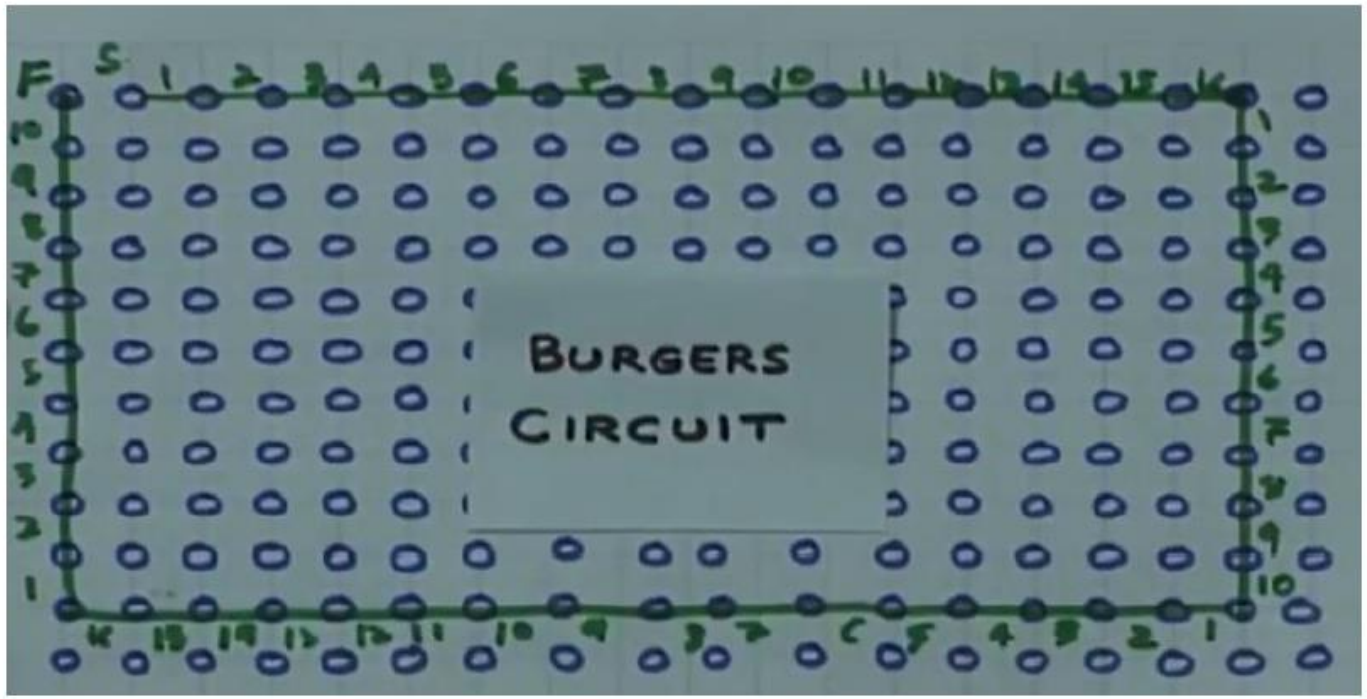
DISLOCATIONS

- Edge dislocation
- Screw dislocation
- Mixed dislocation

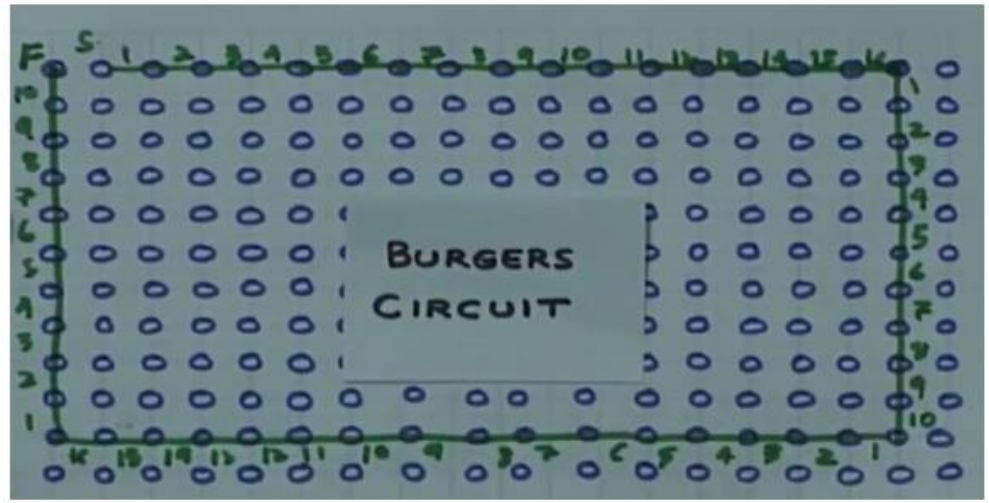
Burgers Circuit

$\vec{b}$ = magnitude and direction of Slip

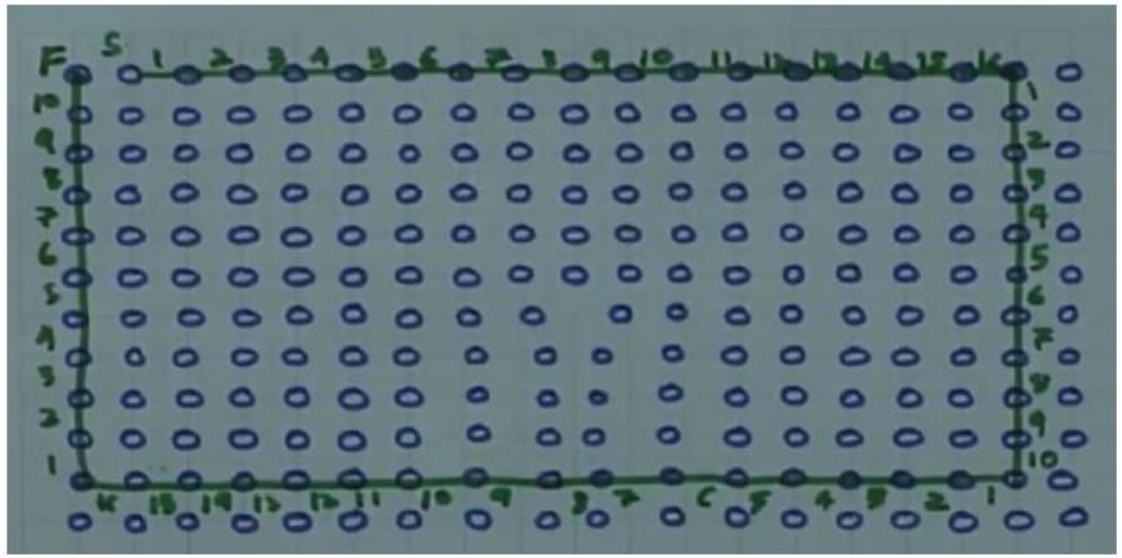


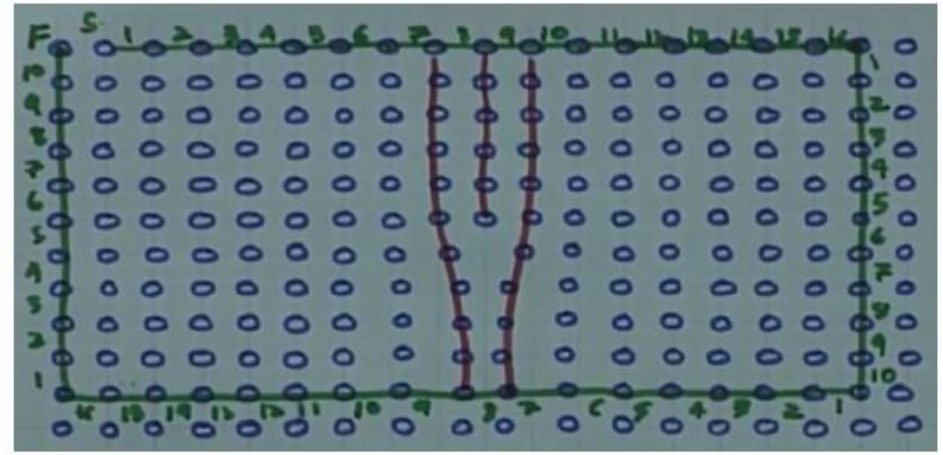


So, there is a closure failure!

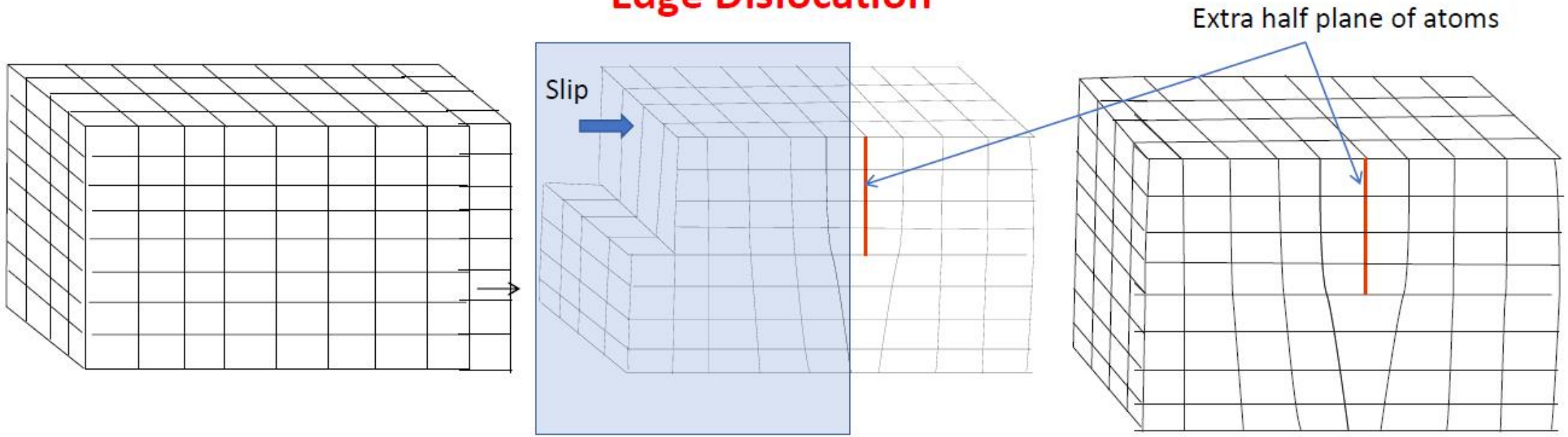


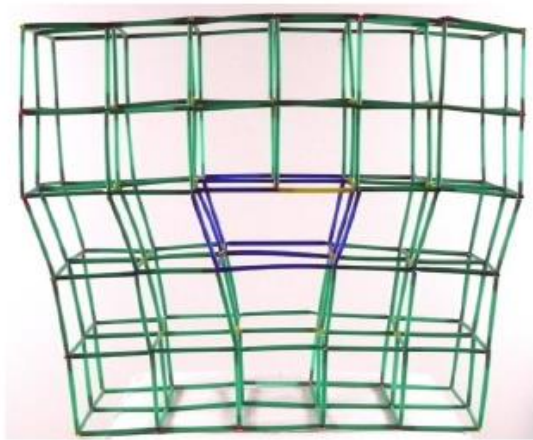
So, there is a closure failure!

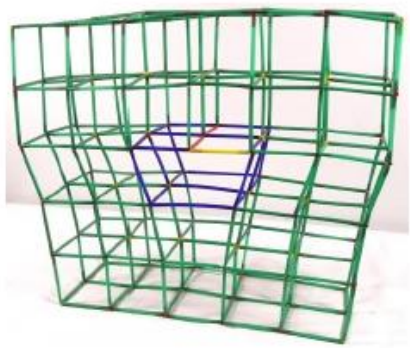




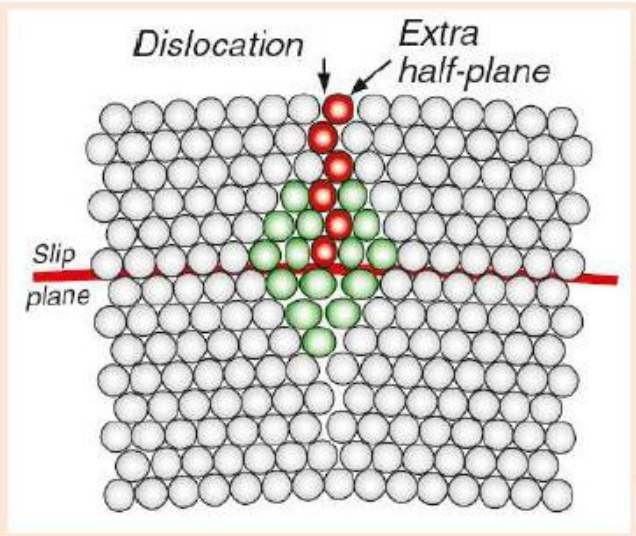
Edge Dislocation

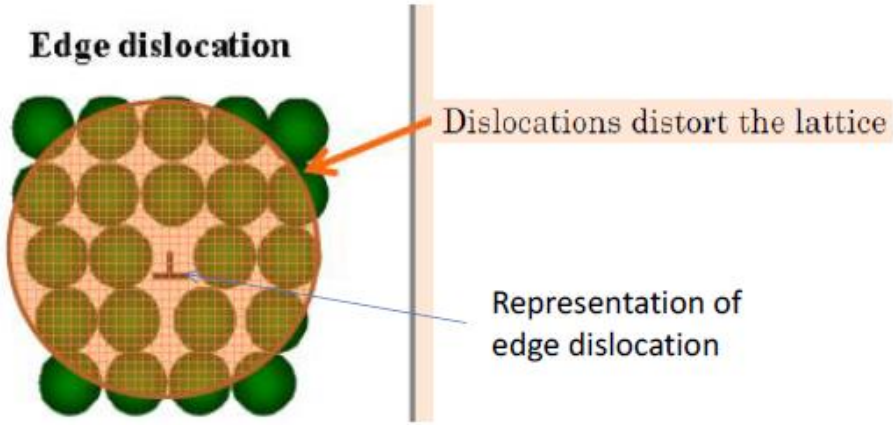


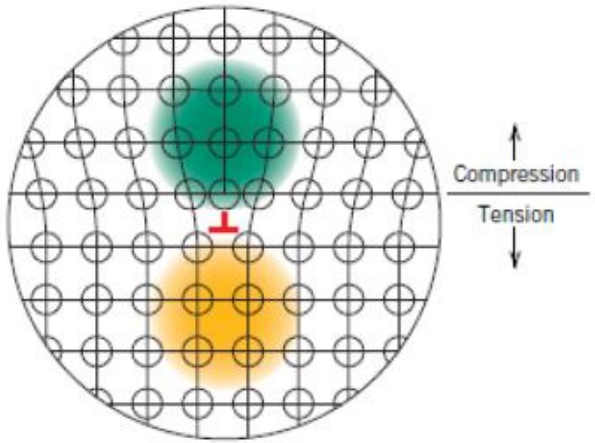


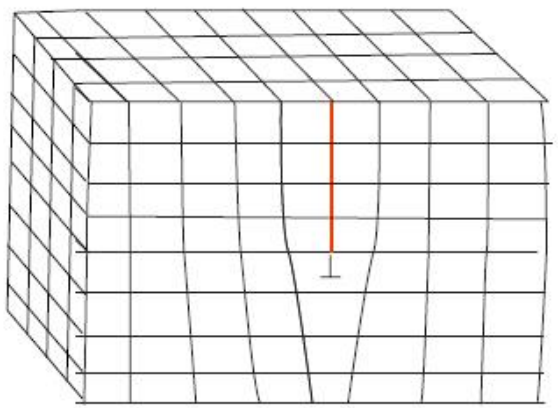


Edge Dislocation

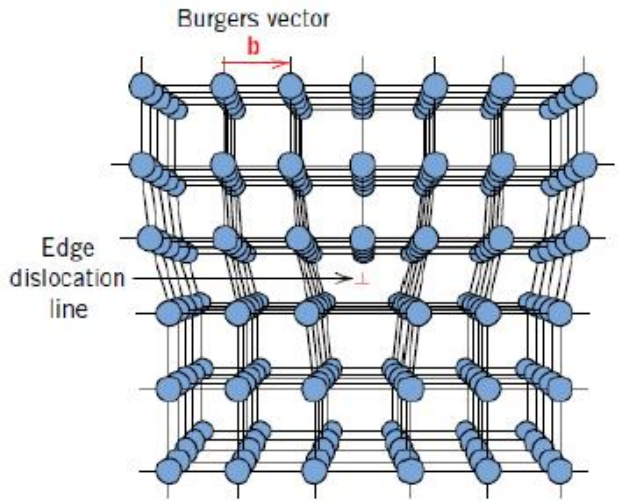




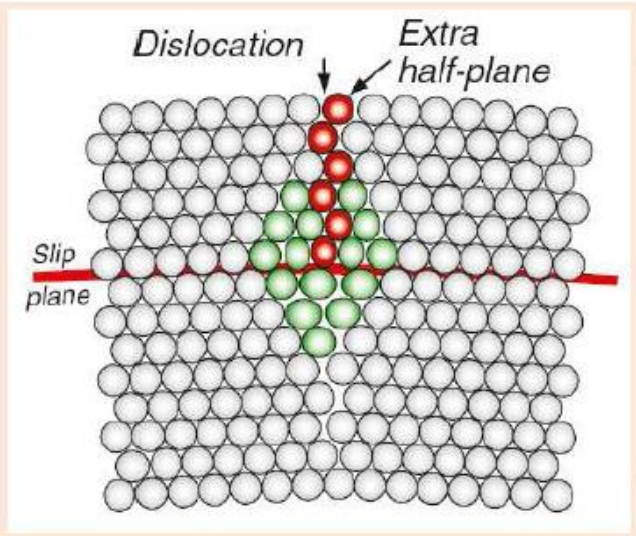


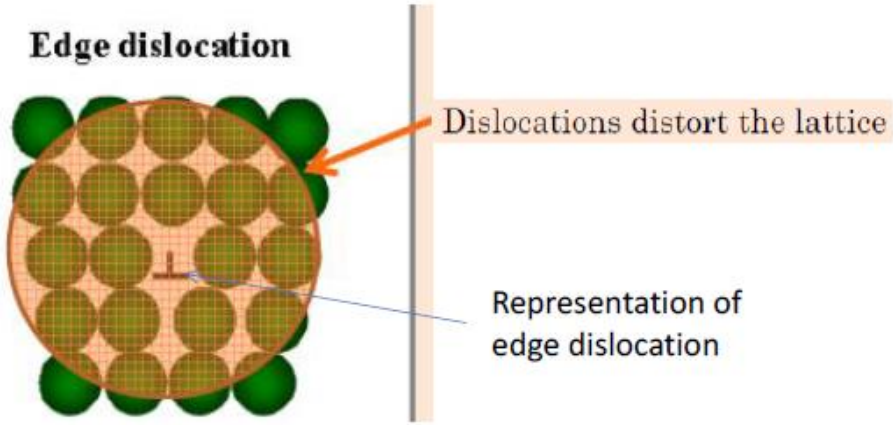


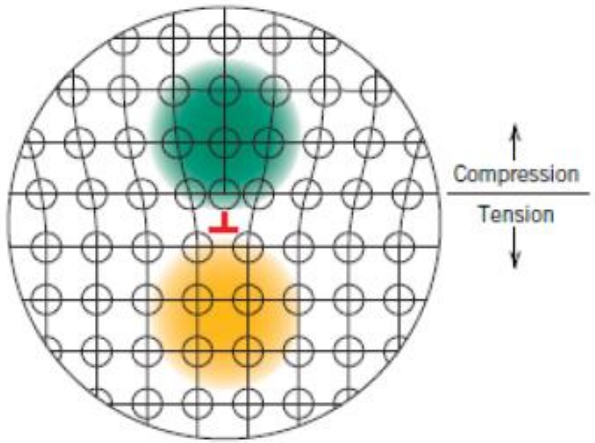
The magnitude of distortion decreases with distance away from the dislocation line

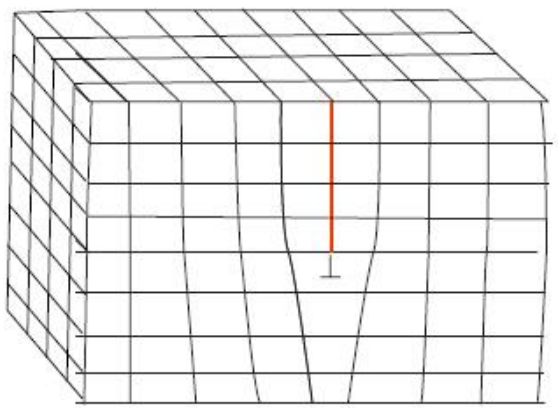


Edge Dislocation

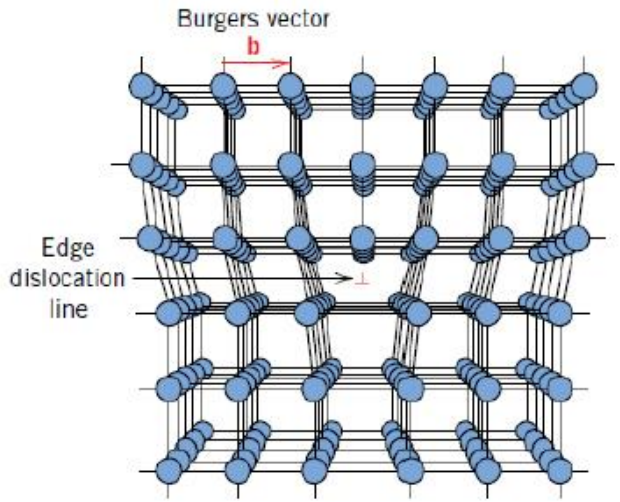






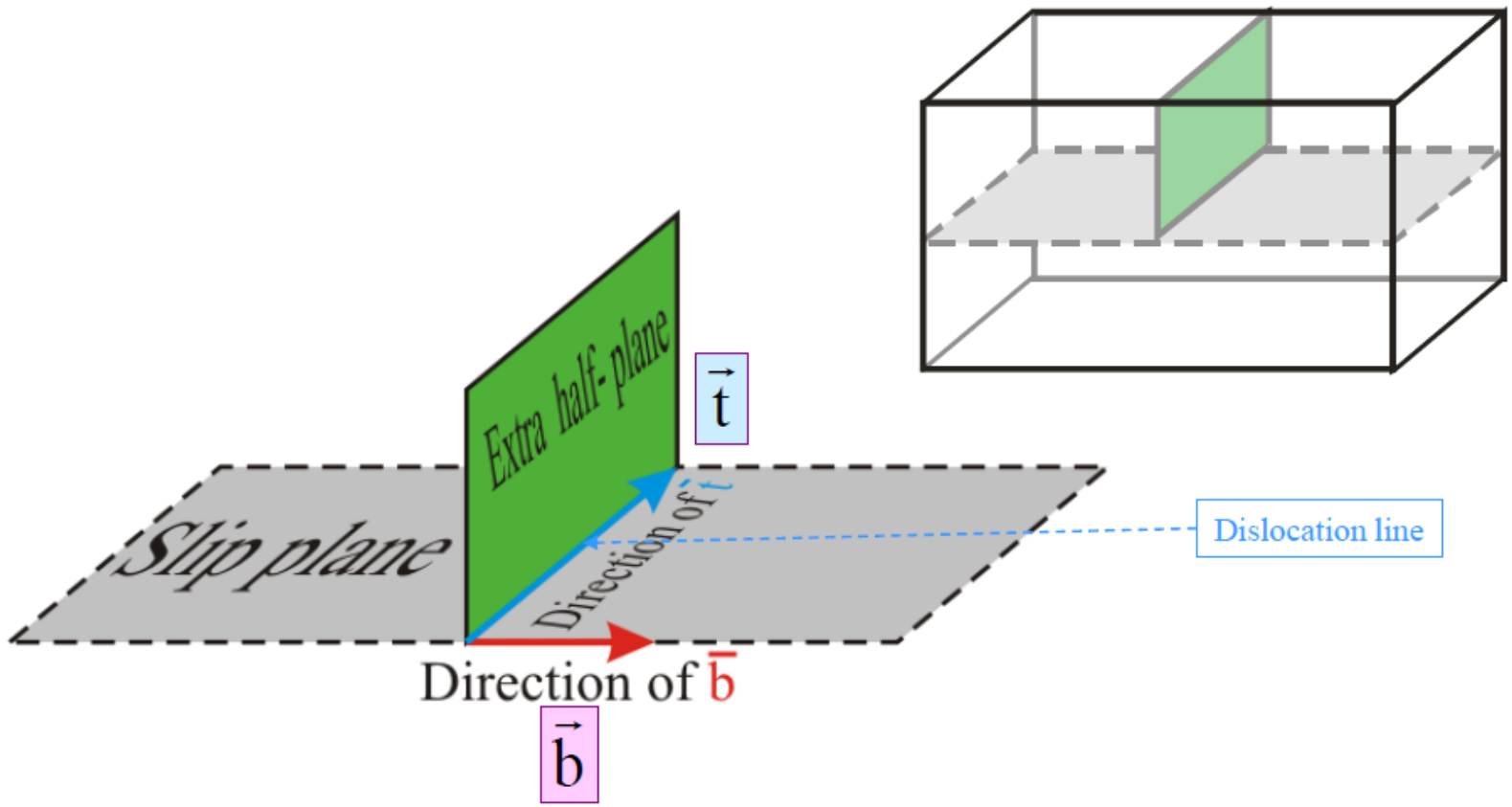


The magnitude of distortion decreases with distance away from the dislocation line



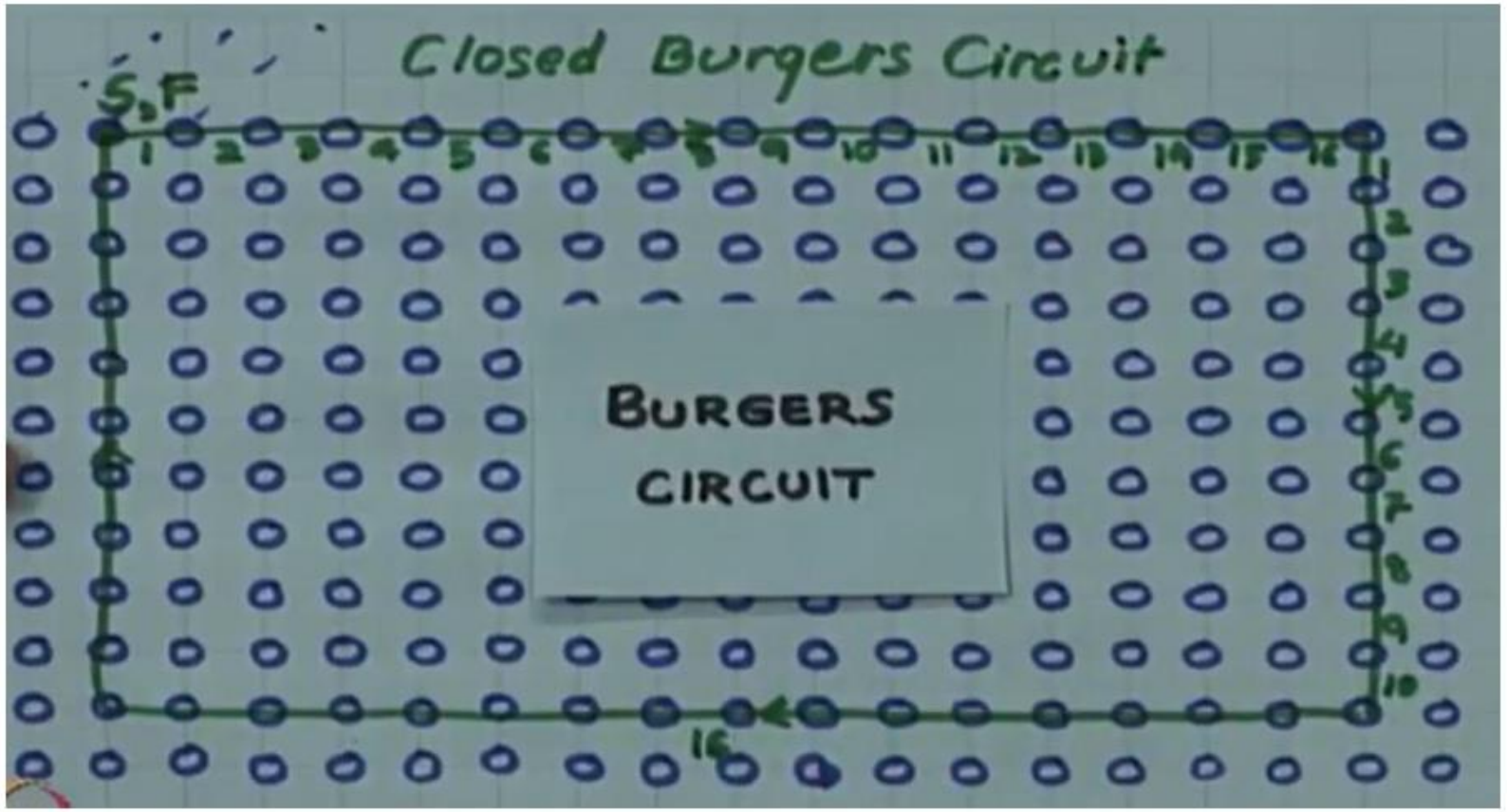
Edge dislocation

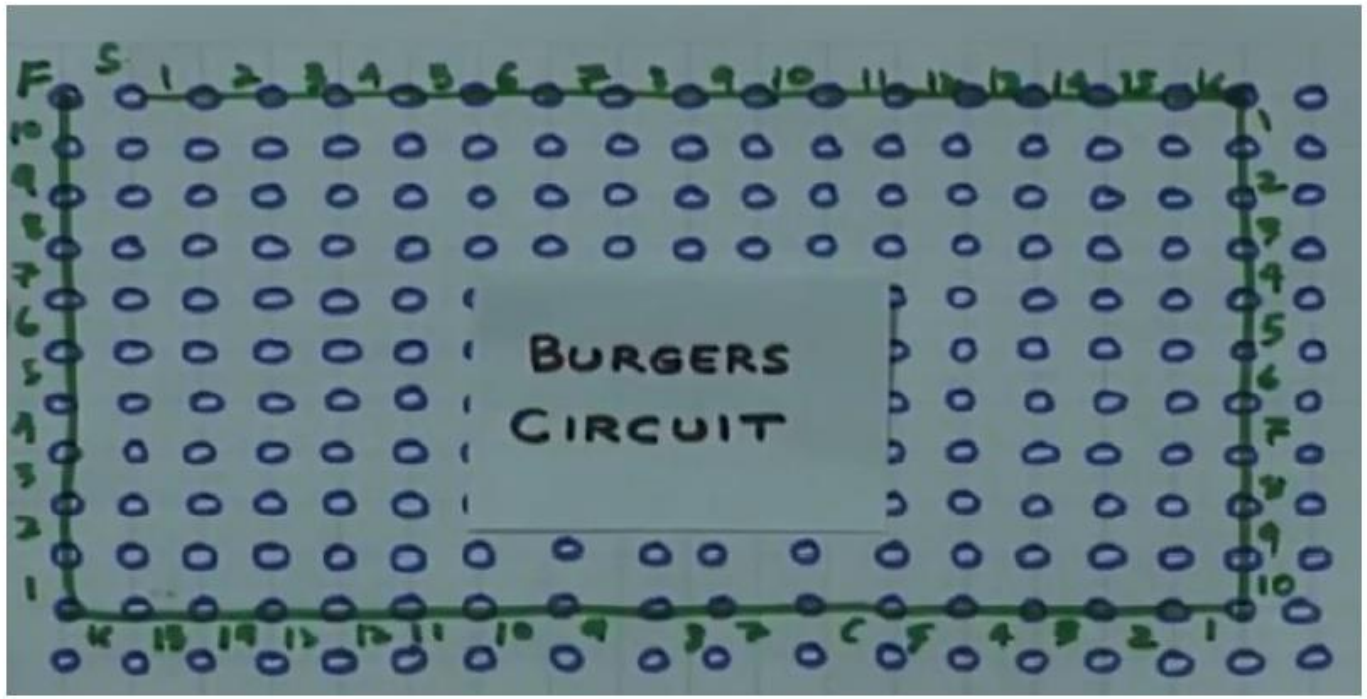
- Often to visualize the edge dislocation, only the extra 'half'-plane and slip plane are shown. The remaining crystal is hidden away.
- The intersection of the extra half-plane and slip plane can be visualized as the dislocation line



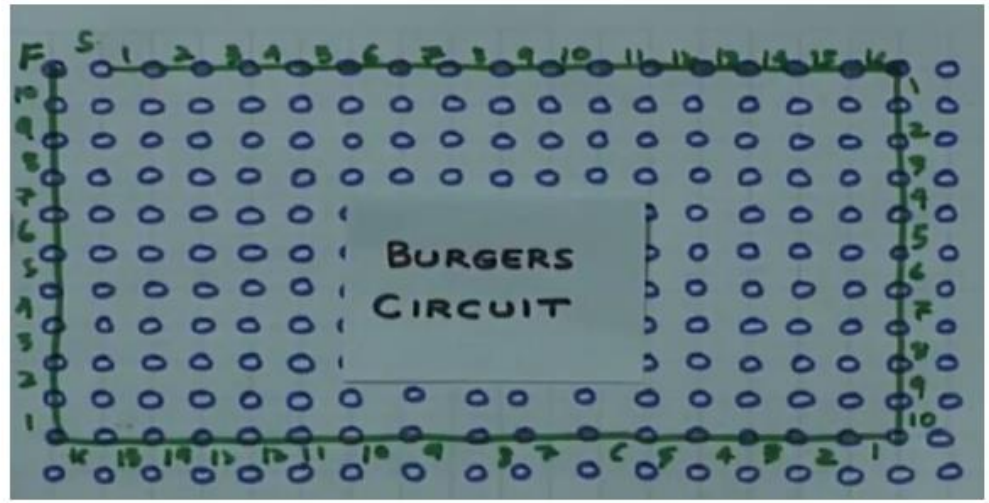
Burgers Circuit

$\vec{b}$ = magnitude and direction of slip

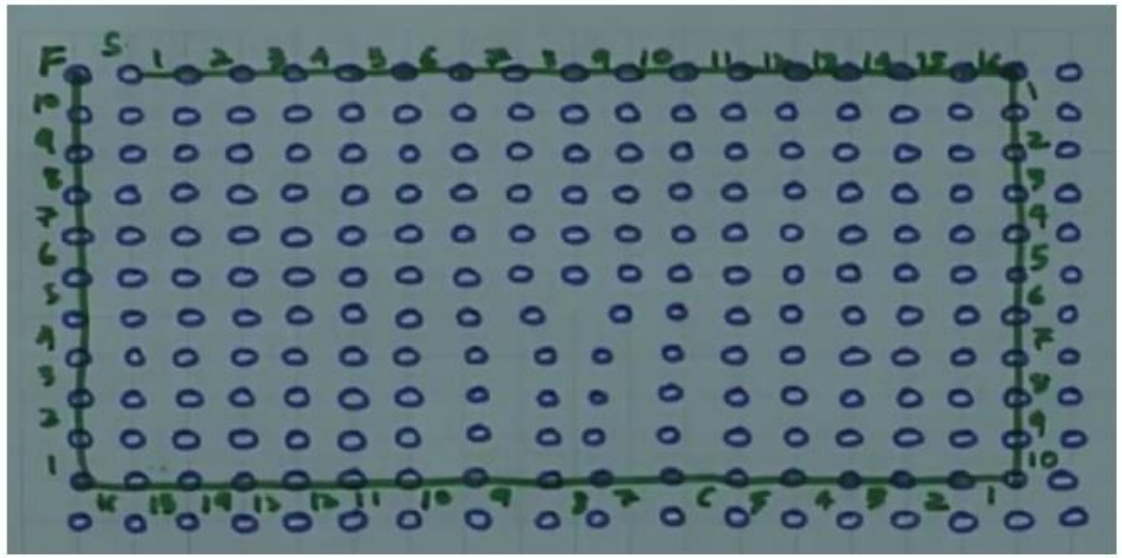


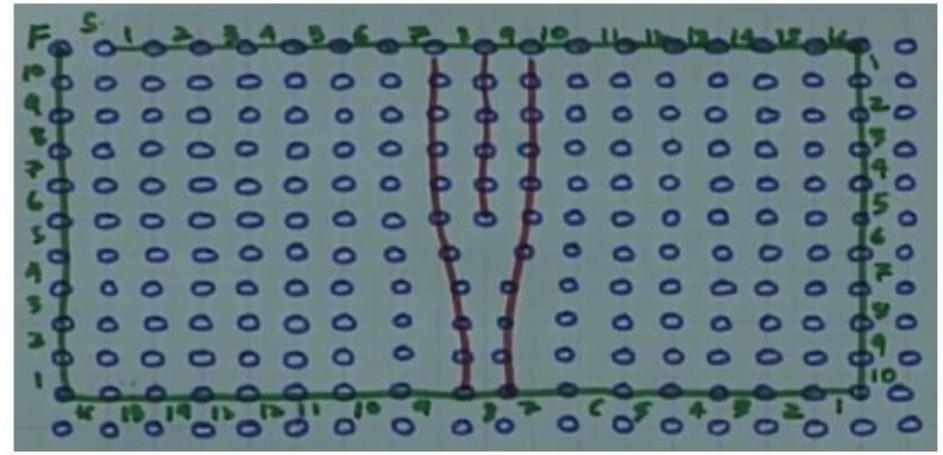


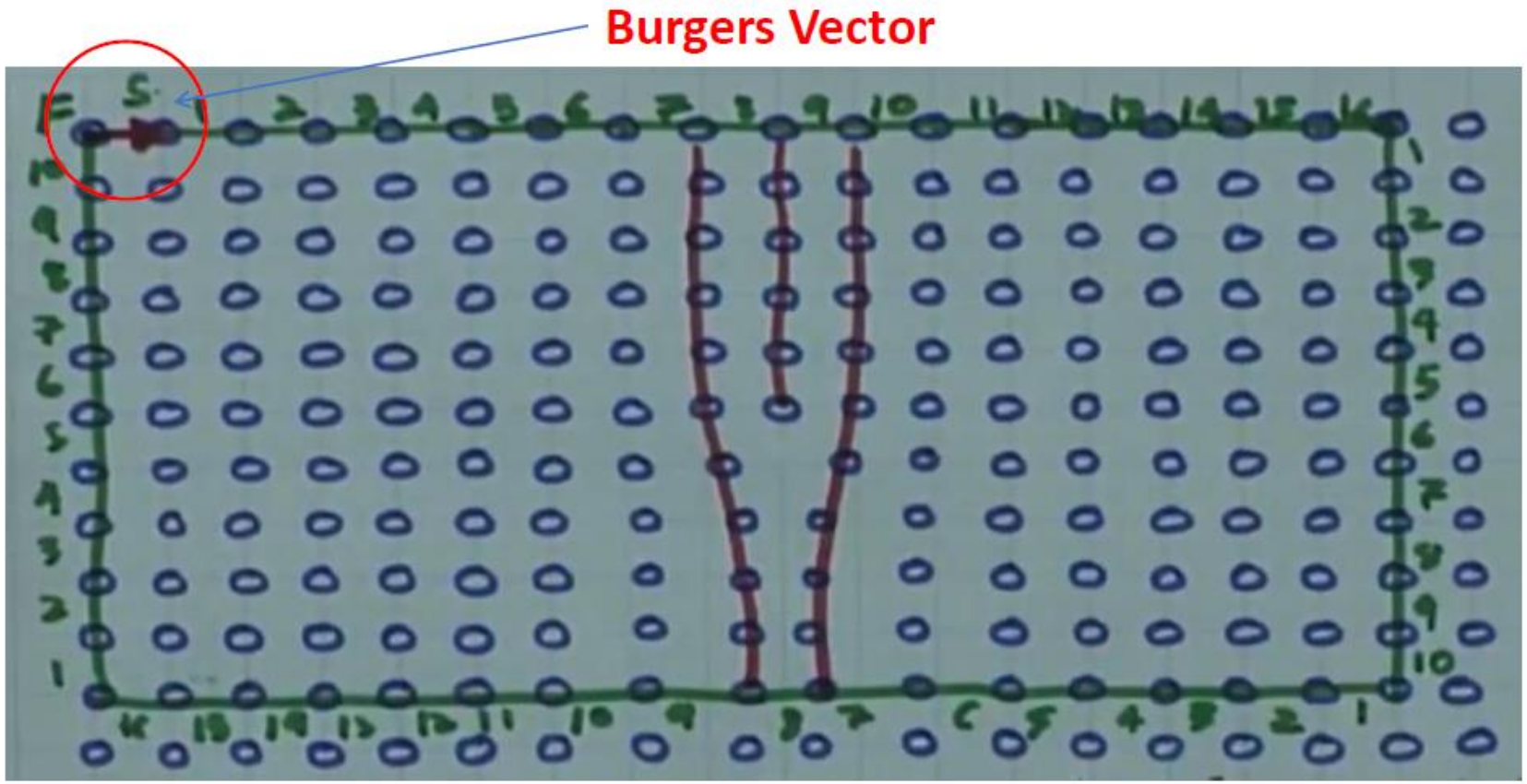
So, there is a closure failure!



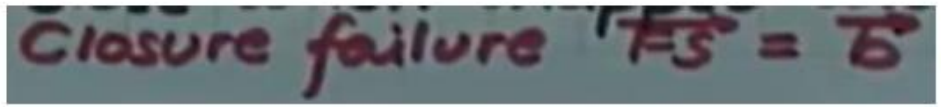
So, there is a closure failure!







A closed Burgers circuit in a perfect crystal fails to close when mapped around a dislocation line



→ **Burgers vector**

Sense could be reversed !!!
Conventionally RHFS followed

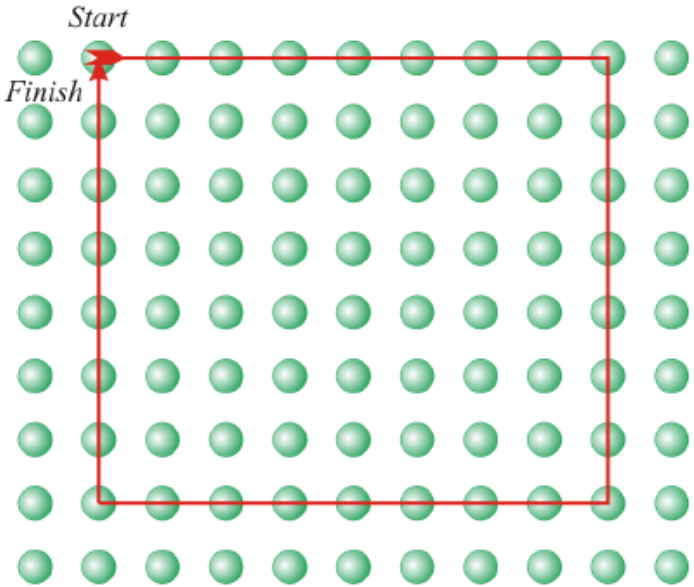
Remain horizontal and perpendicular to edge dislocation line

Burgers vector

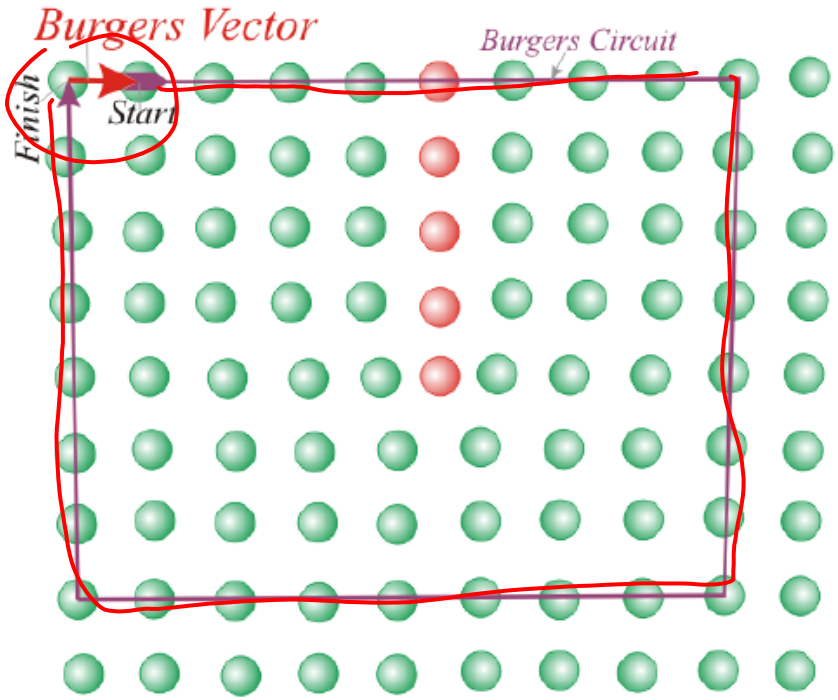
Edge Dislocation

Determination of Burgers vector in a dislocated crystal using Right Hand Finish to Start Rule (RHFS)

- In a perfect crystal make a circuit (e.g. as in the figure shown: 8 atomic steps to right, 7 down, 8 left & 7 up). The circuit is *Right Handed*.
- Follow the same steps in the dislocated crystal. The 'missing link' (using some convention like RHFS) is the Burgers vector.



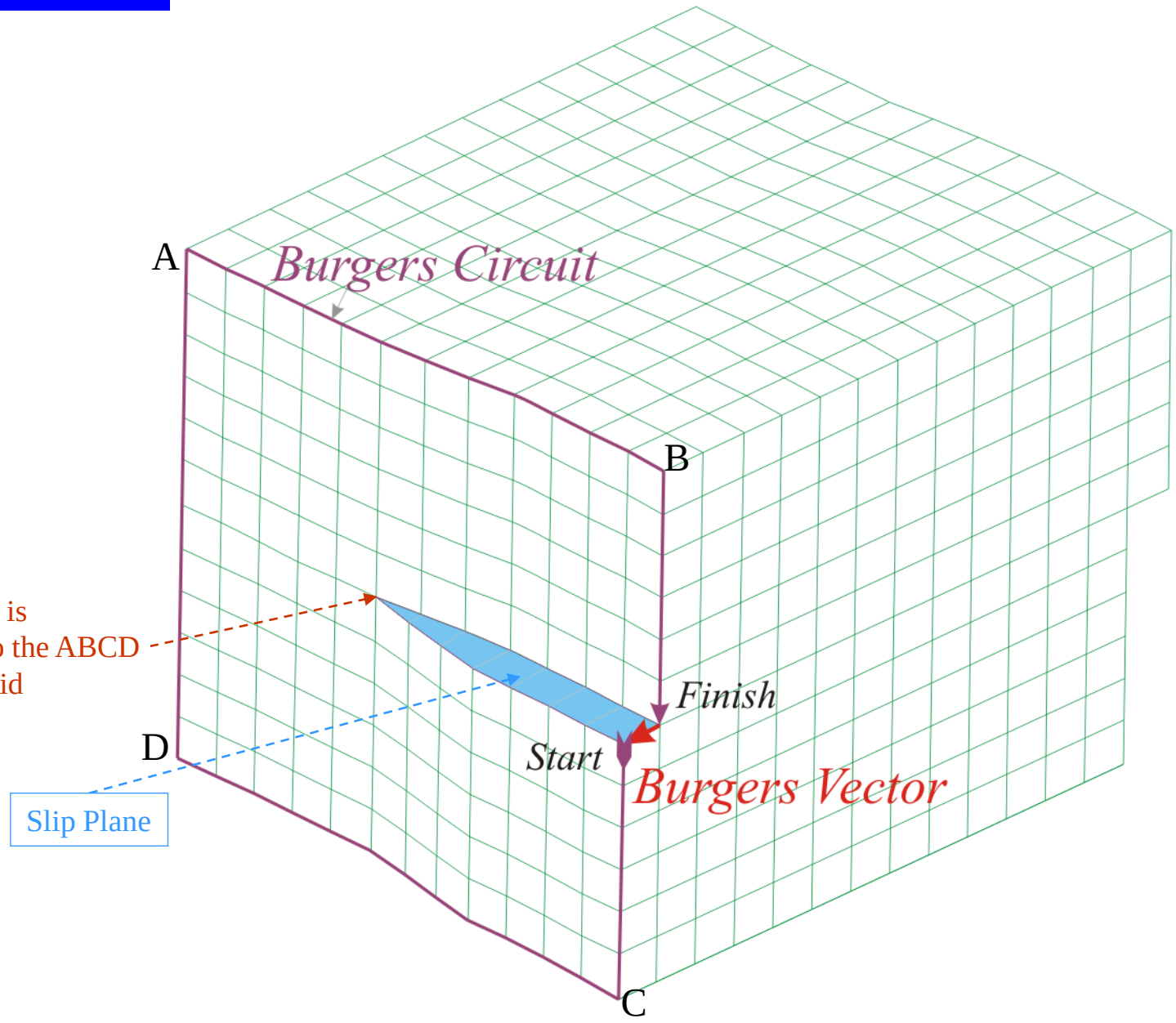
Perfect Crystal



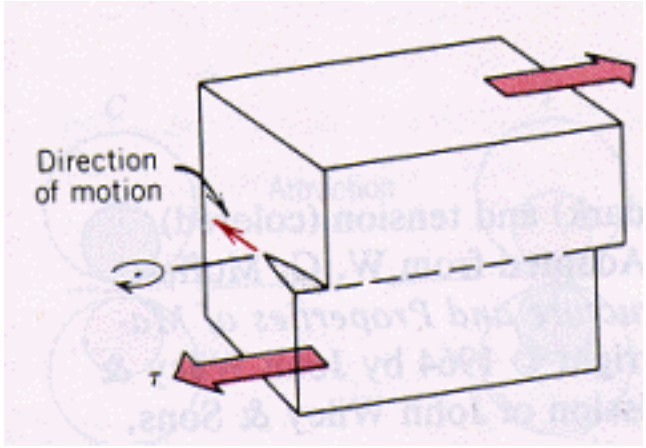
Dislocated Crystal (Edge Dislocation)

RHFS:
Right Hand Finish to Start convention

Screw Dislocation



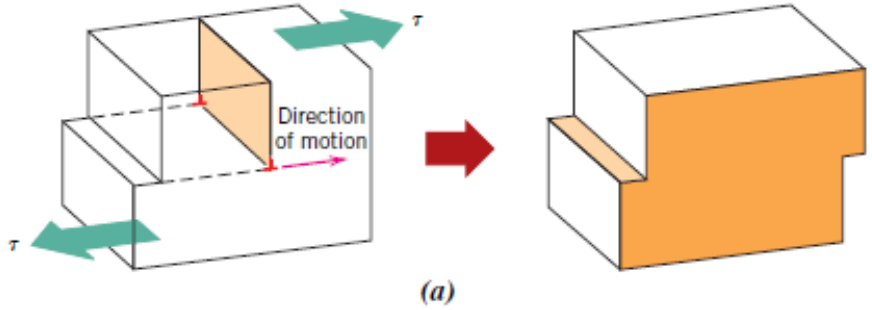
Dislocation line is perpendicular to the ABCD face into the solid



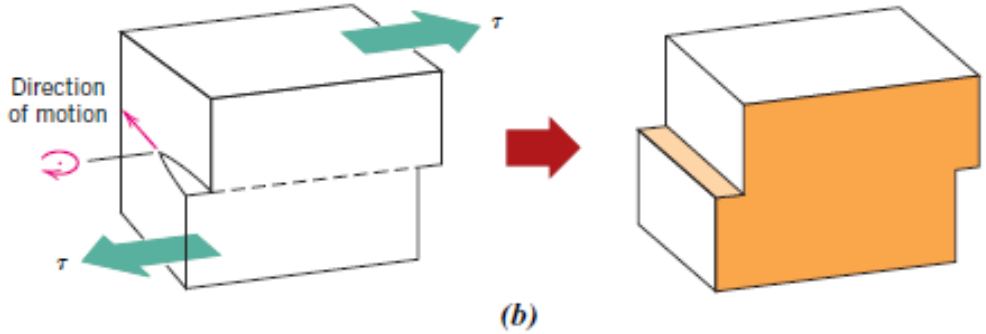
Geometric properties of dislocations

Dislocation Property	Type of dislocation	
	Edge	Screw
Relation between dislocation line (**t**) and **b**	⊥	∥
Direction of dislocation line movement relative to **b**	∥	⊥

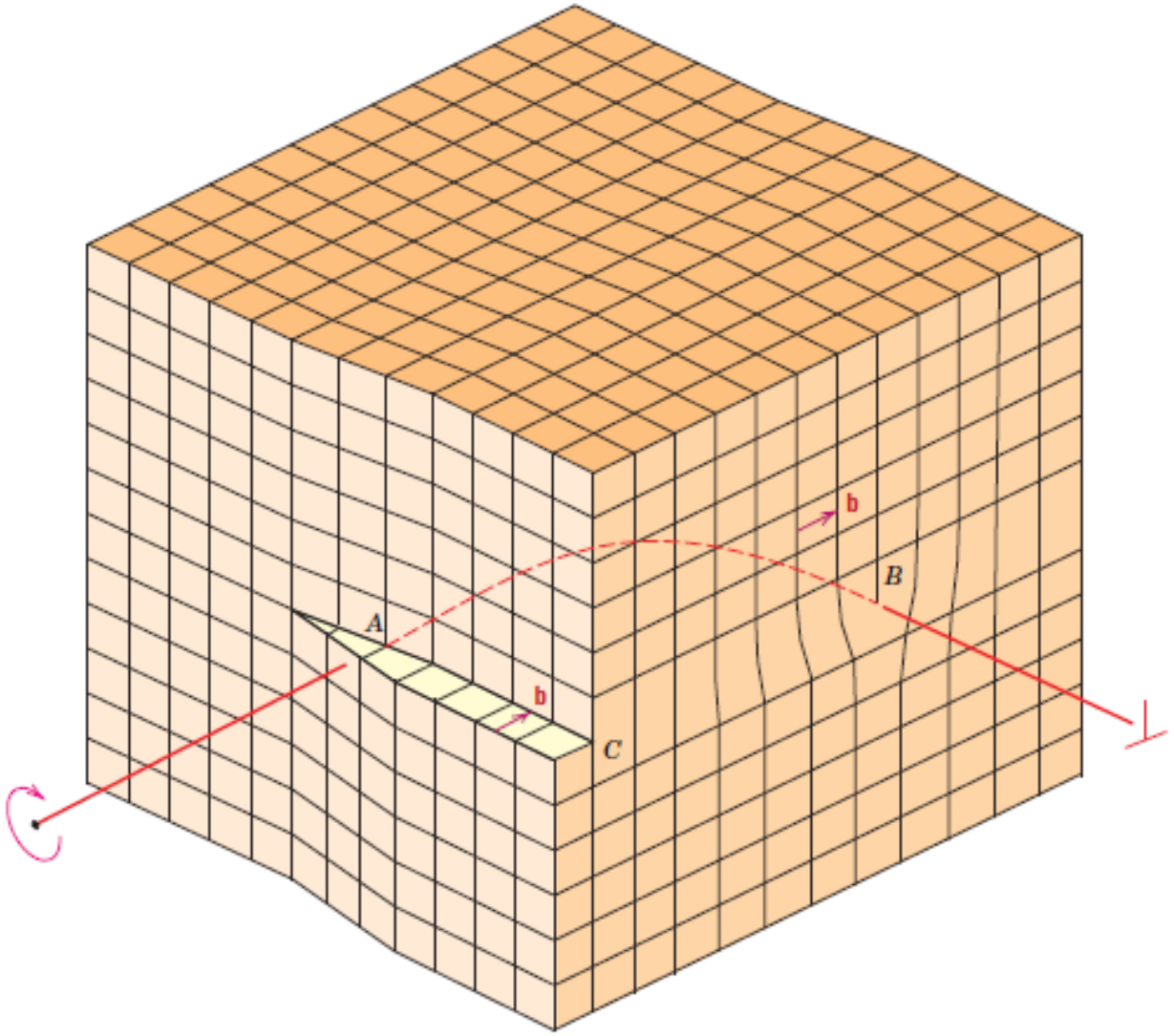
Edge dislocation



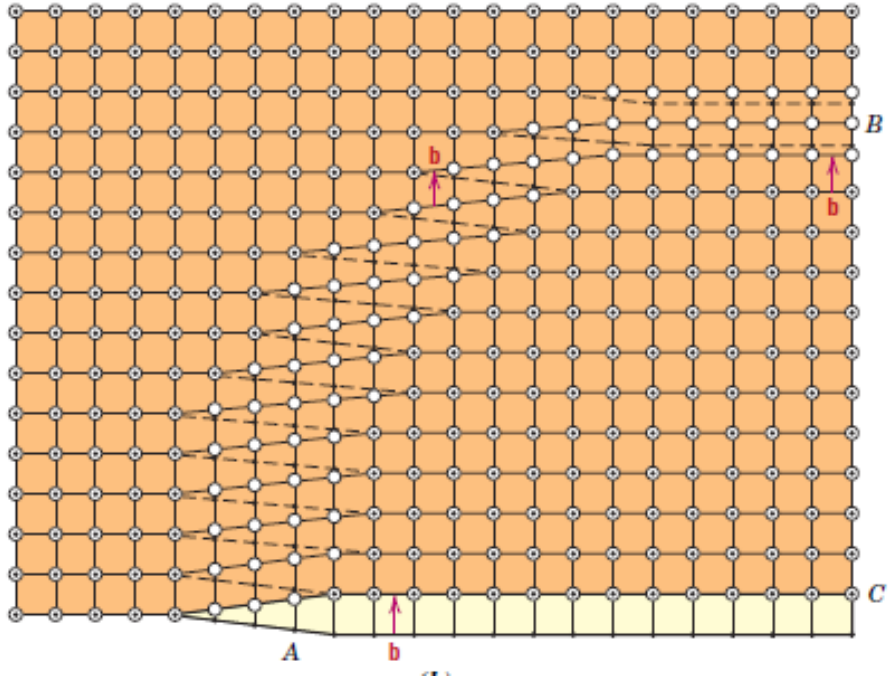
Screw dislocation



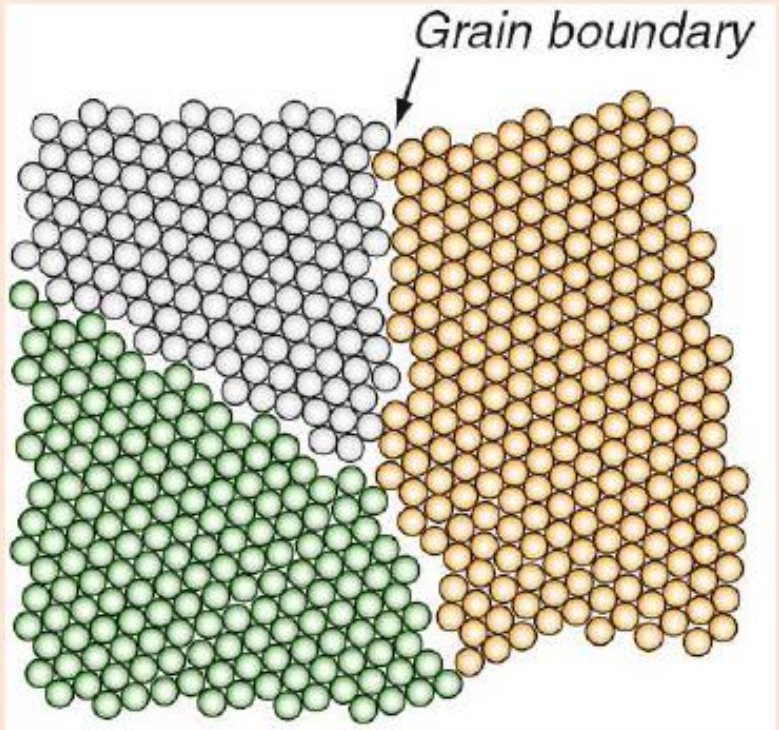
Mixed Dislocation



Top view



Grain Boundaries

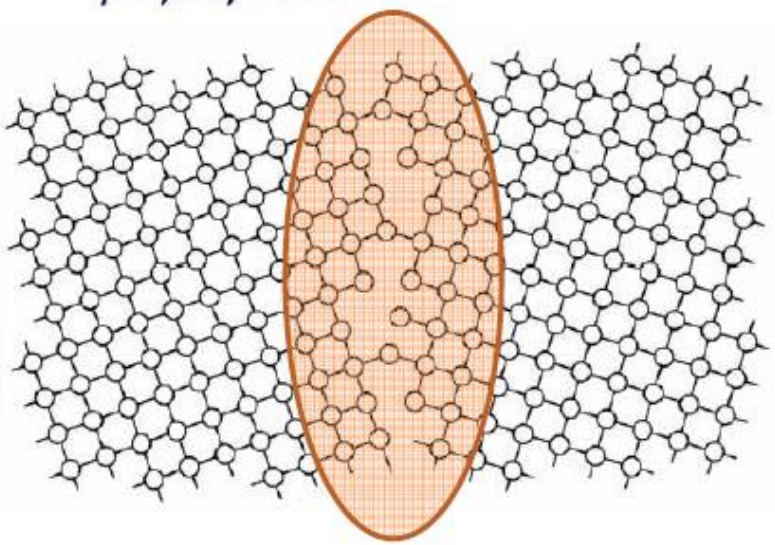


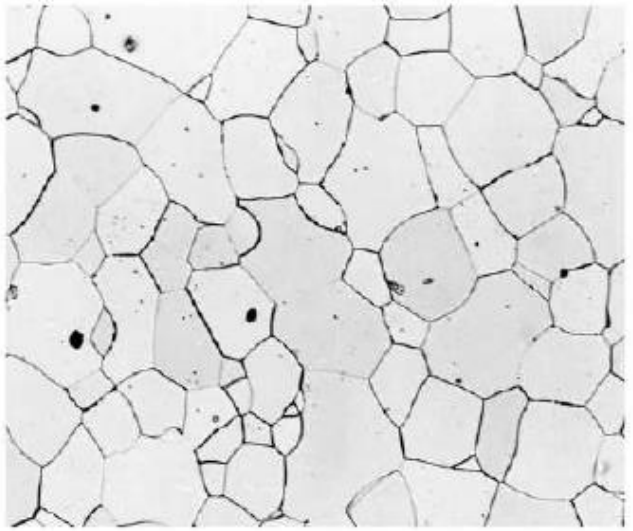
Grain boundaries form when
differently oriented crystals meet –
the individual crystals are called grains, the meeting
surfaces are grain boundaries

POLYCRYSTALLINE STRUCTURE

Grain size: 10 nm – 1 cm (10^{-6} – 10^{-2} m)

Change in crystal orientation at a grain boundary in polycrystals



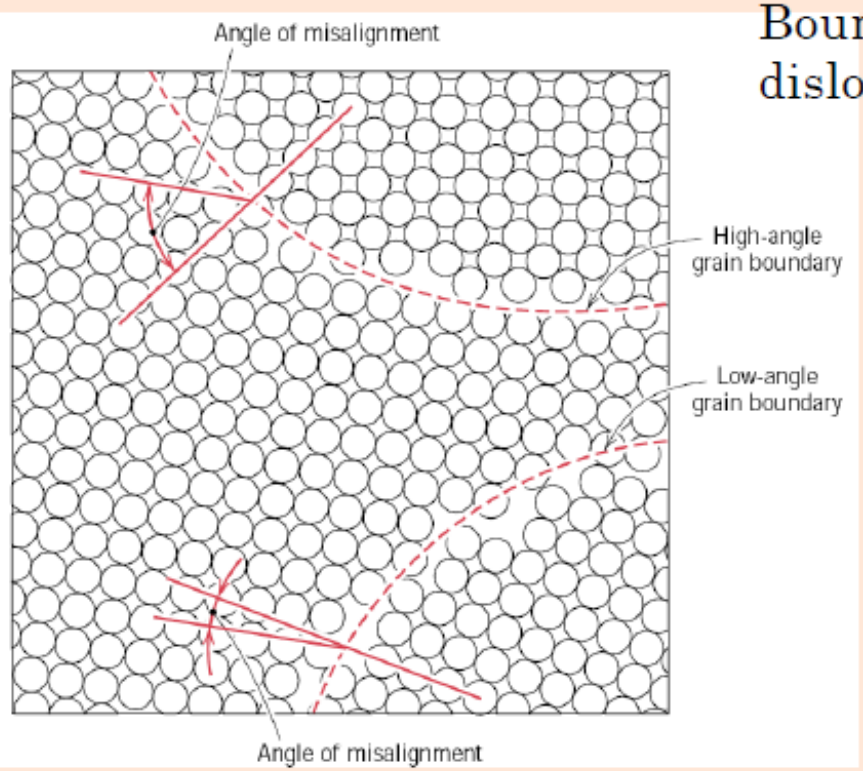


Optical micrograph of an etched polycrystalline sample of an Fe-Cr alloy, showing grain boundaries (dark lines)

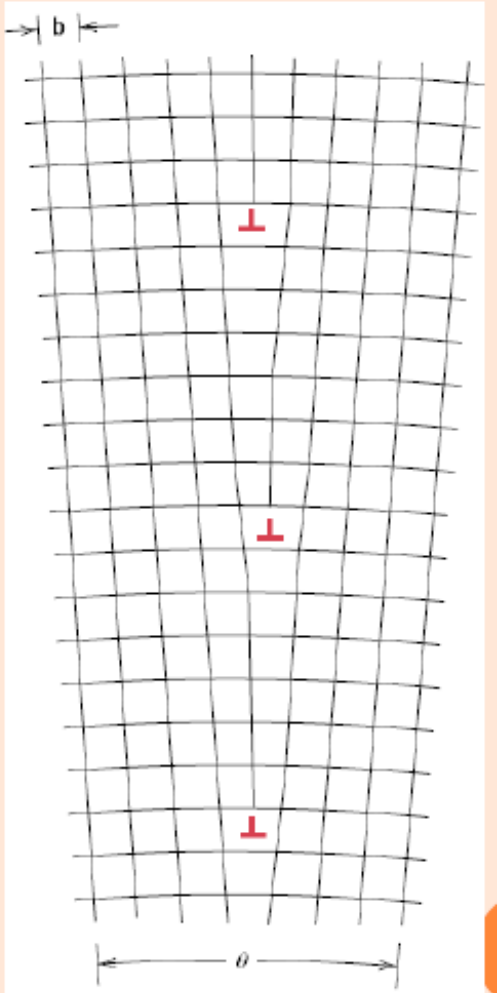
The grain boundaries is a narrow zone where the atoms are not properly spaced

Grain boundaries may be also considered as defects!

GRAIN BOUNDARIES



Boundaries can be described in terms of dislocation arrays



The atoms are bonded less regularly along a grain boundary, there is an interfacial or **grain boundary energy**

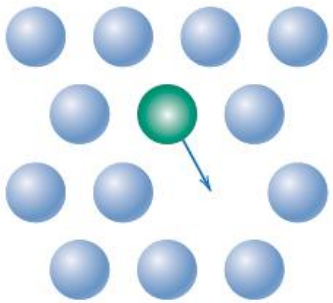
Bulk Defects

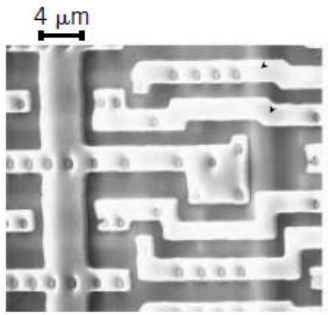
- **Pores** ***(esp. ceramics)*** **-** can greatly affect optical, thermal, mechanical properties
- **Cracks -** can greatly affect mechanical properties
- **Foreign inclusions -** can greatly affect electrical, mechanical, optical properties

Crystal Defects - Processing/Structure/Properties/Performance - Examples

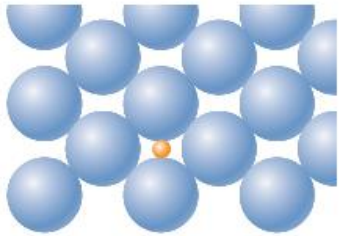
Specification of composition → **Diffusion in semiconductors** → **Fabrication of integrated circuits**

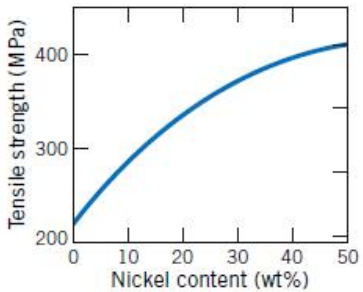
$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$



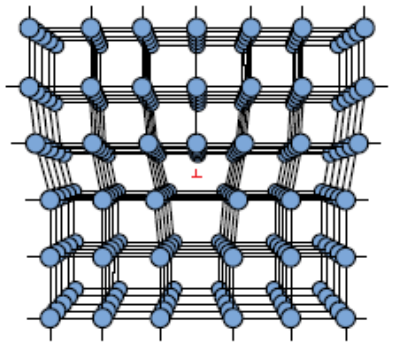


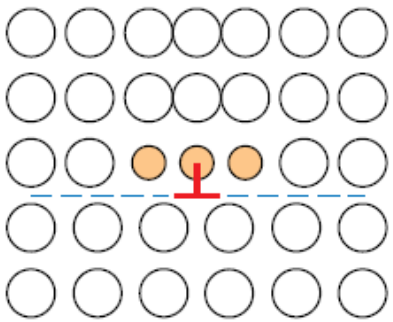
Concept of solid solution → **Solid solution strengthening**



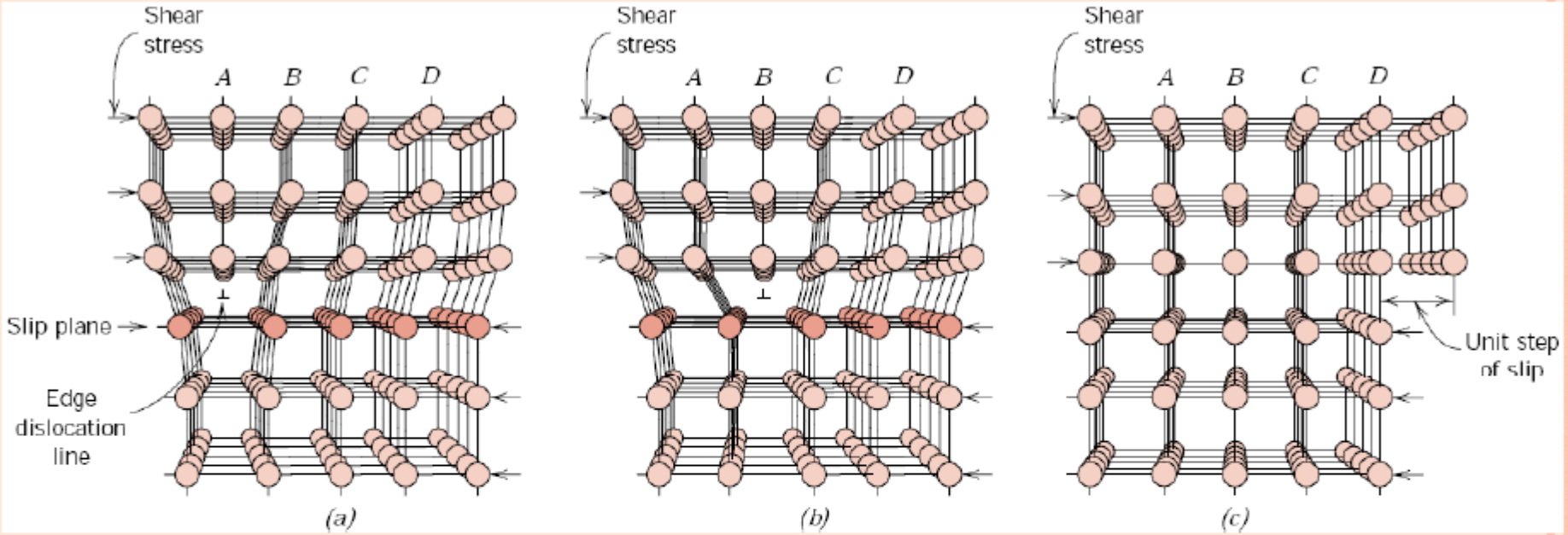


Concept of dislocation defects → **Strengthening mechanisms**





Dislocation movement



When a shear stress is applied to the dislocation, the atoms are displaced, causing the dislocation to move one Burgers vector in the slip direction

Continued movement of the dislocation eventually creates a step

The crystal is deformed

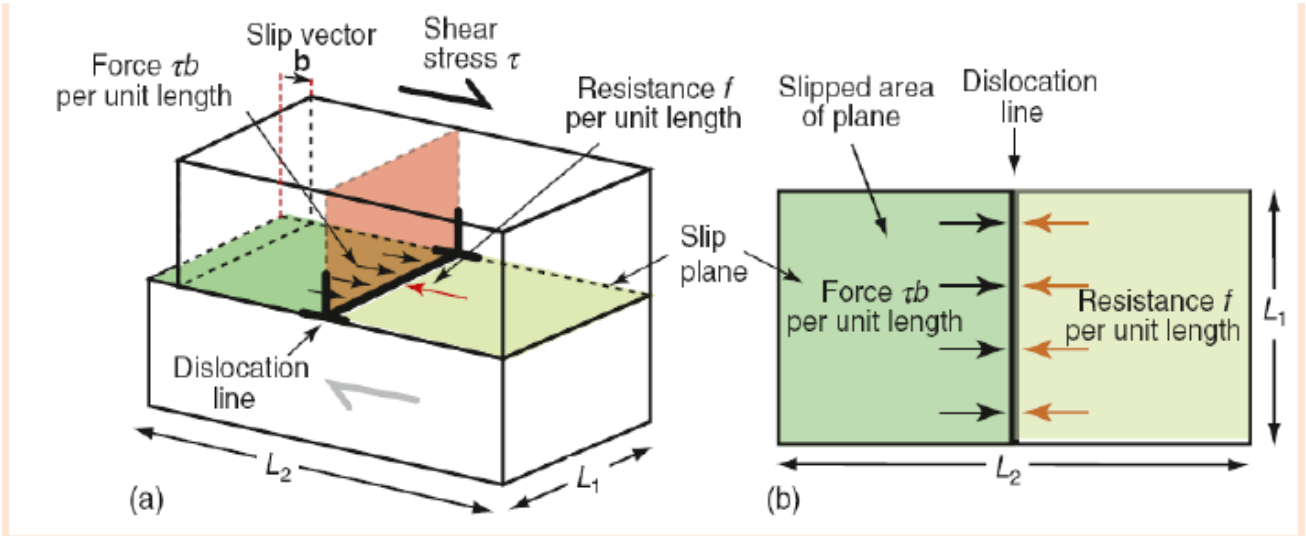
Shear stress for dislocation movement

Crystals resist the motion of dislocations with a friction-like resistance f per unit length

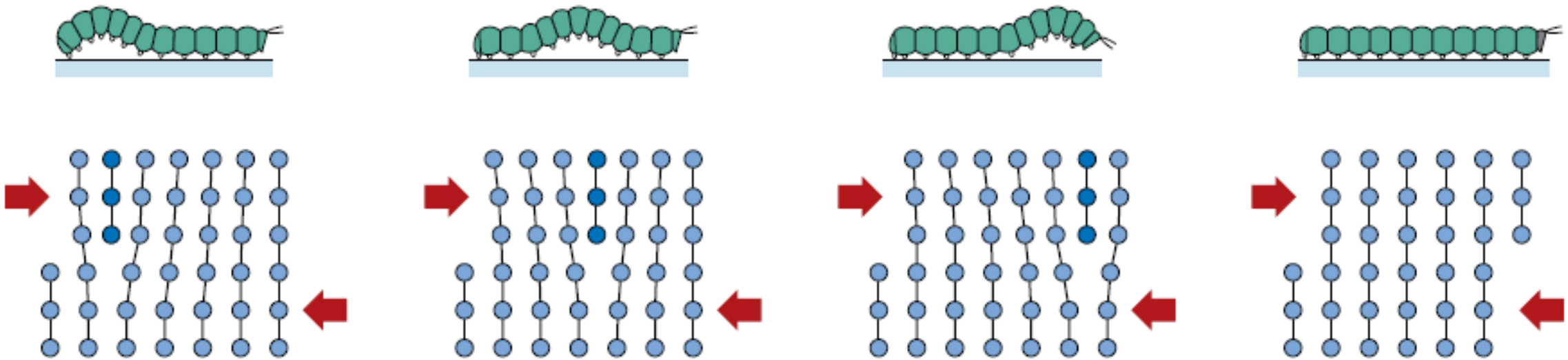
Dislocations move from an applied shear stress τ – as they move the upper half of the crystal shifts relative to the lower half by a distance b

$$\tau b = f$$

Dislocations move if $\boldsymbol{\tau}$ exceeds f/b



The analogy between caterpillar and dislocation motion.



It is easier for dislocation to move through a crystal, by breaking and remaking bonds along the line as its moves (than compared to complete breaking and rearrangement)

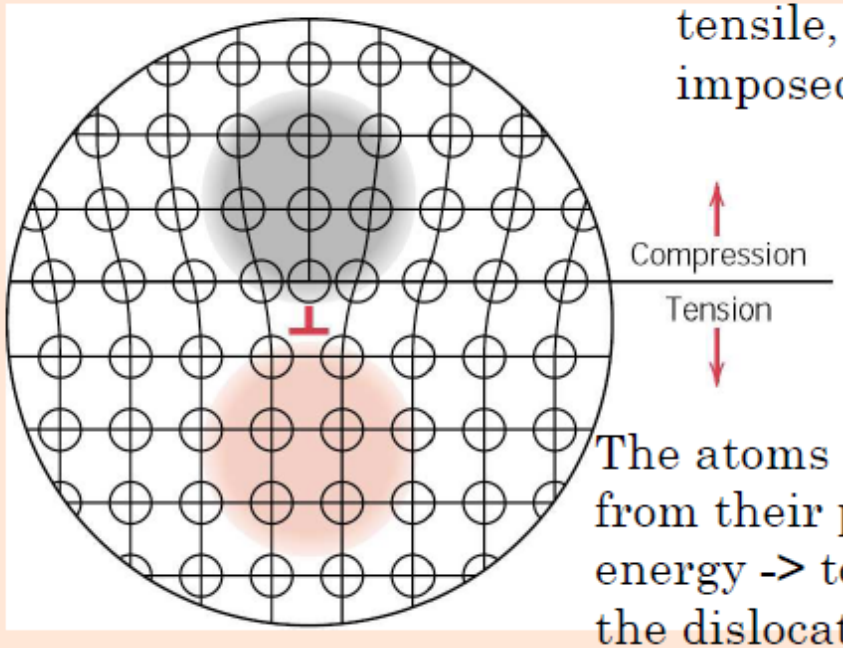
Dislocation density

✓ Total dislocation length per unit volume or,
✓ The number of dislocations in a unit area of a random section.

- The units of dislocation density are millimeters of dislocation per cubic millimeter or just per square millimeter.
- Dislocation densities as low as $10^3/mm^2$ are typically found in carefully solidified metal crystals.
- For heavily deformed metals, the density may run as high as 10^9 to 10^{10} $/mm^2$.
- Heat-treating a deformed metal specimen can diminish the density to on the order of 10^5 to 10^6 $/mm^2$.
- For silicon single crystals used in integrated circuits, the value normally lies between 0.1 and 1 $/mm^2$.

INTERACTION OF DISLOCATIONS

When metals are plastically deformed, some fraction of the deformation energy (~ 5%) is retained internally; the remainder is dissipated as heat. The major portion of this stored energy is as **strain energy** associated with dislocations.



There are regions in which compressive, tensile, and shear **lattice strains** are imposed on the neighboring atoms

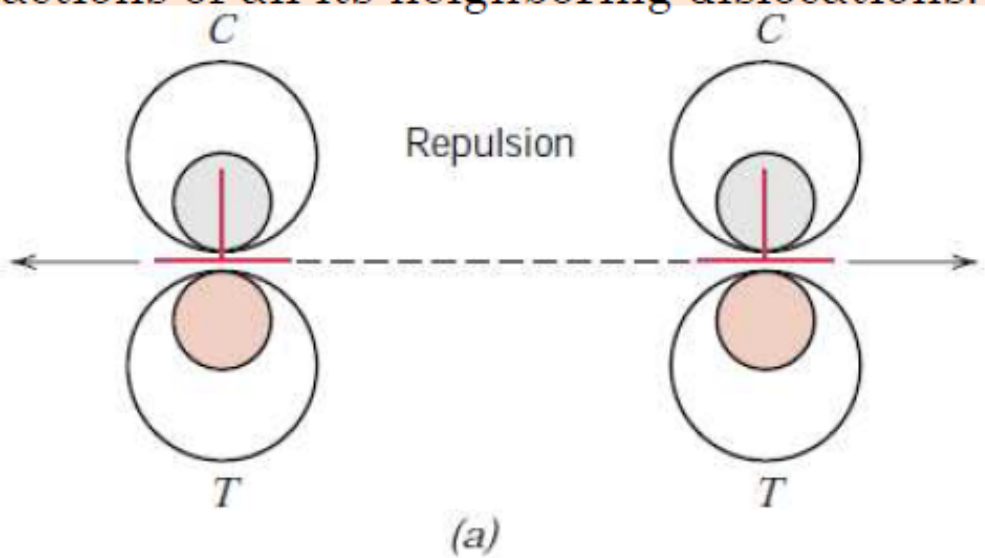
The strains extend into the surrounding atoms, and their magnitudes decrease with radial distance from the dislocation.

The atoms near core of dislocation are displaced from their proper places -> higher potential energy -> to keep the energy as low as possible, the dislocations should be as short as possible

Lattice strains. Slight displacements of atoms relative to their normal lattice positions, normally imposed by crystalline defects such as dislocations, and interstitial and impurity atoms.

DISLOCATIONS INTERACTIONS: REPULSION

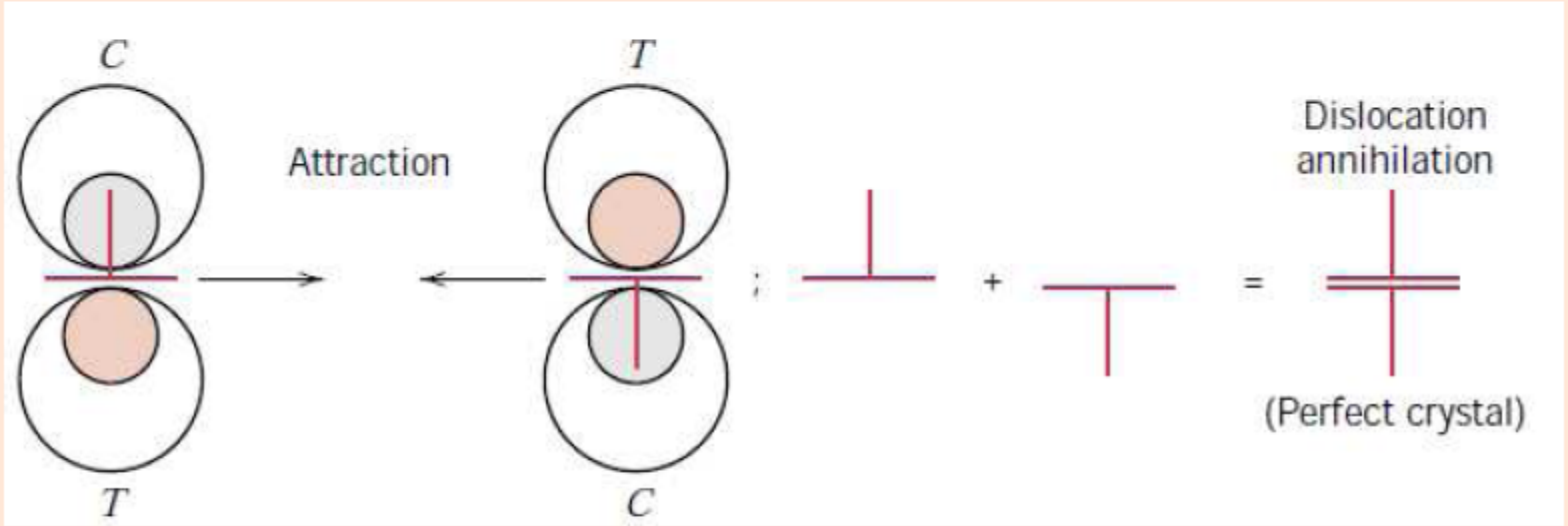
The strain fields surrounding dislocations in close proximity to one another may **interact** such that forces are imposed on each dislocation by the combined interactions of all its neighboring dislocations.



Two edge dislocations of the same sign and lying on the same slip plane exert a repulsive force on each other

C and T denote compression and tensile regions, respectively.

DISLOCATIONS INTERACTIONS: ANNIHILATION



Edge dislocations of opposite sign and lying on the same slip plane exert an attractive force on each other.

Upon meeting, they annihilate each other and leave a region of perfect crystal.

DISLOCATION SLIP (Deformation by Slip)

- **Slip** - The process by which a dislocation moves and deforms a material.
Dislocations **do not** move with the same degree of ease on all crystallographic planes of atoms and in all crystallographic directions.

→ ***Movement of dislocation***

- **Slip direction** - The direction in which a dislocation moves.

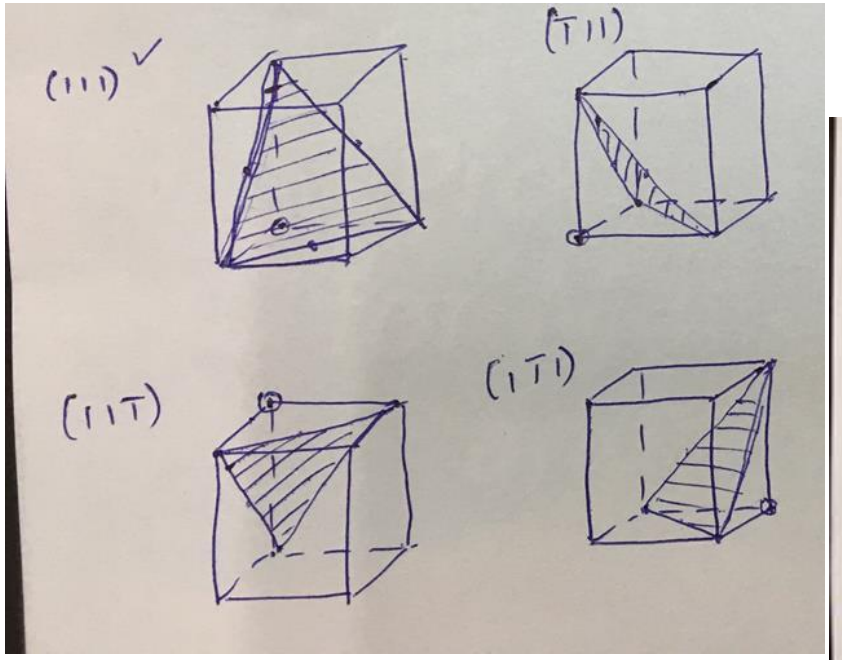
- **Slip plane** - The plane of preferred dislocation movement.

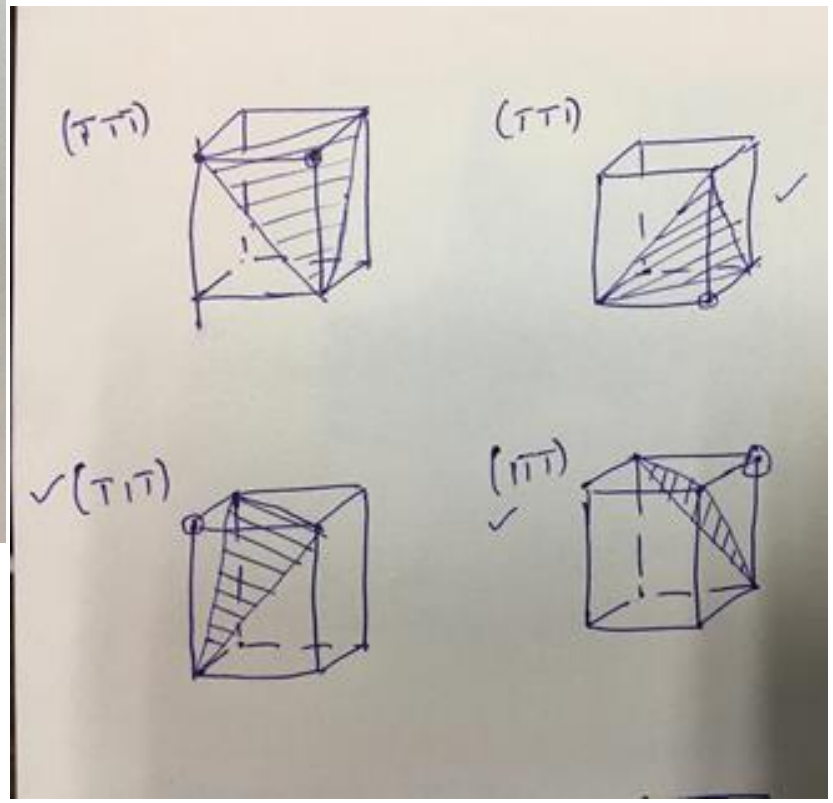
- **Slip systems** - The combination of the slip direction and slip plane makes up the slip system

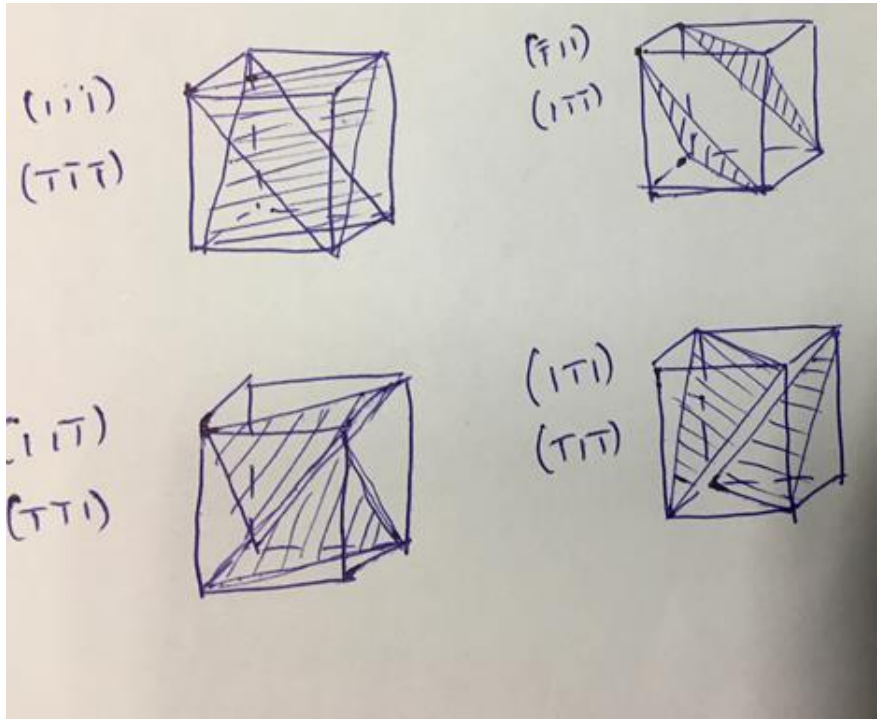
<u>Plastic deformation</u> corresponds to the motion of large numbers of dislocations.

(111) Planes in FCC Crystal

= (111), $(\bar{1}11)$, $(1\bar{1}1)$, $(11\bar{1})$, $(\bar{1}\bar{1}\bar{1})$, $(\bar{1}\bar{1}1)$, $(\bar{1}1\bar{1})$, $(1\bar{1}\bar{1})$

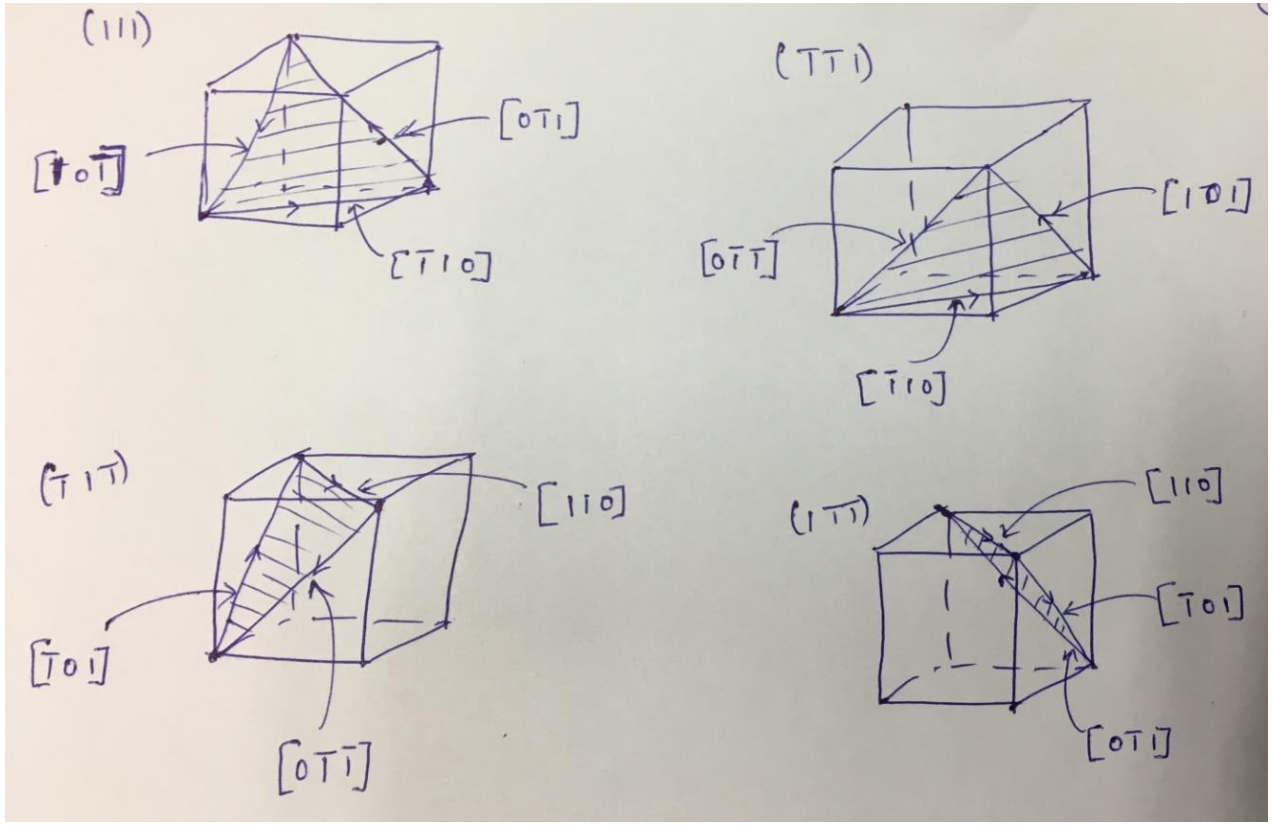






[110] Directions in FCC Crystal

<110> [110], [$\bar{1}$10], [1 $\bar{1}$0], [$\bar{1}$ $\bar{1}$0], [101], [$\bar{1}$01], [10 $\bar{1}$], [$\bar{1}$0 $\bar{1}$], [011], [0 $\bar{1}$1], [01 $\bar{1}$], [0 $\bar{1}$ $\bar{1}$]



I set - (111) <110> , (111) <110>, (111) <110>
II set - (111) <110> , (111) <110>, (111) <110>
III set - (111) <110> , (111) <110>, (111) <110>
IV set - (111) <110> , (111) <110>, (111) <110>

Four unique planes and each contain three unique directions

- The combination of a slip direction and slip plane = slip system
- There are four nonparallel {111} planes
- Each contains three nonparallel directions <110>
- Thus 12 slip systems in FCC are collectively represented as {111}<110>

Family of Planes

- Planes that are crystallographically equivalent have the same atomic packing.
- Also, in cubic systems only, planes having the same indices, regardless of order and sign, are equivalent.

- Ex: {111}

  $= (111), (\bar{1}11), (1\bar{1}1), (11\bar{1}), (\bar{1}\bar{1}\bar{1}), (\bar{1}\bar{1}1), (\bar{1}1\bar{1}), (1\bar{1}\bar{1})$

Ex: $\{100\} = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$

*A family of **planes** contains all the **planes** that are crystallographically **equivalent** —that is, having the same atomic packing*

- **A family of planes contains all the planes that are crystallographically equivalent.**
- **In cubic system, planes with same indices, irrespective of order and sign, are equivalent**
  - $(111), (\bar{1}\bar{1}\bar{1}), (\bar{1}11)$ … belong to {111} family
  - $(\bar{1}00), (100), (010)$, and (001) belong to {100} family
  - $(123), (\bar{1}23), (\bar{3}12)$ in cubic crystals belong to {123} family

Family of directions

Index	Members in family for **cubic lattice**
<100>	[100],[$\bar{1}$00],[010],[0$\bar{1}$0],[001],[00$\bar{1}$]
<110>	[110],[$\bar{1}$10],[1$\bar{1}$0],[$\bar{1}\bar{1}$0],[101],[$\bar{1}$01],[10$\bar{1}$],[$\bar{1}$0$\bar{1}$],[011],[0$\bar{1}$1],[01$\bar{1}$],[0$\bar{1}\bar{1}$]
<111>	[111],[$\bar{1}$11],[1$\bar{1}$1],[11$\bar{1}$],[$\bar{1}\bar{1}$1],[$\bar{1}$1$\bar{1}$],[1$\bar{1}\bar{1}$],[$\bar{1}\bar{1}\bar{1}$]

Linear and planar densities

Linear and planar densities are important considerations relative to the process of slip—that is, the mechanism by which metals plastically deform.

Slip occurs on the most densely packed crystallographic planes and, in those planes, along directions having the greatest atomic packing.

Slip Systems in BCC crystal

There are three types of
planes which are nearly tied for the highest packing density. They are:
near close − packed planes in BCC : {110} {321} {211}
All these planes have been observed to be slip planes. Thus we have the following slip systems in BCC crystals:

- 6 {110} planes each with 2 < 111 > directions
- 24 {321} planes each with 1 < 111 > direction
- 12 {211} planes each with 1 < 111 > direction

$\Rightarrow 6 \times 2 = 12$

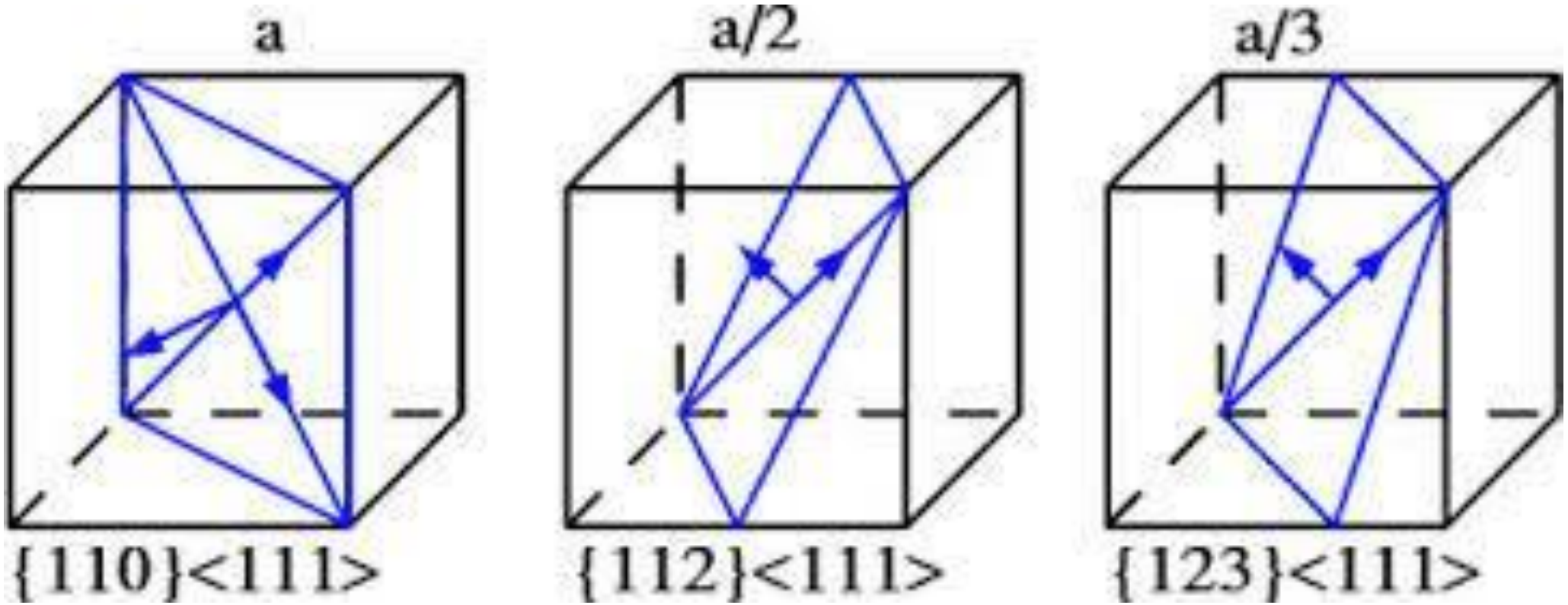
$\rightarrow 24$

$\rightarrow 12$

48

Total = 48 slip systems in BCC

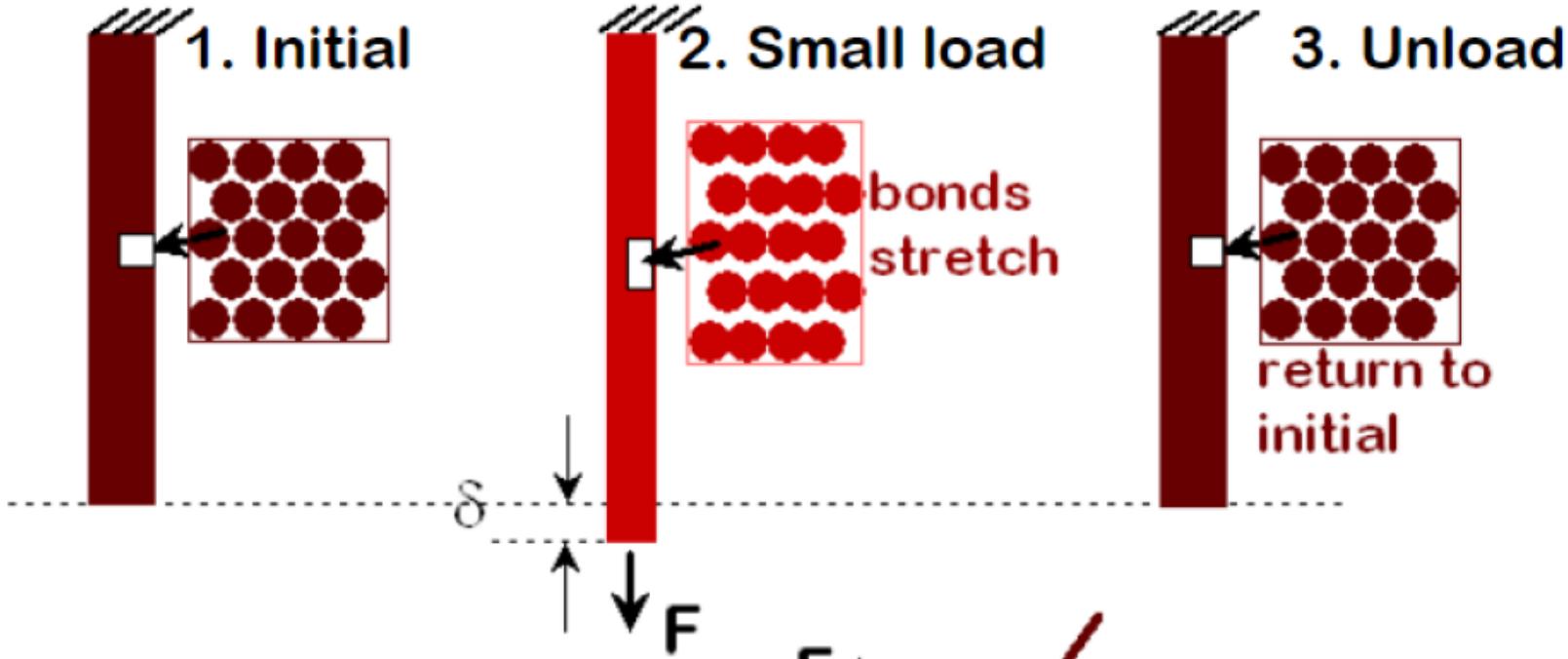
Slip Systems in BCC crystal



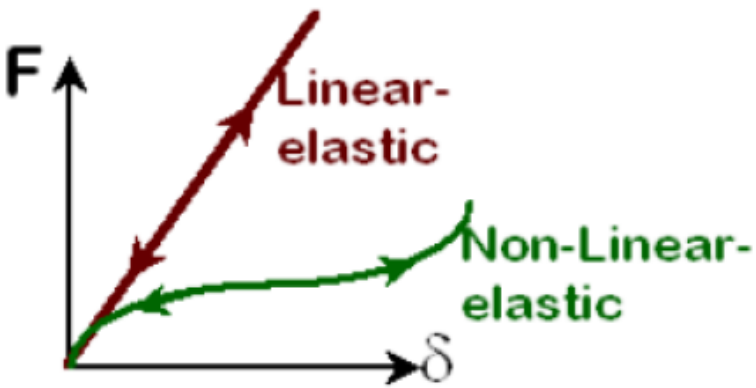
Slip Systems

Metals	*Slip Plane*	*Slip Direction*	*Number of Slip Systems*
	Face-Centered Cubic		
Cu, Al, Ni, Ag, Au	{111}	⟨110⟩	12
	Body-Centered Cubic		
α-Fe, W, Mo	{110}	⟨111⟩	12
α-Fe,W	{211}	⟨111⟩	12
α-Fe, K	{321}	⟨111⟩	24
	Hexagonal Close-Packed		
Cd, Zn, Mg, Ti, Be	{0001}	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg, Zr	$\{10\bar{1}0\}$	$\langle 11\bar{2}0 \rangle$	3
Ti, Mg	$\{10\bar{1}1\}$	$\langle 11\bar{2}0 \rangle$	6

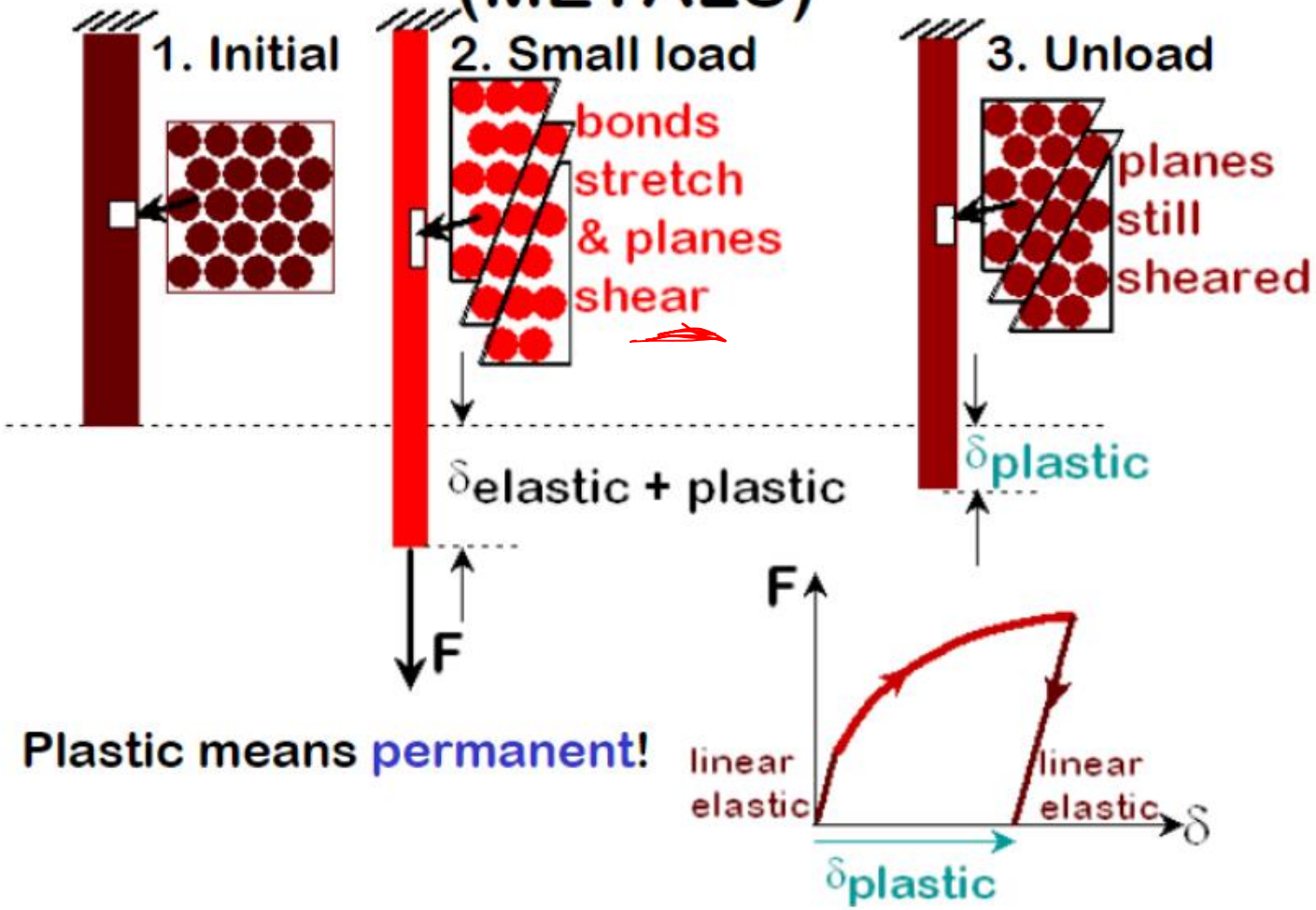
ELASTIC DEFORMATION



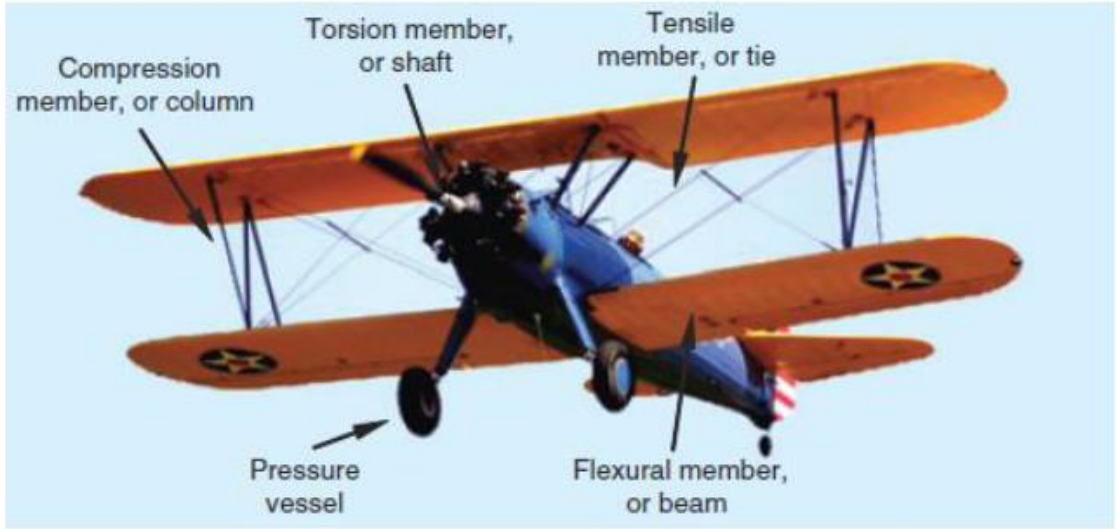
Elastic means reversible!



PLASTIC DEFORMATION (METALS)



Plastic means permanent!

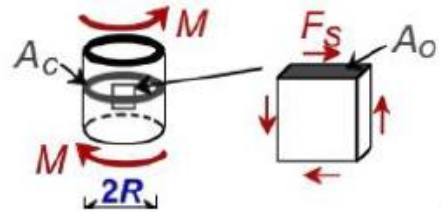


Real Systems of Stress

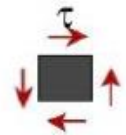
- **Simple tension: cable**

$$\sigma = \frac{F}{A_o}$$

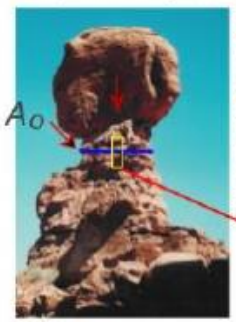
- **Torsion (a form of shear): drive shaft**



$$\tau = \frac{F_s}{A_o}$$



Simple compression:



$$\sigma = \frac{F}{A_o}$$

Materials for Engineers

Lecture 11-12

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Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

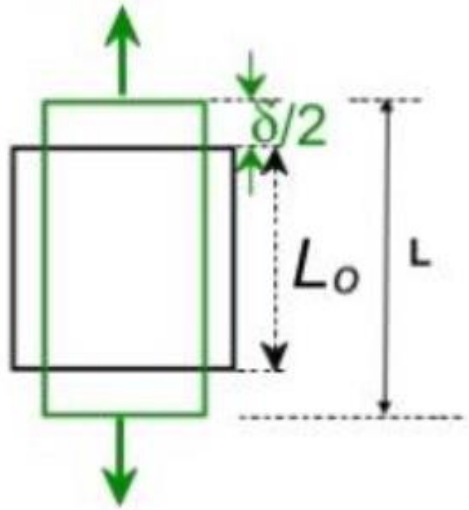
Reference:
1. Materials science and engineering by William Callister, Wiley Publications

Engineering Strain-Definition

- **Tensile strain:**

define

$$d\varepsilon = \frac{dL}{L}$$



Integrate to get ε, from L_o to any L

$$\varepsilon = \frac{\delta \text{ (or } \Delta L)}{L_o}$$

Where $\Delta L = L - L_o$

With L_o (constant) → Engineering Strain*

At any instant $AL = A_oL_o$ initial; (Constant Volume)

Stress Strain Behavior

Two behaviors: low loads versus large loads

Elastic Range

Initially, stress and strain are directly **proportional** to each other

Why: atoms can be thought of as masses connected to each other through a network of **springs** (*Imagine*)

According to **Hooke's law:**

the extension of a spring, x, and the applied force, F, are related by the spring constant, k:

$$F = kx$$

Thus, Stress (F/A) must be linear with strain

This is Stretching of bonds

Plastic Range

Non-Linear relation of stress with strain

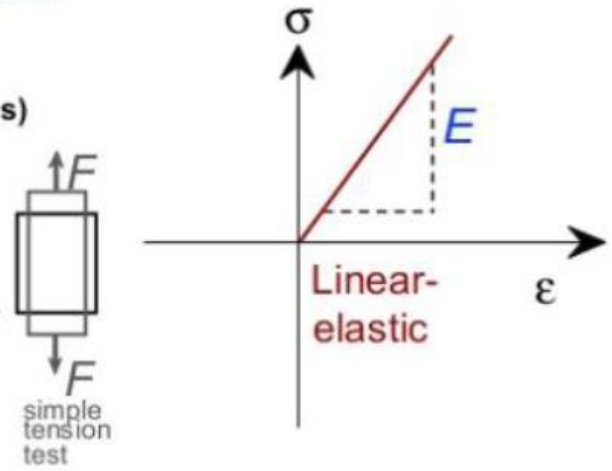
→ breaking of bonds and forming new bonds

Linear Elastic Properties

At low levels of stress: the shape is recoverable
The deformation is reversible

- **Modulus of Elasticity, *E*:**
  (also known as Young's modulus)
- **Hooke's Law:**

$$\sigma = E\,\varepsilon$$

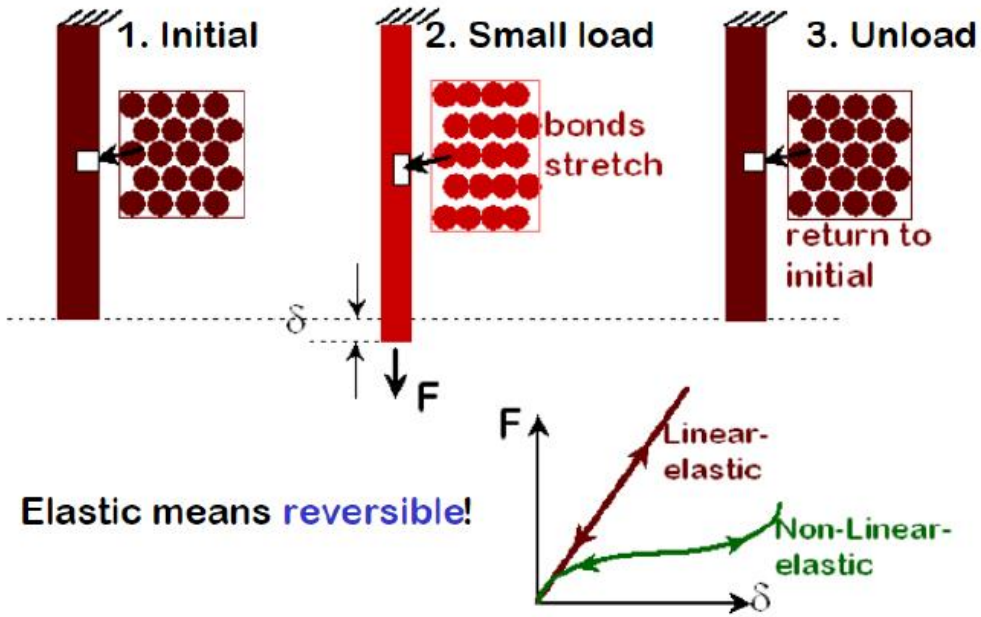


Linearity for Tension

$$\text{Young's Modulus} = E = \frac{\Delta\sigma}{\Delta\varepsilon}$$

E for steel ~ 200 Gpa
E for Al ~ 70 GPa

ELASTIC DEFORMATION



Elastic means reversible!

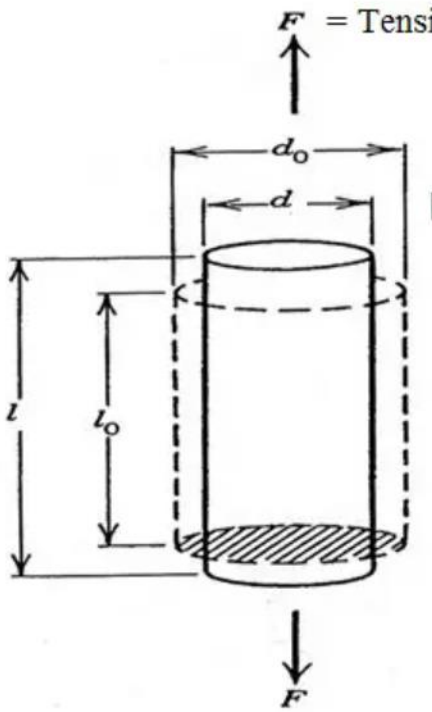
Poisson's ratio, ν

→ Ratio of transverse strain to longitudinal strain

Upon Elongation in one direction (z) i.e. ε_z is +ve

→ contraction occurs in the other two directions (ε_x and ε_y)

Define Poisson's ratio, ν: : how much strain occurs in the lateral directions (x& y) when strained in the (z) direction:



Poisson Ratio = $\frac{\text{Lateral Strain}}{\text{Linear Strain}}$

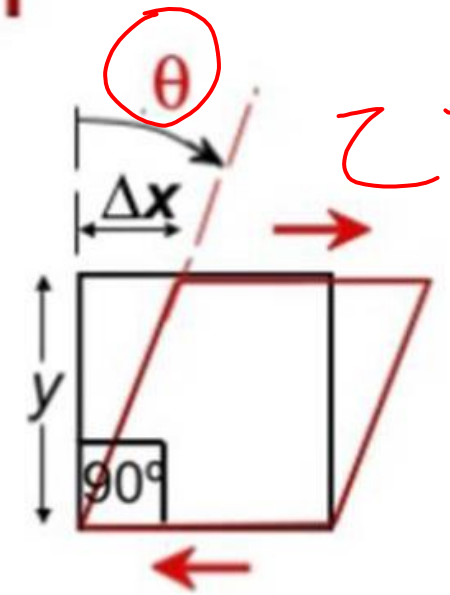
$$\text{Linear strain} = \frac{\Delta l}{l} = \frac{l - l_o}{l_o}$$

$$\text{Lateral Strain} = \frac{\Delta d}{d_o} = \frac{d - d_o}{d_o}$$

Shear Strain-Definition

Define $\gamma = \Delta x/y = \tan\theta$

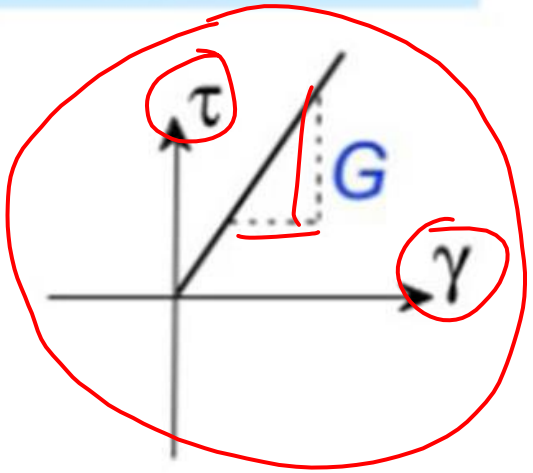
Remember: Strain is always dimensionless.



$\tau = \frac{F}{A}$

Elastic Shear modulus, *G*:

$$\tau = G\gamma$$



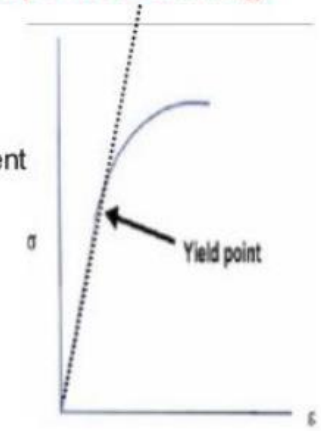
Special relations for isotropic materials:

$$G = \frac{E}{2(1+\nu)}$$

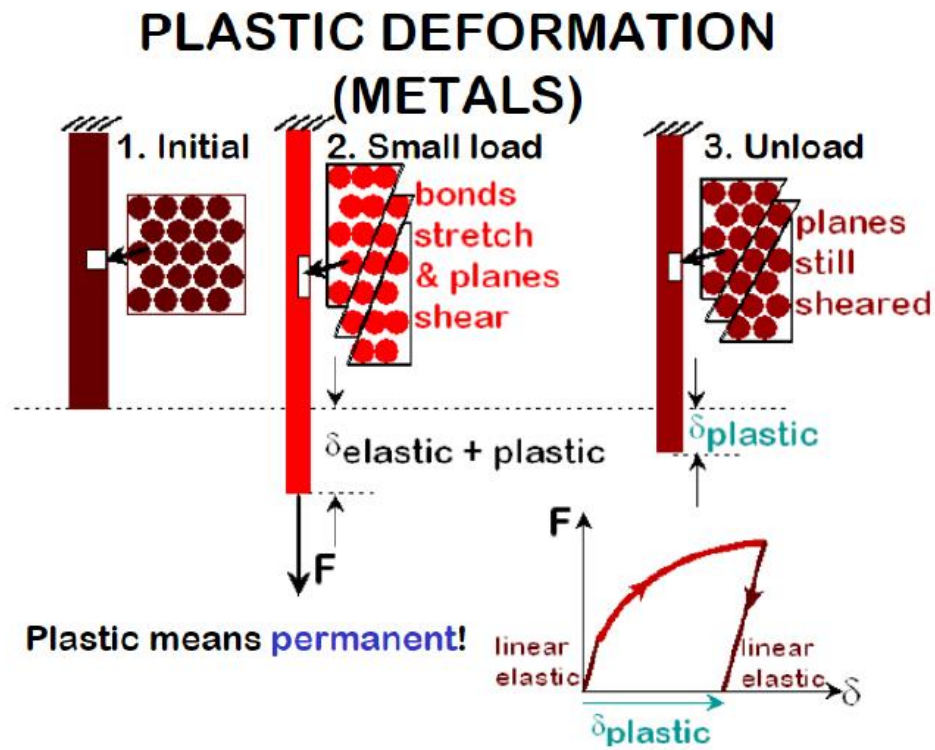
From Elastic to Plastic Behavior

What happens if we continue to apply tensile loading beyond the elastic limit? *(i.e., stretching atomic bonds to the point of breaking)*

Plastic deformation:
- stress and strain are not proportional
- the deformation is not reversible
- deformation occurs by breaking and re-arrangement of atomic bonds (*in*



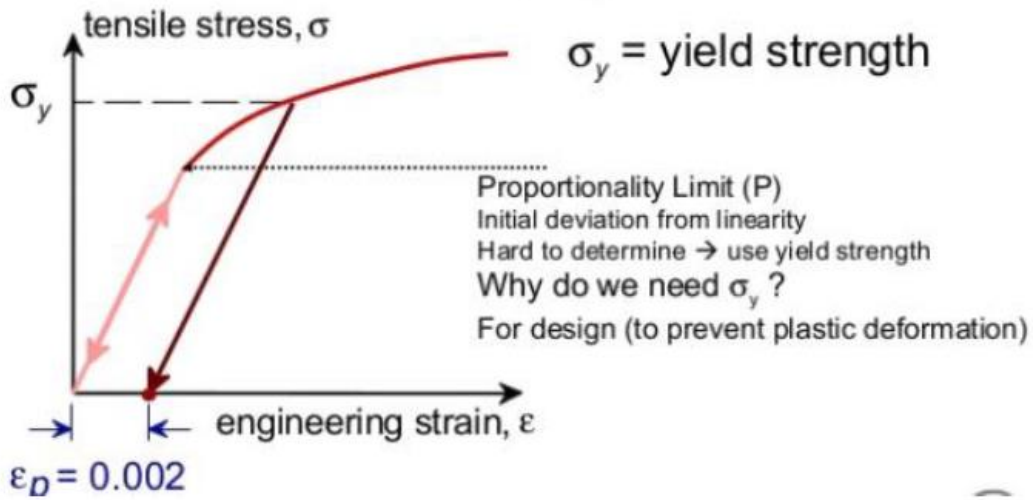
Proportional Limit or *elastic limit*, is the point where
The stress and strain values at this point are known as the proportional-limit stress and strain, respectively.
This is the point beyond which **Hooke's law** can no longer be used – no spring



Yield Strength, σ_y

- **Stress at which *noticeable* plastic deformation has occurred.**

when $\varepsilon_p = 0.002$



σ_y = yield strength

Proportionality Limit (P)
Initial deviation from linearity
Hard to determine → use yield strength

Why do we need σ_y ?

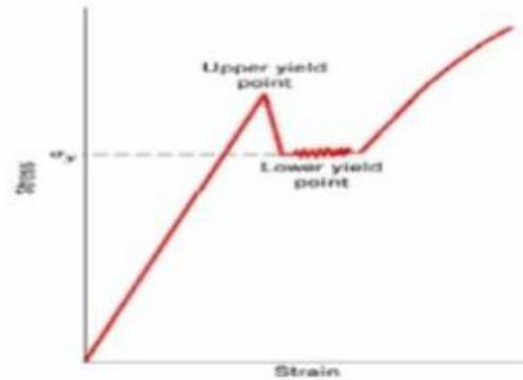
For design (to prevent plastic deformation)

Yield Strength – Clear Case

For a low-carbon steel

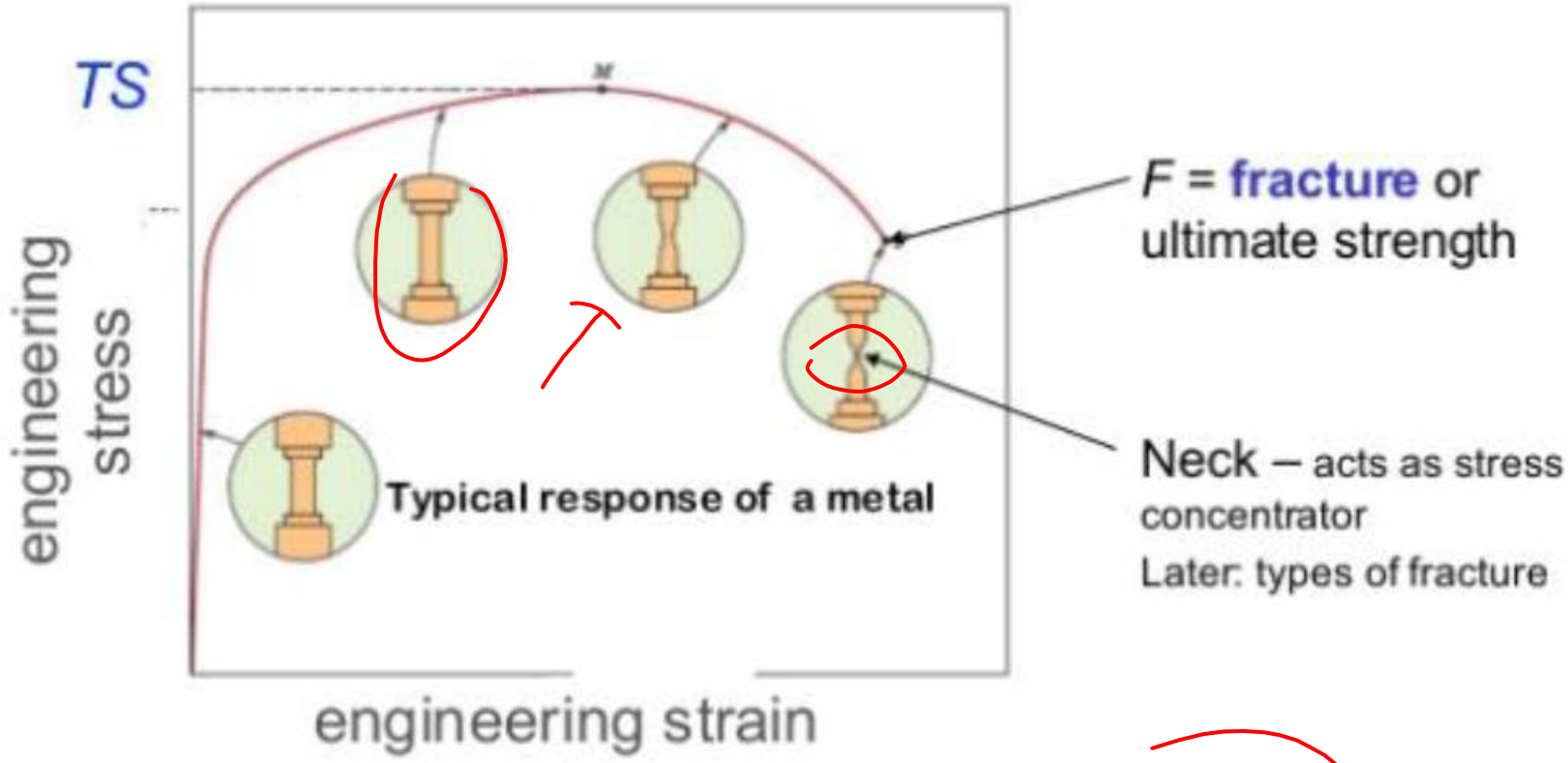
- The stress vs. strain curve includes both an **upper** and **lower** yield point.
- The yield strength is defined in this case as

the **average** stress at the **lower yield** point



Tensile Strength, TS

- **Maximum** stress on engineering stress-strain curve.



For Metals: TS occurs when noticeable necking starts.

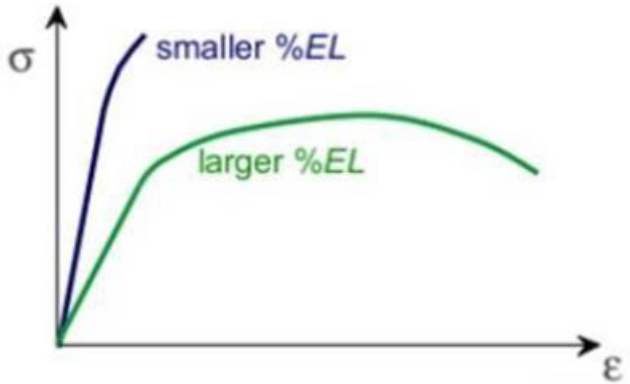
Ductility

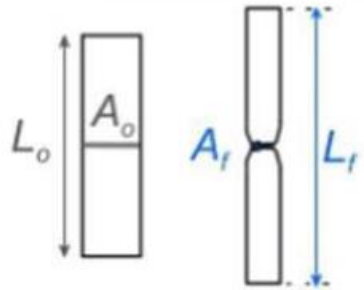
Distinguish Behavior ? ductile versus brittle ?

Define a parameter - ductility: *measures the amount of plastic deformation that a material goes through by the time it breaks.*

1- Elongation:

Percentage tensile strain at failure: $\%EL = \frac{L_f - L_o}{L_o} \times 100$





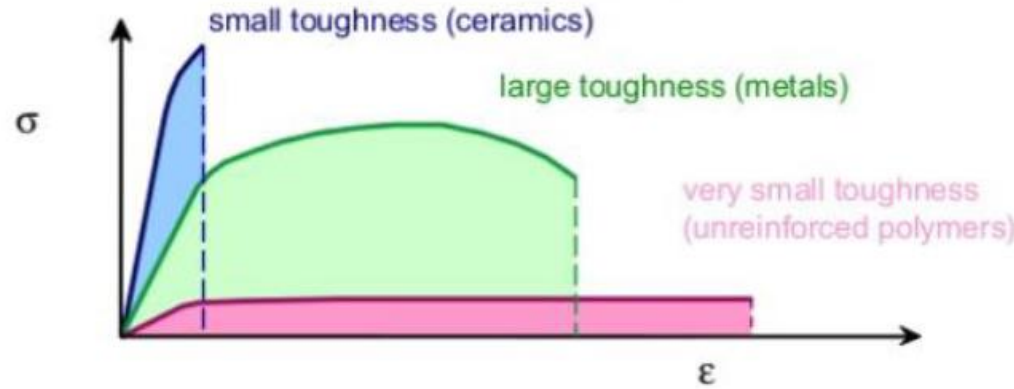
2- Percentage Area Reduction $\%RA = \frac{A_o - A_f}{A_o} \times 100$

Toughness

Energy absorbed by material up to fracture per unit volume

(Energy to break a unit volume of material at low strain rate)

- Approximation: the area under the stress-strain curve.



Brittle fracture: elastic energy

Ductile fracture: elastic + plastic energy

Resilience & Toughness

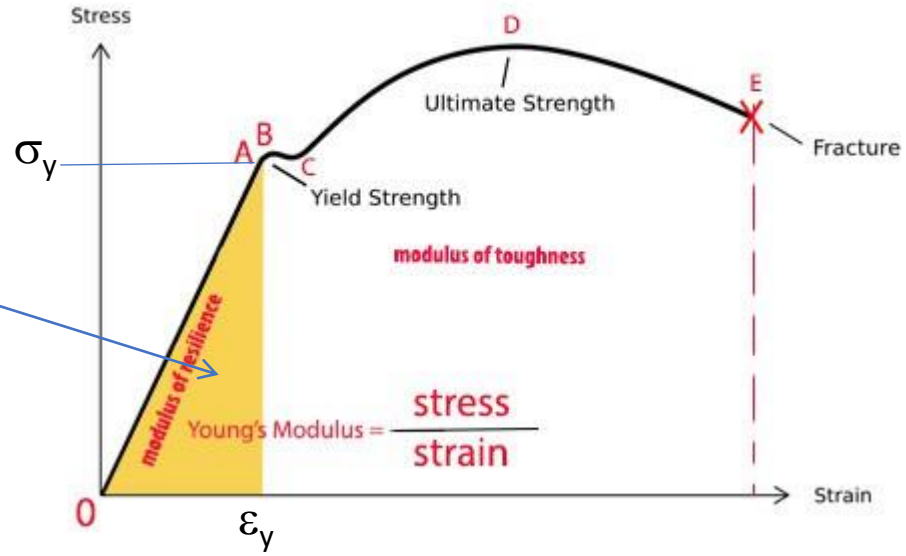
Resilience is the capacity of a material to absorb energy when it is deformed elastically and then, upon unloading, to have this energy recovered.

Assuming a linear elastic region, we have

Energy absorbed (Area under the curve)

$$U_r = \frac{1}{2}\sigma_y \varepsilon_y$$

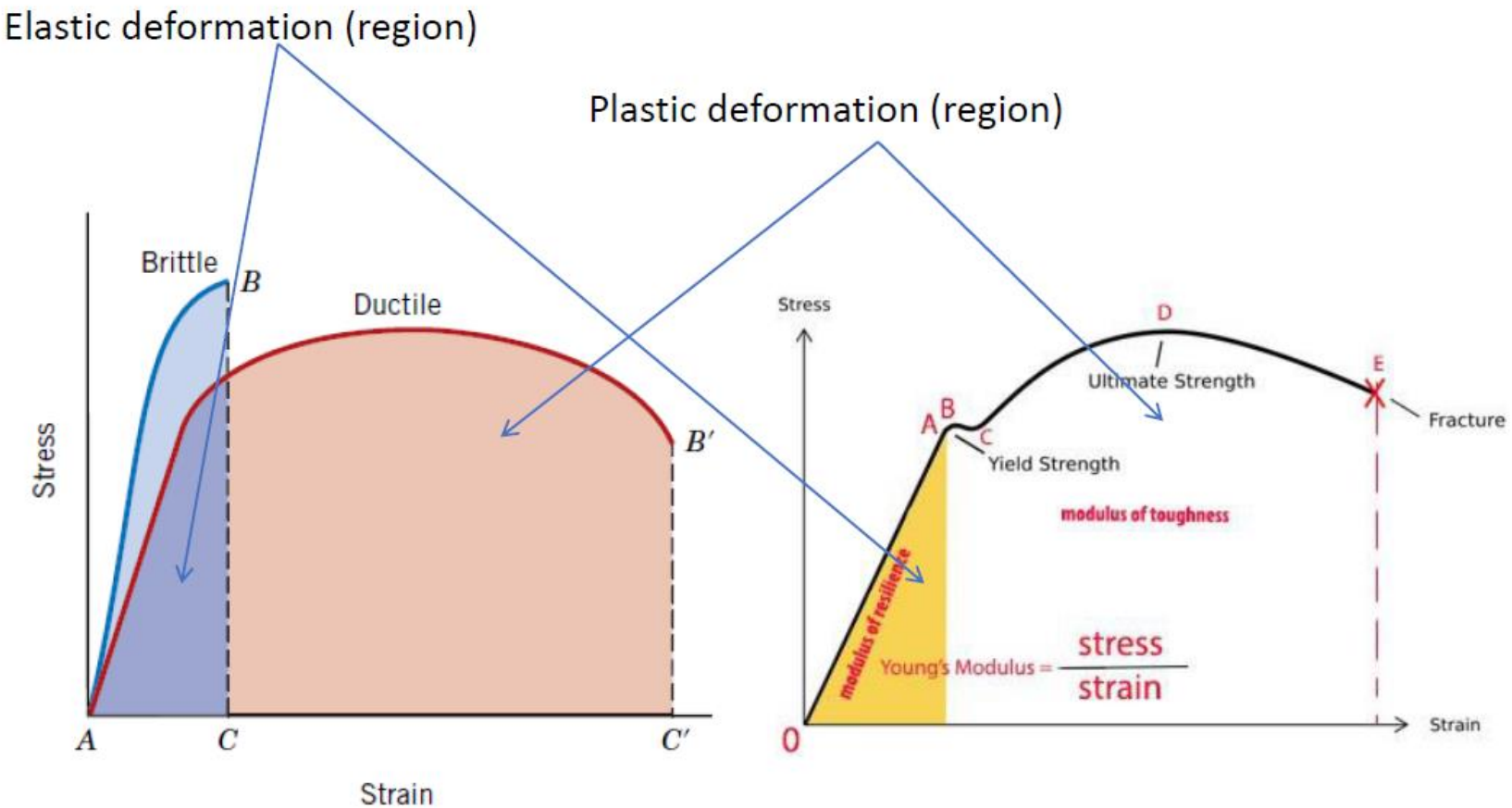
$$U_r = \frac{1}{2}\sigma_y \varepsilon_y = \frac{1}{2}\sigma_y \left(\frac{\sigma_y}{E}\right) = \frac{\sigma_y^2}{2E}$$



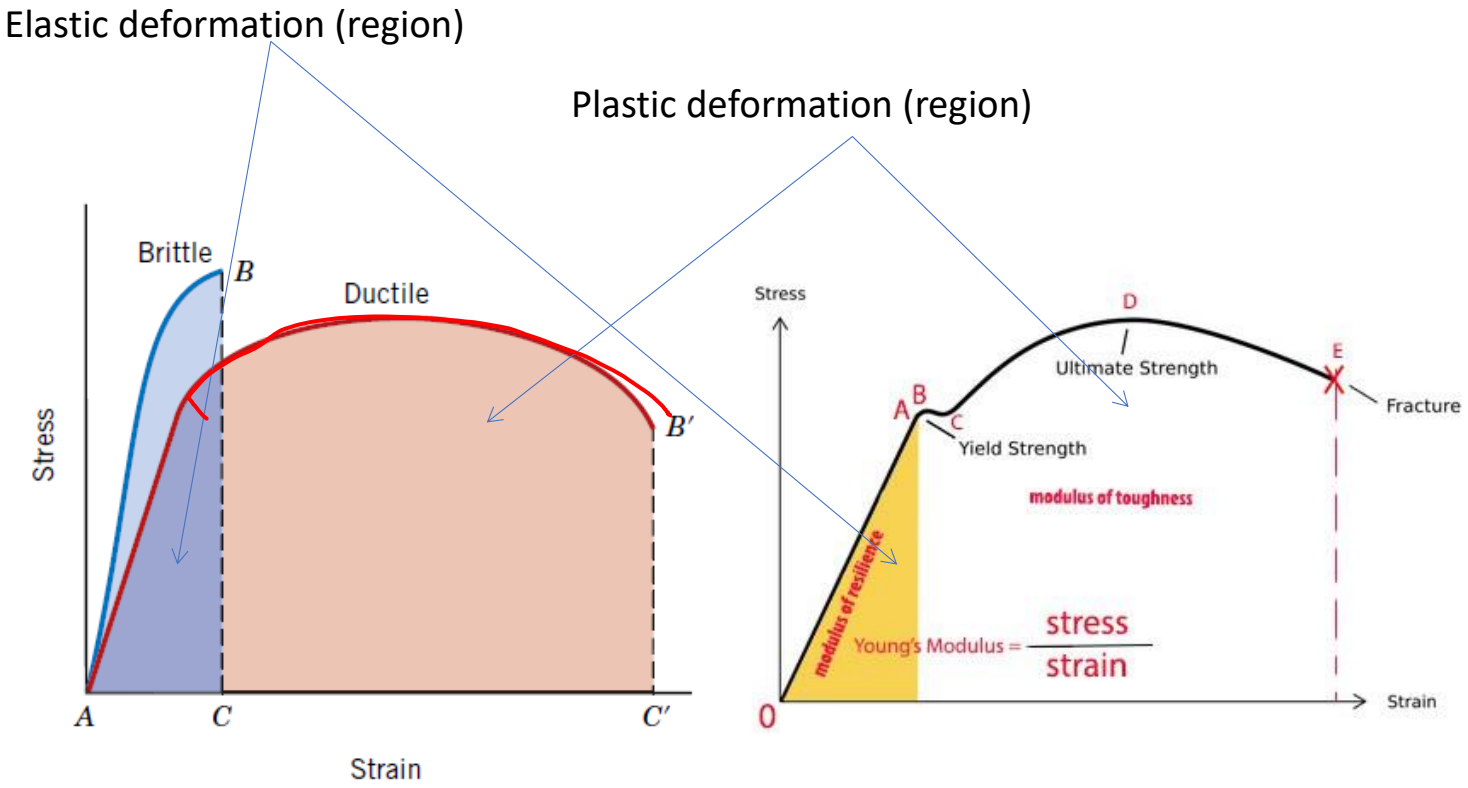
<u>Resilient</u> materials are those *having high yield strengths and low moduli of elasticity*;
such alloys are used in spring applications.

Toughness is as the ability of a material to absorb energy and plastically deform before fracturing.

Summary


Elastic deformation (region)
Plastic deformation (region)
Brittle
Ductile
B
B′
Stress
A
C
C′
Strain
Stress
D
Ultimate Strength
E
Fracture
A
B
C
Yield Strength
modulus of toughness
modulus of resilience
Young's Modulus = stress / strain
0
Strain

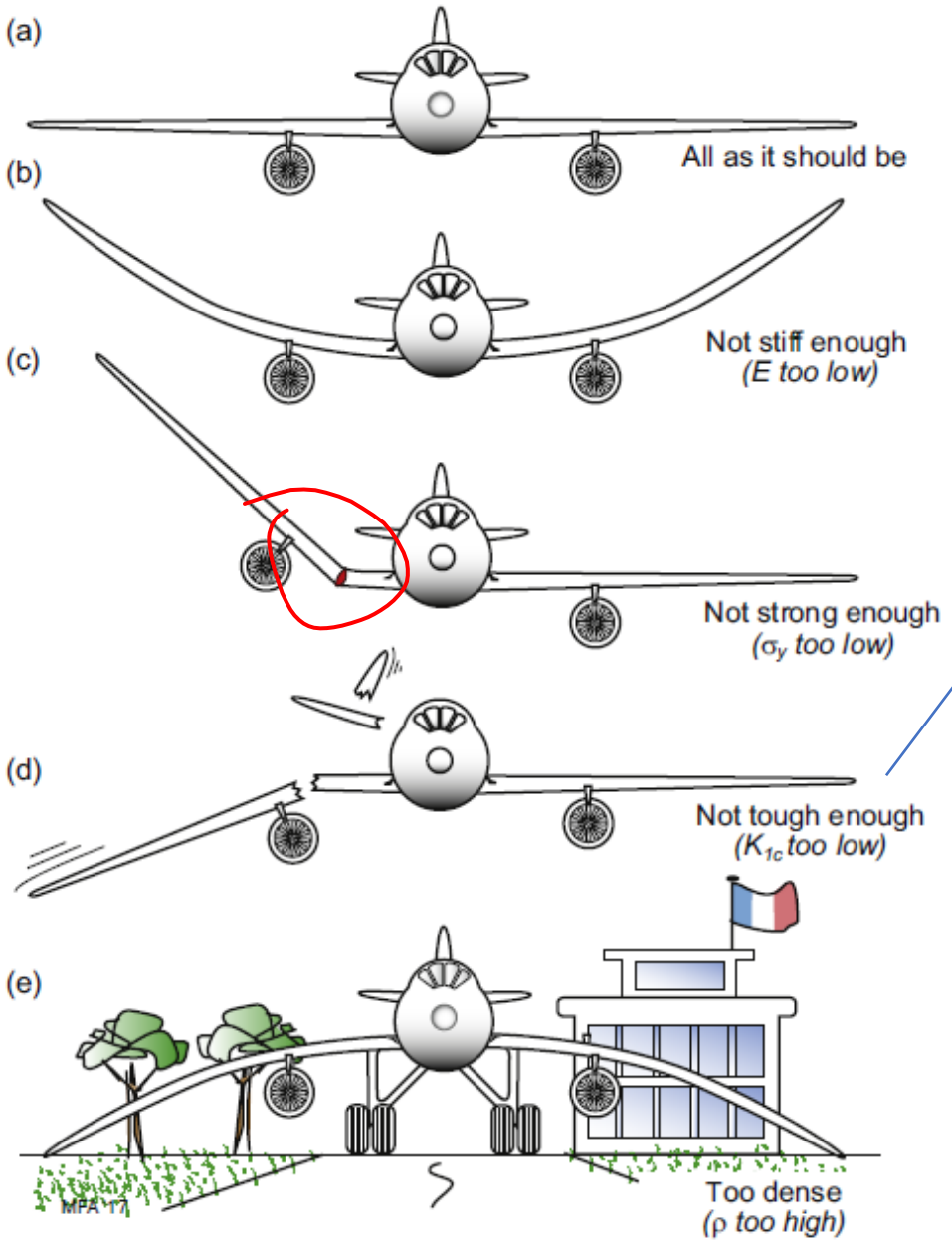
Summary of Mechanical Properties



Definitions of….

- ✓ Tensile load
- ✓ Shear load
- ✓ Compression load
- ✓ Stress
- ✓ Strain
- ✓ Young's modulus
- ✓ Shear modulus
- ✓ Yield strength
- ✓ Tensile (or maximum) strength
- ✓ Fracture strength
- ✓ Longitudinal strain
- ✓ Transverse strain
- ✓ Poisson's ratio
- ✓ Resilience
- ✓ Toughness
- ✓ Ductility
- ✓ Brittle
- ✓ Hardness

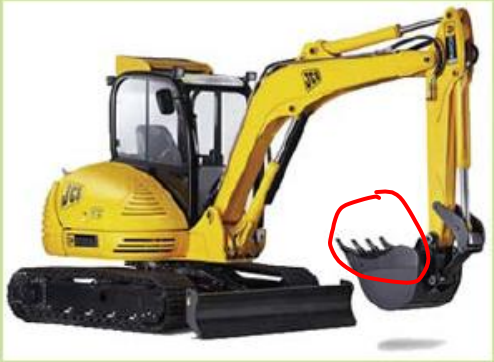
Mechanical Properties – Design Requirements Examples (Selection of Material)



- If there is no permanent deformation, then σ_y is not the right property.
- The resistance of materials to cracking and fracture is measured instead by the fracture toughness, K_{1c}.
- Steels are tough - they have a high K_{1c}.
- Glass is brittle - it has a very low K_{1c}.

The resistance of materials to cracking and fracture is measured by the fracture toughness, K_{1c}.

You are asked to select a material for the teeth of the scoop of a digger truck. To do so, you need to prioritise the material properties that matter. What are they?



Answer. The teeth will be used in a brutal way to cut earth, scoop stones, crunch rock, often in unpleasant environments (ditches, sewers, fresh and salt water and worse), and their maintenance will be neglected. These translate into a need for high hardness, H, to resist wear, and high fracture toughness, K_{1c}, so they don't snap off. Does the cost of the material matter? Not much – it is worth paying for good teeth to avoid expensive downtime.

Materials for Engineers

Lecture 13

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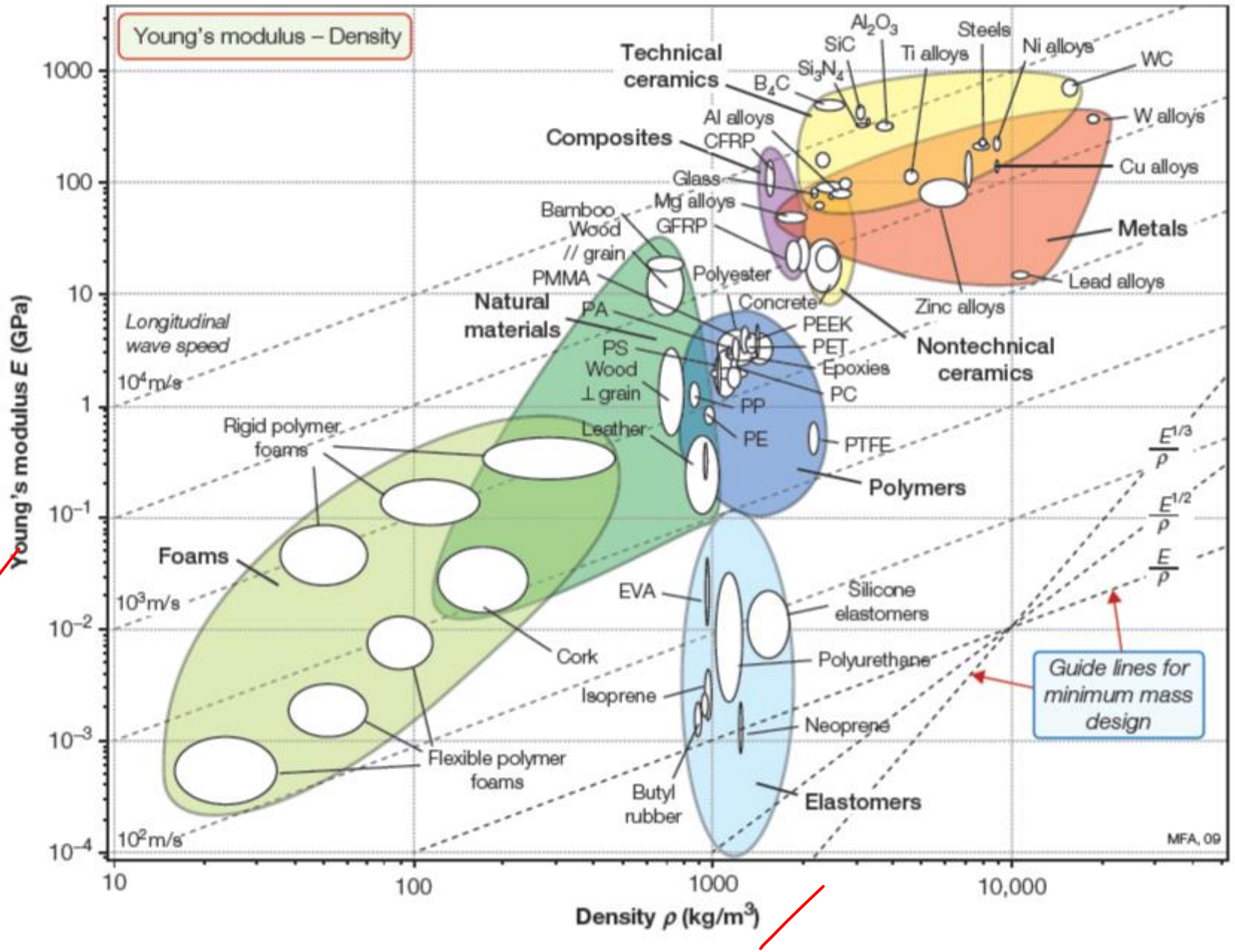
Chennai – 600127

Reference:

1. Materials science and engineering by William Callister, Wiley Publications

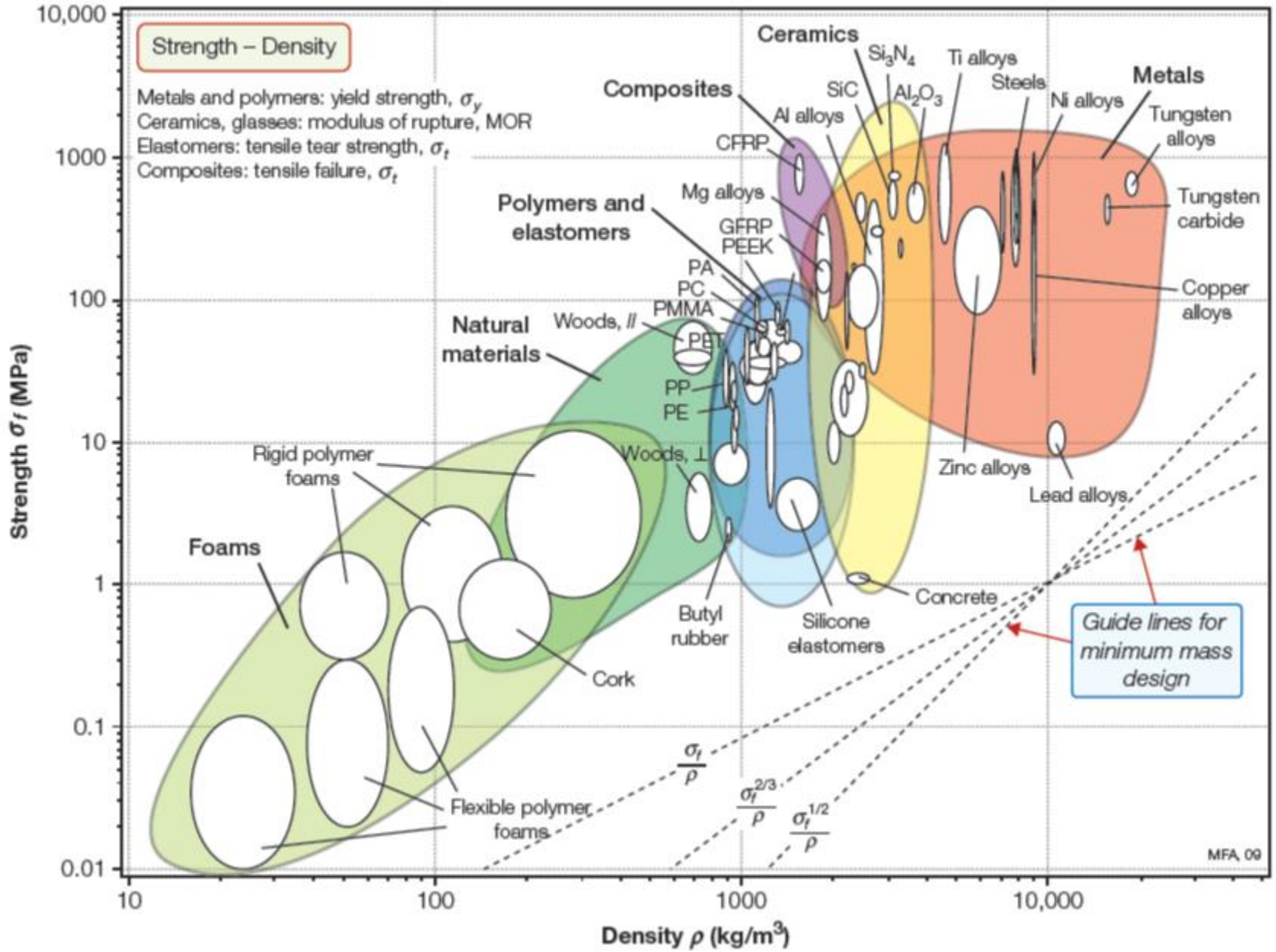
Material Property Charts

Young's Modulus vs Density



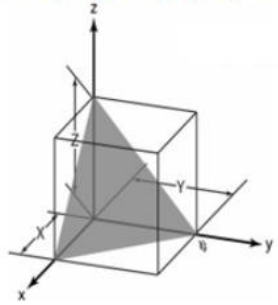
Material Property Charts

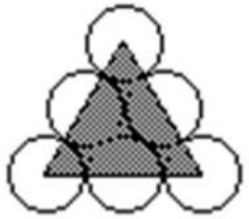
Strength vs Density

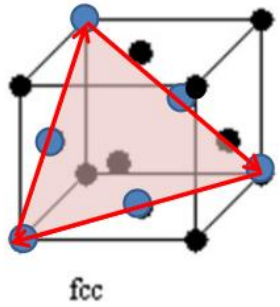


Close-packed planes: FCC

Primary Slip Plane: [111] = normal vector

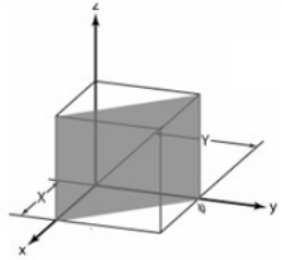


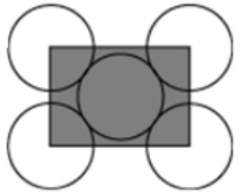


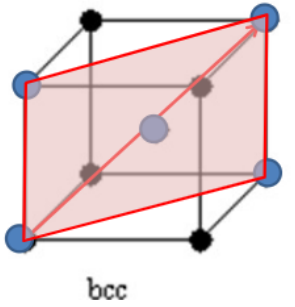


Close-packed planes: BCC

Primary Slip Plane: [110] = normal vector







Structure property correlation

Al: **ductile** — **FCC**

Fe: **not ductile** — **BCC**

Plastic deformation via SLIP
Sliding of crystal planes over one another!

SLIP occurs on most densely packed planes in the most closely packed planes!

slip plane + slip direction: **SLIP SYSTEM**

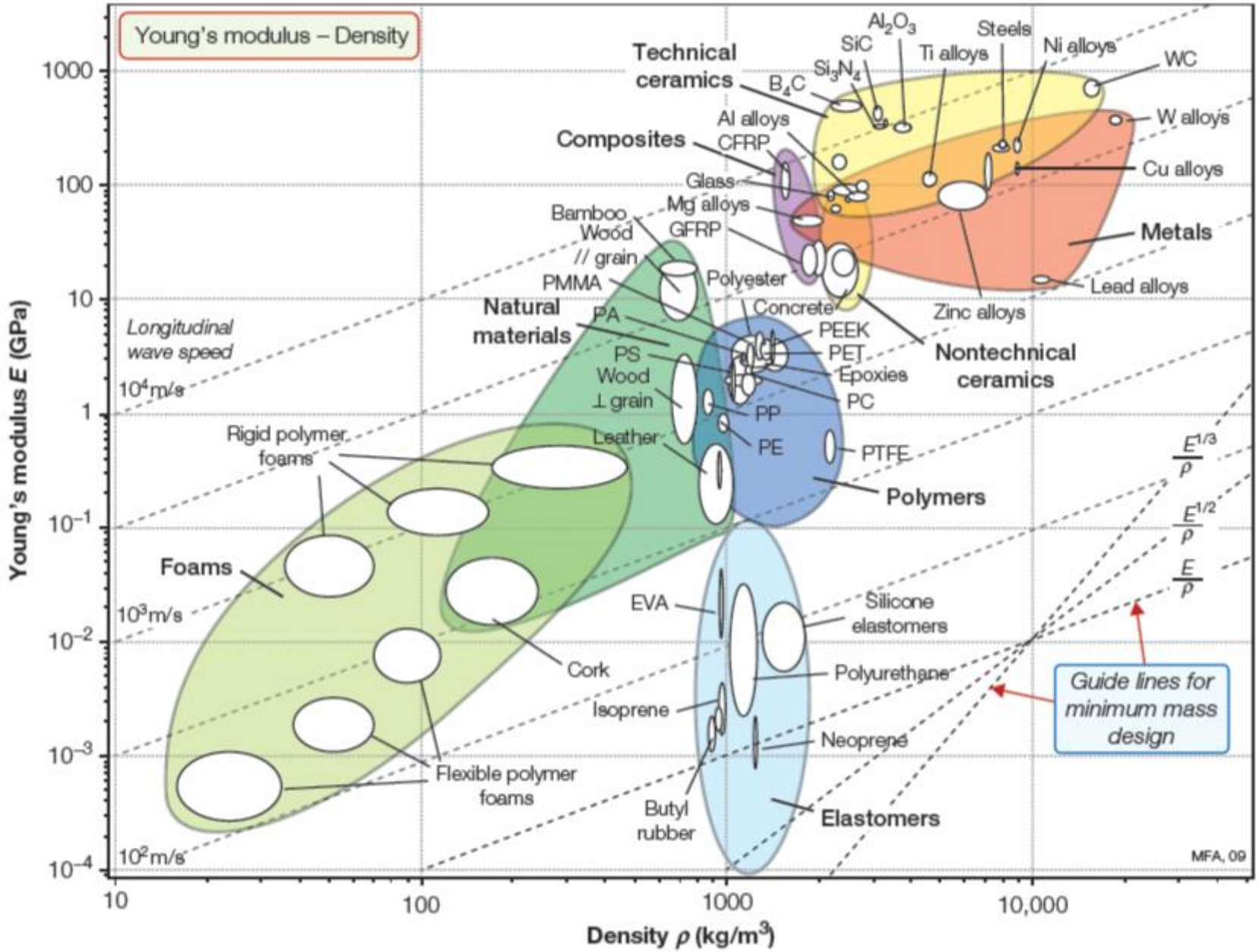
Material Property Charts

Young's Modulus vs Density

Need some polymer rope to float in water and be flexible (E~ 1 GPa)

Climbing rope?

- Modulus
- Yield strength
- Ultimate strength
- Toughness
- Hardness



For a bronze alloy, the stress at which plastic deformation begins is 275 MPa, and the modulus of elasticity is 115 GPa.

(a) What is the maximum load that may be applied to a specimen with a cross-sectional area of 325 mm^2 without plastic deformation?

(b) (b) If the original specimen length is 115 mm, what is the maximum length to which it may be stretched without causing plastic deformation?

For a bronze alloy, the stress at which plastic deformation begins is 275 MPa, and the modulus of elasticity is 115 GPa.

(a) What is the maximum load that may be applied to a specimen with a cross-sectional area of 325 mm^2 without plastic deformation?

(b) (b) If the original specimen length is 115 mm, what is the maximum length to which it may be stretched without causing plastic deformation?

$$F_y = \sigma_y A_0 = (275 \times 10^6\ \text{N/m}^2)(325 \times 10^{-6}\ \text{m}^2)$$

$$= 89{,}375\ \text{N} \quad (20{,}000\ \text{lb}_f)$$

$$l_i = l_0\left(1 + \frac{\sigma}{E}\right)$$

$$= (115\ \text{mm})\left[1 + \frac{275\ \text{MPa}}{115 \times 10^3\ \text{MPa}}\right] = 115.28\ \text{mm} \quad (4.51\ \text{in.})$$

A cylindrical specimen of aluminum having a diameter of 19 mm and length of 200 mm is deformed elastically in tension with a force of 48,800 N.
Using the below data, determine the following:

The amount by which this specimen will elongate in the direction of the applied stress.
The change in diameter of the specimen. Will the diameter increase or decrease?

E = 70 Gpa
Poisson's ratio = 0.33

A cylindrical specimen of aluminum having a diameter of 19 mm (0.75 in.) and length of 200 mm (8.0 in.) is deformed elastically in tension with a force of 48,800 N (11,000 lb^f). Using the data in book, determine the following:

The amount by which this specimen will elongate in the direction of the applied stress.
The change in diameter of the specimen. Will the diameter increase or decrease?

$$\sigma = E\varepsilon$$

$$\frac{F}{\pi\left(\frac{d_0^2}{4}\right)} = E\,\frac{\Delta l}{l_0}$$

$$\Delta l = \frac{4F\,l_0}{\pi\,d_0^2 E}$$

$$= \frac{(4)(48{,}800\ \text{N})(200 \times 10^{-3}\ \text{m})}{(\pi)(19 \times 10^{-3}\ \text{m})^2(69 \times 10^9\ \text{N/m}^2)} = 5 \times 10^{-4}\ \text{m} = 0.50\ \text{mm}\ (0.02\ \text{in.})$$

$$\nu = -\frac{\varepsilon_x}{\varepsilon_z} = -\frac{\Delta d / d_0}{\Delta l / l_0}$$

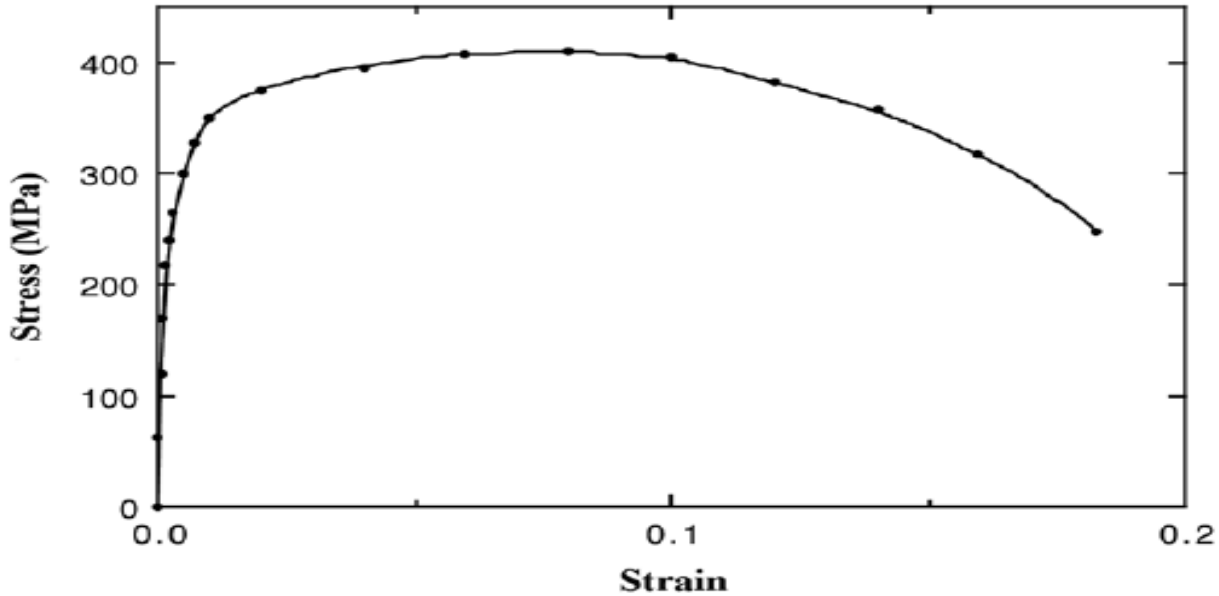
$$\Delta d = -\frac{\nu\,\Delta l\,d_0}{l_0} = -\frac{(0.33)(0.50\ \text{mm})(19\ \text{mm})}{200\ \text{mm}}$$

$$= -1.6 \times 10^{-2}\ \text{mm}\ \ (-6.2 \times 10^{-4}\ \text{in.})$$

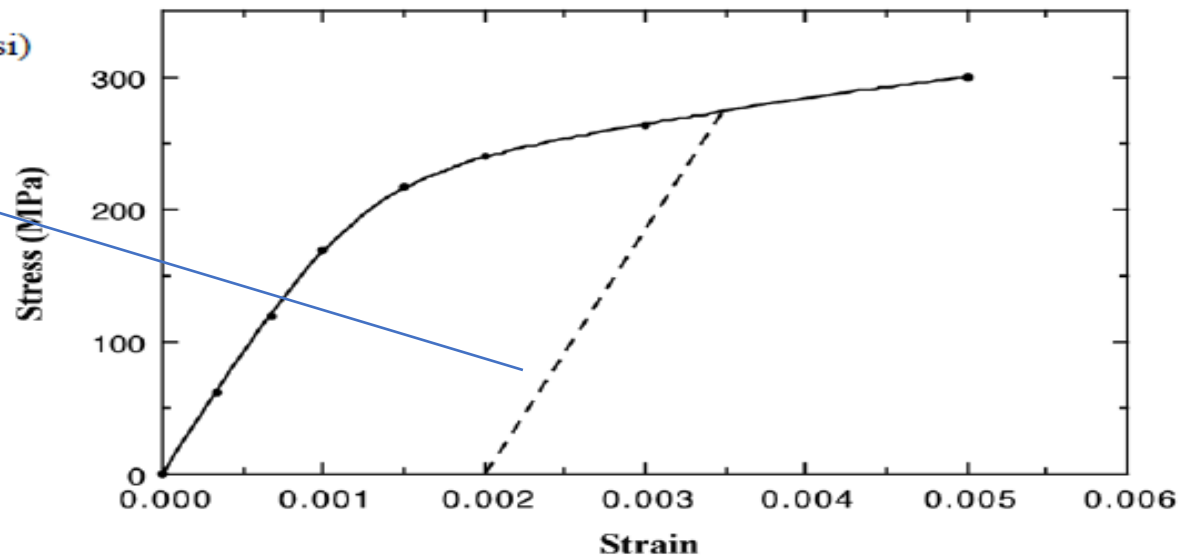
A specimen of ductile cast iron having a rectangular cross section of dimensions 4.8 mm × 15.9 mm is deformed in tension. Using the load-elongation data tabulated below, complete problems (a) through (f).

Load		Length	
N	lb_f	mm	in.
0	0	75.000	2.953
4,740	1,065	75.025	2.954
9,140	2,055	75.050	2.955
12,920	2,900	75.075	2.956
16,540	3,720	75.113	2.957
18,300	4,110	75.150	2.959
20,170	4,530	75.225	2.962
22,900	5,145	75.375	2.968
25,070	5,635	75.525	2.973
26,800	6,025	75.750	2.982
28,640	6,440	76.500	3.012
30,240	6,800	78.000	3.071
31,100	7,000	79.500	3.130
31,280	7,030	81.000	3.189
30,820	6,930	82.500	3.248
29,180	6,560	84.000	3.307
27,190	6,110	85.500	3.366
24,140	5,430	87.000	3.425
18,970	4,265	88.725	3.493
Fracture			

(a) Plot the data as engineering stress versus engineering strain.
(b) Compute the modulus of elasticity.
(c) Determine the yield strength at a strain offset of 0.002.
(d) Determine the tensile strength of this alloy.
(e) Compute the modulus of resilience.
(f) What is the ductility, in percent elongation?



$$E = \frac{\Delta\sigma}{\Delta\varepsilon} = \frac{100\ \text{MPa} - 0\ \text{MPa}}{0.0005 - 0} = 200 \times 10^3\ \text{MPa} = 200\ \text{GPa} \quad (29 \times 10^6\ \text{psi})$$



(c) For the yield strength, the 0.002 strain offset line is drawn dashed. It intersects the stress-strain curve at approximately 280 MPa (40,500 psi).

(d) The tensile strength is approximately 410 MPa (59,500 psi), corresponding to the maximum stress on the complete stress-strain plot.

(e) the modulus of resilience is just

$$U_r = \frac{\sigma_y^2}{2E}$$

which, using data computed above, yields a value of

$$U_r = \frac{(280 \times 10^6 \text{ N/m}^2)^2}{(2)(200 \times 10^9 \text{ N/m}^2)} = 1.96 \times 10^5 \text{ J/m}^3 \quad (28.3 \text{ in.-lb}_f/\text{in.}^3)$$

(f) The ductility, in percent elongation, is just the plastic strain at fracture, multiplied by one-hundred. The total fracture strain at fracture is 0.185; subtracting out the elastic strain (which is about 0.001) leaves a plastic strain of 0.184. Thus, the ductility is about 18.4%EL.

A large tower is to be supported by a series of steel wires. It is estimated that the load on each wire will be 11,100 N (2500 lb^f). Determine the minimum required wire diameter assuming a yield strength of 1030 MPa (150,000 psi).

If a factor of safety of 2 used, what is the diameter?

A large tower is to be supported by a series of steel wires. It is estimated that the load on each wire will be 11,100 N (2500 lb_f). Determine the minimum required wire diameter assuming a factor of safety of 2 and a yield strength of 1030 MPa (150,000 psi).

For this problem the working stress is computed using with $N = 2$, as

$$\sigma_w = \frac{\sigma_y}{2} = \frac{1030\ \text{MPa}}{2} = 515\ \text{MPa}\ \ (75{,}000\ \text{psi})$$

Since the force is given, the area may be determined , and subsequently the original diameter d_0 may be calculated as

$$A_0 = \frac{F}{\sigma_w} = \pi\left(\frac{d_0}{2}\right)^2$$

And

$$d_0 = \sqrt{\frac{4F}{\pi\sigma_w}} = \sqrt{\frac{(4)(11{,}100\ \text{N})}{\pi(515 \times 10^6\ \text{N/m}^2)}}$$

$$= 5.23 \times 10^{-3}\ \text{m} = 5.23\ \text{mm}\ (0.206\ \text{in.})$$

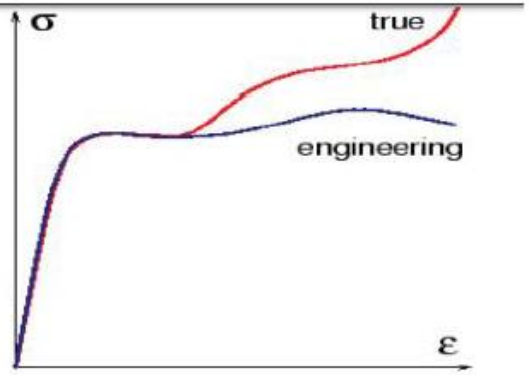
Materials for Engineers

Lecture 14-17

Dr S. Gowthaman
Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

Reference:
1. Materials science and engineering by William Callister, Wiley Publications

- For **engineering stress (σ)** and **engineering strain (ε)**, the original (gauge) dimensions of specimen are employed. However, length and cross-sectional area change in plastic region. **True stress (σ')** and **true strain (ε')** are used for accurate definition of plastic behaviour of ductile materials by considering the actual (instantaneous) dimensions.



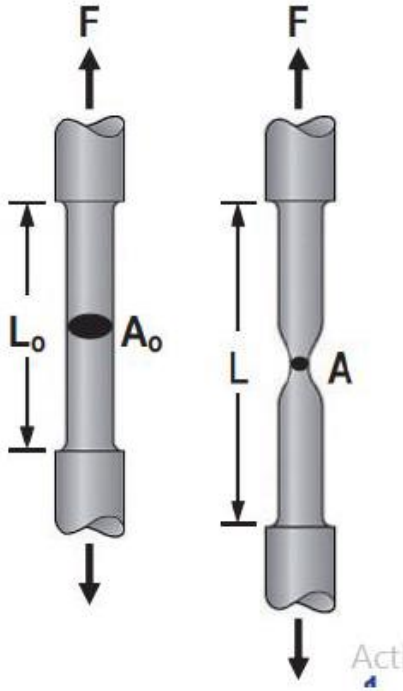
- **True stress (σ')** is the force divided by the actual area: $\sigma' = F/A$

- By the constancy of volume: $V = A * L = A_o * L_o$, we can obtain:

$$\sigma' = \frac{F}{A} = \frac{F * L}{A_o * L_o} \quad \& \quad \varepsilon = \frac{L - L_o}{L_o} = \frac{L}{L_o} - 1 \quad \Rightarrow \quad \sigma' = \frac{F}{A_o}(1+\varepsilon) = \sigma(1+\varepsilon)$$

- **True strain (ε')** is change in length with respect to the instant length:

$$\varepsilon' = \int_{L_0}^{L} dL/L = \ln(L/L_o) \quad \& \quad L = (1+\varepsilon)L_o \quad \Rightarrow \quad \varepsilon' = \ln(1+\varepsilon)$$



Strengthening Mechanisms

Recall….

Applications of Defects / Imperfections in Solids

Recall….

Importance of defects depends upon the material, type of defect, and properties, which are being considered

Dislocation – Strength/toughness of materials

Vacancies – Electrical and thermal conductivity of materials

Impurities – Mechanical/Electronic properties of materials

Grain boundaries – Strength of materials

Recall….

Recall….

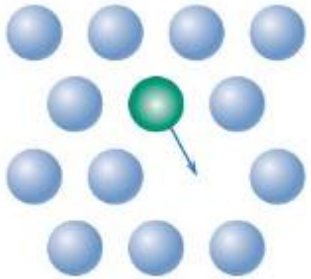
Recall....

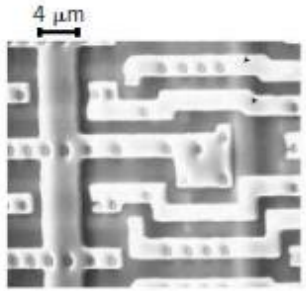
Recall....

Crystal Defects - Processing/Structure/Properties/Performance - Examples

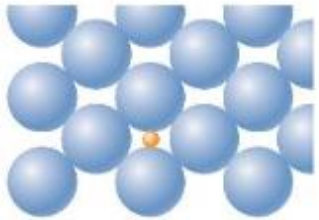
Specification of composition → **Diffusion in semiconductors** → **Fabrication of integrated circuits**

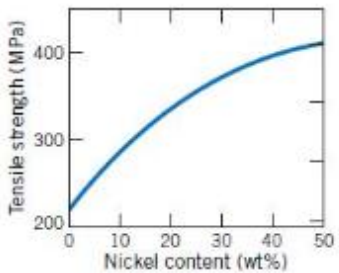
$$C_1 = \frac{m_1}{m_1 + m_2} \times 100$$



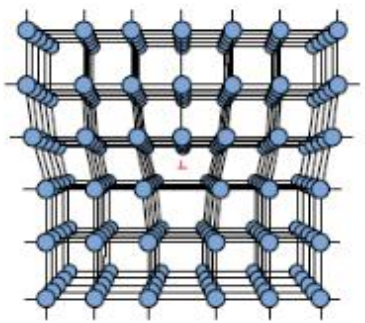


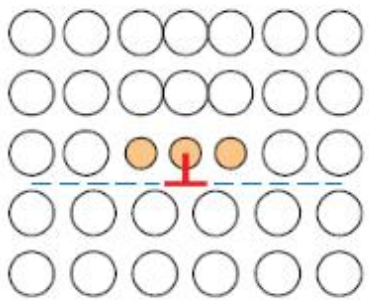
Concept of solid solution → **Solid solution strengthening**





Concept of dislocation defects → **Strengthening mechanisms**



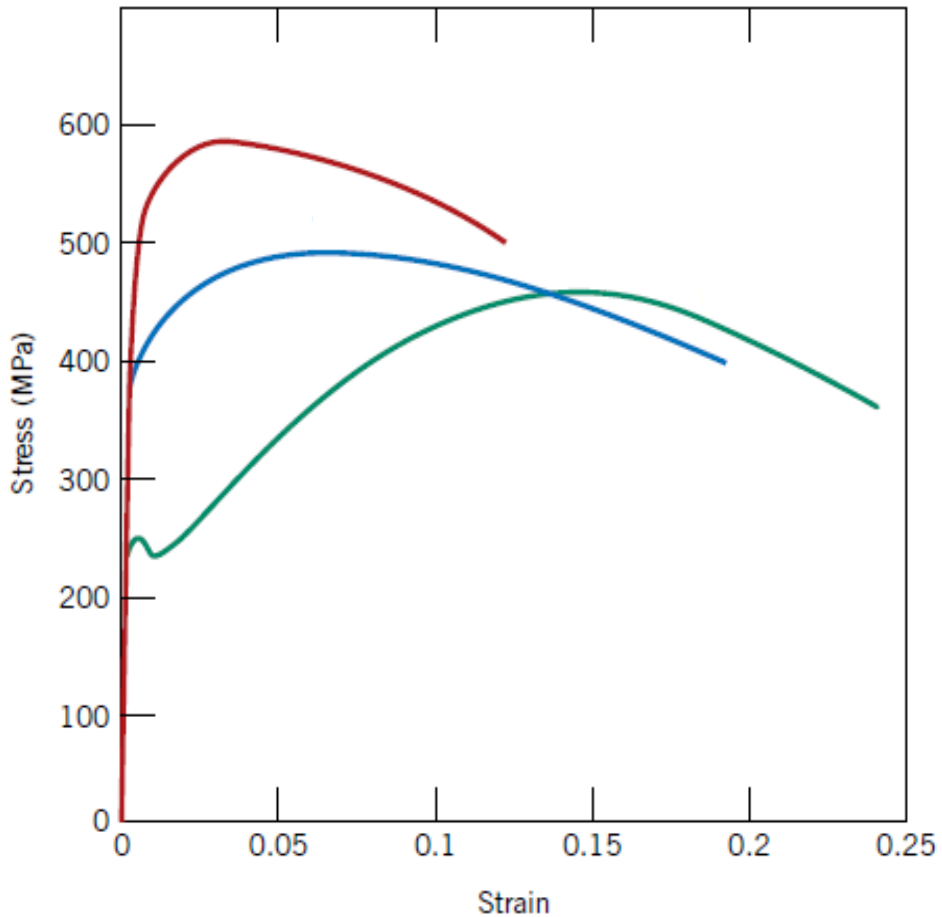


Recall....

Recall....

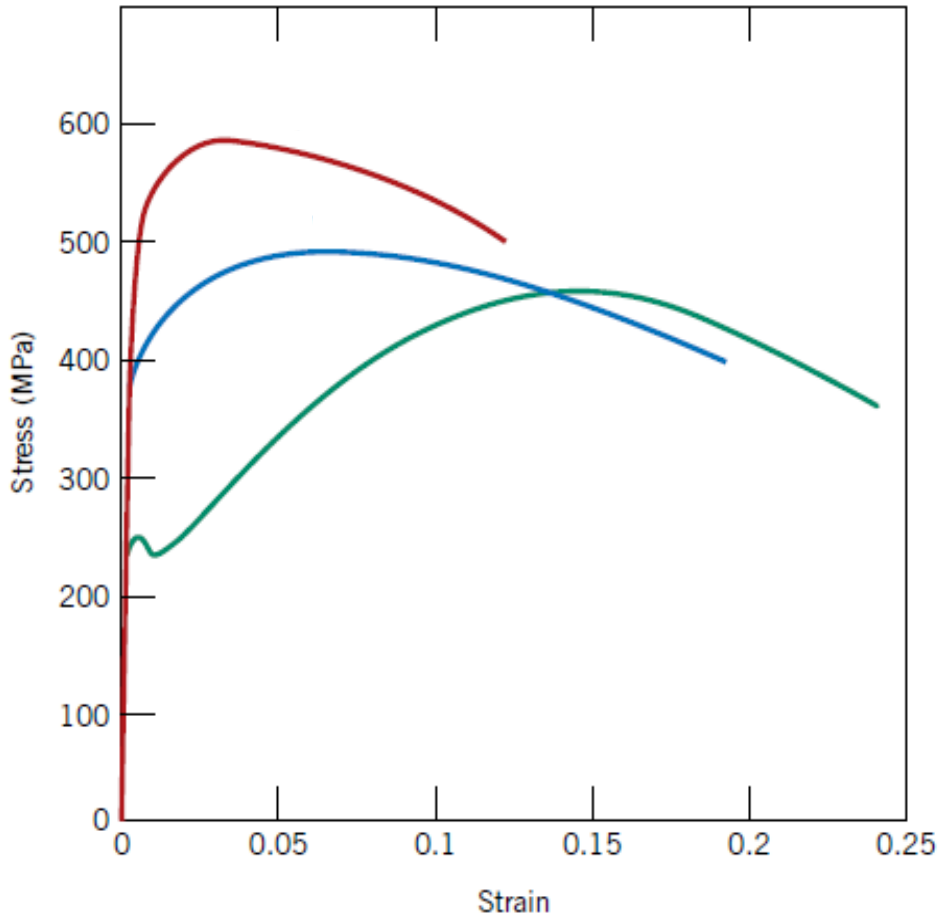
Strengthening Mechanisms

All are same material…



Strengthening Mechanisms

All are same material…



What is Strengthening?

- Plastic deformation occurs when large numbers of dislocations move and multiply so as to result in macroscopic deformation.
- In other words, it is the movement of dislocations in the material which allows for deformation.
- If we want to enhance a material's mechanical properties (i.e. increase the yield and tensile strength), we simply need to introduce a mechanism which prohibits the mobility of these dislocations.

So, what are the mechanisms to restrict dislocation movement??

Strengthening Mechanisms

- ✓ Grain Size Reduction
- ✓ Solid Solution Strengthening
- ✓ Strain Hardening or Work Hardening
- ✓ Precipitation Hardening

All strengthening techniques rely on simple principle:

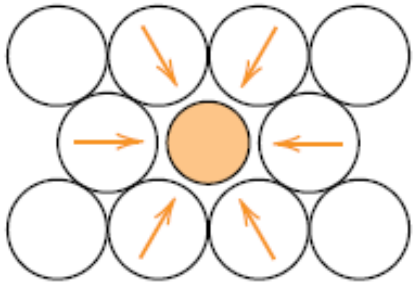
Restricting or *Hindering* dislocation motion renders a material stronger.

Solid Solution Strengthening

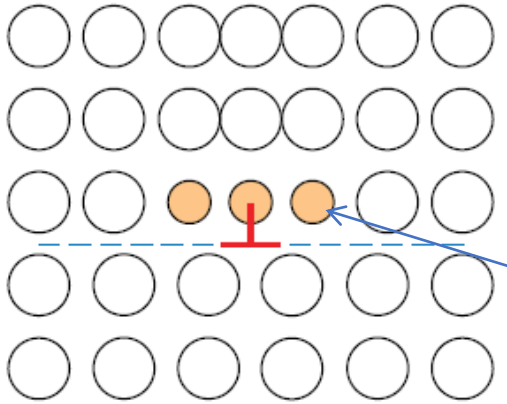
- Add impurity (foreign) atom to metals – make alloys
- The impurity atoms that go into solid solution typically impose lattice strains on the surrounding host atoms.
- These lattice strain field will interact with dislocations to reduce strain energy, and consequently, ***dislocation movement is restricted. Then more force (stress) is needed to move the dislocations.***
- Thus, **alloys become stronger**

Substitute (foreign) atom smaller than host atom

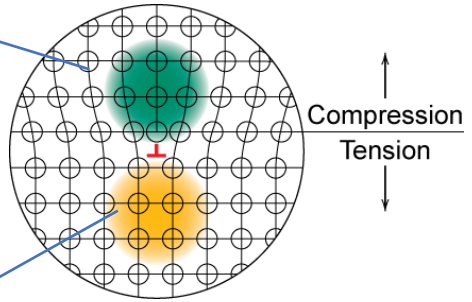
(atoms under tension)



To reduce strain energy will diffuse and segregate around dislocations

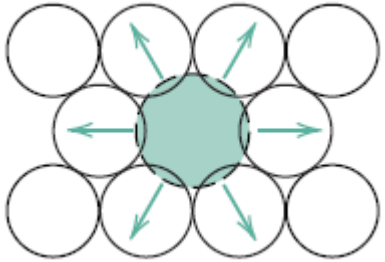


Recall…..

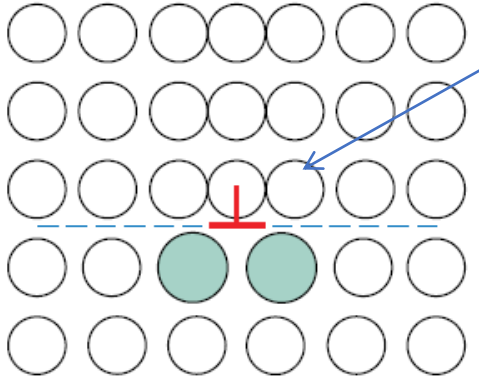


Substitute (foreign) atom larger than host atom

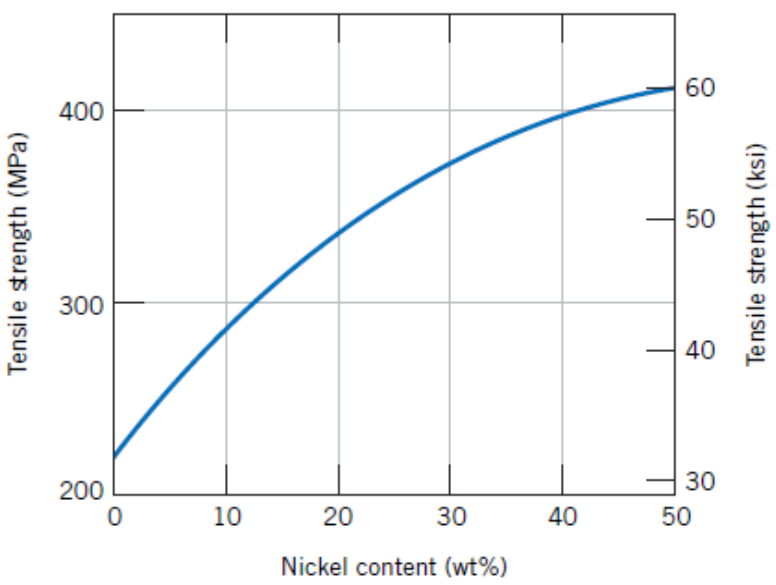
(atoms under compression)

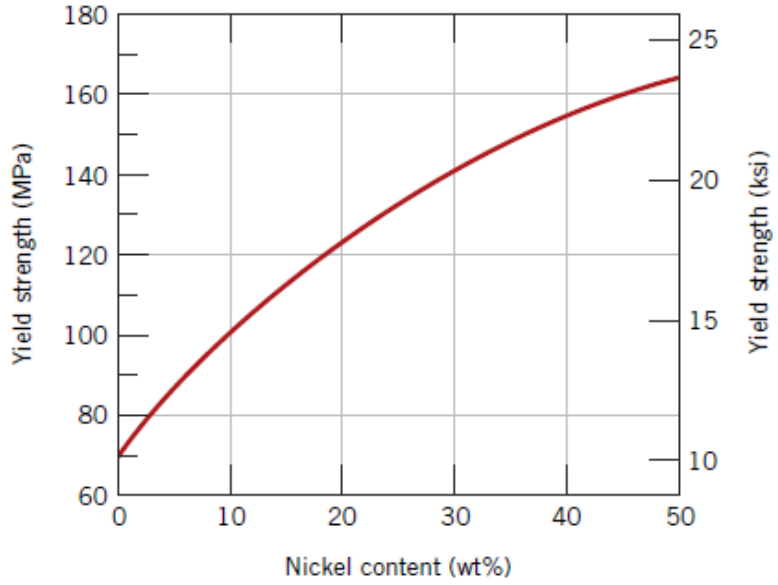


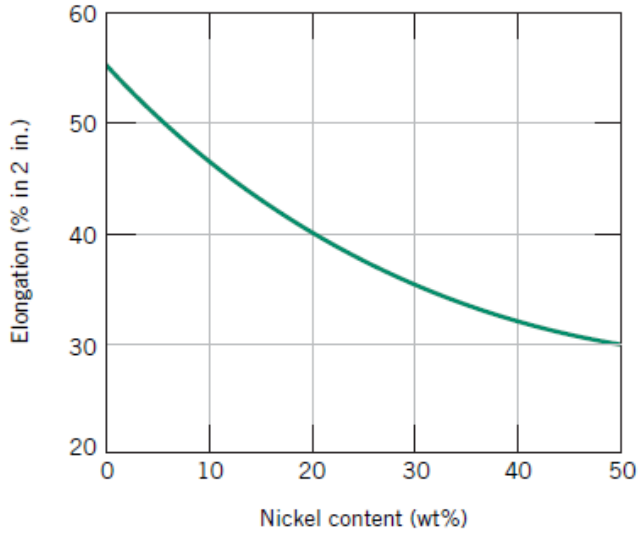
To reduce strain energy will diffuse and segregate around dislocations



Solid Solution Strengthening – Example – Cu Ni Alloys

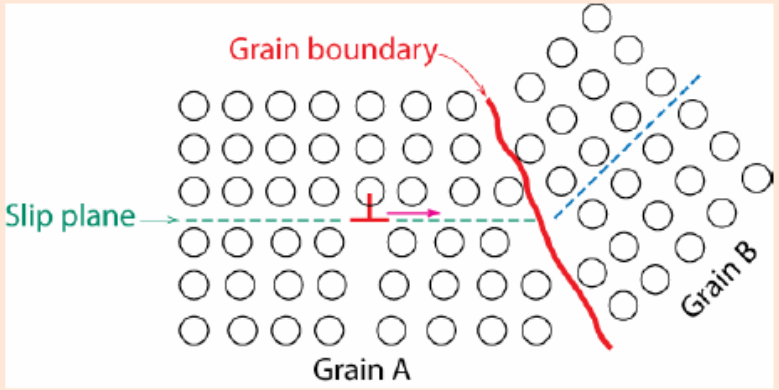






Grain Size Reduction

Is a grain boundary a barrier for dislocation motion?



The grain boundary acts as a barrier to dislocation motion for two reasons:

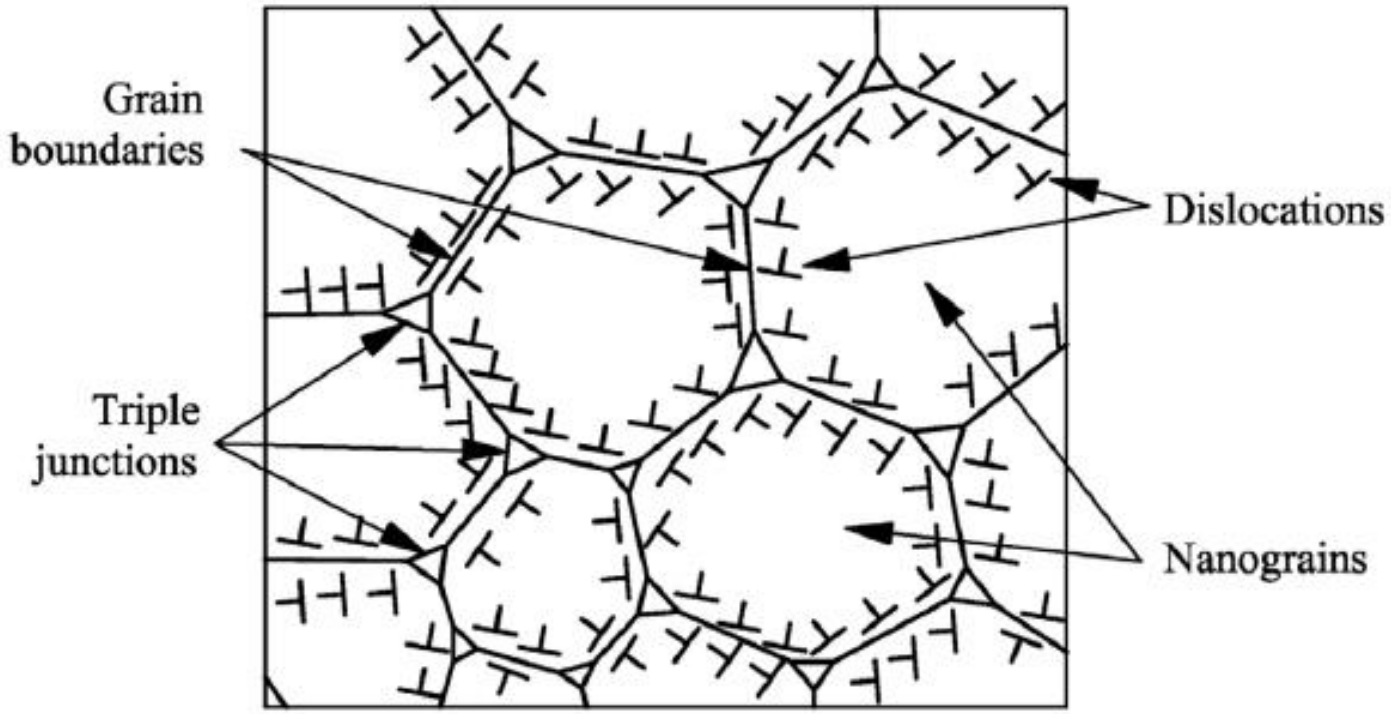
1. Since the two grains are of different orientations, a dislocation passing into grain B will have to change its direction of motion; this becomes more difficult as the crystallographic misorientation increases.
2. The atomic disorder within a grain boundary region will result in a discontinuity of slip planes from one grain into the other.

Grain Size Reduction

GRAIN BOUNDARY ENGINEERING

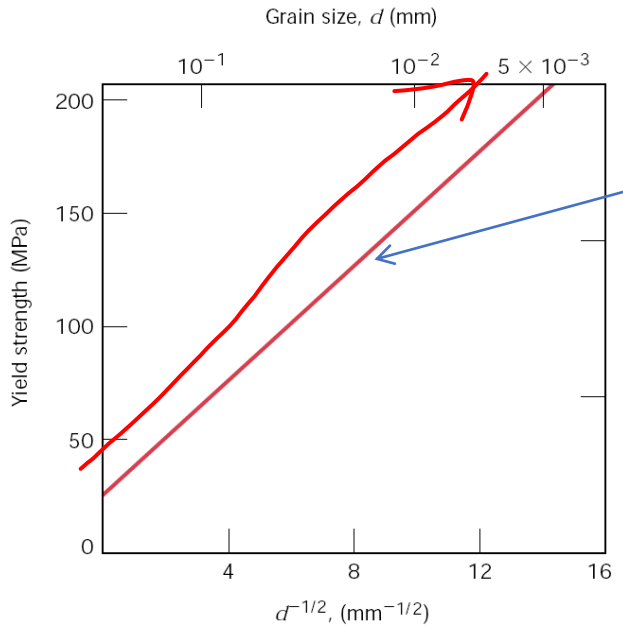
Grain boundary strengthening is a method of strengthening materials by changing their average grain size.

It is based on the observation that grain boundaries **restrict** dislocation movement and that the number of dislocations within a grain have an effect on how easily dislocations can traverse grain boundaries and travel from grain to grain.



Grain Size Reduction ➡

The influence of grain size on the yield strength of a 70 Cu–30 Zn brass alloy.



HALL-PETCH EQUATION

- Why is a fine-grained material is harder and stronger than coarse grained?

The relation between yield stress and grain size is described mathematically by **the Hall-Petch equation**

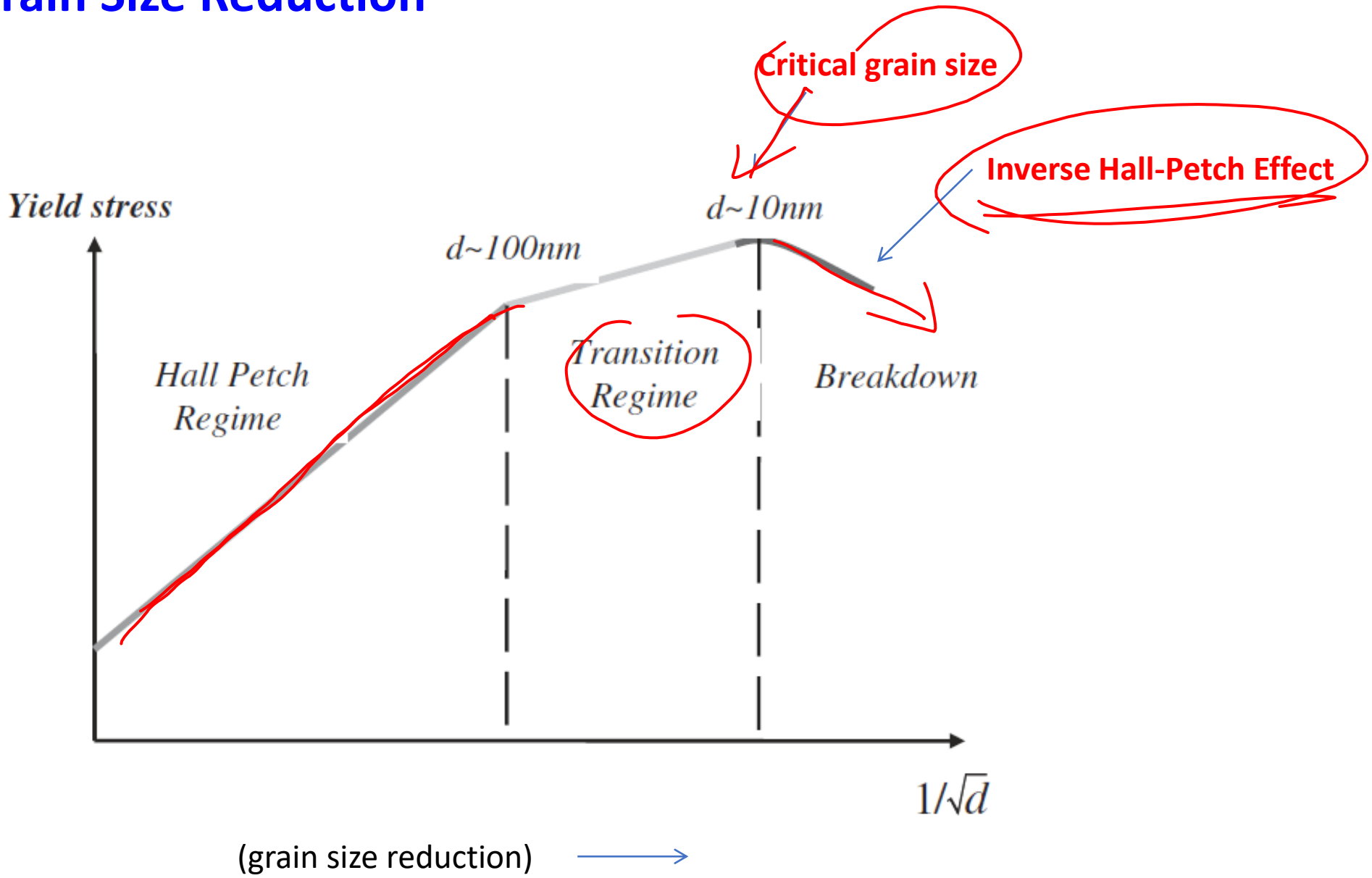
$$\sigma_y = \sigma_0 + \frac{k_y}{\sqrt{d}}$$

where k_y is the strengthening coefficient (a constant unique to each material), σ_o is a materials constant for the starting stress for dislocation movement (or the resistance of the lattice to dislocation motion), d is the grain diameter, and σ_y is the yield stress.

Hall-Petch constants

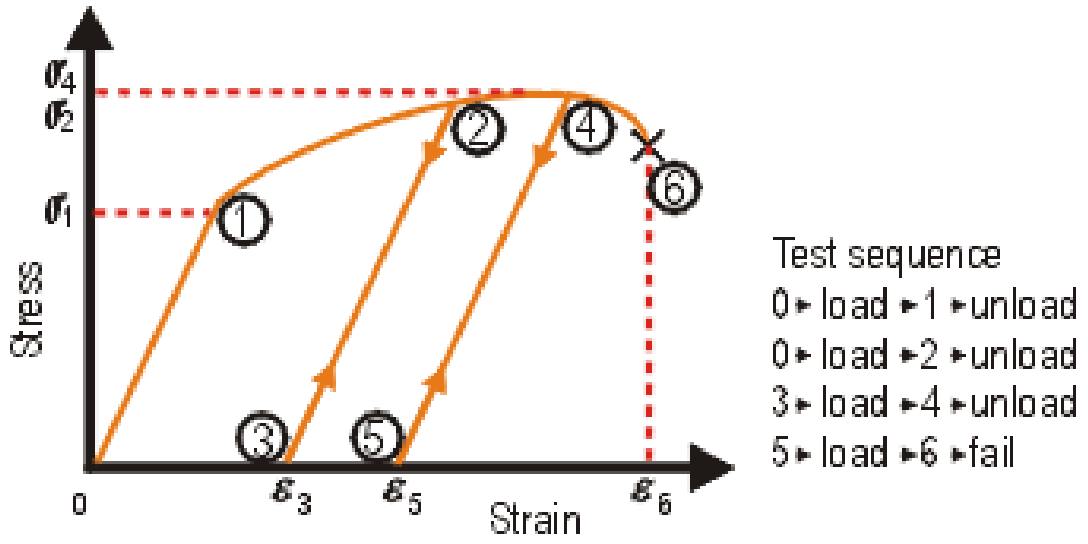
Material	σ_o [MPa]	k [MPa m$^{1/2}$]
Copper	25	0.11
Titanium	80	0.40
Mild steel	70	0.74
Ni_3Al	300	1.70

Grain Size Reduction



(grain size reduction) ⟶

Strain Hardening (or Work Hardening or Cold Working)



The material becomes stronger in this process as $\sigma_4 > \sigma_2 > \sigma_1 >$

Strain hardening is the phenomenon by which a ductile metal becomes harder and stronger as it is plastically deformed.

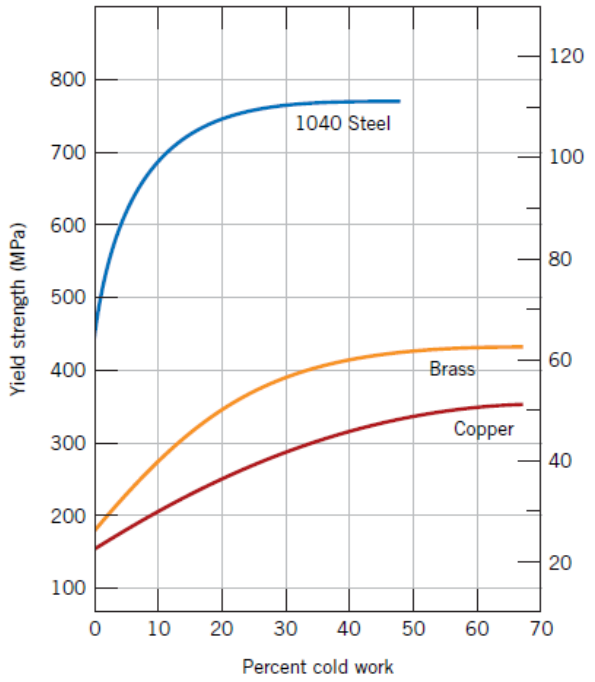
Cold working – done at low temperatures (below 0.3Tm)

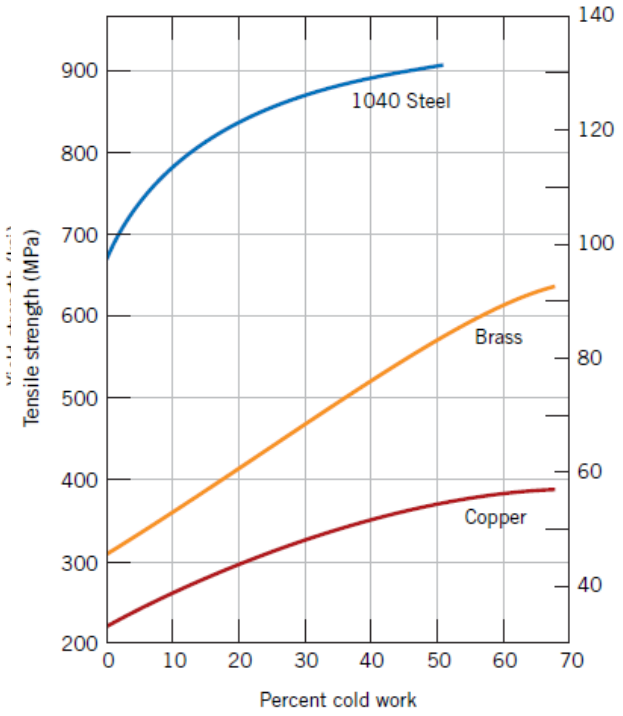
- As the material is deformed, number of dislocations increases . As a result, dislocation become closer and dislocations of same sign repel each other.
- The net result is that the motion of a dislocation is hindered by the presence of other dislocations.
- As the dislocation density increases, this resistance to dislocation motion by other dislocations becomes more pronounced.
- Thus, the imposed stress necessary to deform a metal increases with strain (hardening).

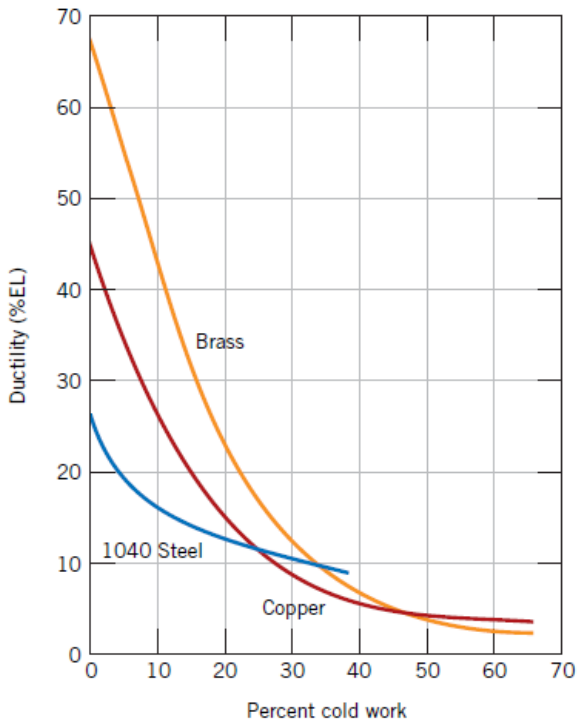
% Cold Work

$$\%\text{CW} = \left(\frac{A_0 - A_d}{A_0}\right) \times 100$$

where A_0 is the original area of the cross section that experiences deformation and A_d is the area after deformation.





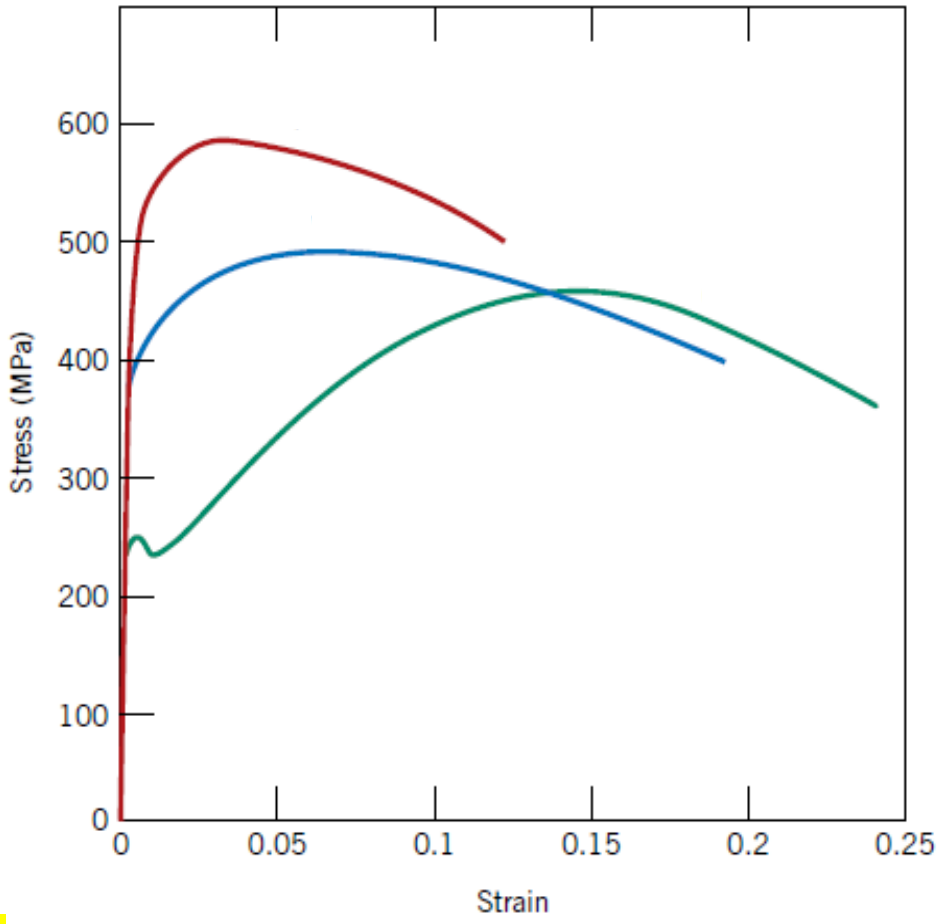


Cold working generally carried at lower temperatures (below $0.3T_m$)

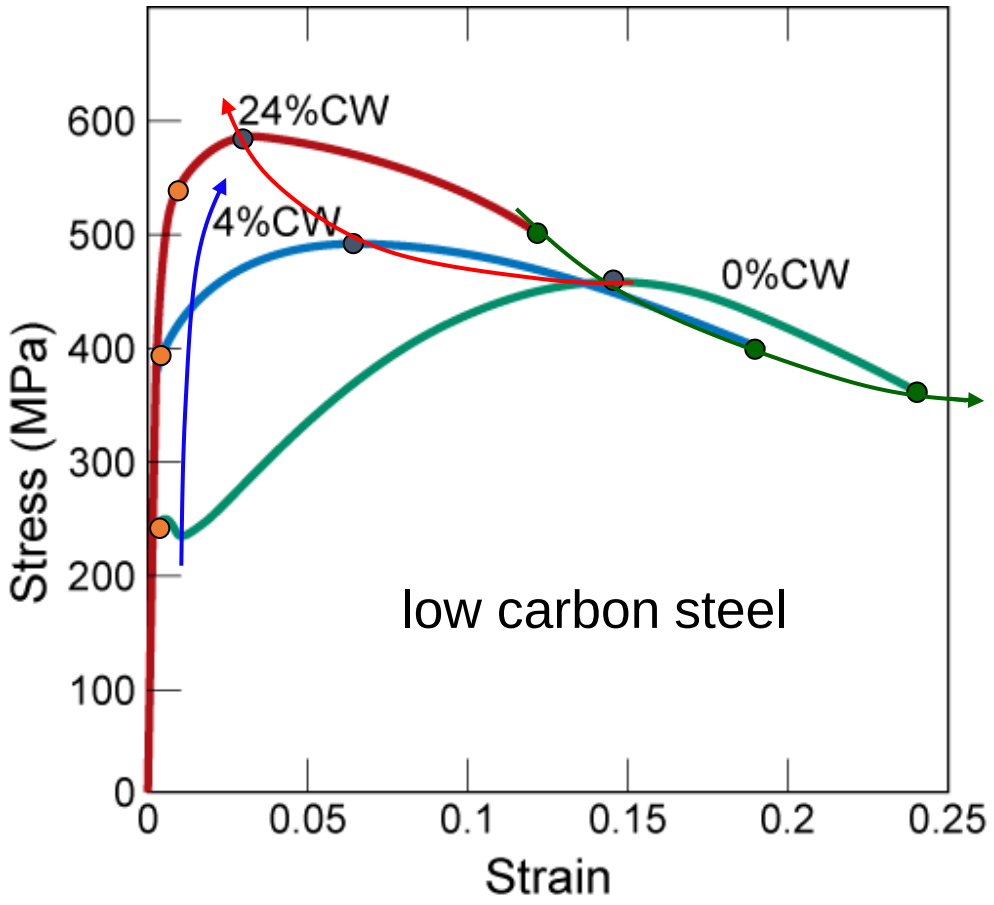
Recall….

Recall….

All are same material…



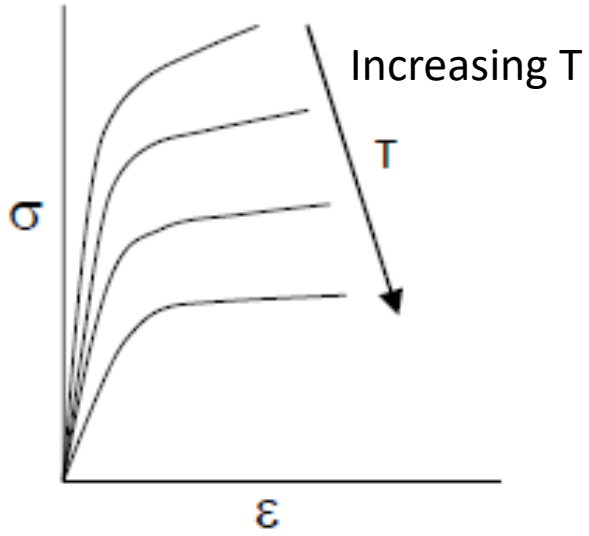
Impact of cold work



Recall….

Recall….

Effect of temperature on plastic deformation metal:



With **increasing temperature,** the stress at which the metal yields and its strain hardening (slope of plastic deformation) keeps decreases. But, plastic deformation increases.

- Because of Recovery, Recrystallization and Grain Growth

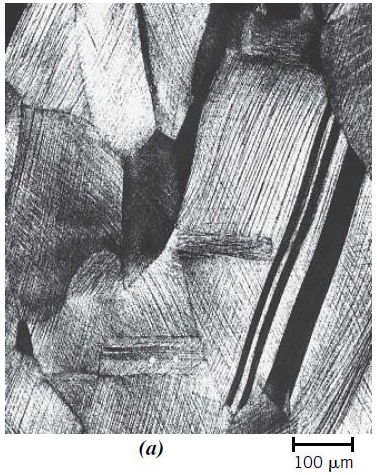
Recovery

- Heating to certain temperature
- Atomic diffusion increases (thermal effect)
- Dislocations move and internal strain energy decreases
- Change in dislocation configuration having low strain energies

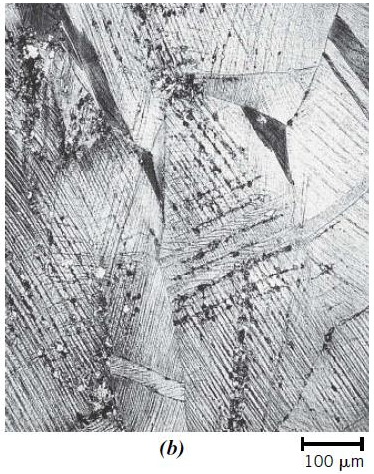
Recrystallization

- Even after recovery is complete, the grains are still in a relatively high strain energy state.
- **Recrystallization** is the formation of a new set of strain-free and equiaxed grains that have low dislocation densities and are characteristic of the precold-worked condition.

Example – for Brass



Cold worked

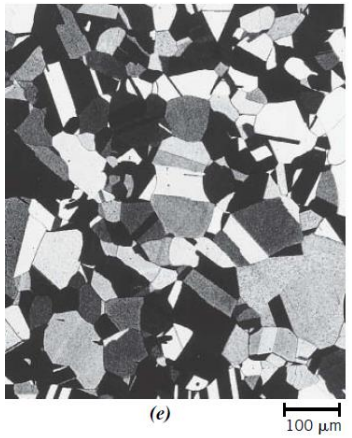


After 3 s of heating at 580°C

After 4 s of heating at 580°C (partial replacement with new grains)

After 8 s of heating at 580°C (complete recrystallization)

Grain Growth

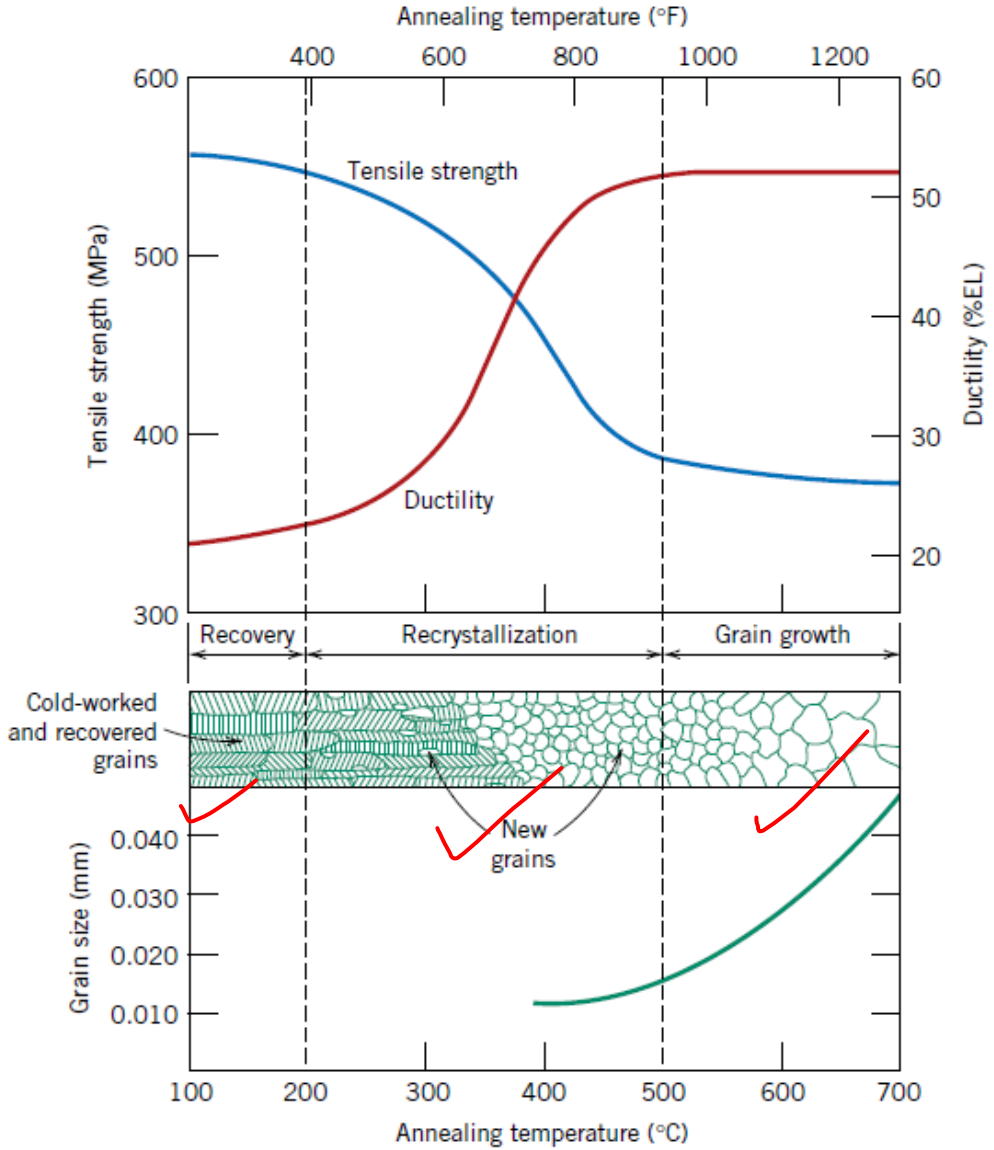


After 15 m of heating at 580°C

After 10 m of heating at 700°C

- At longer times, average grain size increases.
  - -- Small grains shrink (and ultimately disappear)
  - -- Large grains continue to grow

Recovery, Recrystallization and Grain Growth (for Brass)

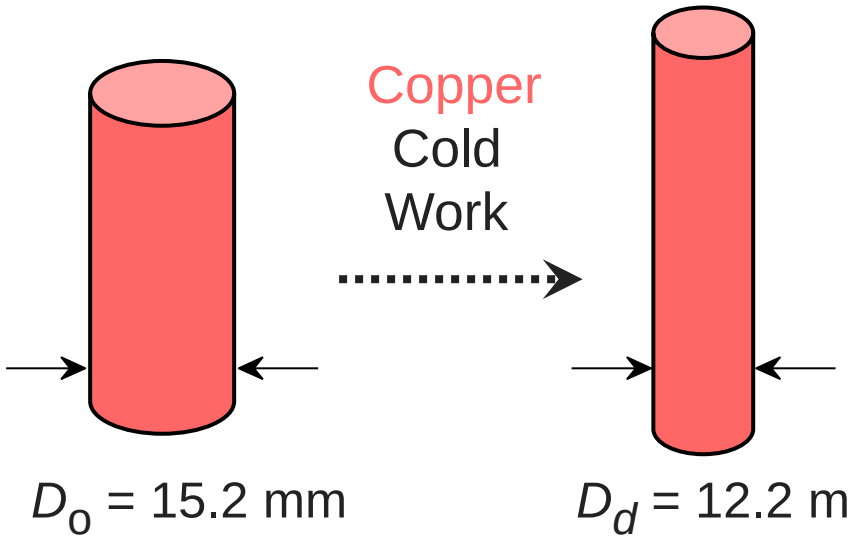


Recrystallization temperature - the temperature at which recrystallization just reaches completion in 1 h
(for Brass it is ~ 450°C)

For pure metals, it is < 0.3Tm
For alloys, as high as 0.7Tm

Mechanical Property Alterations
Due to Cold Working

- What are the values of yield strength, tensile strength & ductility after cold working Cu?



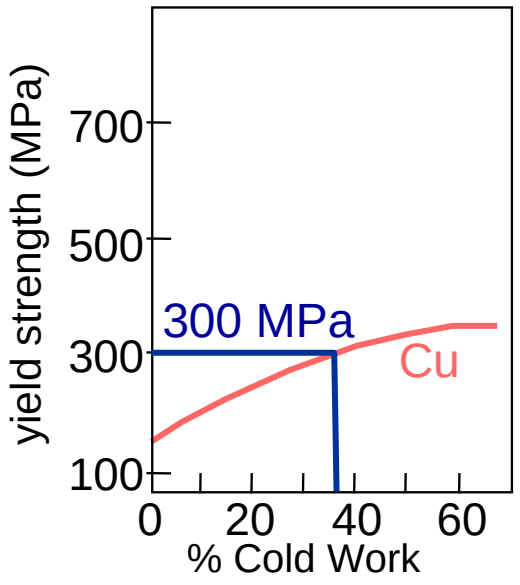
$$\%CW = \frac{\frac{\pi D_o^2}{4} - \frac{\pi D_d^2}{4}}{\frac{\pi D_o^2}{4}} \times 100$$

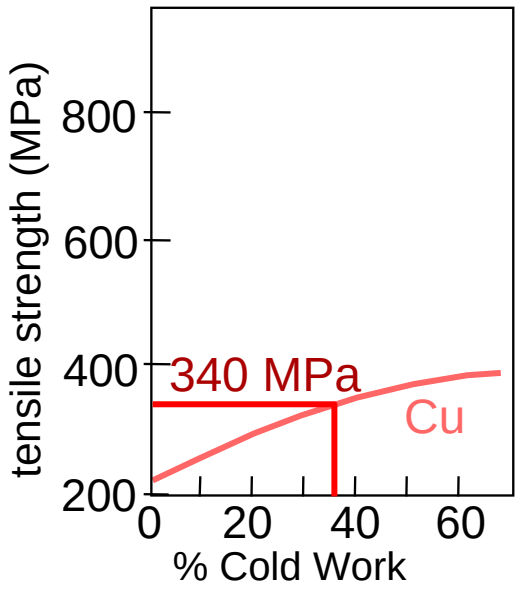
$$= \frac{D_o^2 - D_d^2}{D_o^2} \times 100$$

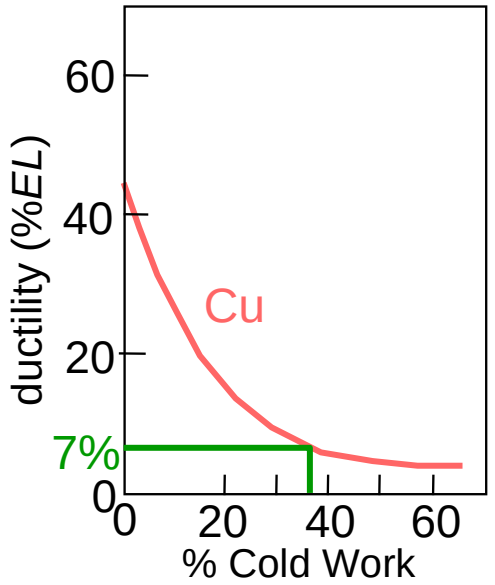
$$\%CW = \frac{(15.2\,\text{mm})^2 - (12.2\,\text{mm})^2}{(15.2\,\text{mm})^2} \times 100 = 35.6\%$$

Mechanical Property Alterations Due to Cold Working

- What are the values of yield strength, tensile strength & ductility for Cu for %CW = 35.6%?







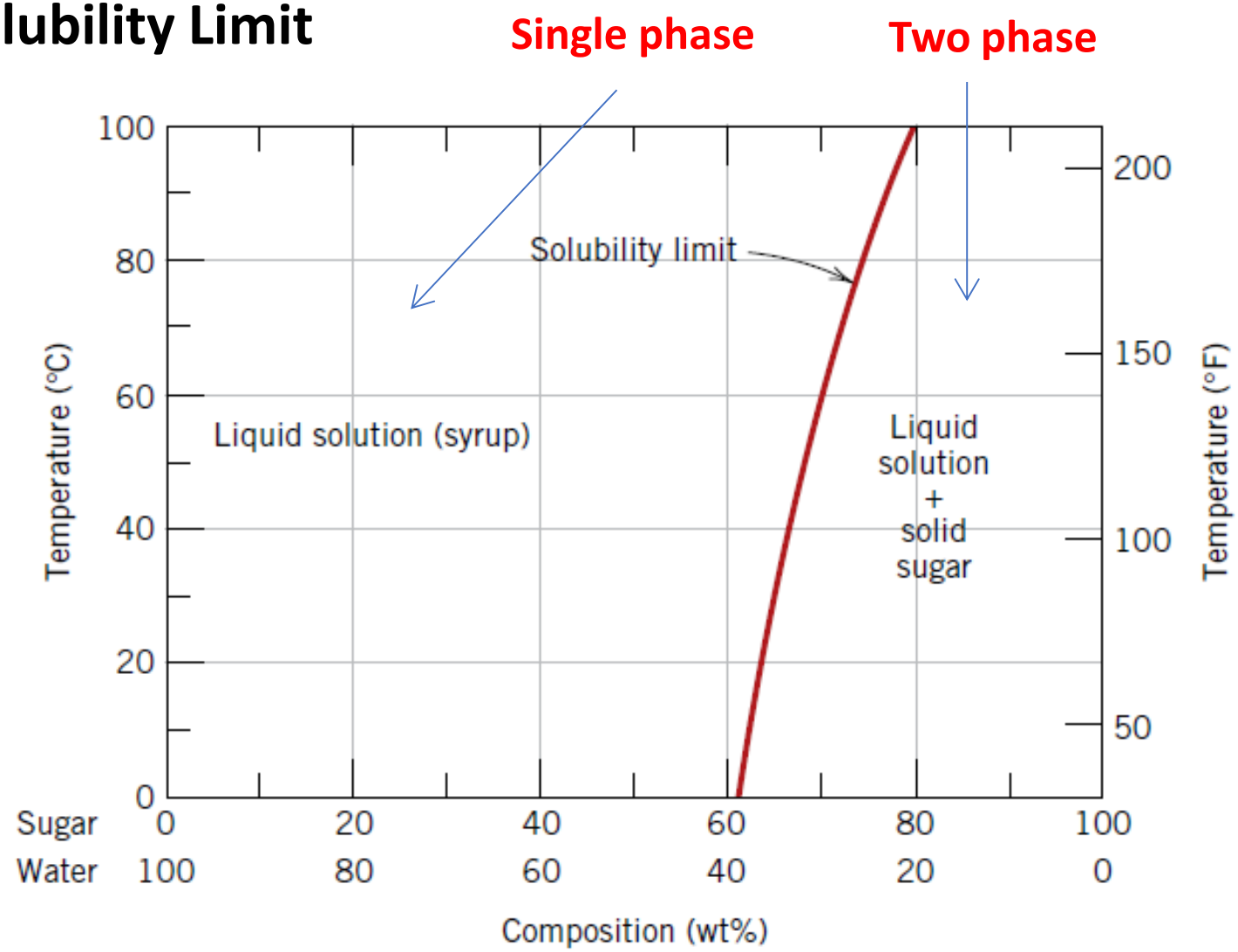
σ_y = 300 MPa

TS = 340 MPa

%*EL* = 7%

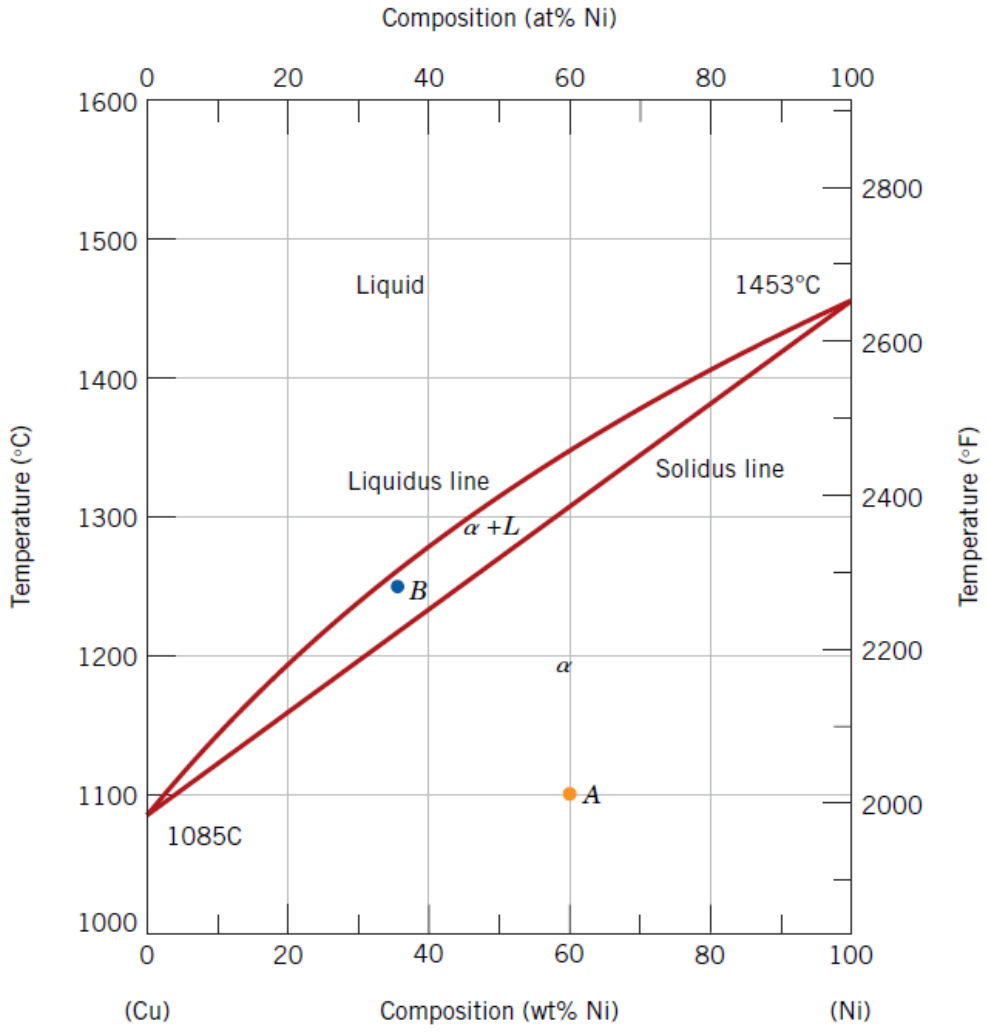
Phase Diagrams

Solubility Limit



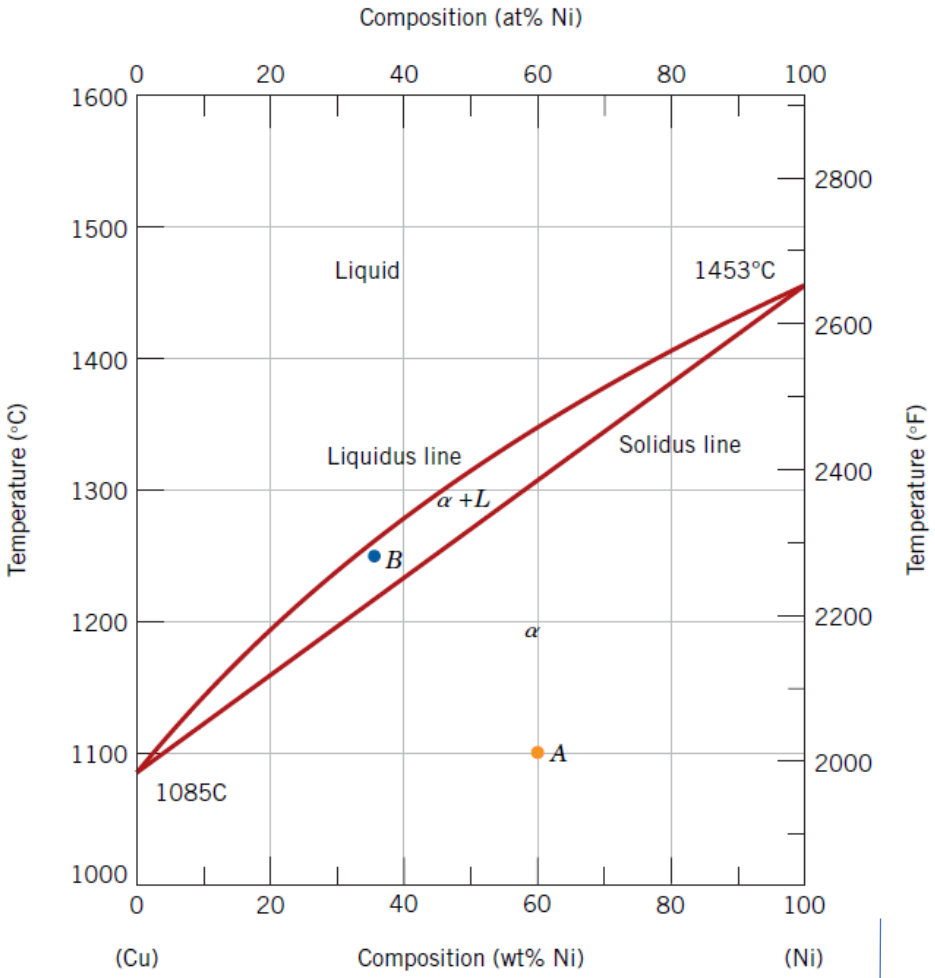
Binary Phase Diagrams – Isomorphous System

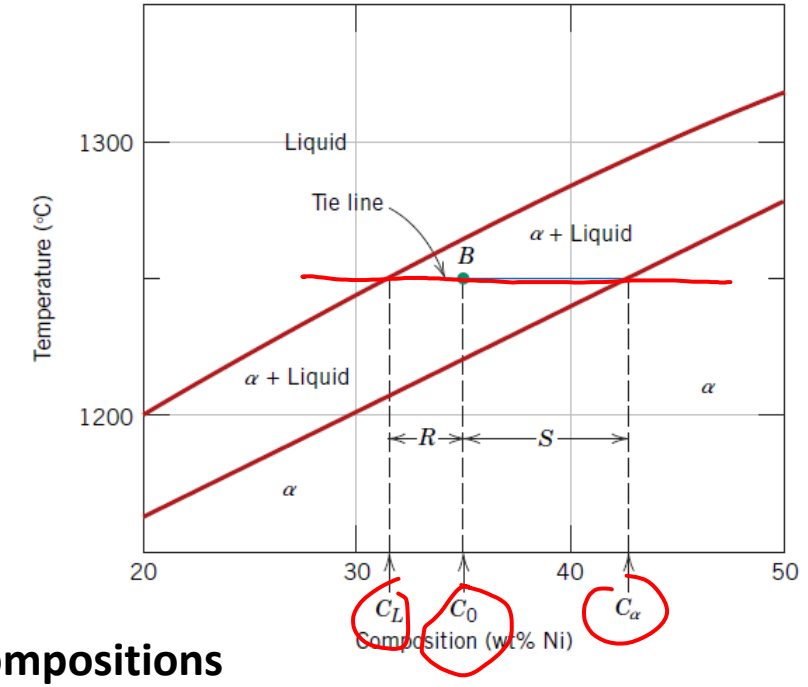
- Two component phase diagram (Binary alloy)
- T&C are variables, P held constant at 1 atm



Binary Isomorphous System

Binary Phase Diagrams – Isomorphous System





What Phases present at A and at B??

Solid at A; Liquid + solid at B

Determination of phase compositions

At A (single phase region):
60 wt% Ni - 40 wt% Cu (all solid)

At B (two phase region):

- Draw tie line (isotherm) at particular temperature
- Drop verticals from the point of intersection of line with phase boundaries
- Determine the composition

C_L, the concentration of liquid is 31.5 wt% Ni – 68.5 wt% Cu
C_α, the concentration of solid is 42.5 wt% Ni – 57.5 wt% Cu

Lever Rule Derivation

Let @ Temperature T, wt. fraction of Ni (for example Nickel) = $C_{O\ (Ni)}$

Let @ Temperature T, we have α and L phases, such that

Wt. fraction of Ni in α phase = $C_{\alpha\ (Ni)}$ and Wt. fraction of Ni in L phase = $C_{L\ (Ni)}$

Let total mass of α phase = w_α
Let total mass of L phase = w_L
Let total mass together = w

Mass of Ni alone in α phase is, $w_{\alpha\ (Ni)} = C_{\alpha\ (Ni).}\ w_\alpha$

Mass of Ni alone in L phase is, $w_{L\ (Ni)} = C_{L\ (Ni).}\ w_L$

Then,

$$w_{\alpha\ (Ni)} + w_{L\ (Ni)} = \text{total mass of Ni} = C_{O\ (Ni)}.\ w$$

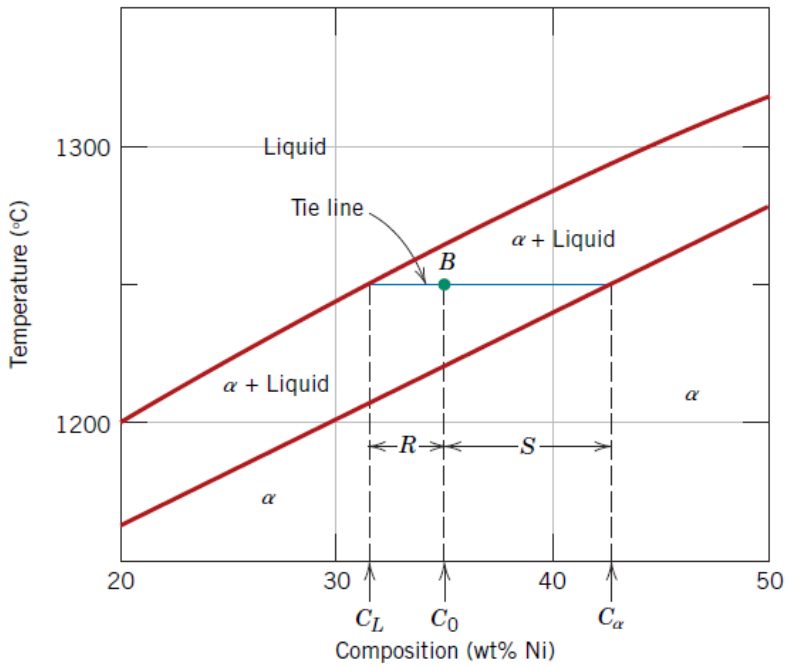
$$C_{\alpha\ (Ni).}\ w_\alpha + C_{L\ (Ni).}\ w_L = C_{O\ (Ni)}.\ w$$

$$C_{\alpha\ (Ni).}\ W_\alpha + C_{L\ (Ni).}\ W_L = C_{O\ (Ni)}$$

$$W_\alpha + W_L = 1$$

$$W_\alpha\ C_\alpha + W_L\ C_L = C_o$$

Lever Rule Derivation



$W_\alpha + W_L = 1$

$W_\alpha C_\alpha + W_L C_L = C_o$

$W_\alpha C_\alpha + W_L C_L = C_o$

$W_\alpha C_\alpha + (1-W_\alpha)C_L = C_o$

$W_\alpha C_\alpha + C_L - W_\alpha C_L = C_o$

$W_\alpha (C_\alpha - C_L) = C_o - C_L$

$\mathbf{W_\alpha = C_o - C_L / (C_\alpha - C_L)}$

$W_\alpha C_\alpha + W_L C_L = C_o$

$(1-W_L)C_\alpha + W_L C_L = C_o$

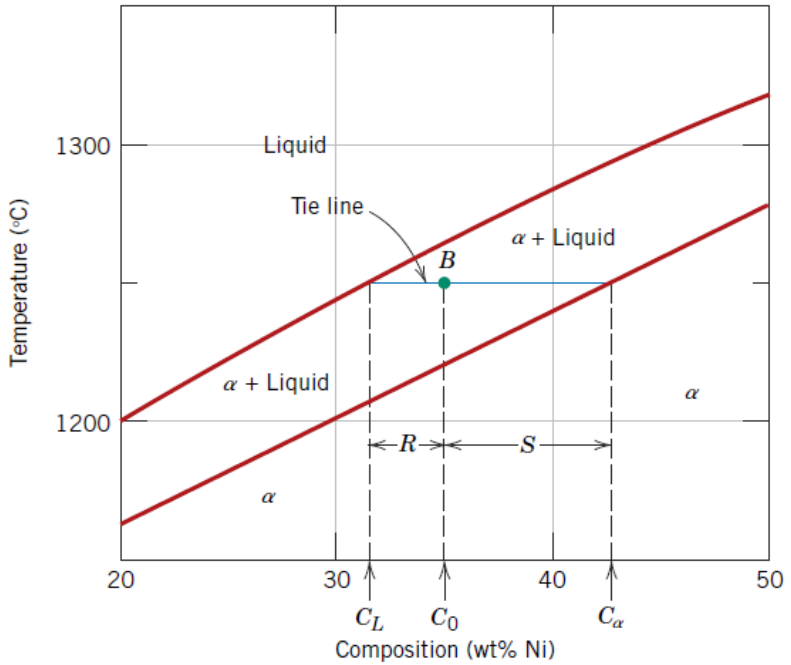
$C_\alpha - W_L C_\alpha + W_L C_L = C_o$

$W_L (C_L - C_\alpha) = C_o - C_\alpha$

$W_L = C_o - C_\alpha / (C_L - C_\alpha)$

$\mathbf{W_L = C_\alpha - C_o / (C_\alpha - C_L)}$

Binary Phase Diagrams



Determination of phase amounts

At A (single phase region):
100 % solid

At B (two phase region): - **Lever Rule**

- Draw tie line (isotherm) at particular temperature
- Drop verticals from the point of intersection of line with phase boundaries
- Determine the composition

C_L, the concentration of liquid is 31.5 wt% Ni – 68.5 wt% Cu
C_α, the concentration of solid is 42.5 wt% Ni – 57.5 wt% Cu

$W_\alpha C_\alpha + W_L C_L =$
$0.32(42.5) + 0.68(31.5) =$
35% Ni = C_o

$W_\alpha C_\alpha + W_L C_L = C_o$
$W_\alpha + W_L = 1$

Amount of liquid (Weight fraction of liquid):

$$W_L = \frac{S}{R+S}$$

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$$

$$W_L = \frac{42.5 - 35}{42.5 - 31.5} = 0.68$$

Amount of solid (Weight fraction of solid):

$$W_\alpha = \frac{R}{R+S} = \frac{C_0 - C_L}{C_\alpha - C_L}$$

$$= \frac{35 - 31.5}{42.5 - 31.5} = 0.32$$

(Similarly % of Cu can also be considered, will give same result)

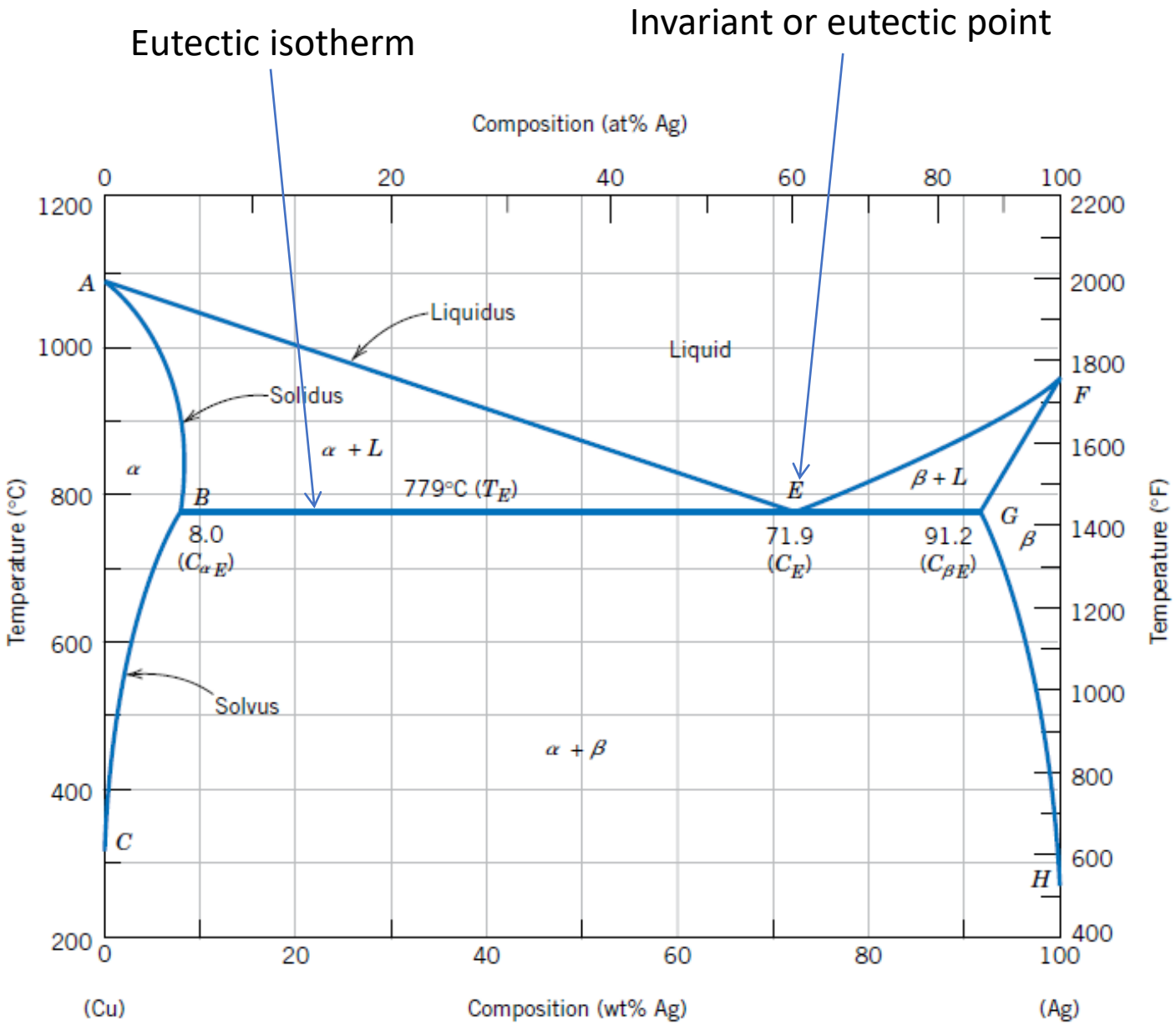
Binary eutectic system

Cu-Ag Phase Diagram

Eutectic reation:

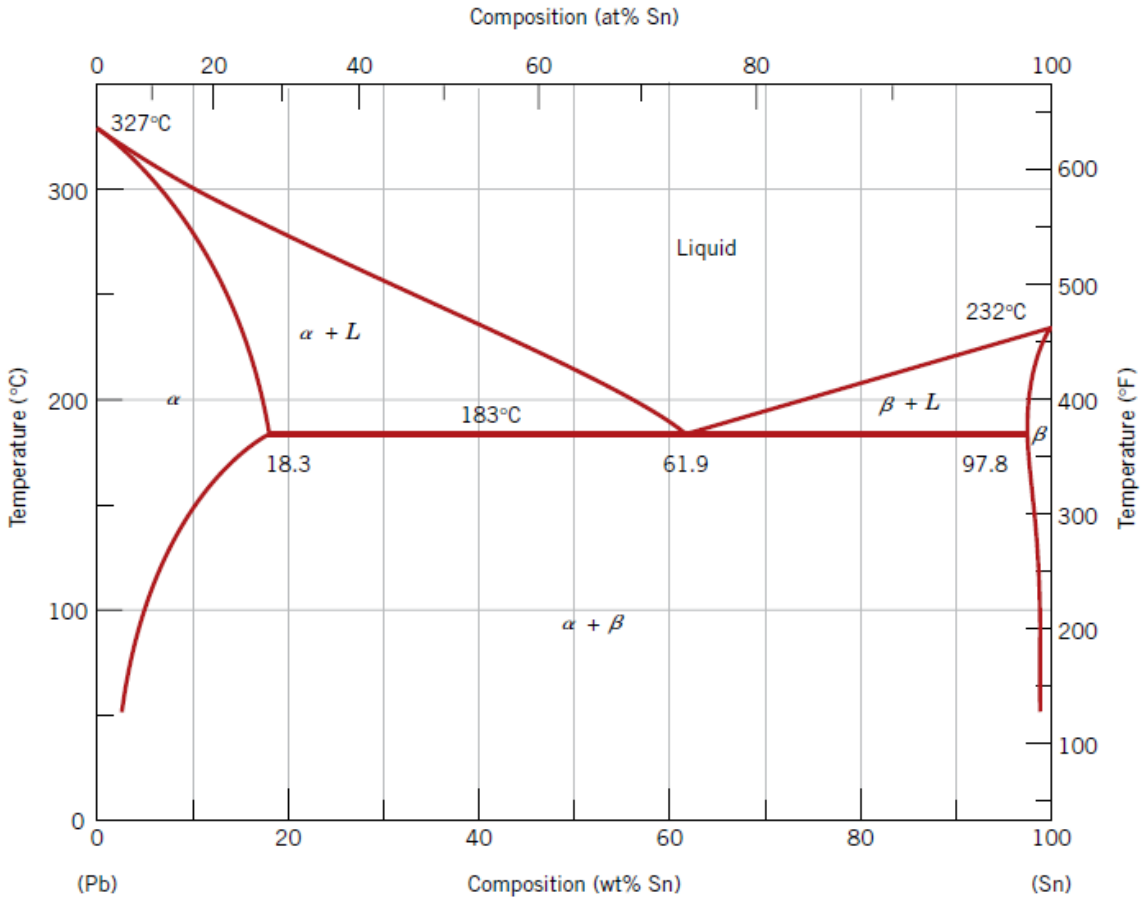
$$L \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha + \beta$$

$$L(71.9\ \text{wt\% Ag}) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha(8.0\ \text{wt\% Ag}) + \beta(91.2\ \text{wt\% Ag})$$

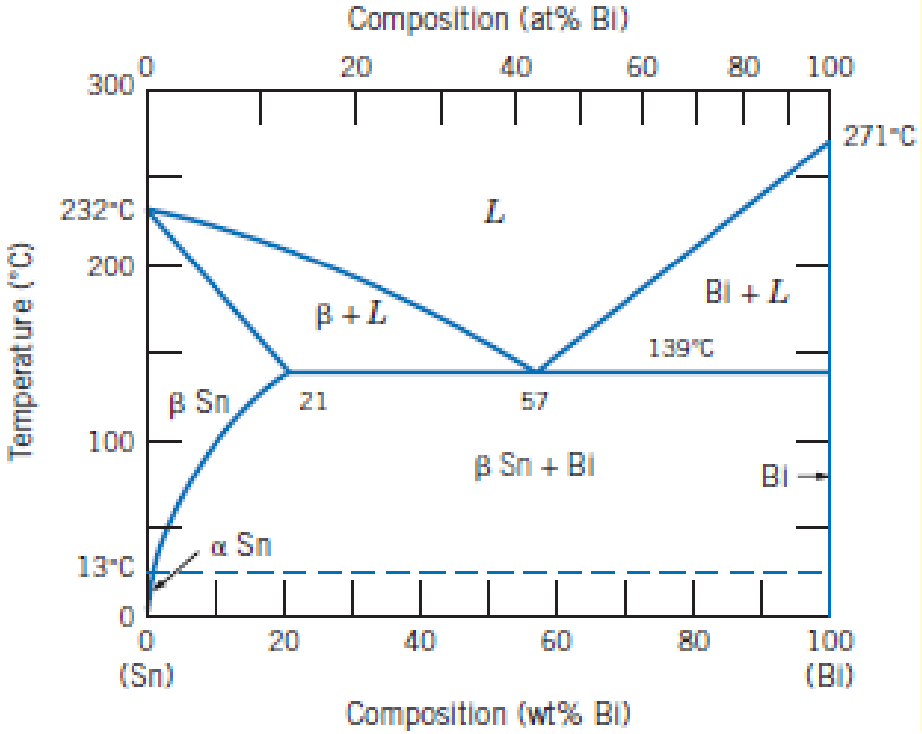


Binary Eutectic System

Pb-Sn phase diagram



Sn-Bi phase diagram
(Lead free solder)



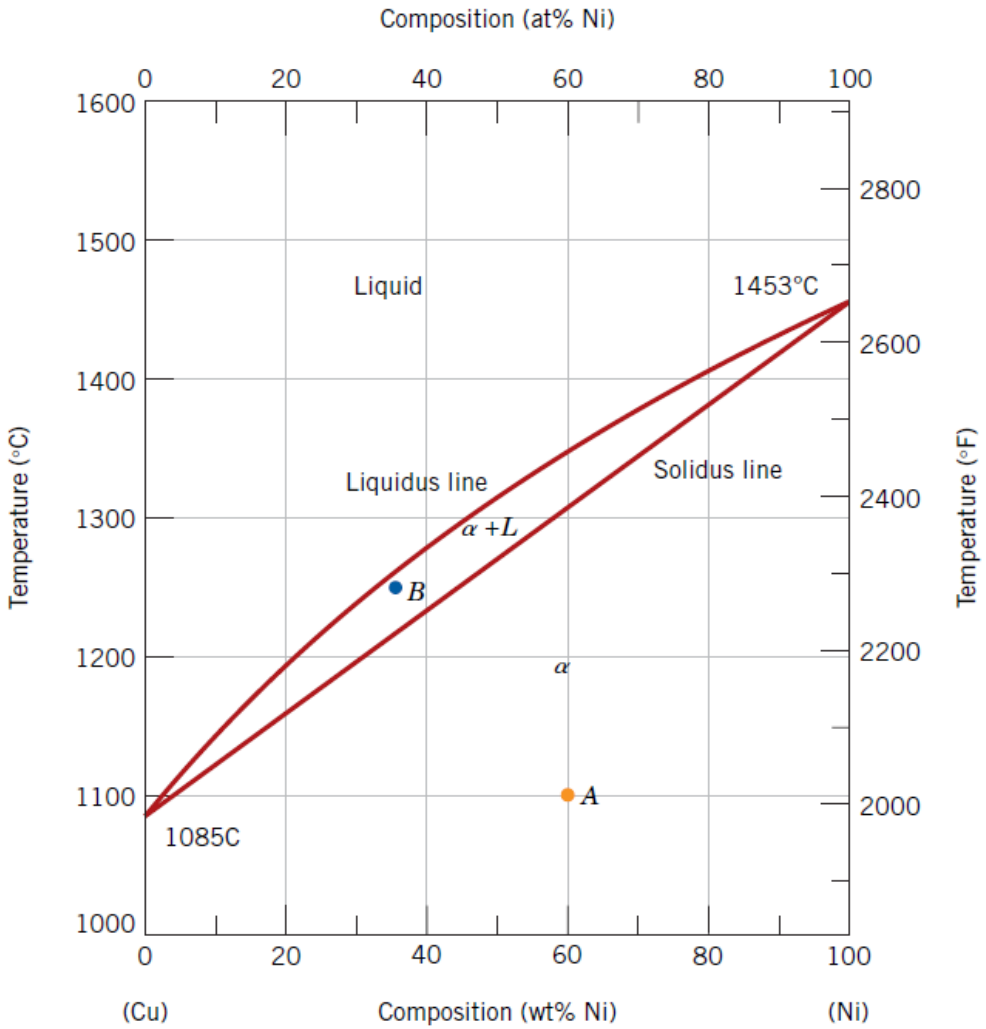
A 50 wt% Ni–50 wt% Cu alloy is slowly cooled from 1400°C (2550°F) to 1200°C (2190°F).

(a) At what temperature does the first solid phase form?

(b) What is the composition of this solid phase?

(c) At what temperature does the liquid solidify?

(d) What is the composition of this last remaining liquid phase?

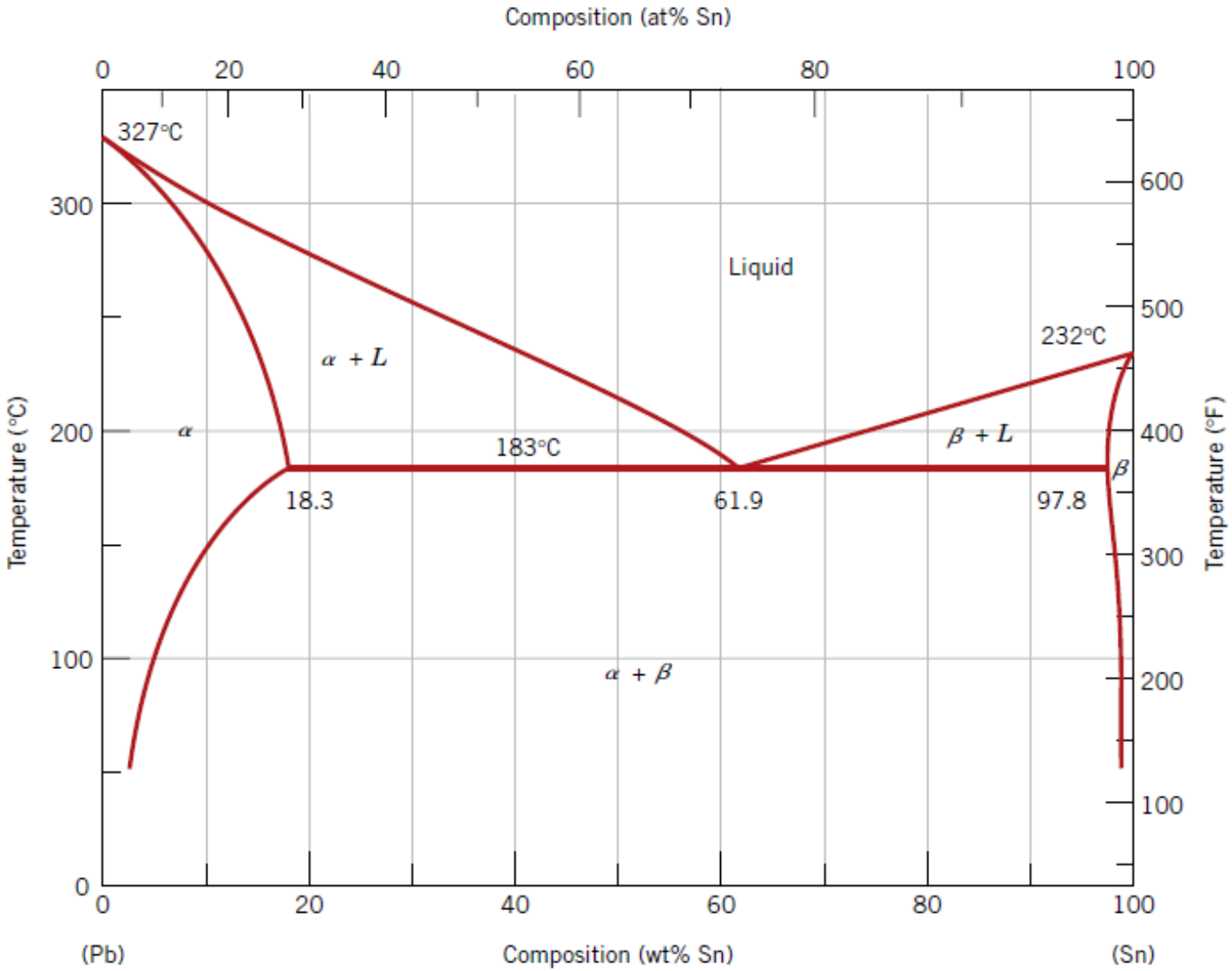


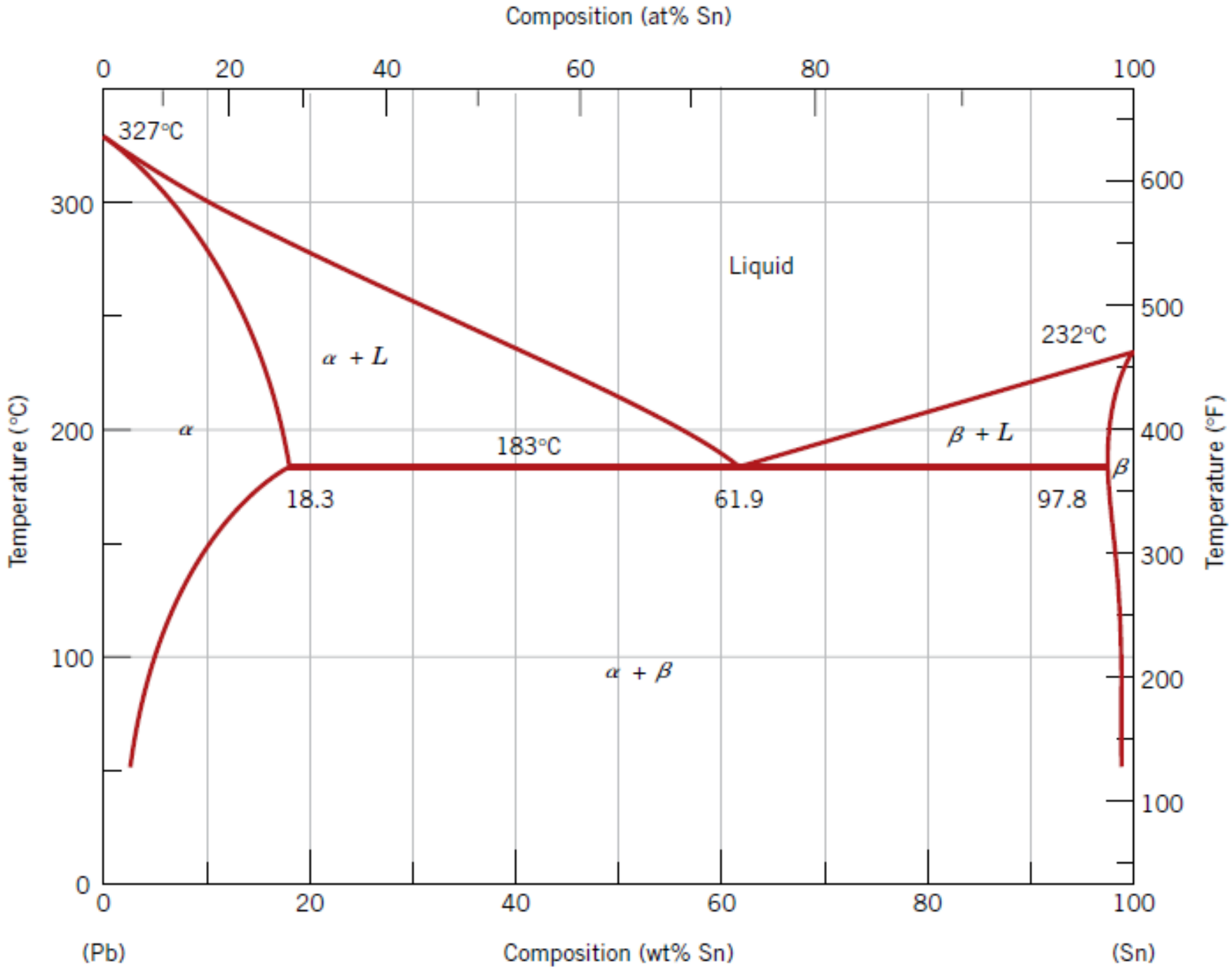
Given here are the solidus and liquidus temperatures for the copper–gold system. Construct the phase diagram for this system and label each region.

Composition (wt% Au)	*Solidus Temperature (°C)*	*Liquidus Temperature (°C)*
0	1085	1085
20	1019	1042
40	972	996
60	934	946
80	911	911
90	928	942
95	974	984
100	1064	1064

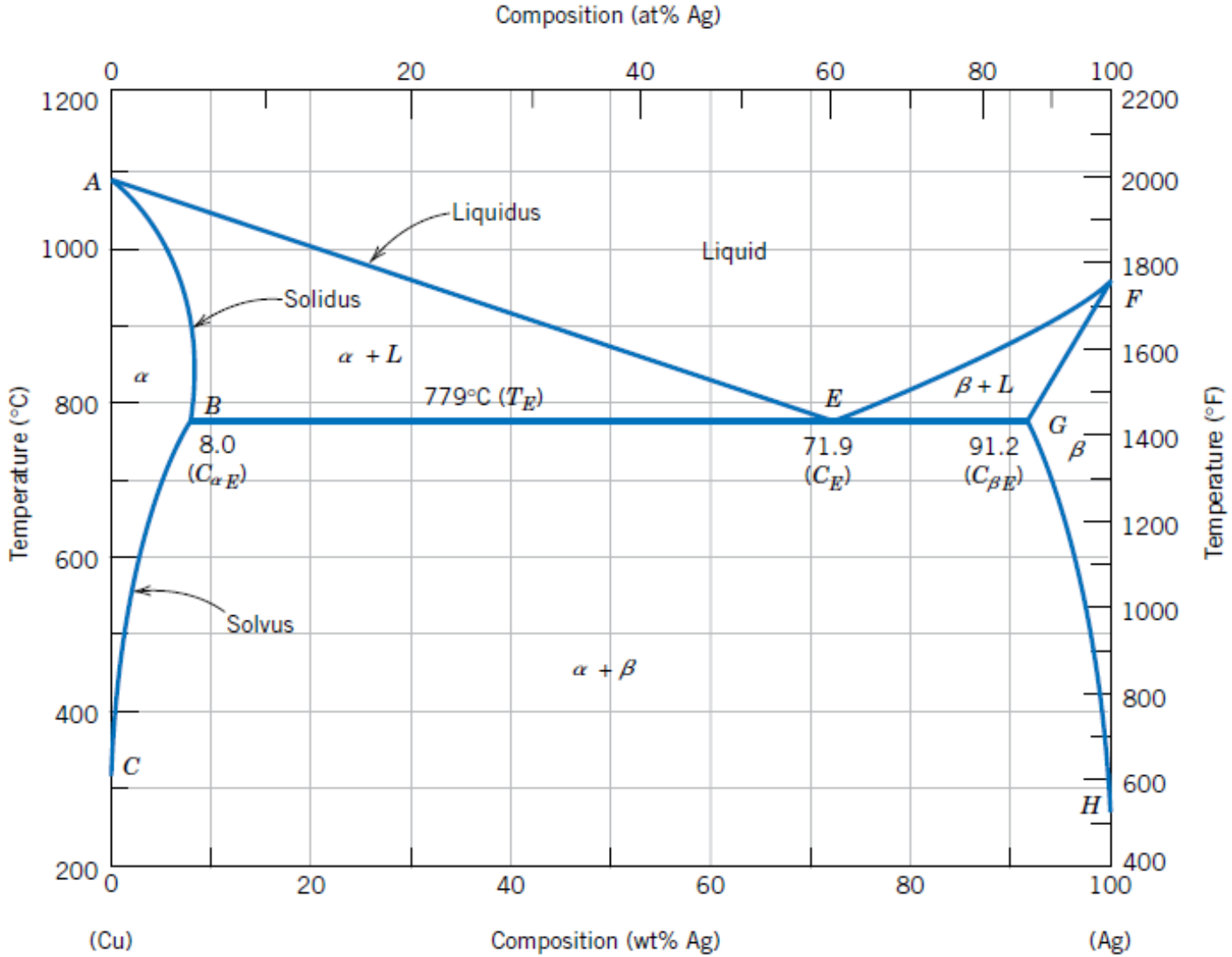
@ particular temperature & composition, can determine phases and their respective compositions?

30 wt% Sn-70 wt% Pb alloy is heated to a temperature within the α + liquid phase region. If the mass fraction of each phase is 0.5, estimate: (a) The temperature of the alloy (b) The compositions of the two phases




Composition (at% Sn)
0
20
40
60
80
100
327°C
Liquid
232°C
α + L
α
183°C
β + L
β
18.3
61.9
97.8
α + β
Temperature (°C)
Temperature (°F)
300
200
100
0
600
500
400
300
200
100
0
20
40
60
80
100
(Pb)
Composition (wt% Sn)
(Sn)

A 90 wt% Ag-10 wt% Cu alloy is heated to a temperature within the β + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine: (a) The temperature of the alloy (b) The composition of the β phase (c) The mass fractions of both phases



A 90 wt% Ag-10 wt% Cu alloy is heated to a temperature within the β + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine: (a) The temperature of the alloy (b) The composition of the β phase (c) The mass fractions of both phases

To determine the temperature of a 90 wt% Ag-10 wt% Cu alloy for which β and liquid phases are present with the liquid phase of composition 85 wt% Ag, we need to construct a tie line across the β + L phase region of Figure that intersects the liquidus line at 85 wt% Ag; this is possible at about 850°C.

The composition of the β phase at this temperature is determined from the intersection of this same tie line with solidus line, which corresponds to about 95 wt% Ag

The mass fractions of the two phases are determined using the lever rule, with C0 = 90 wt% Ag, CL = 85 wt% Ag, and Cβ = 95 wt% Ag.

A hypothetical A–B alloy of composition 55 wt% B–45 wt% A at some temperature is found to consist of mass fractions of 0.5 for both α and β phases. If the composition of the β phase is 90 wt% B–10 wt% A, what is the composition of the α phase?

A hypothetical A–B alloy of composition 55 wt% B–45 wt% A at some temperature is found to consist of mass fractions of 0.5 for both α and β phases. If the composition of the β phase is 90 wt% B–10 wt% A, what is the composition of the α phase?

For this problem, we are asked to determine the composition of the alpha phase given that

$C_0 = 55$ (or 55 wt% B-45 wt% A)

$C_\beta = 90$ (or 90 wt% B-10 wt% A)

$W_\alpha = W_\beta = 0.5$

If we set up the lever rule for W_α,

$$W_\alpha = 0.5 = C_\beta - C_0 / C_\beta - C_\alpha$$
$$= 90 - 55 / 90 - C_\alpha$$

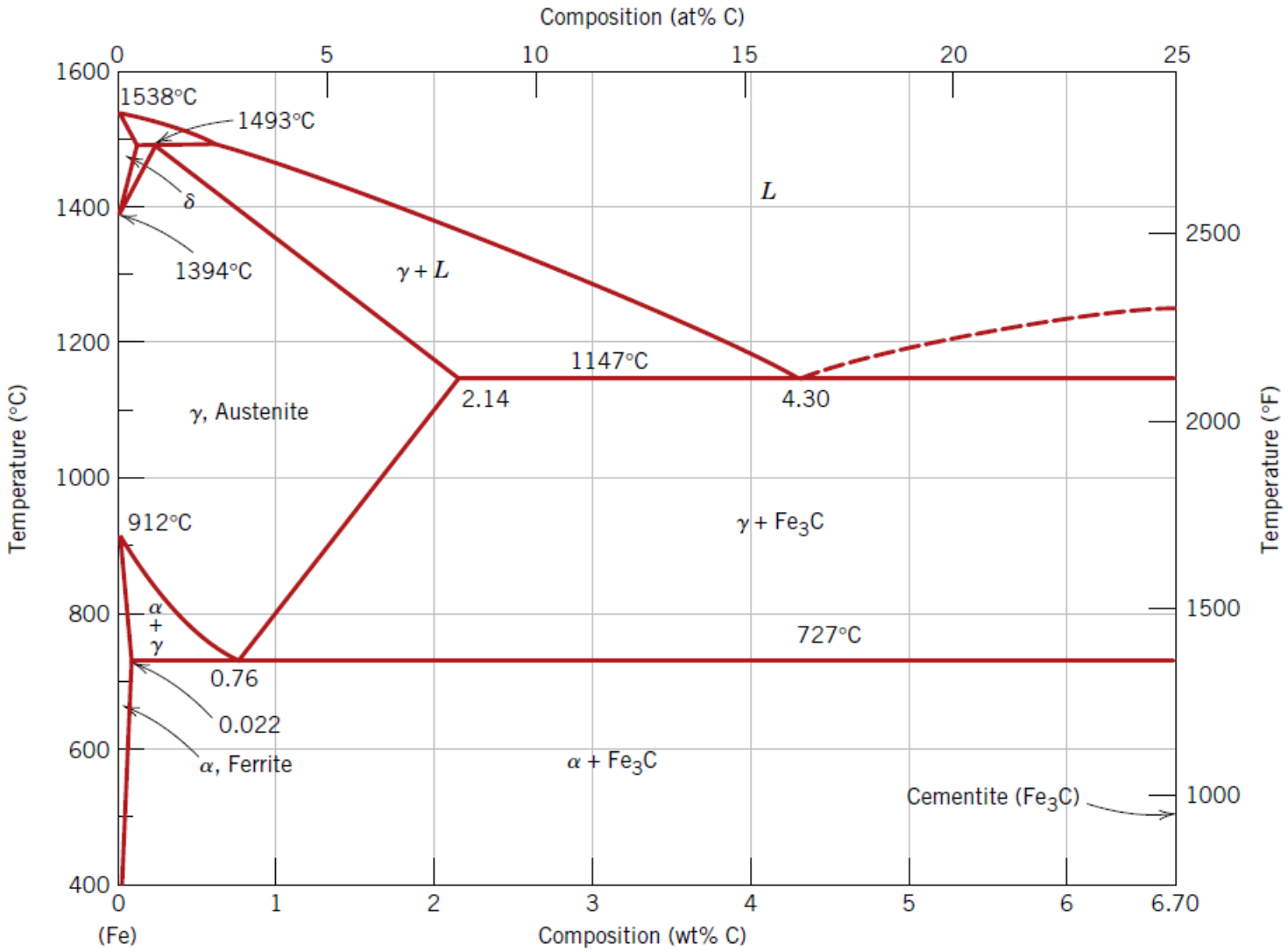
And solving for C,

$C_\alpha = 20$ (or 20 wt% B-80 wt% A

Iron – Carbon System

(Only iron rich portion shown, which is for all steels and CI)

Iron – Iron Carbide Phase Diagram



Eutectic reaction (@ 4.3% wt, 1147 deg cel.)

$$L \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \gamma + Fe_3C$$

Eutectoid reaction (@ 0.76% wt, 727 deg cel.)

$$\gamma \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha + Fe_3C$$

Peritectic reaction (@~0.35 wt, 1493 deg cel.)

$$\delta + L \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \gamma$$

- ✓ Ferrite, Austenite, Cementite
- ✓ Solubility limits
- ✓ Iron (<0.008% wt. C)
- ✓ Steel (0.008% to 2.14% wt. C)
- ✓ Cast iron (between 2.14% to 4.5% wt. C)

Materials for Engineers

Lecture until Oct 20

Dr S. Gowthaman
Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

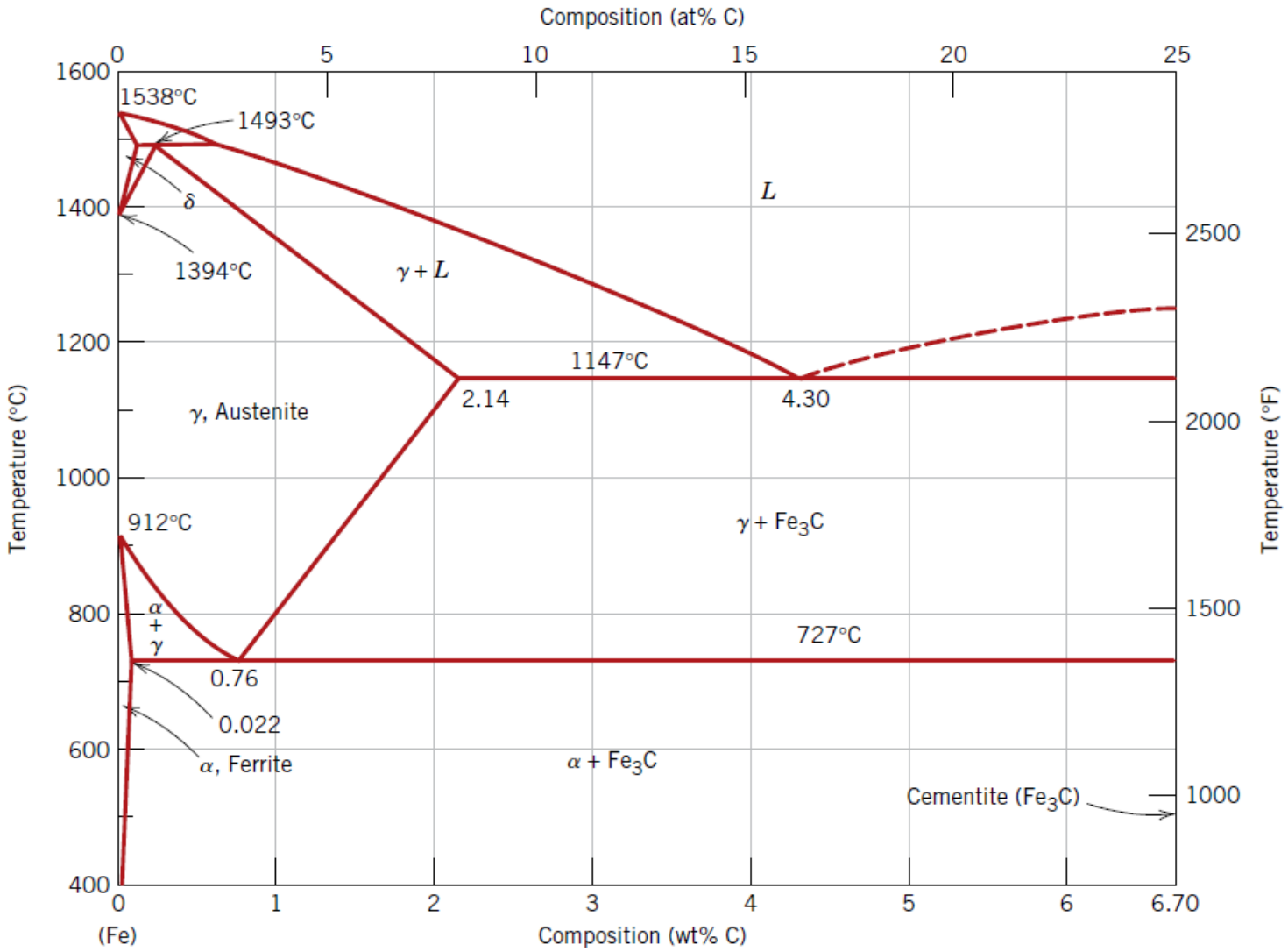
Reference:

1. Materials science and engineering by William Callister, Wiley Publications

Iron – Carbon System

(Only iron rich portion shown, which is for all steels and CI)

Iron – Iron Carbide Phase Diagram



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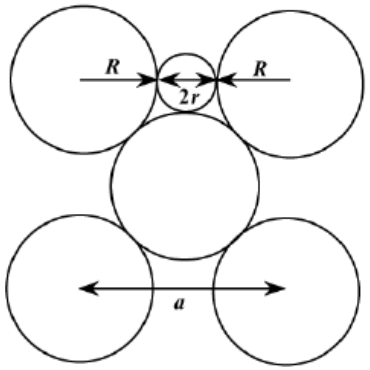
$$\delta + L \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \gamma$$

- ✓ Ferrite, Austenite, Cementite
- ✓ Solubility limits
- ✓ Iron (<0.008% wt. C)
- ✓ Steel (0.008% to 2.14% wt. C)
- ✓ Cast iron (between 2.14% to 4.5% wt. C)

Interstitial C in FCC iron

(100) face of an FCC unit cell; the interstitial site is at the center of the edge.



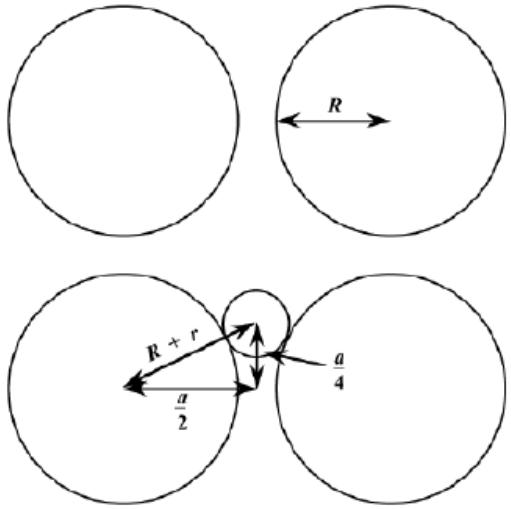
The diameter of an atom that will just fit into this site ($2r$) is just the difference between that unit cell edge length (a) and the radii of the two host atoms that are located on either side of the site (R); that is

$$2r = a - 2R$$

However, for FCC a is related to R as $a = 2R\sqrt{2}$; therefore, solving for r from the above equation gives

$$r = \frac{a - 2R}{2} = \frac{2R\sqrt{2} - 2R}{2} = 0.41R$$

Interstitial C in BCC iron



The interstitial atom that just fits into this interstitial site is shown by the small circle. It is situated in the plane of this (100) face, midway between the two vertical unit cell edges, and one quarter of the distance between the bottom and top cell edges. From the right triangle that is defined by the three arrows we may write

$$\left(\frac{a}{2}\right)^2 + \left(\frac{a}{4}\right)^2 = (R + r)^2$$

However, $a = \frac{4R}{\sqrt{3}}$, and, therefore, making this substitution, the above equation takes the form

$$\left(\frac{4R}{2\sqrt{3}}\right)^2 + \left(\frac{4R}{4\sqrt{3}}\right)^2 = R^2 + 2Rr + r^2$$

After rearrangement the following quadratic equation results:

$$r^2 + 2Rr - 0.667R^2 = 0$$

And upon solving for r, $r = 0.291R$.

Thus, for a host atom of radius R, the size of an interstitial site for FCC is approximately 1.4 times that for BCC.

Ex.

$r_v / r_{FCC} = 0.41$

FCC

$r_v = 0.41 \times 1.29\,\text{Å} = 0.53\,\text{Å}$

$r_c = 0.70\,\text{Å}$ → So Comp strain

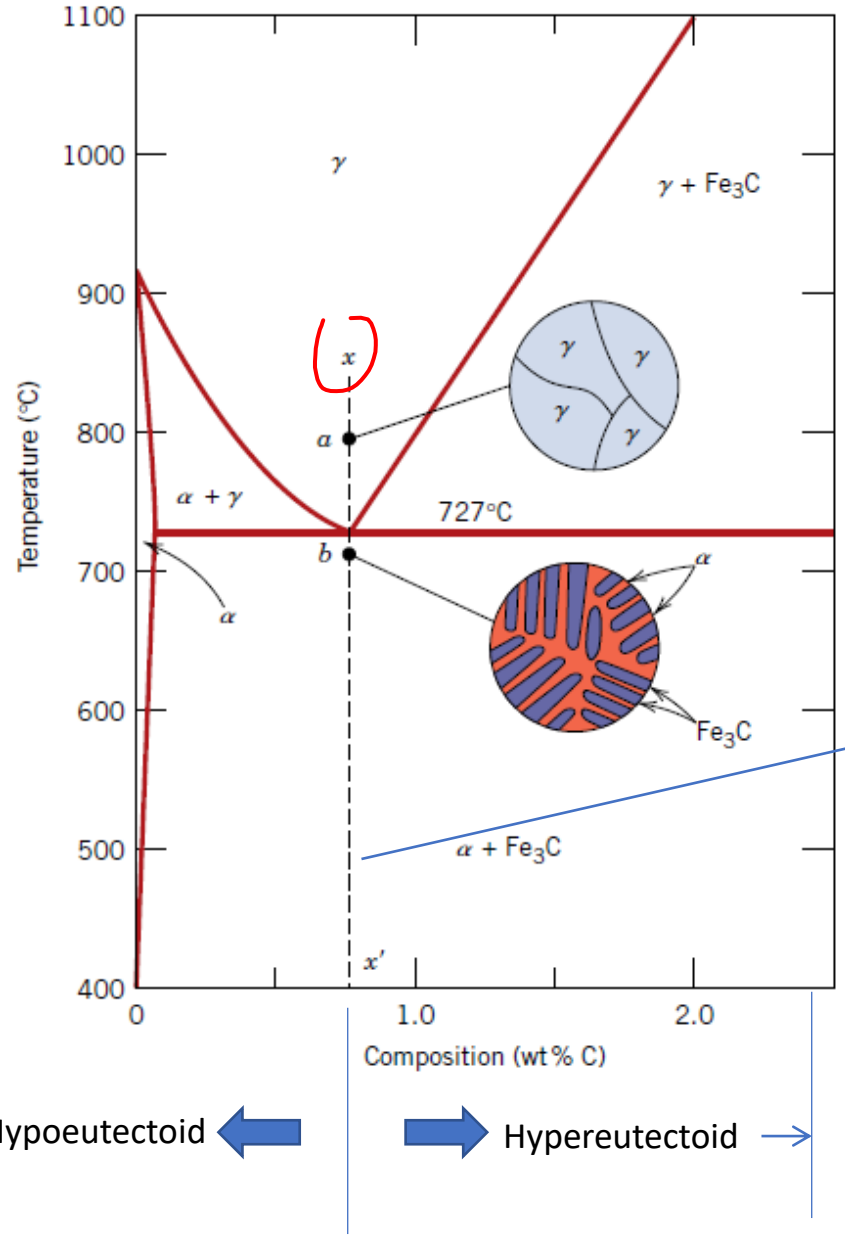
BCC

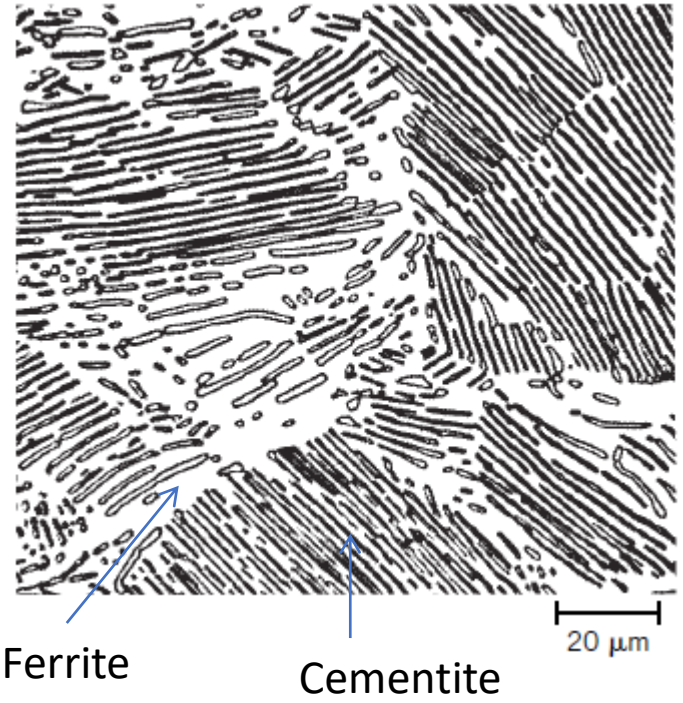
$r_v / r_{FCC} = 0.29$

$r_v = 0.29 \times 1.29\,\text{Å} = 0.37\,\text{Å}$

$r_c = 0.70\,\text{Å}$ → Sever Comp strain

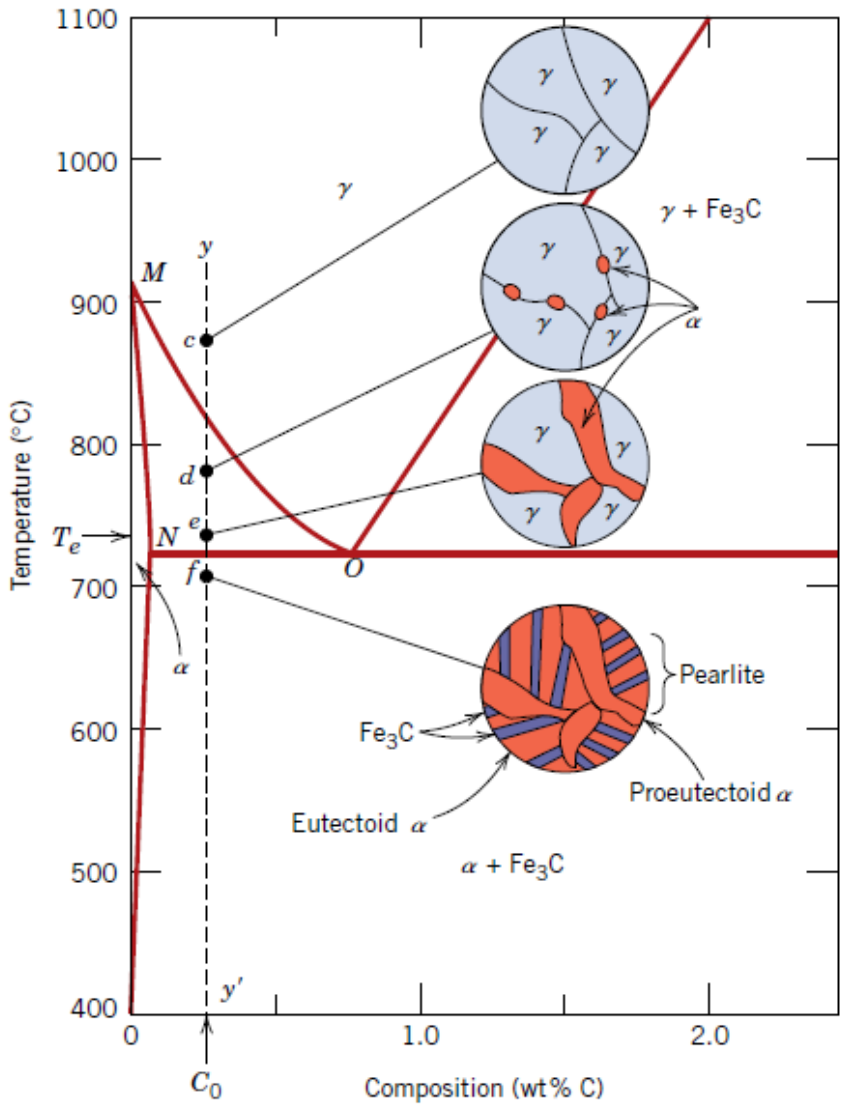
Development of microstructure in iron-carbon alloys





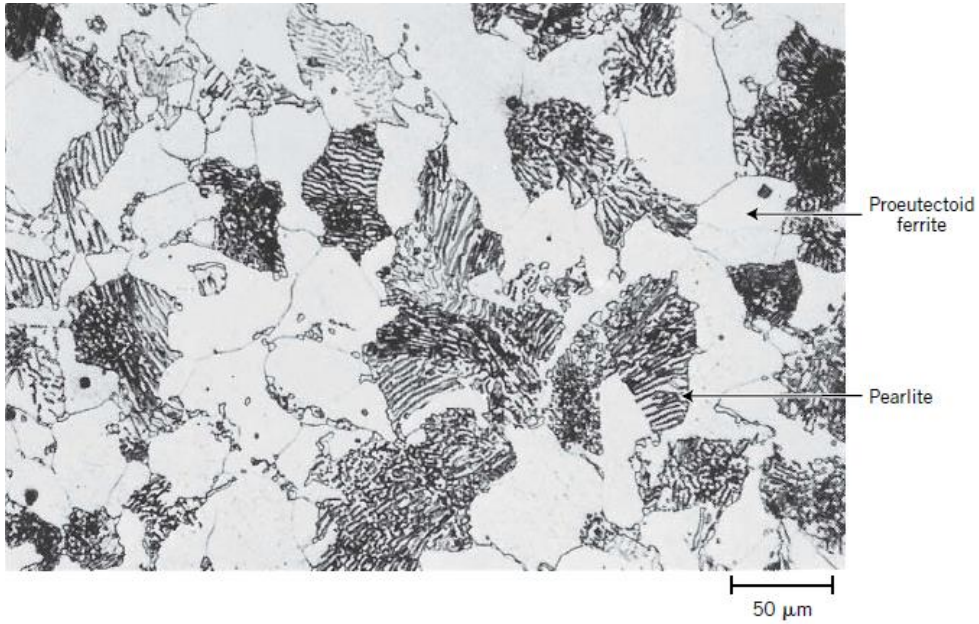
Eutectoid steel (0.76% wt. C)
Pearlite microstructure

Development of microstructure in iron-carbon alloys

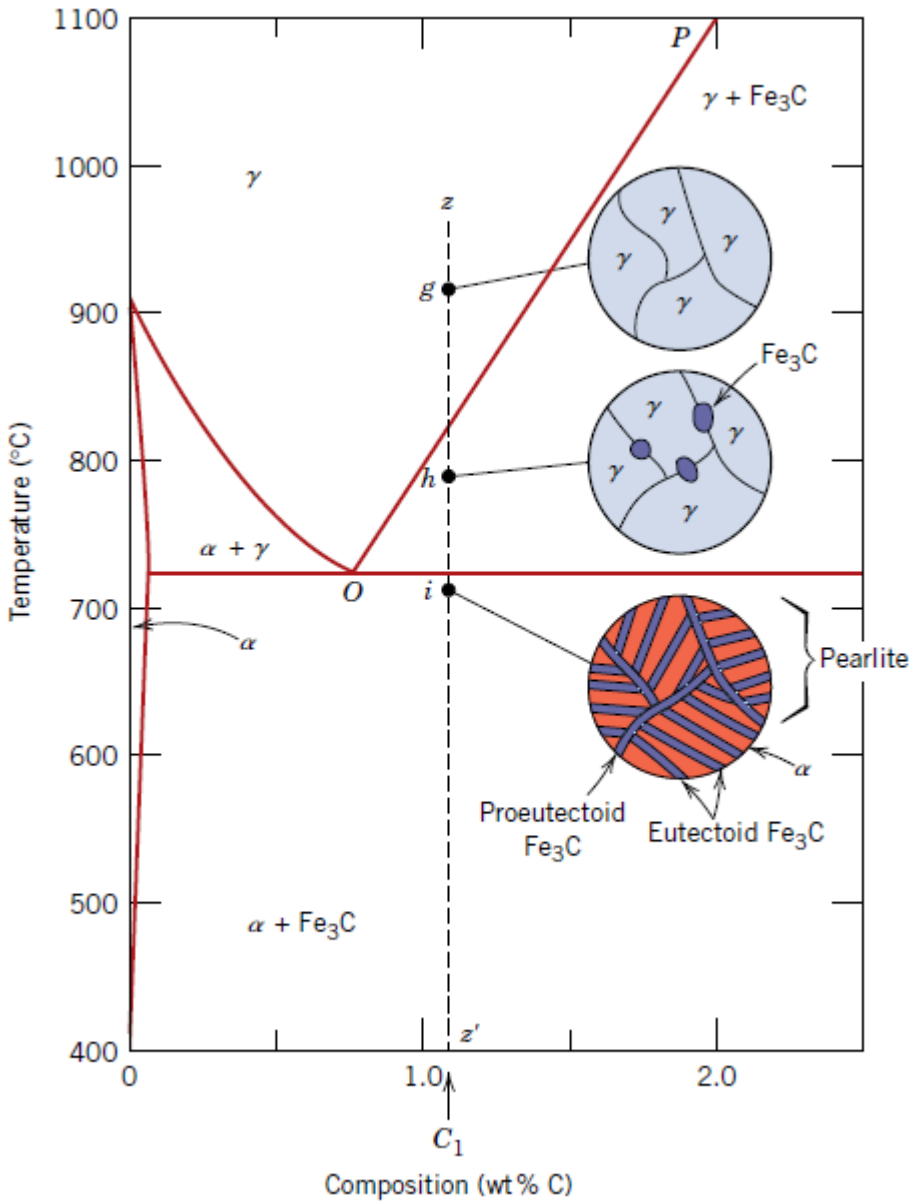


Hypoeutectoid alloy

Proeutectoid ferrite + Eutectoid ferrite = Total ferrite

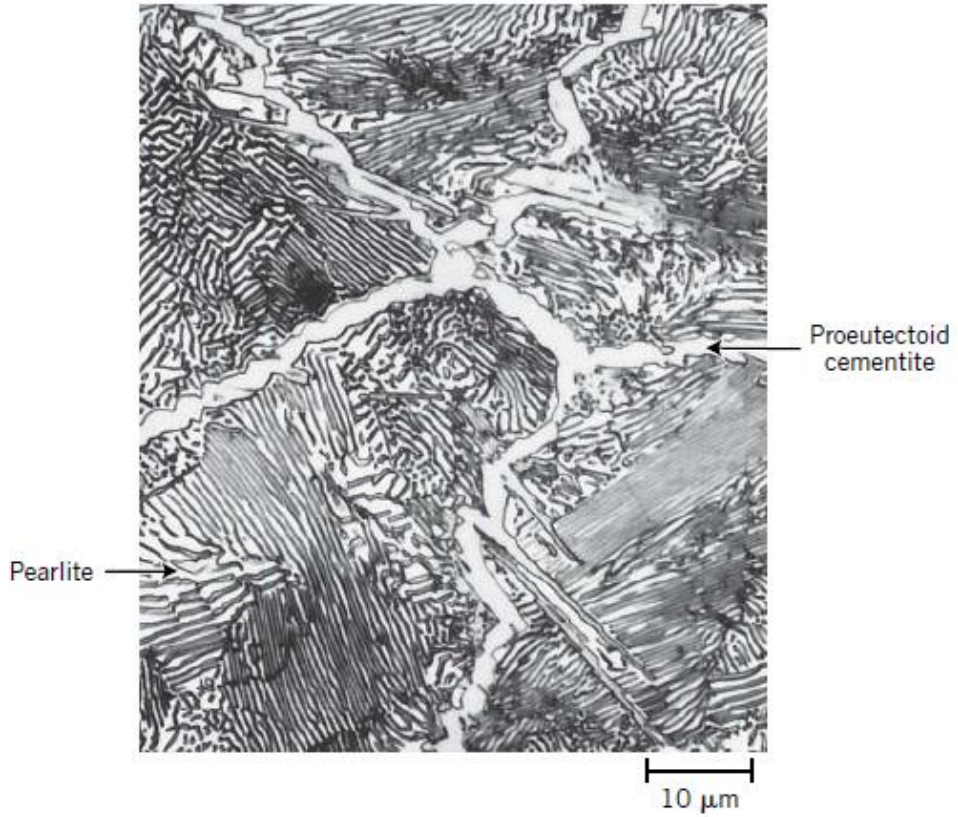


Development of microstructure in iron-carbon alloys

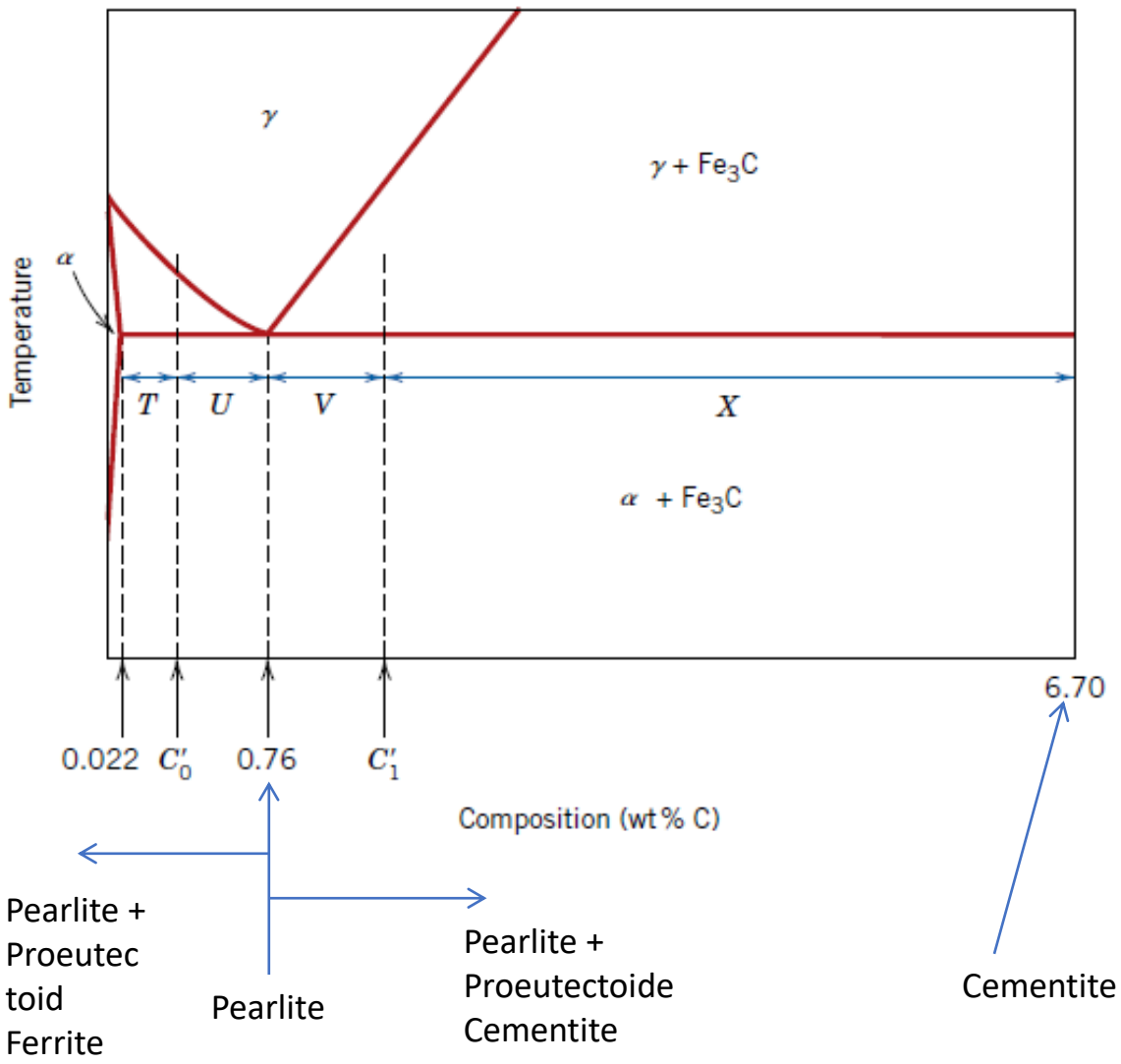


Hypereutectoid alloy

Proeutectoid cementite + Eutectoid cementite = Total cementite



Development of microstructure in iron-carbon alloys



Let us consider an alloy of composition C_o'

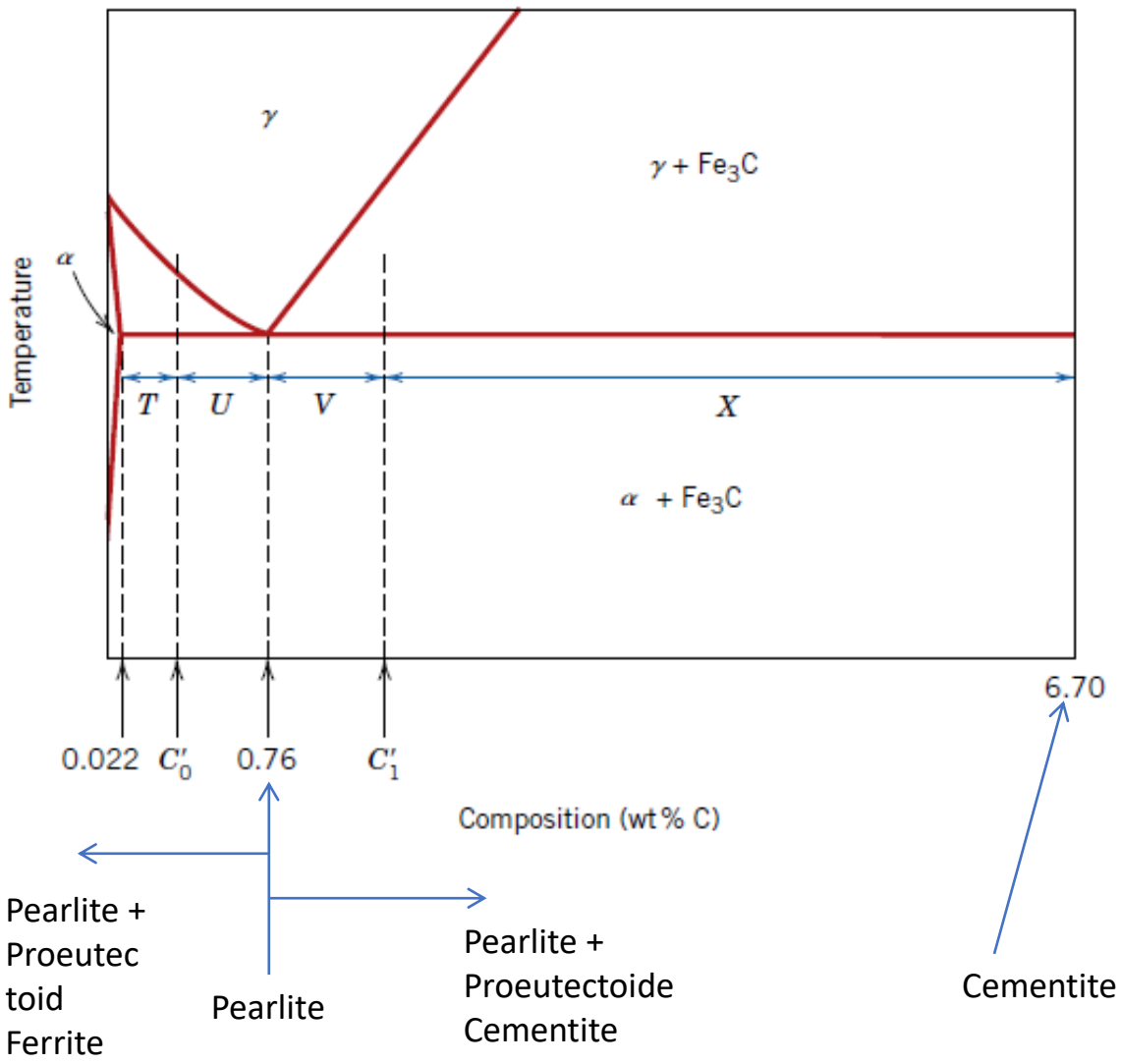
The fraction of pearlite, W_p, may be determined according to

$$W_p = \frac{T}{T+U} = \frac{C_0' - 0.022}{0.76 - 0.022} = \frac{C_0' - 0.022}{0.74}$$

The fraction of proeutectoid α, $W_{\alpha'}$, is computed as follows:

$$W_{\alpha'} = \frac{U}{T+U} = \frac{0.76 - C_0'}{0.76 - 0.022} = \frac{0.76 - C_0'}{0.74}$$

Development of microstructure in iron-carbon alloys



Let us consider an alloy of coposition C_1'

The fraction of pearlite, W_p, may be determined according to

$$W_p = \frac{X}{V + X} = \frac{6.70 - C_1'}{6.70 - 0.76} = \frac{6.70 - C_1'}{5.94}$$

The fraction of proeutectoid cementite, $W_{Fe3C'}$, may be determined according to

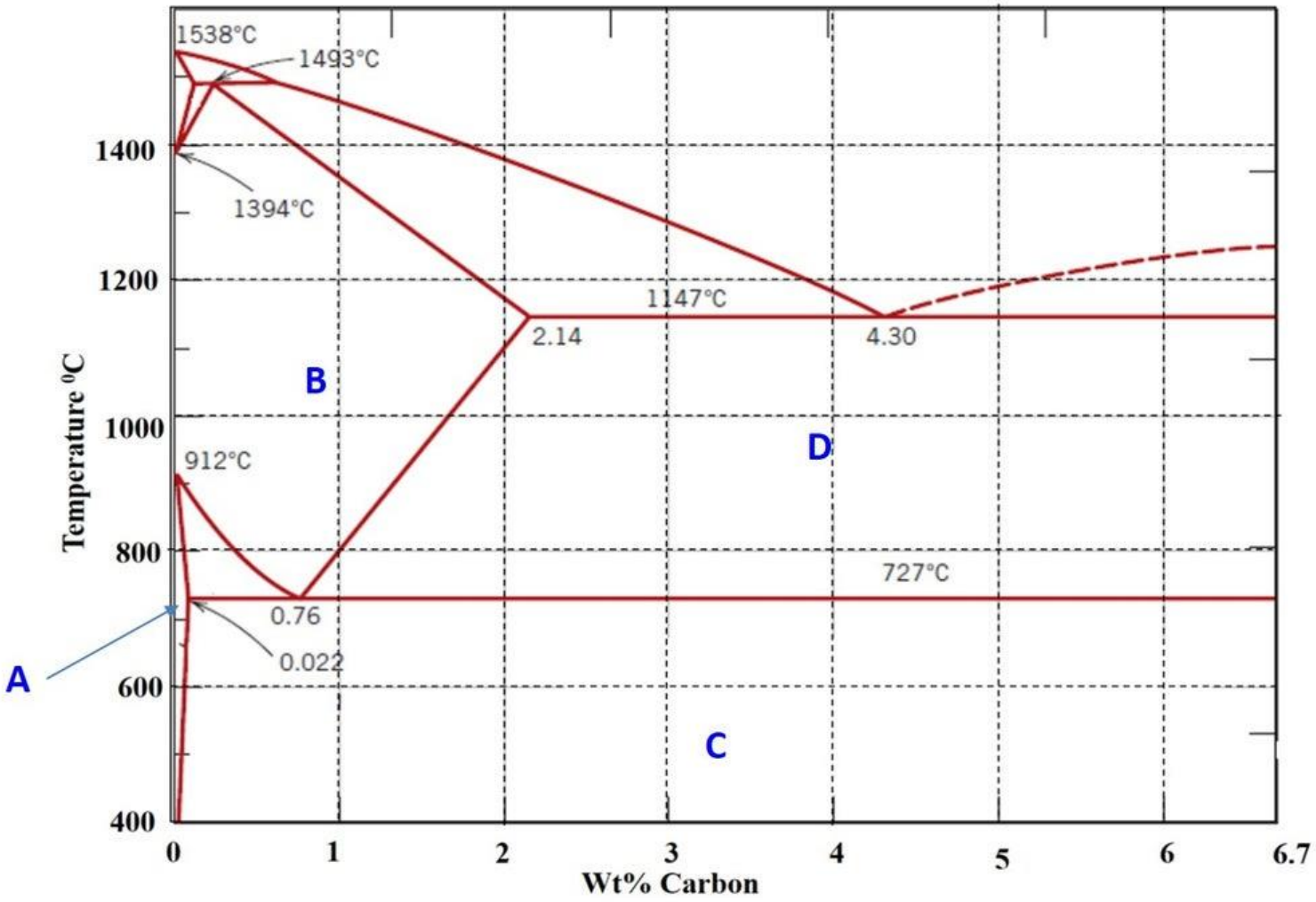
$$W_{Fe_3C'} = \frac{V}{V + X} = \frac{C_1' - 0.76}{6.70 - 0.76} = \frac{C_1' - 0.76}{5.94}$$

For a 99.65 wt% Fe–0.35 wt% C alloy at a temperature just below the eutectoid, determine the following:

(a) The fractions of total ferrite and cementite phases

(b) The fractions of the proeutectoid ferrite and pearlite

(c) The fraction of eutectoid ferrite



For a 99.65 wt% Fe–0.35 wt% C alloy at a temperature just below the eutectoid, determine the following:

(a) The fractions of total ferrite and cementite phases

(b) The fractions of the proeutectoid ferrite and pearlite

(c) The fraction of eutectoid ferrite

(a) This part of the problem is solved by applying the lever rule expressions using a tie line that extends all the way across the $\alpha + Fe_3C$ phase field. Thus, C_0' is 0.35 wt% C, and

$$W_\alpha = \frac{6.70 - 0.35}{6.70 - 0.022} = 0.95$$

and

$$W_{Fe_3C} = \frac{0.35 - 0.022}{6.70 - 0.022} = 0.05$$

(b) The fractions of proeutectoid ferrite and pearlite are determined by using the lever rule and a tie line that extends only to the eutectoid composition (i.e., Equations 9.20 and 9.21). We have

$$W_p = \frac{0.35 - 0.022}{0.76 - 0.022} = 0.44$$

and

$$W_{\alpha'} = \frac{0.76 - 0.35}{0.76 - 0.022} = 0.56$$

(c) All ferrite is either as proeutectoid or eutectoid (in the pearlite). Therefore, the sum of these two ferrite fractions equals the fraction of total ferrite; that is,

$$W_{\alpha'} + W_{\alpha e} = W_\alpha$$

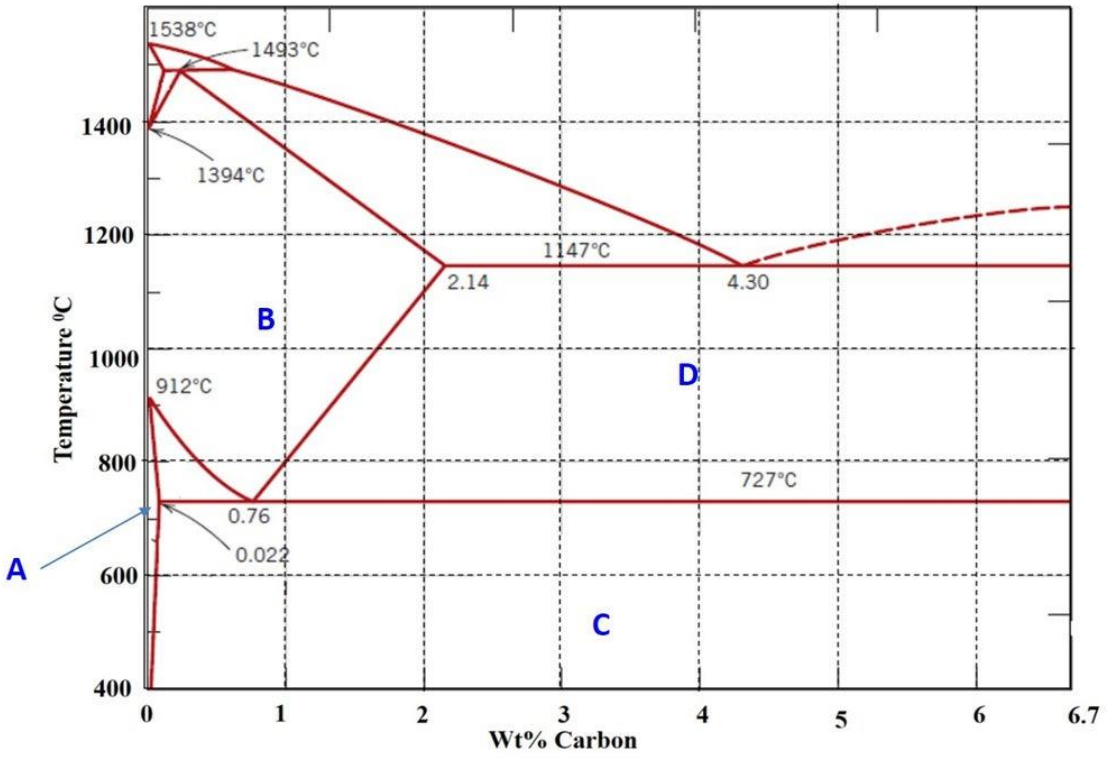
where $W_{\alpha e}$ denotes the fraction of the total alloy that is eutectoid ferrite. Values for W_α and $W_{\alpha'}$ were determined in parts (a) and (b) as 0.95 and 0.56, respectively. Therefore,

$$W_{\alpha e} = W_\alpha - W_{\alpha'} = 0.95 - 0.56 = 0.39$$

Microstructure of 0.76% C steel after undergoing eutectoid transformation (during cooling) is

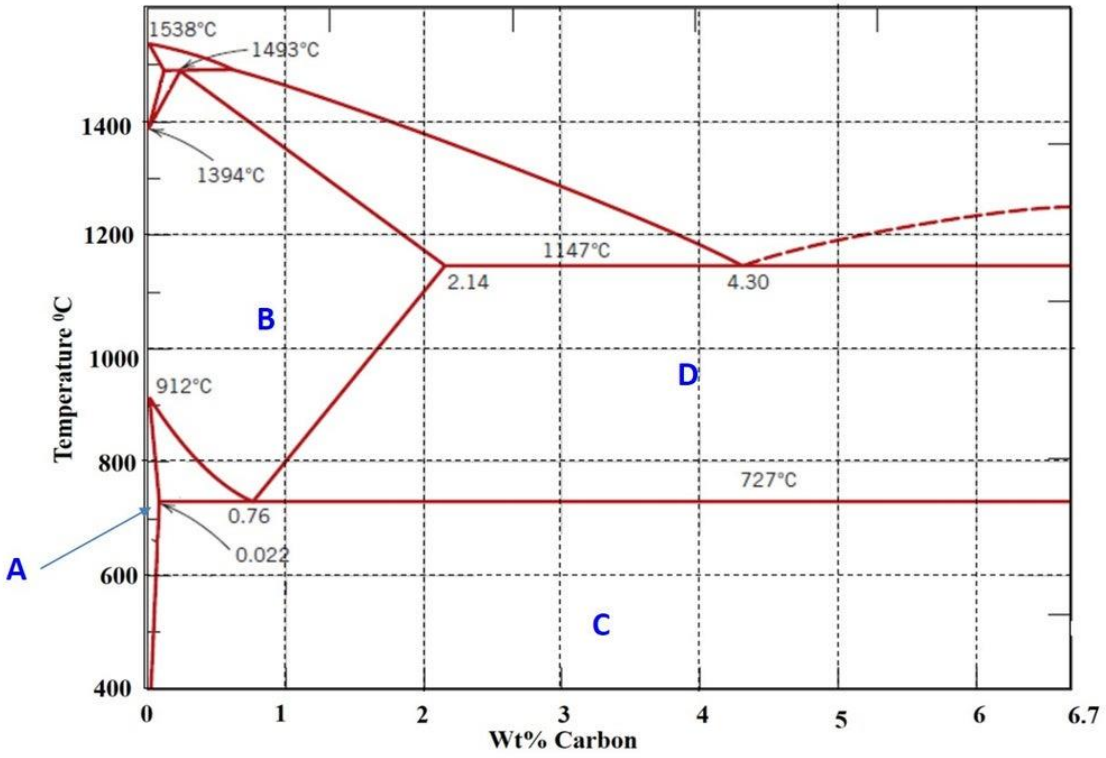
Austenite
Ferrite
Cementite
Pearlite

Microstructures at point A & B



Ferrite (BCC) & Austenite (BCC)
Austenite (BCC) & Ferrite (FCC)
Ferrite (BCC) & Austenite (FCC)
Austenite (FCC) & Ferrite (BCC)

Phases at point D & C



Austenite + cementite and Ferrite + cementite
Austenite + Ferrite and Austenite + cementite
Ferrite + cementite and Pearlite
Austenite + cementite and Ferrite + austenite

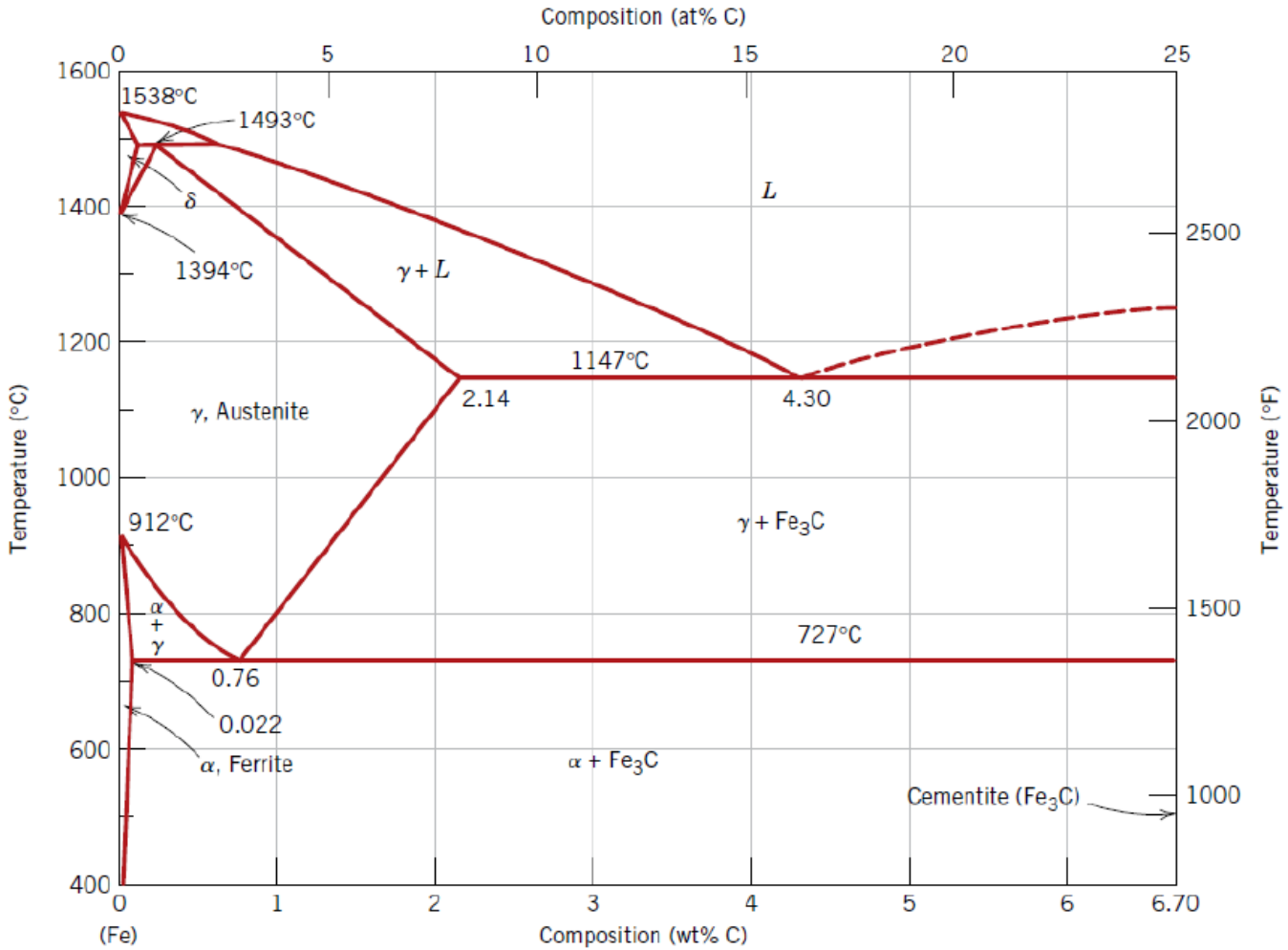
Isothermal Transformation Diagrams

Pearlite

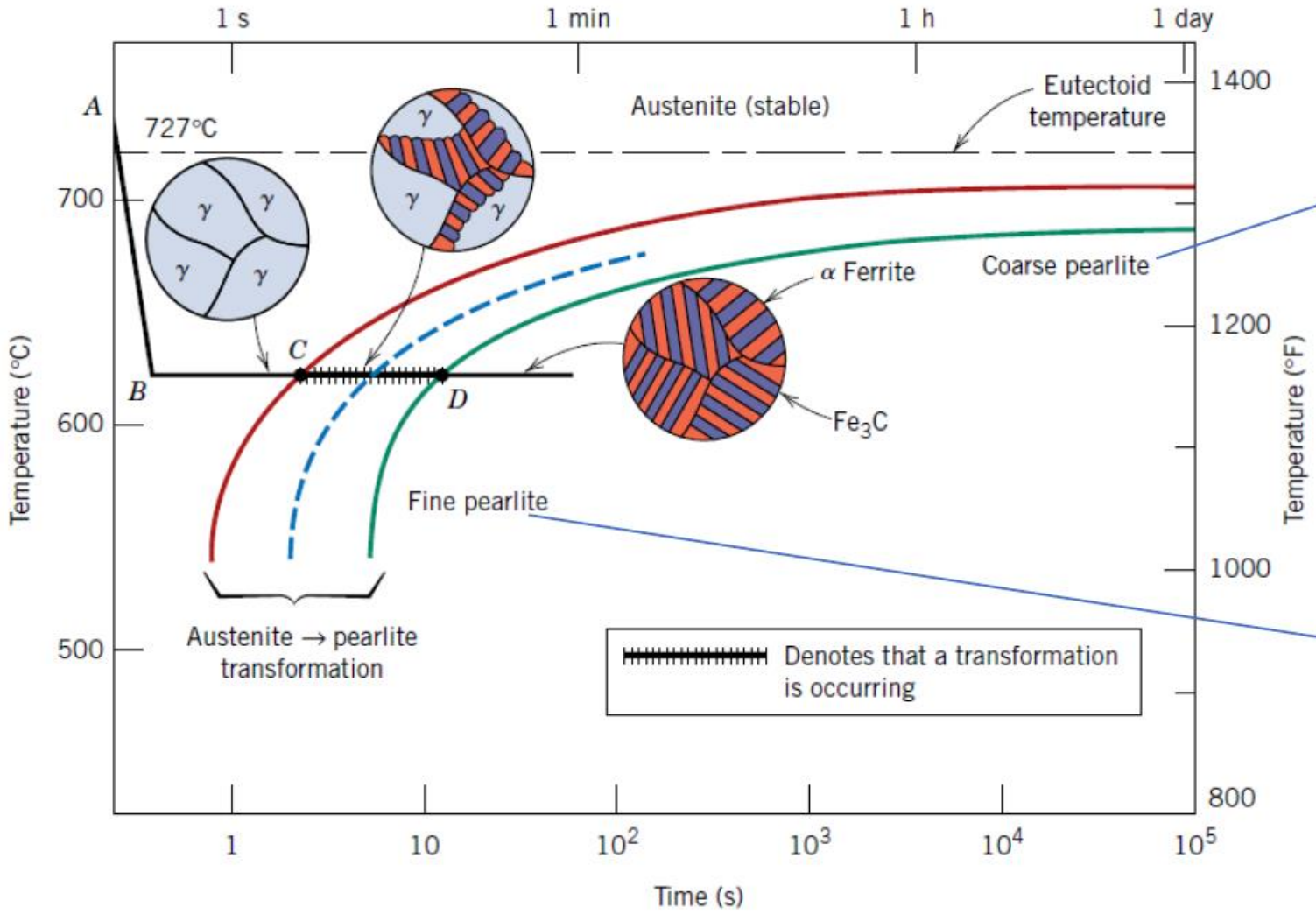
Consider again the iron–iron carbide eutectoid reaction

$$\gamma(0.76\ \text{wt\%C}) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \alpha(0.022\ \text{wt\%}\ \text{C}) + \text{Fe}_3\text{C}\ (6.70\ \text{wt\%}$$

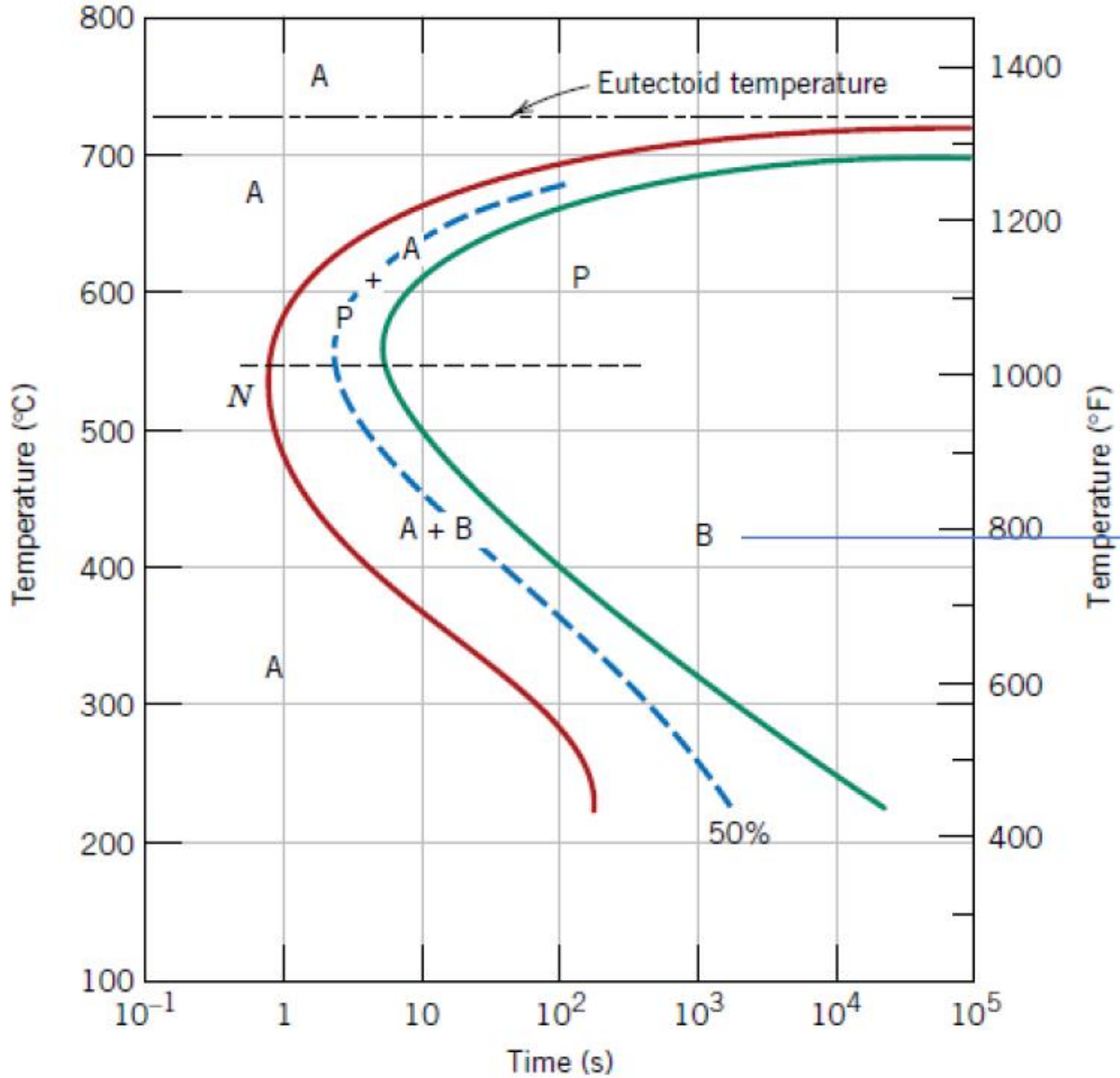
Consider Eutectoid steel having Pearlite microstructure



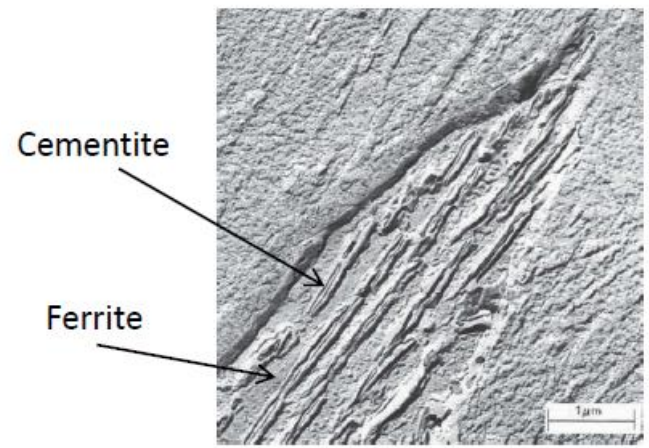
Isothermal transformation diagram for a eutectoid iron–carbon alloy



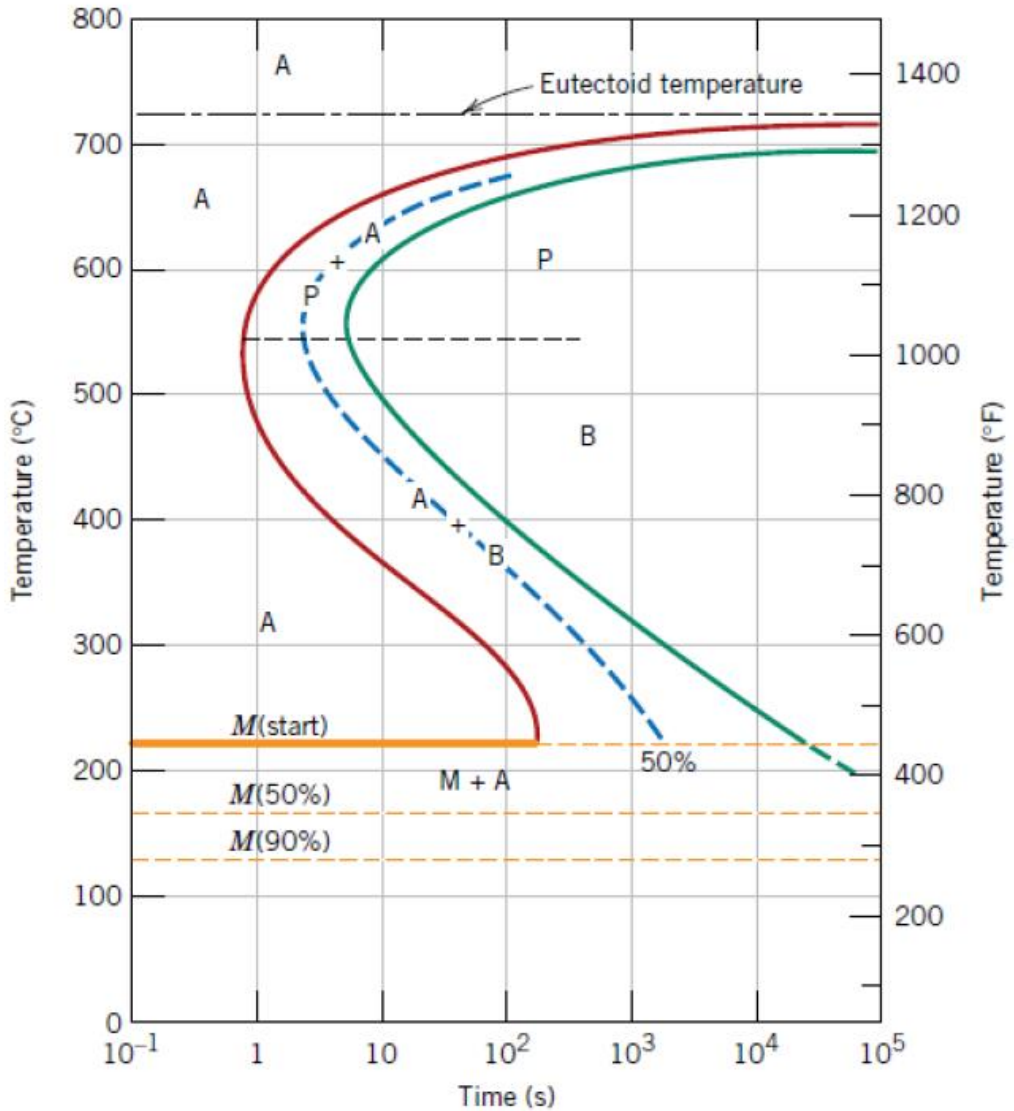
Isothermal transformation diagram for a eutectoid iron–carbon alloy

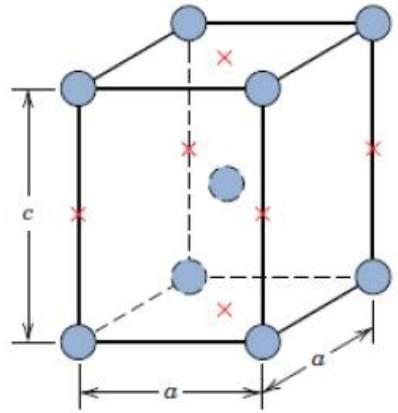


Bainite (forms as needles or plates)



Isothermal transformation diagram for a eutectoid iron–carbon alloy

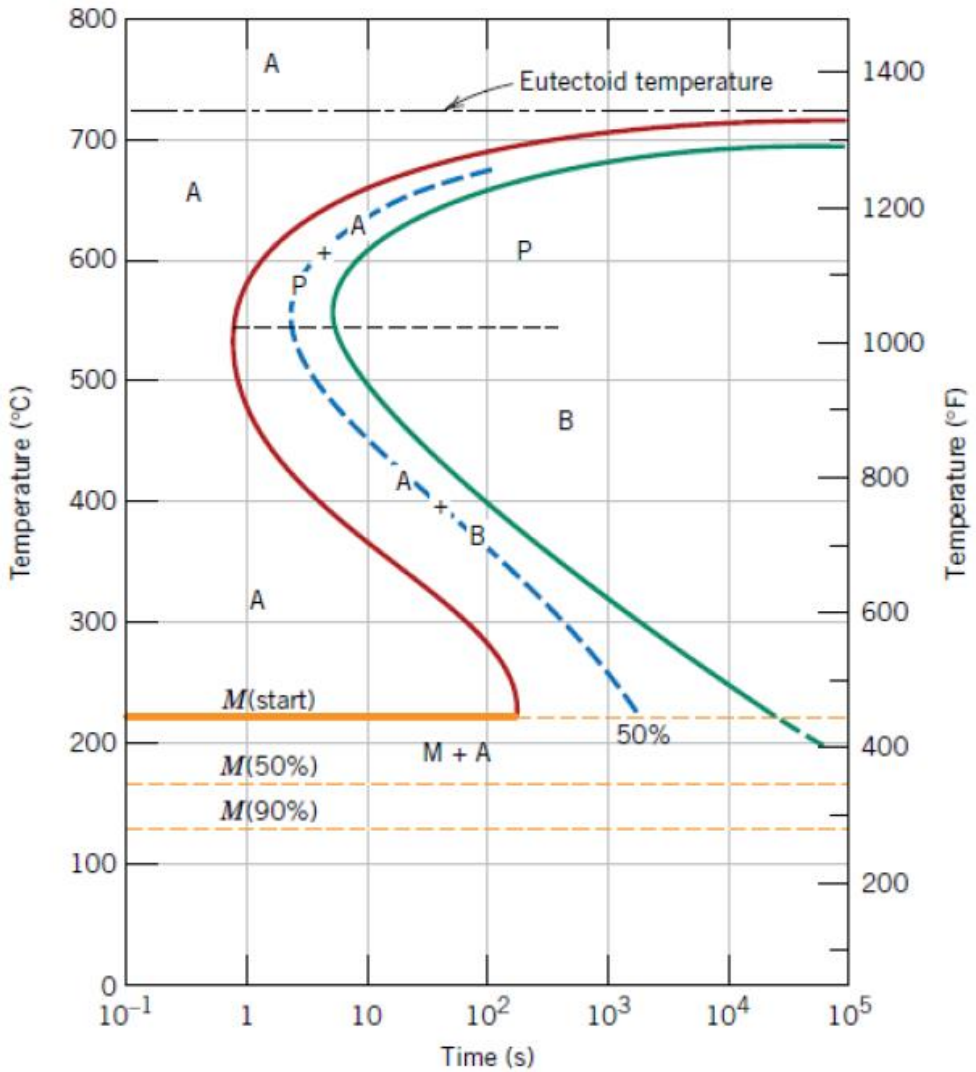




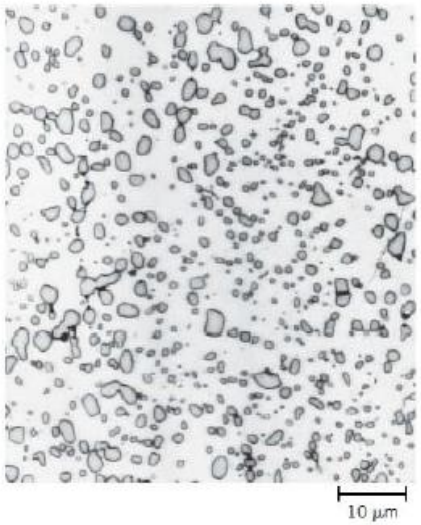
BCT structure of martensite

Martensite (sharp grains) in untransformed austenite (white region)

Isothermal transformation diagram for a eutectoid iron–carbon alloy



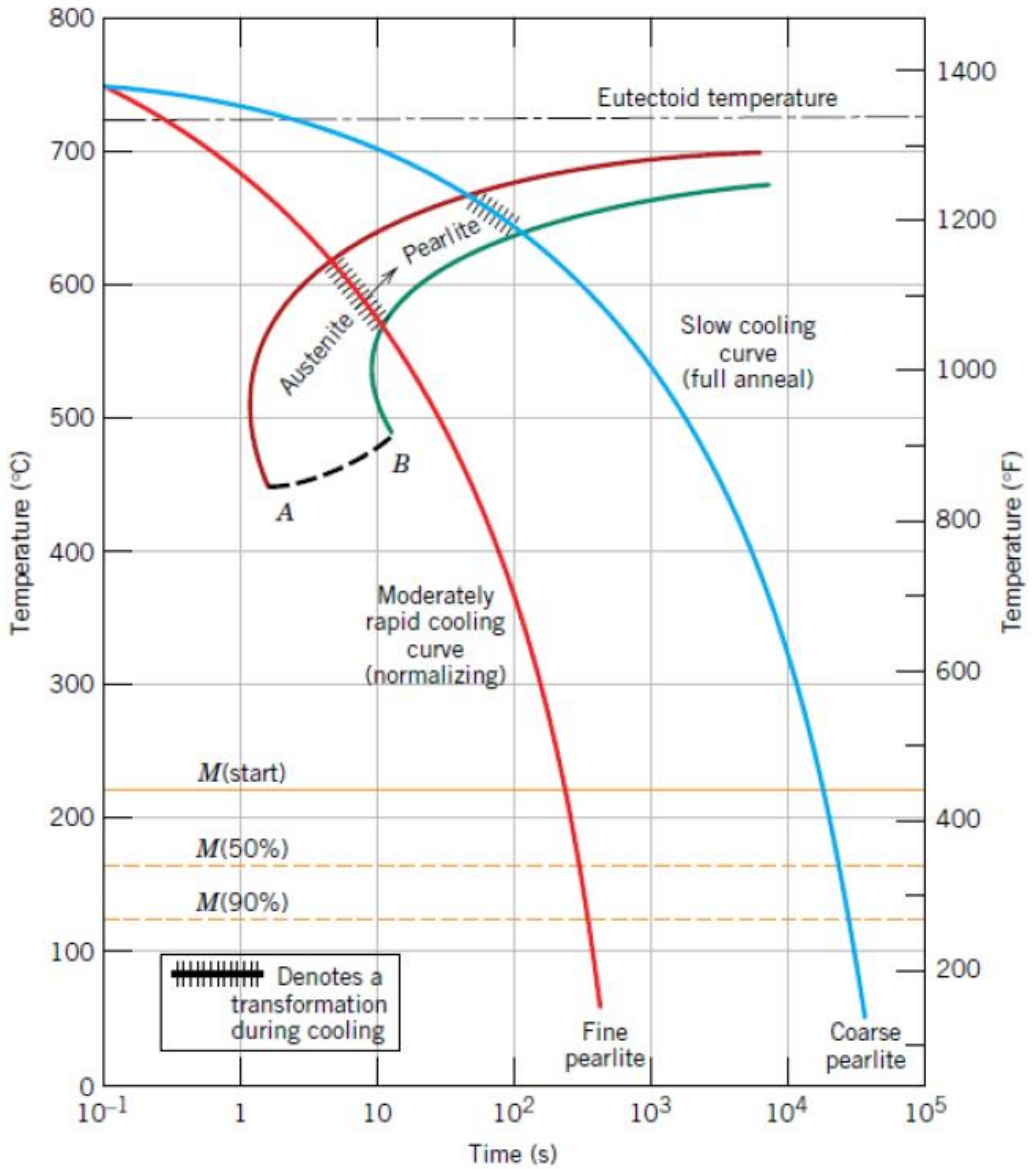
Spheroidite

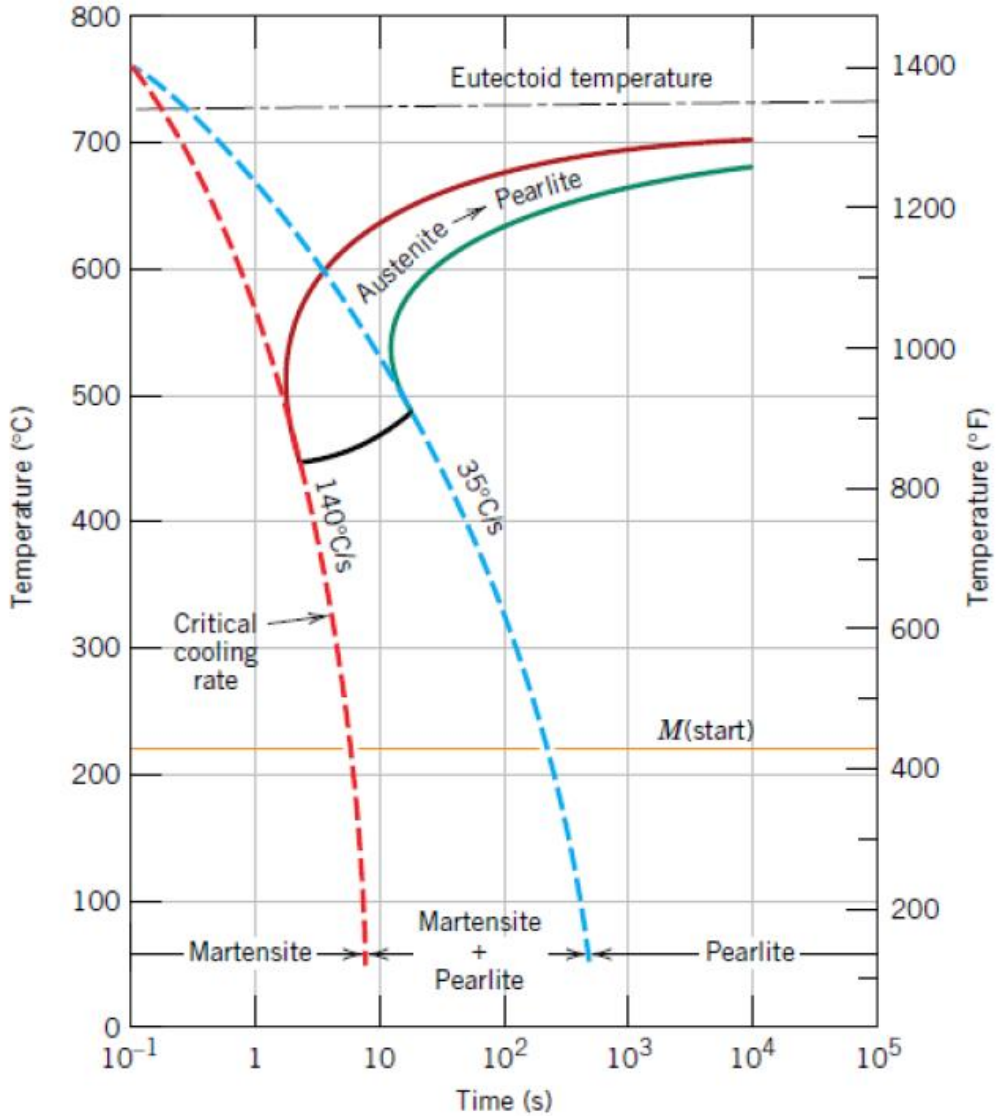


If a steel alloy having either pearlitic or bainitic microstrctures is heated to, and left at, a temperature below the eutectoid for a sufficiently long period of time—for example, at about 700°C for between 18 and 24 h—yet another microstructure will form called **spheroidite**

(dark cementite particles in ferrite matrix)

Continuous Cooling Transformation (CCT) diagram for a eutectoid iron–carbon alloy

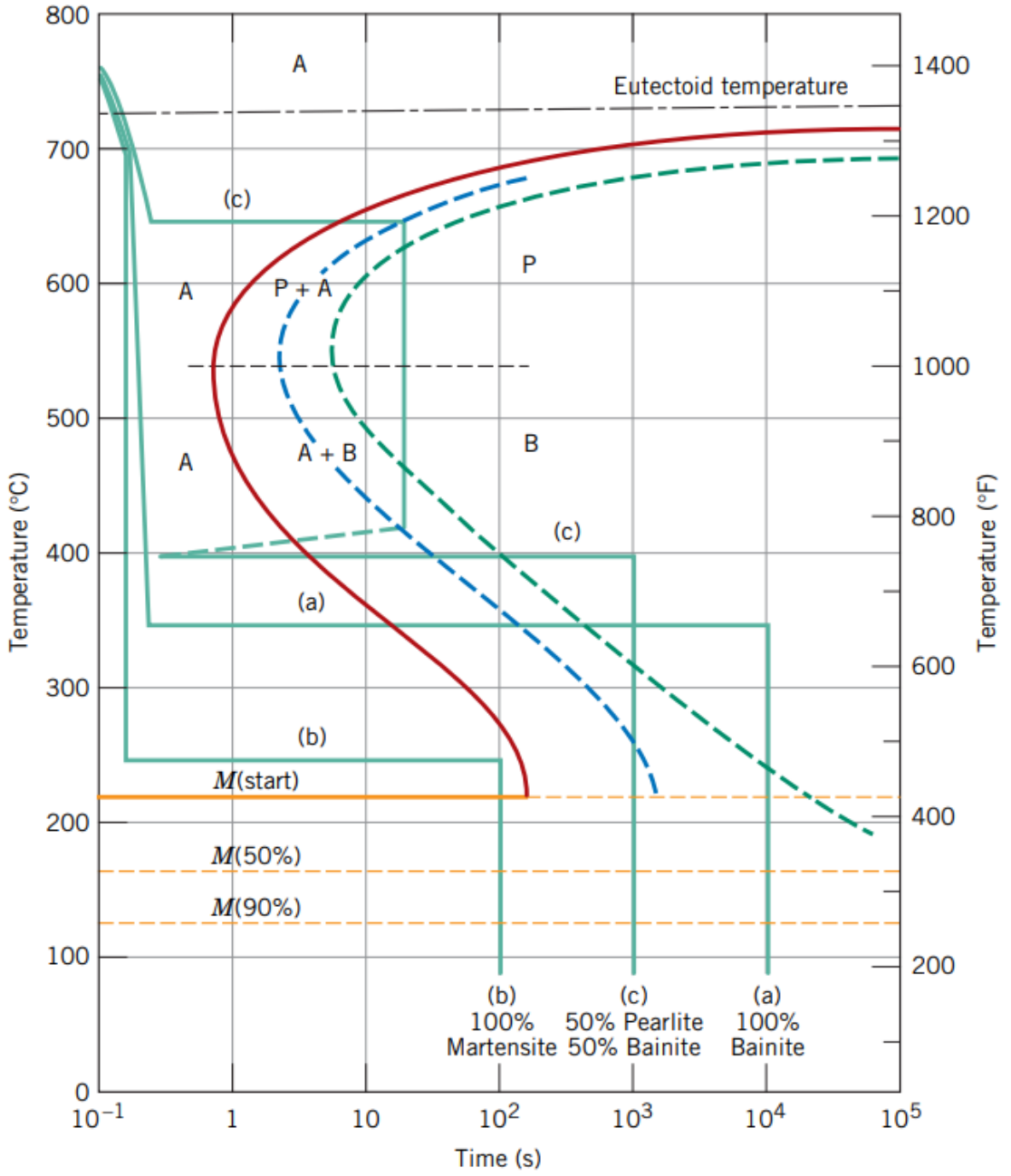


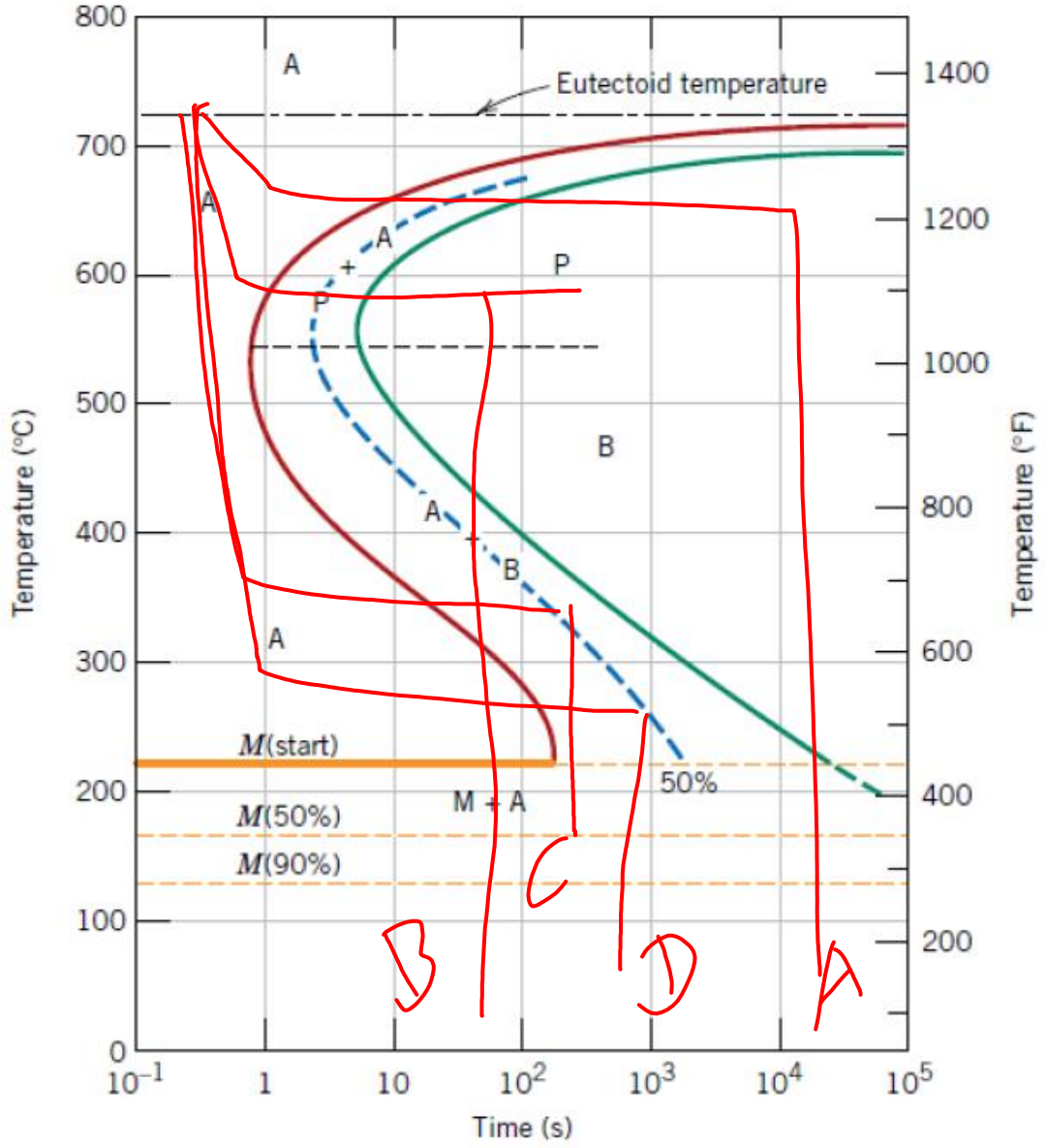


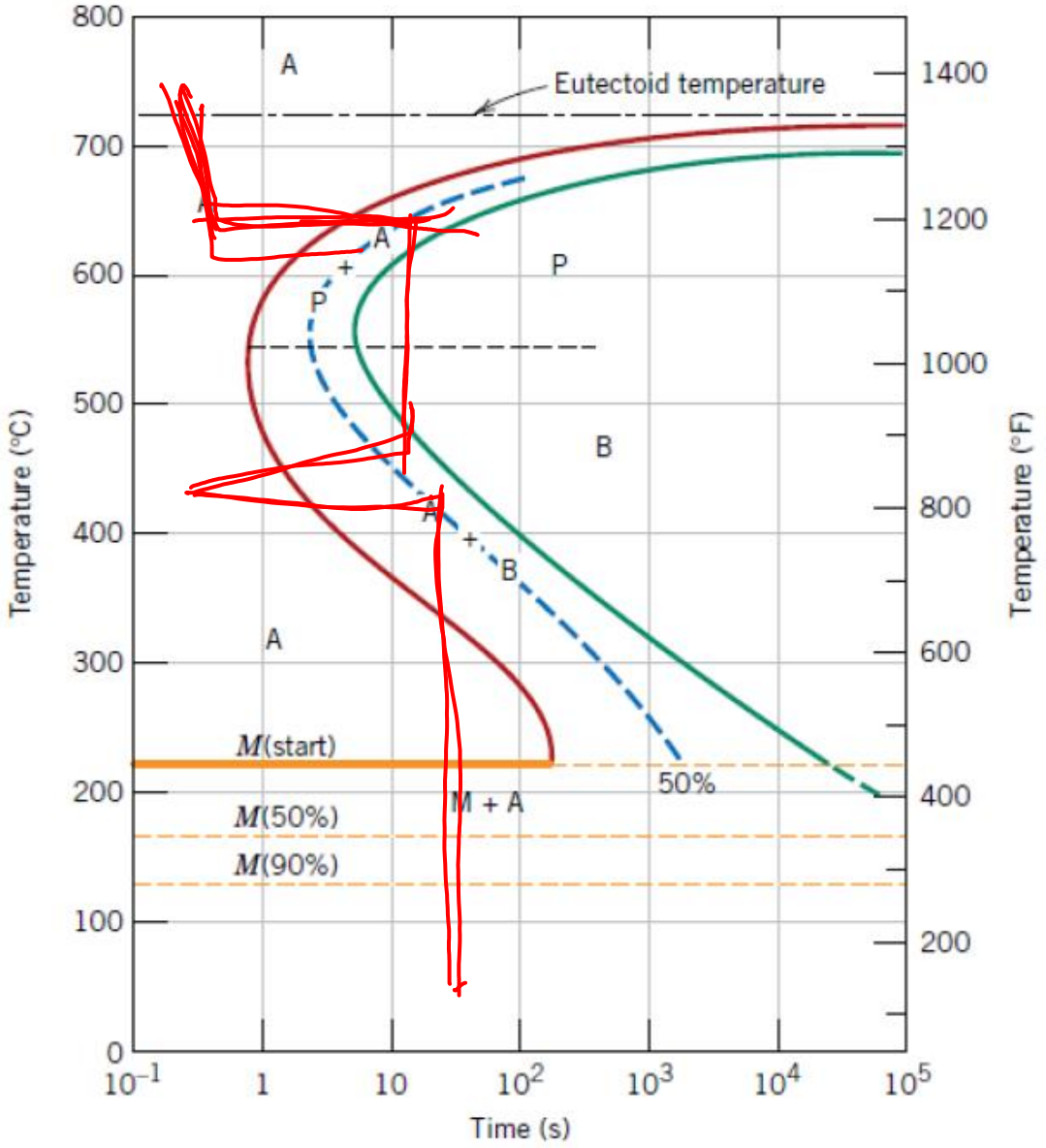
a) Rapidly cool to 350°C (660°F), hold for 104 s, and quench to room temperature.

(b) Rapidly cool to 250°C (480°F), hold for 100 s, and quench to room temperature.

(c) Rapidly cool to 650°C (1200°F), hold for 20 s, rapidly cool to 400°C (750°F), hold for 103 s, and quench to room temperature.



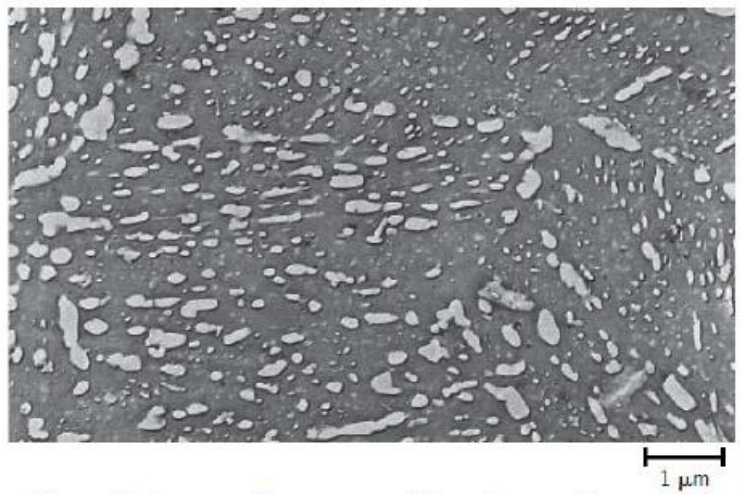

A
Eutectoid temperature
A
A
+
P
P
B
A
+
B
A
M(start)
M(50%)
M(90%)
M + A
50%
B
C
D
A
Temperature (°C)
Temperature (°F)
Time (s)
800
700
600
500
400
300
200
100
0
1400
1200
1000
800
600
400
200
10^{-1}
1
10
10^2
10^3
10^4
10^5


A
Eutectoid temperature
A
+
P
P
B
A
+
B
A
M(start)
M(50%)
M(90%)
M + A
50%
Temperature (°C)
Temperature (°F)
Time (s)
800
700
600
500
400
300
200
100
0
1400
1200
1000
800
600
400
200
10^{-1}
1
10
10^2
10^3
10^4
10^5

Tempered Martensite

- Tempering is accomplished by heating a martensitic steel to a temperature below the eutectoid for a specified time period.
- Normally, tempering is carried out at temperatures between 250°C and 650°C
- With increasing time the hardness decreases, which corresponds to the growth and coalescence of the cementite particles.
- At temperatures approaching the eutectoid [700°C] and after several hours, the microstructure will become spheroiditic, with large cementite spheroids embedded within the continuous ferrite phase. Correspondingly, overtempered martensite is relatively soft and ductile.

martensite (BCT, single phase) $\rightarrow$ tempered martensite (α + Fe_3C phases)



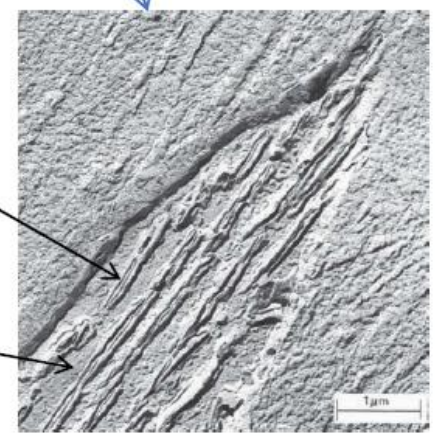
The small particles are the cementite phase; the matrix phase is a-ferrite.

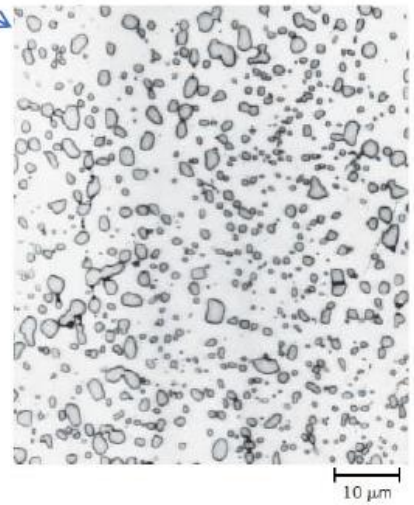
Microstructures of Iron-Carbon Alloys

Different microstructures: Fine & Coarse Pearlite, Bainite, Martensite, Spheroidite

Cementite

Ferrite





(tempered martensite)

Microstructures and Mechanical Properties for Iron–Carbon Alloys

Microconstituent	*Phases Present*	*Arrangement of Phases*	*Mechanical Properties (Relative)*
Spheroidite	α-Ferrite + Fe_3C	Relatively small Fe_3C spherelike particles in an α-ferrite matrix	Soft and ductile
Coarse pearlite	α-Ferrite + Fe_3C	Alternating layers of α-ferrite and Fe_3C that are relatively thick	Harder and stronger than spheroidite, but not as ductile as spheroidite
Fine pearlite	α-Ferrite + Fe_3C	Alternating layers of α-ferrite and Fe_3C that are relatively thin	Harder and stronger than coarse pearlite, but not as ductile as coarse pearlite
Bainite	α-Ferrite + Fe_3C	Very fine and elongated particles of Fe_3C in an α-ferrite matrix	Harder and stronger than fine pearlite; less hard than martensite; more ductile than martensite
Tempered martensite	α-Ferrite + Fe_3C	Very small Fe_3C spherelike particles in an α-ferrite matrix	Strong; not as hard as martensite, but much more ductile than martensite
Martensite	Body-centered, tetragonal, single phase	Needle-shaped grains	Very hard and very brittle

Consider 1.0 kg of austenite containing 1.15 wt% C, cooled to below 727°C (1341°F).

(a)What is the proeutectoid phase?

(b) How many kilograms each of total ferrite and cementite form?

(c) How many kilograms each of pearlite and the proeutectoid phase form?

(d) Schematically sketch and label the resulting microstructure.

(a) The proeutectoid phase will be Fe_3C since 1.15 wt% C is greater than the eutectoid composition (0.76 wt% C).

(b) For this portion of the problem, we are asked to determine how much total ferrite and cementite form. Application of the appropriate lever rule expression yields

$$W_\alpha = \frac{C_{Fe_3C} - C_0}{C_{Fe_3C} - C_\alpha} = \frac{6.70 - 1.15}{6.70 - 0.022} = 0.83$$

which, when multiplied by the total mass of the alloy (1.0 kg), gives 0.83 kg of total ferrite.

Similarly, for total cementite,

$$W_{Fe_3C} = \frac{C_0 - C_\alpha}{C_{Fe_3C} - C_\alpha} = \frac{1.15 - 0.022}{6.70 - 0.022} = 0.17$$

And the mass of total cementite that forms is (0.17)(1.0 kg) = 0.17 kg.

(c) Now we are asked to calculate how much pearlite and the proeutectoid phase (cementite) form. Applying Equation 9.22, in which $C_1' = 1.15$ wt% C

$$W_p = \frac{6.70 - C_1'}{6.70 - 0.76} = \frac{6.70 - 1.15}{6.70 - 0.76} = 0.93$$

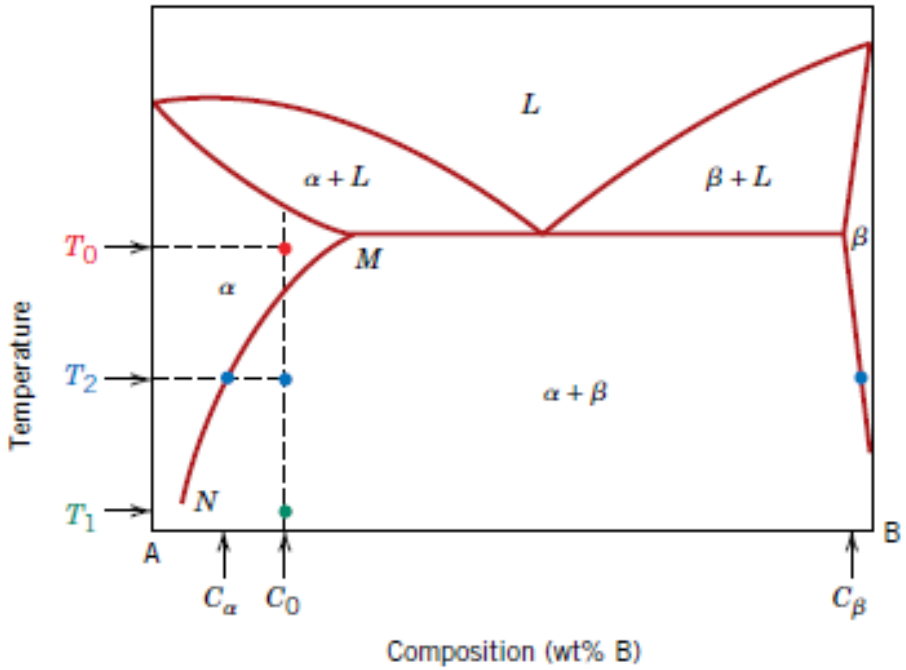
which corresponds to a mass of 0.93 kg. Likewise, from Equation 9.23

$$W_{Fe_3C'} = \frac{C_1' - 0.76}{5.94} = \frac{1.15 - 0.76}{5.94} = 0.07$$

which is equivalent to 0.07 kg of the total 1.0 kg mass.

Precipitation Strengthening

- The strength of some metal alloys may be enhanced by the formation of extremely small, uniformly dispersed particles of a second phase within the original phase matrix.
- This must be accomplished by phase transformations that are induced by appropriate heat treatments.
- The process is called **precipitation hardening** because the small particles of the new phase are termed *precipitates*.



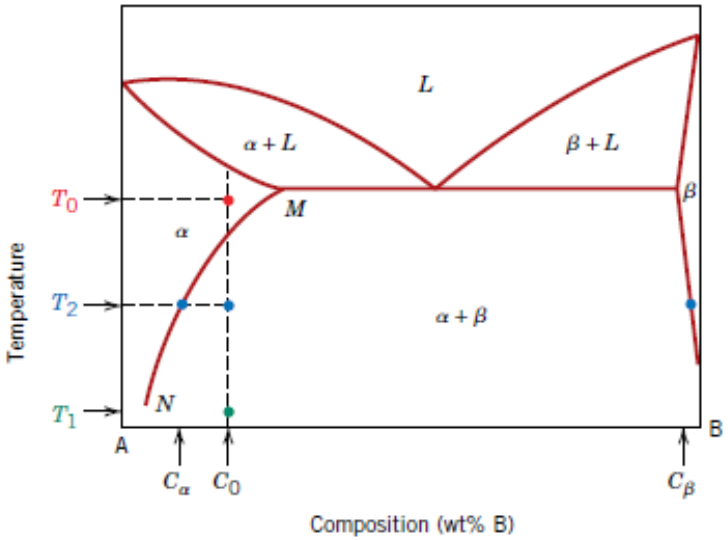
Requirements (Necessary but not sufficient):

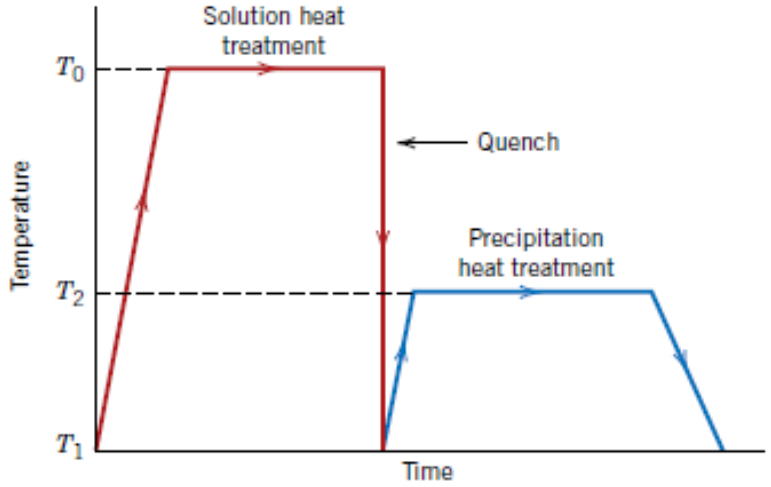
1. An appreciable maximum solubility of one component in the other
2. A solubility limit that rapidly decreases in concentration with temperature reduction
3. The composition of a precipitation-hardenable alloy must be less than the maximum solubility

Examples:
Cu-Al, Cu-Be, Cu-Sn alloys etc

Precipitation Strengthening

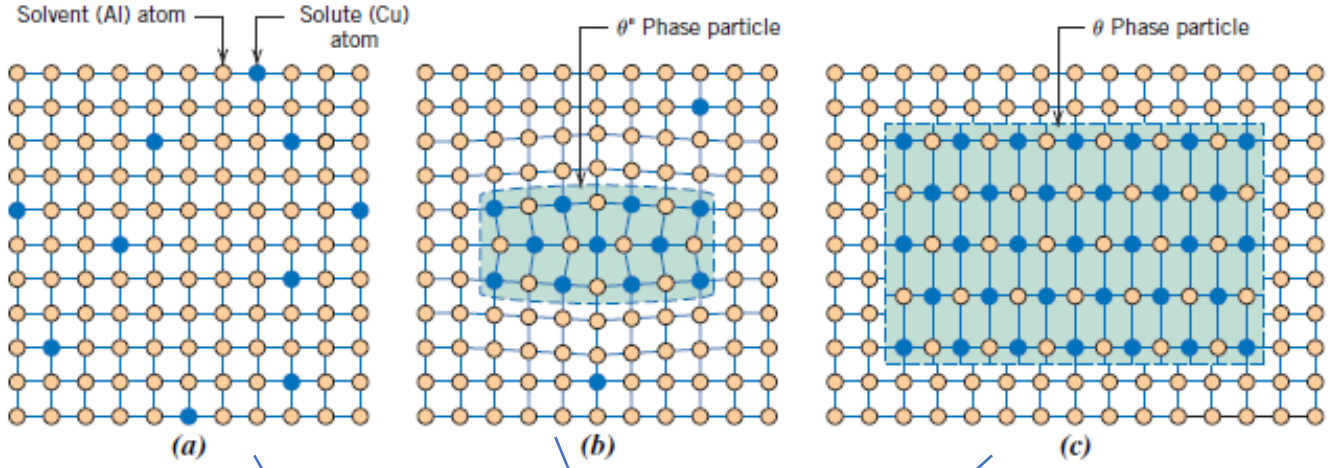
Heat Treatments



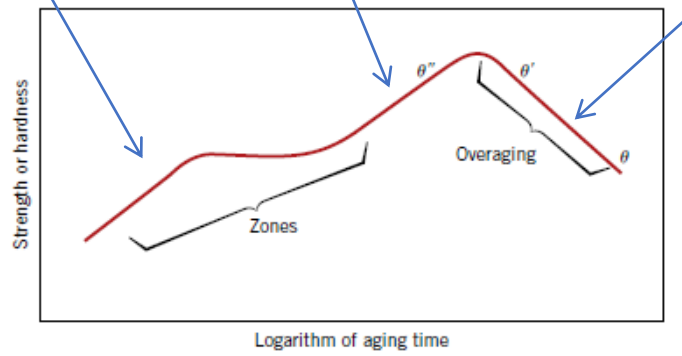


At T1 (after quenching)

At T2 (diffusion increases and β particles precipitate)



At T_2 →



There is a distortion of the crystal lattice structure around and within the vicinity of particles of these transition phases. During plastic deformation, dislocation motions are effectively impeded as a result of these distortions, and, consequently, the alloy becomes harder and stronger. As the θ phase forms, the resultant overaging (softening and weakening) is explained by a reduction in the resistance to slip that is offered by these precipitate particles

Processing / Microstructure / Property / Performance – What we learned so far?

Crystallography – crystal systems

(Processing)

Defects in crystal structures – vacancy, solid solution, dislocations

Slip systems and Slip

Strengthening mechanisms

Phase diagrams

(Processing – cooling rates)

Microstructures of steel

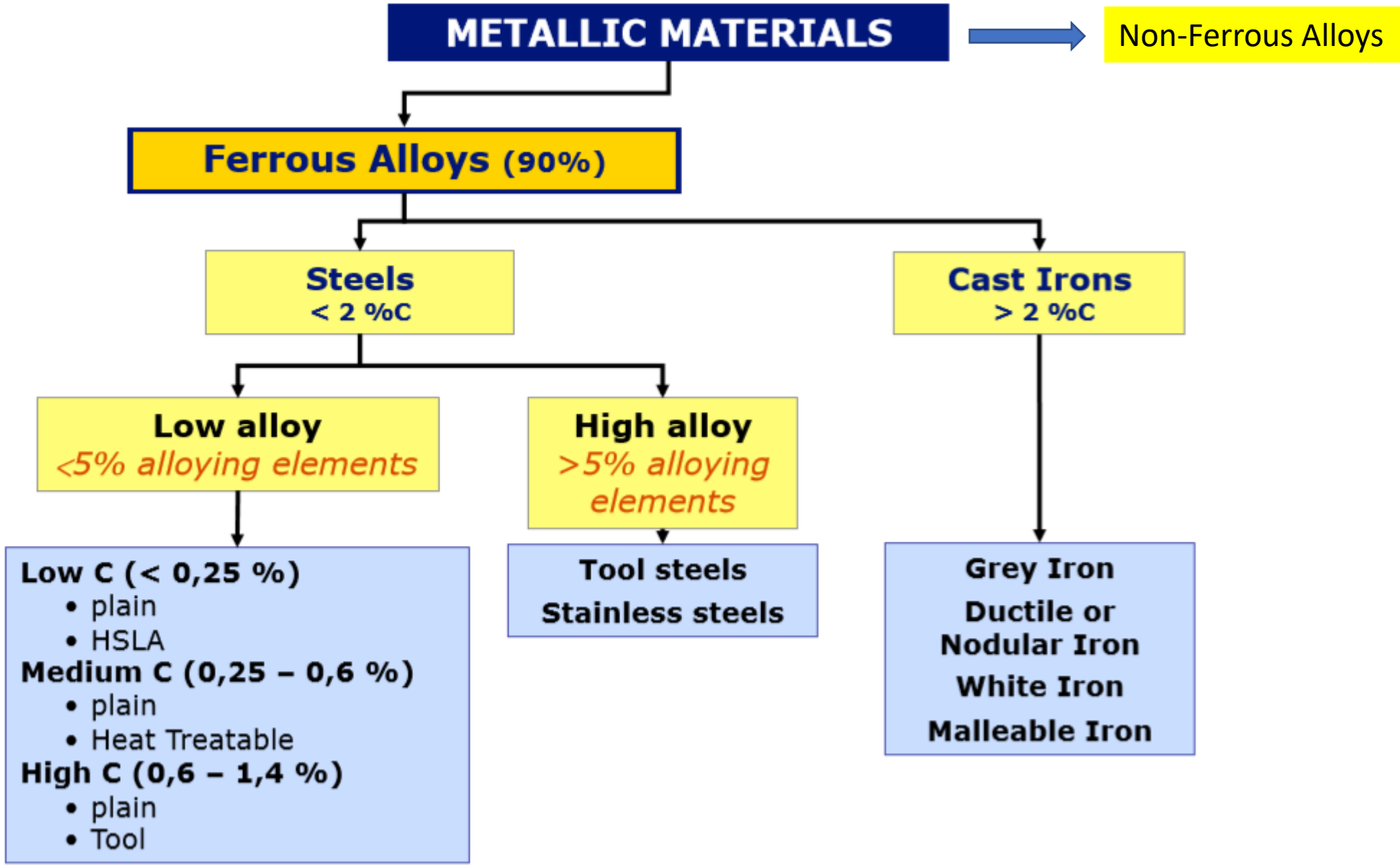
Mechanical properties

Yield strength, Young's modulus, tensile strength, strain, toughness, resilience, Poisson's ratio, ductility, plastic deformation, brittleness, specific strength, specific modulus

Selection of Materials for specific applications (Performance in particular application)

(Quick examples that we discussed – aircraft parts, teeth of digger, helmet visor, climbing rope, floating rope…)

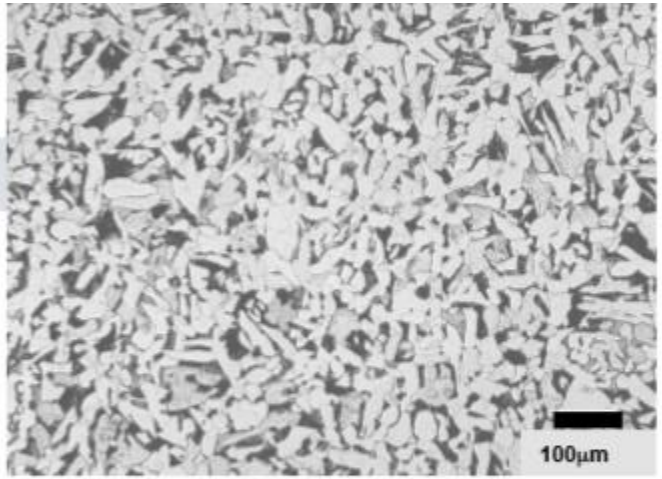
More later in the course…..


METALLIC MATERIALS
Non-Ferrous Alloys
Ferrous Alloys (90%)
Steels
< 2 %C
Cast Irons
> 2 %C
Low alloy
<5% alloying elements
High alloy
>5% alloying elements
Low C (< 0,25 %)
• plain
• HSLA
Medium C (0,25 – 0,6 %)
• plain
• Heat Treatable
High C (0,6 – 1,4 %)
• plain
• Tool
Tool steels
Stainless steels
Grey Iron
Ductile or Nodular Iron
White Iron
Malleable Iron

Ferrous Alloys: **Low alloy steels**

LOW-CARBON STEEL (Mild Steel)

0.1 - 0.25 % C → Ferrite proeutectoid
+ Pearlite (small quantities)



Ferrite (light gray) and perlite (dark gray)

(0.18% carbon)

- High formability, high ductility: **deformation: ~30%**
- Relatively low strength; **yield strength 200~400MPa**
- **Good weldability**
- **Can NOT** be hardened by **heat treatment**
- Normally **hardened** by **cold working**
- **Typical applications: pipelines, sheets, car body components, cans, structural shapes,**

http://commons.wikimedia.org/wiki/File:Hyundai_car_assembly_line.jpg

http://commons.wikimedia.org/wiki/File:114_inch_pipe_installation.jpg

http://en.wikipedia.org/wiki/File:Steel_tower.jpg

http://en.wikipedia.org/wiki/File:I-Beam_002.JPG

Ferrous Alloys: **Low alloy steels**

MEDIUM-CARBON STEEL (STRUCTURAL STEELS)

0.25-0.55%C → Ferrite proeutectoide + pearlite

- Good combination of strength and ductility
- **Yield strength: 300~ 600MPa**
- **Tensile strength: 400~ 800MPa**
- **Deformation: ~25%**
- **Strengthened by heat treatment**
- Weldable (weldability ↓ as % C ↑)
- **Used for structural applications, railway wheels and tracks, crankshafts, gears, heavy machinery, mining, cranes**

http://commons.wikimedia.org/wiki/File:Reduction_Gear.jpg

http://commons.wikimedia.org/wiki/File:Cranes.jpg

http://commons.wikimedia.org/wiki/File:DrillingMachine_Drill_bits.jpg

Ferrous Alloys: **Low alloy steels**

HIGH-CARBON STEELS

springs

http://
commons.wikimedia.org/
wiki/File:Spring05.jpg

Steels for springs: 0.6~0.8%C → Pearlite (predominantly)

Normally strengthened by heat treatment
High strength / moderate toughness

Tool steels :

0.8~1.2%C → Cementite proeutectoide + pearlite

High hardness, low toughness, difficult to machine

Used for chisel, hammer, knifes,
saw blades, drill bits, dies, punch,
cutlery and wear applications

low weldability and machinability

Punches and dies

http://
www.winstonandallan.co.uk/
punches_dies.html

Ferrous Alloys: **Low alloy steels**

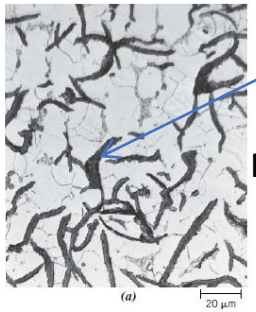
MICROALLOYED STEELS: High Strength Low Alloy, HSLA

They contain other alloying elements such as copper, vanadium, nickel, and molybdenum in combined concentrations as high as 10 wt%, and they possess higher strengths than the plain low-carbon steels.

Most may be strengthened by heat treatment, giving tensile strengths in excess of 480 MPa (70,000 psi); in addition, they are ductile, formable, and machinable.

Applications: big welded structures (Sydney bridge), marine laminated sheets, marine platforms, tubes for pipelines, pressure vessels and storage tanks.

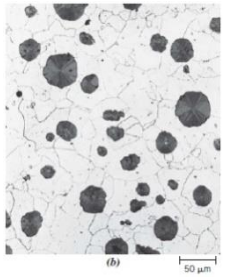
Cast Irons



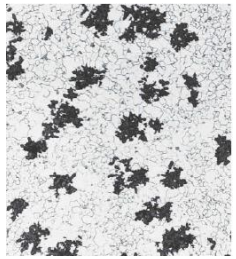
Graphite

Flakes

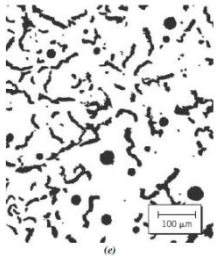
Spheres or nodular



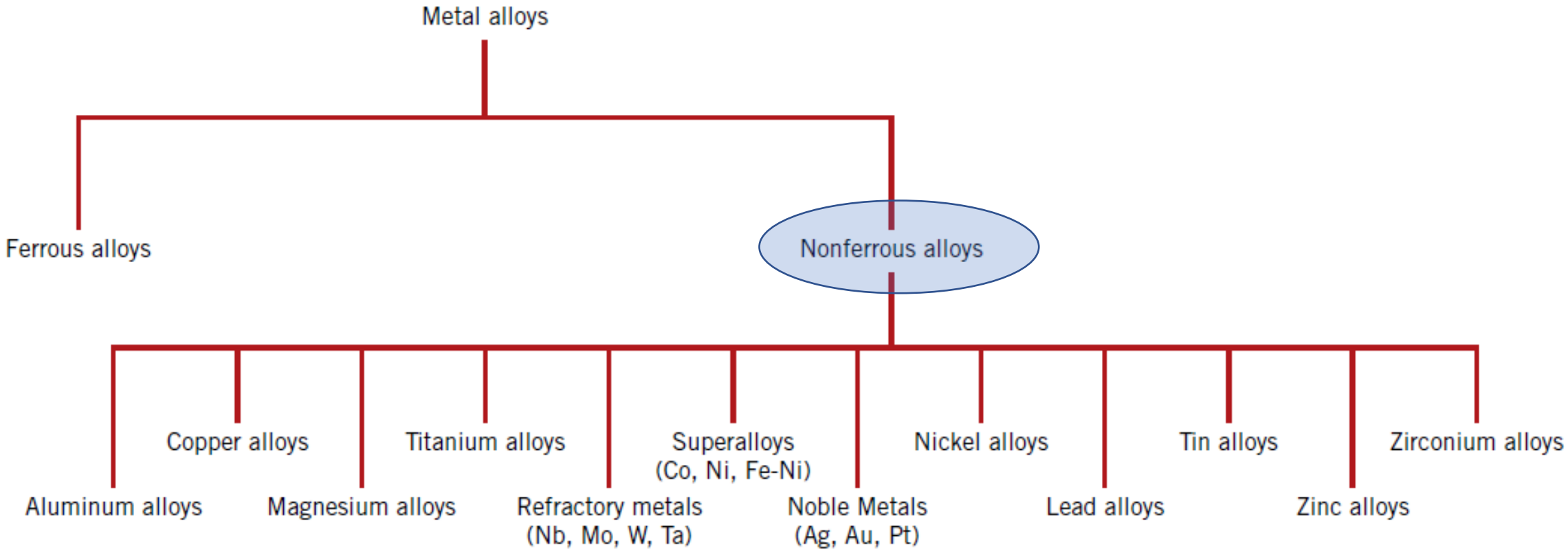
Clusters

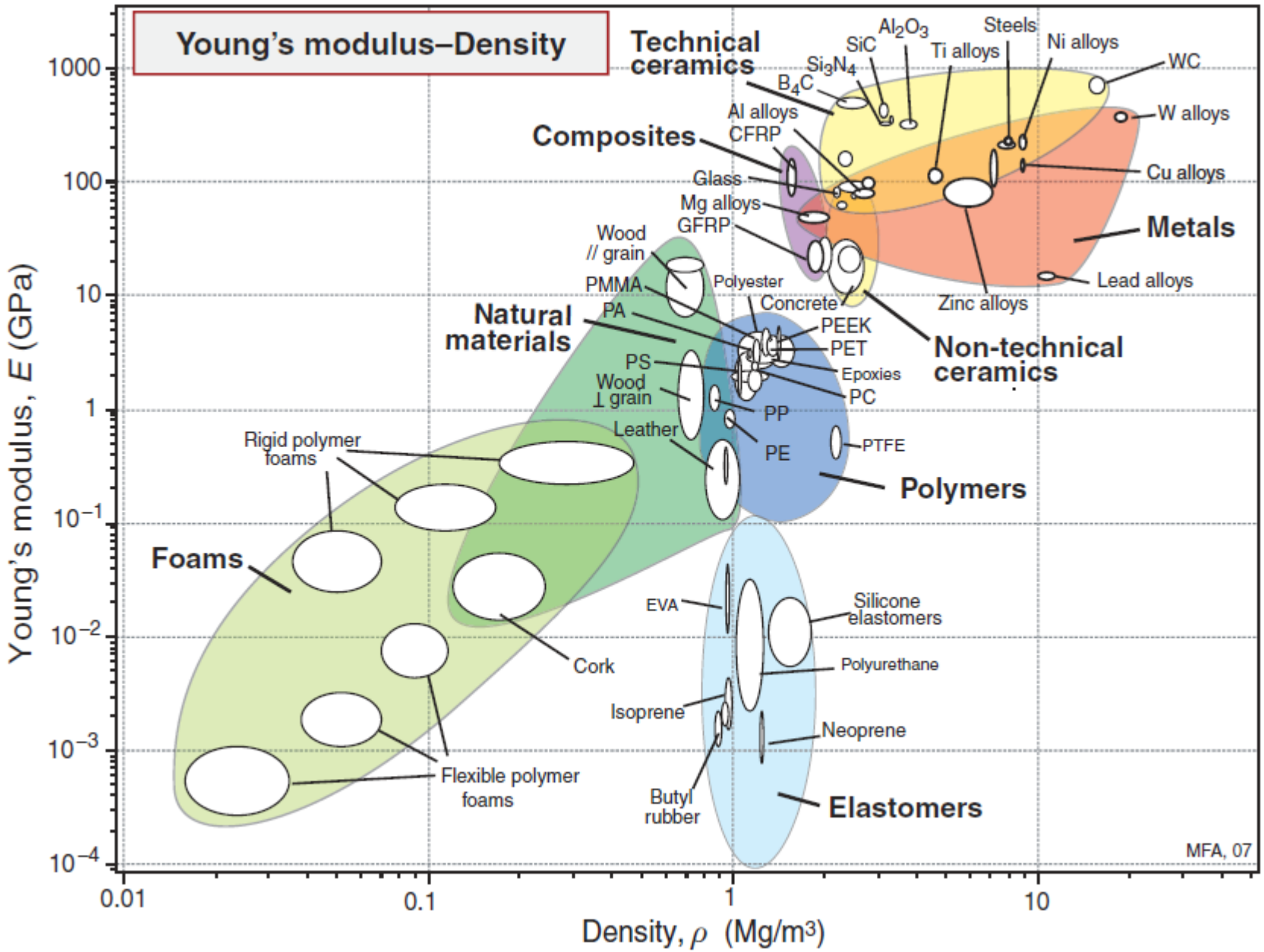


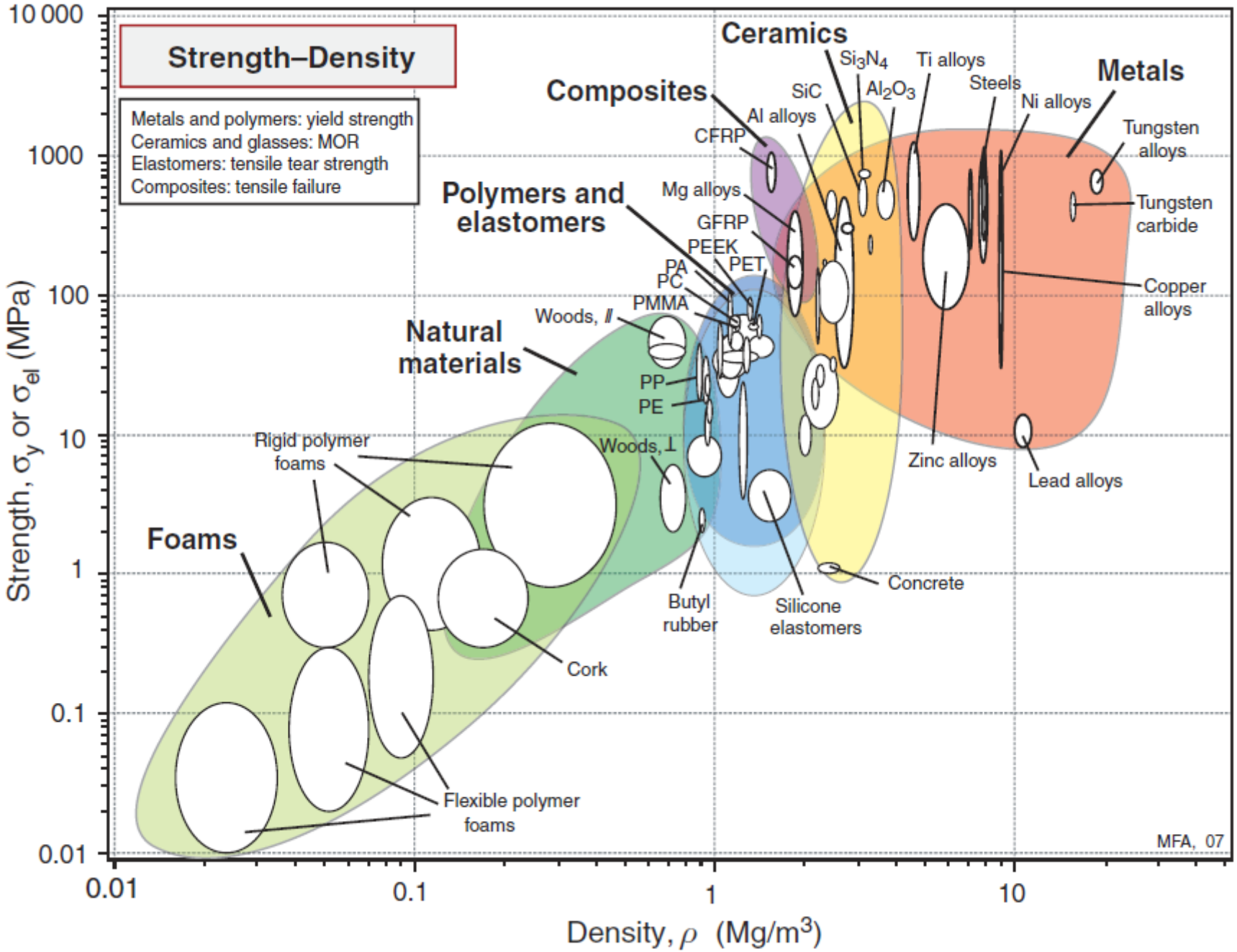
Worm like

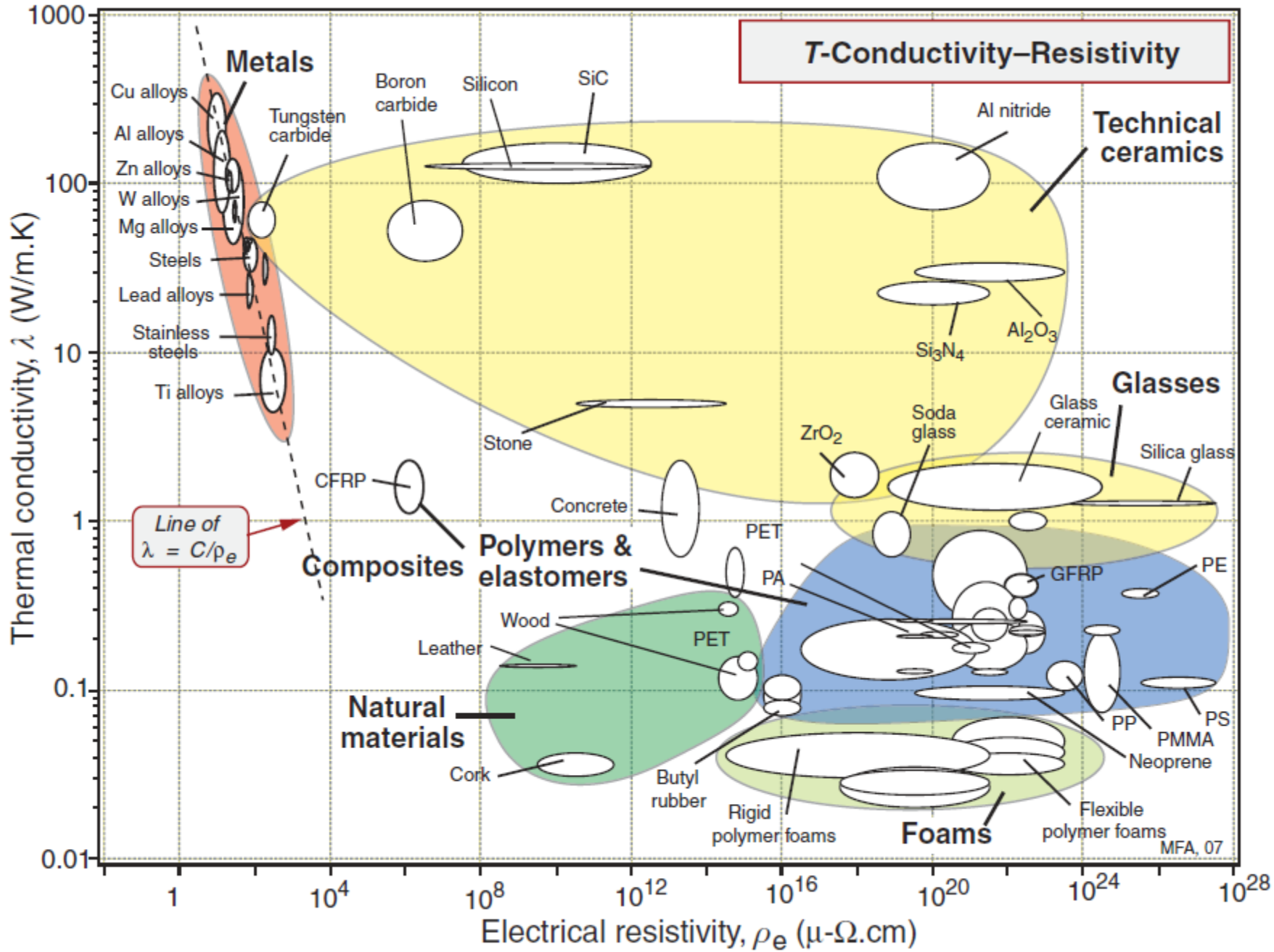


Grade	UNS Number	Composition (wt%)[a]	Matrix Structure	Mechanical Properties: Tensile Strength [MPa (ksi)]	Mechanical Properties: Yield Strength [MPa (ksi)]	Mechanical Properties: Ductility [%EL in 50 mm (2 in.)]	Typical Applications
Gray Iron							
SAE G1800	F10004	3.40–3.7 C, 2.55 Si, 0.7 Mn	Ferrite + pearlite	124 (18)	—	—	Miscellaneous soft iron castings in which strength is not a primary consideration
SAE G2500	F10005	3.2–3.5 C, 2.20 Si, 0.8 Mn	Ferrite + pearlite	173 (25)	—	—	Small cylinder blocks, cylinder heads, pistons, clutch plates, transmission cases
SAE G4000	F10008	3.0–3.3 C, 2.0 Si, 0.8 Mn	Pearlite	276 (40)	—	—	Diesel engine castings, liners, cylinders, and pistons
Ductile (Nodular) Iron							
ASTM A536							
60–40–18	F32800	3.5–3.8 C, 2.0–2.8 Si, 0.05 Mg, <0.20 Ni, <0.10 Mo	Ferrite	414 (60)	276 (40)	18	Pressure-containing parts such as valve and pump bodies
100–70–03	F34800		Pearlite	689 (100)	483 (70)	3	High-strength gears and machine components
120–90–02	F36200		Tempered martensite	827 (120)	621 (90)	2	Pinions, gears, rollers, slides
Malleable Iron							
32510	F22200	2.3–2.7 C, 1.0–1.75 Si, <0.55 Mn	Ferrite	345 (50)	224 (32)	10	General engineering service at normal and elevated temperatures
45006	F23131	2.4–2.7 C, 1.25–1.55 Si, <0.55 Mn	Ferrite + pearlite	448 (65)	310 (45)	6	
Compacted Graphite Iron							
ASTM A842							
Grade 250	—	3.1–4.0 C, 1.7–3.0 Si, 0.015–0.035 Mg, 0.06–0.13 Ti	Ferrite	250 (36)	175 (25)	3	Diesel engine blocks, exhaust manifolds, brake discs for high-speed trains
Grade 450	—		Pearlite	450 (65)	315 (46)	1	

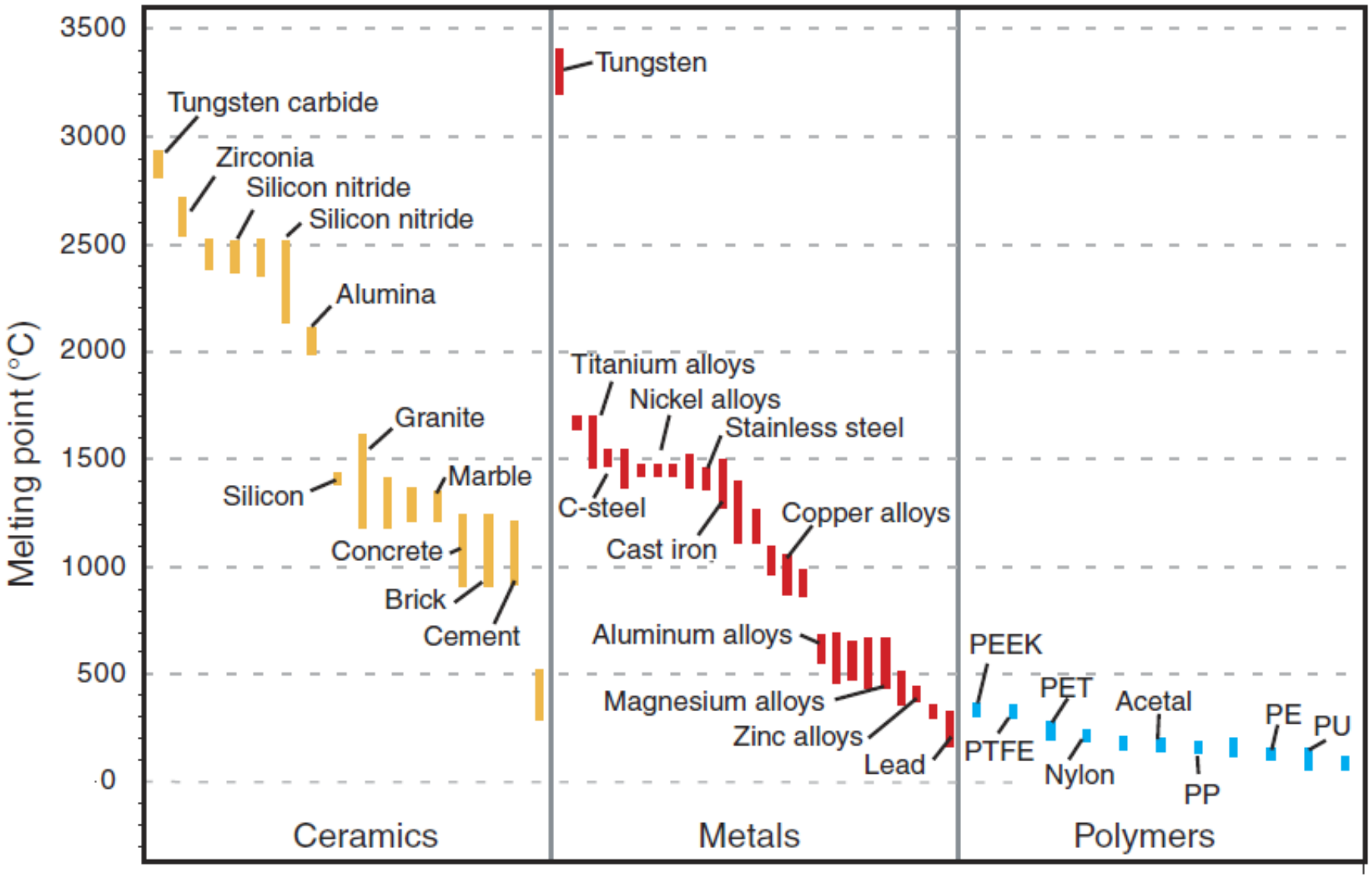

Metal alloys
Ferrous alloys
Nonferrous alloys
Aluminum alloys
Copper alloys
Magnesium alloys
Titanium alloys
Refractory metals
(Nb, Mo, W, Ta)
Superalloys
(Co, Ni, Fe-Ni)
Noble Metals
(Ag, Au, Pt)
Nickel alloys
Lead alloys
Tin alloys
Zinc alloys
Zirconium alloys


Young's modulus–Density
Technical ceramics
Composites
Natural materials
Foams
Metals
Non-technical ceramics
Polymers
Elastomers
Al_2O_3
Steels
SiC
Ti alloys
Ni alloys
Si_3N_4
WC
B_4C
W alloys
Al alloys
CFRP
Cu alloys
Glass
Mg alloys
GFRP
Wood // grain
PMMA
Polyester
Lead alloys
PA
Concrete
Zinc alloys
PEEK
PET
PS
Epoxies
Wood ⊥ grain
PC
PP
Leather
PE
PTFE
Rigid polymer foams
EVA
Silicone elastomers
Cork
Polyurethane
Isoprene
Neoprene
Flexible polymer foams
Butyl rubber
MFA, 07
Young's modulus, E (GPa)
Density, ρ (Mg/m³)
1000
100
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0.01
0.1
1
10

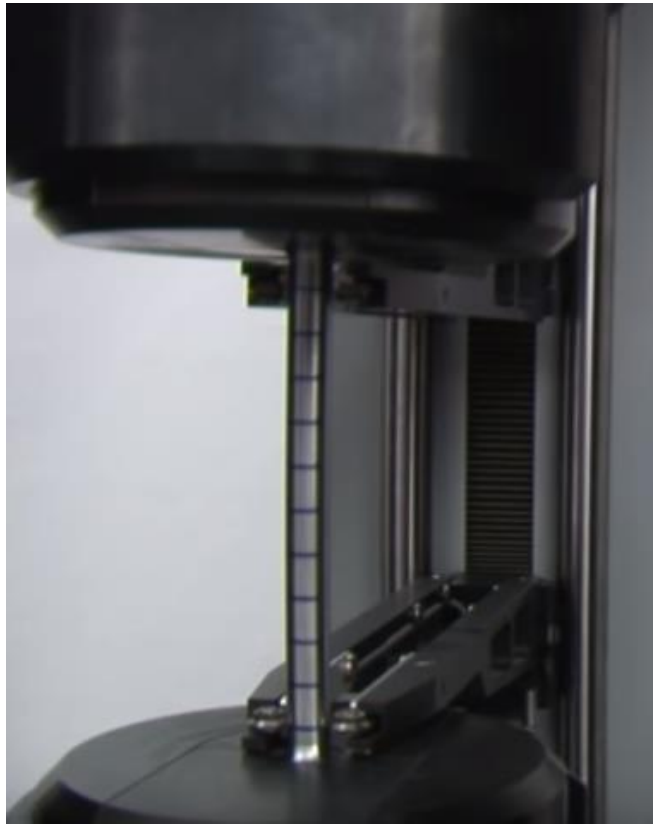

Strength–Density
Metals and polymers: yield strength
Ceramics and glasses: MOR
Elastomers: tensile tear strength
Composites: tensile failure
Ceramics
Composites
Polymers and elastomers
Natural materials
Foams
Metals
Si_3N_4
Ti alloys
SiC
Al_2O_3
Steels
Ni alloys
Al alloys
CFRP
Tungsten alloys
Tungsten carbide
Mg alloys
GFRP
PEEK
PET
PA
PC
PMMA
Copper alloys
Woods, //
PP
PE
Woods, ⊥
Rigid polymer foams
Zinc alloys
Lead alloys
Concrete
Butyl rubber
Silicone elastomers
Cork
Flexible polymer foams
MFA, 07
10 000
1000
100
10
1
0.1
0.01
0.01
0.1
1
10
Strength, σ_y or σ_{el} (MPa)
Density, ρ (Mg/m^3)

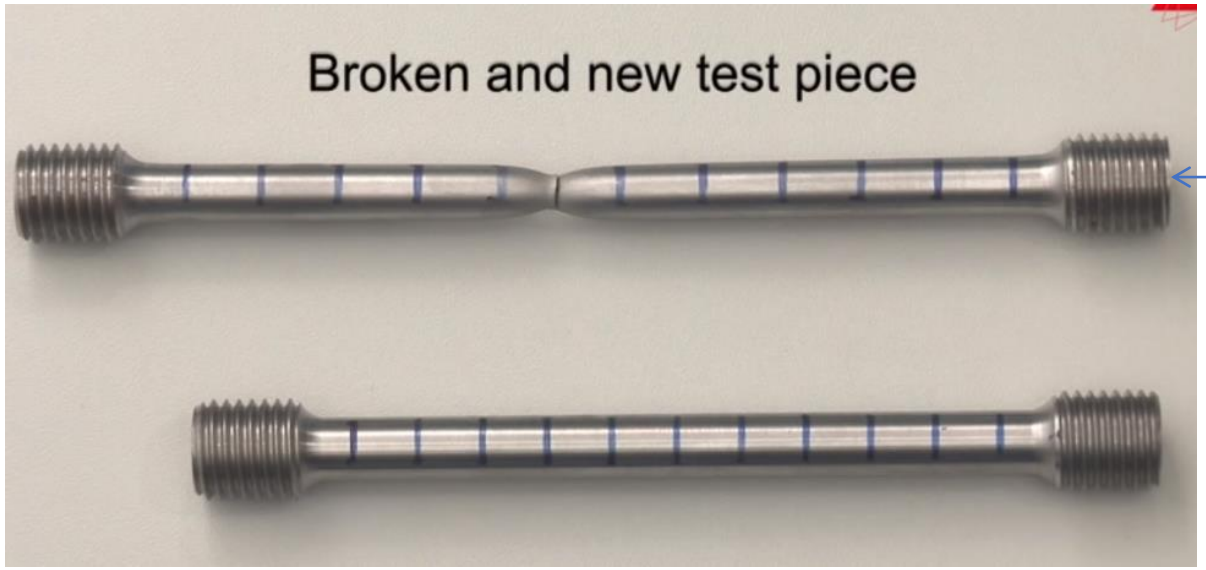

T-Conductivity–Resistivity
Thermal conductivity, λ (W/m.K)
Electrical resistivity, ρe (μ-Ω.cm)
1000
100
10
1
0.1
0.01
1
10^4
10^8
10^12
10^16
10^20
10^24
10^28
Metals
Cu alloys
Al alloys
Zn alloys
W alloys
Mg alloys
Steels
Lead alloys
Stainless steels
Ti alloys
Tungsten carbide
Boron carbide
Silicon
SiC
Al nitride
Technical ceramics
Si3N4
Al2O3
Stone
Glasses
Glass ceramic
Soda glass
ZrO2
Silica glass
CFRP
Concrete
Line of λ = C/ρe
Composites
Polymers & elastomers
PET
PA
GFRP
PE
Wood
Leather
PET
Natural materials
Cork
Butyl rubber
Rigid polymer foams
Foams
Flexible polymer foams
Neoprene
PP
PMMA
PS
MFA, 07

Melting point of materials

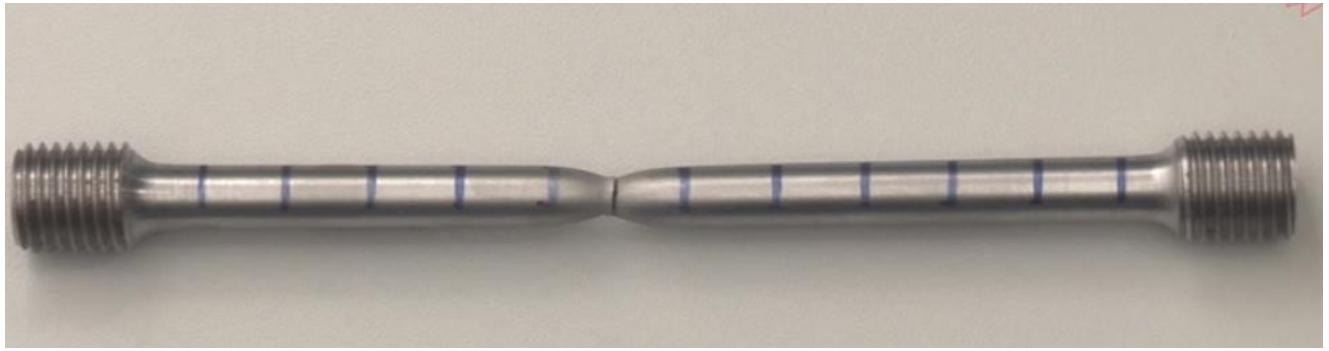


Tensile Test

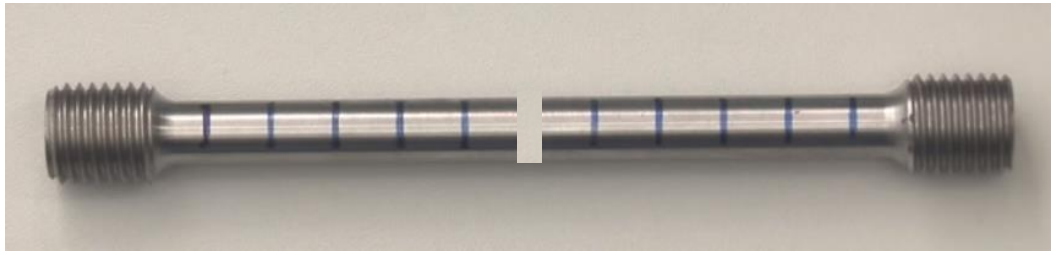




Ductile Failure



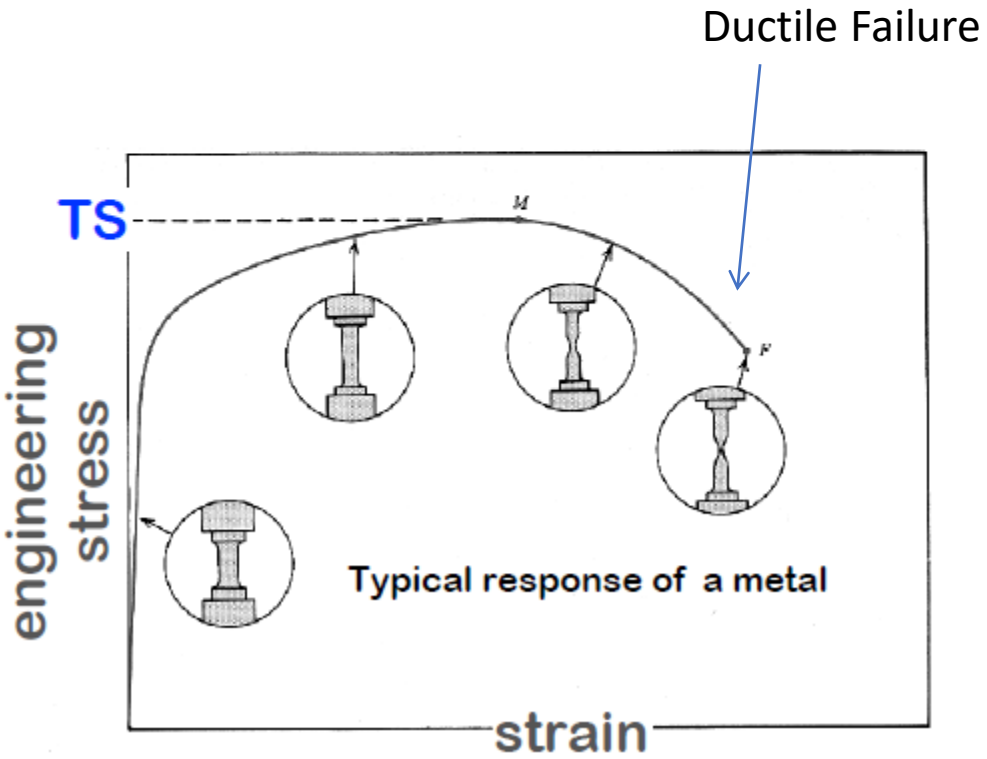
Ductile Failure



Brittle Failure

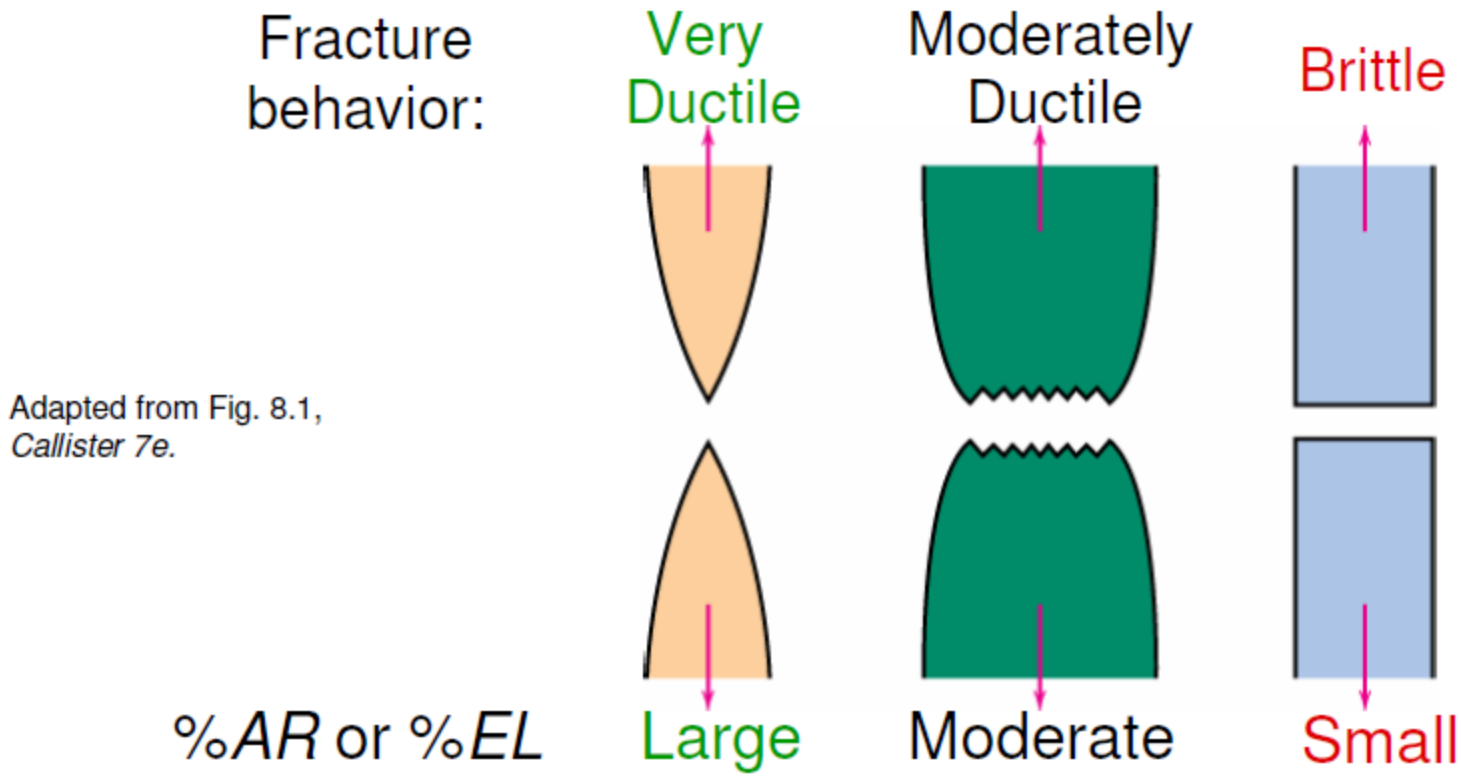
Fracture mechanisms

- Ductile fracture
  - Occurs with plastic deformation
- Brittle fracture
  - Little or no plastic deformation
  - Catastrophic



Ductile vs Brittle Failure

- Classification:



Adapted from Fig. 8.1, *Callister 7e.*

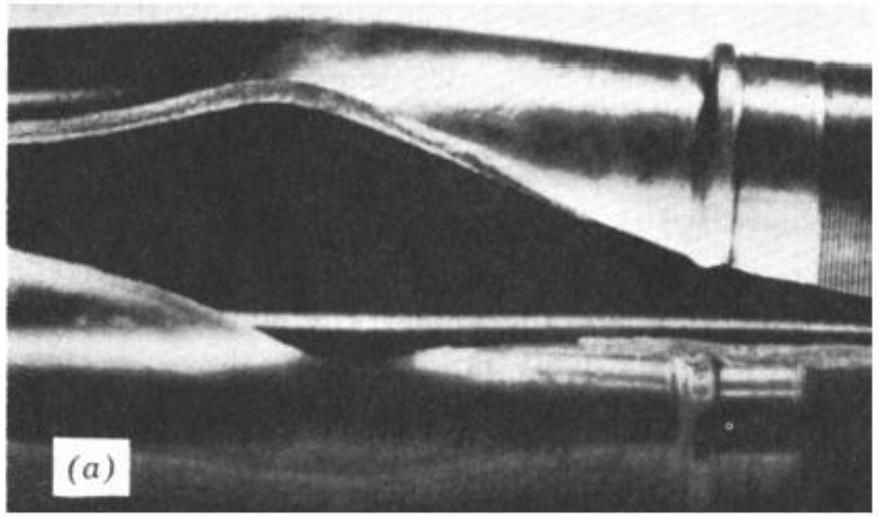
- Ductile fracture is usually desirable!

Ductile: warning before fracture

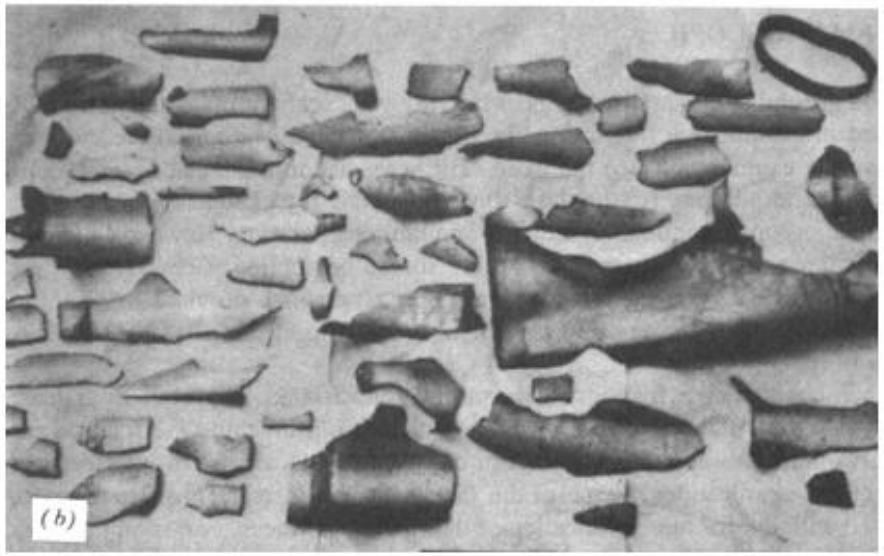
Brittle: No warning

Example: Failure of a Pipe

- **Ductile failure:**
  --one piece
  --large deformation



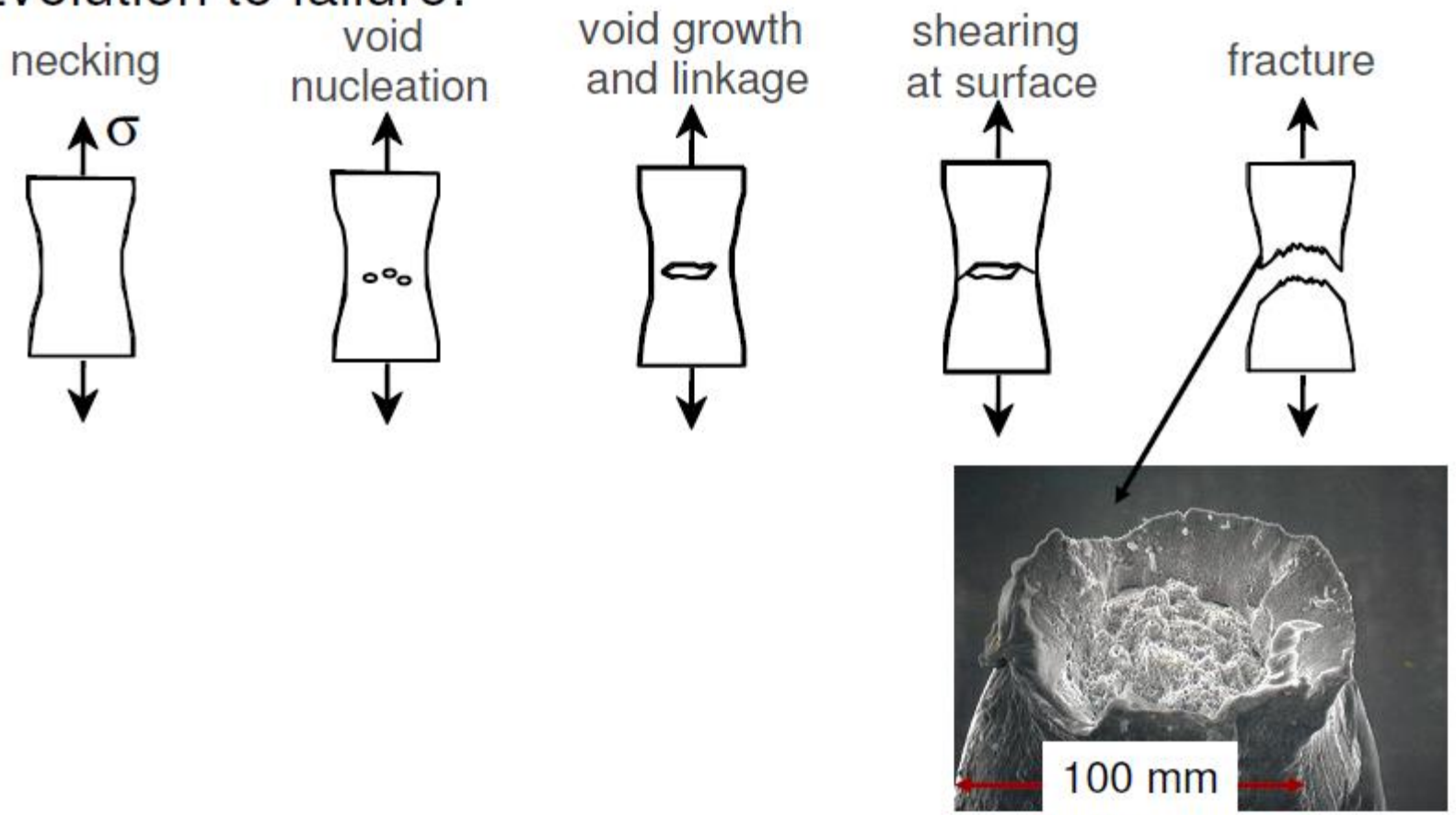
- **Brittle failure:**
  --many pieces
  --small deformation



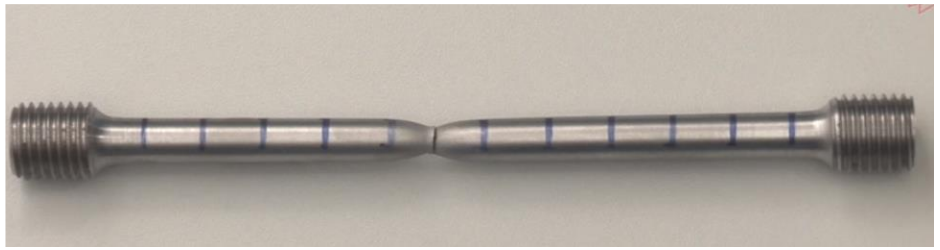
Figures from V.J. Colangelo and F.A. Heiser, *Analysis of Metallurgical Failures* (2nd ed.), Fig. 4.1(a) and (b), p. 66 John Wiley and Sons, Inc., 1987. Used with permission.

Ductile Fracture – Stages in the cup-and-cone fracture

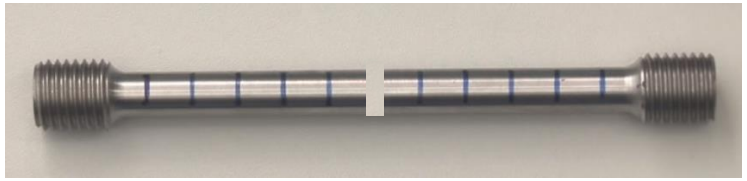
- Evolution to failure:



Ductile vs. Brittle Failure

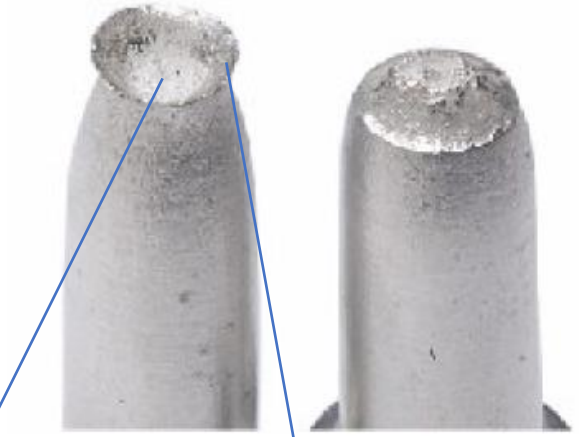


Ductile Failure



Brittle Failure

Cup-and-cone fracture in aluminum

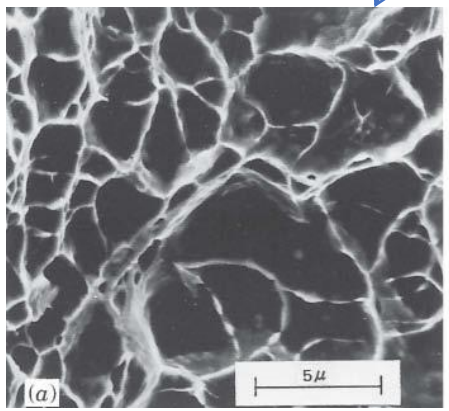


cup-and-cone fracture

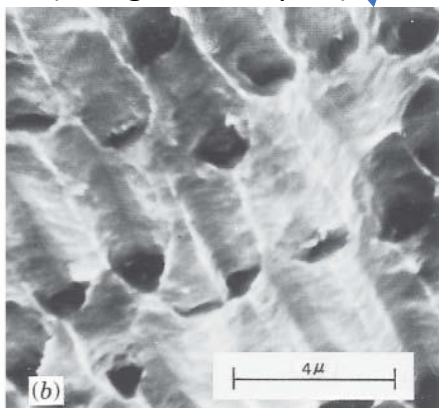
Brittle fracture in a mild steel.

brittle fracture

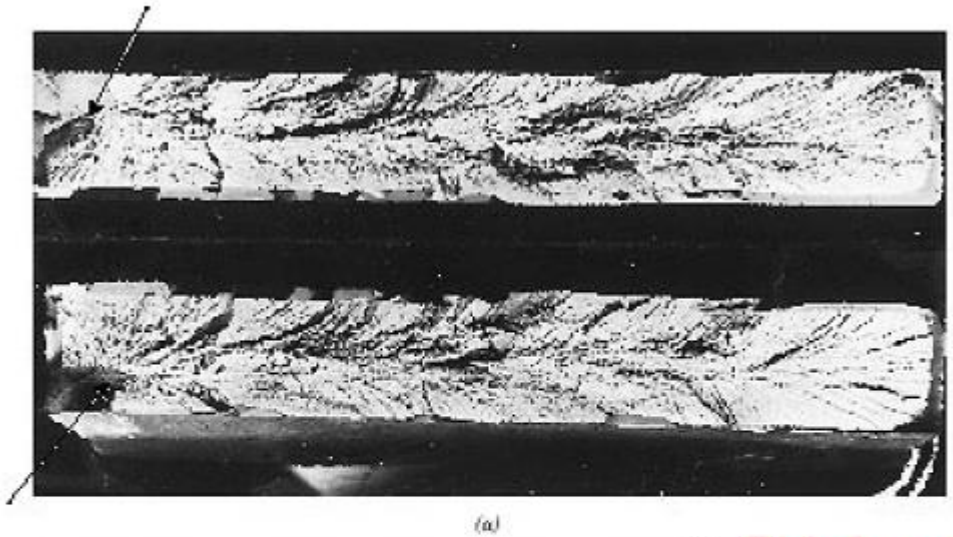
Ductile under tension (Spherical dimples)



Ductile under shear (Elongated dimples)

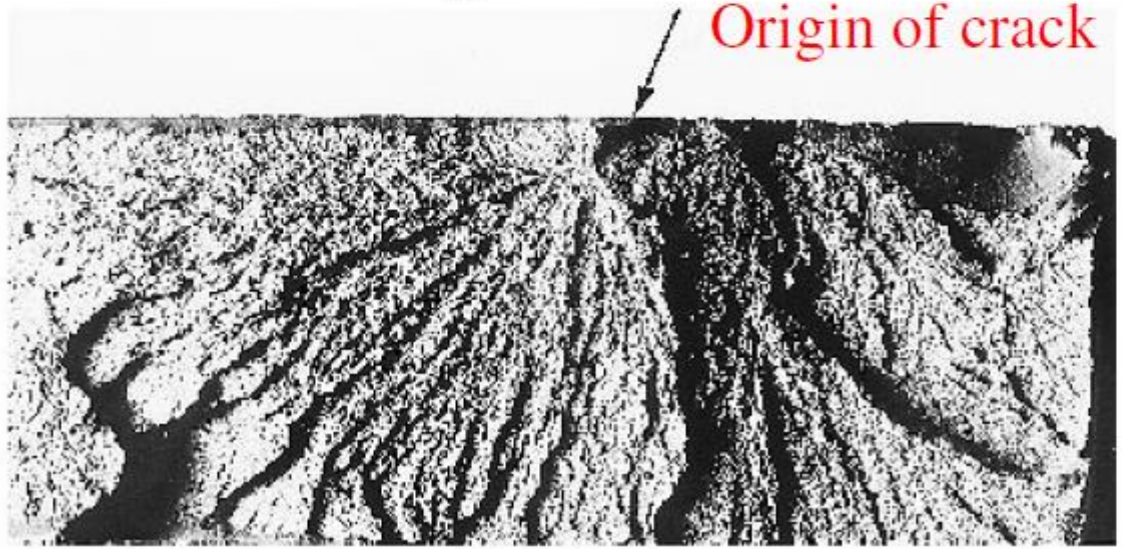


Brittle Fracture Surface



(a)

Chevron marks
From brittle fracture

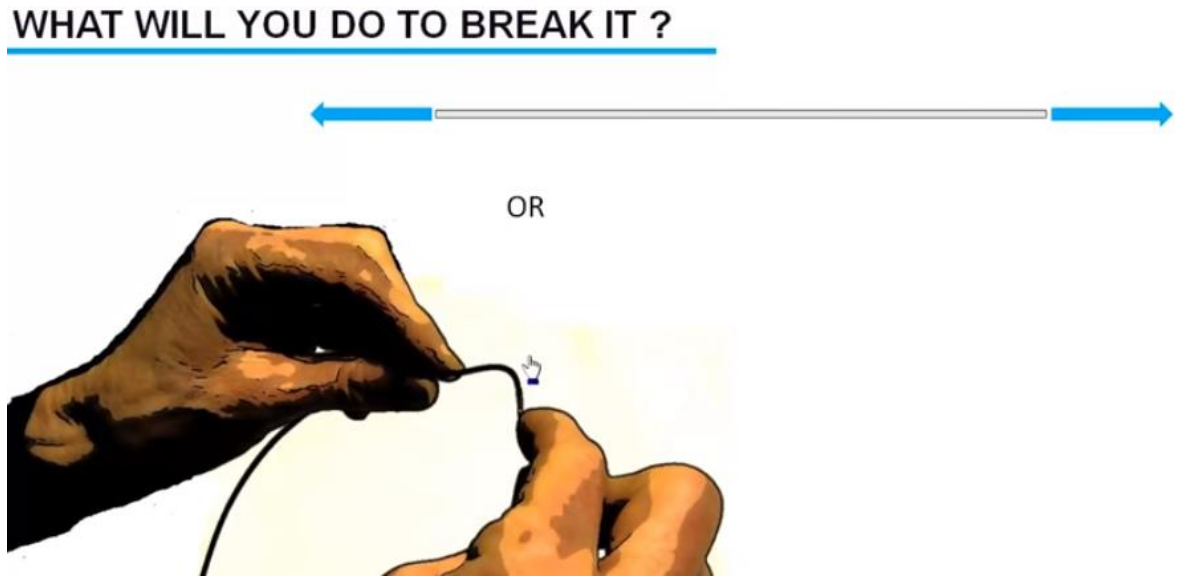


Fan-shaped ridges
coming from crack

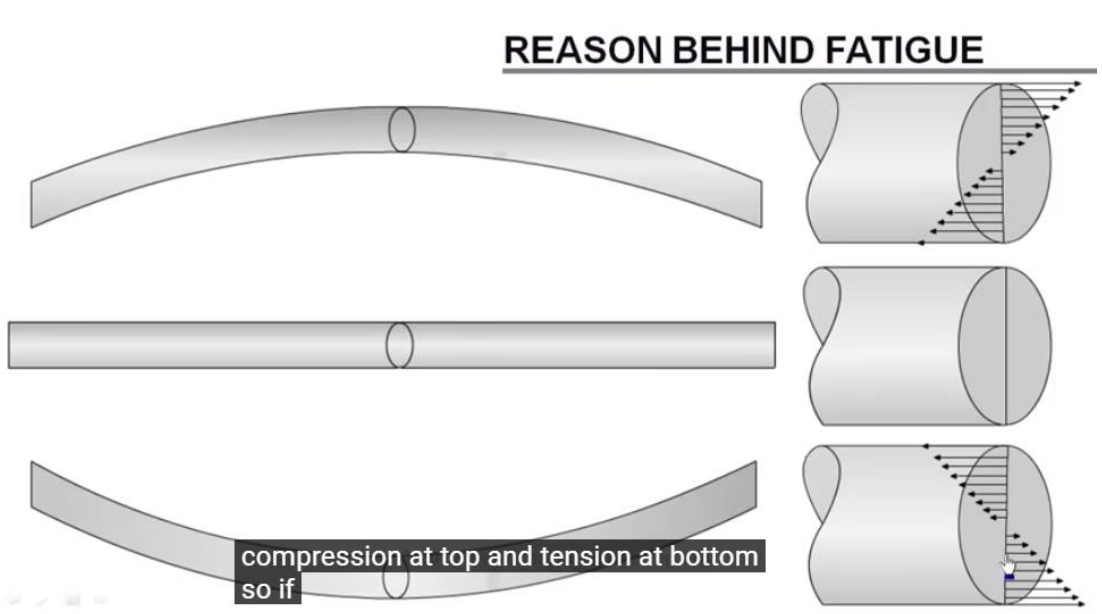
Anything peculiar in this image !

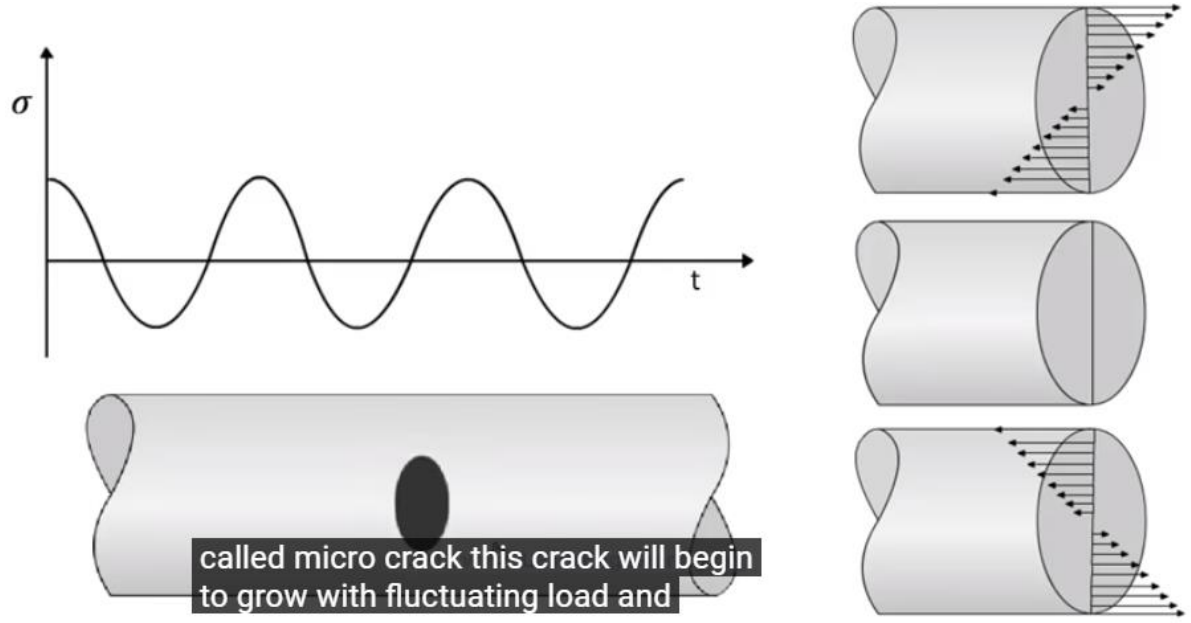
What about here????

Fatigue Failure



What about breaking the credit card?...

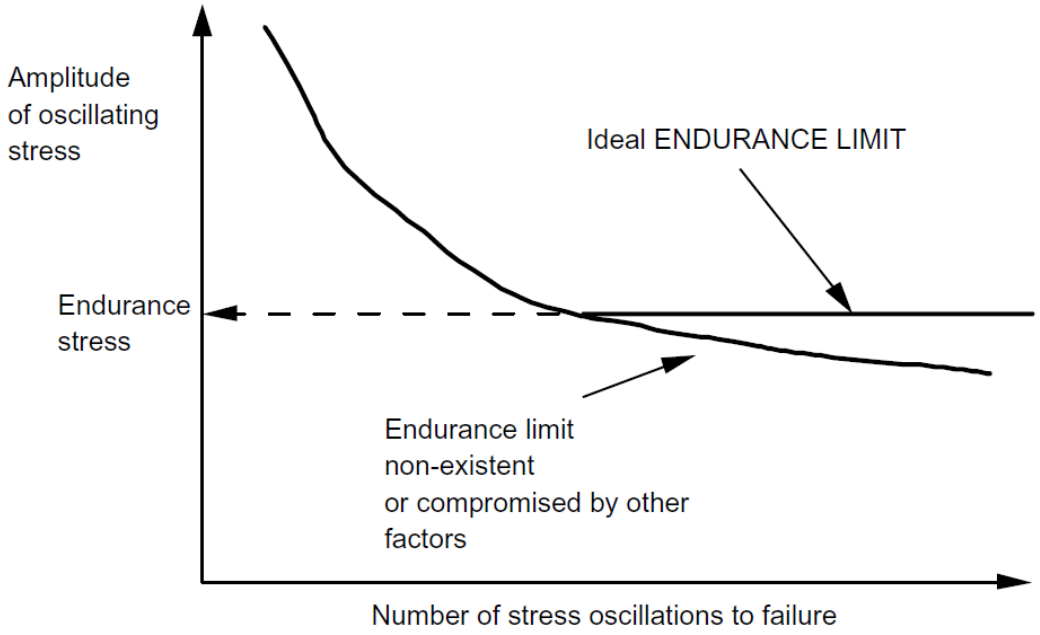




Fatigue Induced Fracture

Many factors have a controlling effect on fatigue failure of materials, which are listed below:

- Loading effects
- Surface finish
- Surface treatments
- Temperature
- Environment.

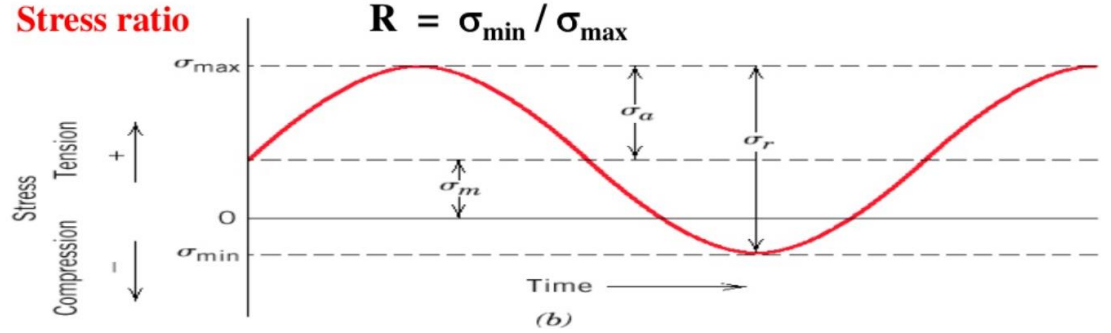


Mean stress $\sigma_m = (\sigma_{max} + \sigma_{min}) / 2$

Range of stress $\sigma_r = (\sigma_{max} - \sigma_{min})$

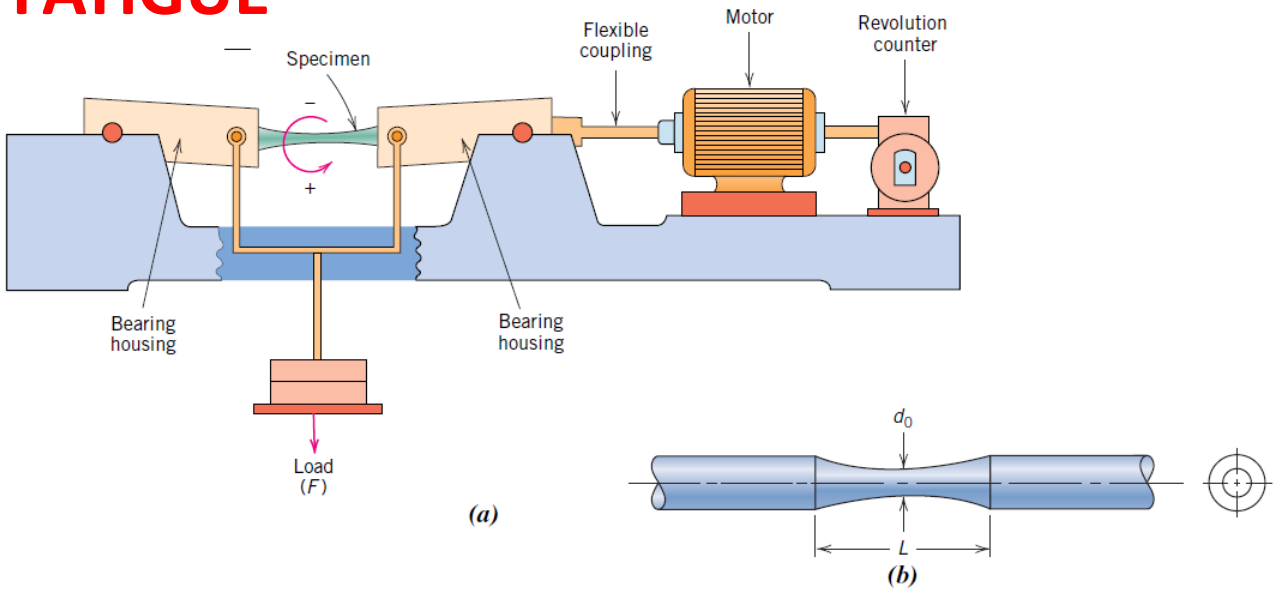
Stress amplitude $\sigma_a = \sigma_r/2 = (\sigma_{max} - \sigma_{min}) / 2$

Stress ratio $R = \sigma_{min} / \sigma_{max}$

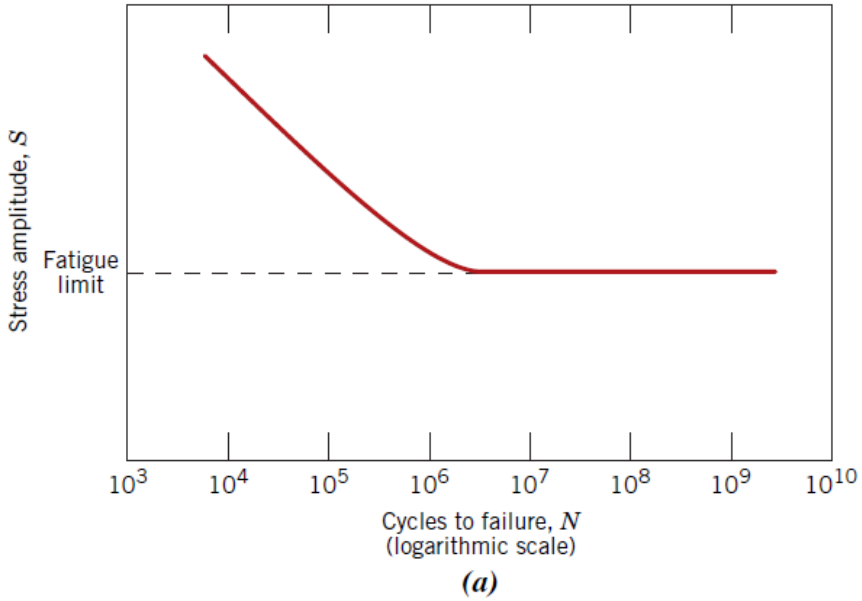


FATIGUE

THE S–N CURVE



Fatigue limit or Endurance limit



mean stress σ_m

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2}$$

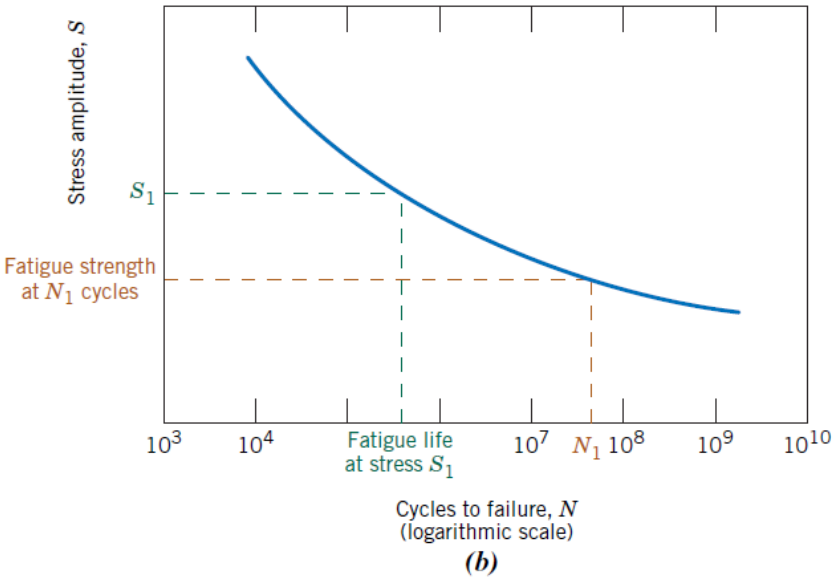
range of stress σ_r

$$\sigma_r = \sigma_{max} - \sigma_{min}$$

Stress amplitude σ_a

$$\sigma_a = \frac{\sigma_r}{2} = \frac{\sigma_{max} - \sigma_{min}}{2}$$

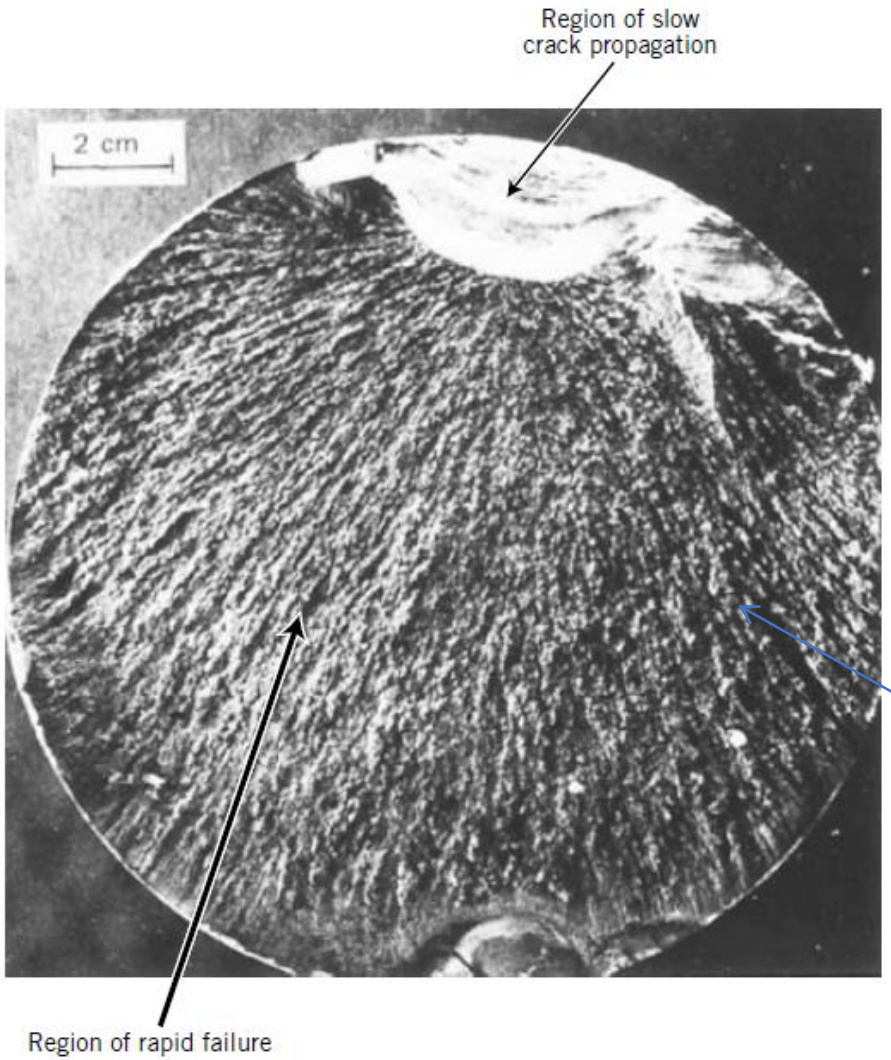
Fatigue strength



Endurance or Fatigue Limit – Stress amplitude below which fatigue failure will not occur

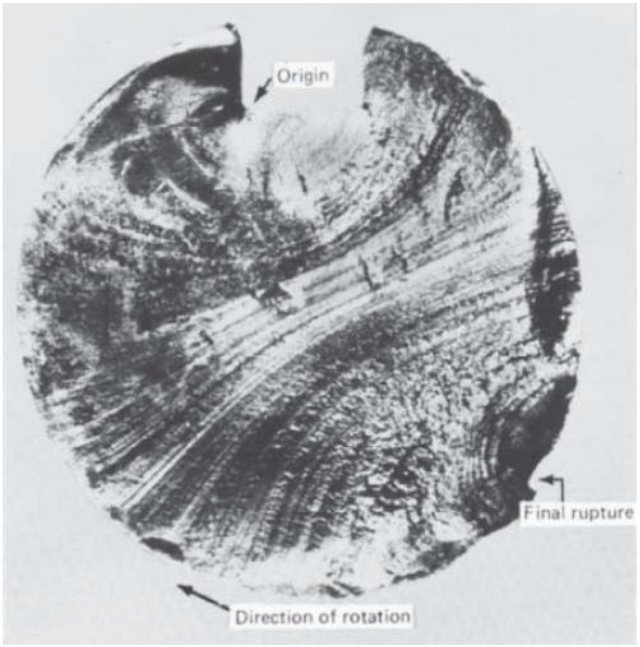
Fatigue strength – Stress amplitude at which the fatigue failure will occur at specified number of cycles – example 10^7 cycles

CRACK INITIATION AND PROPAGATION



Could be ductile or brittle

Beachmarks



Striations

Plastic

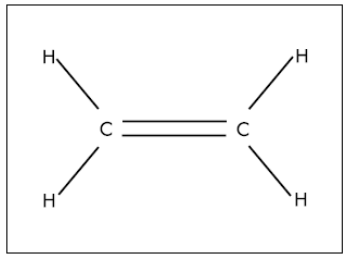
The word plastic comes from the Greek word *Plastikos*, meaning "*able to be shaped and molded*"

- Plastics
  - Thermoplastics
  - Thermosets

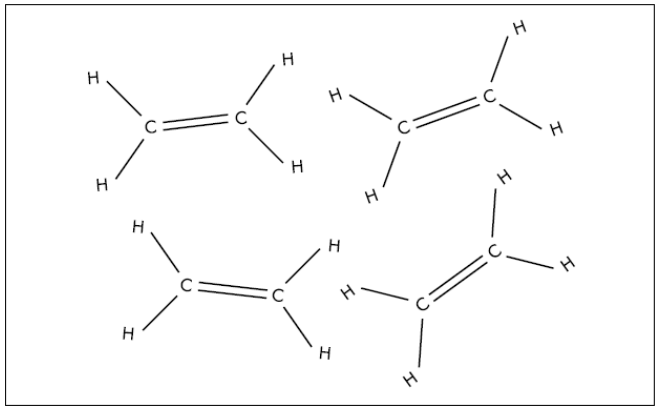
Polymerized compound

Polymerization, generally involves combination of **monomers** in the presence of catalyst, heat & pressure

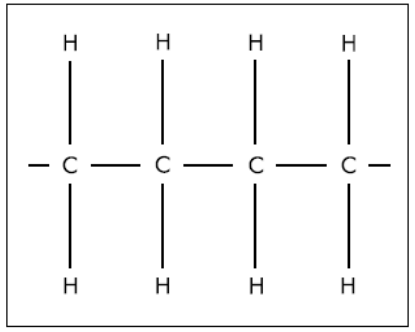
Monomer – a single unit composed of molecules
Polymers – combination of many units of monomers



Ethylene monomer molecule

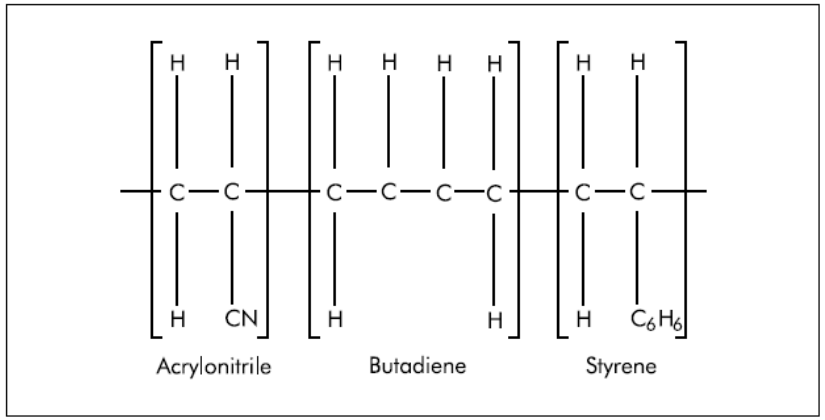


Multiple monomer molecules



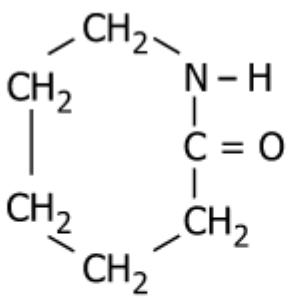
polyethylene

Homopolymer



Copolymer molecular structure (ABS)

Examples of some common plastics and their monomers

Monomer		Polymer	
Ethylene	$CH_2 = CH_2$	Polyethylene (PE)	$-[CH_2 - CH_2]_n-$
Propylene	$\begin{matrix} CH = CH_2 \\ \vert \\ CH_3 \end{matrix}$	Polypropylene (PP)	$-[\begin{matrix} CH - CH_2 \\ \vert \\ CH_3 \end{matrix}]_n-$
Vinylchloride	$CH_2 - C\begin{matrix} H \\ \diagdown \\ \diagup \\ Cl \end{matrix}$	Polyvinylchloride (PVC)	$-[\begin{matrix} CH - CH_2 \\ \vert \\ Cl \end{matrix}]_n-$
Caprolactame		Poly(E-Caprolactame) (PA6)	$-[NH - (CH_2)_5 - \overset{\overset{O}{\Vert}}{C}]_n-$
Tetraflourethylene $CF_2 = CF_2$	Polytetraflourethylene (PTFE)	$-[CF_2 - CF_2]-$	$-[CF_2 - CF_2]_n-$

Molecular Weight

Molecular weight is the sum of the atomic weights of the atoms that make up the molecule.

Carbon has an atomic weight of 12.011 grams/mol.

A mol is **6.0221415 × 10^{23}** atoms (or units)

Because of their extremely long molecules (8,000-10,000 mers long), polymers can have extremely high molecular weights.

A mole of polypropylene (3 carbons [12] and 6 hydrogen [1]) weighs 378,000 grams/mol (assuming 9,000 mers have linked up)

Propylene $CH = CH_2$ (with CH_3 below)

Polypropylene (PP) $-[CH - CH_2]_n-$ (with CH_3 below)

Molecular Weight

The polymerization (formation of polymers) process is subject to variation so there is no single chain length, there is actually a wide range of lengths. So the molecular weight in polymers - mean the average molecular weight of the material.

There are two different categories of molecular weight average that are commonly used:

The first is the Number Average Molecular Weight (M_n)

The second is the Weight Average Molecular Weight (M_w)

Number Average Molecular Weight (M_n)

- Amount fraction distribution of different sized molecules in a sample

Example:

We have:

10 units of Polyethylene (PE) that are 500 monomers long

5 units of PE that are 100 monomers long

5 units of PE that are 800 monomers long

What is M_n?

Different sized molecules are:

500 x 28 = 14000 g/mol

100 x 28 = 2800 g/mol

800 x 28 = 22400 g/mol

- **Now amount fraction distribution of different sized molecules are:**

$\frac{10}{20}$ (500 x 28) + $\frac{5}{20}$ (100 x 28) + $\frac{5}{20}$ (800 x 28) = 13,300 g/mol

$$Mn = \frac{\sum N_i M_i}{\sum N_i}$$

Materials for Engineers

Lecture Oct 27

Dr S. Gowthaman
Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

Reference:

1. Materials science and engineering by William Callister, Wiley Publications

Number Average Molecular Weight (M_n)

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$$Mn = \frac{\sum N_i M_i}{\sum N_i}$$

Weight Average Molecular Weight (M_w)

- Weight fraction distribution of different sized molecules in a sample

Example:

We have:

10 units of Polyethylene (PE) that are 500 monomers long

5 units of PE that are 100 monomers long

5 units of PE that are 800 monomers long

What is M_n?

Different sized molecules are:
500 x 28 = 14000 g/mol
100 x 28 = 2800 g/mol
800 x 28 = 22400 g/mol

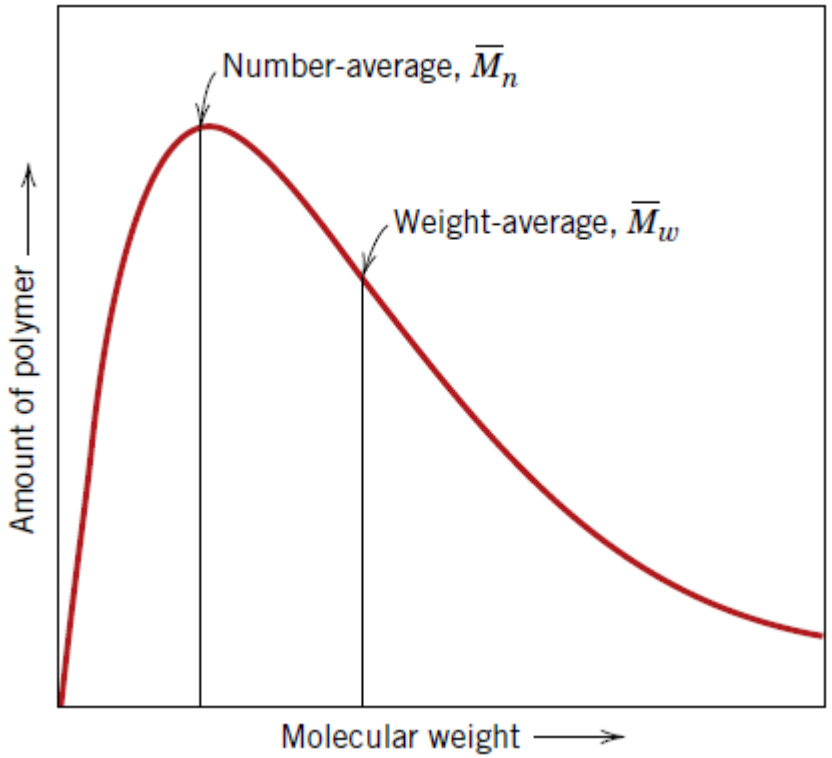
Total weight of different sized molecules are:
10 x 500 x 28 = 140000 g/mol
5 x 100 x 28 = 14000 g/mol
5 x 800 x 28 = 112000 g/mol

- **Now weight fraction distribution of different sized molecules are:**

$$\frac{140000}{(140000+14000+112000)}(500 \times 28) + \frac{14000}{(140000+14000+112000)}(100 \times 28) + \frac{112000}{(140000+14000+112000)}(800 \times 28) = 16940 \text{ g/mol}$$

$$Mw = \frac{\sum N_i M_i^2}{\sum N_i M_i}$$

Molecular weight Distribution



One way of expressing average chain size of a polymer is as the **degree of polymerization,** *DP*, which represents the average number of repeat units in a chain

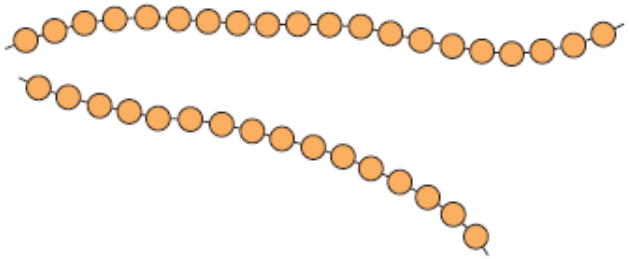
$$DP = \frac{\overline{M}_n}{m}$$

m is the molecular weight of repeating unit

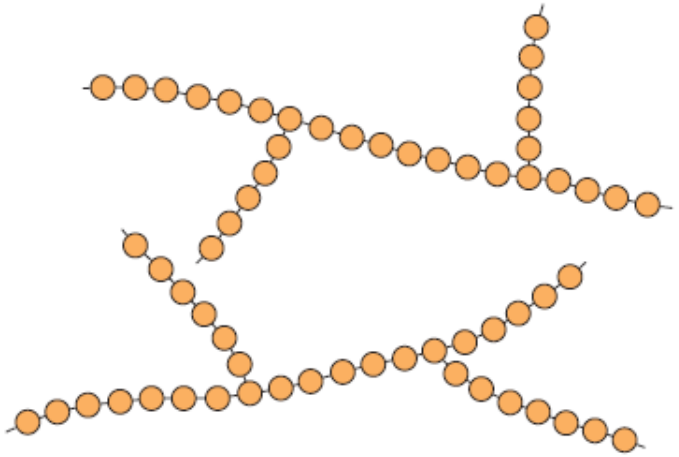
In previous example for polyethylene:
DP = 13300 / 28 = 475

MOLECULAR STRUCTURE

PVC, HDPE, PMMA, PS, Nylon etc



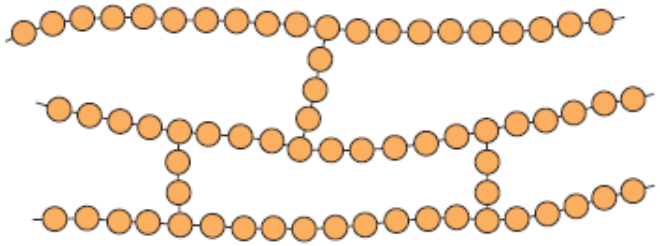
(a)



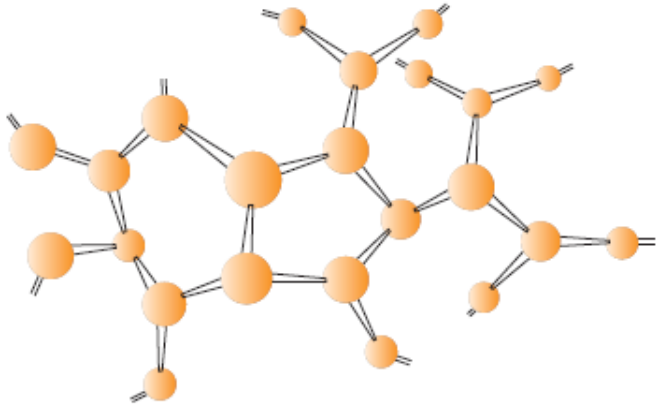
(b)

LDPE, etc (lower density)

Elastic rubbers, etc



(c)



(d)

Epoxy, Phenolics, PU etc (Highly cross linked)

(*a*) linear, (*b*) branched, (*c*) crosslinked, and (*d*) network (three-dimensional) molecular structures

Thermoplastics

A material that cures reversibly and softens when heated above the glass transition temperature or melting point and becomes hard after cooling is called thermoplastic.

Thermoplastics have a linear or branched molecular structure

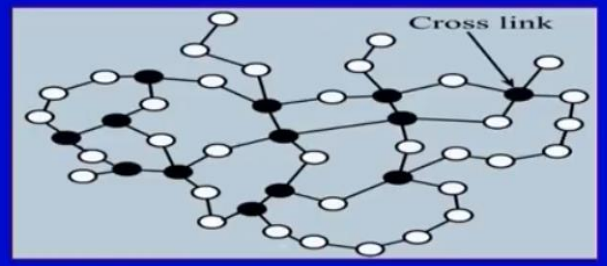
Molecular structure of thermoplastics

Types of Thermoplastic Materials

Different types of thermoplastic materials are:

- Acrylonitrile Butadiene Styrene
- Acrylics
- Polycarbonates, Polyethylene
- Polypropylenes Polystyrenes
- Polyetheretherketone
- Polyvinyl Chloride
- Liquid Crystal Polymers
- Polyphenylene Sulphide
- Polyamides
- Vinyls
- Fluorocarbons
- Acetals

Molecular Structure of Thermosets



Thermosets have three dimensional network of covalent intermolecular bonds (cross-linked structures)

Thermosets

The materials which cures irreversibly and become permanently hard and rigid after curing are called thermosets.

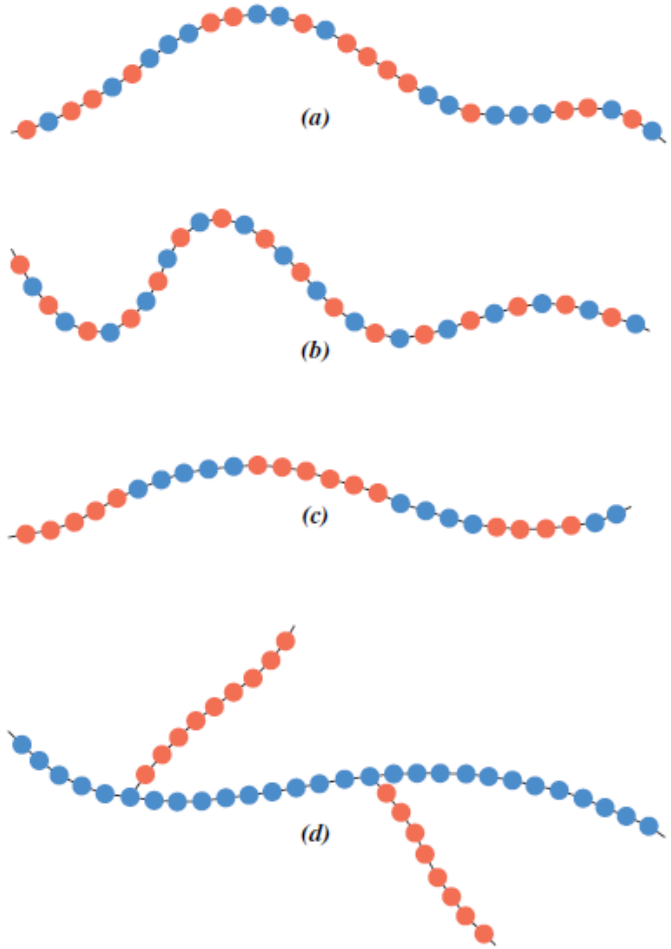
Thermoset plastics cannot be re-melted, continued heating for a long time leads to degradation or decomposition.

Types of Thermosets Materials

Different types of thermosets materials are Alkyds, Allylics, Amine, Bakelite, Epoxies, Phenolic, Polyester, Silicone, Polyurethane, and Vinyl Ester.

COPOLYMERS

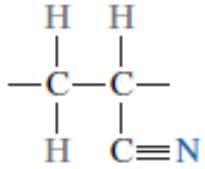
Consider a copolymer that is composed of two repeat units as represented by ● and ●



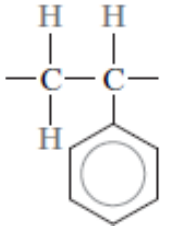
Schematic representations of (*a*) random, (*b*) alternating, (*c*) block, and (*d*) graft copolymers.

Some repeat units in rubbers

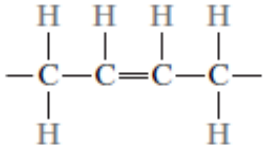
Acrylonitrile



Styrene



Butadiene

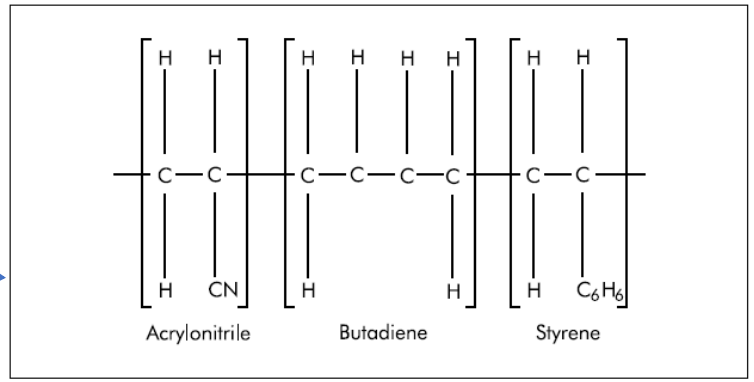


Copolymer rubbers

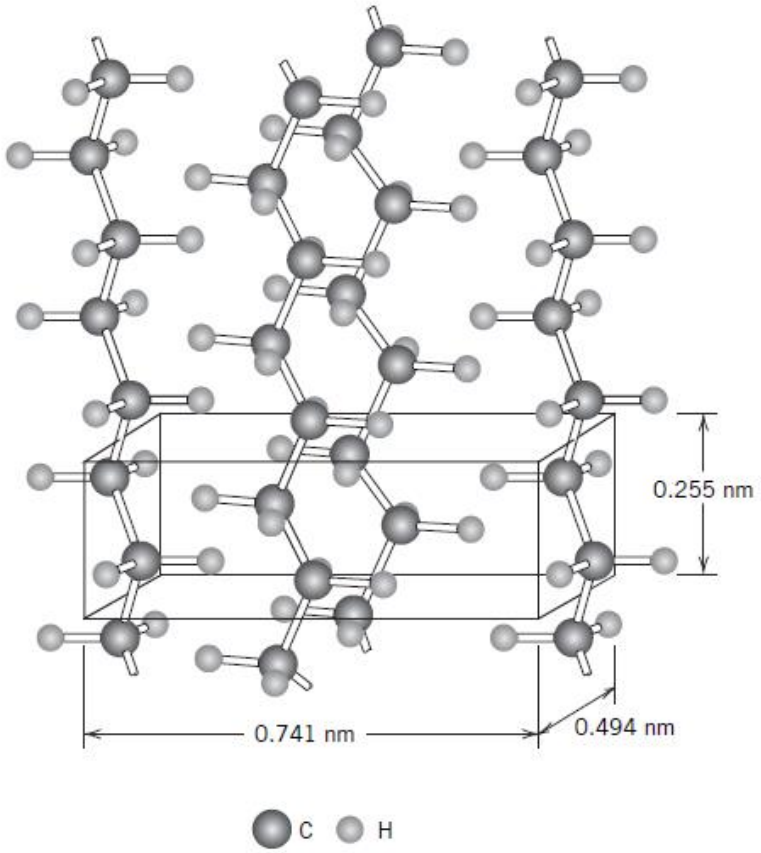
SBR – styrene butadiene

NBR – acrylonitrile butadiene

ABS

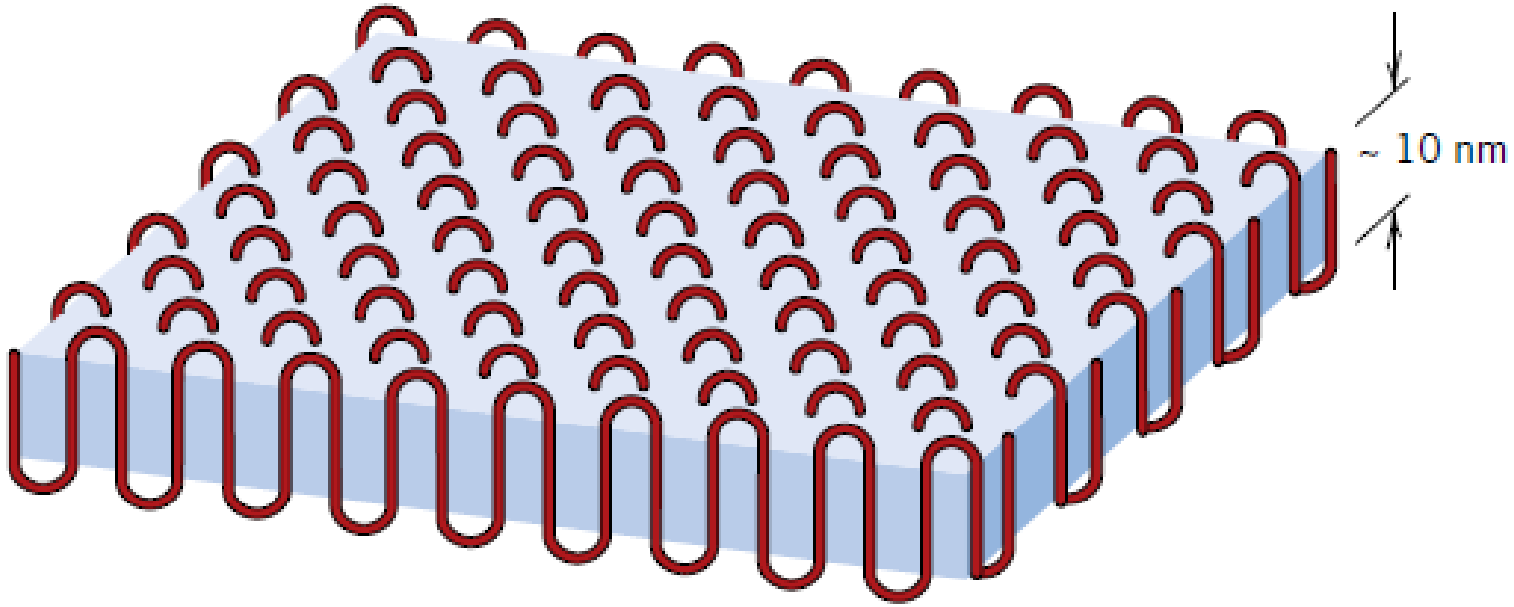


POLYMER CRYSTALLINITY



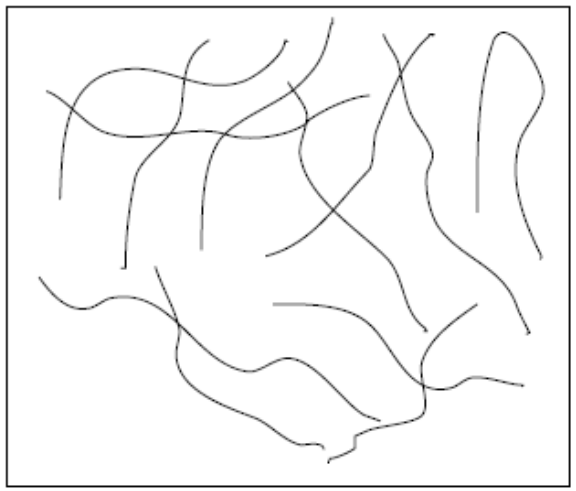
Polymer crystals

The chain-folded model for a plate-shaped polymer crystallite

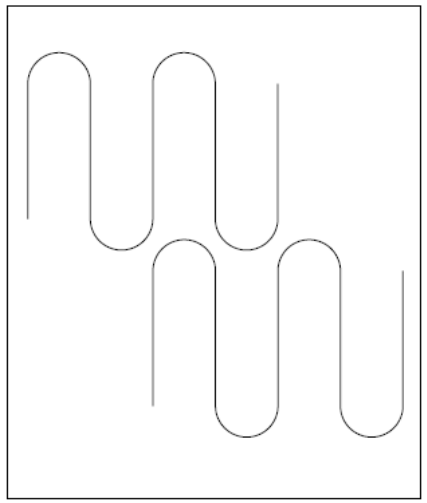


The degree of crystallinity may range from completely ***amorphous*** *to almost entirely (up to about 95%)* ***crystalline*** *(Semi crystalline)*

Amorphous and Crystalline Polymers



Amorphous molecular chains



Crystalline molecular chains

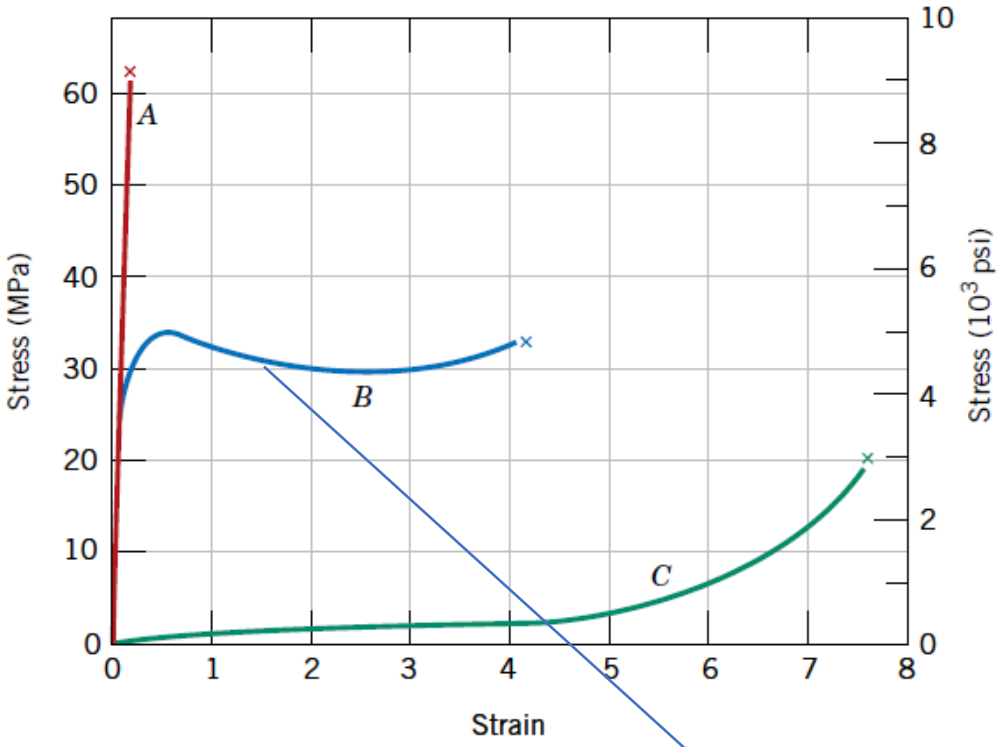
Amorphous

- *Broad softening range* – thermal agitation of the molecules breaks down the weak secondary bonds. The rate at which this occurs throughout the formless structure varies producing broad temperature range for softening.
- *Usually transparent* – the looser structure transmits light so the material appears transparent.
- *Low shrinkage* – all thermoplastics are processed in the amorphous state. On solidification, the random arrangement of molecules produces little volume change and hence low shrinkage.
- *Low chemical resistance* – the more open random structure enables chemicals to penetrate deep into the material and to destroy many of the secondary bonds.
- *Poor fatigue and wear resistance* – the random structure contributes little to fatigue or wear properties.

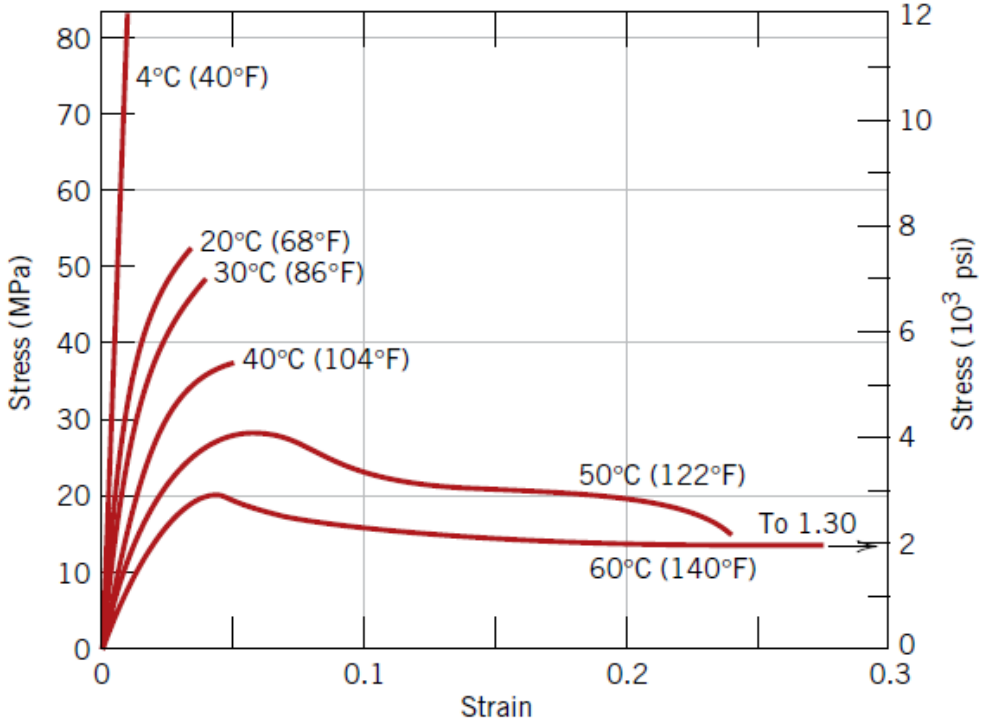
Crystalline

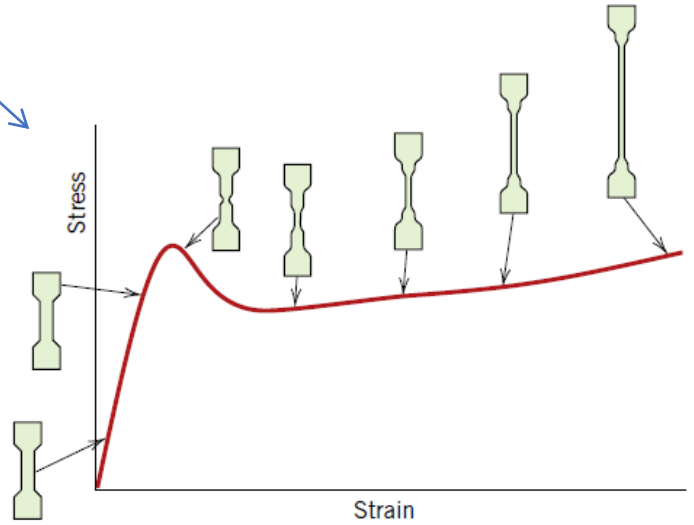
- *Sharp melting point* – the regular close-packed structure results in most of the secondary bonds being broken down at the same time.
- *Usually opaque* – the difference in refractive indices between the two phases (amorphous and crystalline) causes interference so the material appears translucent or opaque.
- *High shrinkage* – as the material solidifies from the amorphous state the polymers take up a closely packed, highly aligned structure. This produces a significant volume change manifested as high shrinkage.
- *High chemical resistance* – the tightly packed structure prevents chemical attack deep within the material.
- *Good fatigue and wear resistance* – the uniform structure is responsible for good fatigue and wear properties.

Stress – Strain Behaviour of Polymers

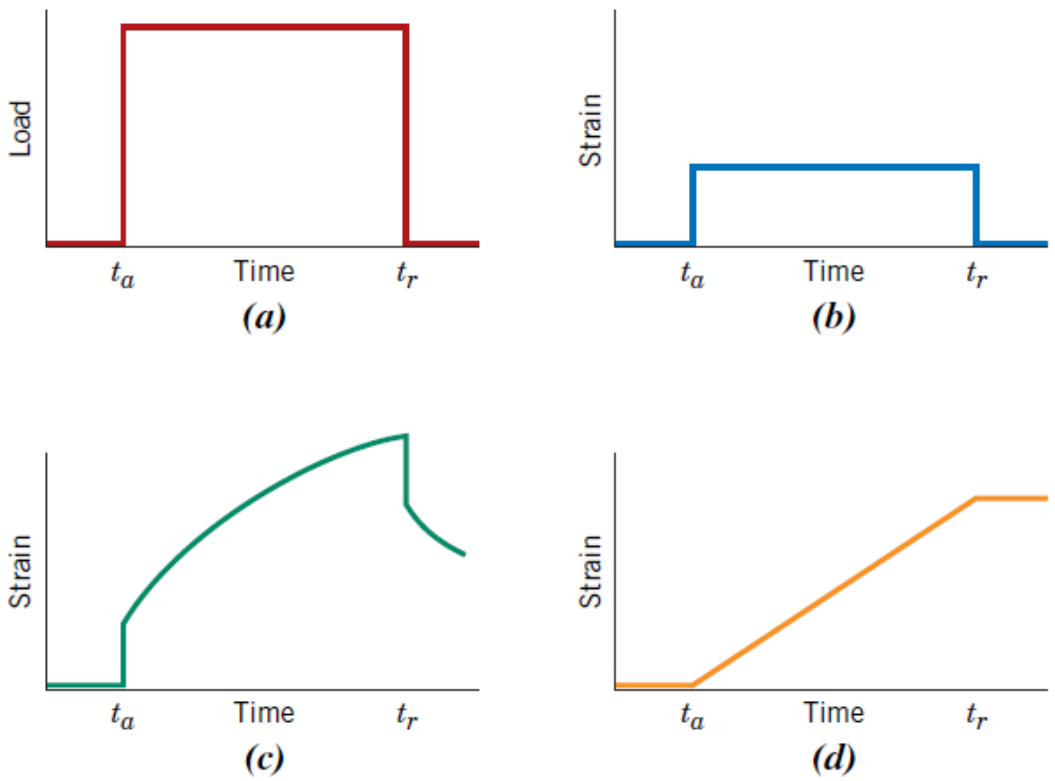


Influence of temperature





Visco elastic deformation

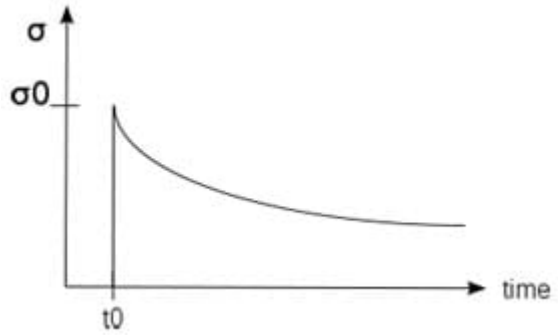


(*a*) Load versus time, where load is applied instantaneously at time t_a and released at t_r. For the load–time cycle in (*a*), the strain-versus-time responses are for totally elastic (*b*), viscoelastic (*c*), and viscous (*d*) behaviors.

- Elastic deformation is instantaneous; apply a force and the deformation happens instantaneously. Also, elastic deformation is linear, i.e. strain is proportional to stress (Figure b)
- Viscous deformation is not instantaneous; that is, in response to an applied stress, deformation is delayed or dependent on time (more time, more strain). Also, this deformation is not reversible (permanent strain).(Figure d)
- In elastic behaviour strain is (more or less) proportional the stress. In viscous behaviour strain rate is (more or less) proportional to stress.
- Polymers exhibit a combination of elastic and viscous behaviour. The total elastic deformation is thus comprised of two components; instantaneous and delayed (or time-dependent).(Figure c)

Stress Relaxation

If you apply a constant displacement to a viscoelastic material, then the force to hold the material in this configuration decreases over time.



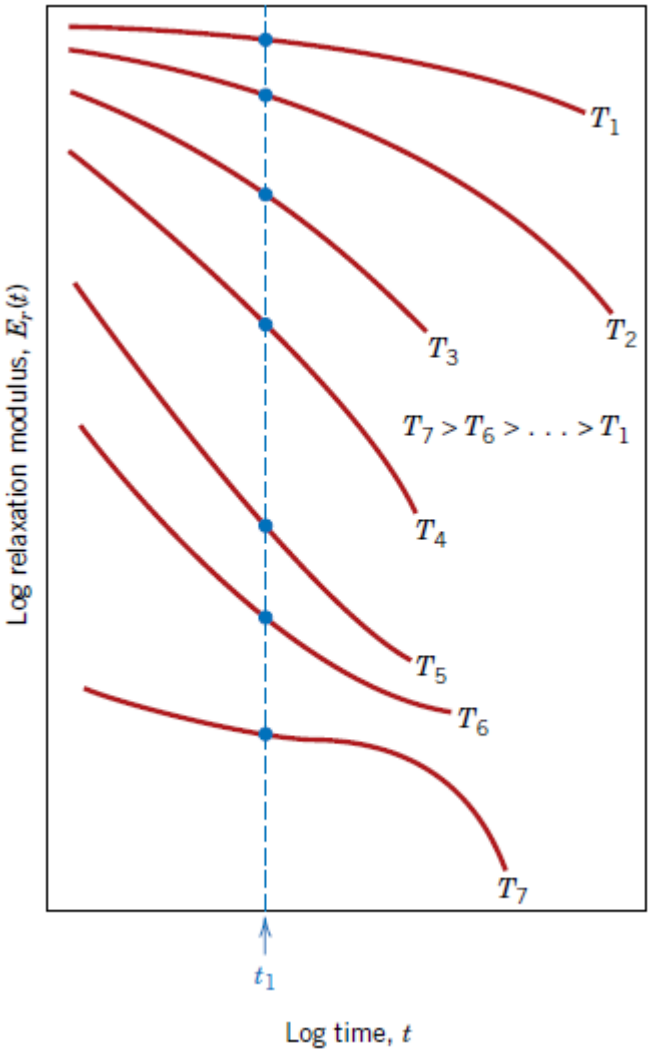
The initial vertical line represents the amount of force or stress it took to displace the material. As the material is held at that displacement, the stress in the material (or force it takes to hold the material at that displacement) decreases over time. Towards the end of the graph, the line becomes horizontal, indicating that the stress is no longer changing and the material has reached equilibrium.

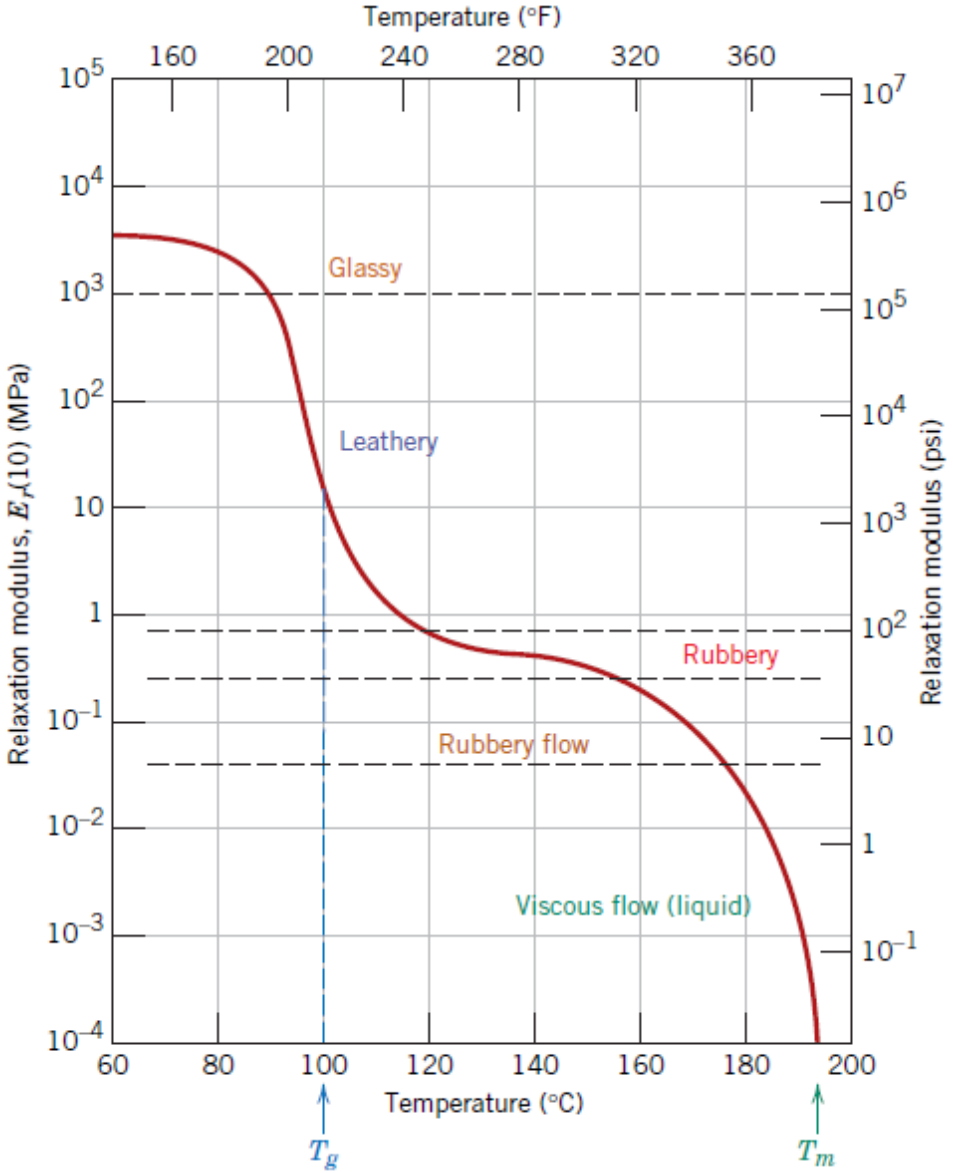
So, you have **viscoelastic relaxation modulus** corresponding to stress at any time (at some constant strain)

$$E_r(t) = \frac{\sigma(t)}{\epsilon_0}$$

Stress Relaxation

Furthermore, relaxation in polymers is a function of temperature



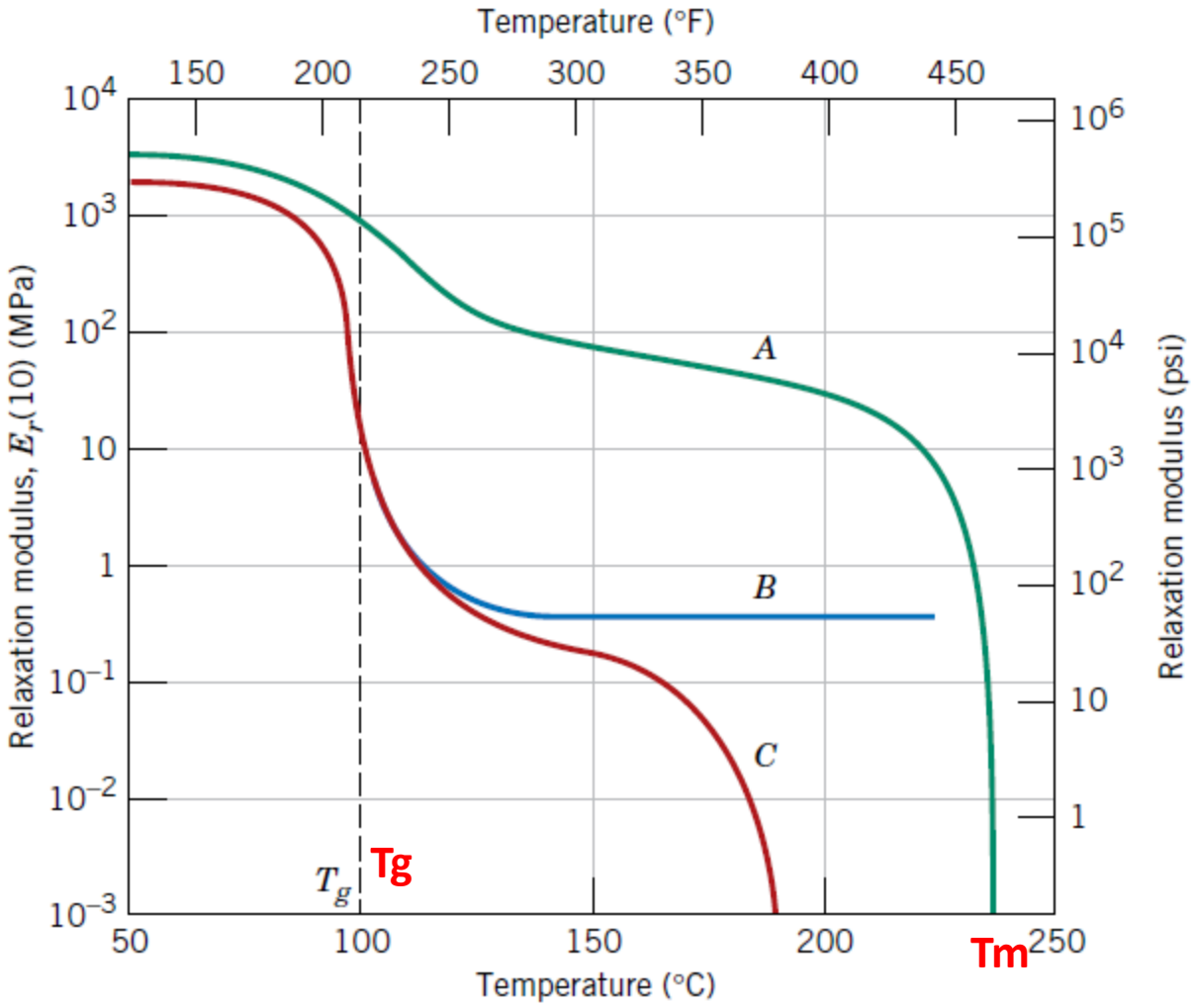


The temperature at which the material transforms from glass state to rubbery state (and vice versa) is called Glass Transition temperature.

It can be measured in several ways and techniques. One example shown in figure.

Stress Relaxation

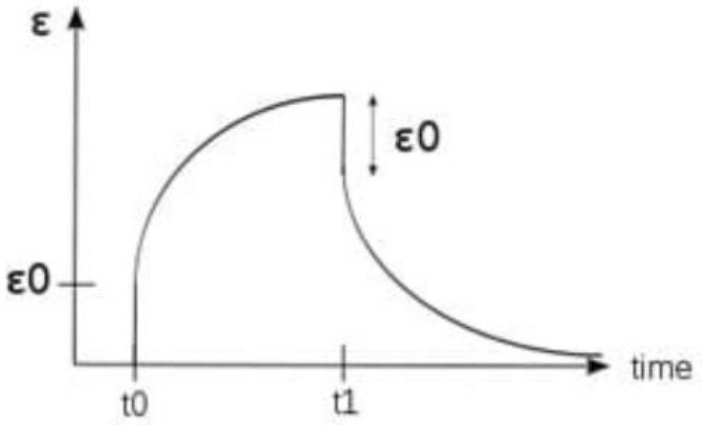
Furthermore, relaxation in polymers is a function of temperature



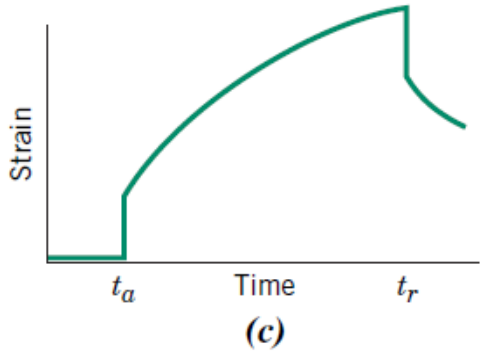
Logarithm of the relaxation modulus versus temperature for crystalline (curve *A*), lightly crosslinked (curve *B*), and amorphous (curve *C*) polystyrene.

Viscoelastic Creep

Material that was exposed to a constant force over time. The response is termed creep.



When the force is removed, the initial decrease in strain is equal to the amount of strain the material instantaneously experienced when it first had the force applied. Over time, the material returns to its original configuration and its strain becomes zero.



So, you have **viscoelastic creep modulus** corresponding to strain at any time (at some constant stress)

$$E_c(t) = \frac{\sigma_0}{\epsilon(t)}$$

The creep modulus is also temperature sensitive and decreases with increasing temperature.

Materials for Engineers

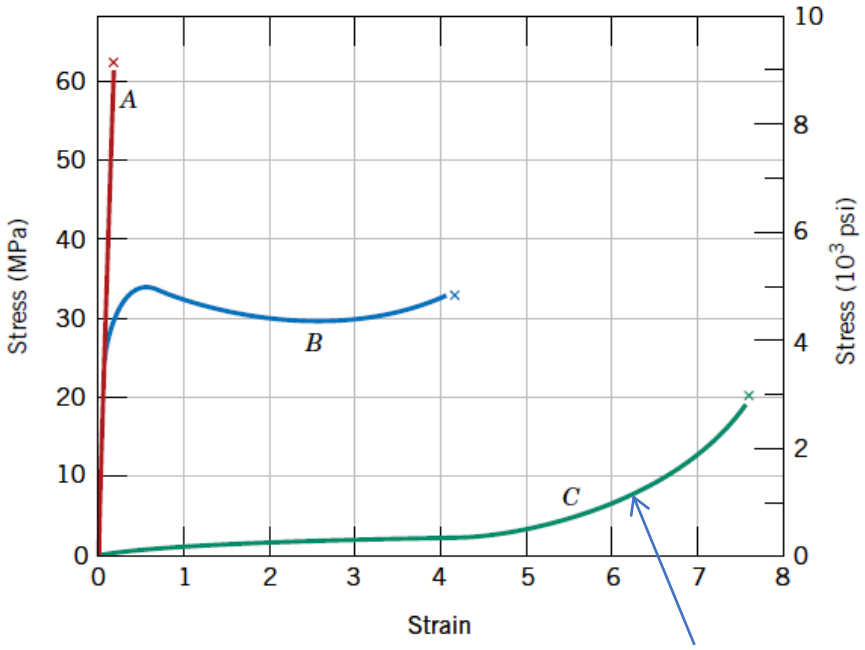
Lecture until Nov 10

Dr S. Gowthaman
Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

Reference:

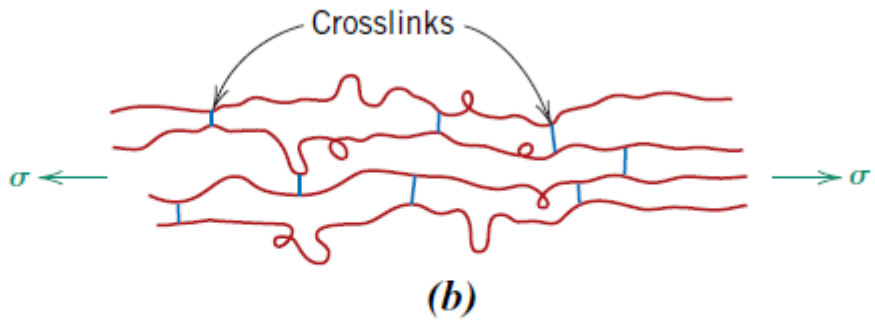
1. Materials science and engineering by William Callister, Wiley Publications

Elastomers



Elastomers

(a)



(b)

Crosslinked polymer chain molecules (*a*) in an unstressed state and (*b*) during elastic deformation in response to an applied tensile stress.

- Upon release of the stress, the chains spring back to their prestressed conformations, and the macroscopic piece returns to its original shape.

- Elastic deformation depends on the degree of disorder in unstressed state

Elastomers

Criteria for a polymer to be elastomeric:

- It must not easily crystallize; elastomeric materials are amorphous, having molecular chains that are naturally coiled and kinked in the unstressed state.

- For elastomers to experience relatively large elastic deformations, the onset of plastic deformation must be delayed. Restricting the motions of chains past one another by crosslinking accomplishes this objective. The crosslinks act as anchor points between the chains and prevent chain slippage from occurring. Crosslinking in many elastomers is carried out in a process called **vulcanization**

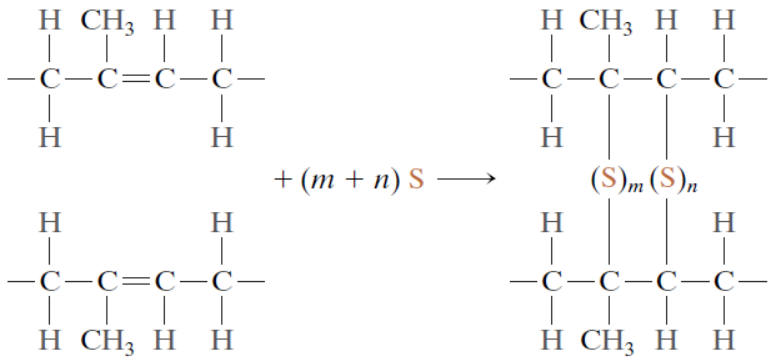
- Finally, the elastomer must be above its glass transition temperature. The lowest temperature at which rubber like behavior persists for many of the common elastomers is between -50°C and -90°C. Below its glass transition temperature, an elastomer becomes brittle.

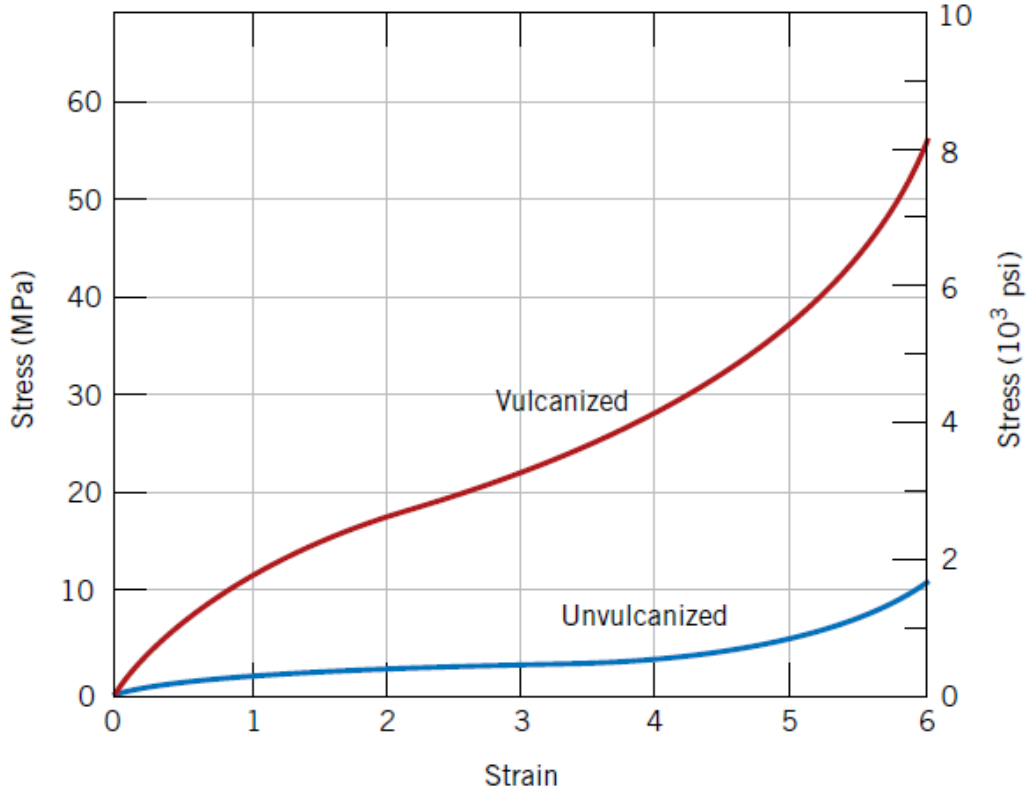
Elastomers

Vulcanization

- The crosslinking process in elastomers is called **vulcanization,** which is achieved by a nonreversible chemical reaction, typically carried out at an elevated temperature.

- In most vulcanizing reactions, sulfur compounds are added to the heated elastomer; chains of sulfur atoms bond with adjacent polymer backbone chains and crosslink them.





- *Useful rubbers result when about 1 to 5 parts (by weight) of sulfur are added to 100 parts of rubber.*

- *Too much sulfur will increase crosslink density, and it will reduce its extensibility.*

Melting and Glass Transition Temperature of Polymers

Material	*Glass Transition Temperature [°C (°F)]*	*Melting Temperature [°C (°F)]*
Polyethylene (low density)	−110 (−165)	115 (240)
Polytetrafluoroethylene	−97 (−140)	327 (620)
Polyethylene (high density)	−90 (−130)	137 (279)
Polypropylene	−18 (0)	175 (347)
Nylon 6,6	57 (135)	265 (510)
Poly(ethylene terephthalate) (PET)	69 (155)	265 (510)
Poly(vinyl chloride)	87 (190)	212 (415)
Polystyrene	100 (212)	240 (465)
Polycarbonate	150 (300)	265 (510)

Factors that affect Tg and Tm

- Main chain stiffness (Double bonds in chain increases stiffness, increases Tg and Tm)
- Size and type of side groups (Methyl group in PP increases its Tg and Tm relative to PE, Cl in PVC increases its Tg and Tm)
- Chain length of polymers (mol. wt) (Higher the chain length, higher is the Tg and Tm)
- Molecular weight distribution (Narrow and higher Mn, higher Tg and Tm)
- Degree of branching (HDPE vs LDPE)

Thermoplastics vs Thermosets

Thermoplastics

Thermoplastic polymers are polymers that can be repeatedly heated and molded without much change in their chemical or physical properties.(of course this will change with recycling)

Thermosets

Thermoset polymers are polymers that once molded and hardened, cannot be reshaped or recycled.

PROPERTIES

Thermoplastics

- The molecules have weak, straight-chain bonds between them that can be broken by heating.
- Thermoplastics are prepolymerized by the manufacturer, and don't require polymerization again during processing.
- They are elastic and flexible in nature.
- They dissolve in organic solvents.
- On heating, they will soften and ultimately melt.
- The melting point of thermoplastics is lower than their degradation point.
- Thermoplastics are in the form of solid pellets before use.

Thermosets

- They have strong chemical bonds between molecules, including cross-linking, due to which they do not separate on heating.
- Thermosets have to undergo a two-stage polymerization during the processing stage.
- They are hard and brittle.
- They do not dissolve in organic solvents.
- On heating, thermoset polymers will char, not melt.
- Their degradation point is lower than their melting point.
- Thermosets are in the liquid state before processing.

Thermoplastics vs Thermosets

ADVANTAGES

Thermoplastics

- High impact resistance (10 X thermosets)
- Can be recycled and reused
- They can be blended with other thermoplastic materials

Thermosets

- High structural rigidity (more strength)
- Heat and chemical resistance
- Easy fiber reinforcement of liquid raw materials
- High durability
- Ability of thick and thin wall formation
- Highly adaptable design process
- Production process is well-established

DISADVANTAGES

Thermoplastics

- Heat and pressure requirement for fiber reinforcement
- High cost
- Heat sensitivity
- Polymer degradation after constant remelting and remolding
- Structural failure on high stress or long-term load application

Thermosets

- Recycling inability
- Refrigerated storage required by liquid raw materials
- Lengthy, complex process stages
- Structural failure on high-force impact

Thermoplastics vs Thermosets

COMMON EXAMPLES AND USES

Thermoplastics

- **Nylon (Polyamide):** Mechanical and automotive parts, clothing, packaging, cupboard hinges, heat-resistant composite manufacture.
- **Polyethylene:** Drums, gas tank coating, milk bottles, squeeze bottles, jugs, movable machine parts, bullet-resistant vests, laundry detergent containers.
- **Acrylic:** Battery covers, lightweight glass alternative, vehicle taillight covers, eye lenses, as bone cement in medicine.
- **Polypropylene:** Toys, sanitary tissues, heat-proof medical equipment, rope, string, plastic seats, laboratory equipment, detergent-proof food containers, automobile components, folders.
- **Polyvinyl chloride:** Cabinets, fume hoods, tanks, electrical insulation, toys, pipes, fittings, flooring, medical devices.
- **Teflon:** Flange spacers, gaskets, non-stick cookware, machine parts, gears, wires, lubricant for sliding doors.

Thermosets

- **Polyurethanes:** Paints, coatings, insulating foams, car parts, print rollers, footwear, sealants, adhesives.
- **Epoxy resin:** Aircraft components, tooling jigs and fixtures, coatings, adhesives for automobiles, marine vessel parts, electronic components.
- **Melamine formaldehyde:** Adhesives, coatings, laminates, dinnerware, electrical components, knobs, household items, construction material, furniture-making.
- **Urea-formaldehyde:** Plywood foam, electrical items, doorknobs, toilet items, adhesives, chemically-inert coatings, paper, plastic molds, decorative articles.
- **Polyester resins:** Casting materials, non-metallic car body fillers, electrical components, reinforced plastic sheets used in restaurants/kitchens, low-care walls, laser printer toners, bonding materials.
- **Bakelite:** Precision-made parts, vehicle disc brake cylinders, knobs, plastic ware, electrical products and insulation, plugs and sockets, automotive parts, light bulb supports, kitchenware handles.

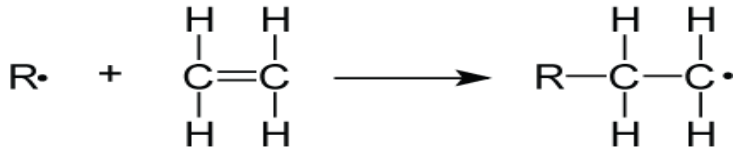
Elastomers

Chemical Type	*Trade (Common) Names*	*Elongation (%)*	*Useful Temperature Range [°C (°F)]*	*Major Application Characteristics*	*Typical Applications*
Natural polyisoprene	Natural rubber (NR)	500–760	−60 to 120 (−75 to 250)	Excellent physical properties; good resistance to cutting, gouging, and abrasion; low heat, ozone, and oil resistance; good electrical properties	Pneumatic tires and tubes; heels and soles; gaskets; extruded hose
Styrene-butadiene copolymer	GRS, Buna S (SBR)	450–500	−60 to 120 (−75 to 250)	Good physical properties; excellent abrasion resistance; not oil, ozone, or weather resistant; electrical properties good, but not outstanding	Same as natural rubber
Acrylonitrile–butadiene copolymer	Buna A, Nitrile (NBR)	400–600	−50 to 150 (−60 to 300)	Excellent resistance to vegetable, animal, and petroleum oils; poor low-temperature properties; electrical properties not outstanding	Gasoline, chemical, and oil hose; seals and O-rings; heels and soles; toys
Chloroprene	Neoprene (CR)	100–800	−50 to 105 (−60 to 225)	Excellent ozone, heat, and weathering resistance; good oil resistance; excellent flame resistance; not as good in electrical applications as natural rubber	Wire and cable; chemical tank linings; belts, hoses, seals, and gaskets
Polysiloxane	Silicone (VMQ)	100–800	−115 to 315 (−175 to 600)	Excellent resistance to high and low temperatures; low strength; excellent electrical properties	High- and low-temperature insulation; seals, diaphragms; tubing for food and medical uses

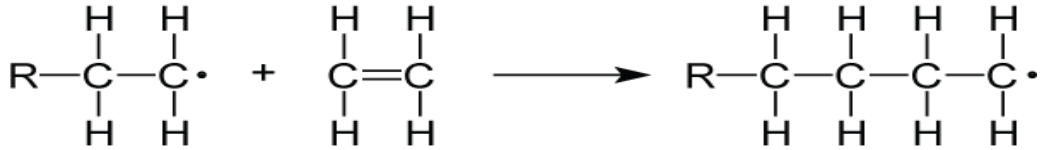
Polymerization

Examples: PE, PP, PVC, PS etc

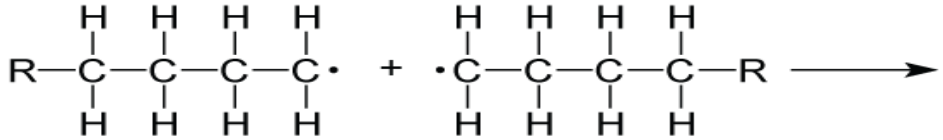
Addition (Chain) Polymerization

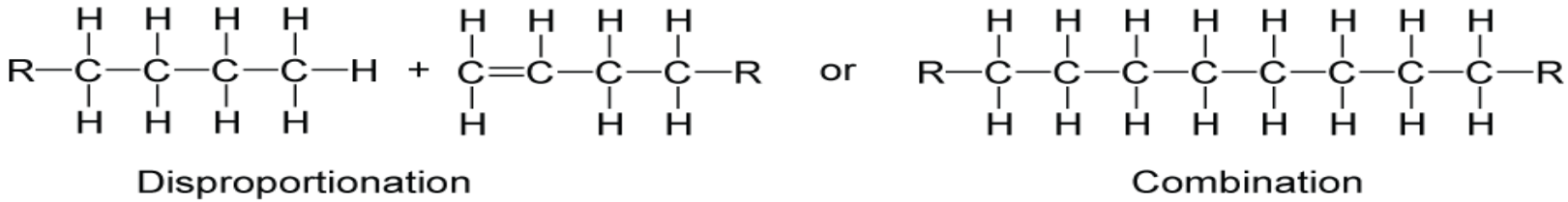


– **Propagation**



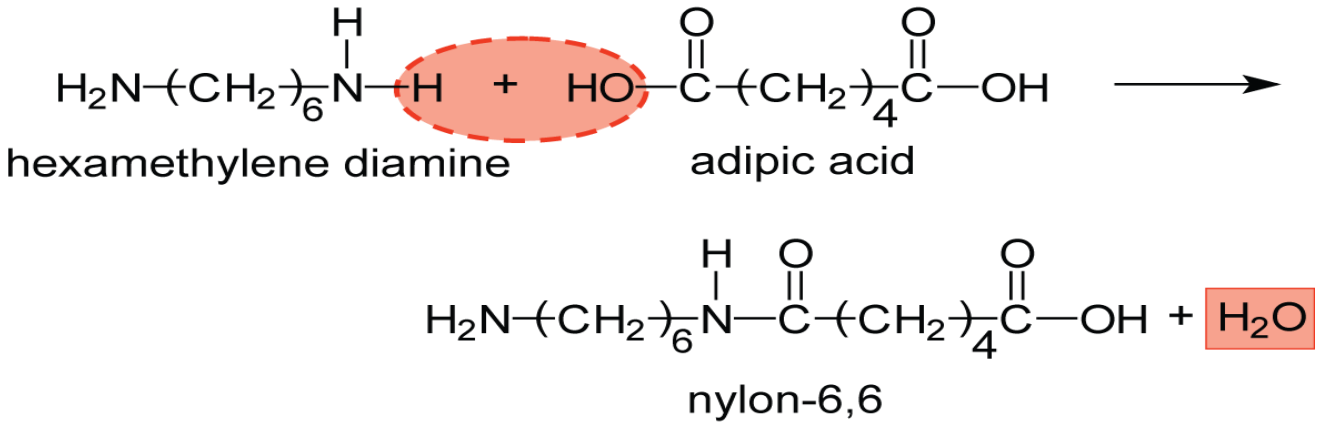
– **Termination**



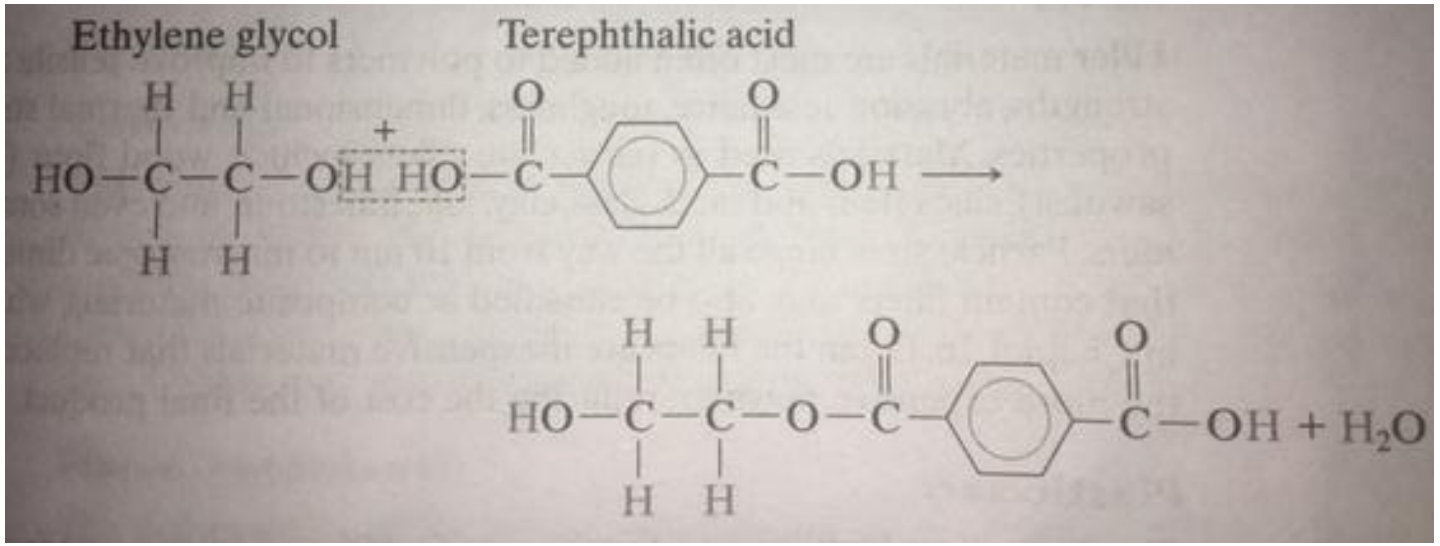


Condensation (Step) Polymerization

(involves more than one monomer)



Nylon – 6,6 (Polyamide)



Polyethylene terephthalate (PET) (Polyester)

Advanced Polymers

Ultrahigh Molecular Weight Polyethylene (UHMWPE)

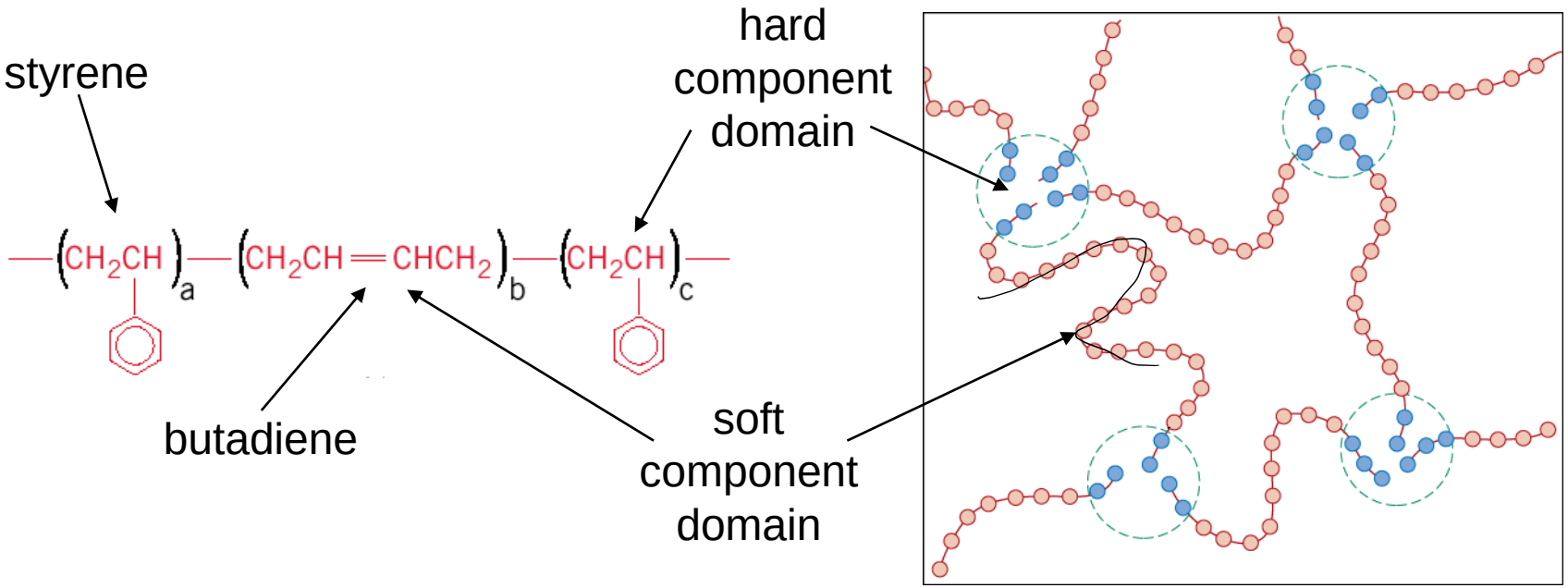
- Outstanding properties
  - Extremely high impact strength
  - resistance to wear/abrasion
  - low coefficient of friction
  - self-lubricating surface
  - Chemical resitance
  - But low Tm (137C)!
- Important applications
  - bullet-proof vests
  - hip implants (acetabular cup)

Kevlar (aramid) is another polymer fiber used

Thermoplastic Elastomers

Styrene-butadiene block copolymer

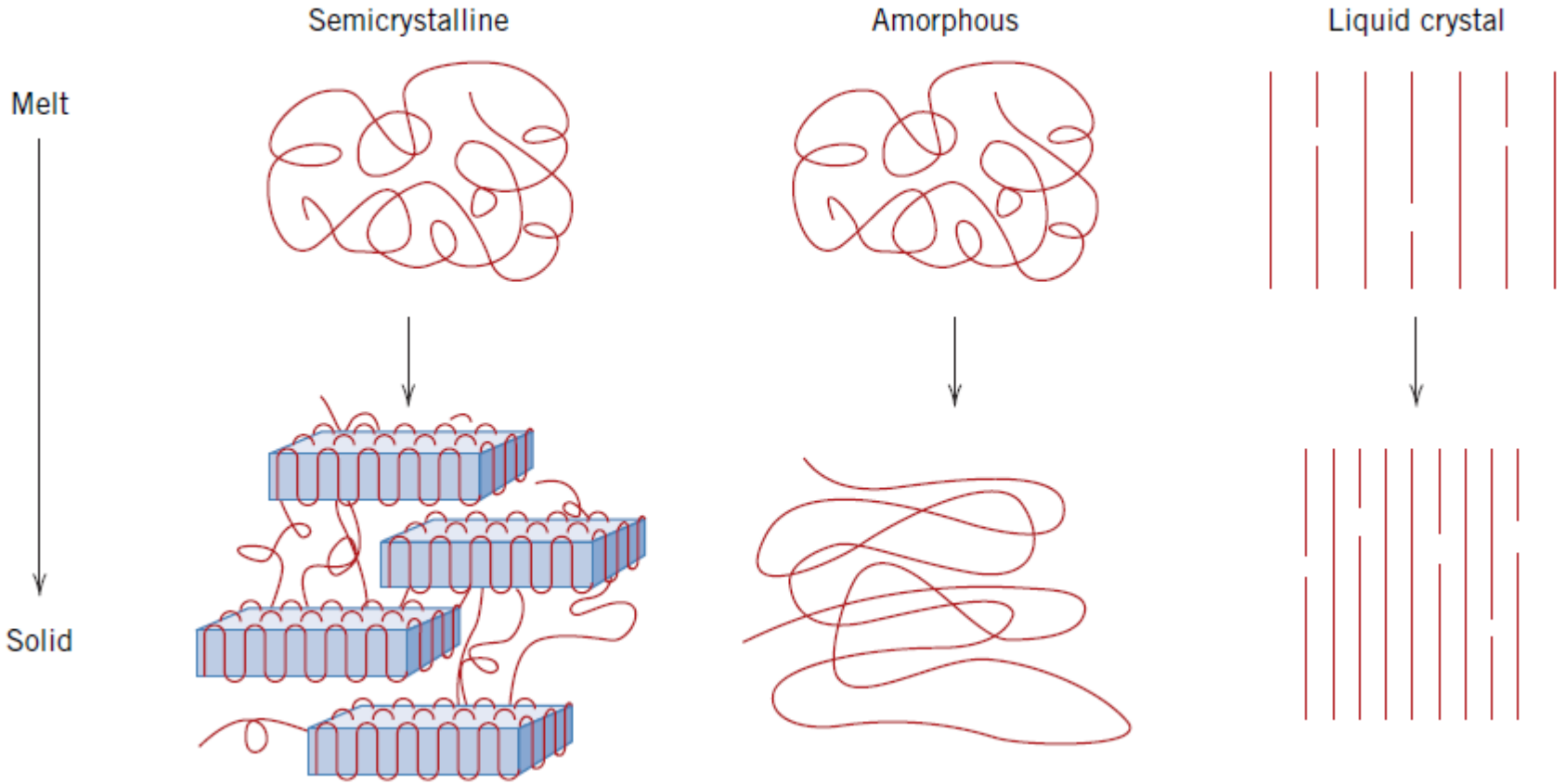
- No chemical crosslinking as in thermosets



- Soft-chain segment motions are restricted by hard segments

The chief advantage of the TPEs over the thermoset elastomers is that upon heating above Tm of the hard phase, they melt (i.e., the physical crosslinks disappear), and, therefore, they may be processed by conventional thermoplastic forming techniques [blow molding, injection molding, etc.

Liquid Crystal Polymers



Flat-panel computer monitors and televisions, and other digital displays

Processing of Plastics

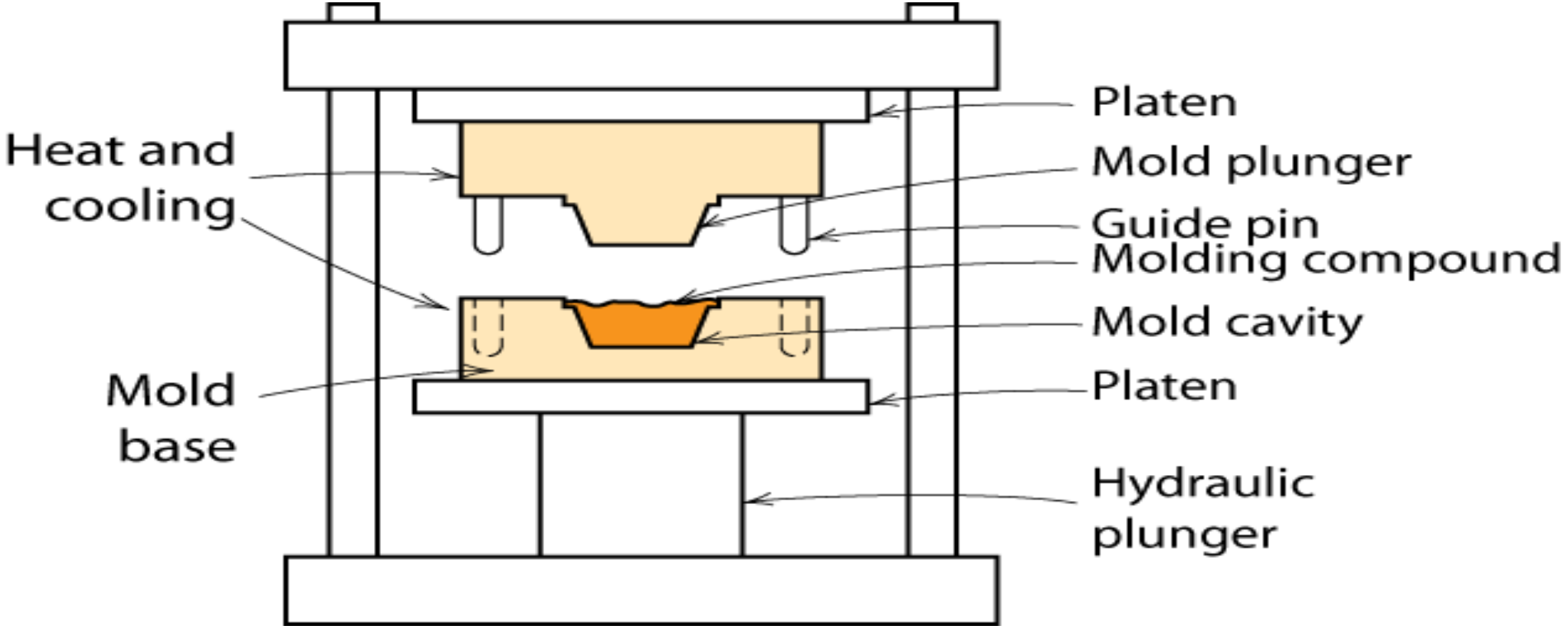
- Thermoplastic
  - can be reversibly cooled & reheated, i.e. recycled
  - heat until soft, shape as desired, then cool
  - ex: polyethylene, polypropylene, polystyrene.

- Thermoset
  - when heated forms a molecular network (chemical reaction)
  - degrades (doesn't melt) when heated
  - a prepolymer molded into desired shape, then chemical reaction occurs
  - ex: urethane, epoxy

Processing Plastics – Compression Molding

Thermoplastics and thermosets

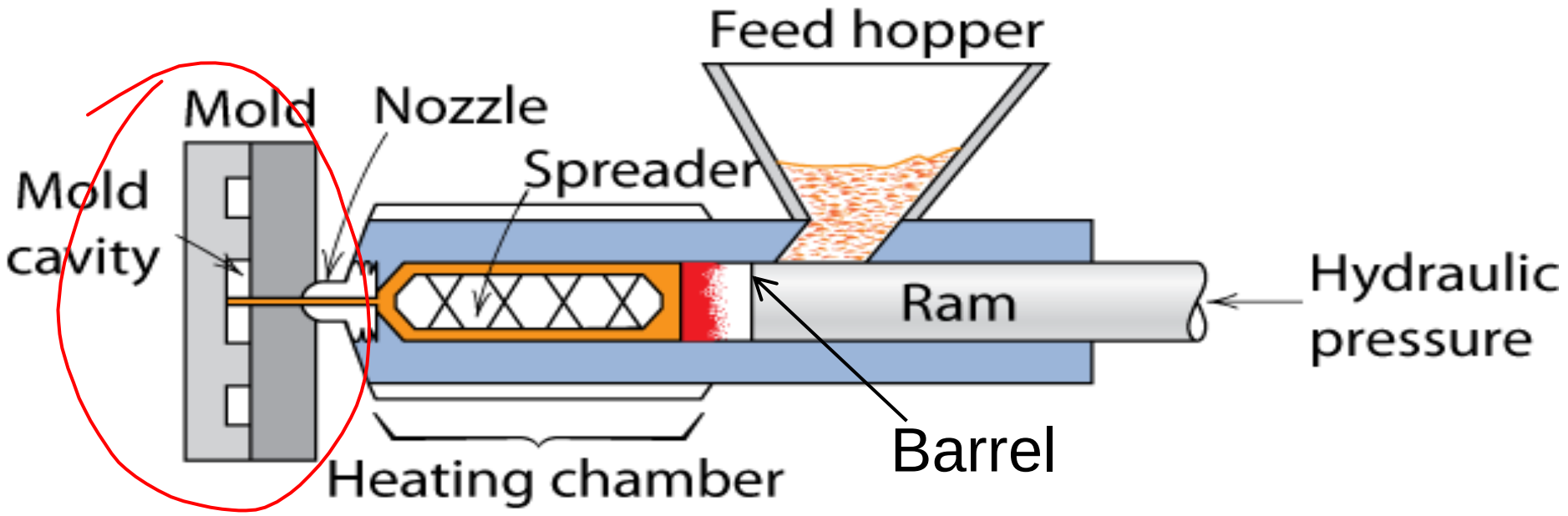
- polymer and additives placed in mold cavity
- mold heated and pressure applied
- fluid polymer assumes shape of mold



Processing Plastics – Injection Molding

Thermoplastics and some thermosets

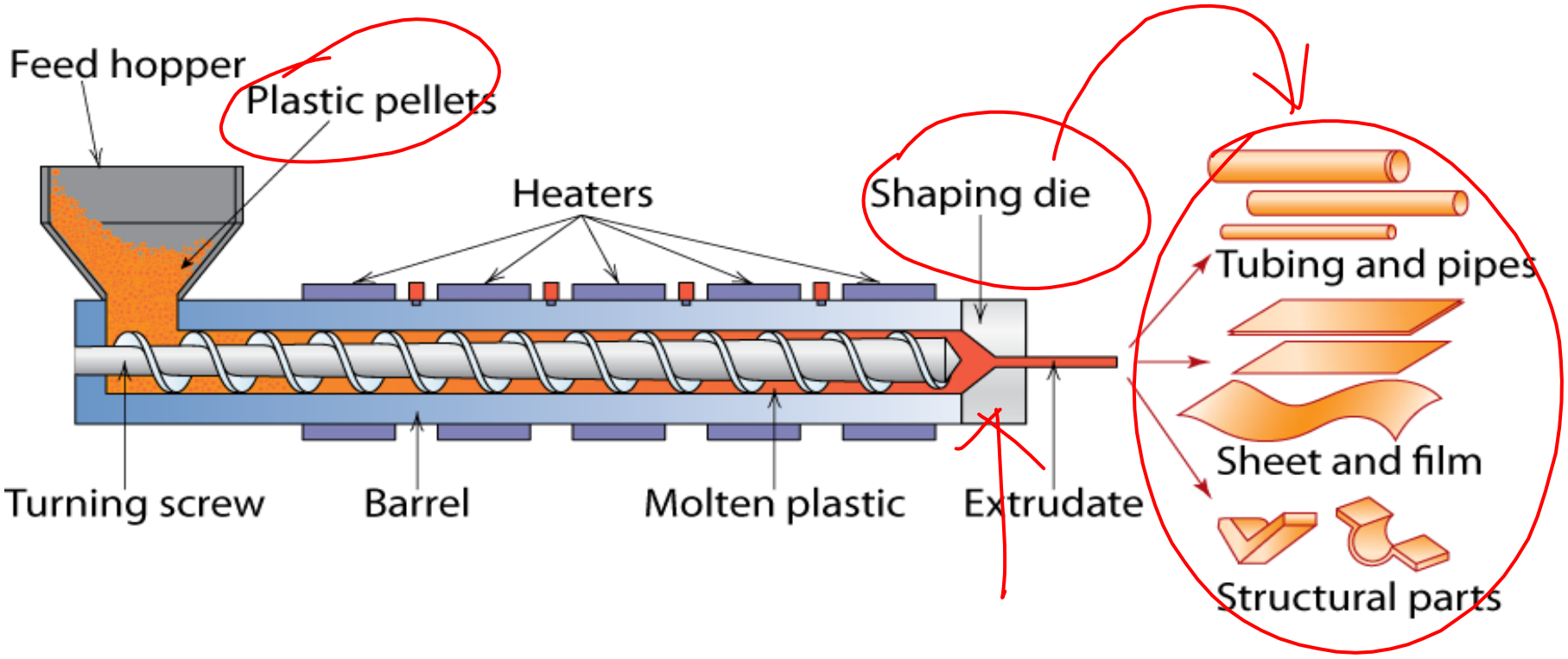
- when ram retracts, plastic pellets drop from hopper into barrel
- ram forces plastic into the heating chamber (around the spreader) where the plastic melts as it moves forward
- molten plastic is forced under pressure (injected) into the mold cavity where it assumes the shape of the mold



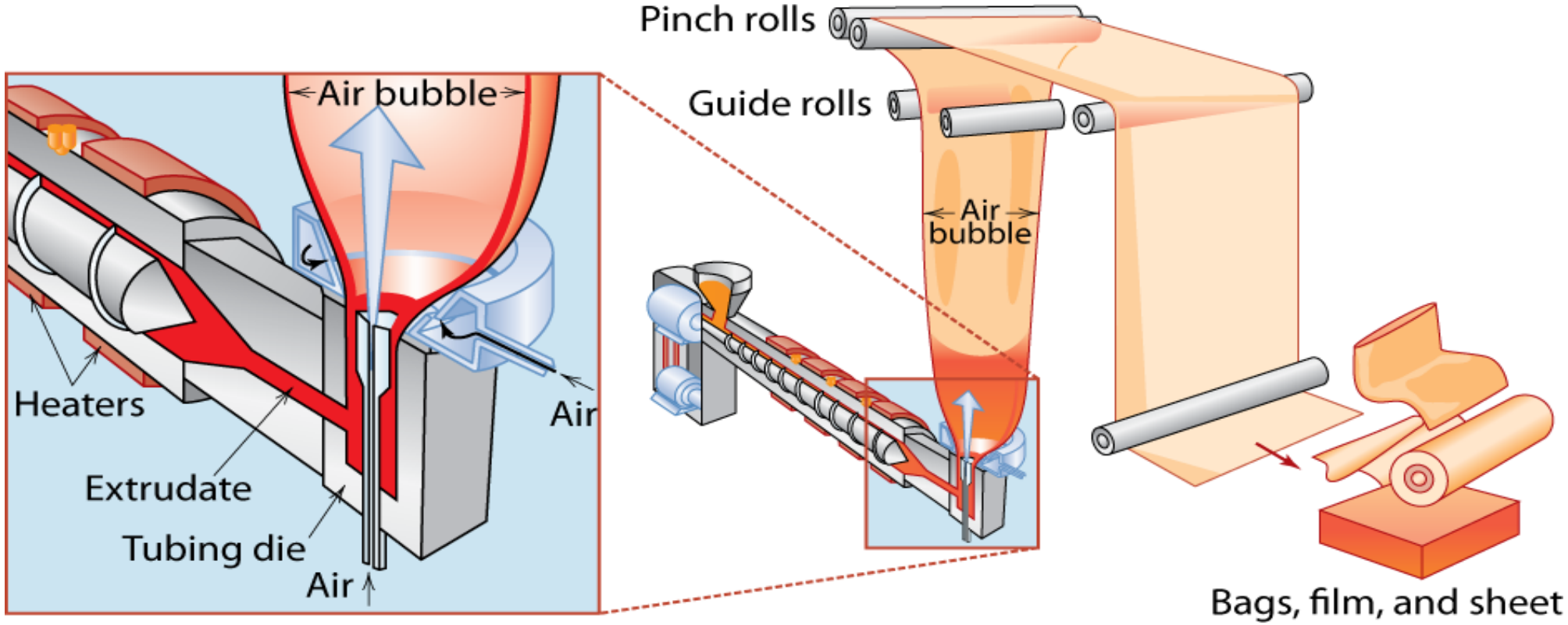
Processing Plastics – Extrusion

Thermoplastics

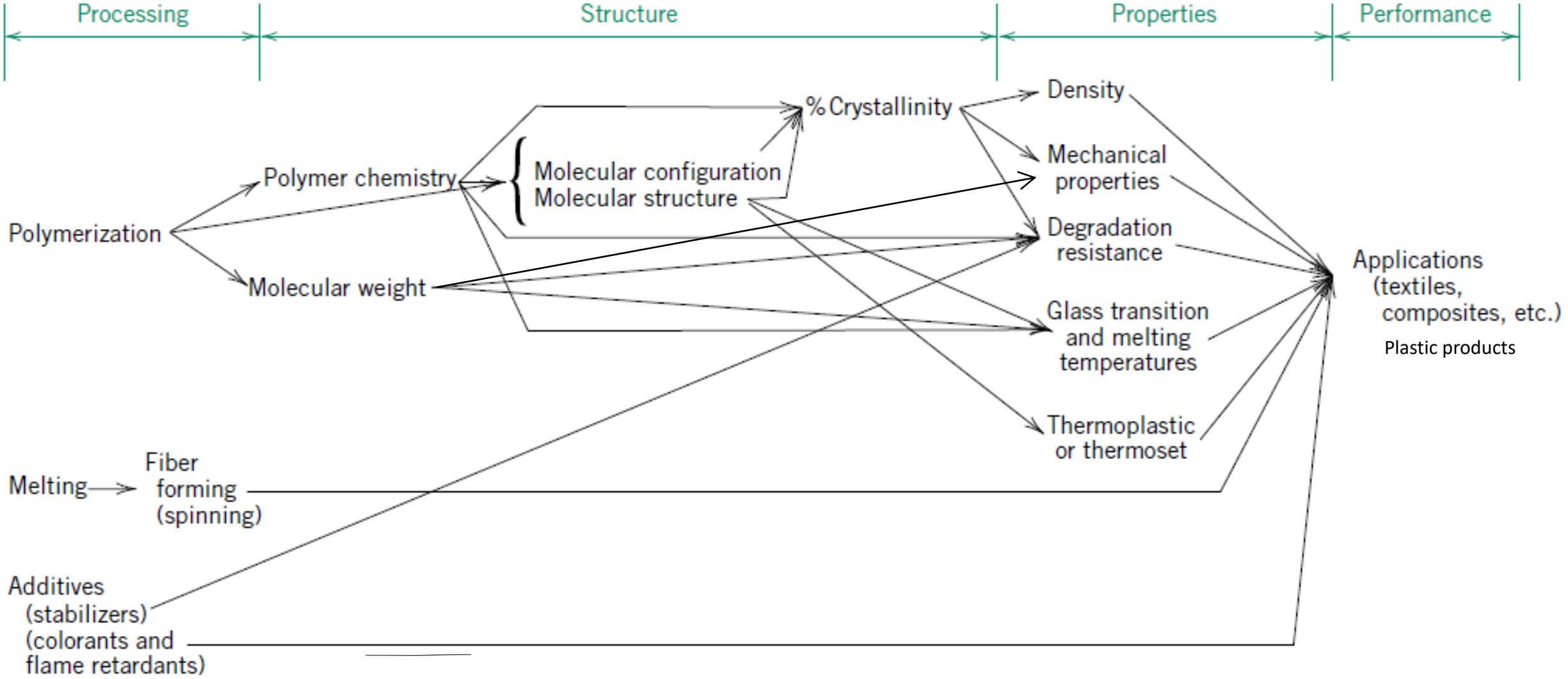
- plastic pellets drop from hopper onto the turning screw
- plastic pellets melt as the turning screw pushes them forward by the heaters
- molten polymer is forced under pressure through the shaping die to form the final product (extrudate)



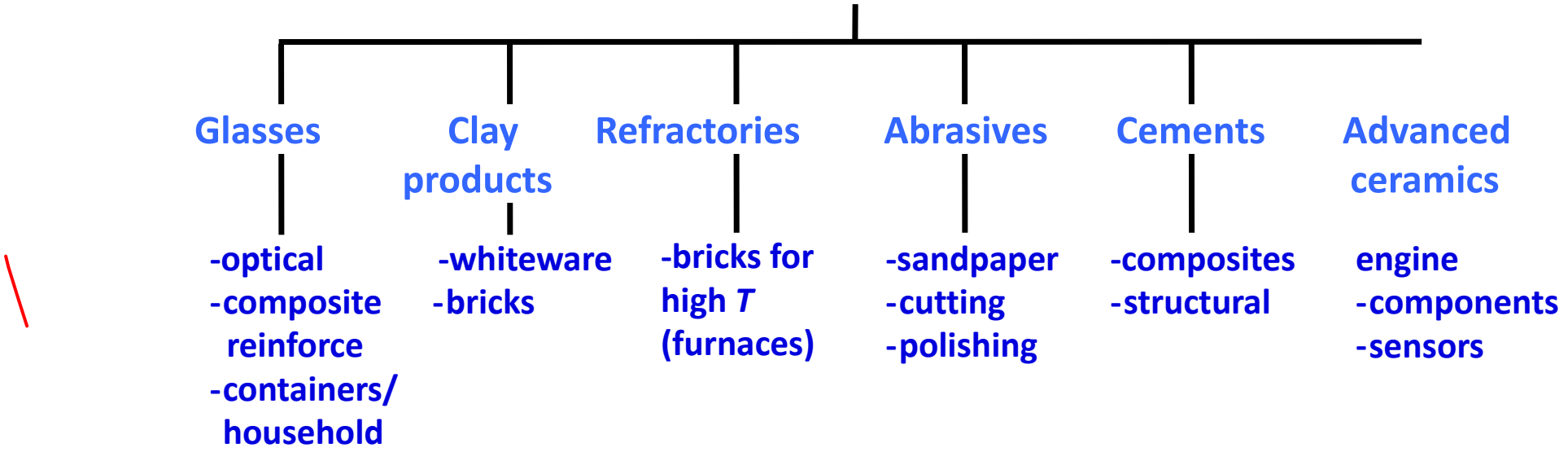
Processing Plastics – Blown-Film Extrusion



Processing / Structure / Property / Performance



Ceramics

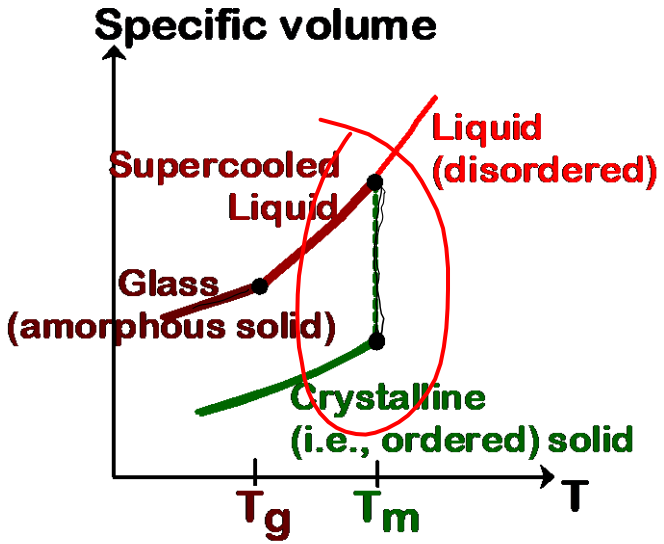


- Properties:
  -- T_m for glass is moderate, but large for other ceramics.
  -- Small toughness, ductility; large moduli & creep resist.
  -- Mostly ionic bonding

- Applications:
  -- High *T*, wear resistant.

GLASS PROPERTIES

- **Specific volume** vs Temperature:

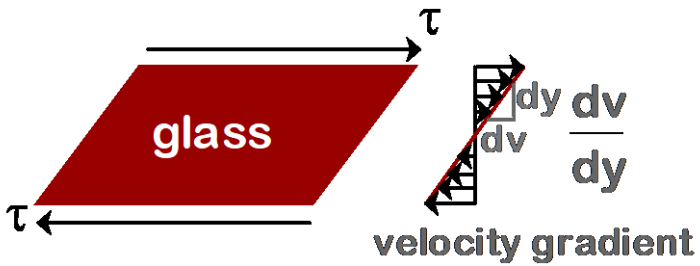


- **Crystalline materials**:
  --crystallize at melting temp, T_m
  --have abrupt change in spec. vol. at T_m

- **Glasses**:
  --do not crystallize
  --spec. vol. varies smoothly with T
  --Glass transition temp, T_g

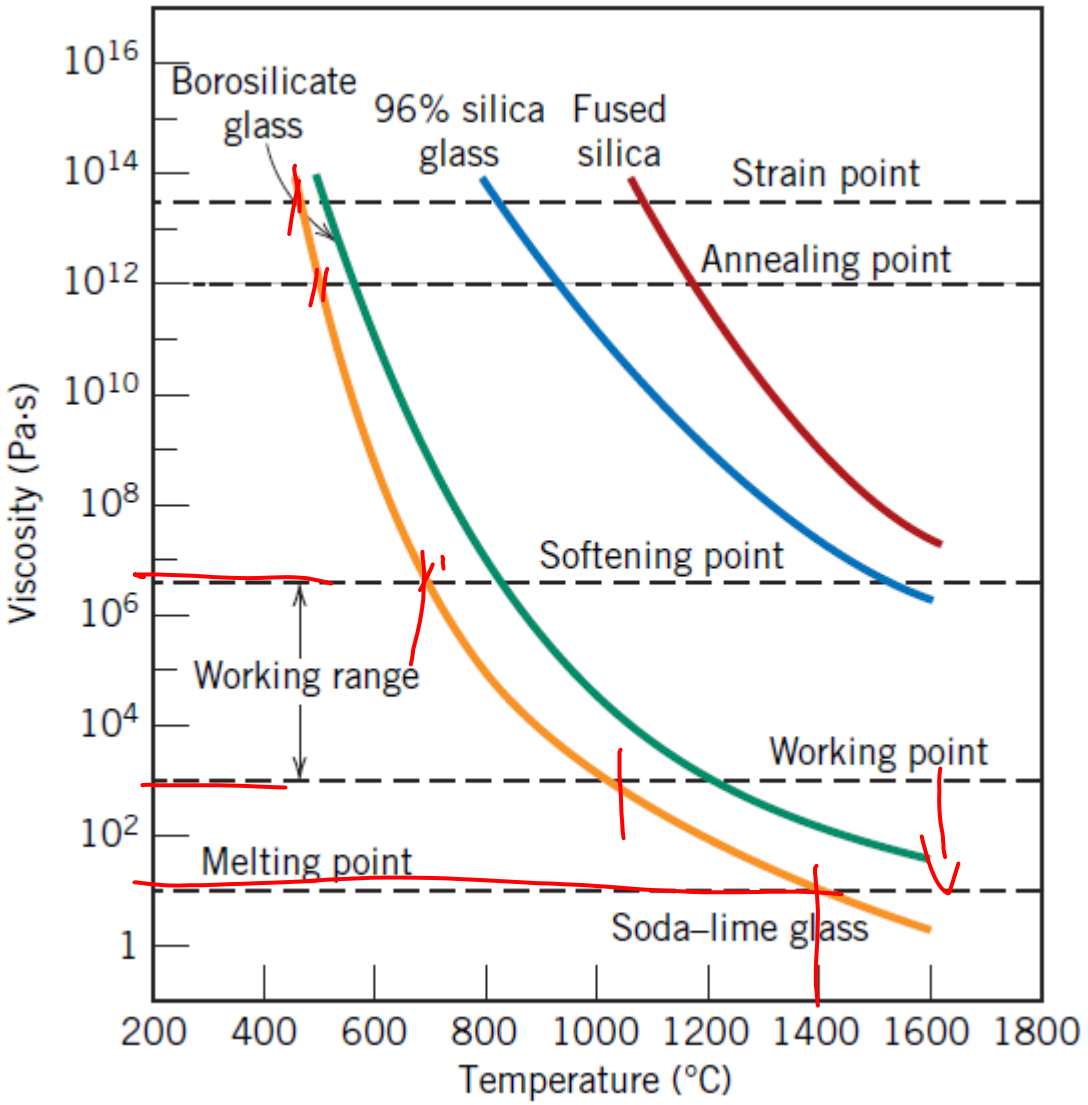
- **Viscosity**:
  --relates shear stress & velocity gradient:
  --has units of (Pa-s)

$$\tau = \eta \frac{dv}{dy}$$



GLASS VISCOSITY VS TEMPERATURE

- **Viscosity decreases with T increase**



•Melting point = viscosity of 10 Pa.s (above which liquid)

•Working point= viscosity of 1000 Pa.s (until which has easy workability)

•Softening point= viscosity of ~10^{7} Pa.s (until which glass can be handled easily)

•Annealing point= viscosity of 10^{12} Pa.s. (at this point residual stresses may be removed within 15 min)

•Strain point = viscosity of $3x10^{13}$Pa.s (*For temperatures below this point, fracture occurs before the onset of plastic deformation)*

The glass transition temperature will be above the strain point.

Most glass working operation are carried out in working range

Annealing –

- When a ceramic material is cooled from an elevated temperature, internal stresses, called thermal stresses, may be introduced as a result of the difference in cooling rate and thermal contraction between the surface and interior regions.

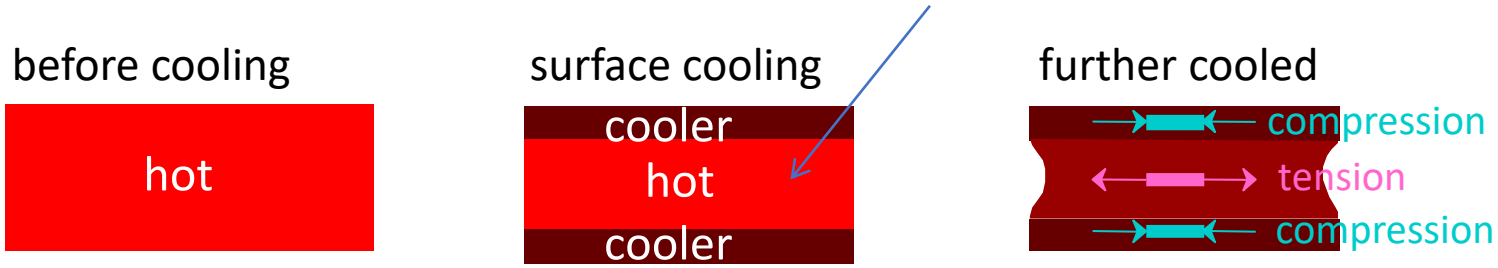
- To relieve the internal stresses, the glassware is heated to the annealing point, then slowly cooled to room temperature.

Heat Treatment of Glasses

Tempering –

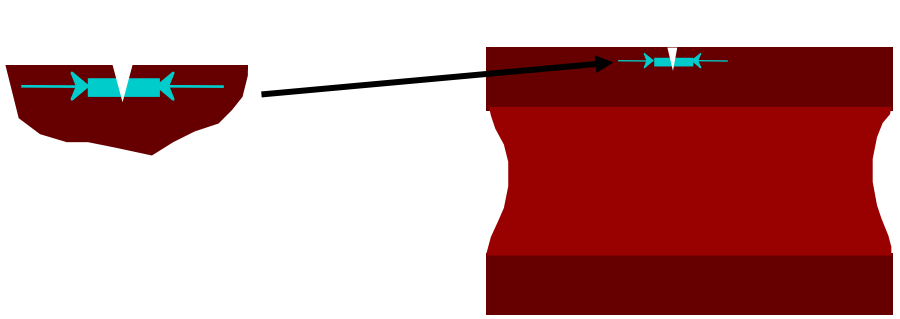
Glassware is heated to a temperature above the glass transition region yet below the softening point, then cooled.

Still plastic (interior temp. above strain point - as material cools)

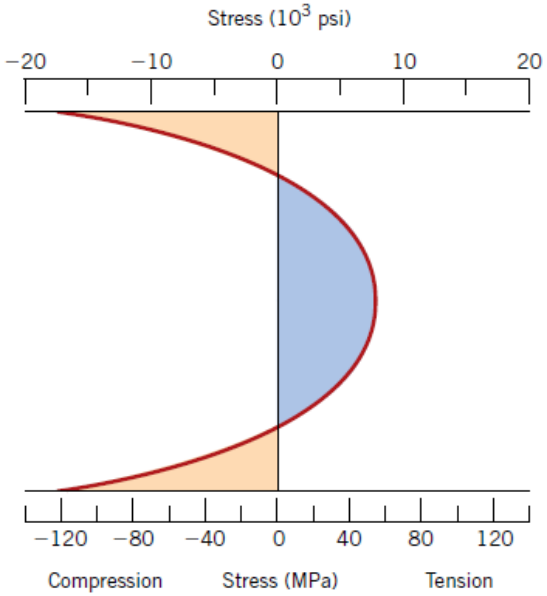


With further cooling, the interior attempts to contract to a higher degree than the rgid exterior will allow. Thus, the inside tends to draw in the outside. As a consequence, after cooled to room temperature, it sustains compressive stresses on the surface and tensile stresses at interior regions

--Result: surface crack growth is suppressed.

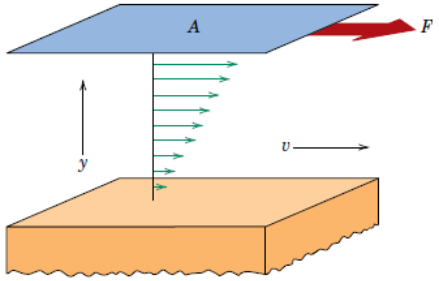


External stress must be high to overcome the residual stresses and make the crack grow



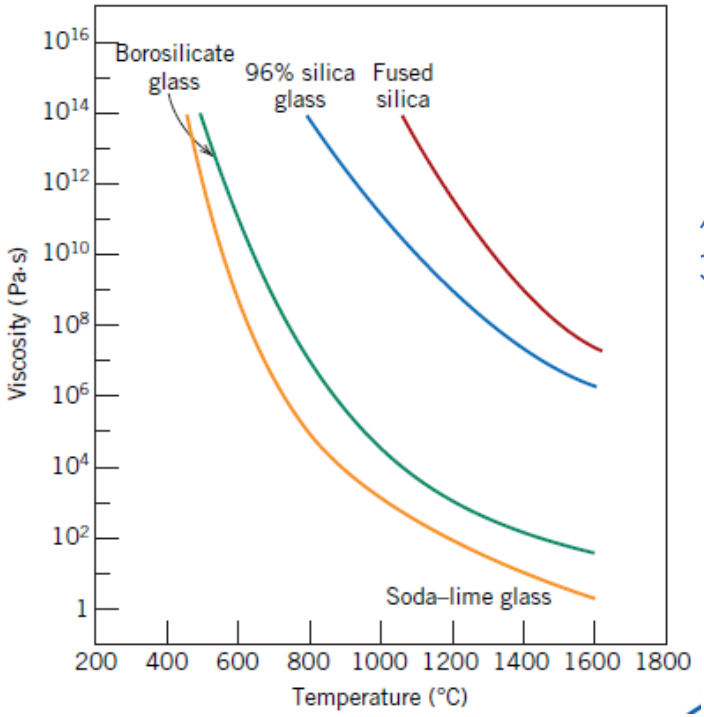
Tempered glass is used for applications in which high strength is important; these include large doors and eyeglass lenses.

Processing/Structure/Properties/Performance - Ceramics

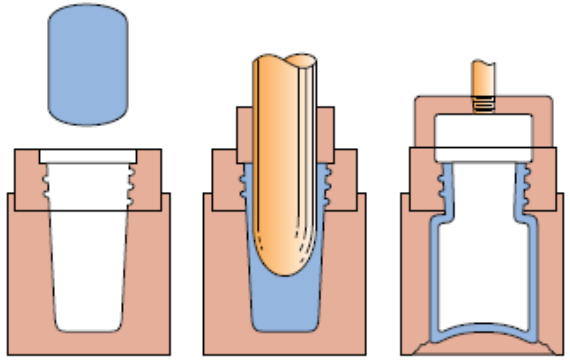


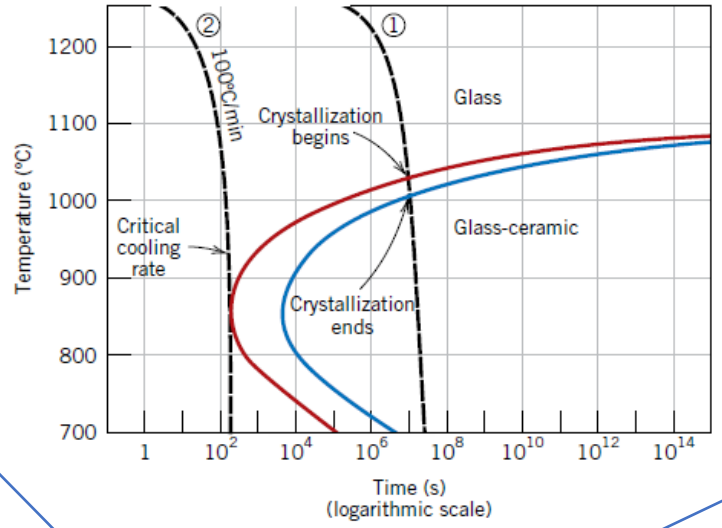
Viscosity

Temperature dependent viscosity



Processing / Structure





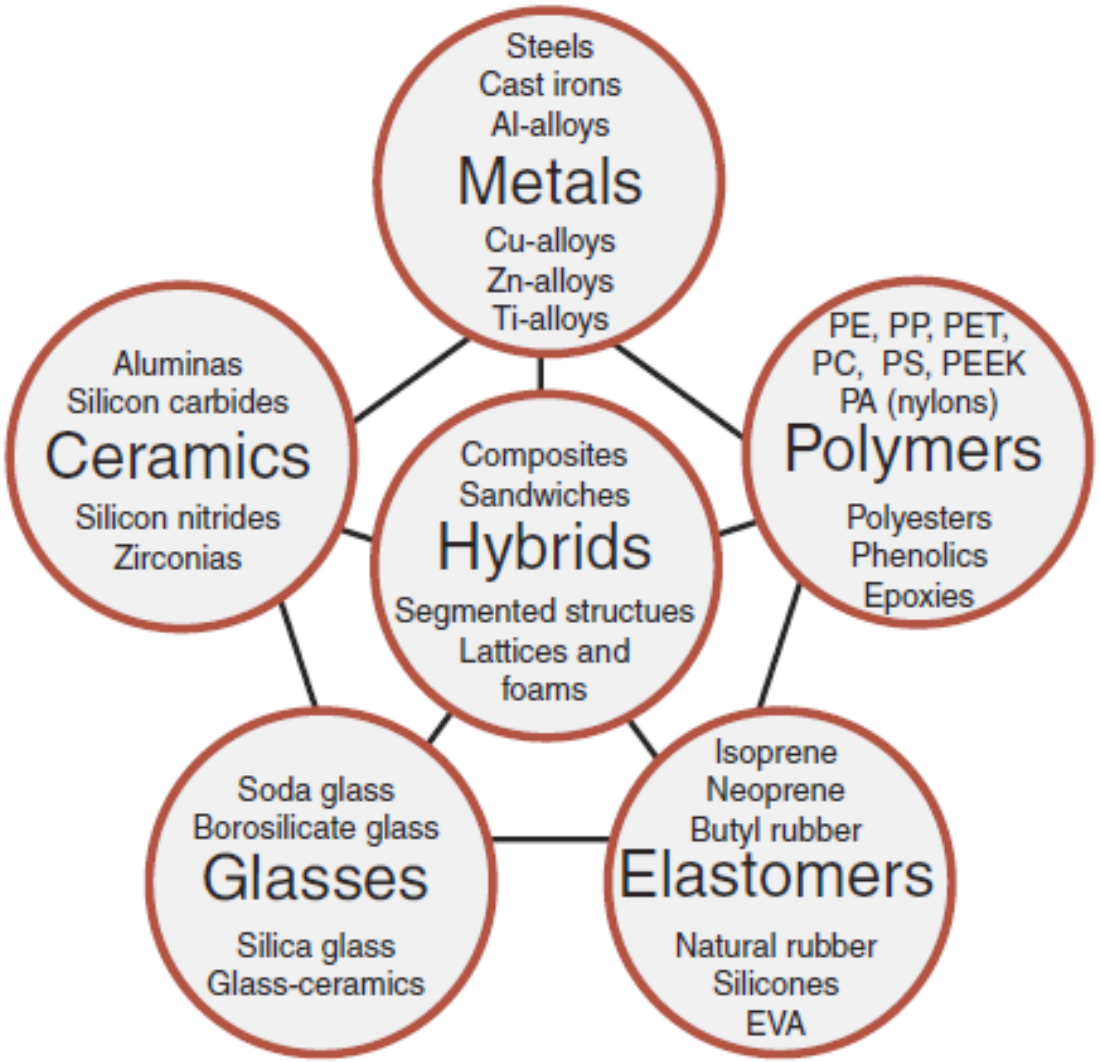
Properties / Performance

Mechanical Properties

Tempering of glass

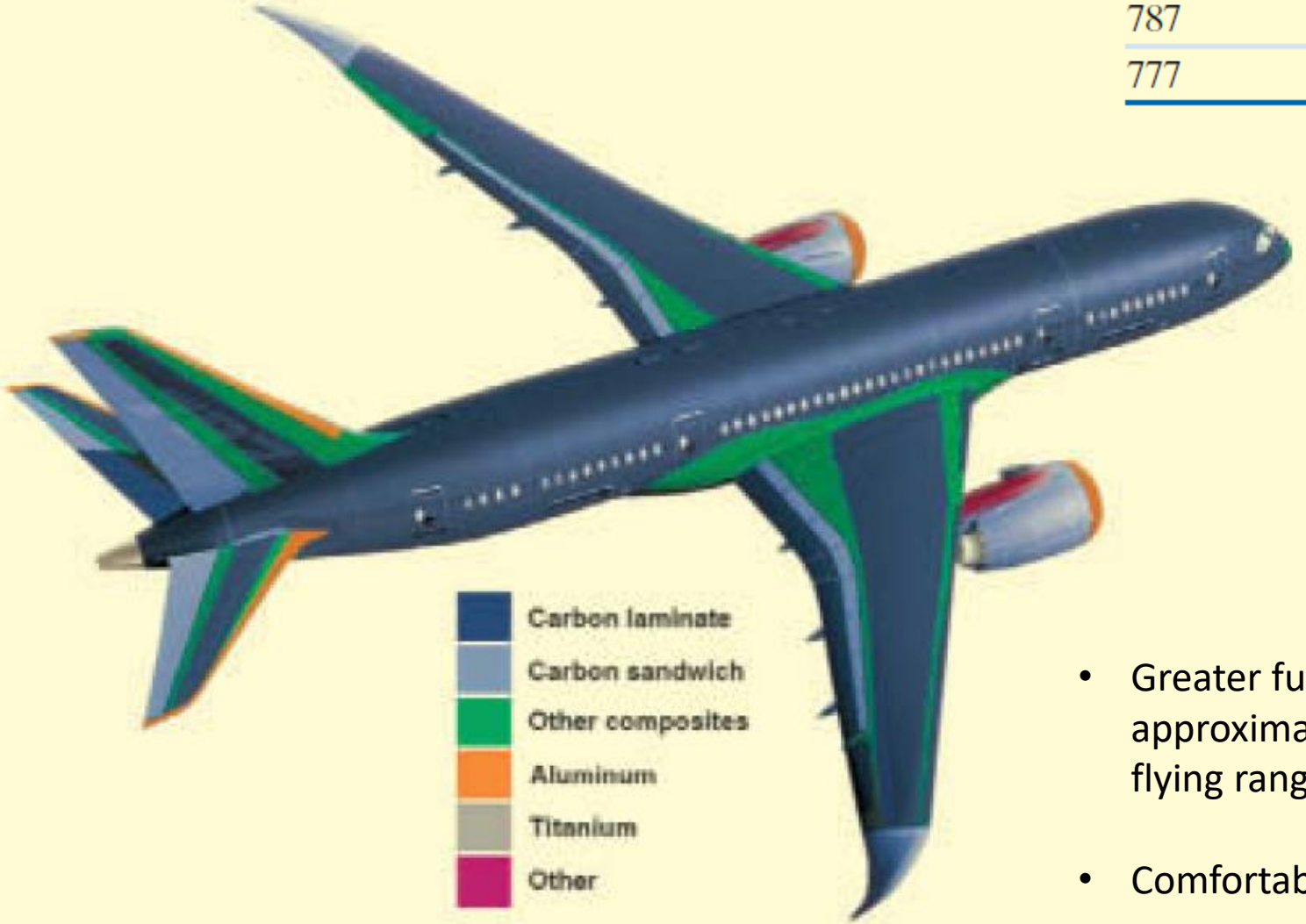
Composite Materials

Menu of Engineering Materials



Use of Composites in the Boeing 787 Dreamliner

Aircraft	Material Content (Weight Percent)				
	Composites	Al Alloys	Ti Alloys	Steel	Other
787	50	20	15	10	5
777	11	70	7	11	1



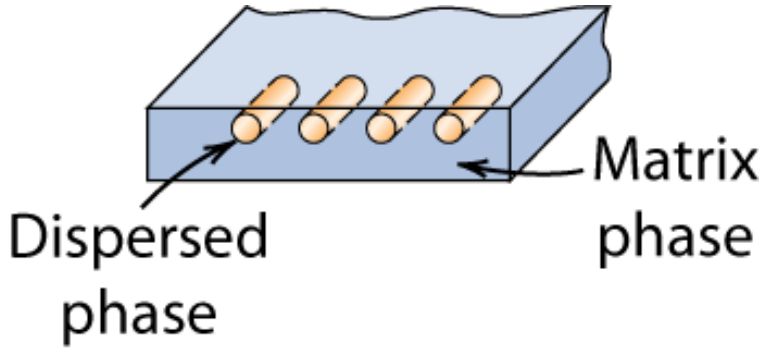
- Greater fuel efficiency (a reduction of approximately 20%), fewer emissions, and longer flying ranges.
- Comfortable flying experience

Composite

- Combination of two or more individual materials

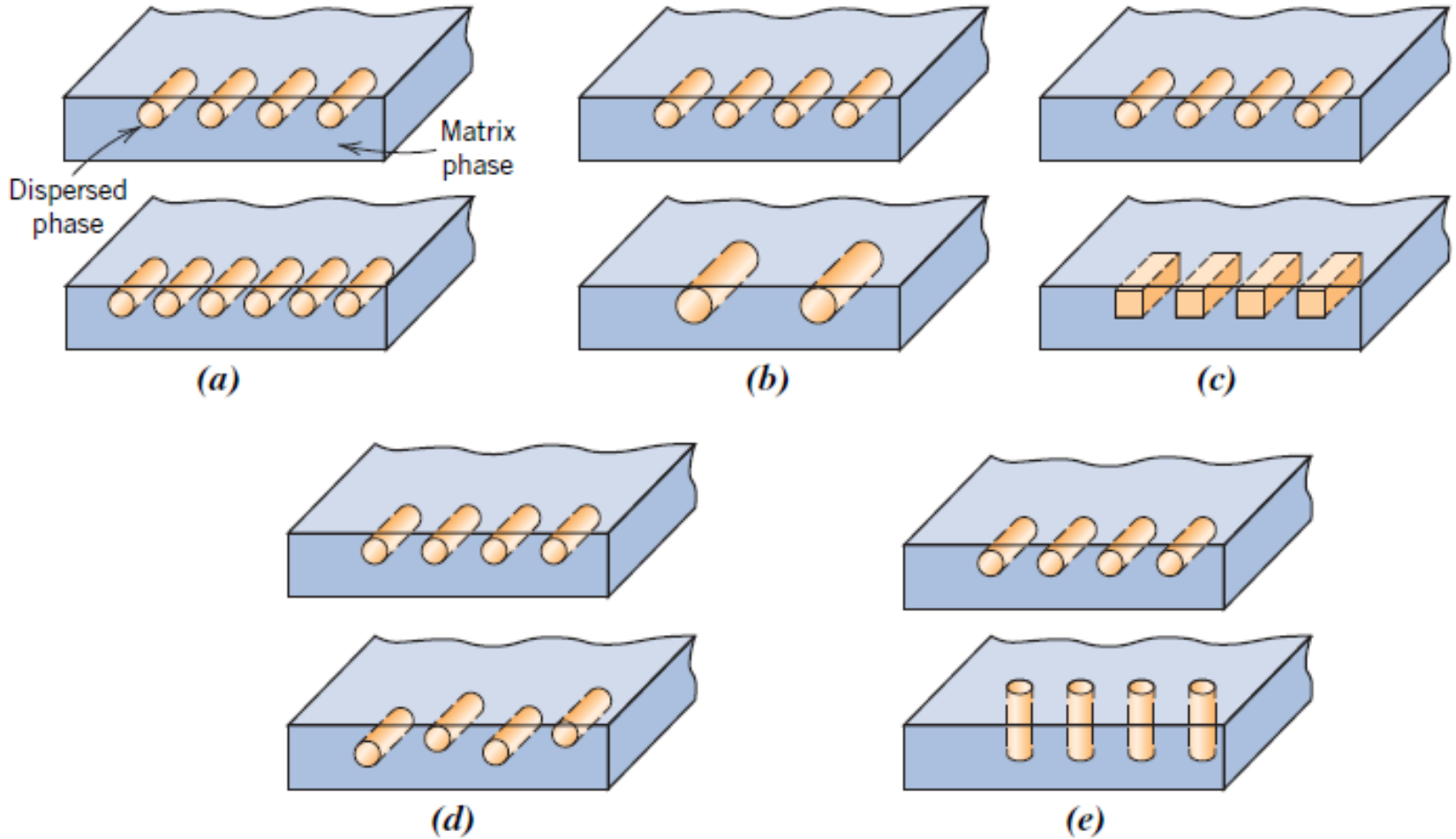
- Design goal: obtain a more desirable combination of properties (principle of combined action)
  - e.g., low density and high strength

- Phase types:
  -- Matrix - is continuous
  -- Dispersed - is discontinuous and surrounded by matrix

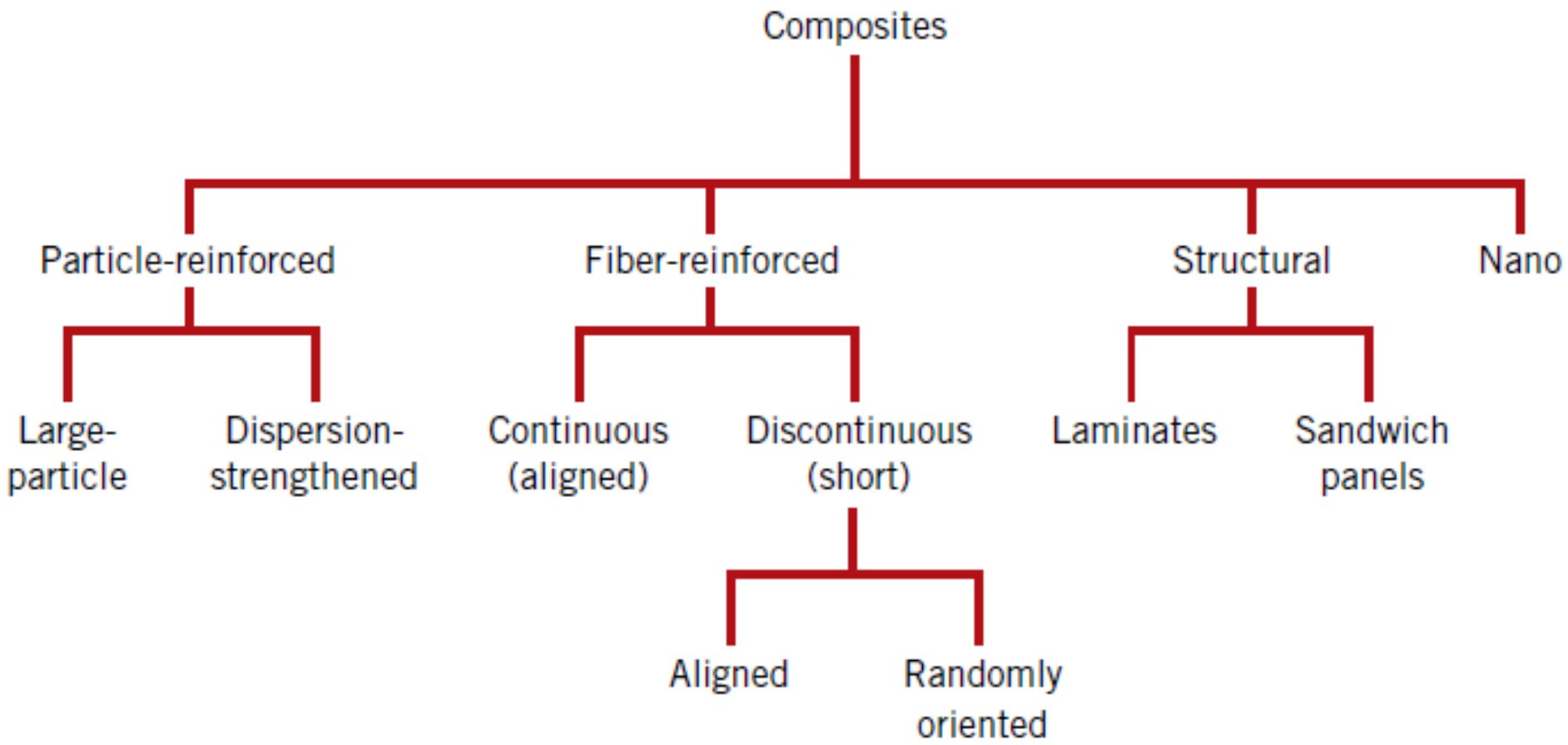


Geometrical and spatial characteristics of particles of the dispersed phase that may influence the properties of composites:

(*a) concentration, (b) size, (c) shape,* (*d) distribution, and (e) orientation.*



Classification of Composites – one simple way

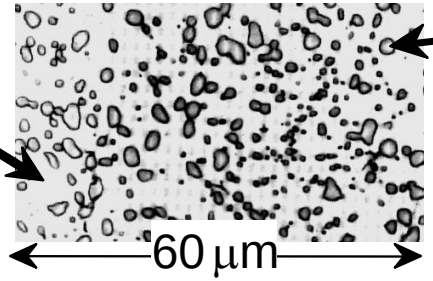


Classification: Particle-Reinforced (i)

Particle-reinforced | Fiber-reinforced | Structural

- Examples:
  - Spheroidite steel

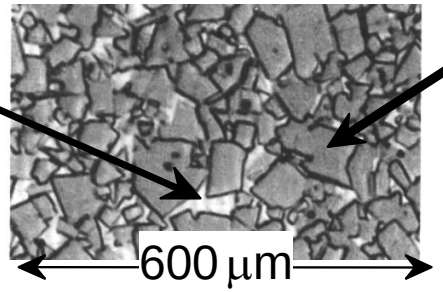
matrix: ferrite (α) (ductile)



particles: cementite (Fe_3C) (brittle)

  - WC/Co cemented carbide

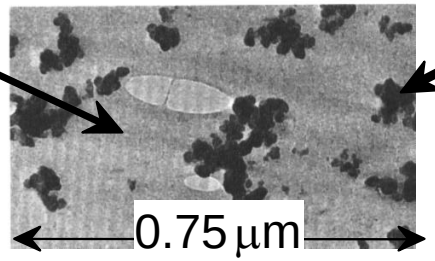
matrix: cobalt (ductile, tough)



particles: WC (brittle, hard)

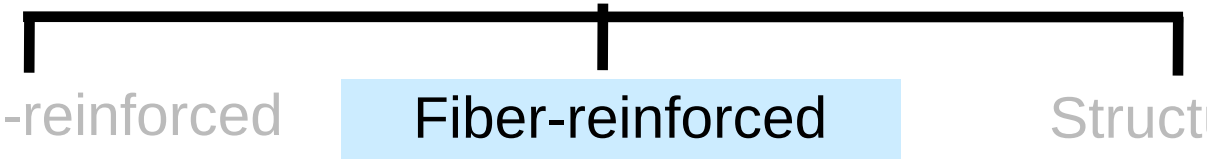
  - Automobile tire rubber

matrix: rubber (compliant)



particles: carbon black (stiff)

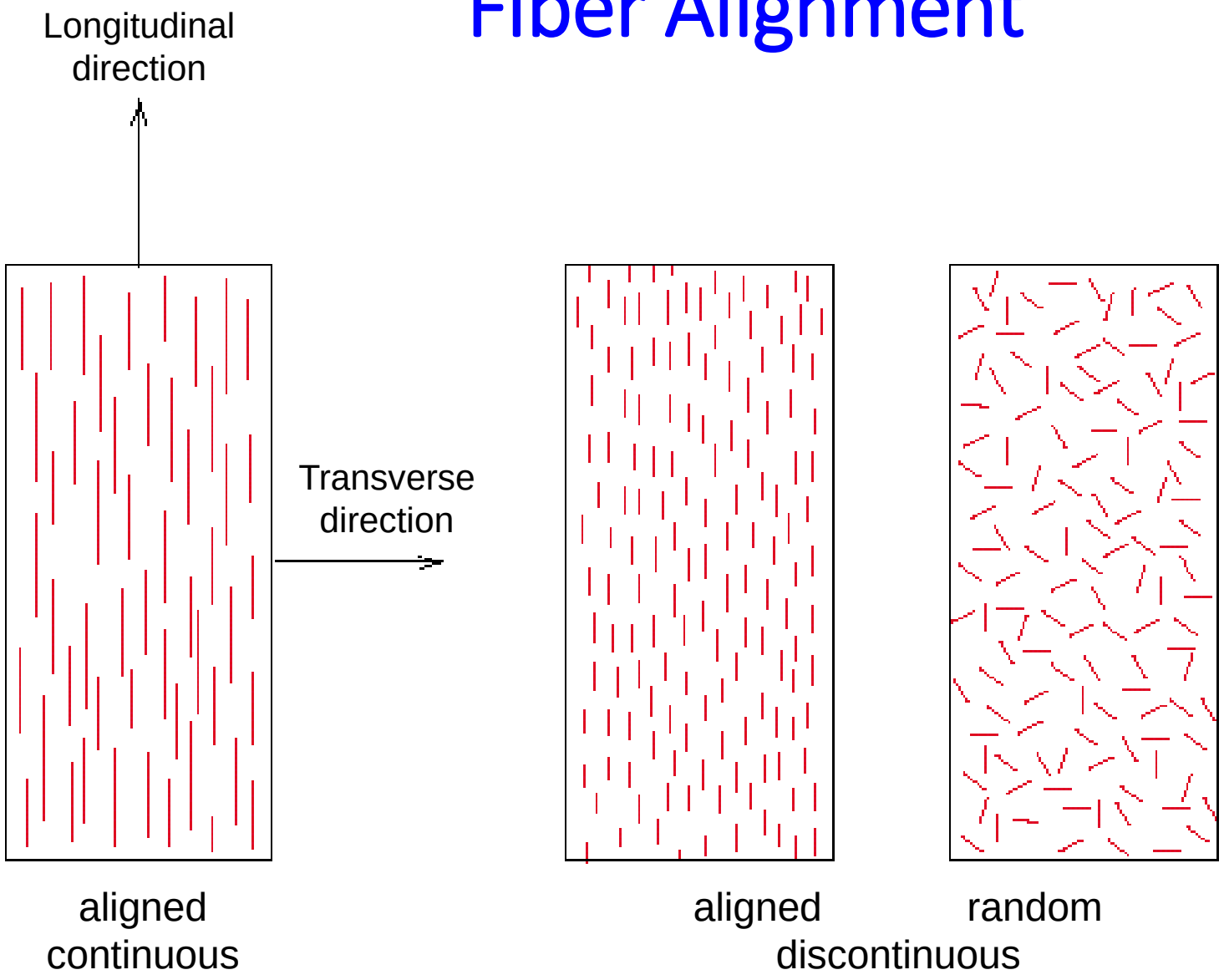
Classification: Fiber-Reinforced (i)



- Fibers very strong in tension
  - Provide significant strength improvement to the composite
  - Ex: fiber-glass - continuous glass filaments in a polymer matrix
    - Glass fibers
      - strength and stiffness
    - Polymer matrix
      - holds fibers in place
      - protects fiber surfaces
      - transfers load to fibers

- Fiber Types
  - Whiskers - thin - large length to diameter ratios
    - graphite, silicon nitride, silicon carbide
    - High perfection – extremely strong, strongest known
    - very expensive and difficult to disperse
  - Fibers
    - polycrystalline or amorphous
    - generally polymers or ceramics
    - Ex: alumina, aramid, E-glass, boron, UHMWPE
  - Wires
    - metals – steel, molybdenum, tungsten

Fiber Alignment



Polymer Matrix Composites (PMCs)

Example of properties for epoxy based composites

Property	*Glass (E-Glass)*	*Carbon (High Strength)*	*Aramid (Kevlar 49)*
Specific gravity	2.1	1.6	1.4
Tensile modulus			
Longitudinal [GPa (10^6 psi)]	45 (6.5)	145 (21)	76 (11)
Transverse [GPa (10^6 psi)]	12 (1.8)	10 (1.5)	5.5 (0.8)
Tensile strength			
Longitudinal [MPa (ksi)]	1020 (150)	1240 (180)	1380 (200)
Transverse [MPa (ksi)]	40 (5.8)	41 (6)	30 (4.3)
Ultimate tensile strain			
Longitudinal	2.3	0.9	1.8
Transverse	0.4	0.4	0.5

Applications:

Glass fiber polymer composites - automotive and marine bodies, plastic pipes, storage containers, and industrial floorings, infrastructure applications

Carbon fiber polymer composites - sports and recreational equipment (fishing rods, golf clubs), filament-wound rocket motor cases, pressure vessels, and aircraft structural components—both military and commercial, both fixed-wing aircraft and helicopters (e.g., as wing, body, stabilizer, and rudder components).

Aramid fiber composites – Bullterproof vests and armor), sporting goods, tires, ropes, missile cases, and as a replacement for asbestos in automotive brake and clutch linings and gaskets.

Metal Matrix Composites (MMCs)

Properties of several metal-matrix composites

Fiber	*Matrix*	*Fiber Content (vol%)*	*Density (g/cm^3)*	*Longitudinal Tensile Modulus (GPa)*	*Longitudinal Tensile Strength (MPa)*
Carbon	6061 Al	41	2.44	320	620
Boron	6061 Al	48	—	207	1515
SiC	6061 Al	50	2.93	230	1480
Alumina	380.0 Al	24	—	120	340
Carbon	AZ31 Mg	38	1.83	300	510
Borsic	Ti	45	3.68	220	1270

Compared to PMCs, the advantages of CMCs include higher operating temperatures, nonflammability, good electrical and thermal conductivities, and greater resistance to degradation by organic fluids.

Applications:

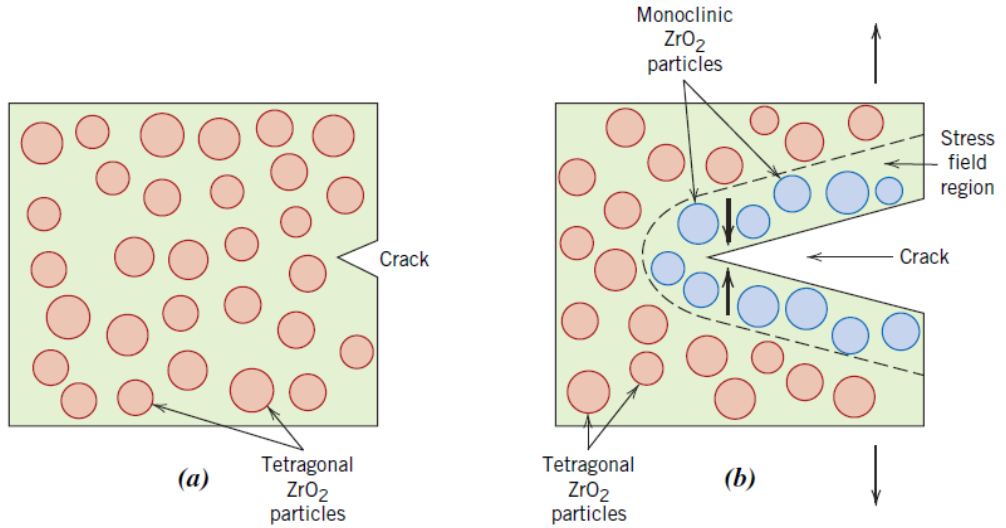
- Engine components (Ex: Aluminumalloy matrix that is reinforced with aluminum oxide and carbon fibers)
- Aerospace industry (Ex: Al alloy metal matrix composites)
- High temperature components in turbine engine (Ni and Co based superalloy composites reinforced with ceramics)

Ceramic Matrix Composites (CMCs)

Low fracture toughness values of ceramic materials (1 - 5 MPa.m$^{1/2}$) are significantly improved by addition of ceramic particulates, fibers, or whiskers. Thus CMCs have extended fracture toughnesses of about 6 – 20 MPa.m$^{1/2}$

Toughening mechanisms:

- Transformation toughening
- Deflecting crack tips
- Forming bridges across crack faces
- Absorbing energy during pullout as the whiskers debond from the matrix



Room Temperature Fracture Strengths and Fracture Toughnesses for Various SiC Whisker Contents in Al_2O_3

Whisker Content (vol%)	*Fracture Strength (MPa)*	*Fracture Toughness (MPa$\sqrt{m}$)*
0	—	4.5
10	455 ± 55	7.1
20	655 ± 135	7.5–9.0
40	850 ± 130	6.0

Applications:

- C/SiC or SiC/SiC composites in aerojet or rocket engines
- CMC braking systems (**C/C for aircraft** and C/C – SiC for cars)

Basic Properties

Fiber volume ratio: $$V_f = \frac{\text{volume of fibers}}{\text{volume of composite}}$$

Fiber weight ratio: $$W_f = \frac{\text{weight of fibers}}{\text{weight of composite}}$$

Matrix volume ratio: $$V_m = \frac{\text{volume of matrix}}{\text{volume of composite}}$$

Matrix weight ratio: $$W_m = 1 - W_f = \frac{\text{weight of matrix}}{\text{weight of composite}}$$

Void volume ratio: $$V_v = 1 - V_f - V_m = \frac{\text{volume of voids}}{\text{volume of composite}}$$

Density of Composite

$$v_c = v_f + v_m$$

$V_f = \frac{v_f}{v_c}$ and

$V_m = \frac{v_m}{v_c}$

v_c - volume of composite material

v_f - volume of fiber

v_m - volume of matrix

$$V_f + V_m = 1$$

$$w_c = w_f + w_m$$

$W_f = \frac{w_f}{w_c}$ and

$W_m = \frac{w_m}{w_c}$

w_c - weight of composite material

w_f - weight of fiber

w_m - weight of matrix

$$W_f + W_m = 1$$

$$v_c = v_f + v_m \quad \text{(Eq. 1)}$$

Since $v = \frac{w}{\rho}$

Eq. 1 can be written as,

$$\frac{w_c}{\rho_c} = \frac{w_f}{\rho_f} + \frac{w_m}{\rho_m}$$

$$\frac{1}{\rho_c} = \frac{1}{\rho_f}\left(\frac{w_f}{w_c}\right) + \frac{1}{\rho_m}\left(\frac{w_m}{w_c}\right)$$

$$\frac{1}{\rho_c} = \frac{W_f}{\rho_f} + \frac{W_m}{\rho_m}$$

$$w_c = w_f + w_m \quad \text{(Eq. 2)}$$

Since $w = \rho v$

Eq. 2 can be written as,

$$\rho_c v_c = \rho_f v_f + \rho_m v_m$$

$$\rho_c = \rho_f\left(\frac{v_f}{v_c}\right) + \rho_m\left(\frac{v_f}{v_c}\right)$$

$$\rho_c = \rho_f V_f + \rho_m V_m$$

Fiber Volume and Weight Fraction of Composite

$$W_f = \frac{w_f}{w_c} = \frac{\rho_f v_f}{\rho_c v_c} = V_f \frac{\rho_f}{\rho_c}$$ (Weight fraction in terms of volume fraction)

$$V_f = \frac{v_f}{v_c} = \frac{w_f/\rho_f}{w_c/\rho_c} = W_f \frac{\rho_c}{\rho_f}$$ (Volume fraction in terms of weight fraction)

Density of composite

The density of a composite made from unidirectional glass fibres in an epoxy matrix is 1950 kg/m^3. If the densities of the glass and epoxy are known to be 2540 kg/m^3 and 1300 kg/m^3, calculate the weight fraction of fibres in the composite.

Density of composite

The density of a composite made from unidirectional glass fibres in an epoxy matrix is 1950 kg/m^3. If the densities of the glass and epoxy are known to be 2540 kg/m^3 and 1300 kg/m^3, calculate the weight fraction of fibres in the composite.

From the rule of mixtures

$$\rho_c = \rho_f V_f + \rho_m V_m = \rho_f V_f + \rho_m (1 - V_f)$$

$$1950 = 2540 \cdot V_f + 1300 \cdot (1 - V_f)$$

$$V_f = 0.52$$

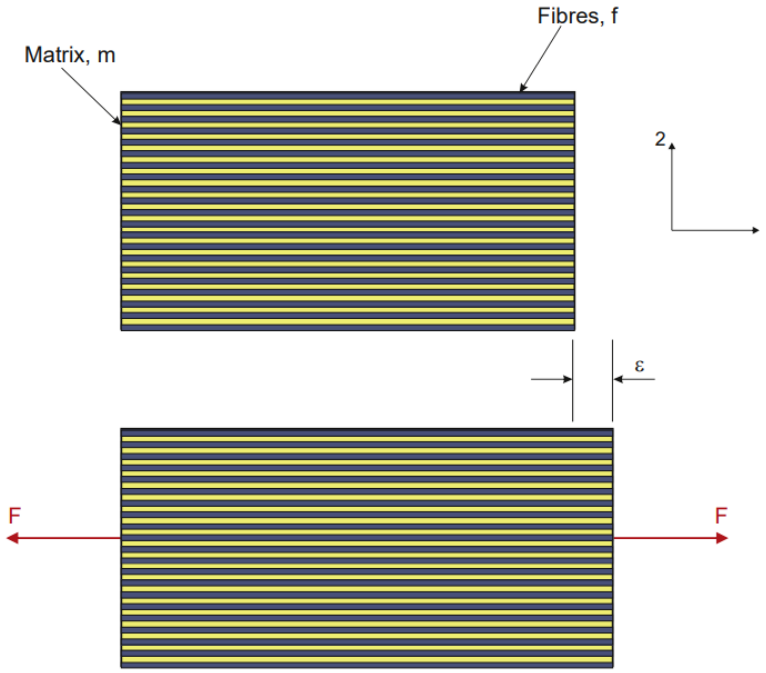
$$W_f = \frac{\rho_f}{\rho_c} V_f = \frac{2540}{1950} (0.52) = 0.68$$

Composite Modulus:

Continuous fibers - Estimate fiber-reinforced composite modulus of elasticity for continuous fibers

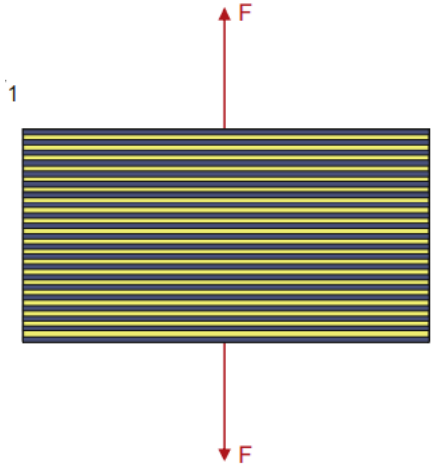
$E_{cl} = E_m V_m + E_f V_f$

E_{cl} = longitudinal modulus



E = Modulus
V = Volume fraction

c = composite
f = fiber
m = matrix



$$\frac{1}{E_{ct}} = \frac{V_m}{E_m} + \frac{V_f}{E_f}$$

$$E_{ct} = \frac{E_m E_f}{V_m E_f + V_f E_m}$$

E_{ct} = transverse modulus

Modulus of composite

PEEK is to be reinforced with 30% by volume of unidirectional carbon fibres and the properties of the individual materials are given below. Calculate the density, modulus of the composite in the fibre and perpendicular to fibre direction.

Material	Density (kg/m^3)	Tensile strength (GPa)	Modulus (GPa)
PEEK	1300	0.058[†]	3.8
Carbon fibre (HM)	1800	2.1	400

Modulus of composite

PEEK is to be reinforced with 30% by volume of unidirectional carbon fibres and the properties of the individual materials are given below. Calculate the density, modulus of the composite in the fibre and perpendicular to fibre direction.

Material	Density (kg/m^3)	Tensile strength (GPa)	Modulus (GPa)
PEEK	1300	0.058†	3.8
Carbon fibre (HM)	1800	2.1	400

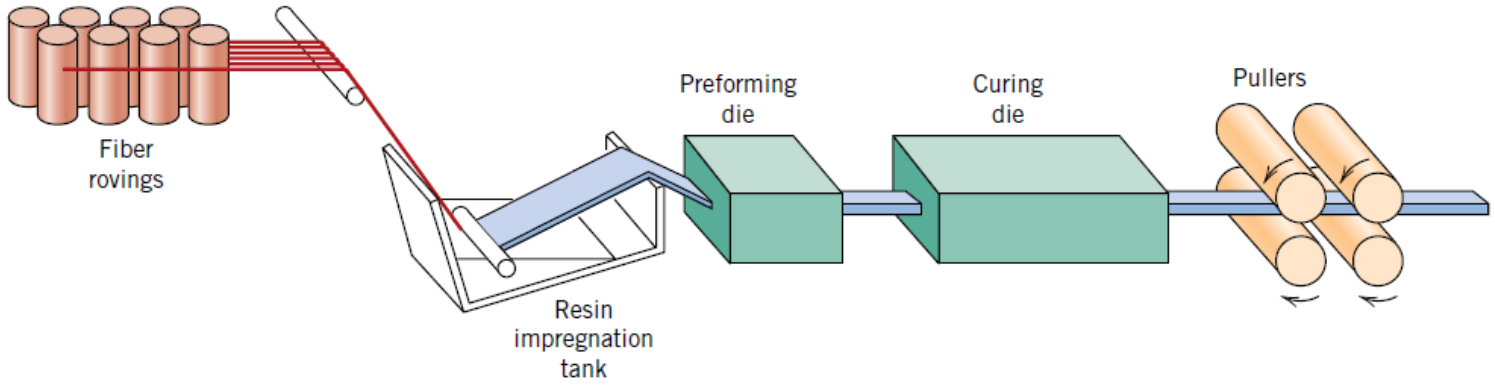
$$\rho_c = \rho_f V_f + \rho_m V_m = 0.13(1800) + 0.7(1300) = 1450 \text{ kg/m}^3$$

$$E_{cL} = E_f V_f + E_m V_m = 0.3(400) + 0.7(3.8) = 122.7 \text{ GPa}$$

$$E_2 = \frac{3.8(400)}{0.3(3.8) + 0.7(400)} = 5.4 \text{ GPa}$$

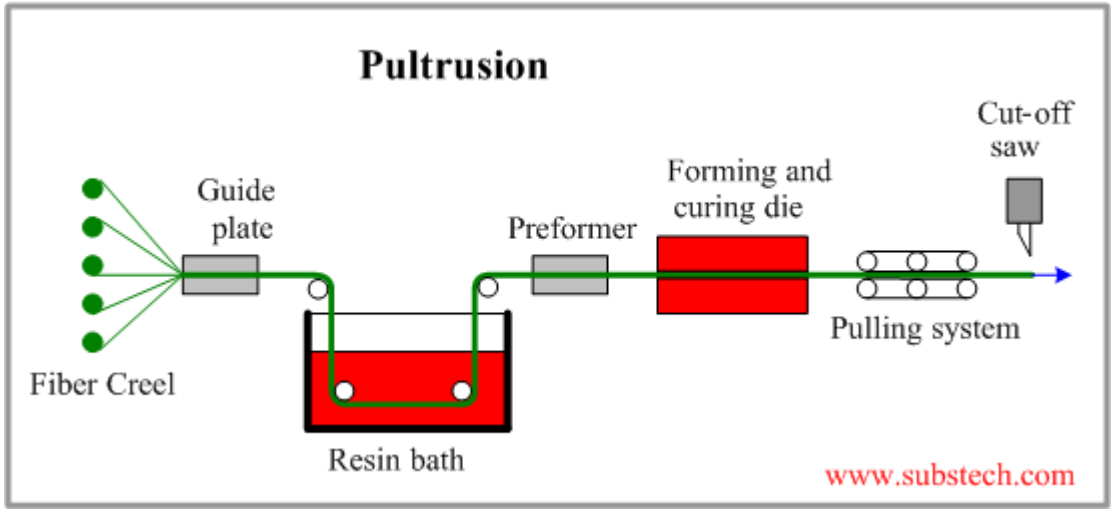
Processing of Composites

Pultrusion



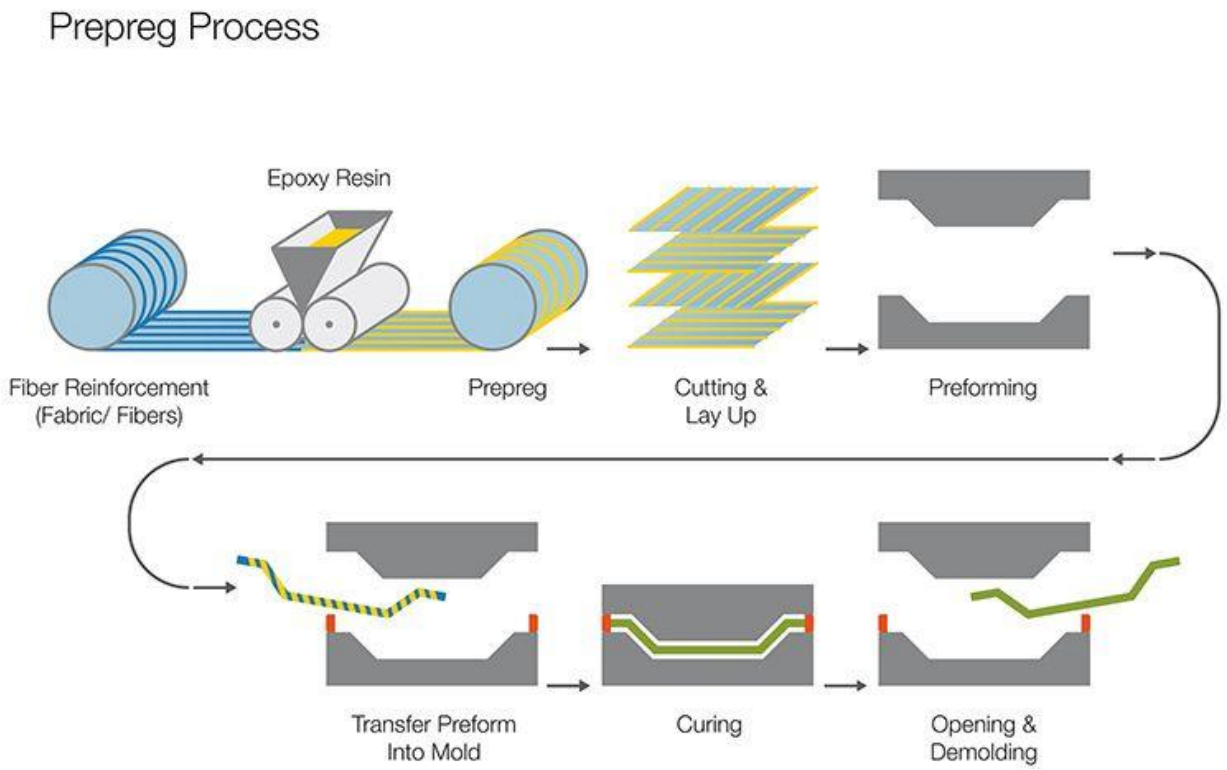
- Used for components having continuous lengths and a constant cross-sectional shape (rods, tubes, beams, etc.).

- Tubes and hollow sections are made possible by using center mandrels or inserted hollow cores.

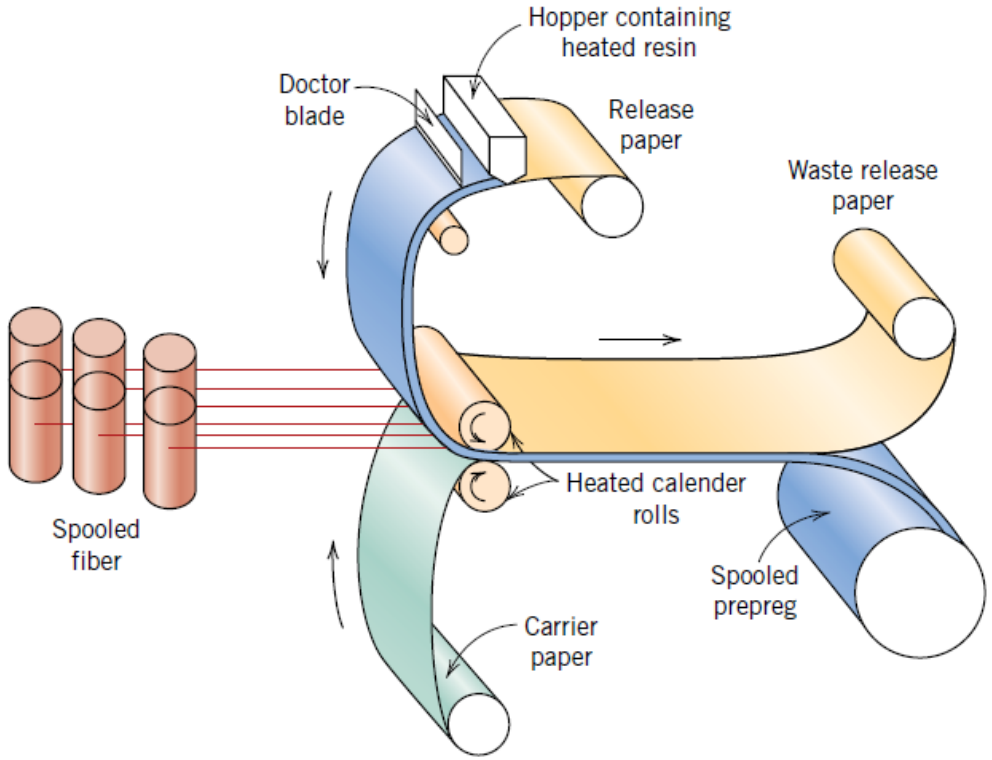


Processing of Composites

- **Prepreg** is the composite industry's term for continuous-fiber reinforcement pre-impregnated with a polymer resin that is only partially cured

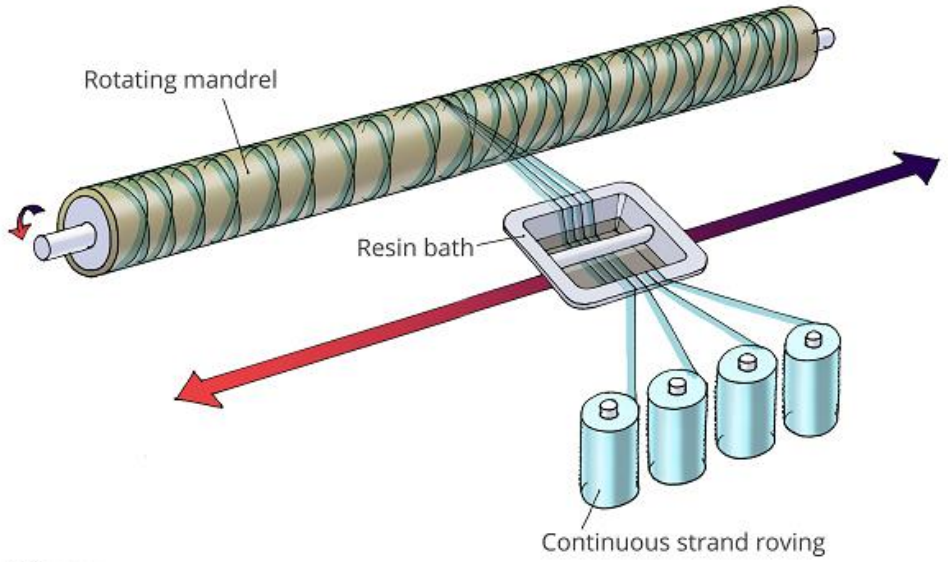


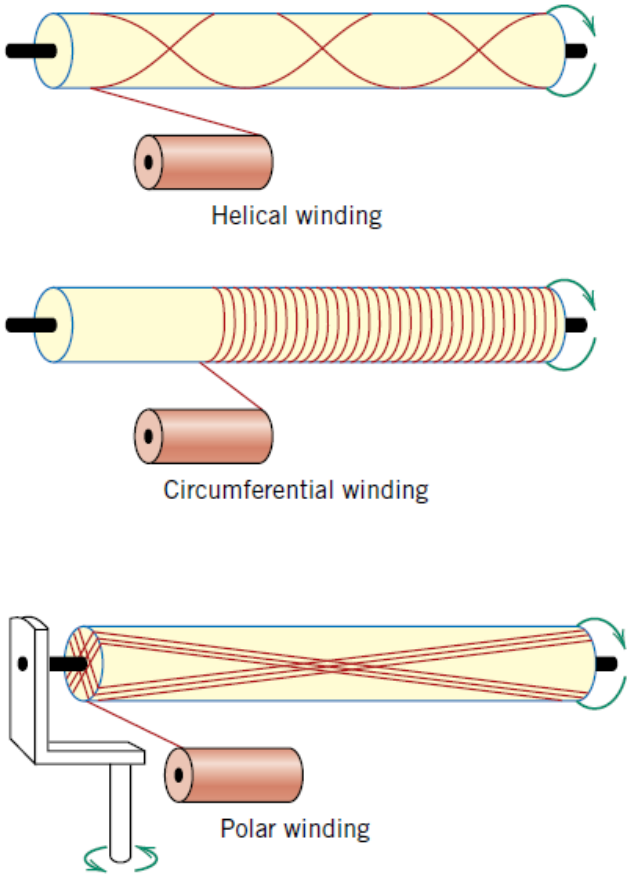
Prepreg Production Processes



Processing of Composite

Filament Winding





© Aliancys

- Usually used for cylindrical shape components (hollow)
- Not necessarily limited to surfaces of revolution

Polymer Matrix Composites

Compositions influence the type of processing and resulting microstructure / properties……

Possible Compositions

Fiber reinforcements

Inorganic

Glass
Boron/tungsten wire
Silicon carbide

Organic

Aramid (Kevlar)
Carbon
Graphite

Fillers

Powder
Silica
Carbon
Microballoon
Phenolic
Carbon
Glass

Matrix materials

Thermoplastic

Polyester
Polyamide
Polysulfone

Thermoset

Epoxy
Phenolic
Polyester
Polyimide
Bismaleimide
Pitch

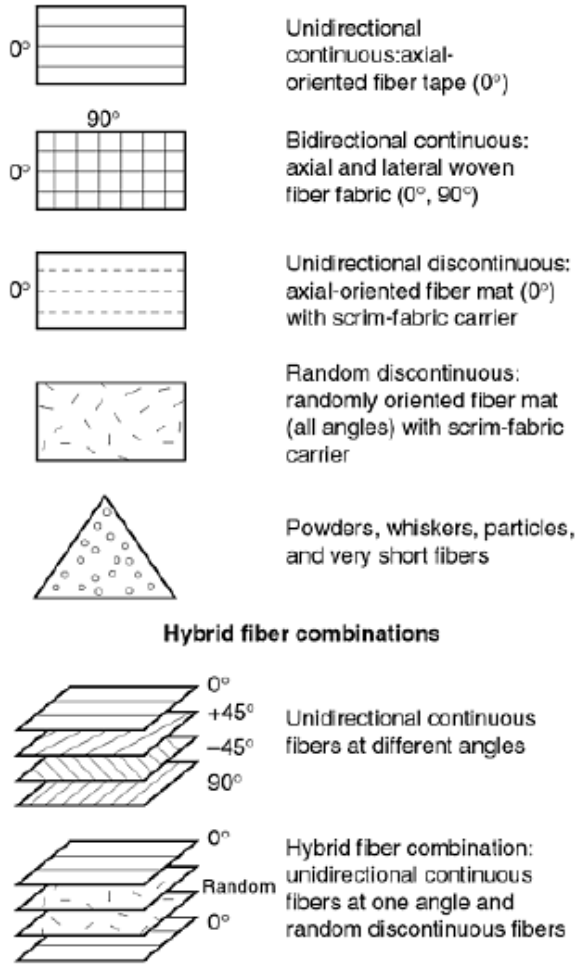
Metal

Stainless steel alloy
Aluminum alloy
Titanium alloy

Carbon

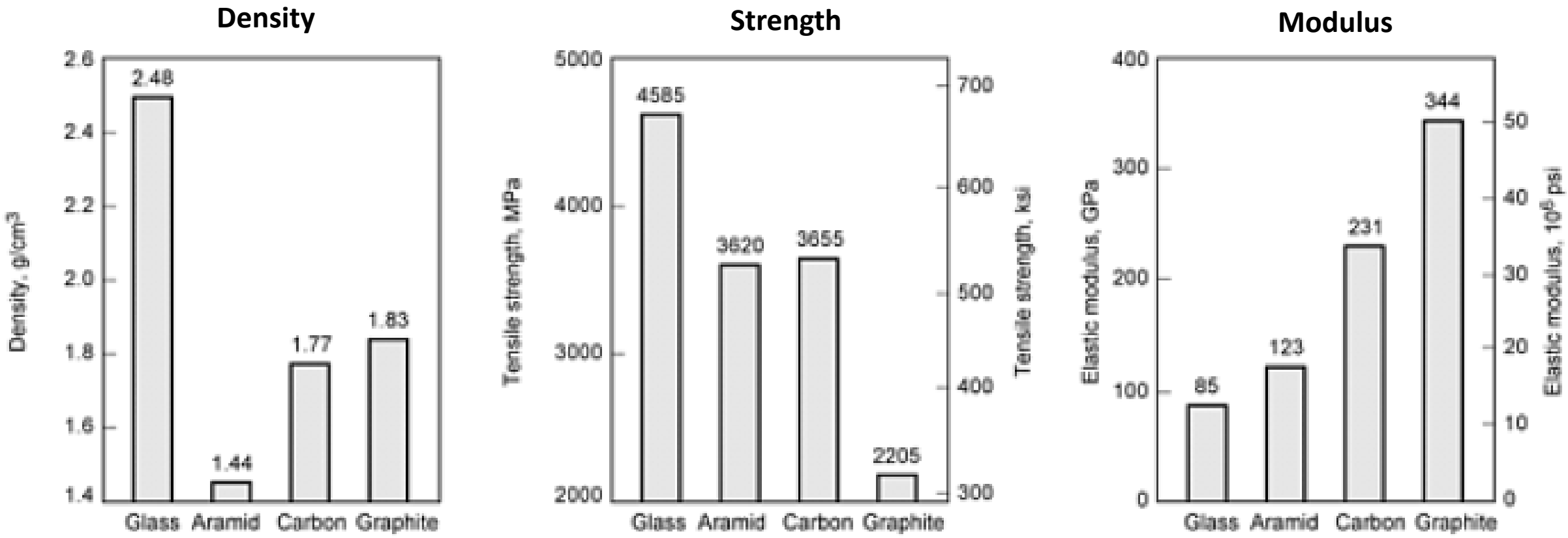
Carbonized resin
CVD carbon or graphite
Carbon powder

Possible Reinforcement Forms



Polymer Matrix Composites

Compositions *influence the type of processing and resulting microstructure / properties……*



Mechanical properties and service temperatures for selected reinforcement fibers

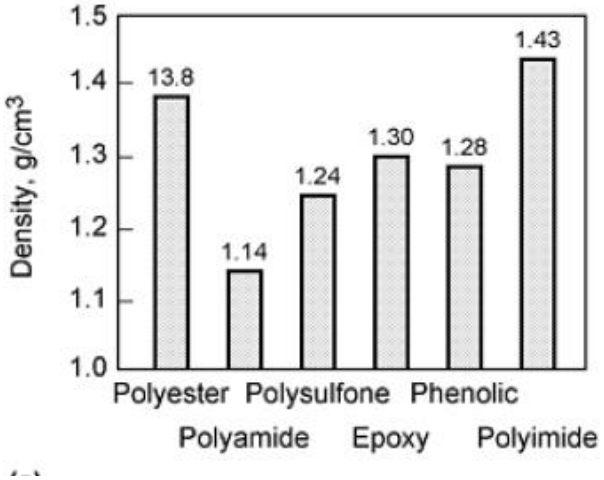
Inorganic fibers: glass (maximum temperature 970 °C) and aramid (maximum temperature 500 °C).

Organic fibers: carbon (maximum temperature 2500 °C) and graphite (maximum temperature 3000 °C

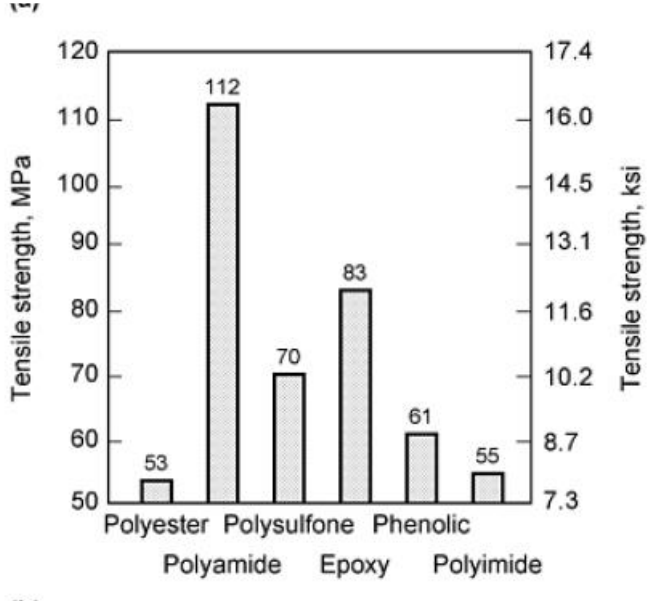
Polymer Matrix Composites

Compositions *influence the type of processing and resulting microstructure and properties……..*

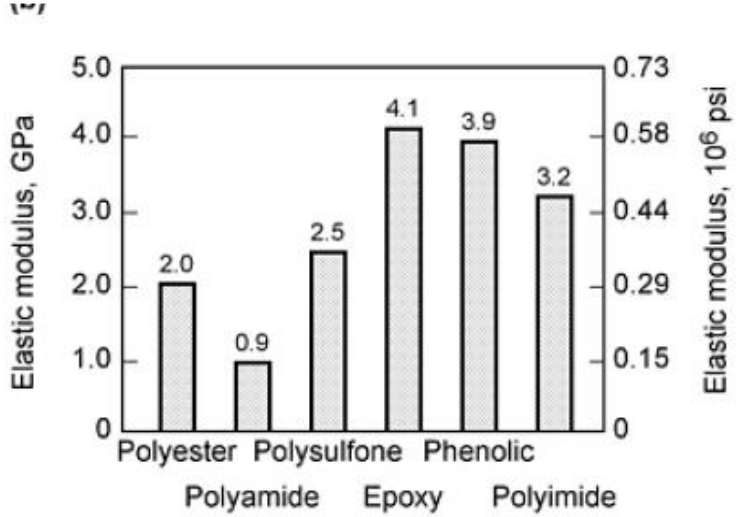
Density



Strength



Modulus



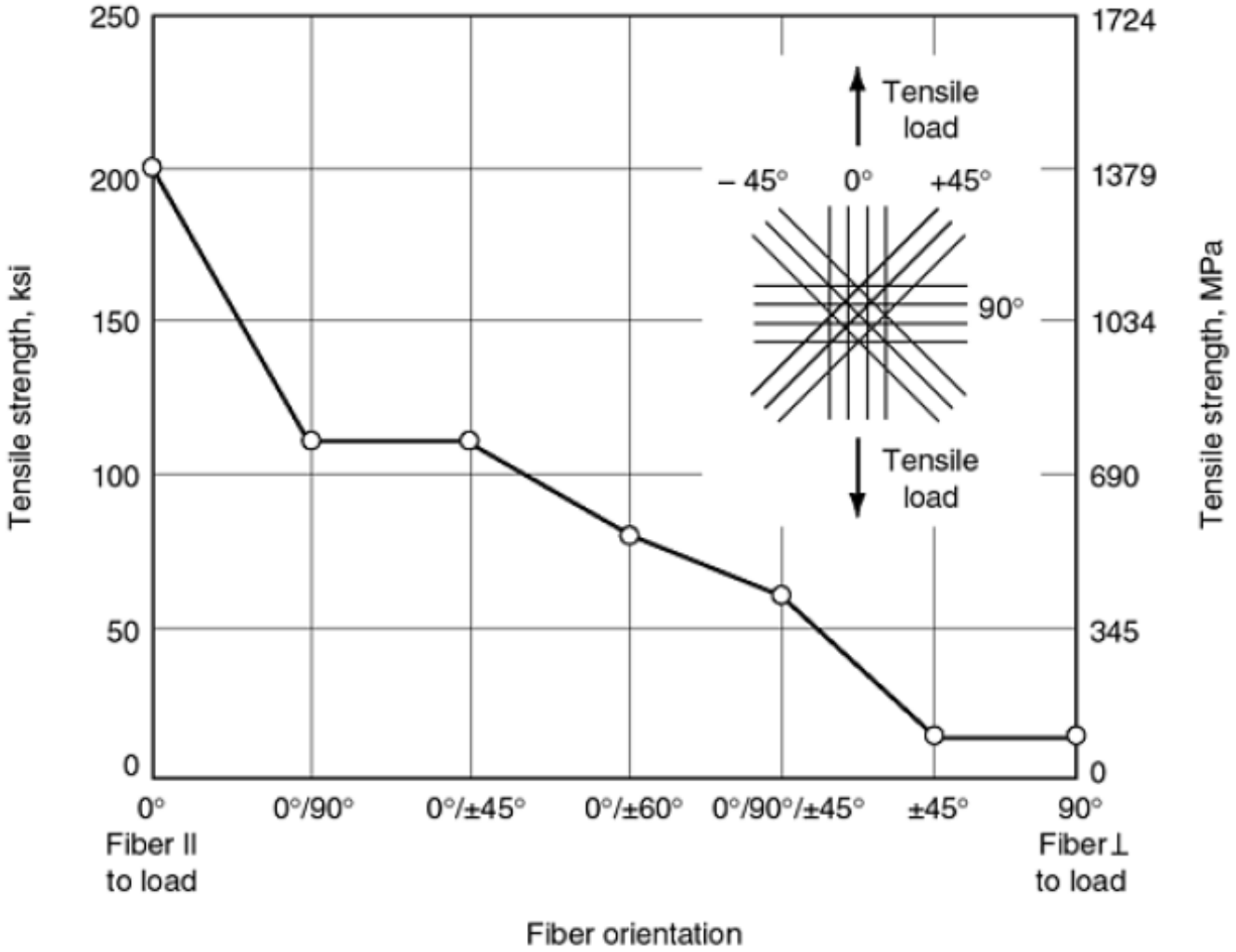
Mechanical properties and service temperatures for selected matrix resins

Thermoplastics: polyester (unfilled; maximum temperature 140 °C), polyamide (nylon 6/6, unfilled; maximum temperature 130 °C), and polysulfone (standard; maximum temperature 160 °C).

Thermosets: epoxy (unfilled; maximum temperature 260 °C), phenolic (unfilled; maximum temperature 230°C) and polyimide (unfilled; maximum temperature 370 °C)

Polymer Matrix Composites

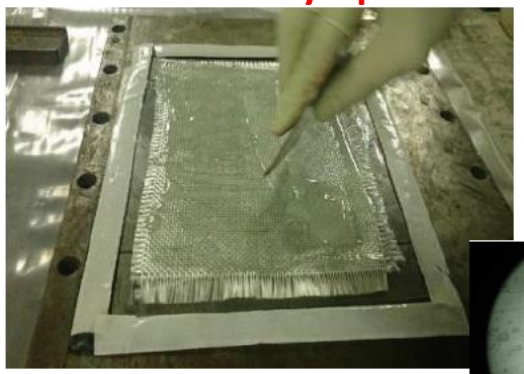
Forms of composition *influences microstructure and properties……Effect of fiber orientation on the strength of carbon-fiber/epoxy composites is shown in the Figure.*



Polymer Matrix Composites

Type of processing influences the microstructure and properties – Glass / epoxy composites

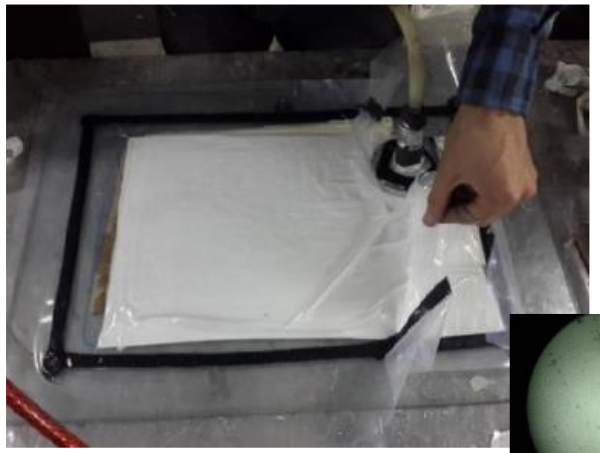
Hand lay-up



More air bubbles

Microstructure

Vacuum bagging



Less air bubbles

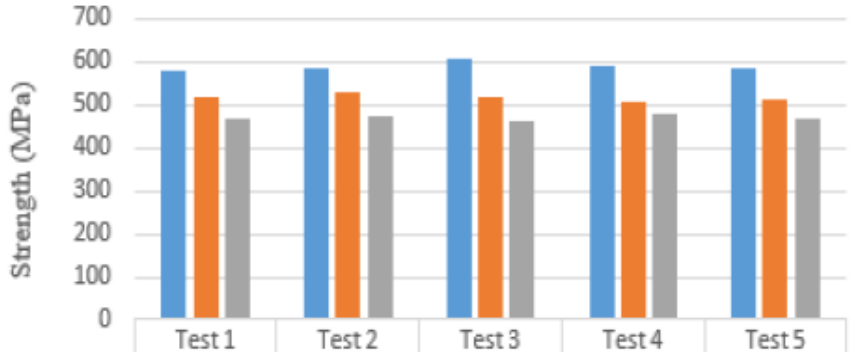
Microstructure

Vacuum infusion of resin

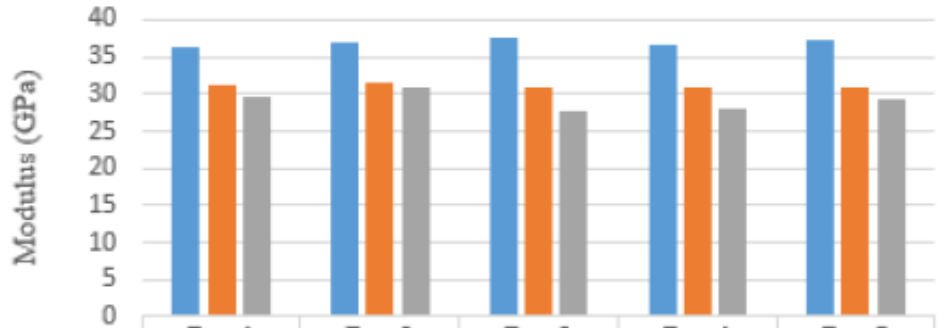
No air bubbles

Microstructure

Properties →

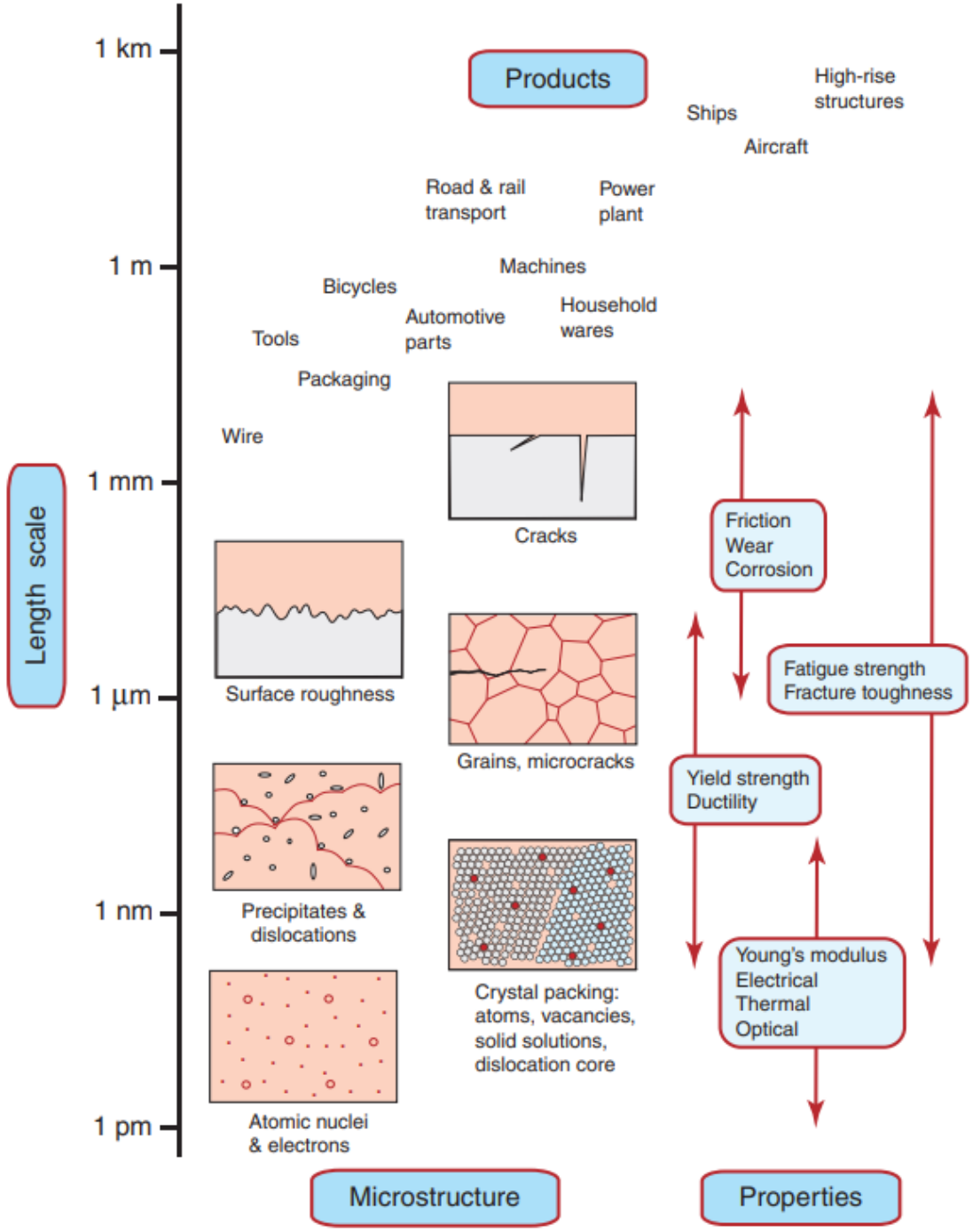


	Test 1	Test 2	Test 3	Test 4	Test 5
Vacuum Infusion	578.3	586.131	605.429	590.2	588.045
Vacuum Bag	521.046	526.9	518.928	507.981	515.212
Hand Lay up	468.356	473.171	465.319	478.601	470.824

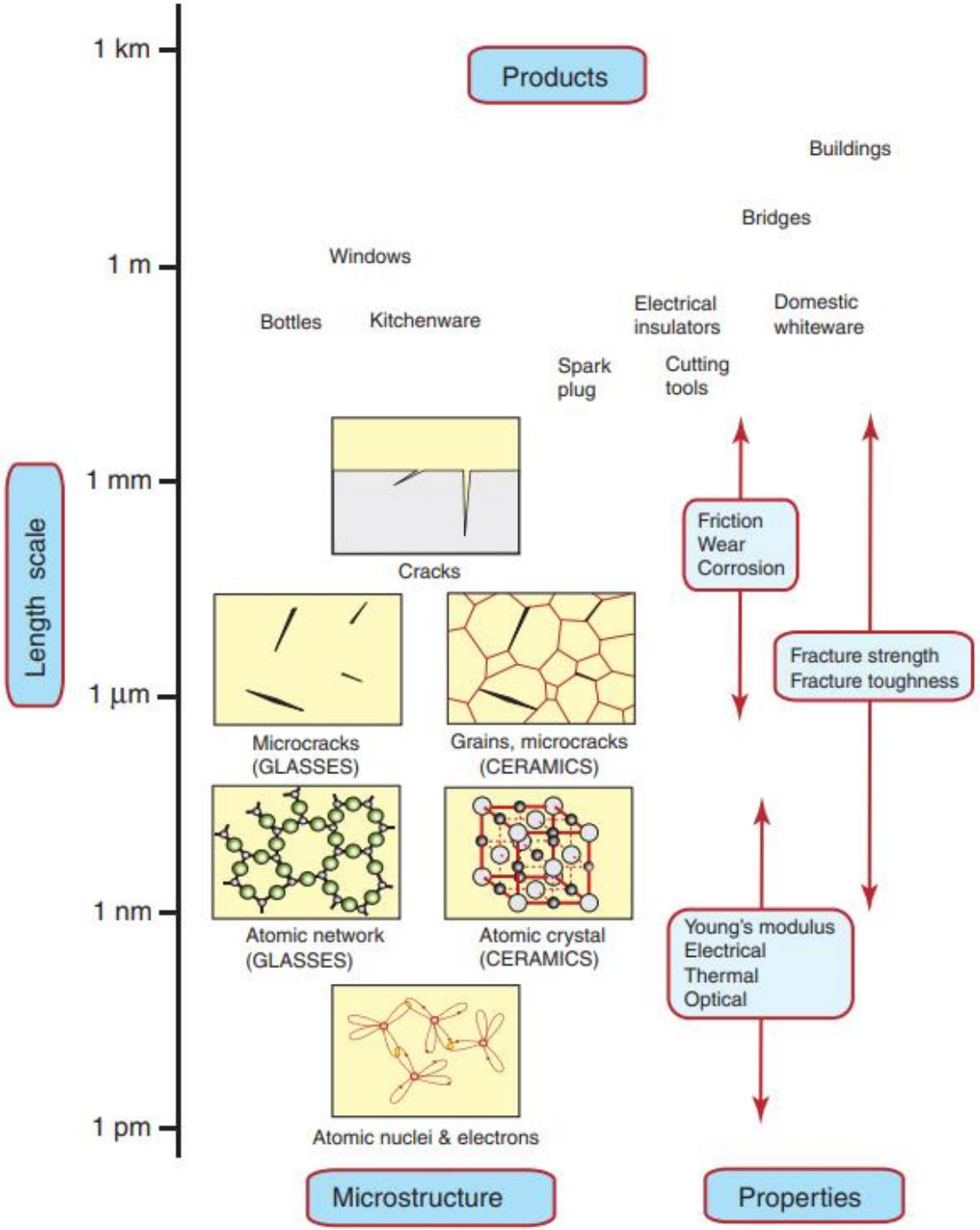


	Test 1	Test 2	Test 3	Test 4	Test 5
Vacuum Infusion	36.224	36.89	37.531	36.719	37.327
Vacuum Bag	31.227	31.653	30.792	30.944	31.054
Hand Lay up	29.719	30.901	27.828	28.065	29.273

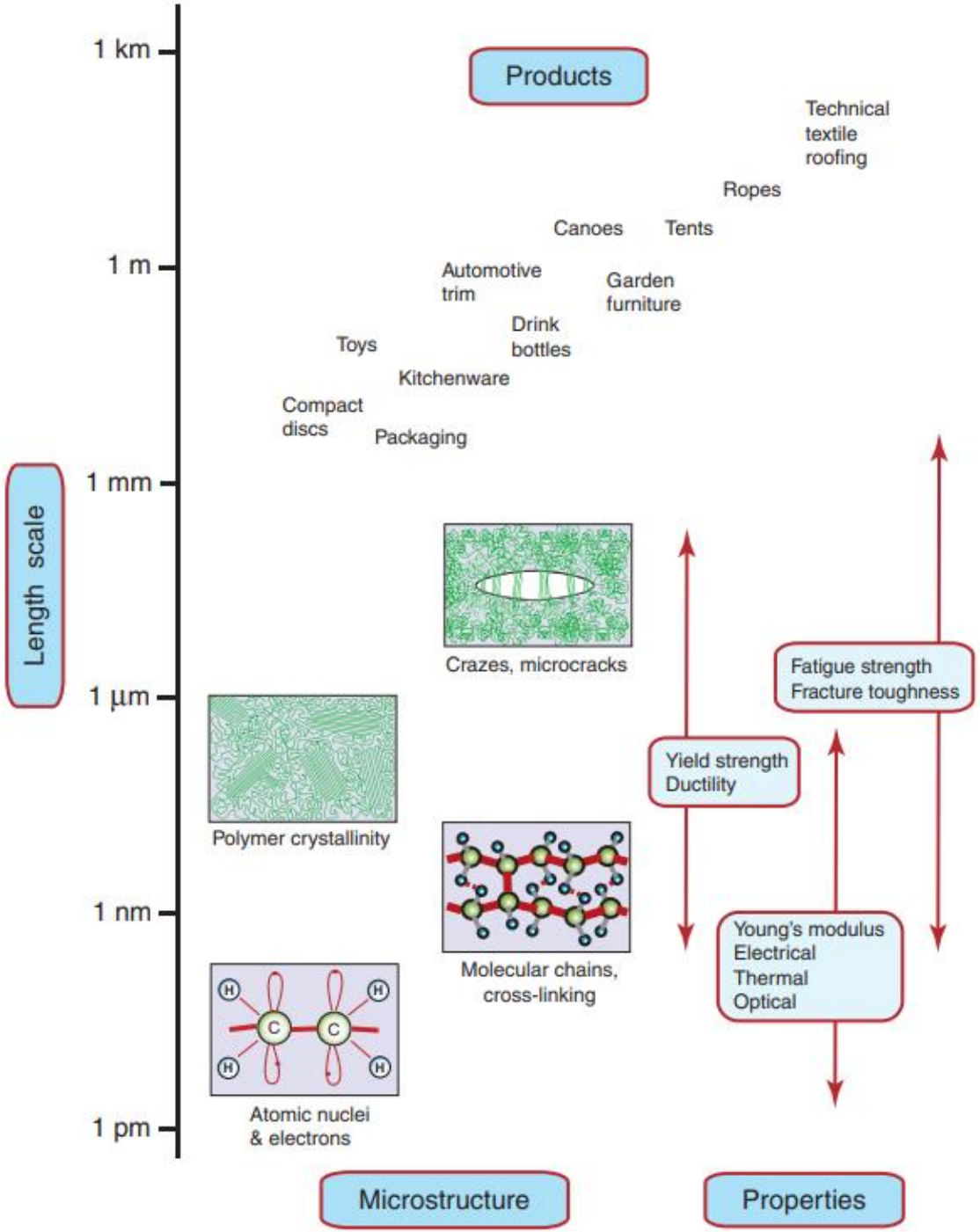
Microstructure and properties - Metals



Microstructure and properties - Ceramics



Microstructure and properties - Polymers



Materials for Engineers

Lecture until Nov. 22

Dr S. Gowthaman
Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

Reference:

1. Materials science and engineering by William Callister, Wiley Publications

Nanocomposite

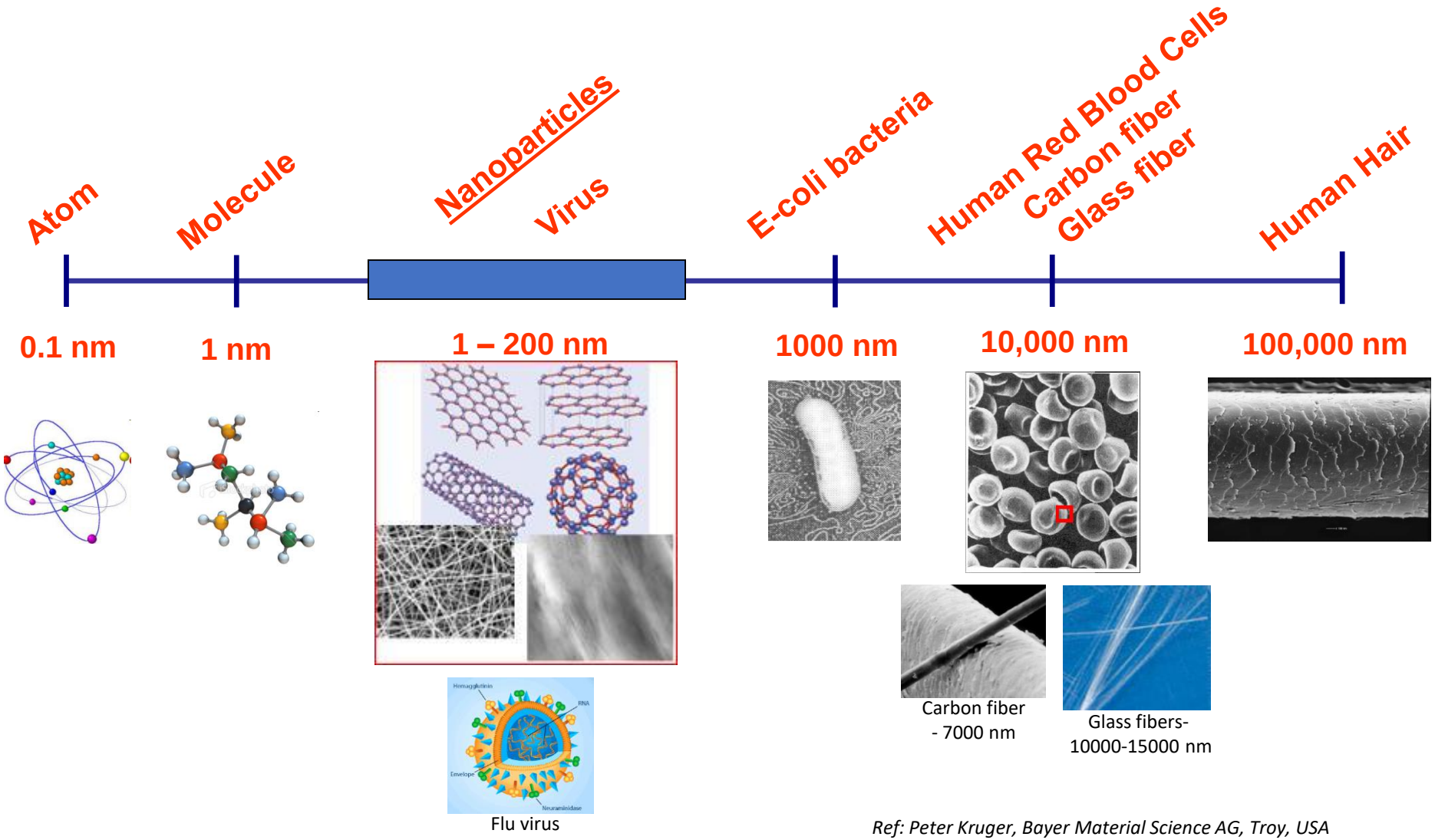
- Nano comes from Greek word nanos which means Dwarf or Extremely small
- One nanometer is one-billionth (10^{-9}) of a meter

Nanocomposites

- Nanocomposites are materials consisting of two or more components, with at least one component having dimensions in the nm regime (i.e. between 1 and 100 nm)

(Ref: Polycorpos Pissis, Nanostructured and Nanocomposite Polymeric Materials, National Technical University of Athens, Greece)

Nanomaterials - Comparison

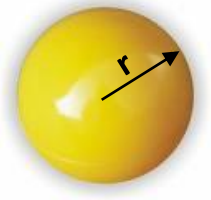


Ref: Peter Kruger, Bayer Material Science AG, Troy, USA

Surface Area to Volume Ratio

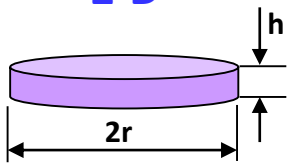
Micro vs Nano

3-D



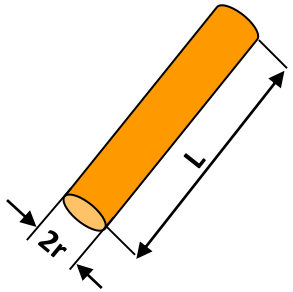
SA / V = 3/r

2-D



SA / V ~ 2/h

1-D

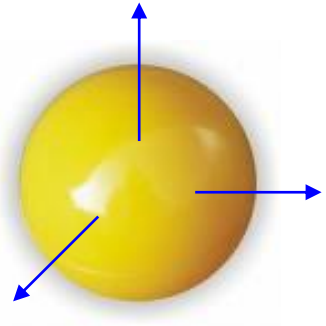


SA / V ~ 2/r

- Surface area per unit volume varies as the reciprocal of the characteristic dimension of the filler
- Surface area from micro to nano increases by 1000 times
- Surface area is critical in chemical reaction and bonding

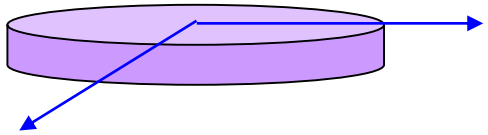
Fillers and Functionalities

Types of Fillers



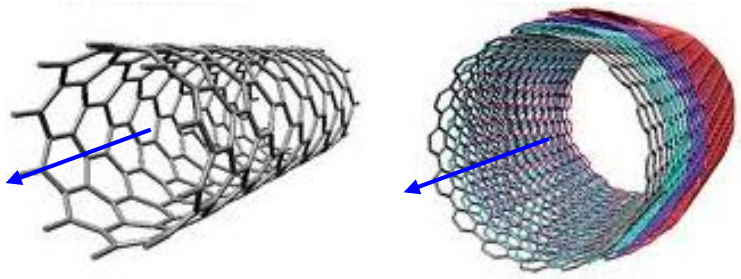
Nanoparticles

Example: Nanosilica, Fullerene



Nanolayers/platelets

Example: Nanoclay, Graphene



Nanotubes/fibers

Example: CNT, Nanofibers

Mechanical Properties

TABLE 1	MECHANICAL PROPERTIES COMPARISON			
Material	Modulus, GPa	T.S., GPa	Density, g/cm^3	Diameter
SWCNT/MWCNT	~1,000	~100-200	~0.7-1.7	1/~20 nm
Carbon Nanofibers	~500	3-7	1.8-2.1	20-200 nm
Graphene	~1,000	~100-400	1.8-2.2	platelet
High Tensile Steel	210	1.3	7.8	--
Carbon Fiber	230	3.5	1.75	5-10 μm
Aramid Fiber	60	3.6	1.44	5-10 μm
Glass Fiber	22	3.4	2.6	5-10 μm

Automobile Applications

Improved mechanical properties, surface finish, ease of processing

Toyota Timing Belt Cover
MMT/Nylon 6

Step assistant component in GMC Safari and Chevrolet Astro, MMT/TPO

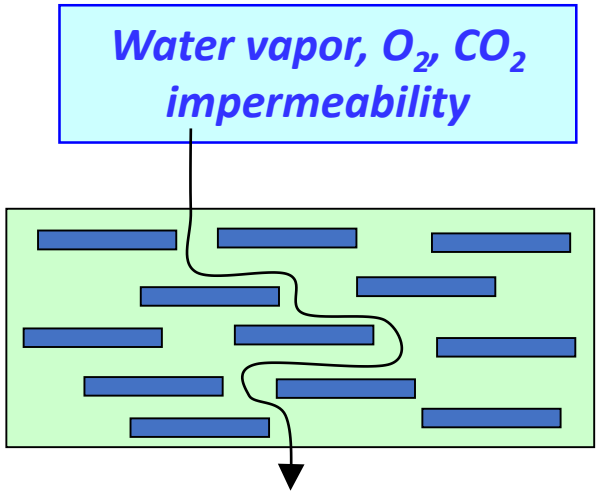
Mitsubishi GDI engine cover
MMT/Nylon-6

Door of Chevrolet Impala
MMT/TPO

Seat backs of Honda Acura, MMT/PP

Packaging Applications

Stand up pouch, MMT/Nylon 6



PET bottles, MMT/Nylon MXD6

*Meal Ready to Eat **(MRE)** Pack, MMT/Nylon MXD6*

No of companies: Ex – Nanocor, MGCC, Kuraray, etc

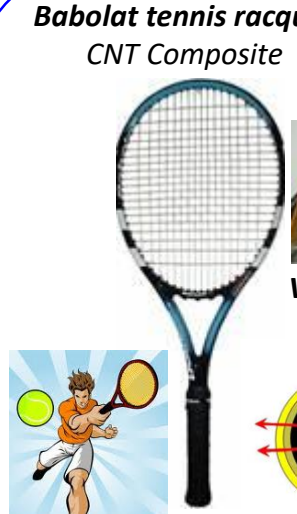
MRE Packs used in Military Applications

- Eliminates foil layer
- Capable of microwave processing
- Reduces stress-cracks, pin-holes
- Reduces processing steps (no lamination)
- Decreases weight

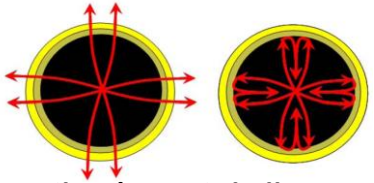
Sporting Goods

CNT composites –
Stiff, strong, lightweight

Babolat tennis racquet
CNT Composite



Vokl DNX tennis racquet
CNT Composite



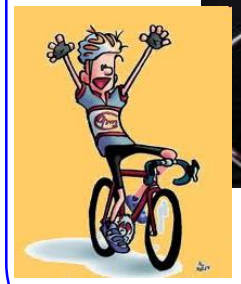
Wilson's tennis ball
nanoclay coated (barrier)

Easton/Zyvex base ball bat
CNT composite

Easton/Zyvex hockey stick
CNT composite

Enlight Earth LLC/Eric Arakawa
Surfboard, CNT Composite

BMC/Easton bicycles
CNT Composite



ABS Nanodesu bowling ball
Fullerene coated (wear)

Materials Needs for Some Aircraft Applications

Still there are needs of new materials for many challenges in aircraft structures….some examples are shown below.

Problem - Structural failure

Property needed **- Strength, stiffness, toughness**

Problem - Lightning

Property needed **- Electrical conductivity**

Problem - Icing

Property needed - Electrical & thermal conductivity

Problem - Structural health monitoring

Property needed - Electrical conductivity, self healing

We will see some materials solutions for these from composition – processing – microstructure – property perspective…

Materials Needs for Aircraft Applications (Mechanical Properties)

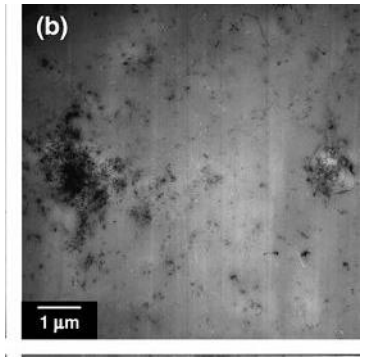
Composition: CNT + Epoxy + Carbon fiber composite

Processing → **Microstructure** → **Processing of composites**

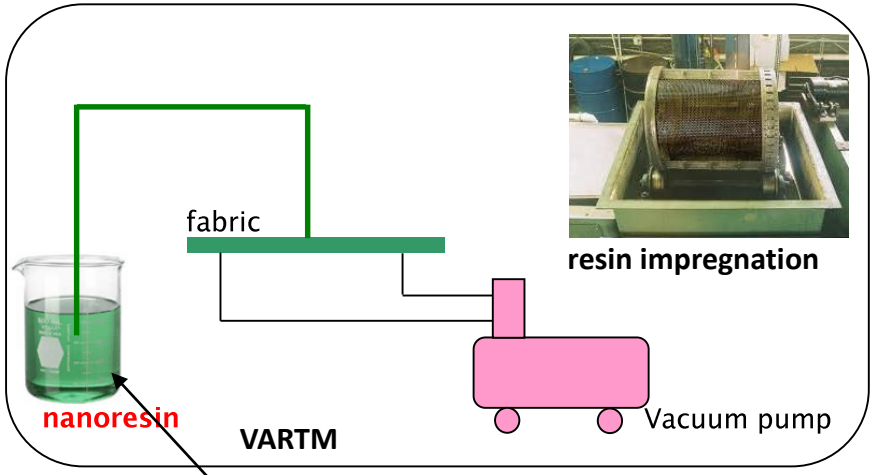
Hand mix epoxy + CNT

↓

Process in three roll mill



Non uniform mixing of CNT



Use nanoresin now to make composite

↓

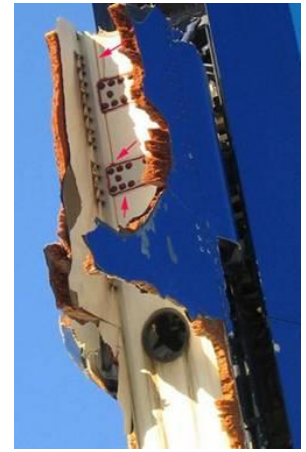
Property

Resulting strength, modulus of composites is not good (because of non uniform mixing of CNTs)

Problem - Structural failure

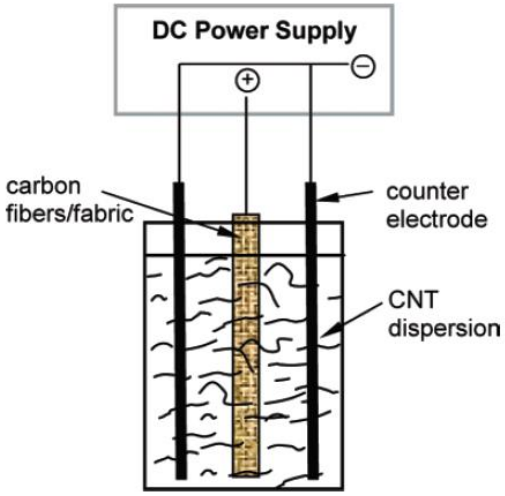
Property needed - Strength, stiffness, toughness

Materials Needs for Aircraft Applications (Mechanical Properties)

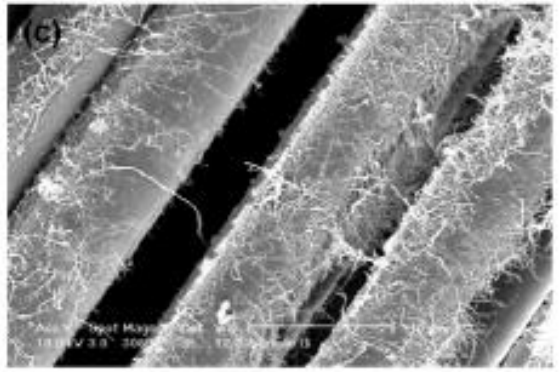


Composition: CNT + PC + Carbon fiber composite

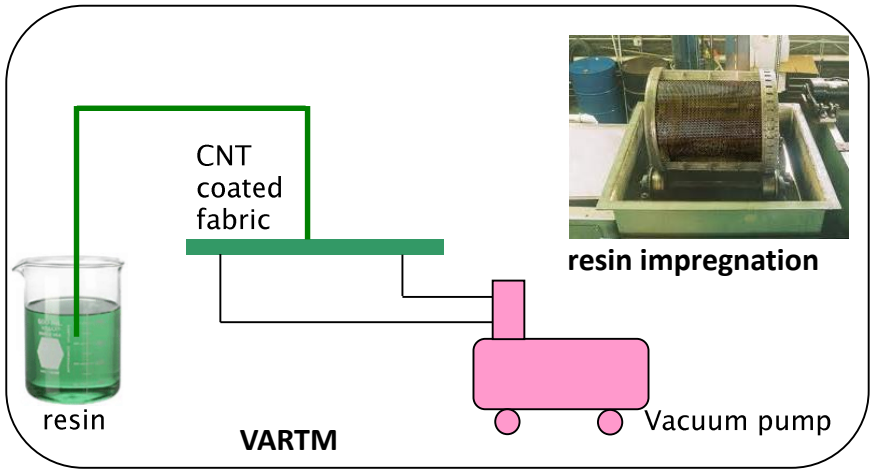
Processing → **Microstructure** → **Processing of composites**



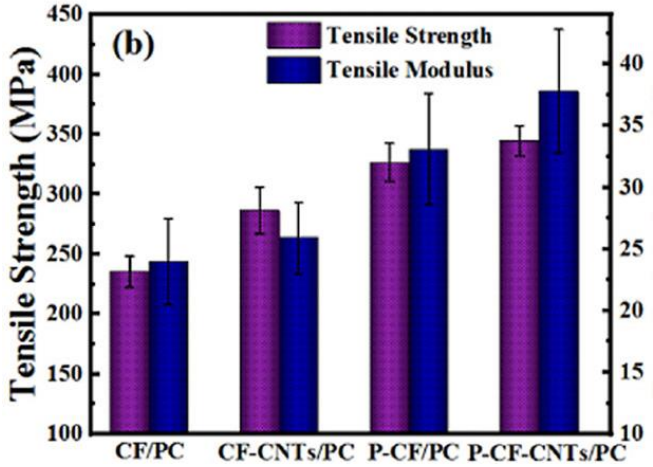
Electrophoretic deposition



Uniform coating of CNT on fabrics



Use CNT coated fabric now to make composite



Property

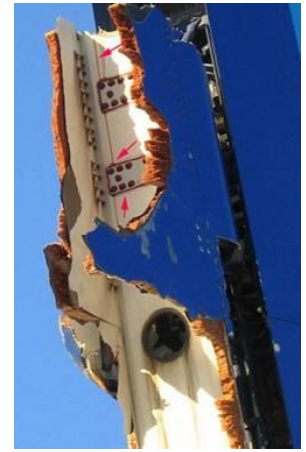
Resulting strength, modulus of composites is good (because uniform deposition of CNTs)

(Note - Figure shows with some chemical treatment along with CNT coatings)

Problem - Structural failure

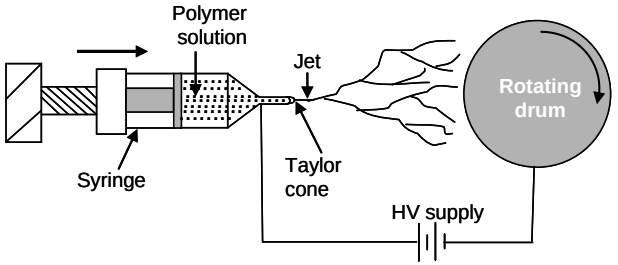
Property needed - Strength, stiffness, toughness

Materials Needs for Aircraft Applications (Fracture Toughness)

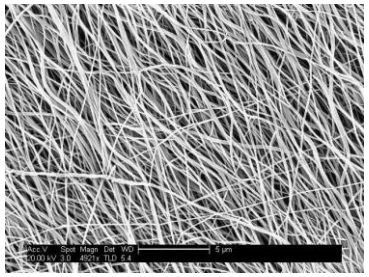


Composition: Nylon nanofiber + Epoxy + Carbon fiber composite

Processing → **Microstructure** → **Processing of composites**

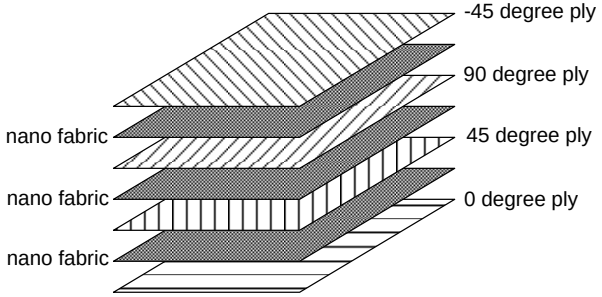


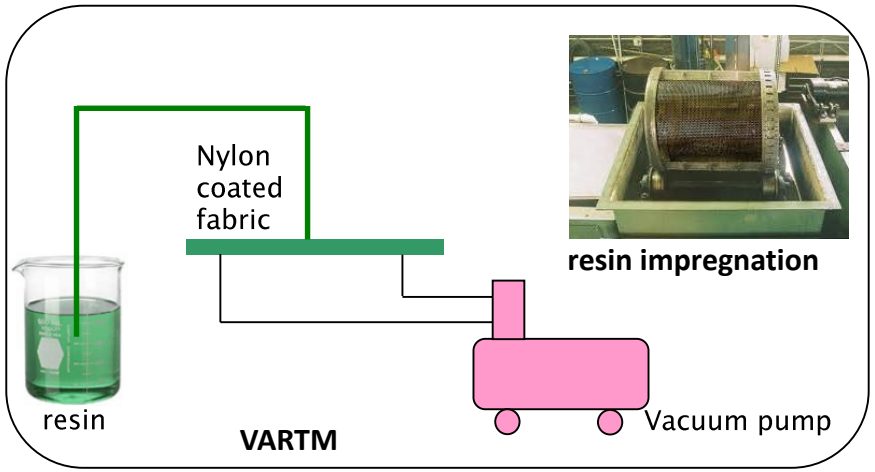
Electrospinning



Nylon nanofibers

Fabric stacking



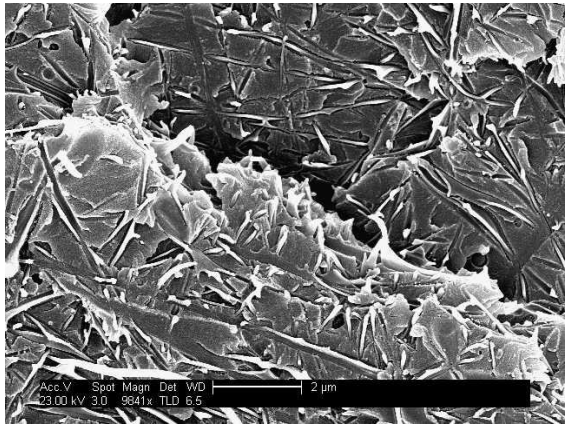


Use Nylon coated fabric now to make composite

Property

Fracture toughness improved by 150%

(nanofiber bridging, pull out are mechanisms)



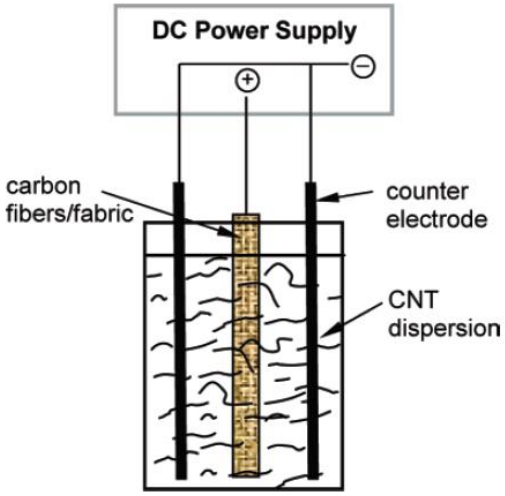
Problem - Structural failure

Property needed - Strength, stiffness, toughness

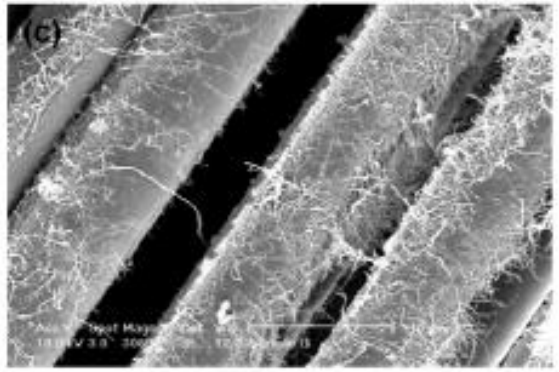
Materials Needs for Aircraft Applications (Lightning Strike)

Composition: CNT + Epoxy + Carbon fiber composite

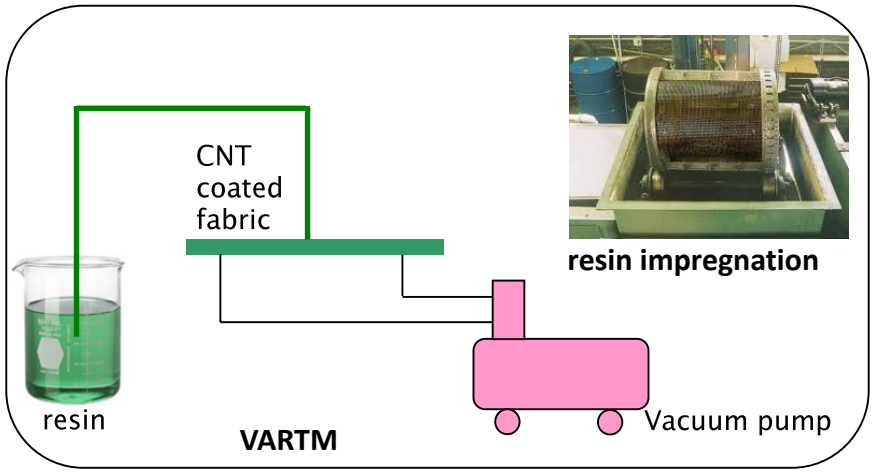
Processing → **Microstructure** → **Processing of composites**



Electrophoretic deposition



Uniform coating of CNT on fabrics

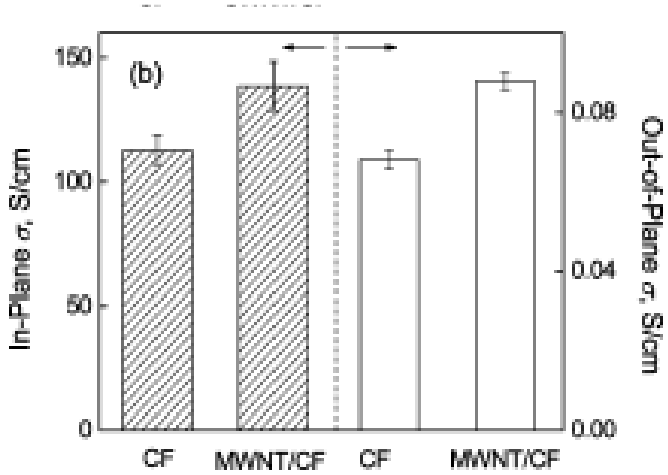


Use CNT coated fabric now to make composite

Problem - Lightning ***Property needed - Electrical conductivity***

Property

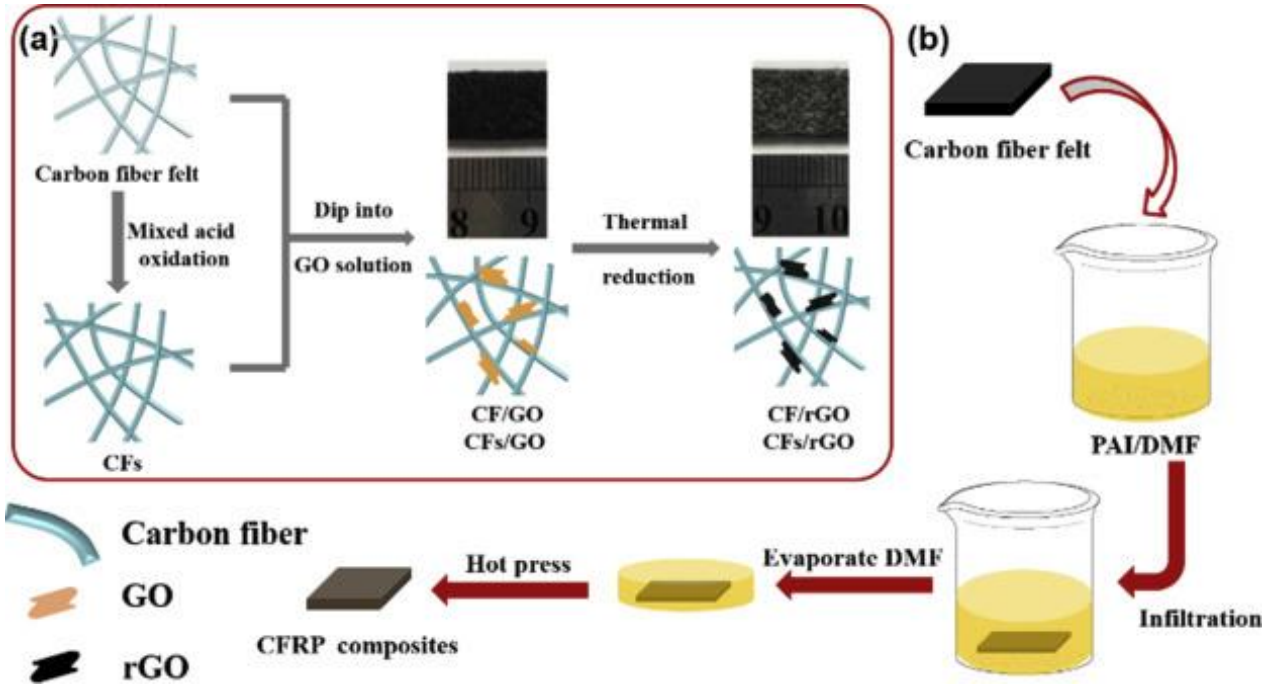
Increase in in-plane and out-of-plane electrical conductivity increased because of CNTs



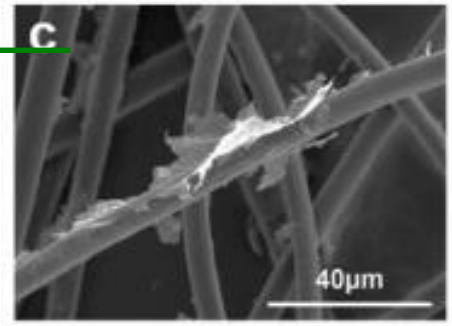
Materials Needs for Aircraft Applications (Icing)

Composition: Graphene Oxide + Polyamide + Carbon fiber composite

Processing of composites

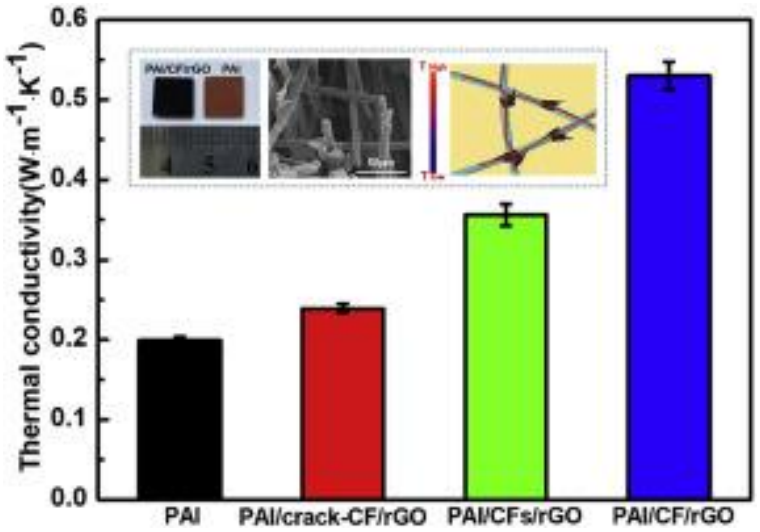


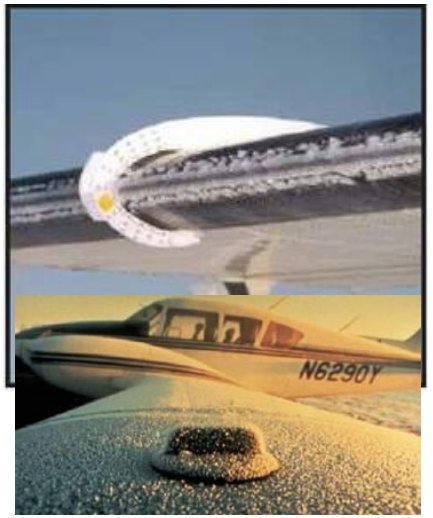
Microstructure



Property

***Significant improvements in thermal conductivity** for composites with various chemical treatments*





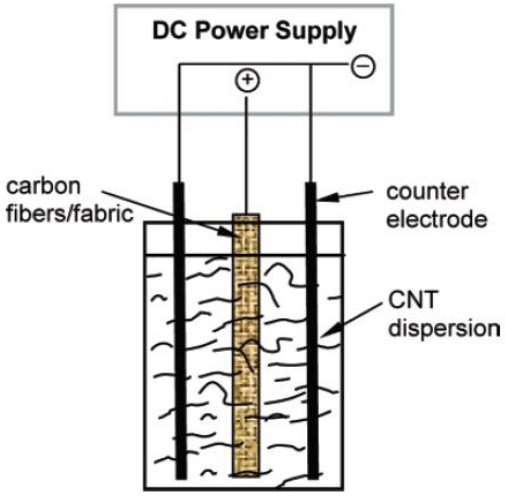
Problem - Icing

Property needed - Electrical & thermal conductivity

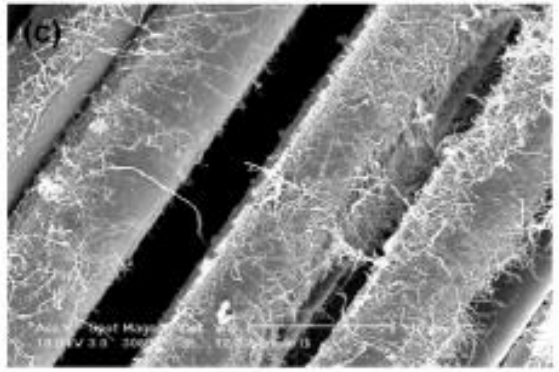
Materials Needs for Aircraft Applications (Health Monitoring)

Composition: CNT + Epoxy + Carbon fiber composite

Processing → **Microstructure** → **Processing of composites**



Electrophoretic deposition



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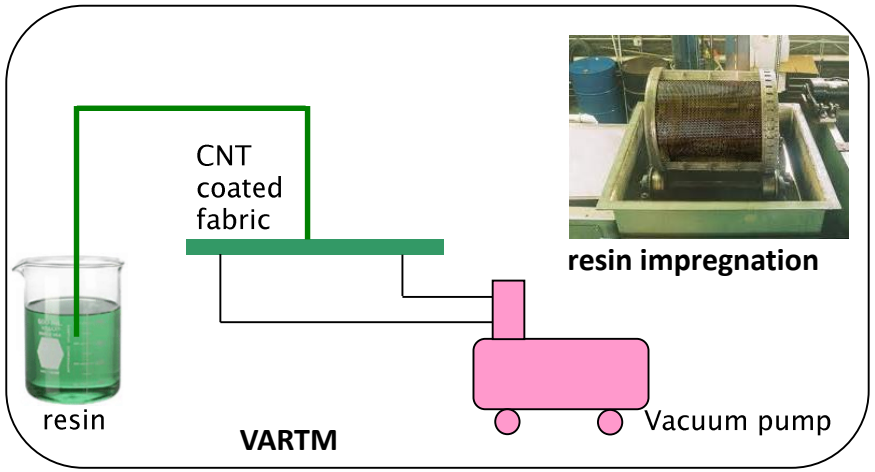
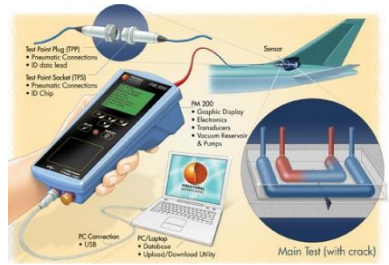


Use CNT coated fabric now to make composite

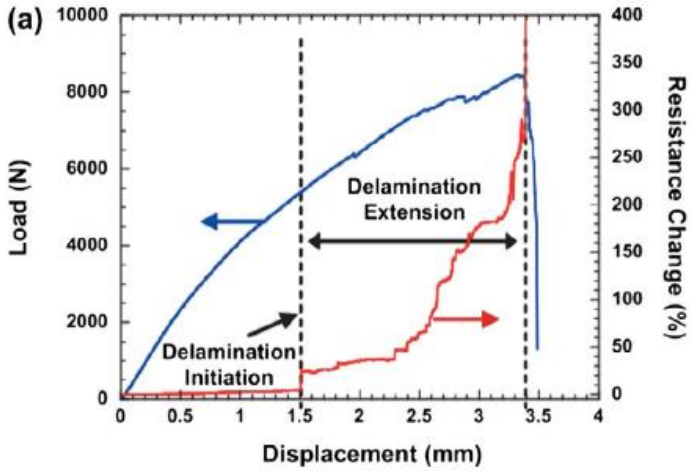


Problem - Structural health monitoring

Property needed - Electrical conductivity

Property

Internal damages which is not shown in load-displacement curves are shown as change in resistances – beneficial to detect and monitor defects in internal structures

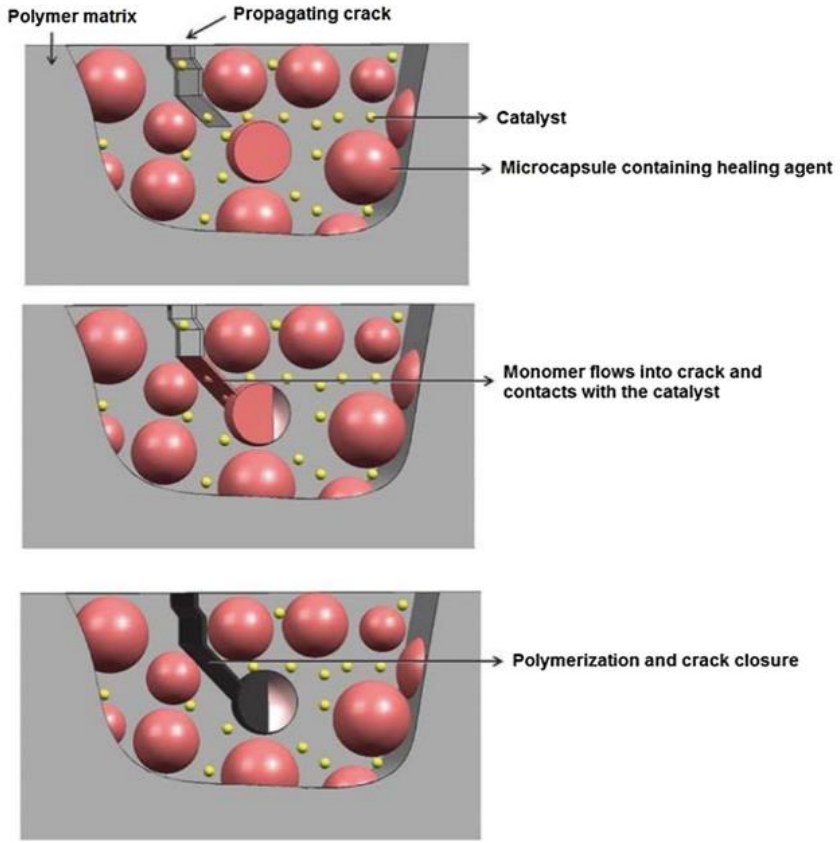


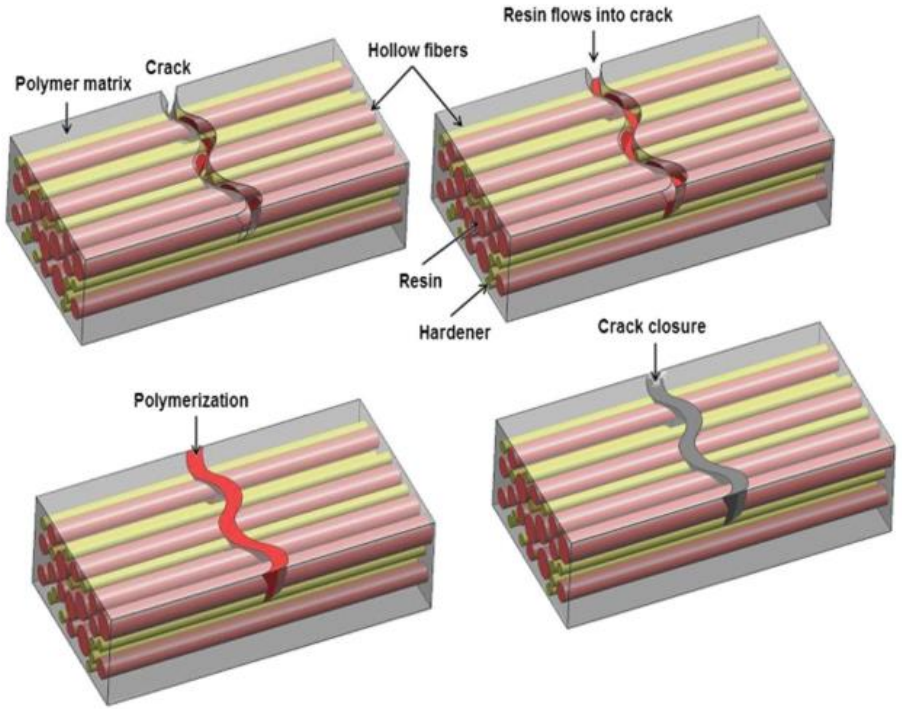
Materials Needs for Aircraft Applications (Self Healing)

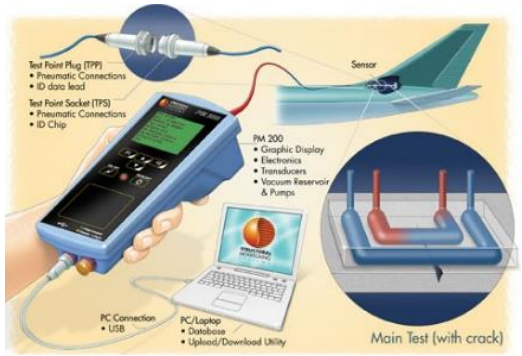
Smart materials

Composition: Polymer + Self healing capsules or fibers + Carbon fibers

Microstructures







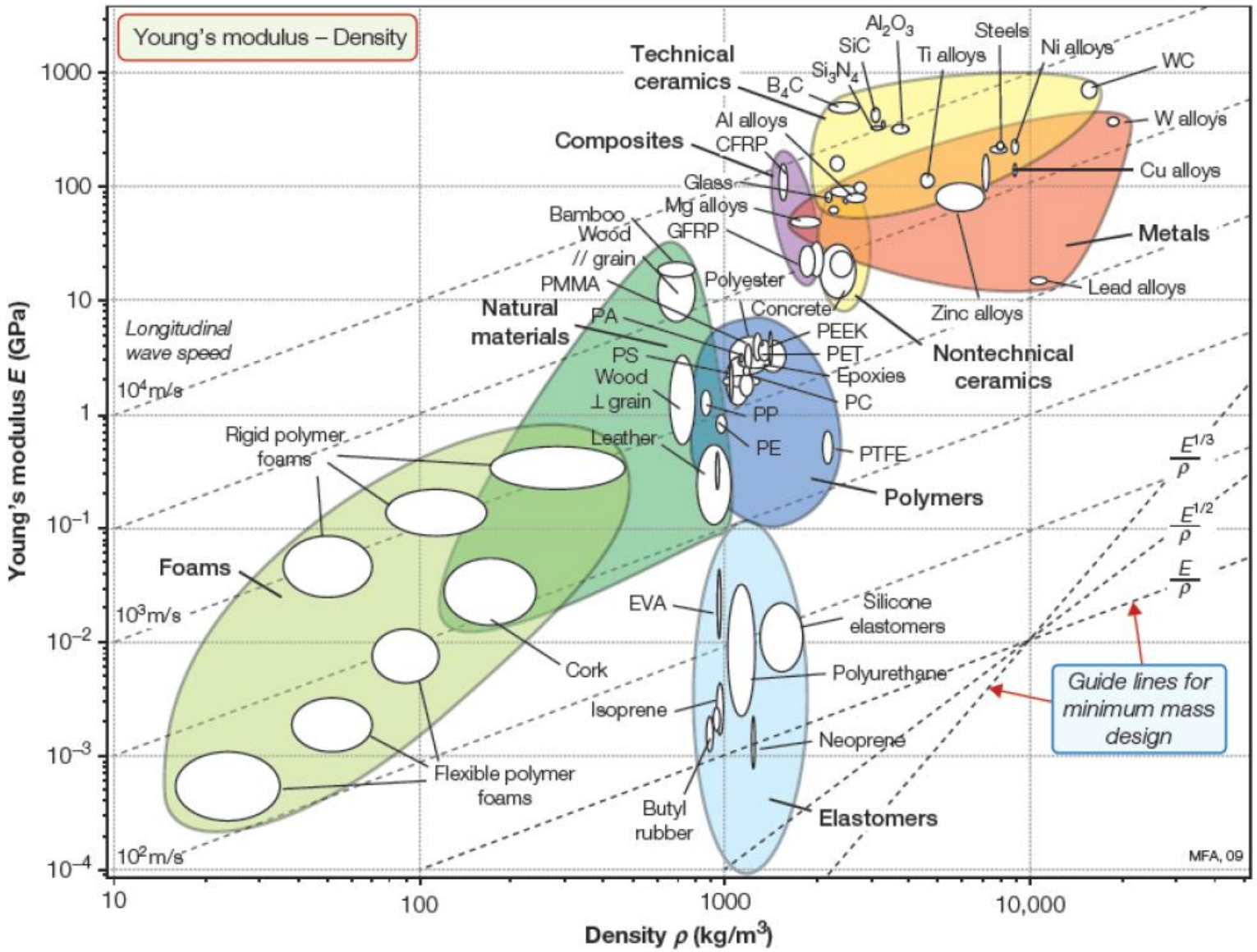
Problem - Structural health monitoring

self healing

Hollow fibers with uncured resin are used. Under cracking of fibers, resin comes out and reacts with catalyst outside. It cures and closes the cracks.

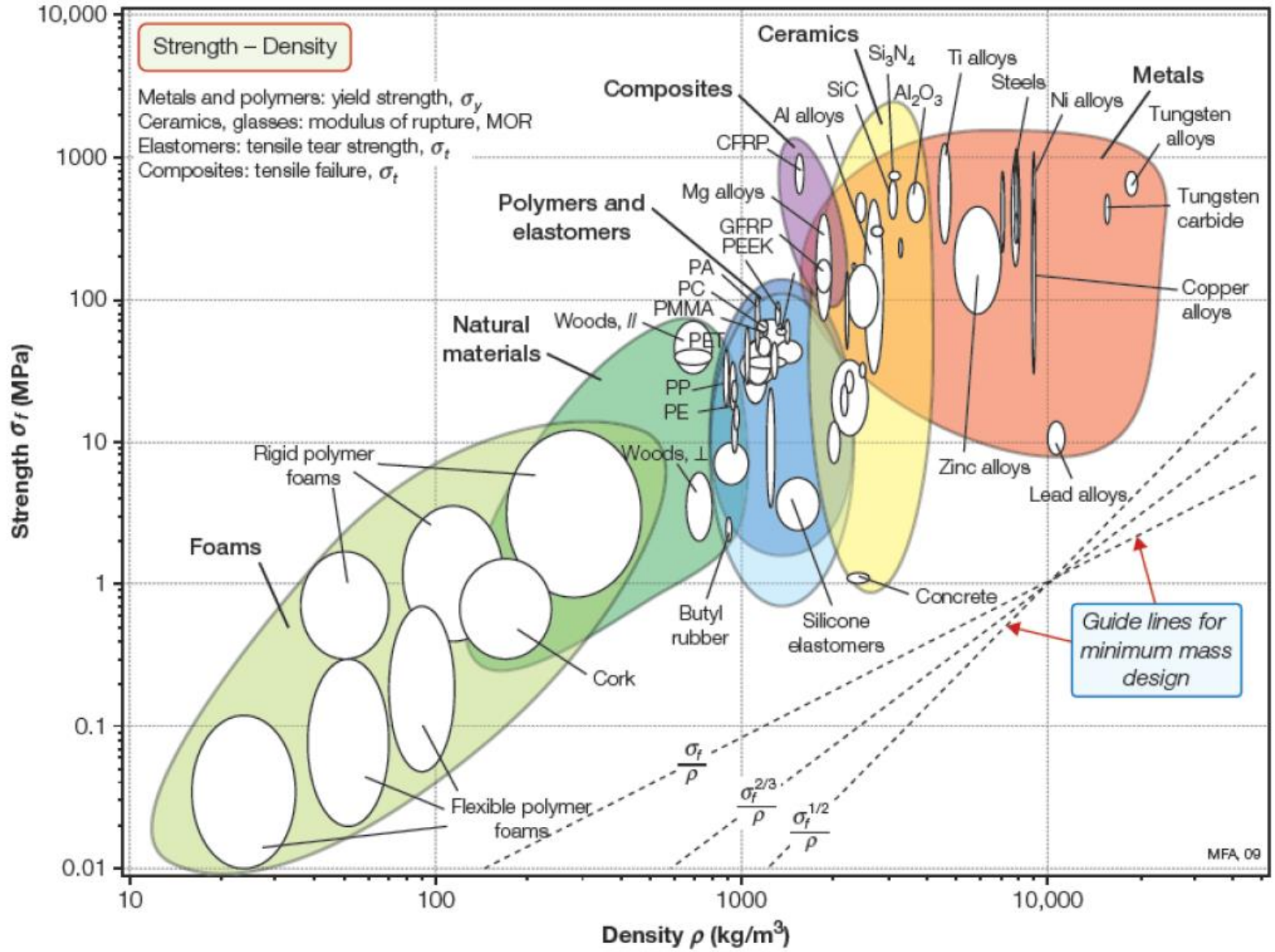
Material Property Charts

Young's Modulus vs Density



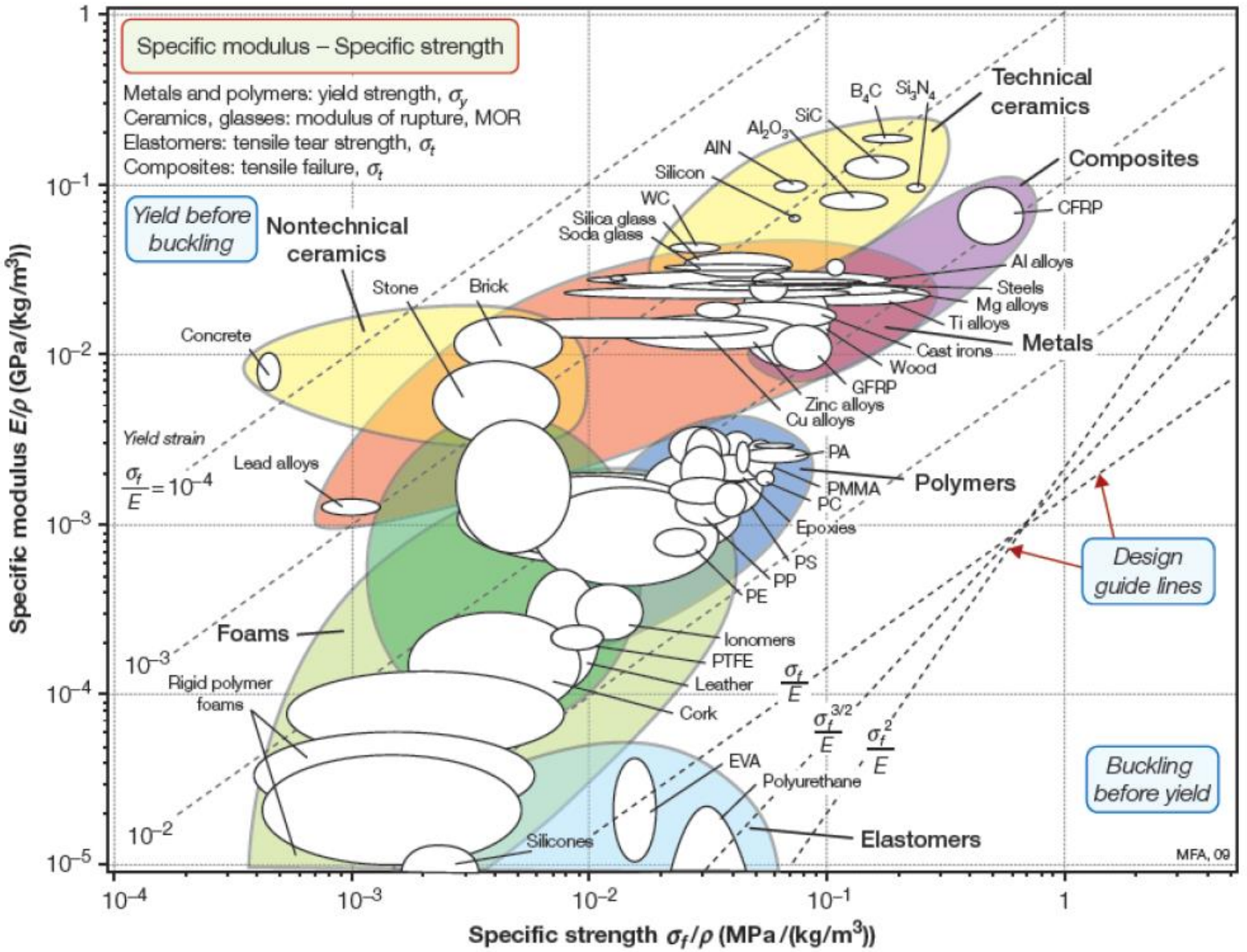
Material Property Charts

Strength vs Density



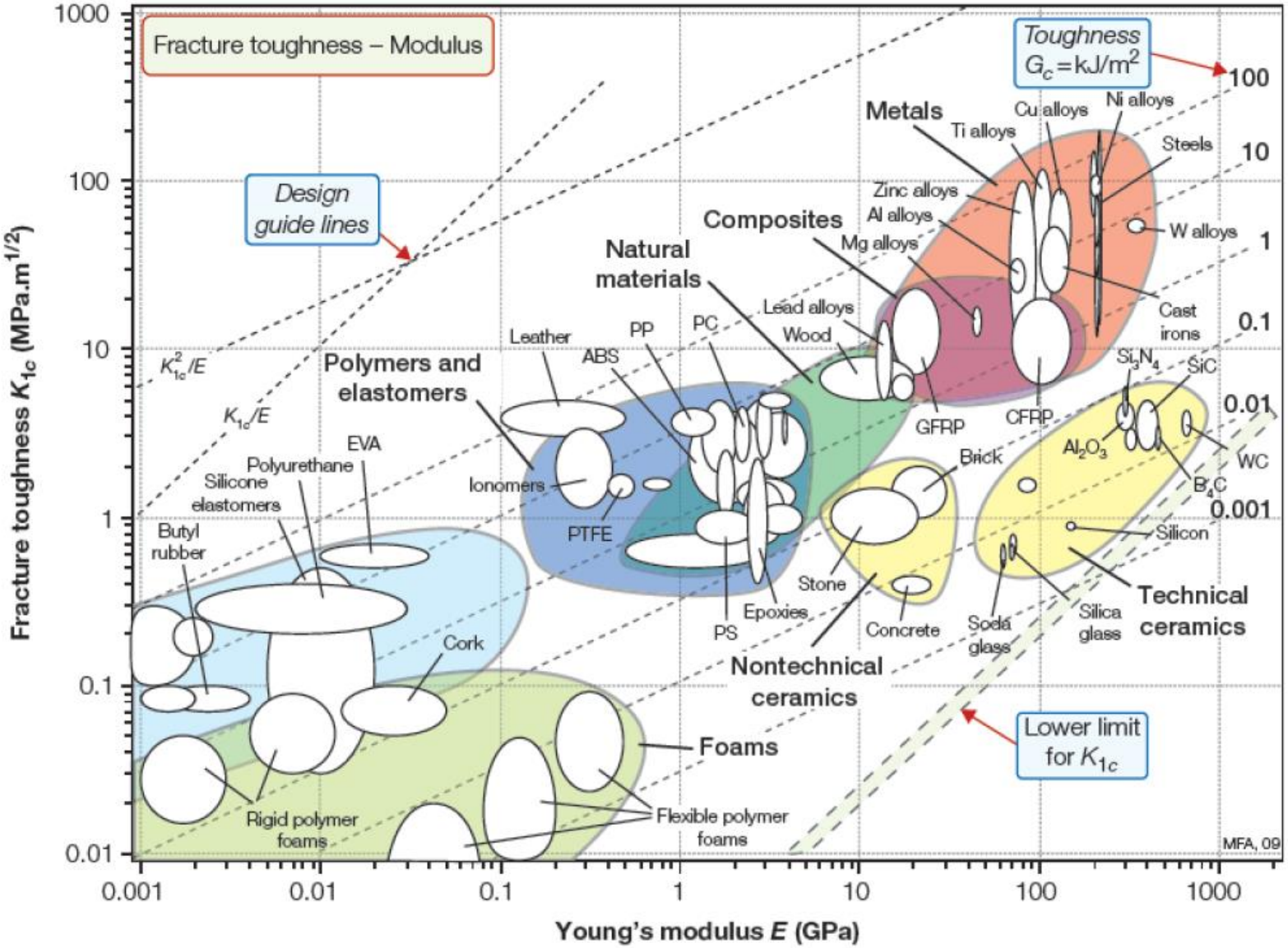
Material Property Charts

Specific modulus vs Specific strength



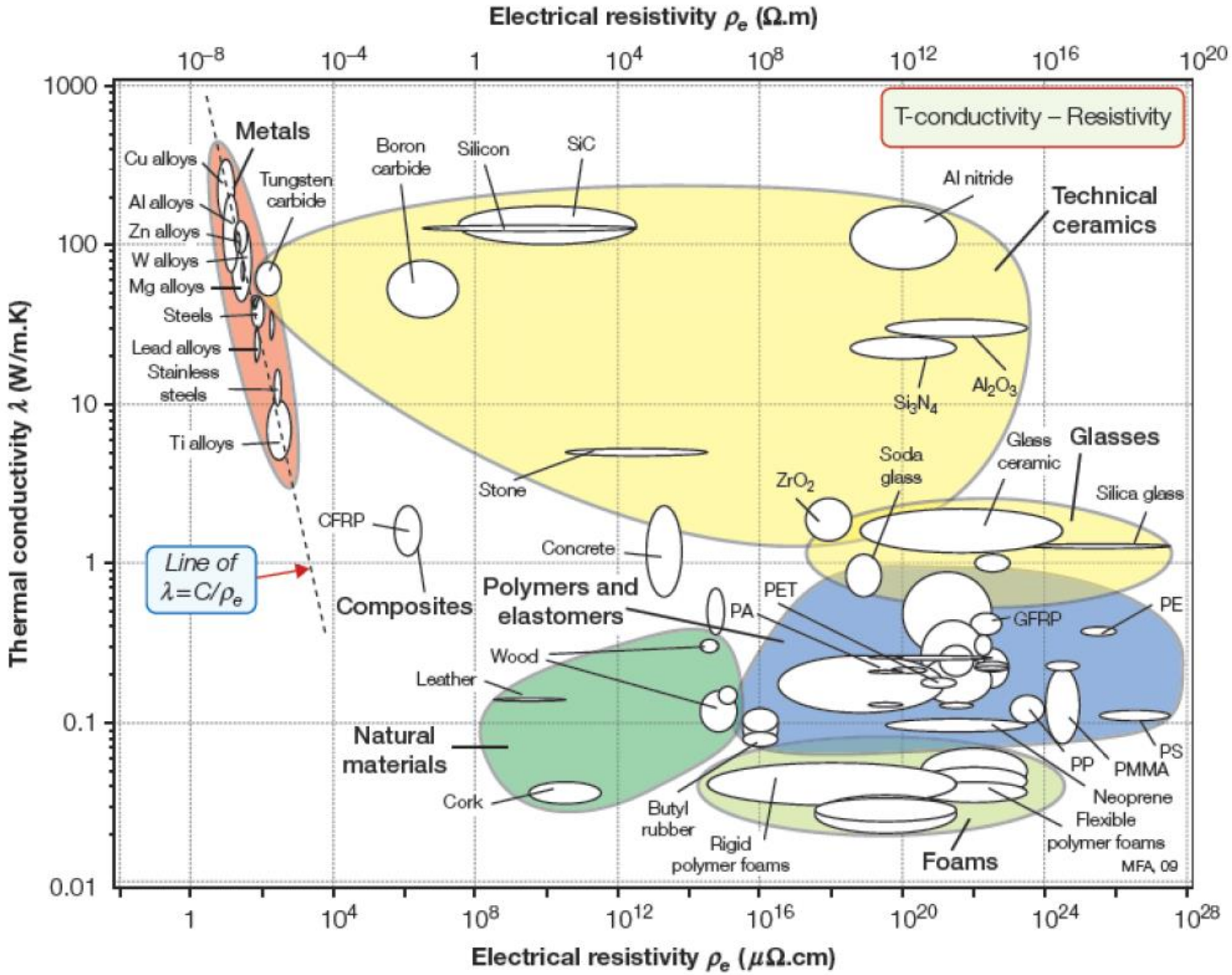
Material Property Charts

Fracture toughness vs Modulus



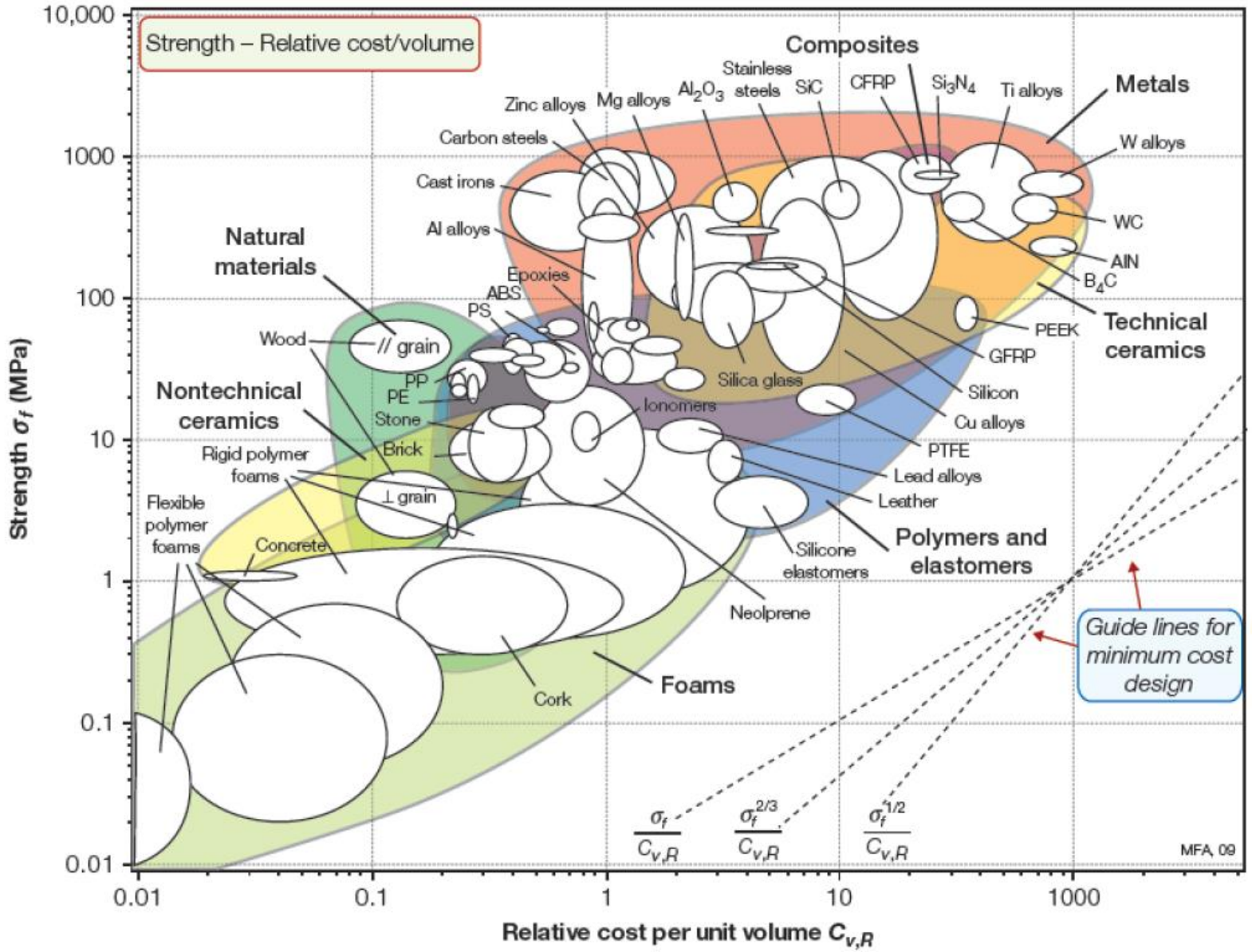
Material Property Charts

Thermal conductivity vs Electrical resistivity

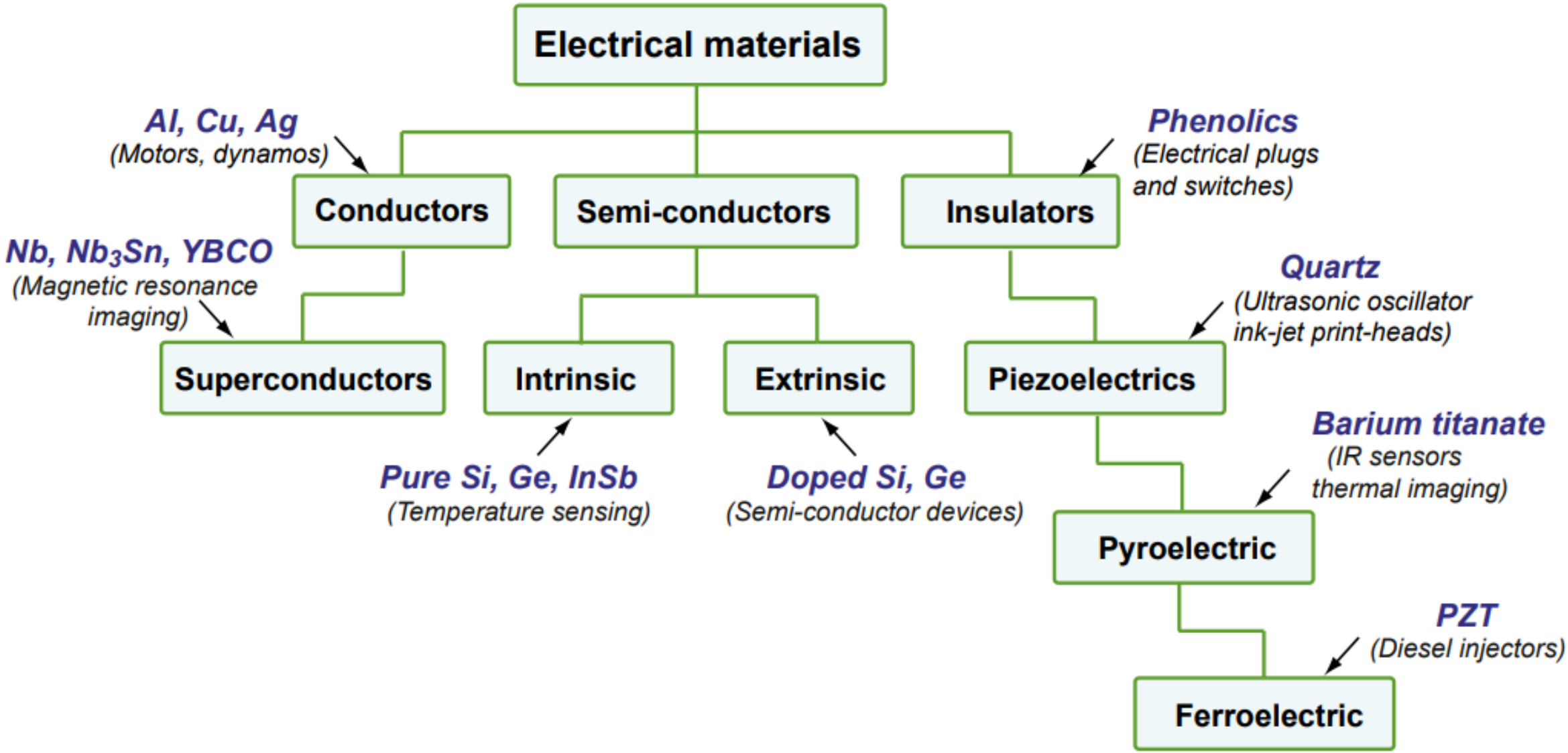


Material Property Charts

Strength vs Cost



Electrical Properties



Electrical Properties

- Ohm's Law:

$$V = I R$$

voltage drop (volts = J/C)
C = Coulomb

current (amps = C/s)

resistance (Ohms)

- Resistivity, ρ:
  -- a material property that is independent of sample size and geometry

$$\rho = \frac{RA}{l}$$

surface area of current flow

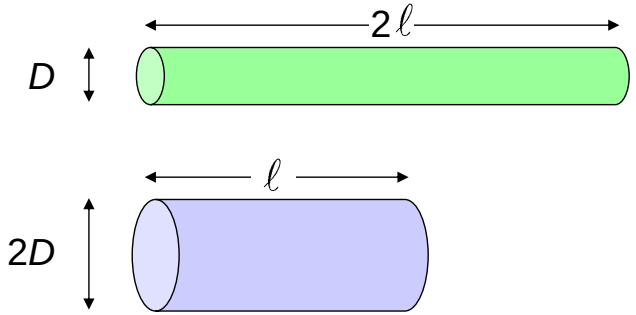
current flow path length

- Conductivity, σ

$$\sigma = \frac{1}{\rho}$$

Electrical Conduction

- Which will have the greater resistance?



$$R_1 = \frac{2\rho\ell}{\pi\left(\frac{D}{2}\right)^2} = \frac{8\rho\ell}{\pi D^2}$$

$$R_2 = \frac{\rho\ell}{\pi\left(\frac{2D}{2}\right)^2} = \frac{\rho\ell}{\pi D^2} = \frac{R_1}{8}$$

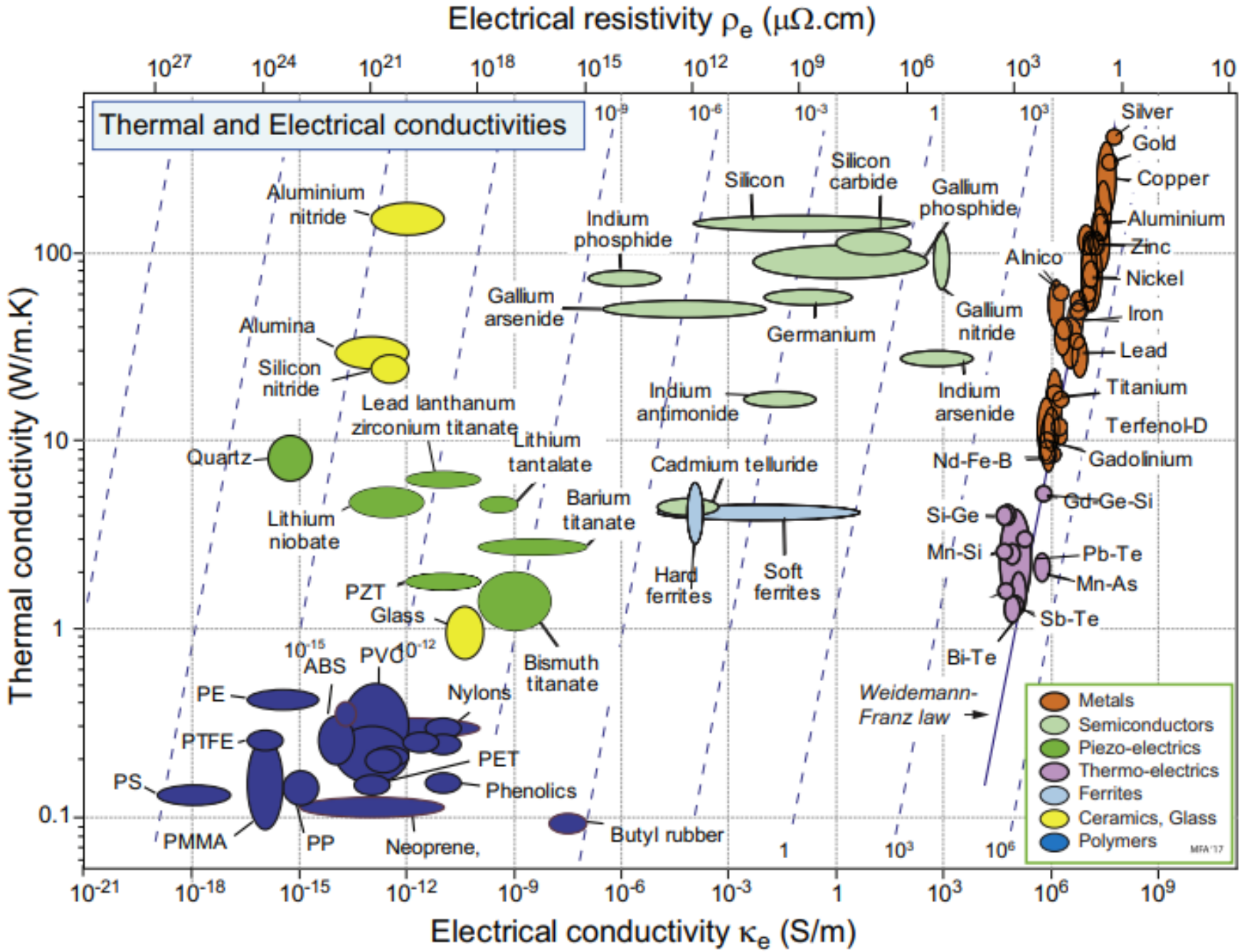
- Resistance depends on sample geometry and size.

Temperature-dependence of resistivity:

The resistivity of metals increases with temperature because thermal vibrations scatter electrons.

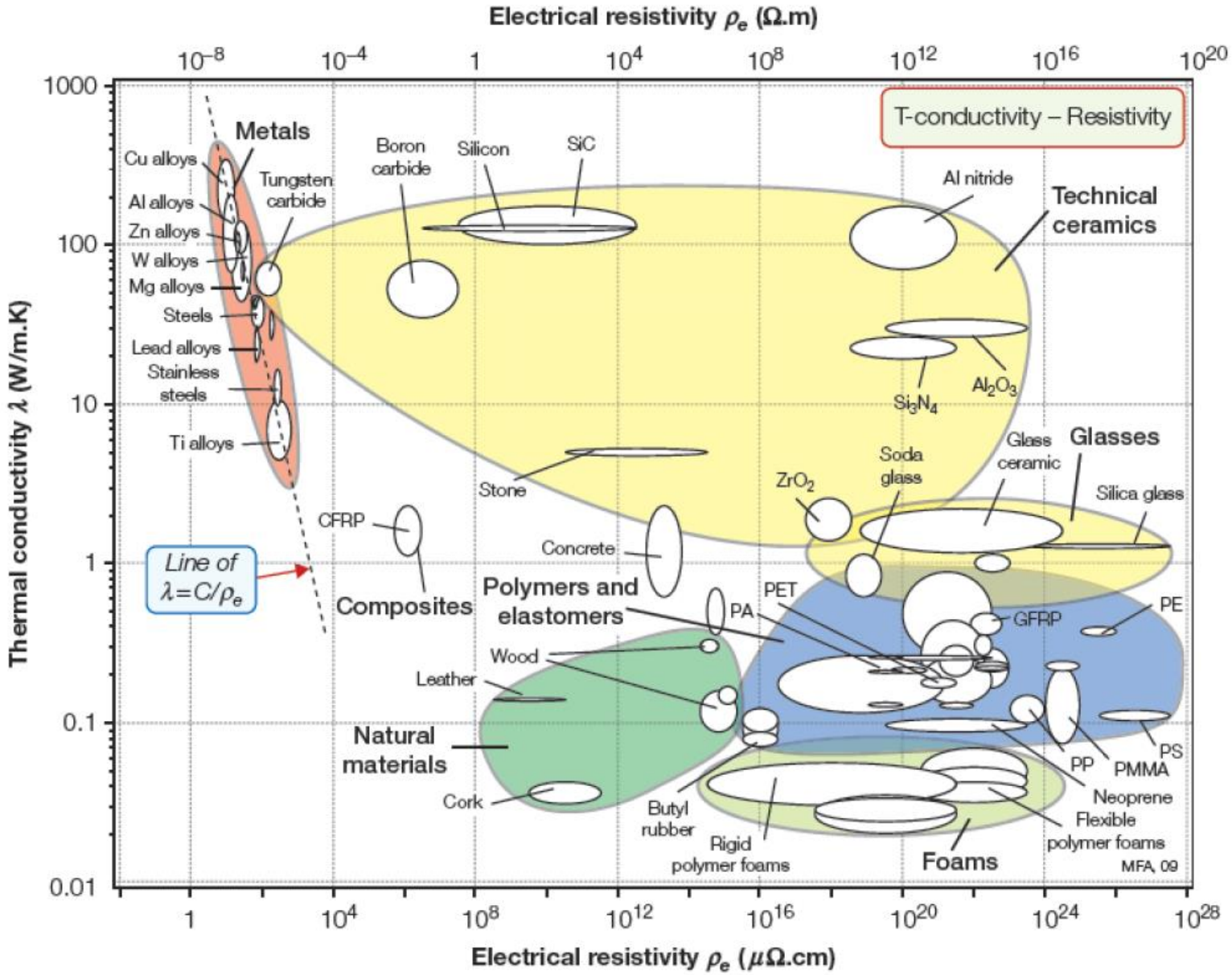
The resistivity of semiconductors, by contrast, decreases as temperature increases, because thermal energy allows more carriers to cross the band gap, entering the conduction band.

Superconducting materials suddenly lose all resistance at a critical temperature, Tc. Below Tc, the current in a superconducting material flows without any resistive loss; above Tc, superconductivity is suppressed.


Thermal and Electrical conductivities
Electrical resistivity ρ_e ($\mu\Omega$.cm)
Electrical conductivity κ_e (S/m)
Thermal conductivity (W/m.K)
Aluminium nitride
Alumina
Silicon nitride
Lead lanthanum zirconium titanate
Quartz
Lithium tantalate
Lithium niobate
Barium titanate
PZT
Glass
Bismuth titanate
ABS
PVC
PE
Nylons
PTFE
PET
PS
Phenolics
PMMA
PP
Neoprene,
Butyl rubber
Indium phosphide
Silicon
Silicon carbide
Gallium phosphide
Gallium arsenide
Germanium
Gallium nitride
Indium antimonide
Indium arsenide
Cadmium telluride
Hard ferrites
Soft ferrites
Silver
Gold
Copper
Aluminium
Zinc
Alnico
Nickel
Iron
Lead
Titanium
Terfenol-D
Nd-Fe-B
Gadolinium
Gd-Ge-Si
Si-Ge
Mn-Si
Pb-Te
Mn-As
Sb-Te
Bi-Te
Weidemann-Franz law
Metals
Semiconductors
Piezo-electrics
Thermo-electrics
Ferrites
Ceramics, Glass
Polymers
MFA'17

Material Property Charts

Thermal conductivity vs Electrical resistivity



Conductivity: Comparison

- Room temperature values $(\text{Ohm-m})^{-1} = (\Omega \text{ - m})^{-1}$

conductors

METALS	
Silver	6.8×10^{7}
Copper	6.0×10^{7}
Iron	1.0×10^{7}

semiconductors

SEMICONDUCTORS	
Silicon	4×10^{-4}
Germanium	2×10^{0}
GaAs	10^{-6}

insulators

CERAMICS	
Soda-lime glass	10^{-10} - 10^{-11}
Concrete	10^{-9}
Aluminum oxide	$<10^{-13}$

POLYMERS	
Polystyrene	$<10^{-14}$
Polyethylene	10^{-15} - 10^{-17}

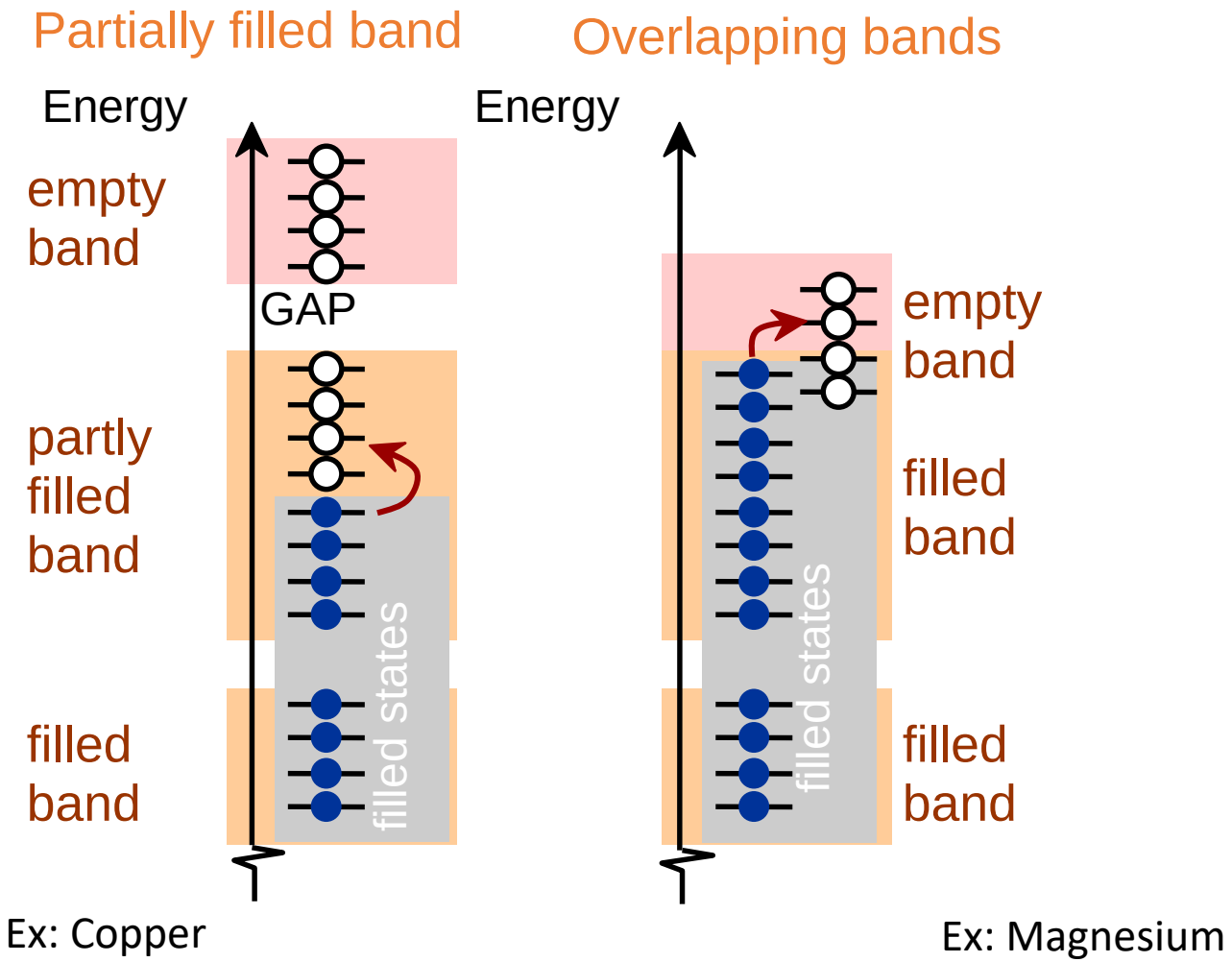
Energy Band Structures: Metals

- **Metals (Conductors):**

-- for metals empty energy states are adjacent to filled states.

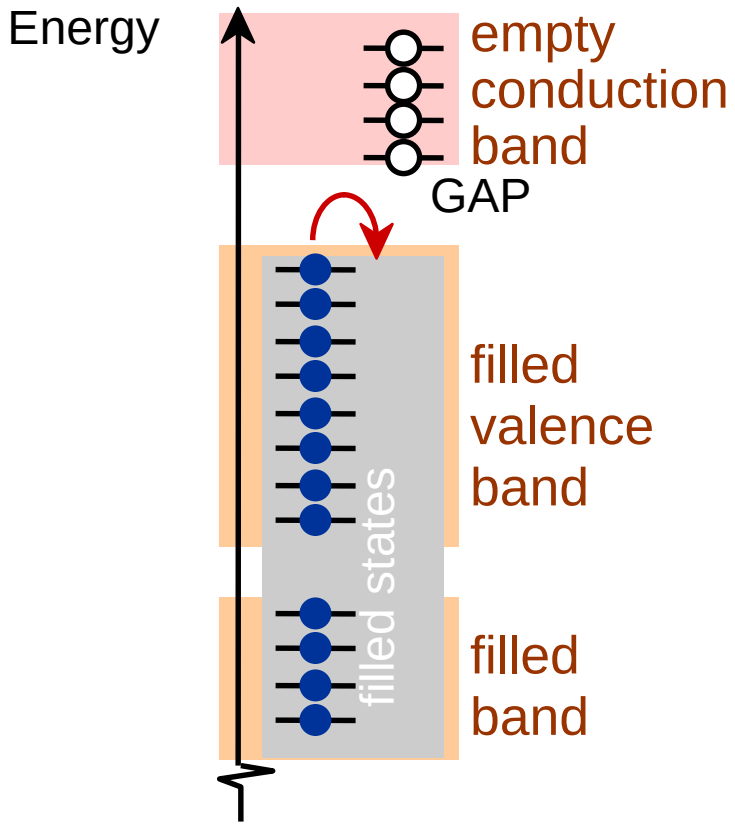
-- thermal energy excites electrons into empty higher energy states.

-- two types of band structures for metals
- partially filled band
- empty band that overlaps filled band

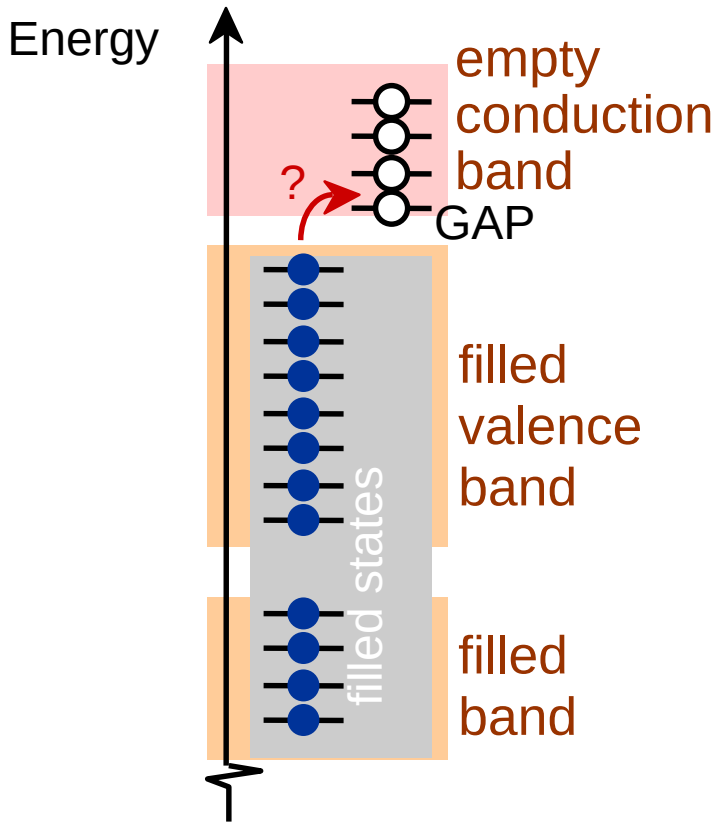


Energy Band Structures: Insulators & Semiconductors

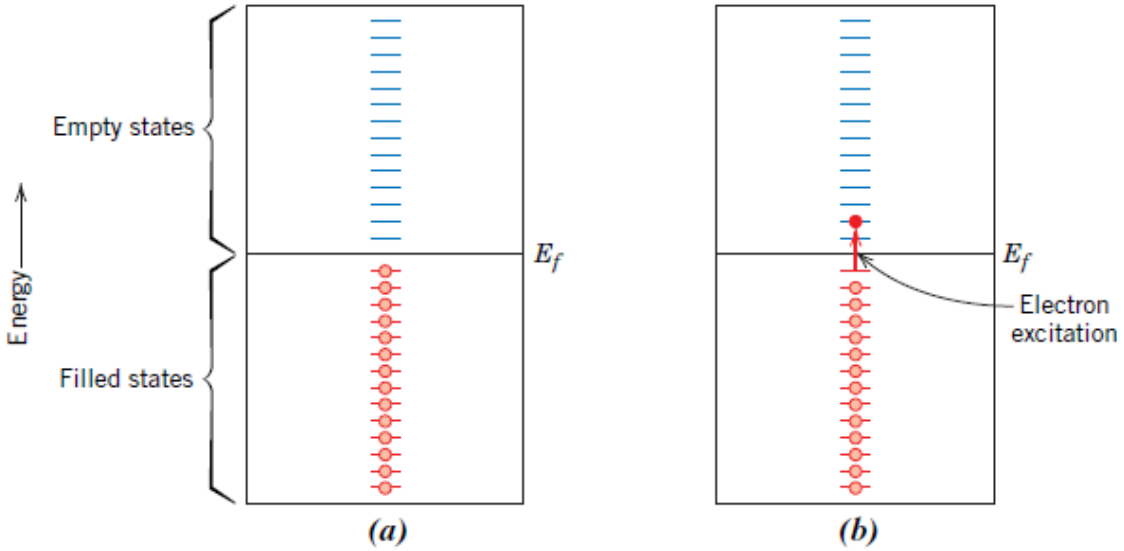
- **Insulators:**
  -- wide band gap (> 2 eV)
  -- few electrons excited across band gap



- **Semiconductors:**
  -- narrow band gap (< 2 eV)
  -- more electrons excited across band gap

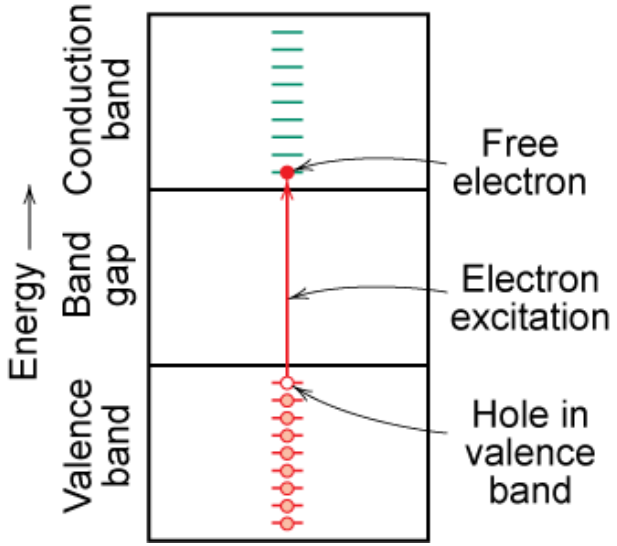


Conduction in Metals



For a metal, occupancy of electron states: (*a) before and (b) after an electron* excitation.

Charge Carriers in Insulators and Semiconductors



Two types of electronic charge carriers:

Free Electron

– negative charge

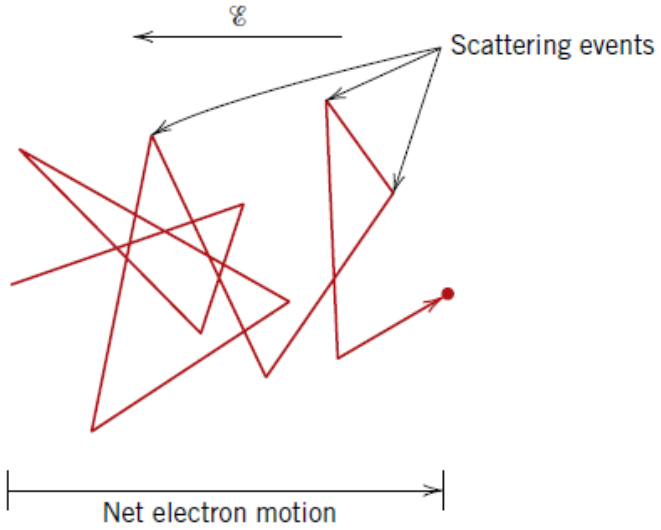
– in conduction band

Hole

– positive charge
– vacant electron state in the valence band

ELECTRON MOBILITY

- Scattering of electrons by imperfections in the crystal lattice, including impurity atoms, vacancies, interstitial atoms, dislocations, and thermal vibrations of the atoms themselves.



Drift velocity – average electron velocity

$$v_d = \mu_e \mathscr{E}$$

Constant is electron mobility – indication of frequency of scattering events

The conductivity σ of most materials may be expressed as

$$\sigma = n|e|\mu_e$$

where n is the number of free or conducting electrons per unit volume (e.g., per cubic meter) and $|e|$ is the absolute magnitude of the electrical charge on an electron (1.6×10^{-19} C). Thus, the electrical conductivity is proportional to both the number of free electrons and the electron mobility.

Resistivity in Metals

- Presence of imperfections increases resistivity
  - -- grain boundaries
  - -- dislocations
  - -- impurity atoms
  - -- vacancies

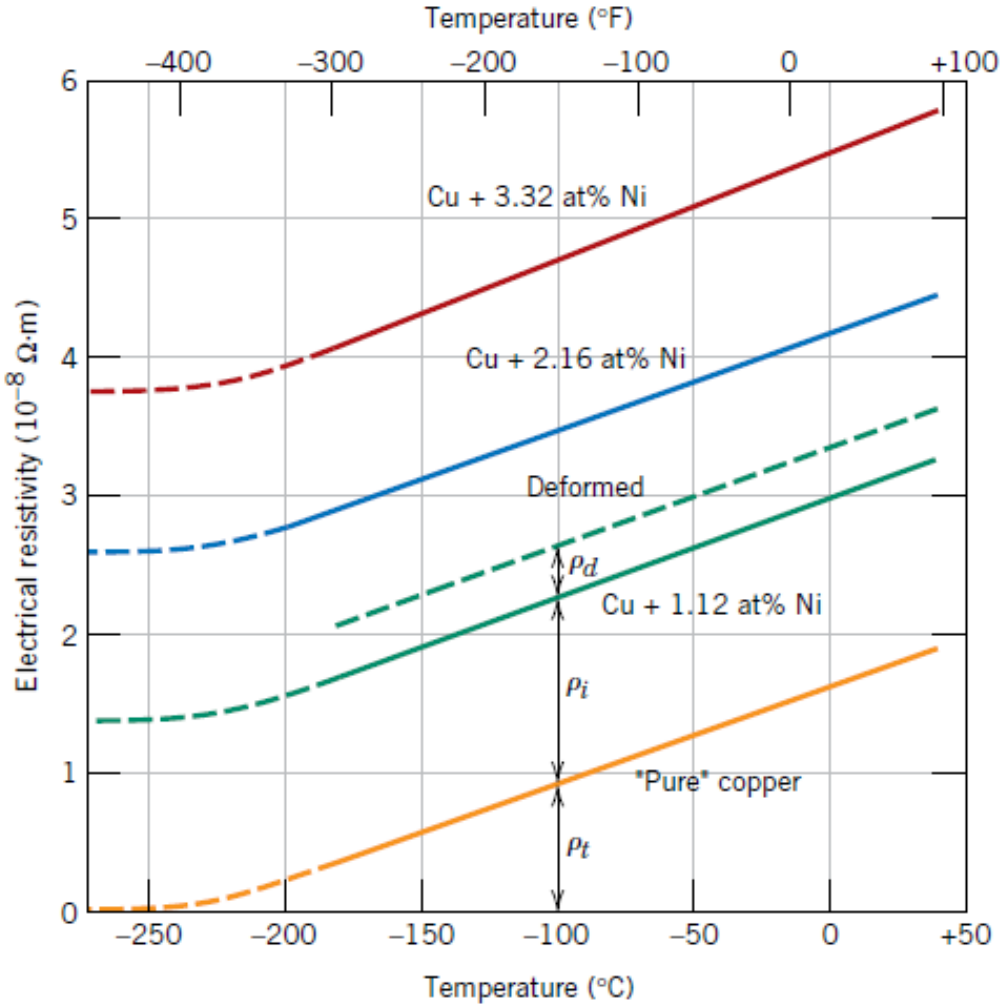
These act to scatter electrons so that they take a less direct path.

- Resistivity increases with:
  - -- temperature
  - -- wt% impurity
  - -- deformation (%CW)

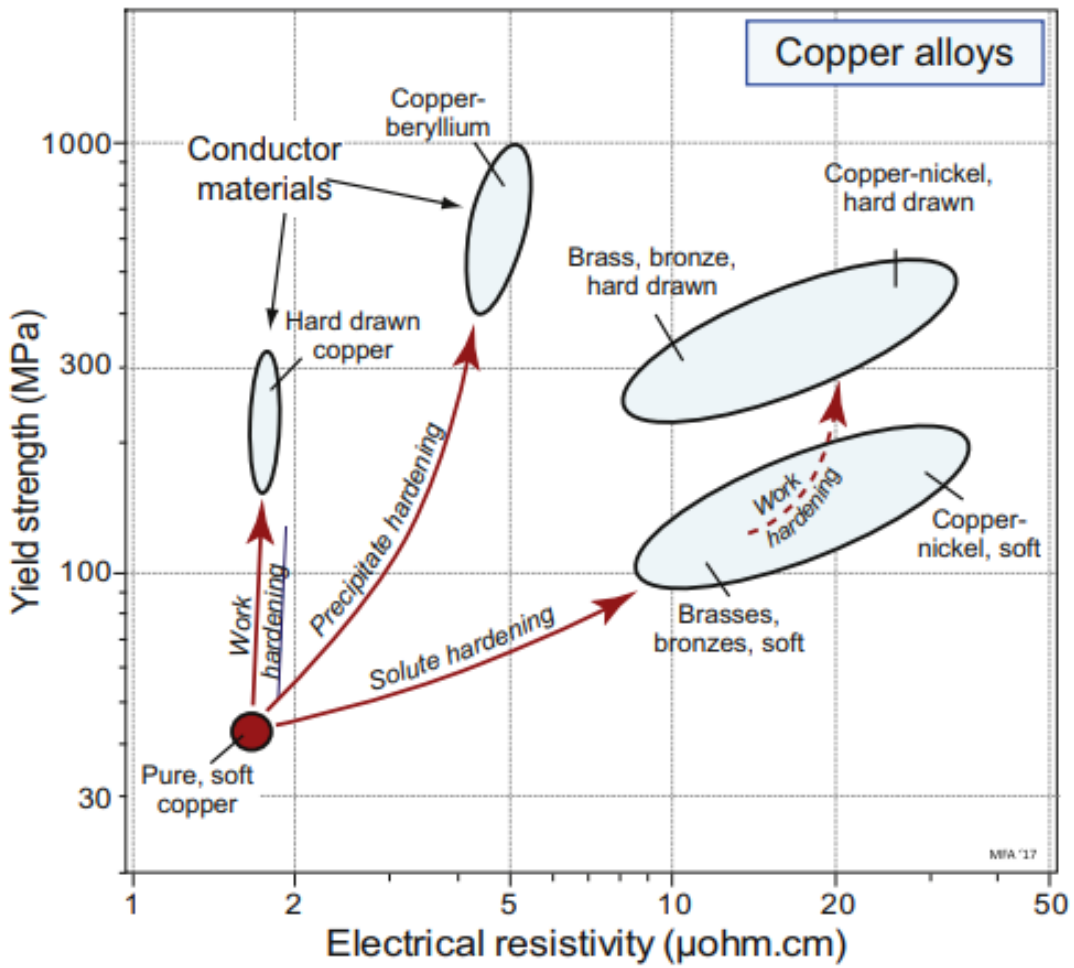
$$\rho = \rho_{thermal} + \rho_{impurity} + \rho_{deformation}$$

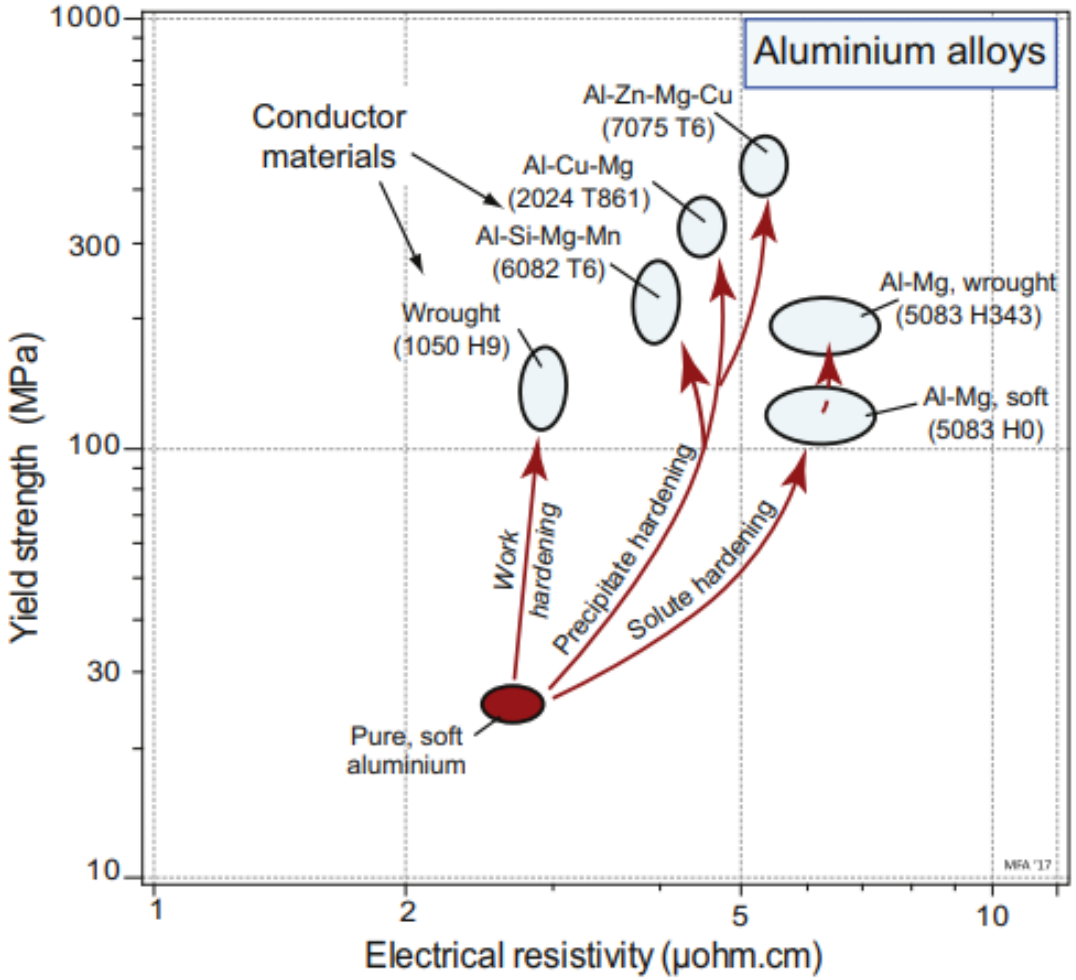
Matthiessen's rule

Metals: Influence of Temperature and Impurities on Resistivity



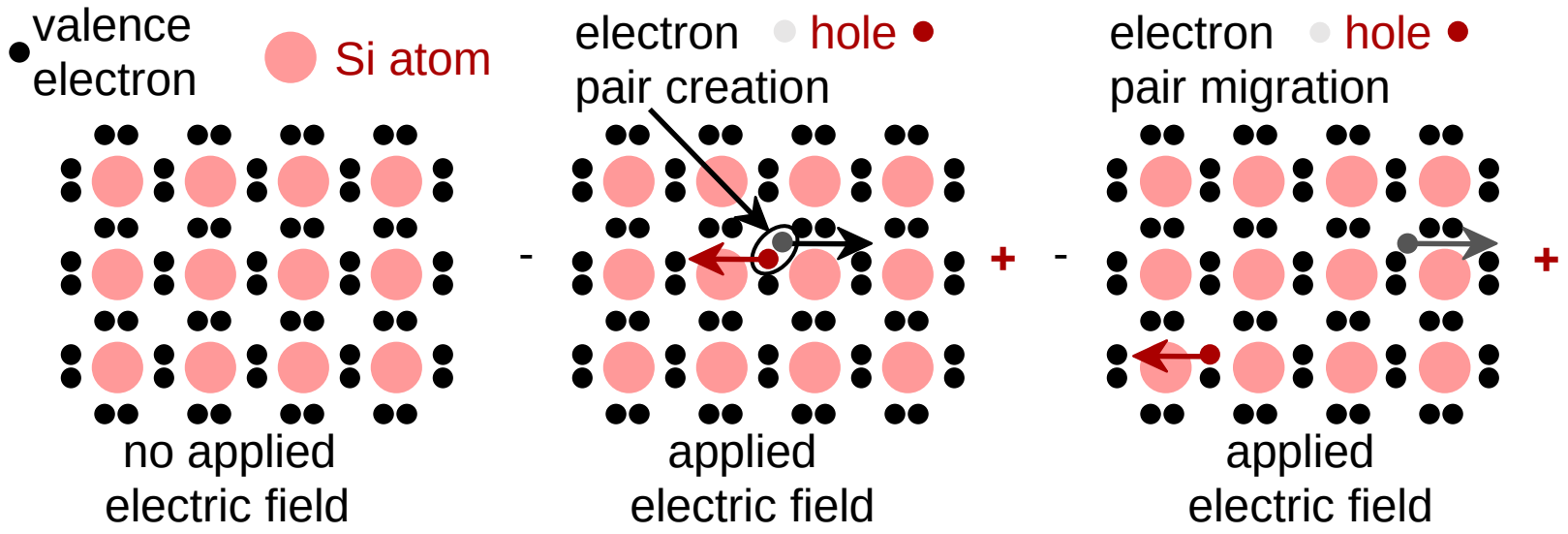
Resistivity in Metals





Semiconductors - Intrinsic Semiconduction in Terms of Electron and Hole Migration

- Concept of electrons and holes:



- Electrical Conductivity given by:

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

holes/m³

hole mobility

electrons/m³

electron mobility

- for intrinsic semiconductor $n = p = n_i$

$$\therefore \quad \sigma = n_i|e|(\mu_e + \mu_h)$$

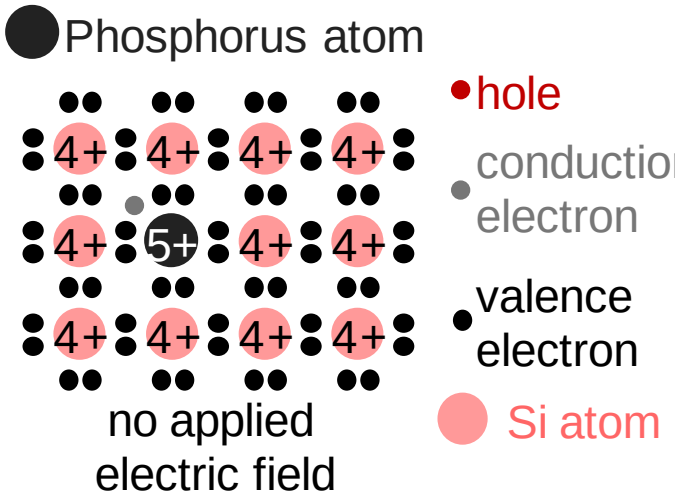
Intrinsic vs Extrinsic Conduction

- Intrinsic:
  -- case for pure Si
  -- # electrons = # holes ($n = p$)

- Extrinsic:
  -- electrical behavior is determined by presence of impurities that introduce excess electrons or holes
  -- $n \neq p$

- *n*-type Extrinsic: ($n >> p$)

$\sigma \approx n|e|\mu_e$



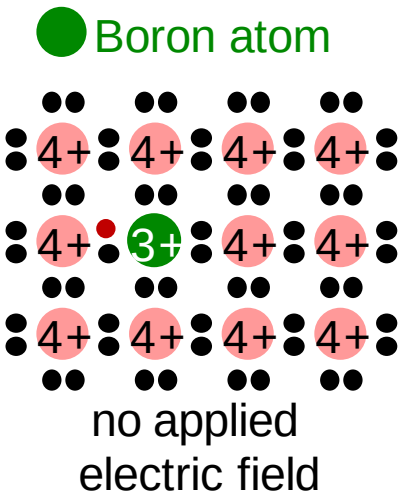
hole

conduction electron

valence electron

Si atom

- *p*-type Extrinsic: ($p >> n$)



$\sigma \approx p|e|\mu_h$

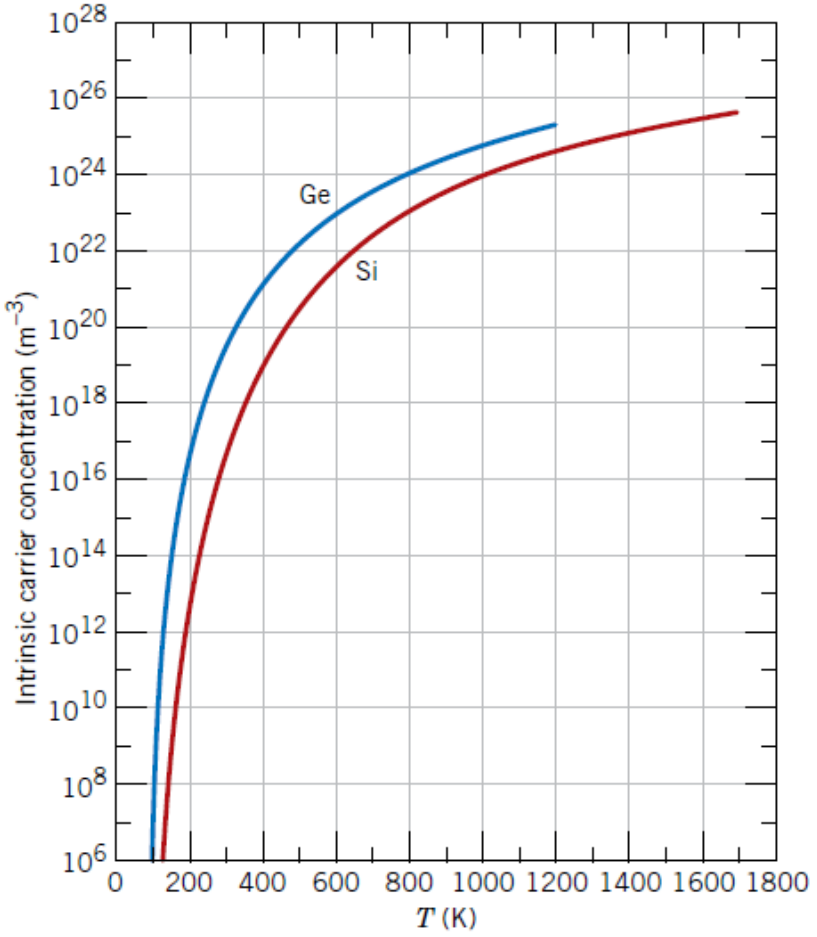
Under electric field, electron moves

Under electric field, hole moves

Carrier concentration with respect to temperature

For both intrinsic and extrinsic semiconduction, carrier concentration is a function of temperature.

Intrinsic



Intrinsic carrier concentration (logarithmic scale) as a function of temperature for germanium and silicon.

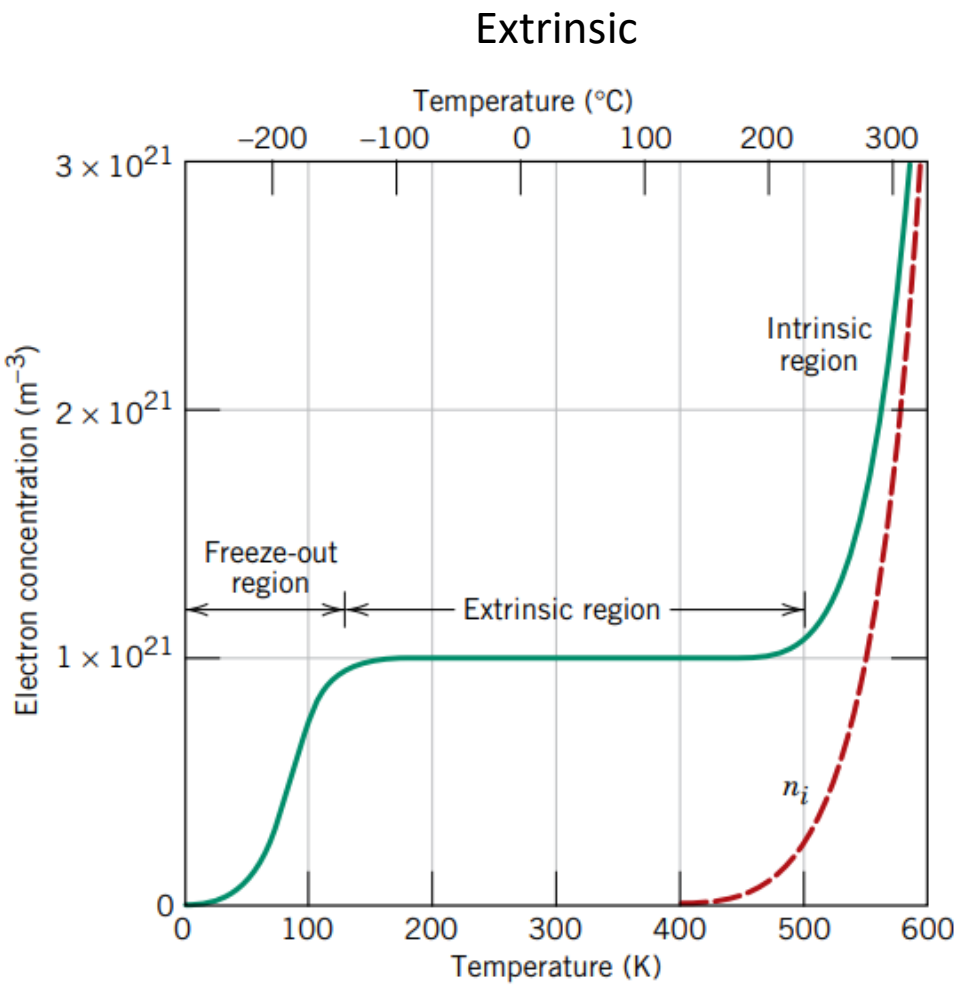
Extrinsic

the concentrations of electrons and holes increase with temperature.

at all temperatures, carrier concentration in Ge is greater than in Si. This effect is due to germanium's smaller band gap (0.67 vs. 1.11 eV)

Carrier concentration with respect to temperature

For both intrinsic and extrinsic semiconduction, carrier concentration is a function of temperature.



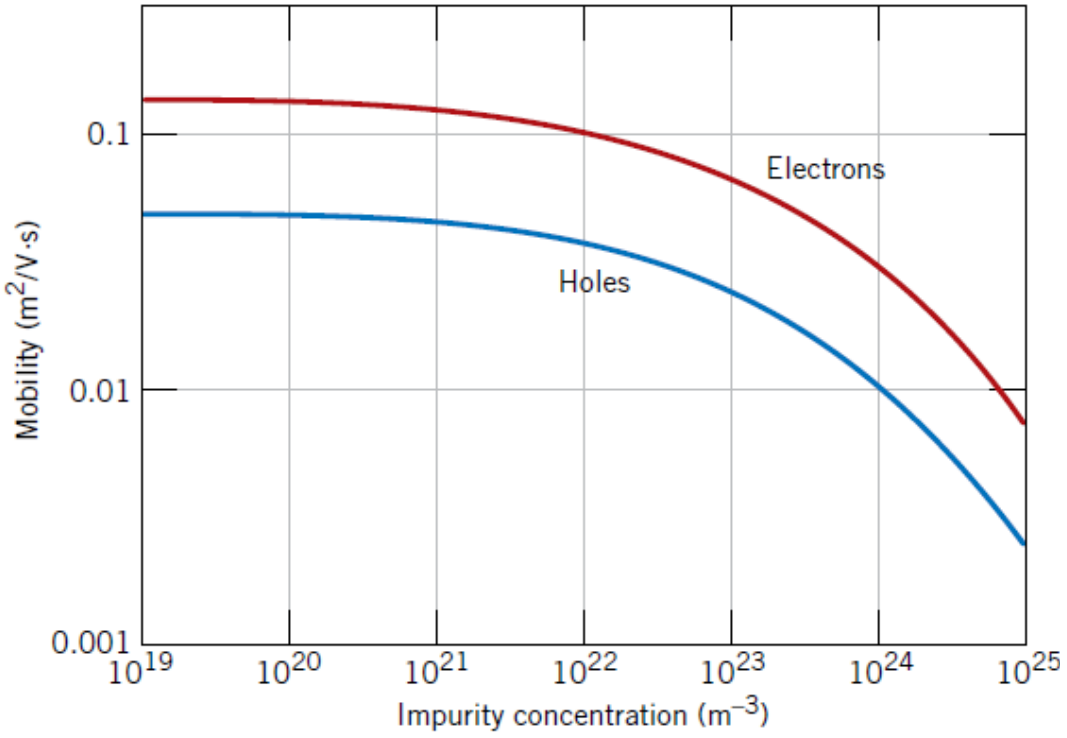
Electron concentration versus temperature for silicon (n-type) that has been doped with 10^{21} m^{-3} of a donor impurity and for intrinsic silicon (dashed line).

At intermediate temperatures (between approximately 150 K and 475 K) the material is n-type (inasmuch as P is a donor impurity), and electron concentration is constant. Also, intrinsic excitations across the band gap are insignificant in relation to these extrinsic donor excitations.

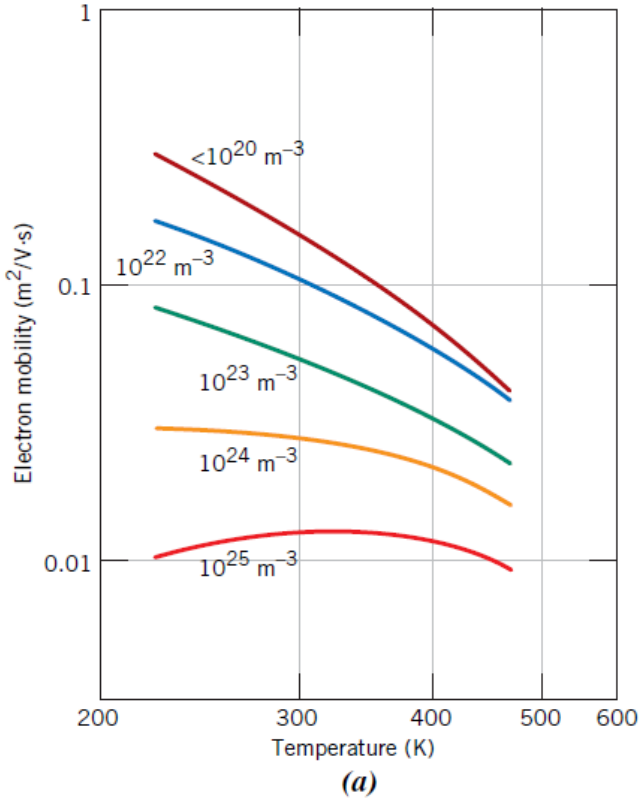
At low temperatures, the thermal energy is insufficient to excite electrons from the P donor level into the conduction band.

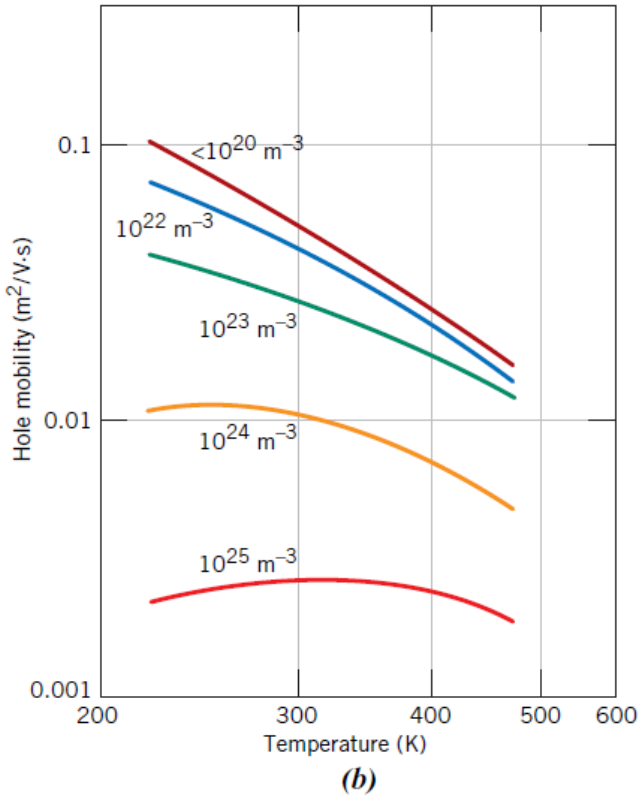
At high temperatures, charge carrier concentrations resulting from electron excitations across the band gap first become equal to and then completely overwhelm the donor carrier contribution with rising temperature.

Mobility of charge carriers with respect to dopant and temperature



For silicon, dependence of room-temperature electron and hole mobilities (logarithmic scale) on dopant concentration (logarithmic scale).





Temperature dependence of (*a*) electron and (*b*) hole mobilities for silicon that has been doped with various donor and acceptor concentrations.

Calculate the electrical conductivity of intrinsic silicon at 150°C (423 K)

Calculate the electrical conductivity of intrinsic silicon at 150°C (423 K)

$$\begin{aligned}\sigma &= n_i|e|(\mu_e + \mu_h)\\ &= (4 \times 10^{19}\ \mathrm{m}^{-3})(1.6 \times 10^{-19}\ \mathrm{C})(0.06\ \mathrm{m}^2/\mathrm{V{\cdot}s} + 0.022\ \mathrm{m}^2/\mathrm{V{\cdot}s})\\ &= 0.52\ (\Omega{\cdot}\mathrm{m})^{-1}\end{aligned}$$

Silicon (n-type) that has been doped with 10^{23} m^{-3} of a phosphorus impurity. Compute conductivity at RT and at 373K.

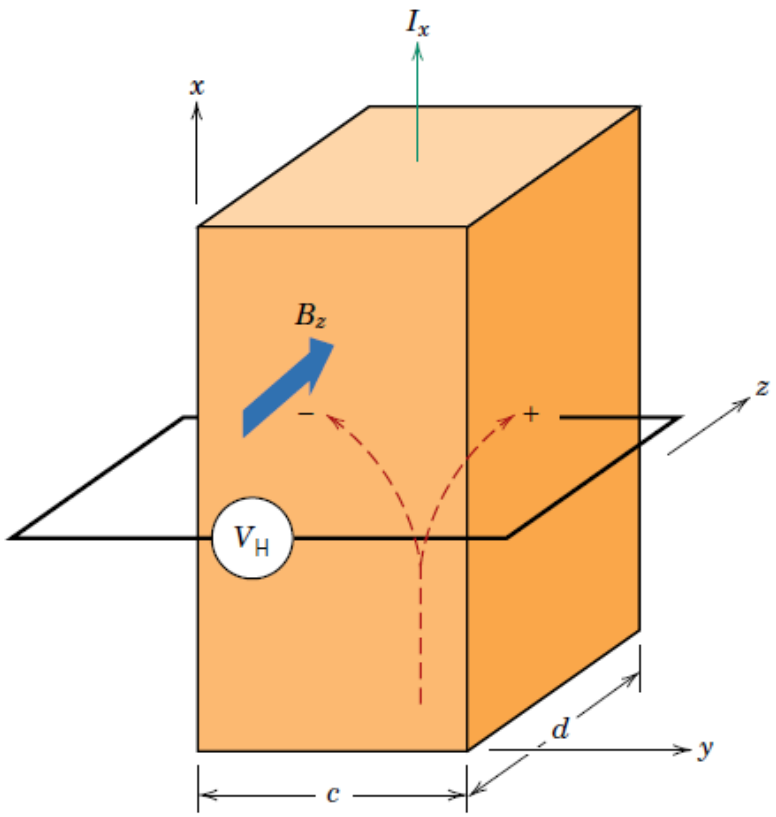
Silicon (n-type) that has been doped with 10^{23} m^{-3} of a phosphorus impurity. Compute conductivity at RT and at 373K.

$$\begin{aligned}\sigma &= n|e|\mu_e \\ &= (10^{23}\,\mathrm{m}^{-3})(1.6\times10^{-19}\,\mathrm{C})(0.07\,\mathrm{m}^2/\mathrm{V\cdot s}) \\ &= 1120\,(\Omega\cdot\mathrm{m})^{-1}\end{aligned}$$

$$\begin{aligned}\sigma &= n|e|\mu_e \\ &= (10^{23}\,\mathrm{m}^{-3})(1.6\times10^{-19}\,\mathrm{C})(0.04\,\mathrm{m}^2/\mathrm{V\cdot s}) \\ &= 640\,(\Omega\cdot\mathrm{m})^{-1}\end{aligned}$$

THE HALL EFFECT

To determine concentration of charge carrier, and mobility



A magnetic field applied perpendicular to the direction of motion of a charged particle exerts a force on the particle perpendicular to both the magnetic field and the particle motion directions.

The magnitude of V_H depends on I_x, B_z, and the specimen thickness d as

$$V_H = \frac{R_H I_x B_z}{d}$$

R_H is termed the *Hall coefficient*

$$R_H = \frac{1}{n|e|}$$

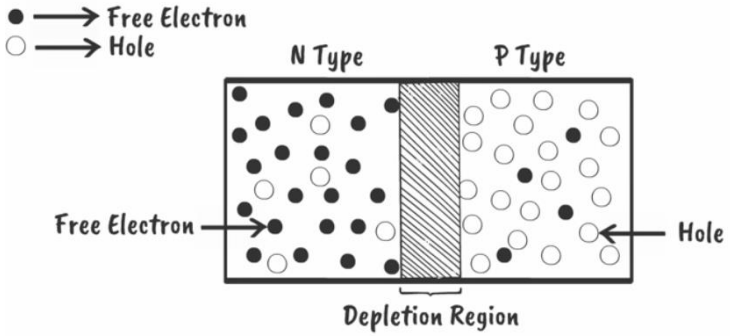
(From this, n can be determined)

Also,

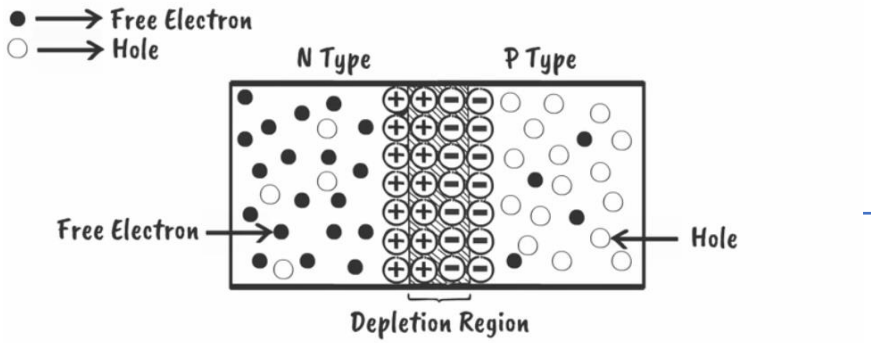
$$\mu_e = \frac{\sigma}{n|e|}$$

Semiconductor Devices

PN Diode



n type material is electrically neutral => free electrons = positive donor ions
p type material is electrically neutral => holes = negative acceptor ions



n type material is electrically neutral => free electrons = positive donor ions
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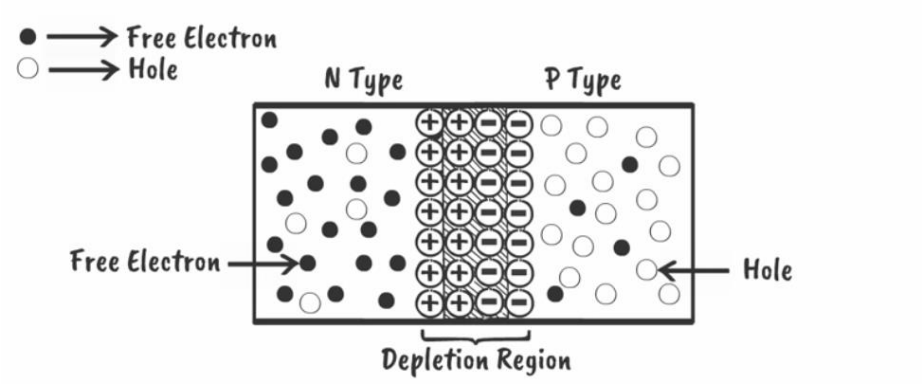
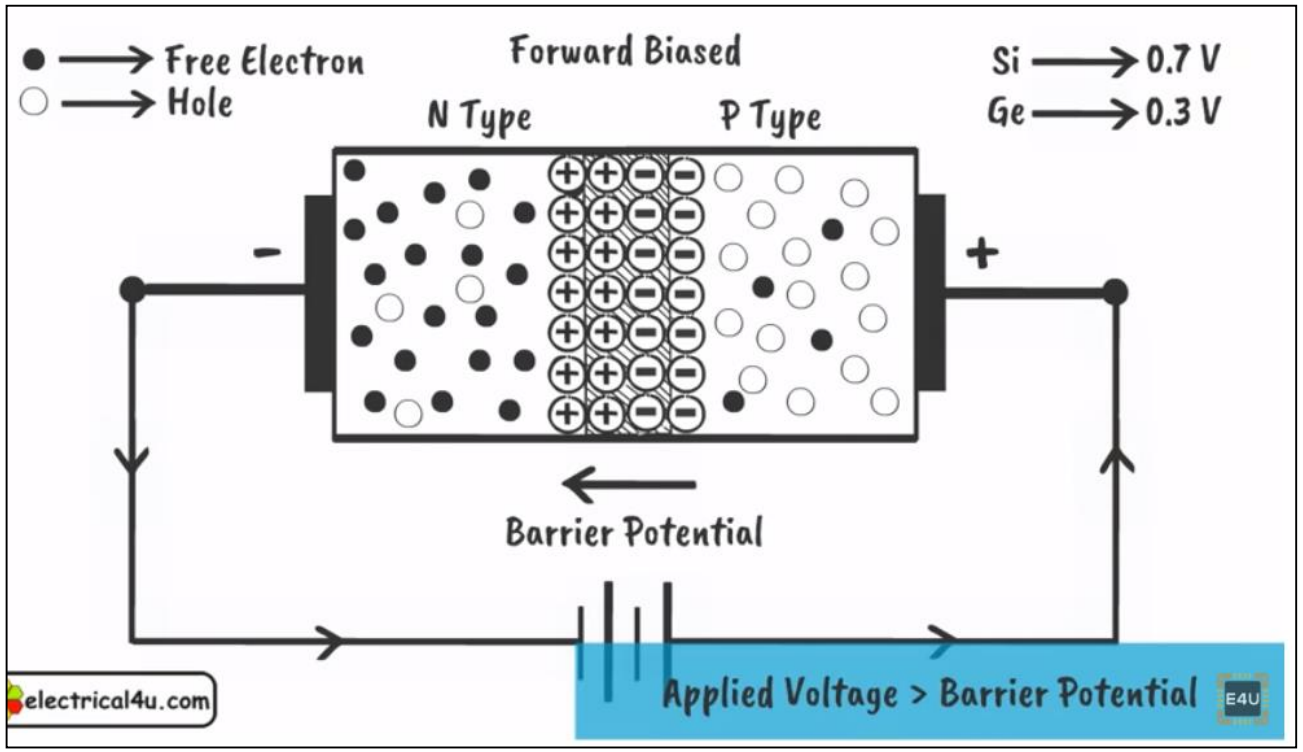


n type material is electrically neutral => free electrons = positive donor ions
p type material is electrically neutral => holes = negative acceptor ions

Depletion region generates static electric field which acts as barrier for further migration of electrons and holes across the junction

Semiconductor Devices

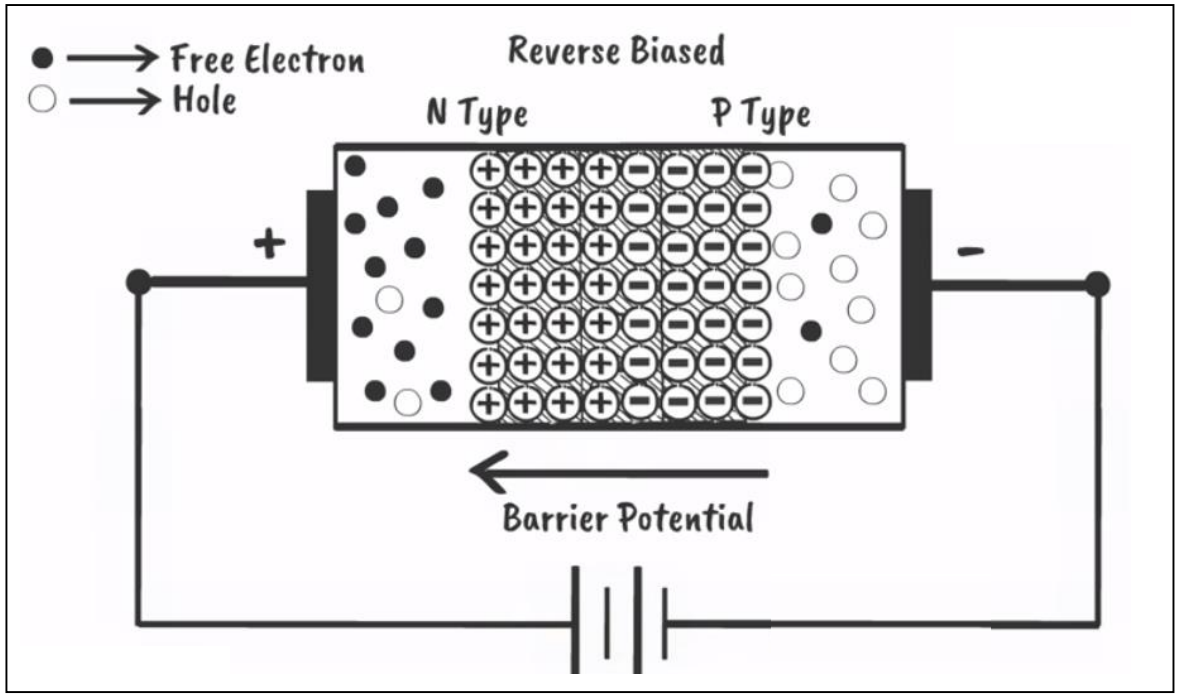
PN Diode



Only minor current flows - due to minority carrier – (electrons in p type and holes in n type) – so reverse bias blocks current

Depletion region generates static electric field which acts as barrier for further migration of electrons and holes across the junction

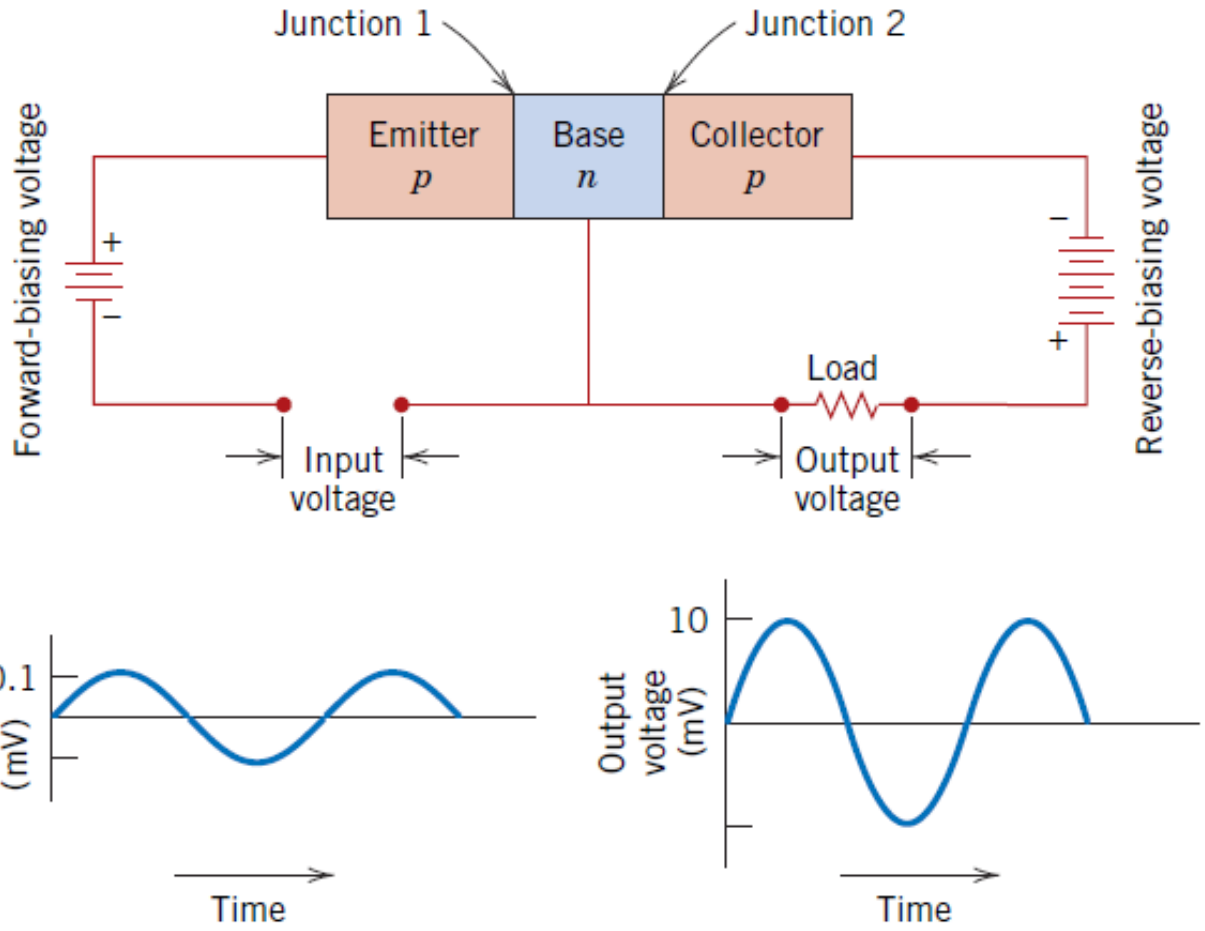
Current flows when applied voltage > barrier potential in forward bias

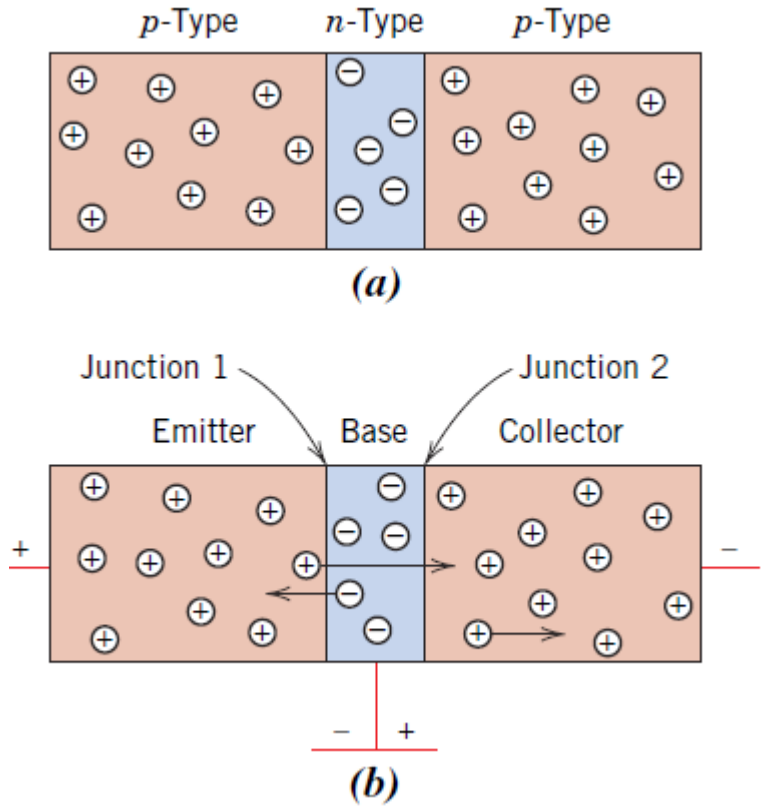


Semiconductor Devices

Transistor - For signal amplifications

PNP Junction Transistor



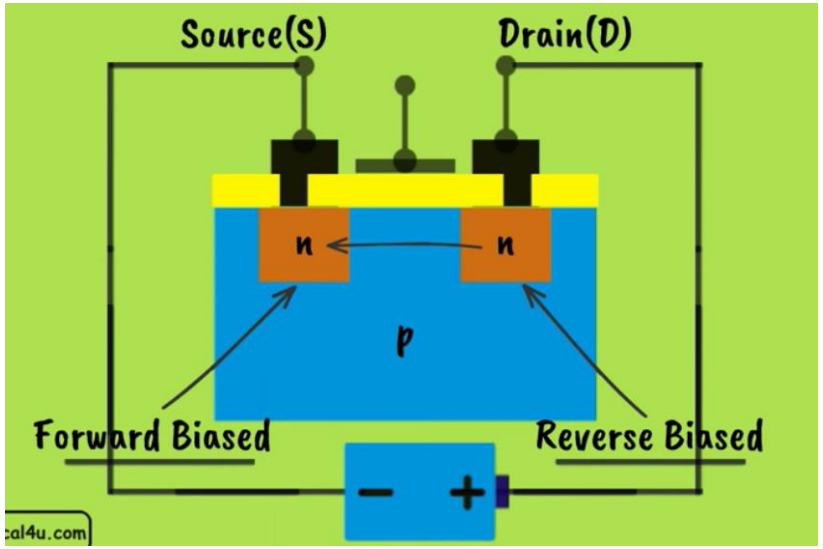


For a junction transistor (p–n–p type), the distributions and directions of electron and hole motion (a) when no potential is applied and (b) with appropriate bias for voltage amplification.

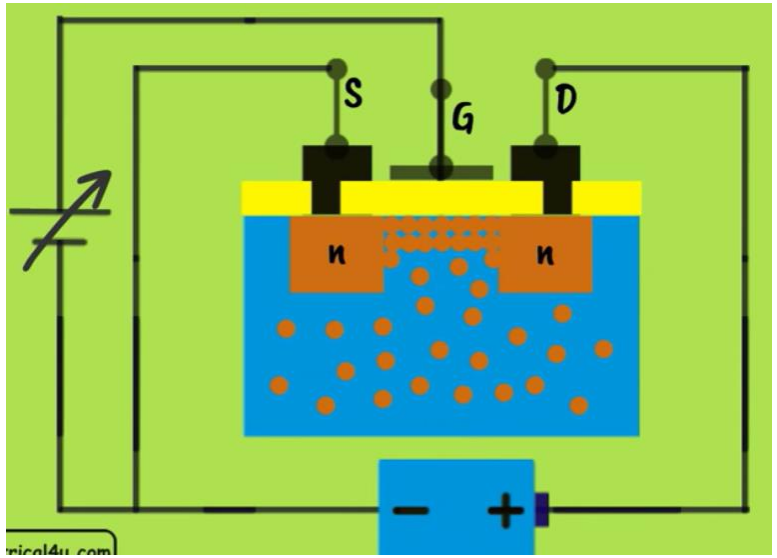
Semiconductor Devices

Transistor

MOSFET *metal-oxide-semiconductor field-effect transistor*



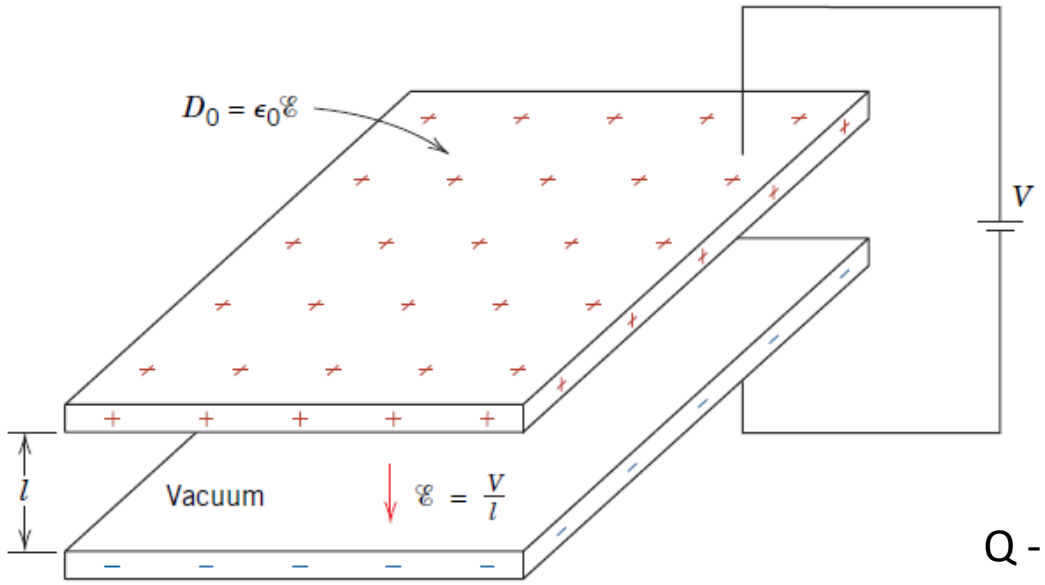
No channel between Drain and source



When gate is connected to suitable positive voltage, accumulation of attracted minority carriers (negative electrons) in P-type as shown takes place – form a channel inside p-type semiconductor – and thus channel behaves as n-type semiconductor – thus the device will conduct current as electron are attracted towards +ve terminal of the drain as the gate voltage is increased – Result is electrons will start flowing from source to drain through the channel.

Capacitor

A parallel-plate capacitor (when a vacuum is present in-between)



Capacitance *C*

$$C = \frac{Q}{V} = \epsilon_0 \frac{A}{l}$$

The units of capacitance are coulombs per volt, or farads (F).

Q - quantity of charge stored on either plate

V - the voltage applied across the capacitor

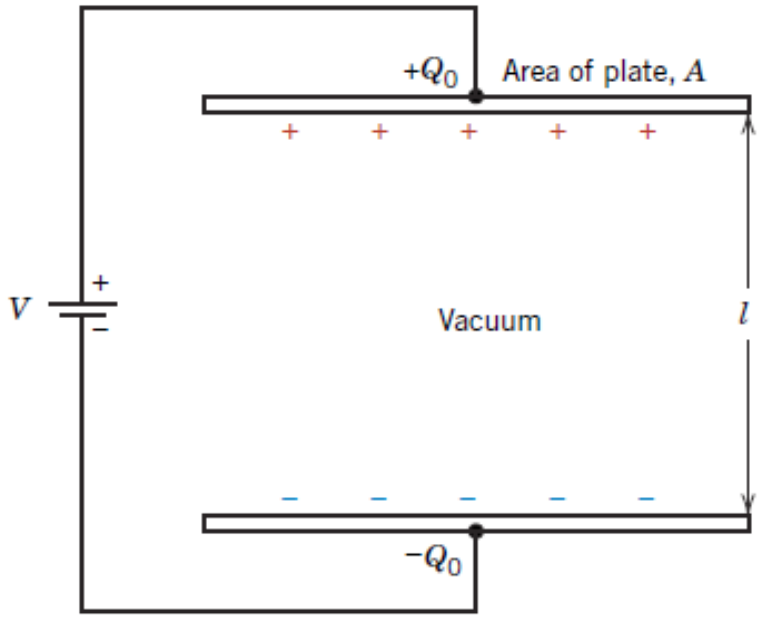
A represents the area of the plates

l is the distance between them.

Parameter ε_0, called the **permittivity** of a vacuum, is a universal constant having the value of 8.85 x10^{-12} F/m.

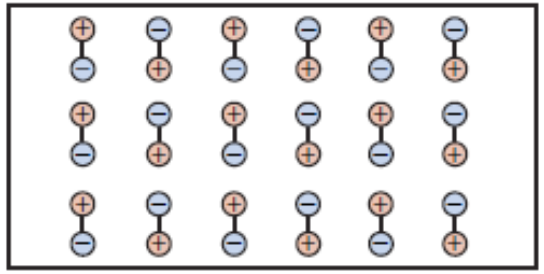
Dielectric Behaviour

Capacitor with vacuum in-between

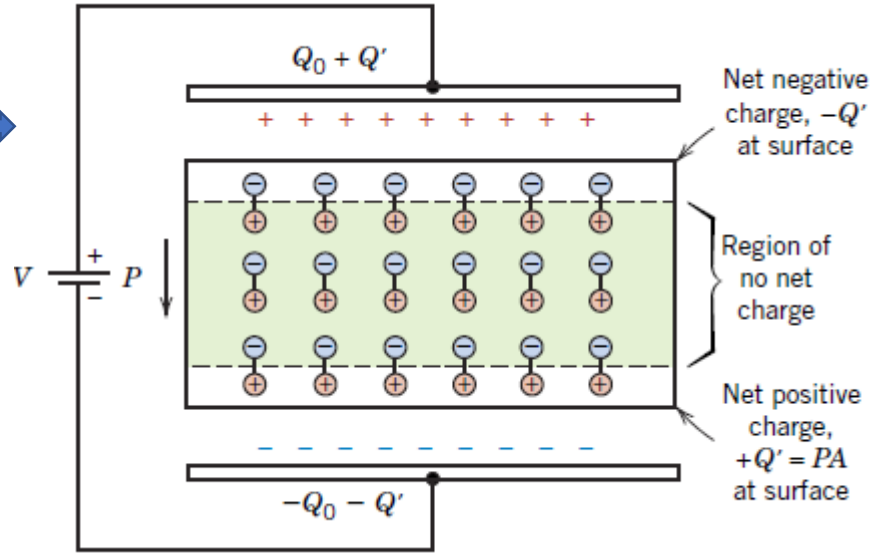


The charge stored on capacitor plates for a vacuum

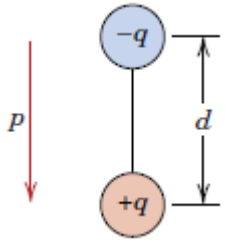
Because of dielectric, the amount of charge stored per unit volt increased



Capacitor with dielectric material in-between



The increased charge-storing capacity resulting from the **polarization of a dielectric material. P is increase in charge density because of dielectric.**

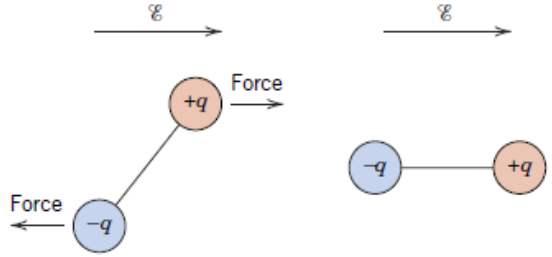


$$p = qd$$

p = dipole moment

A **dielectric** material is one that is electrically insulating (nonmetallic) and exhibits or may be made to exhibit an **electric dipole** structure—that is, there is a separation of positive and negative electrically charged entities.

Dipole under electric field



Dielectric Behaviour

If a dielectric material is inserted into the region within the plates

$$C = \epsilon \frac{A}{l}$$

ε is the **permittivity** dielectric medium

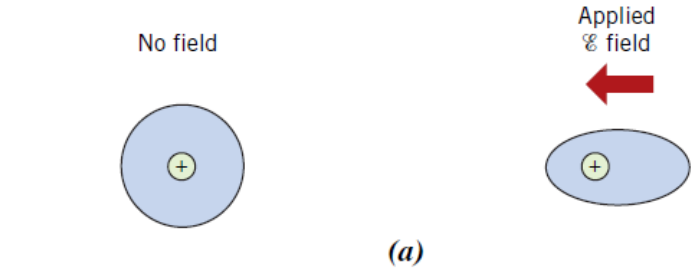
Relative permittivity or Dielectric constant, ε_r, is given by

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

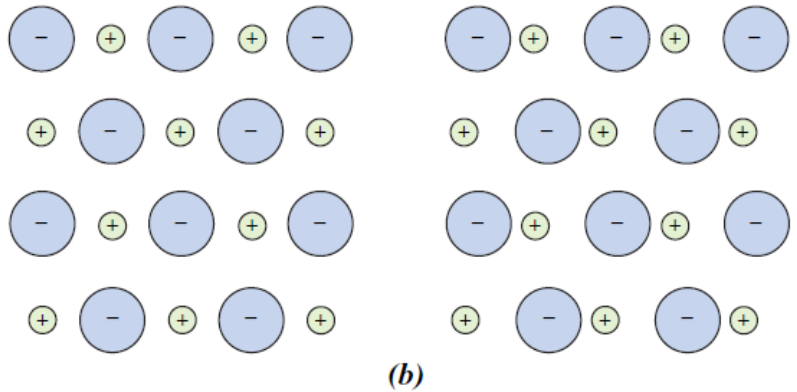
which is greater than unity and represents the **increase in charge-storing capacity upon insertion of the dielectric medium between the plates.**

The dielectric constant is one material property of prime consideration for capacitor design.

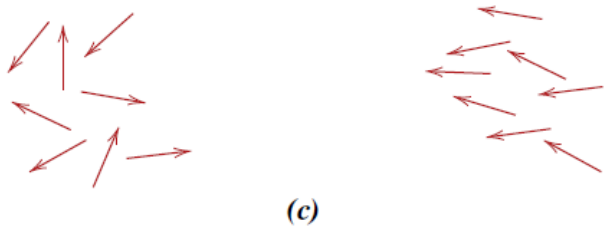
Types of Polarization



Electronic polarization that results from the distortion of an atomic electron cloud by an electric field.



Ionic polarization that results from the relative displacements of electrically charged ions in response to an electric field.

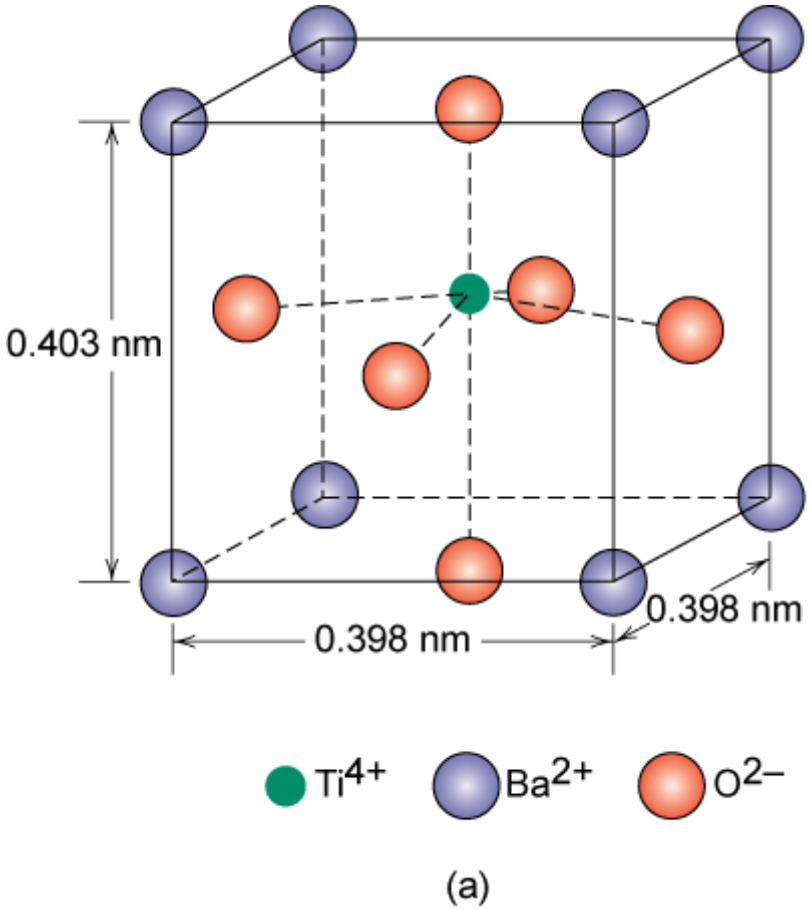


Response of permanent electric dipoles (arrows) to an applied electric field, producing **orientation polarization**.

Total polarization $P = P_e + P_i + P_o$

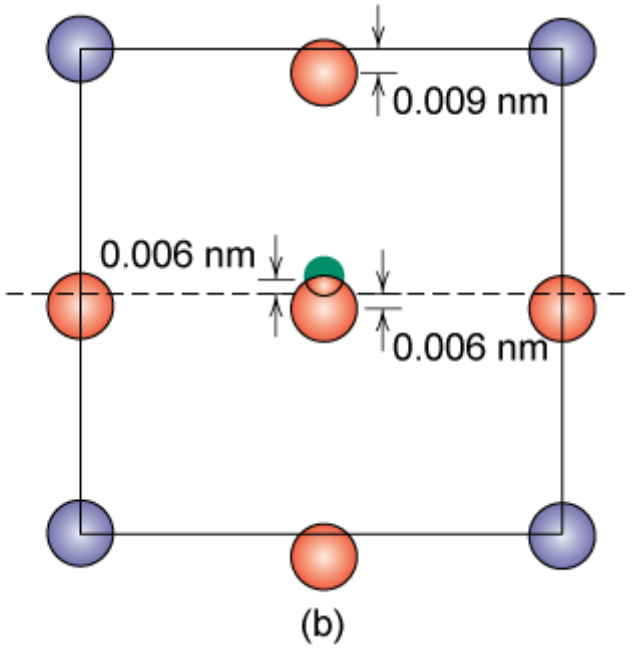
Ferroelectricity

- Experience spontaneous polarization (in the absence of electric field)
- Consists of permanent electric dipoles



Example:

$BaTiO_3$ -- ferroelectric (below its Curie temperature (120ºC))



Spontaneous polarization results because of interactions between adjacent permanent dipoles wherein they mutually align, all in the same direction.

Ferroelectrics have extremely high dielectric constants;

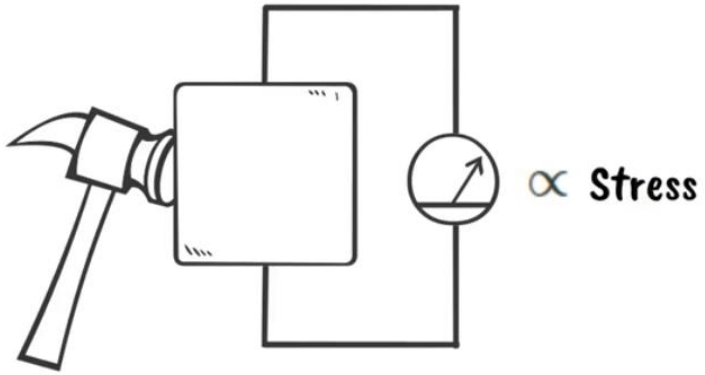
for example, at room temperature, ε_r for barium titanate may be as high as 5000.

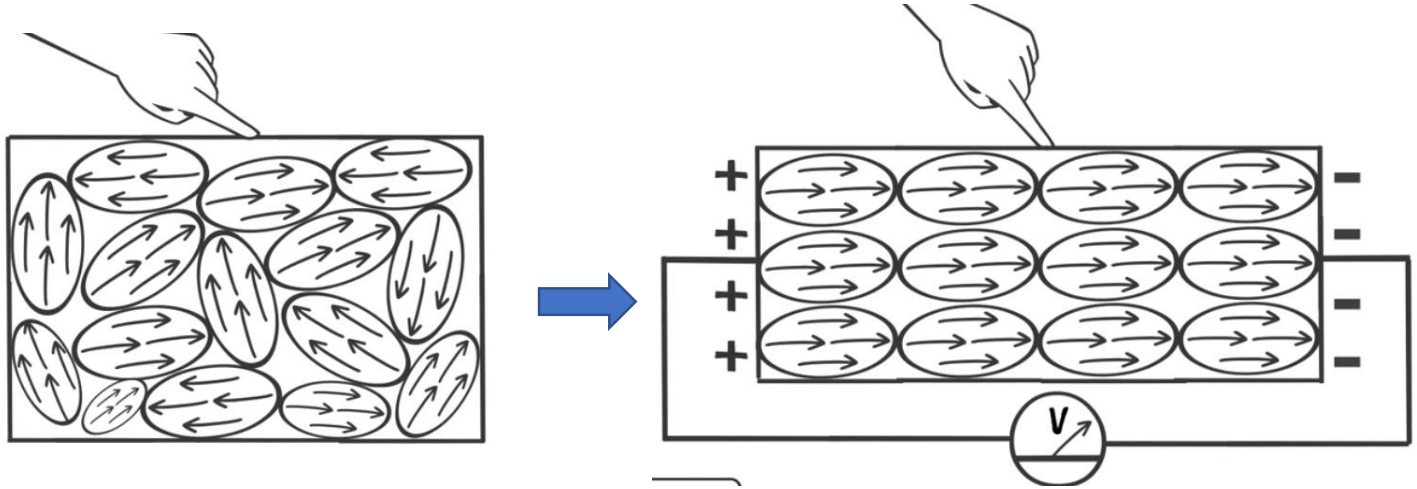
Consequently, capacitors made from these materials can be significantly smaller than capacitors made from other dielectric materials.

PIEZOELECTRICITY

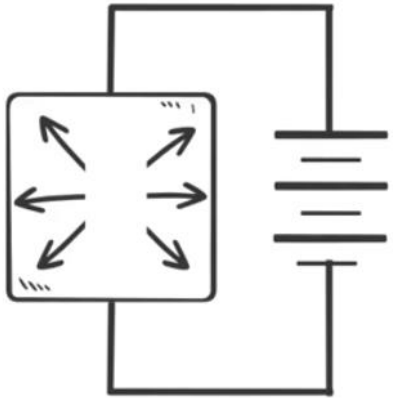
Piezoelectric ceramic materials include Quartz, titanates of barium and lead (BaTiO3 and PbTiO3), lead zirconate (PbZrO3), lead zirconate–titanate (PZT).

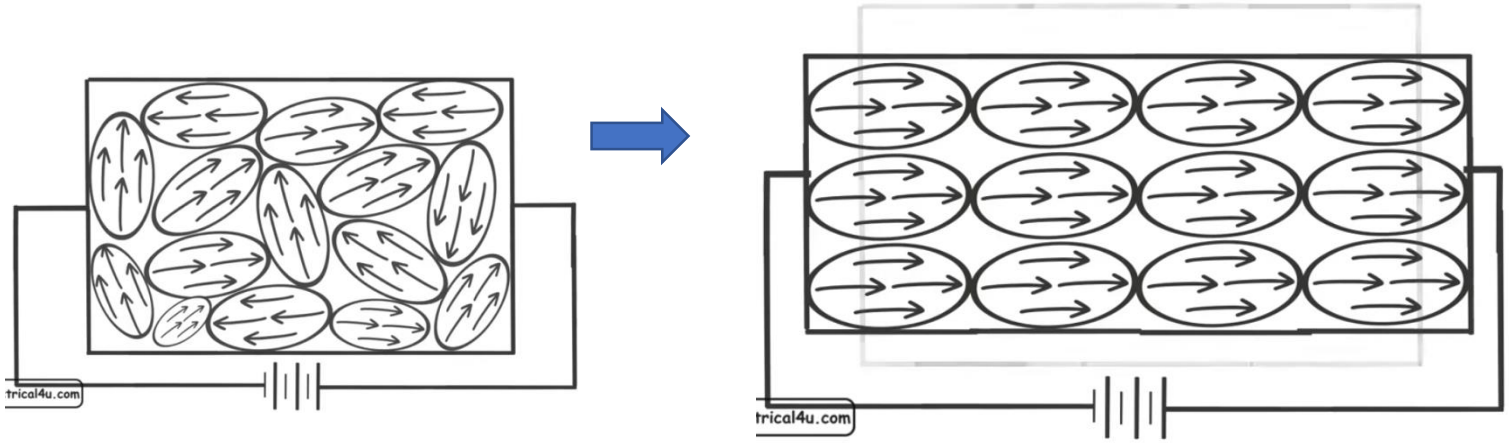
Application of stress (or strain) on a material produces electric potential – piezoelectric effect



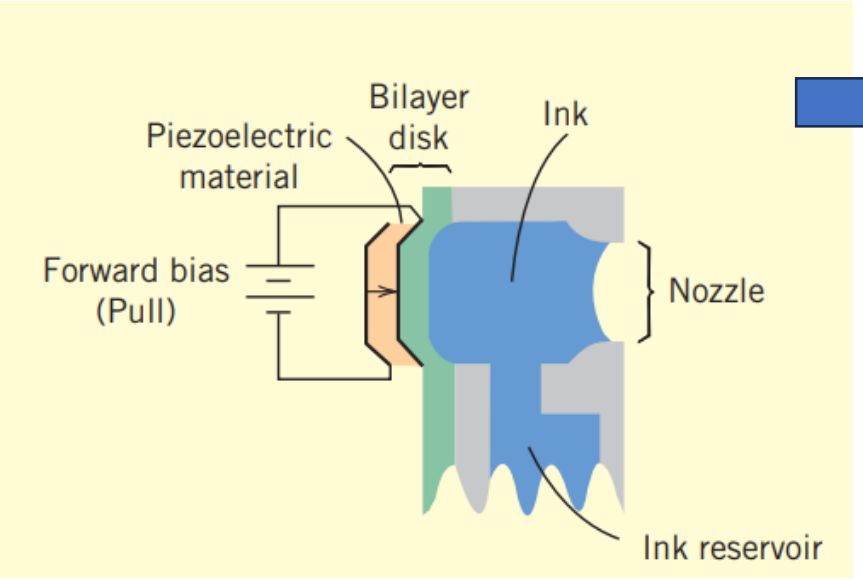


Application of electric filed on a material produces strain – inverse piezoelectric effect

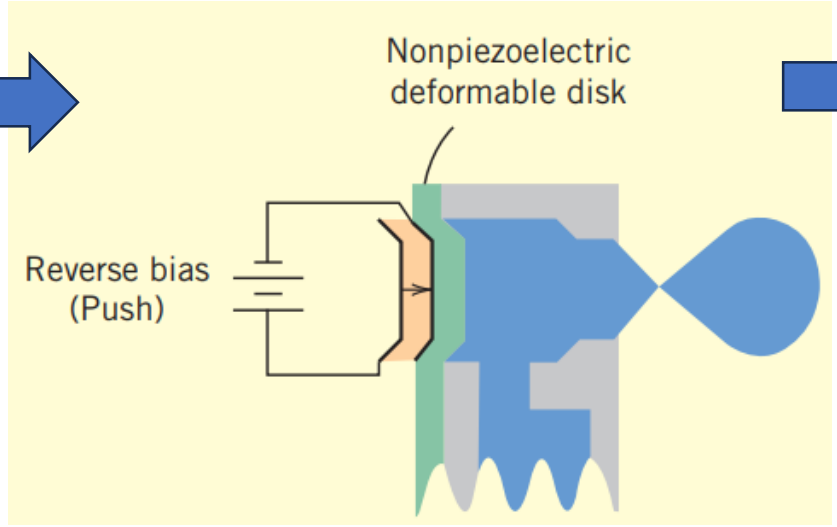




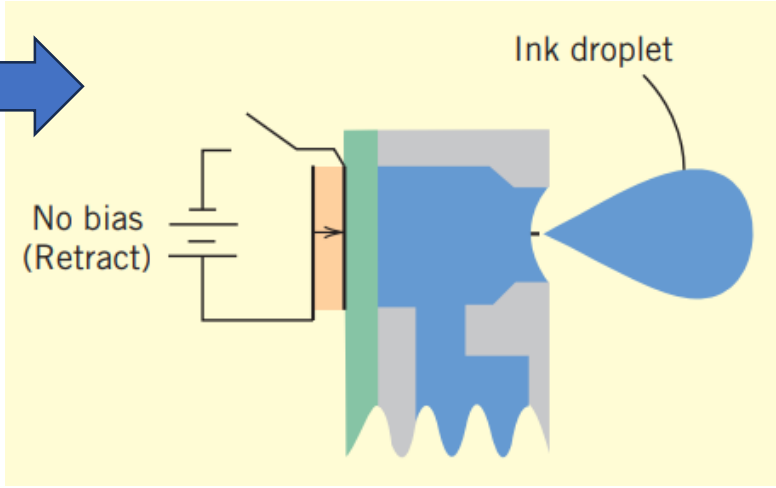
Piezoelectric Ink-Jet Printer Heads



the imposition of forward bias voltage causes the bilayer disk to flex in such a way as to pull (or draw) ink from the reservoir into the nozzle chamber

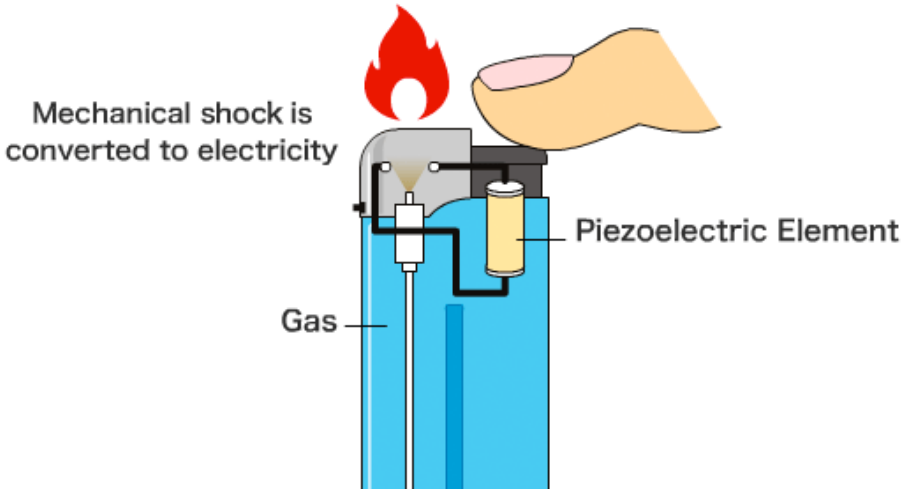


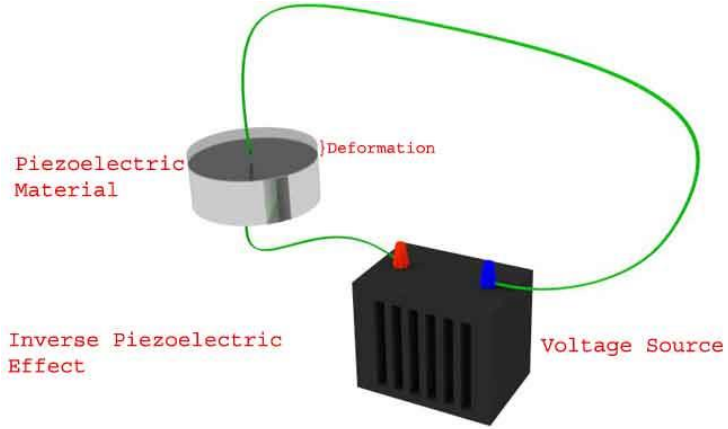
Reversing the voltage bias forces the bilayer disk to bend in the opposite direction—toward the nozzle—so as to eject a drop of ink

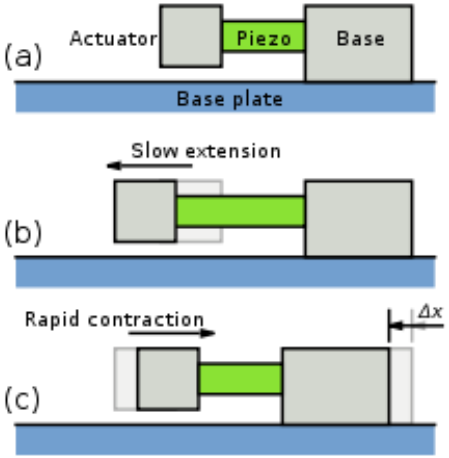


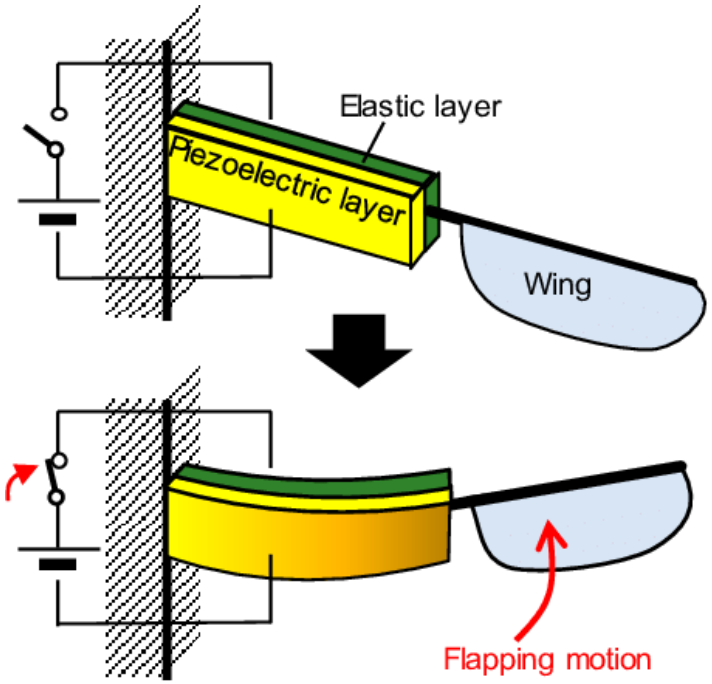
Finally, removal of the voltage causes the disk to return to its unbent configuration in preparation for another ejection sequence.

PIEZOELECTRICITY



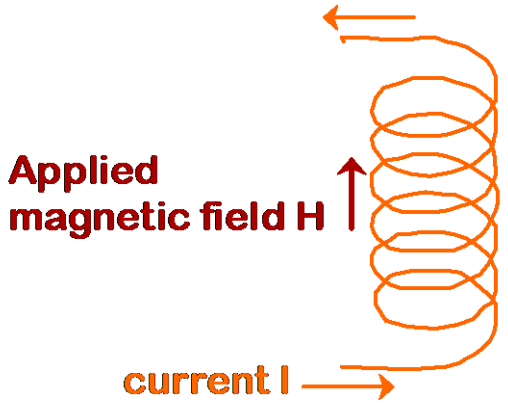






Magnetic Properties

- Created by current through a coil:



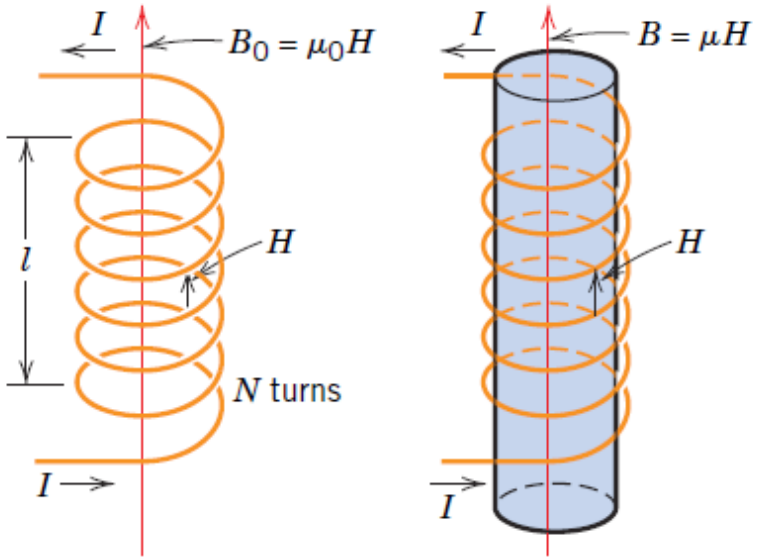
- Relation for the applied magnetic field, H:

$$H = \frac{NI}{L}$$

N = number of turns, I = current, L = length

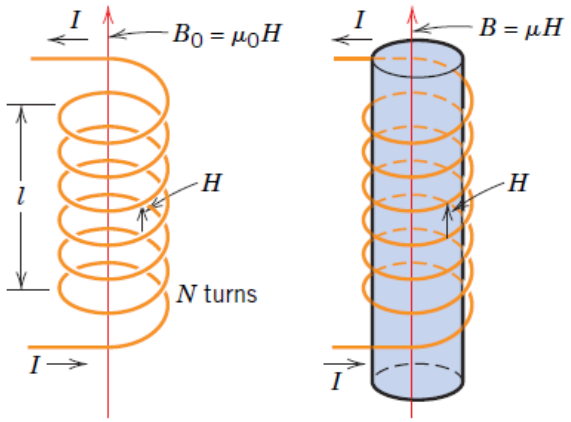
applied magnetic field (Magnetic field strength)

units = (ampere-turns/m)



Magnitude of the internal field strength within a substance that is subjected to an *H is called* **magnetic induction,** or **magnetic flux density (B)**

Magnetic induction



The magnetic field strength and flux density are related according to

$$B = \mu H$$

The parameter μ is called the **permeability,** which is a property of the specific medium. (Units – Wb/A.m or Henries/m)

In a vacuum, $B_0 = \mu_0 H$

where μ_0 is the *permeability of a vacuum,* a universal constant, which has a value of $4\pi \times 10^{-7}$ H/m

μ_r is called the *relative permeability* $\mu_r = \dfrac{\mu}{\mu_0}$

Measure of the degree to which the material can be magnetized, or the ease with which a 'B' field can be induced in the presence of an external 'H' field.

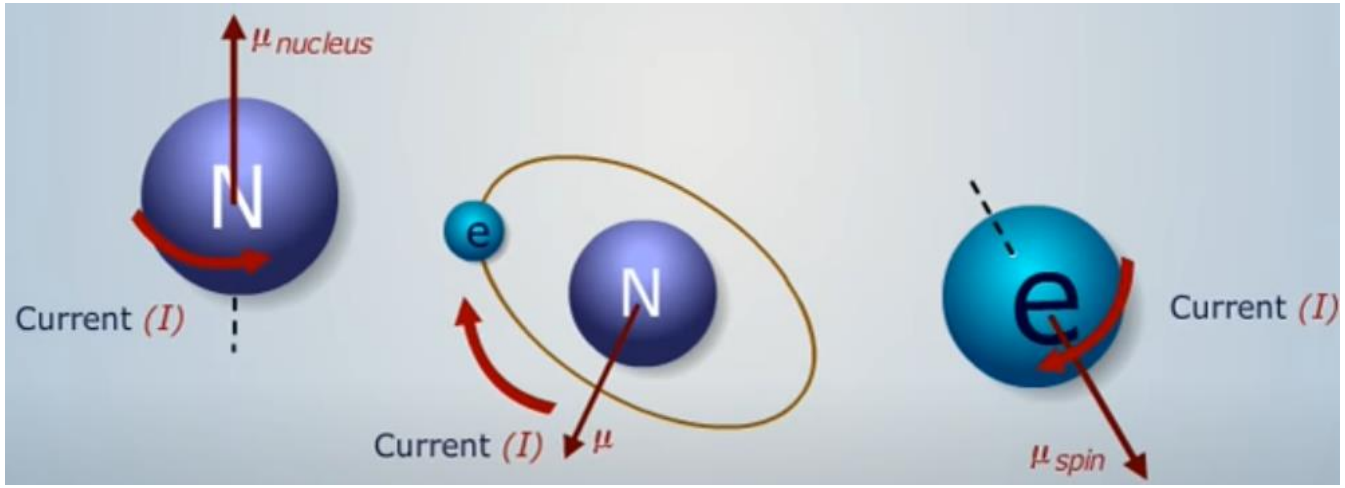
Another property, M, called the magnetization of the solid, is defined by the expression: $B = \mu_0 H + \mu_0 M$

In the presence of an 'H' field, the magnetic moments within a material tend to become aligned with the field and tend to reinforce it; the term $\mu_0 M$ in the Equation is a measure of this contribution.

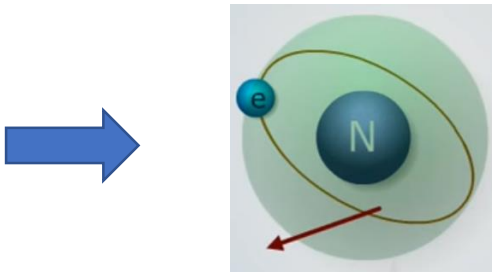
However, the magnitude of M is related to the applied field as follows: $M = \chi_m H$

χ_m is called the magnetic susceptibility and is related to relative permeability as $\chi_m = \mu_r - 1$

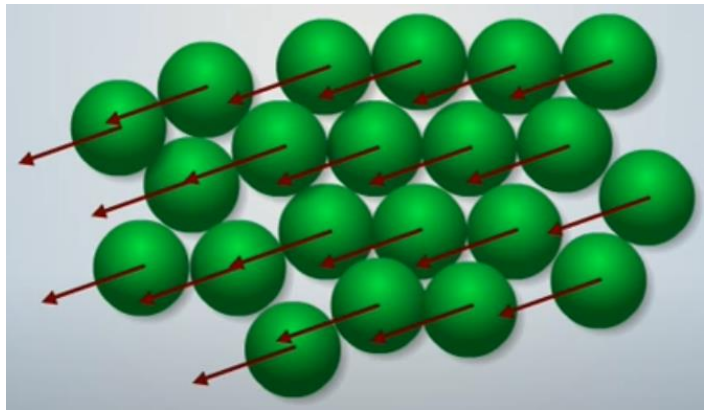
Atomic magnetic moments



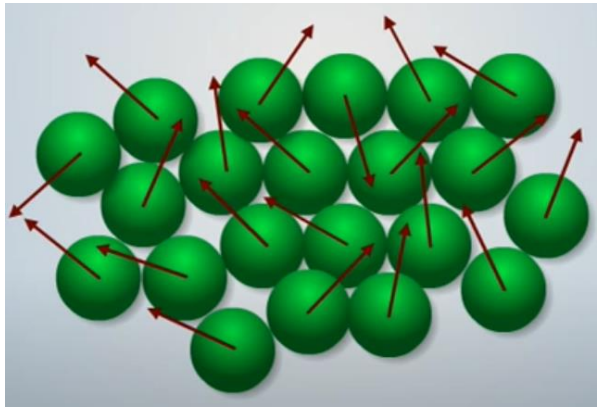
Net atomic magnetic moment



- **Electrons produce magnetic orbital, spin moments, nucleus produce spin moment**

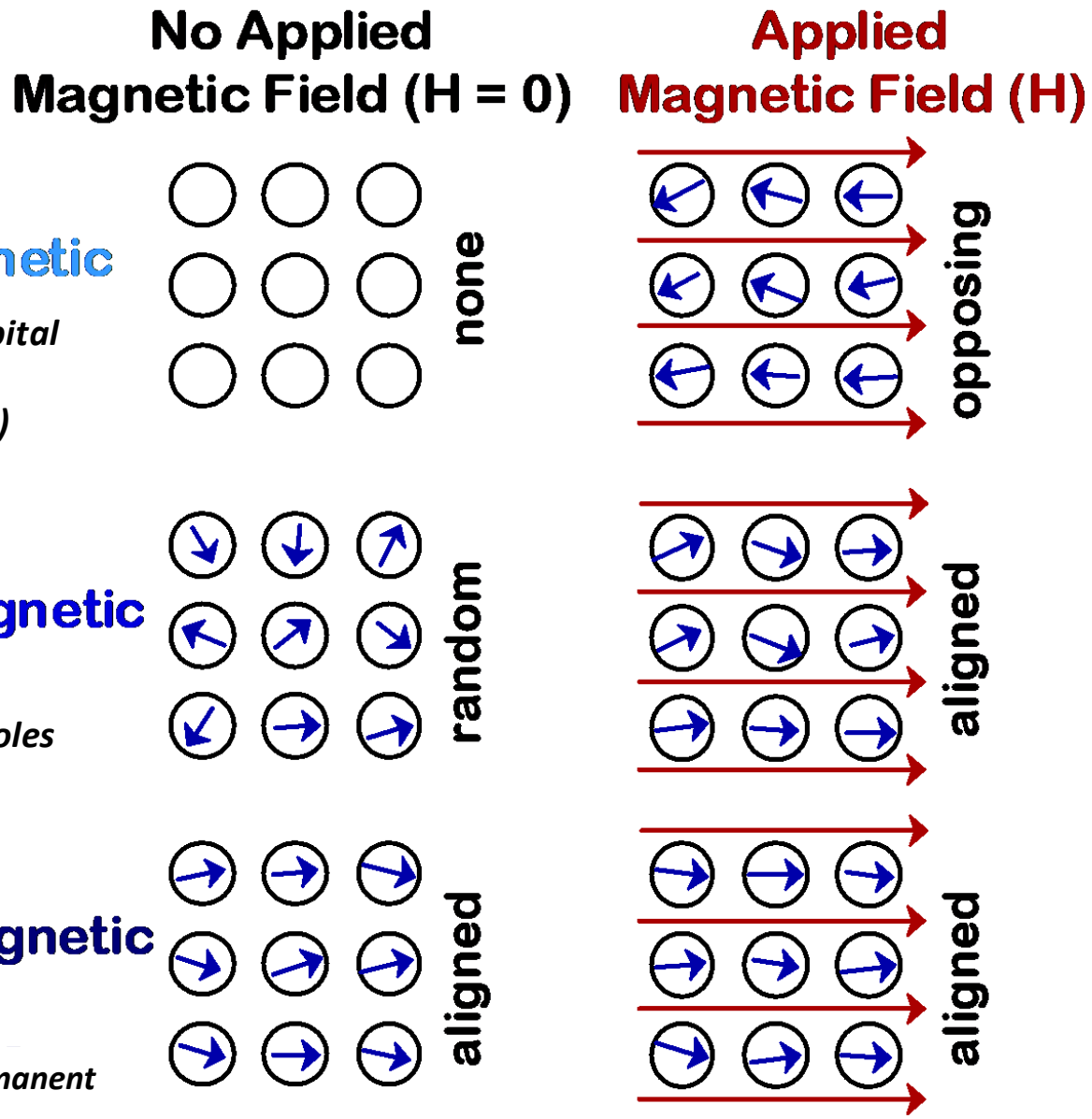


Aligned



Randomly oriented

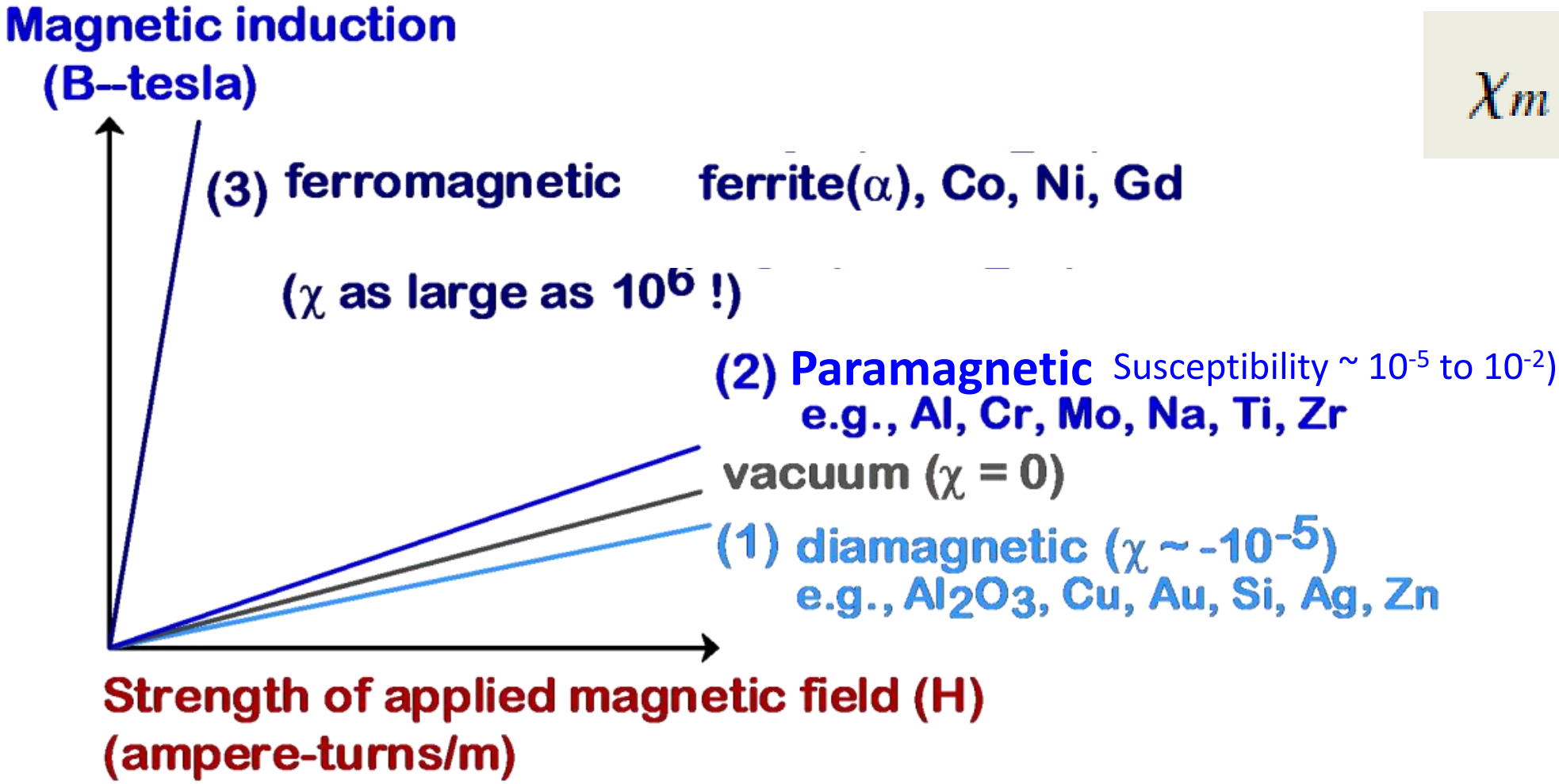
MAGNETIC MOMENTS FOR 3 TYPES



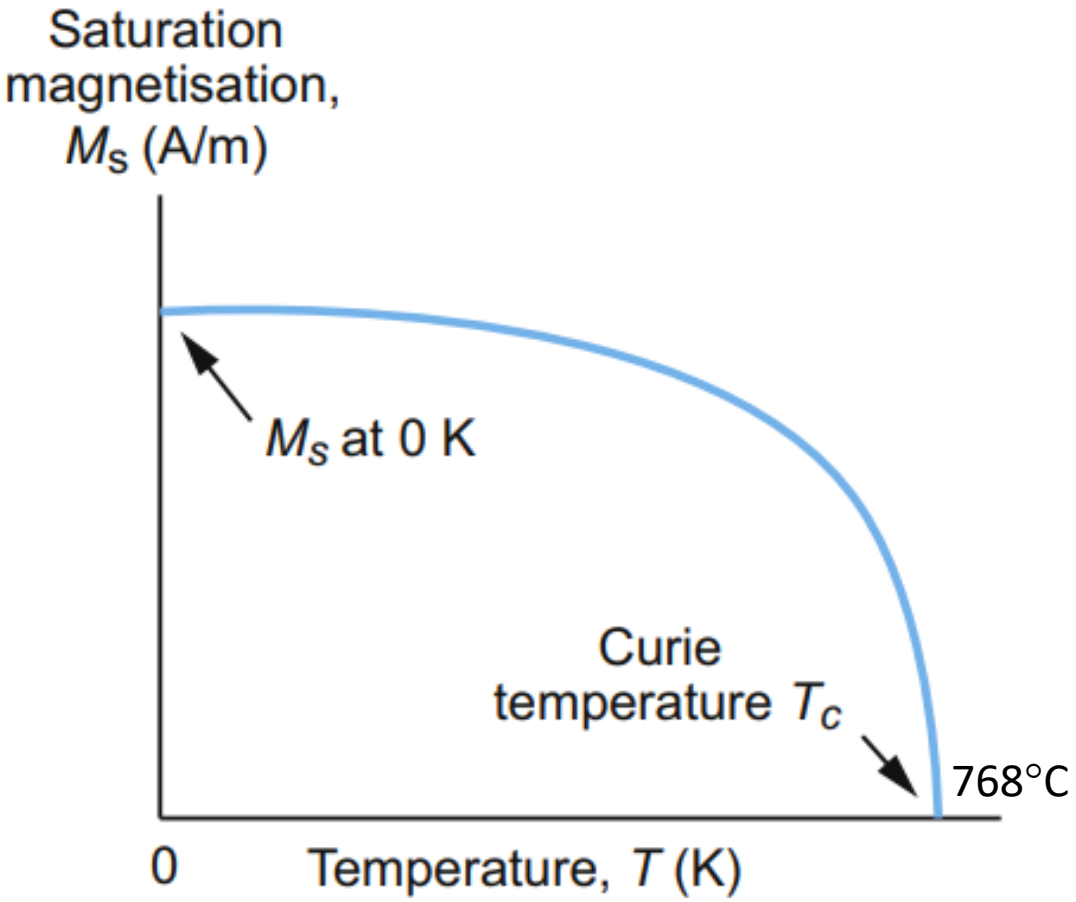
3 TYPES OF MAGNETISM

Magnetic susceptibility

$$\chi_m = \mu_r - 1$$



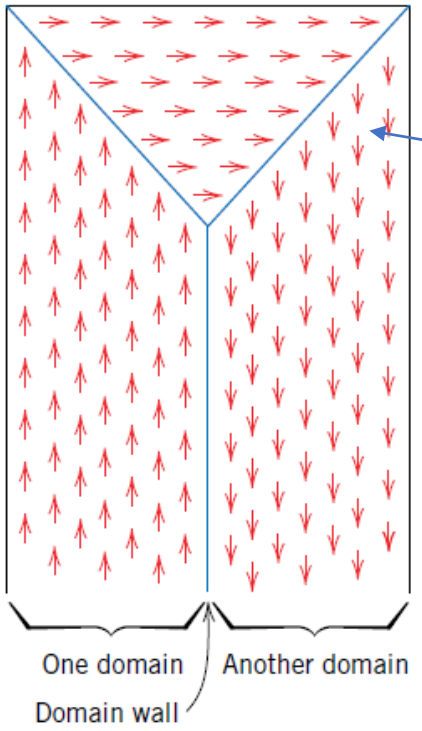
INFLUENCE OF TEMPERATURE ON MAGNETIC BEHAVIOR



With increasing temperature, the increased thermal motion of the atoms tend to randomize the direction of any moments that may be aligned.

With increasing temperature, the saturation magnetization decreases gradually and then abruptly drops to zero at what is called the **Curie temperature** T_c Above this temperature, the material is paramagnetic.

DOMAINS AND HYSTERESIS

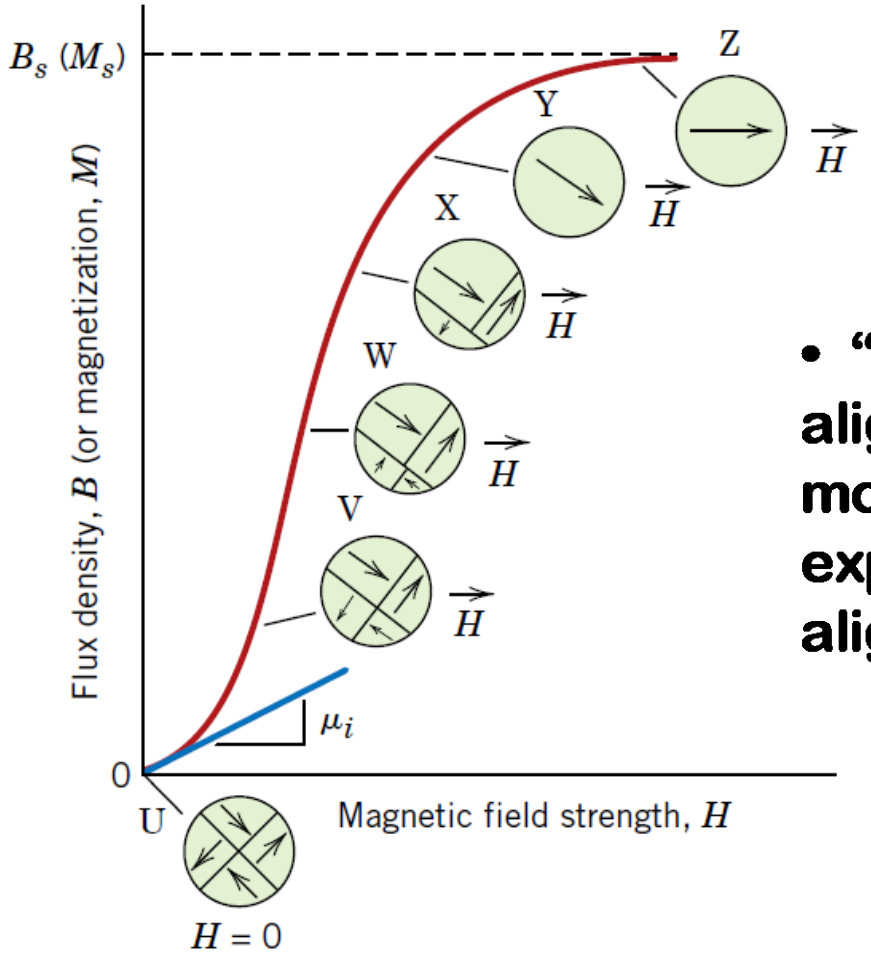


Below Tc, small-volume regions in which there is a mutual alignment in the same direction of all magnetic dipole moments

Schematic depiction of domains in a ferromagnetic material;
arrows represent atomic magnetic dipoles. Within each domain, all dipoles are aligned, whereas the direction of alignment varies from one domain to another.

FERROMAGNETIC MATERIALS

- **As the applied field (H) increases…**
  --the magnetic moment aligns with H.

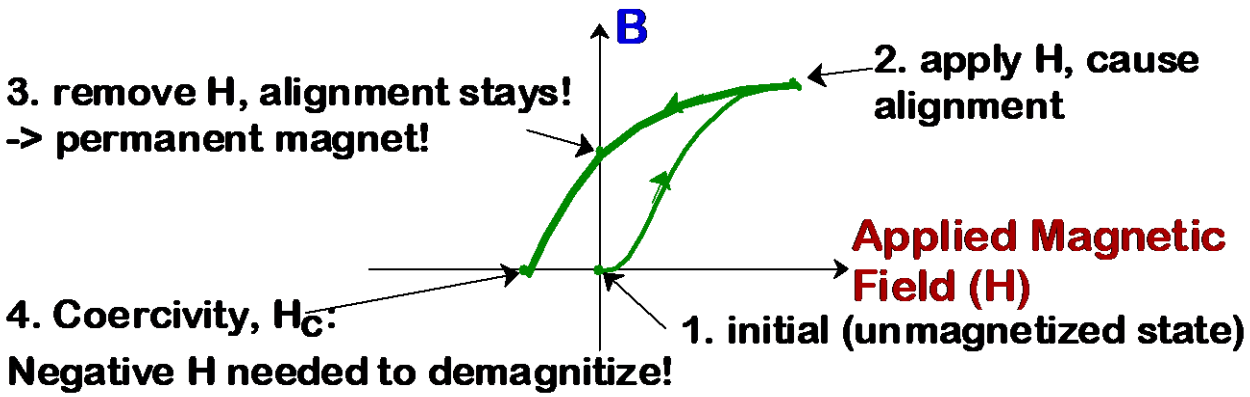


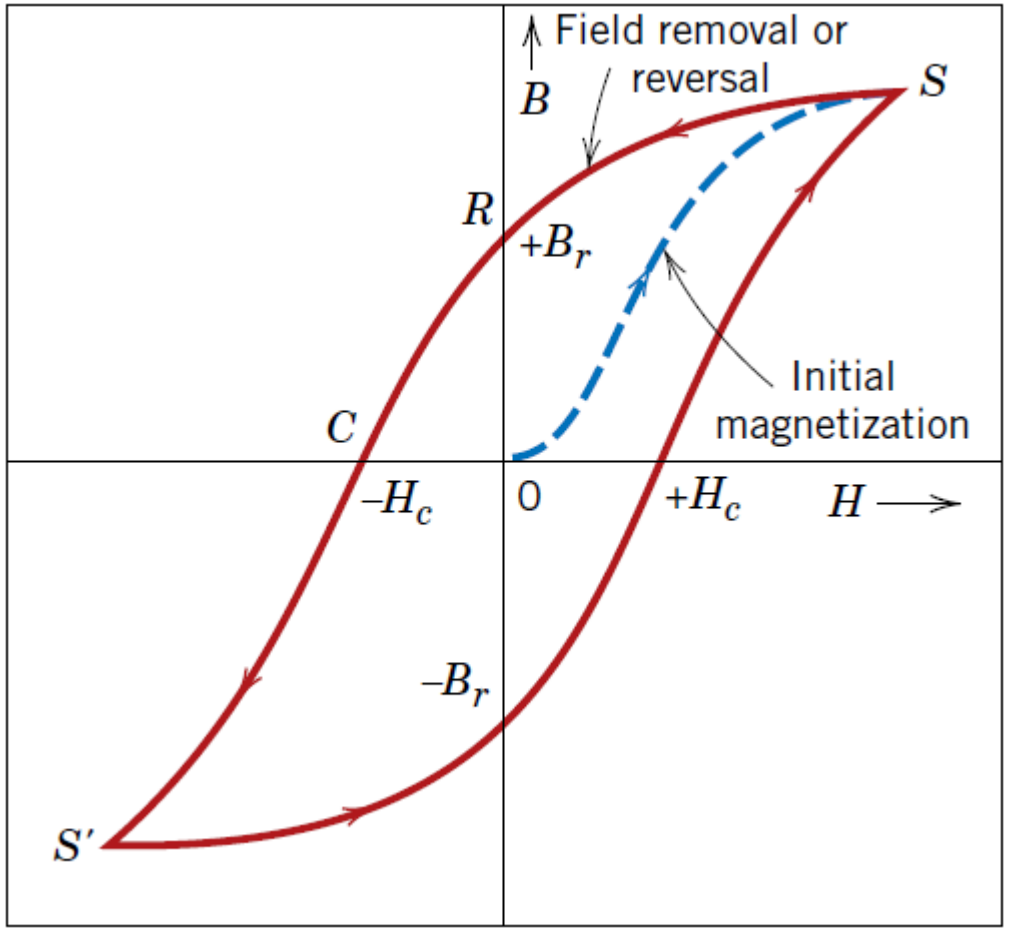
- **"Domains" with aligned magnetic moment grow at expense of poorly aligned ones!**

DOMAINS AND HYSTERESIS

FERROMAGNETIC MATERIALS

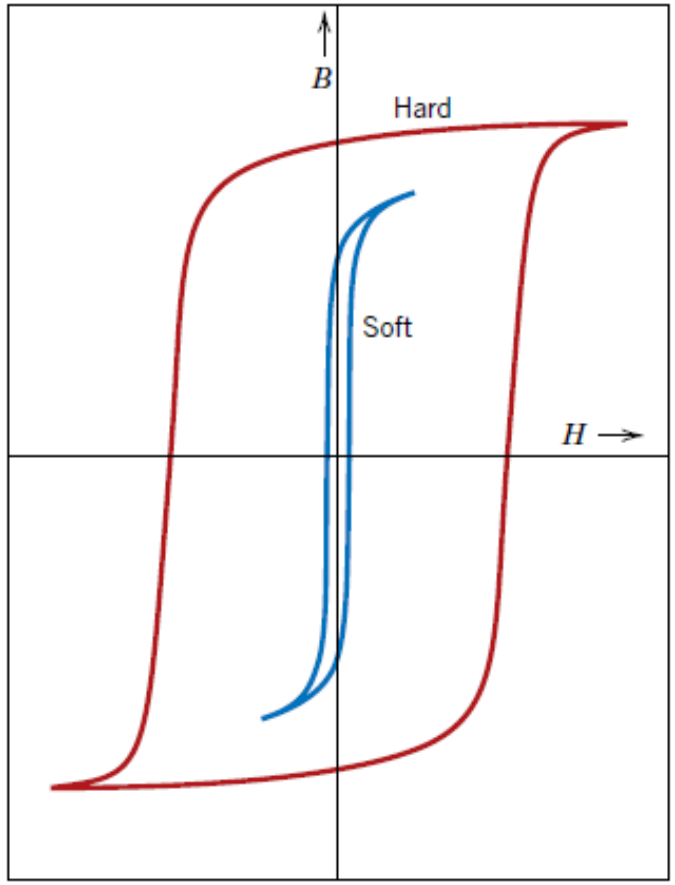
Hysteresis curve (for forward saturation)





Hysteresis curve for forward and reverse saturations

Hard vs Soft Magnets



Ferromaterials are classified as either *soft* or *hard* on the basis of their hysteresis characteristics.

Soft magnetic materials are used in devices that are subjected to alternating magnetic fields and in which energy losses must be low; one familiar example consists of transformer cores.

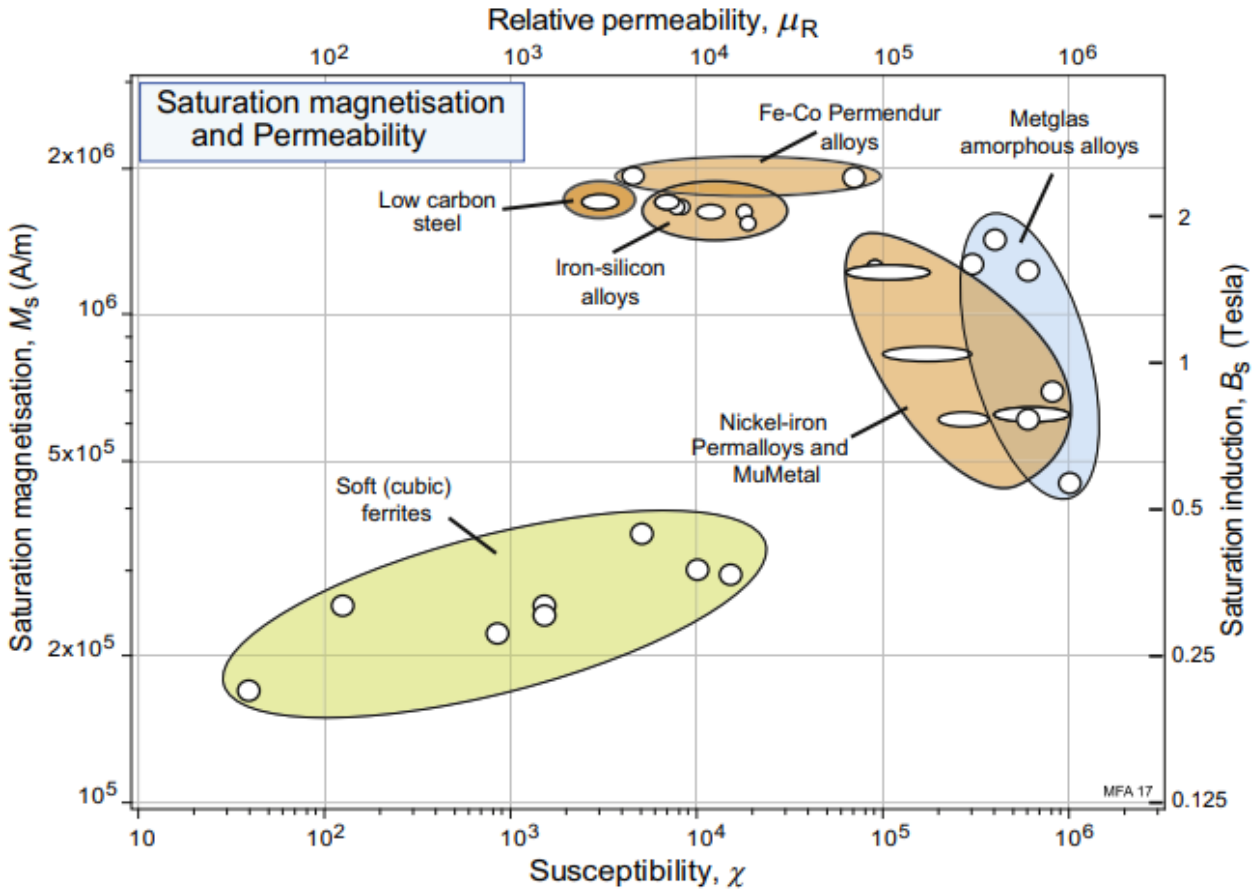
Soft magnetic materials have a low resistance to demagnetization. In terms of hysteresis behavior, a **soft magnetic material** has low coercivity, as well as high initial permeability and low hysteresis energy losses.

Hard magnetic materials are used in permanent magnets, which must have a high resistance to demagnetization.- example application in motors.
In terms of hysteresis behavior, a **hard magnetic material** has high remanence, coercivity and saturation flux density, as well as low initial permeability and high hysteresis energy losses.

Soft magnetic materials

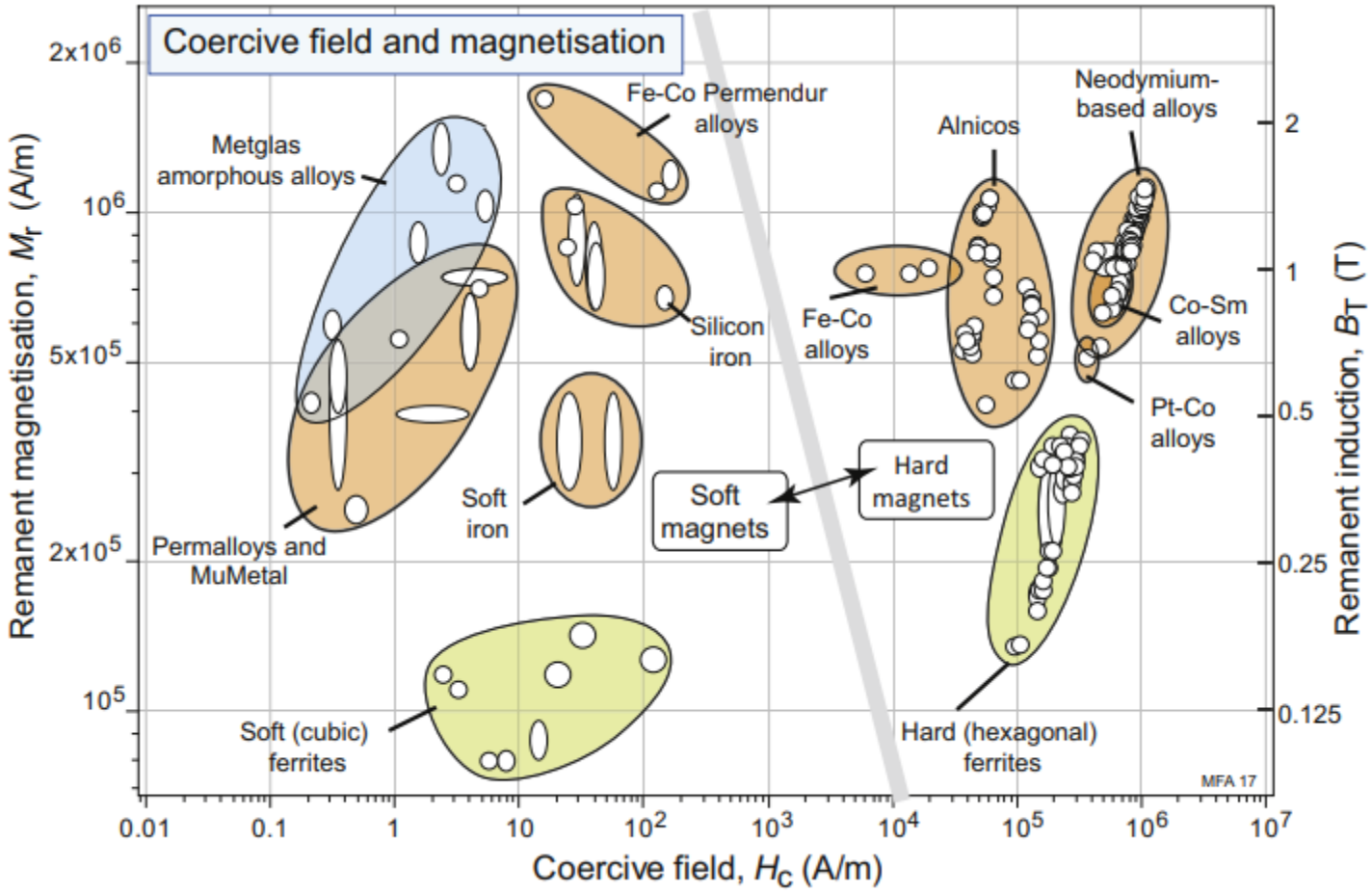
Electromagnets, transformers: - have magnetic cores that must magnetize easily when the field switches on, be capable of creating a high flux density, yet lose their magnetism when the field is switched off.

The higher the susceptibility, the greater is the magnetization.

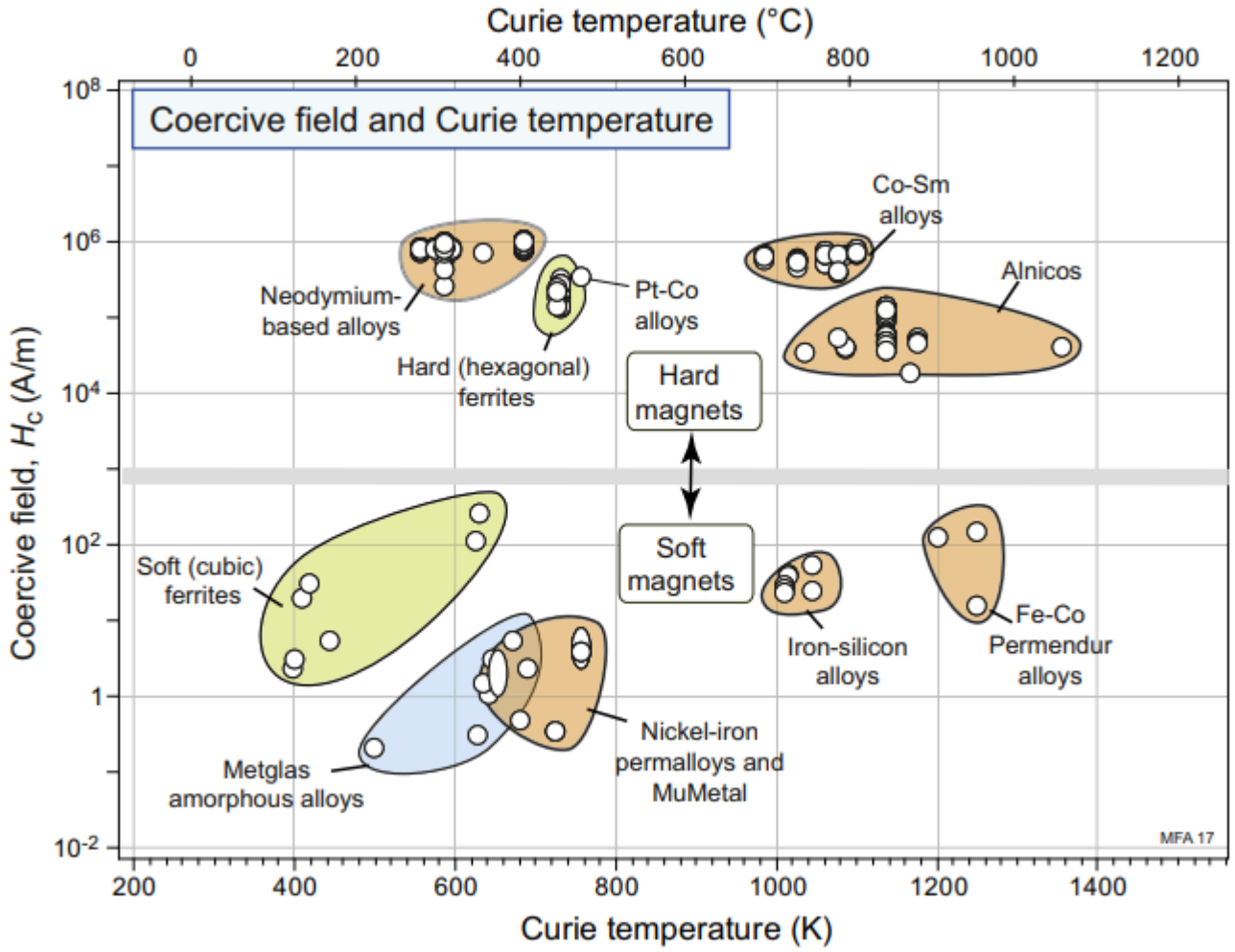


Hard magnetic materials

Motors – require high remanence and resistance to demagnetization (coercivity)



Coercive field - Curie temperature chart



A permanent magnet is needed for an aerospace application. In use the magnet may be exposed to demagnetising fields as high as 3 x 10^5 A/m and temperatures of 600 deg.C. What material would you recommend for the magnet?

Materials for Engineers

Lecture until Dec 6th

Dr S. Gowthaman
Indian Institute of Information Technology Design and Manufacturing Kancheepuram
Chennai – 600127

Reference:

1. Materials science and engineering by William Callister, Wiley Publications

Nanocomposite

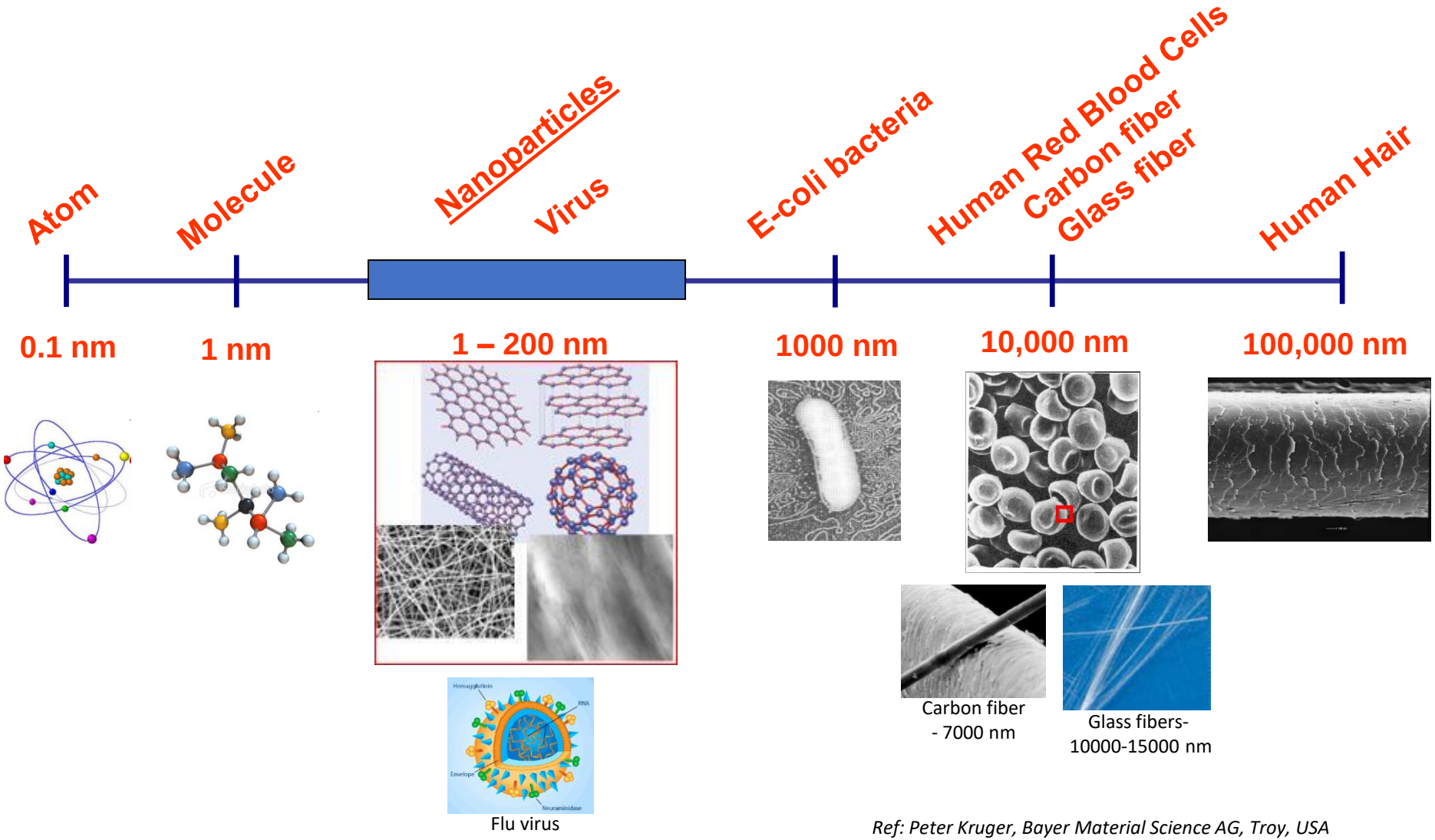
- Nano comes from Greek word nanos which means Dwarf or Extremely small
- One nanometer is one-billionth (10^{-9}) of a meter

Nanocomposites

- Nanocomposites are materials consisting of two or more components, with at least one component having dimensions in the nm regime (i.e. between 1 and 100 nm)

(Ref: Polycorpos Pissis, Nanostructured and Nanocomposite Polymeric Materials, National Technical University of Athens, Greece)

Nanomaterials - Comparison

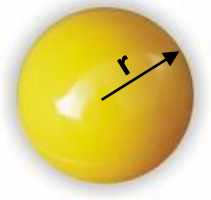


Ref: Peter Kruger, Bayer Material Science AG, Troy, USA

Surface Area to Volume Ratio

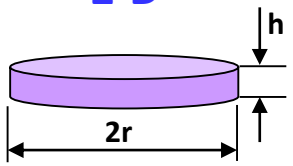
Micro vs Nano

3-D



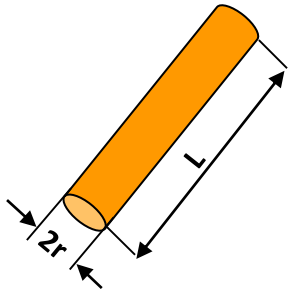
SA / V = 3/r

2-D



SA / V ~ 2/h

1-D

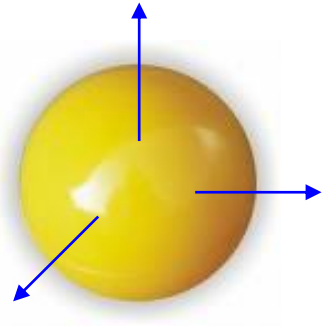


SA / V ~ 2/r

- Surface area per unit volume varies as the reciprocal of the characteristic dimension of the filler
- Surface area from micro to nano increases by 1000 times
- Surface area is critical in chemical reaction and bonding

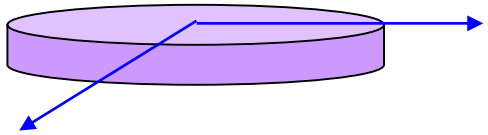
Fillers and Functionalities

Types of Fillers



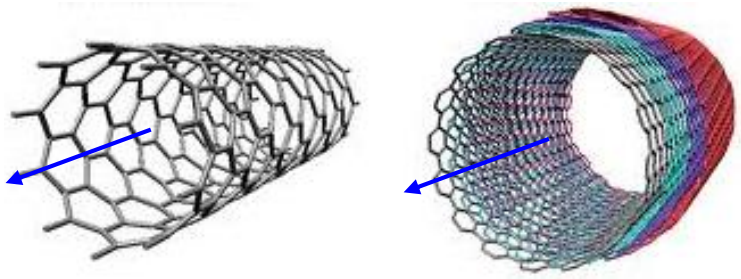
Nanoparticles

Example: Nanosilica, Fullerene



Nanolayers/platelets

Example: Nanoclay, Graphene



Nanotubes/fibers

Example: CNT, Nanofibers

Mechanical Properties

TABLE 1	MECHANICAL PROPERTIES COMPARISON			
Material	Modulus, GPa	T.S., GPa	Density, g/cm^3	Diameter
SWCNT/MWCNT	~1,000	~100-200	~0.7-1.7	1/~20 nm
Carbon Nanofibers	~500	3-7	1.8-2.1	20-200 nm
Graphene	~1,000	~100-400	1.8-2.2	platelet
High Tensile Steel	210	1.3	7.8	--
Carbon Fiber	230	3.5	1.75	5-10 μm
Aramid Fiber	60	3.6	1.44	5-10 μm
Glass Fiber	22	3.4	2.6	5-10 μm

Automobile Applications

Improved mechanical properties, surface finish, ease of processing

Toyota Timing Belt Cover
MMT/Nylon 6

Step assistant component in GMC Safari and Chevrolet Astro, MMT/TPO

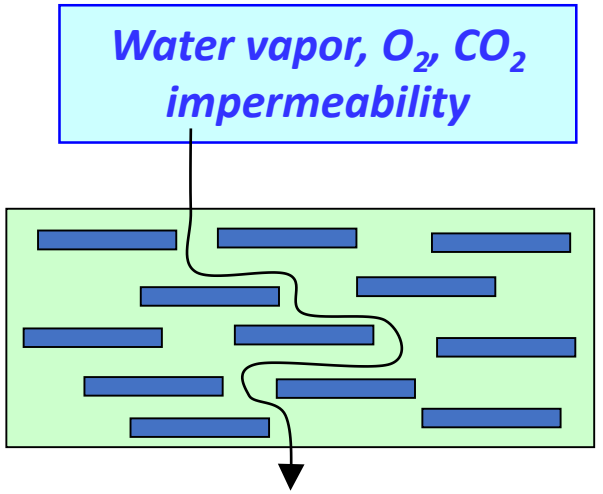
Mitsubishi GDI engine cover
MMT/Nylon-6

Door of Chevrolet Impala
MMT/TPO

Seat backs of Honda Acura, MMT/PP

Packaging Applications

Stand up pouch, MMT/Nylon 6



PET bottles, MMT/Nylon MXD6

*Meal Ready to Eat **(MRE)** Pack, MMT/Nylon MXD6*

No of companies: Ex – Nanocor, MGCC, Kuraray, etc

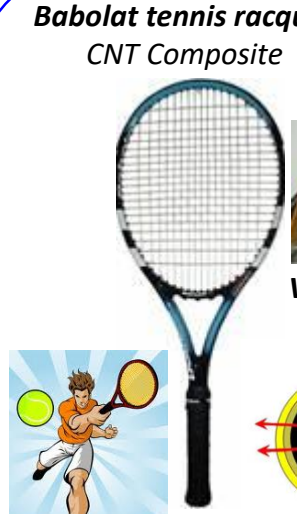
MRE Packs used in Military Applications

- Eliminates foil layer
- Capable of microwave processing
- Reduces stress-cracks, pin-holes
- Reduces processing steps (no lamination)
- Decreases weight

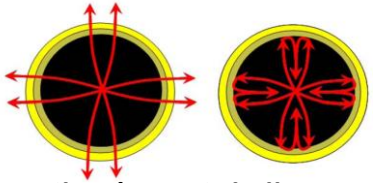
Sporting Goods

CNT composites –
Stiff, strong, lightweight

Babolat tennis racquet
CNT Composite



Vokl DNX tennis racquet
CNT Composite



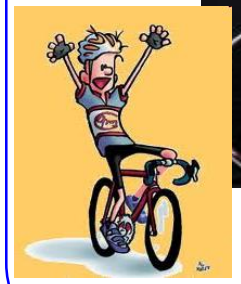
Wilson's tennis ball
nanoclay coated (barrier)

Easton/Zyvex base ball bat
CNT composite

Easton/Zyvex hockey stick
CNT composite

Enlight Earth LLC/Eric Arakawa
Surfboard, CNT Composite

BMC/Easton bicycles
CNT Composite



ABS Nanodesu bowling ball
Fullerene coated (wear)

Materials Needs for Some Aircraft Applications

Still there are needs of new materials for many challenges in aircraft structures….some examples are shown below.

Problem - Structural failure

Property needed **- Strength, stiffness, toughness**

Problem - Lightning

Property needed **- Electrical conductivity**

Problem - Icing

Property needed - Electrical & thermal conductivity

Problem - Structural health monitoring

Property needed - Electrical conductivity, self healing

We will see some materials solutions for these from composition – processing – microstructure – property perspective…

Materials Needs for Aircraft Applications (Mechanical Properties)

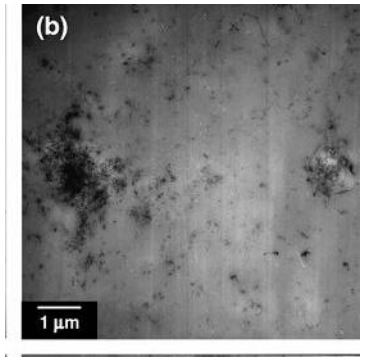
Composition: CNT + Epoxy + Carbon fiber composite

Processing → **Microstructure** → **Processing of composites**

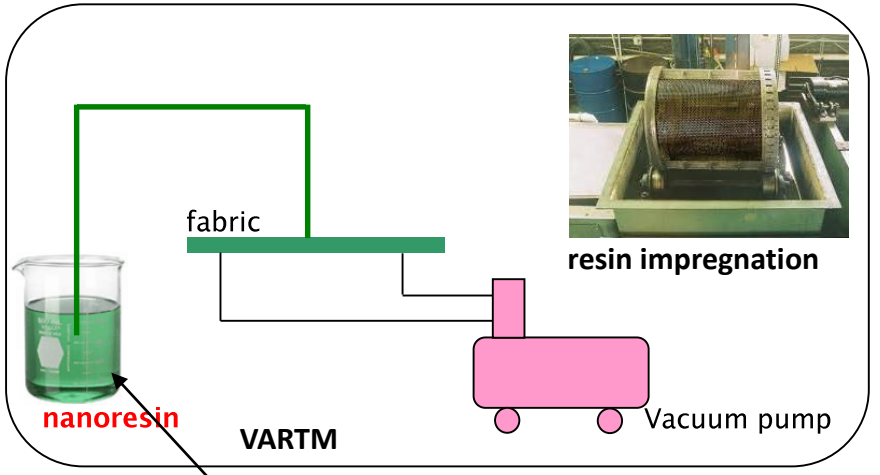
Hand mix epoxy + CNT

↓

Process in three roll mill



Non uniform mixing of CNT



Use nanoresin now to make composite

↓

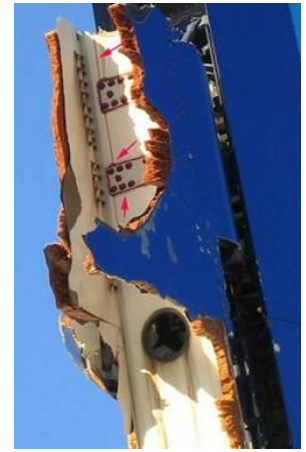
Property

Resulting strength, modulus of composites is not good (because of non uniform mixing of CNTs)

Problem - Structural failure

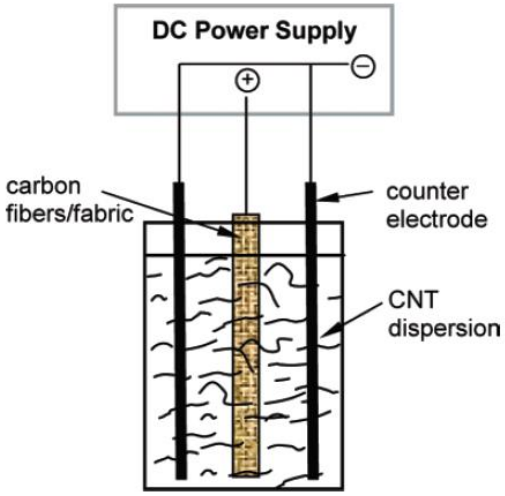
Property needed - Strength, stiffness, toughness

Materials Needs for Aircraft Applications (Mechanical Properties)

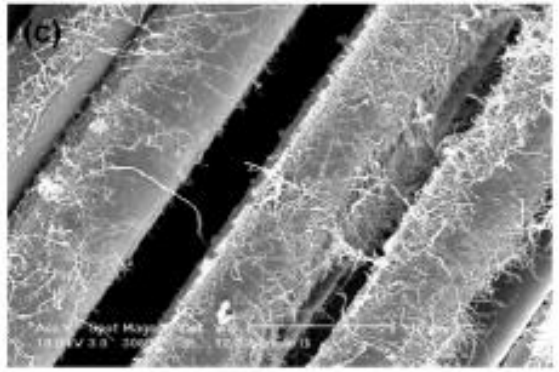


Composition: CNT + PC + Carbon fiber composite

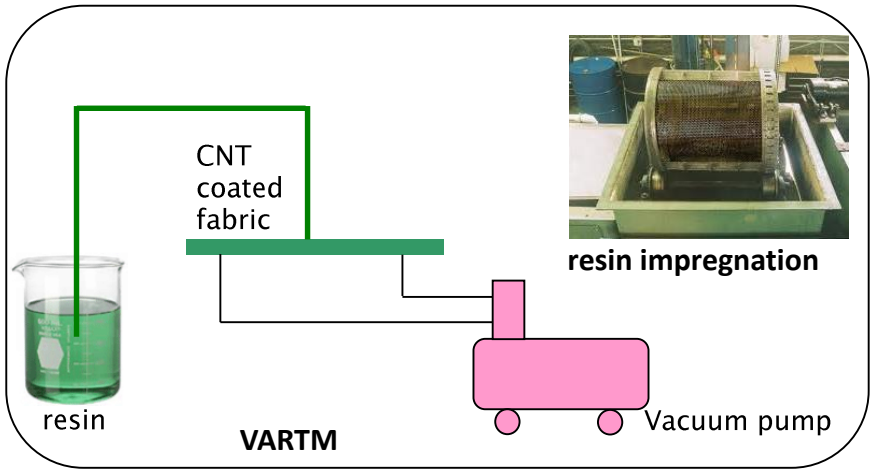
Processing → **Microstructure** → **Processing of composites**



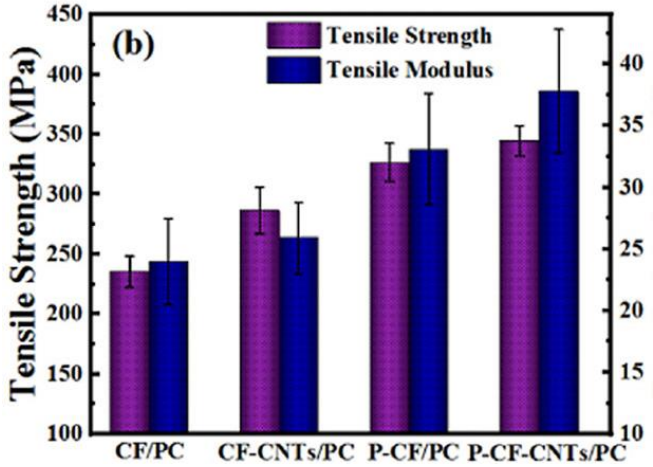
Electrophoretic deposition



Uniform coating of CNT on fabrics



Use CNT coated fabric now to make composite



Property

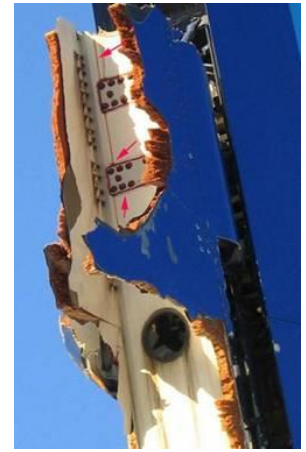
Resulting strength, modulus of composites is good (because uniform deposition of CNTs)

(Note - Figure shows with some chemical treatment along with CNT coatings)

Problem - Structural failure

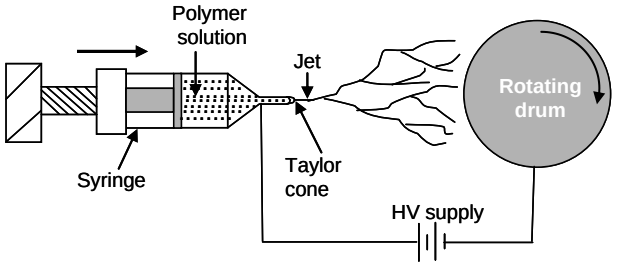
Property needed - Strength, stiffness, toughness

Materials Needs for Aircraft Applications (Fracture Toughness)

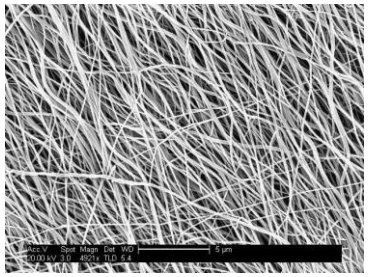


Composition: Nylon nanofiber + Epoxy + Carbon fiber composite

Processing → **Microstructure** → **Processing of composites**

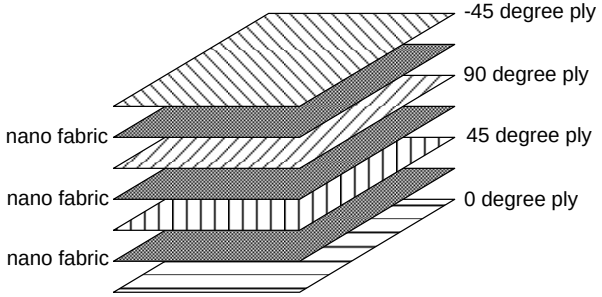


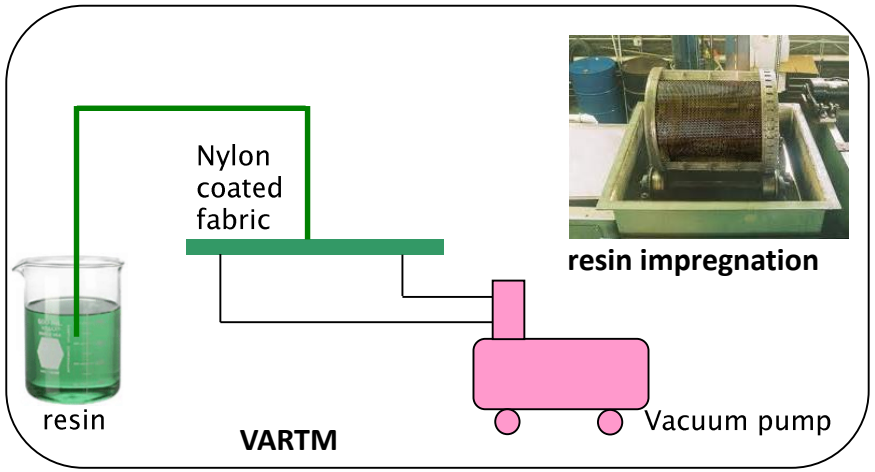
Electrospinning



Nylon nanofibers

Fabric stacking



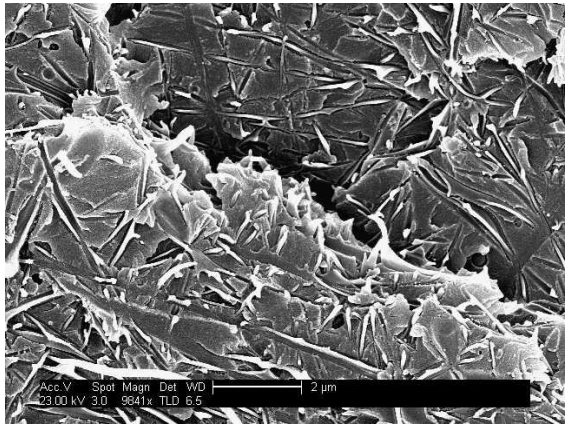


Use Nylon coated fabric now to make composite

Property

Fracture toughness improved by 150%

(nanofiber bridging, pull out are mechanisms)



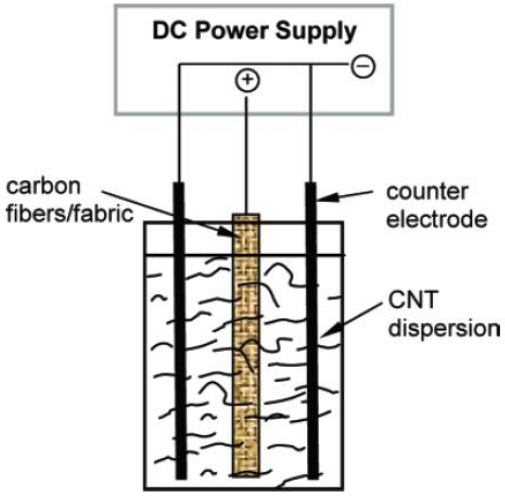
Problem - Structural failure

Property needed - Strength, stiffness, toughness

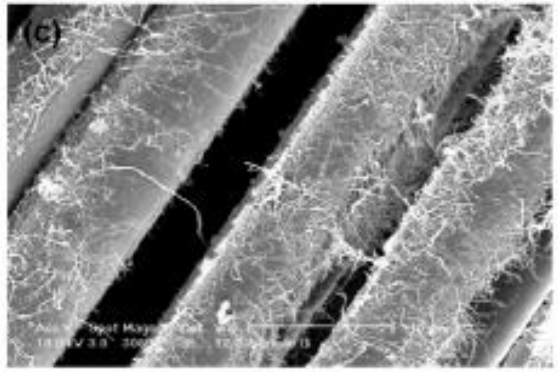
Materials Needs for Aircraft Applications (Lightning Strike)

Composition: CNT + Epoxy + Carbon fiber composite

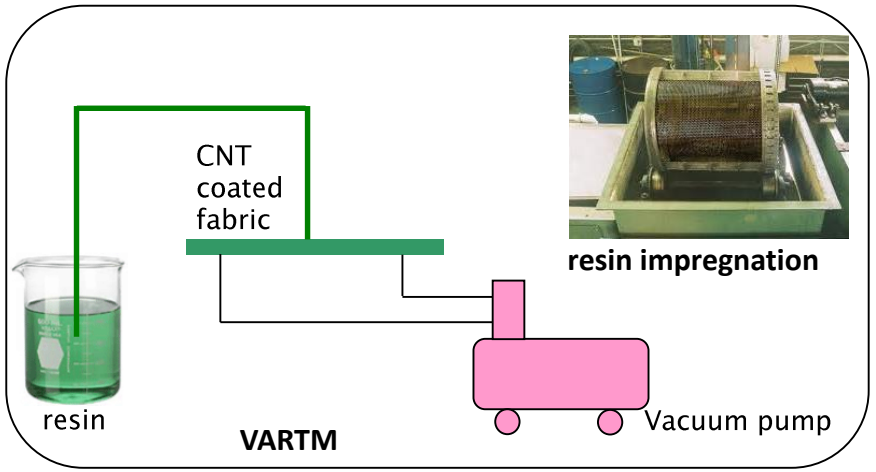
Processing → **Microstructure** → **Processing of composites**



Electrophoretic deposition



Uniform coating of CNT on fabrics

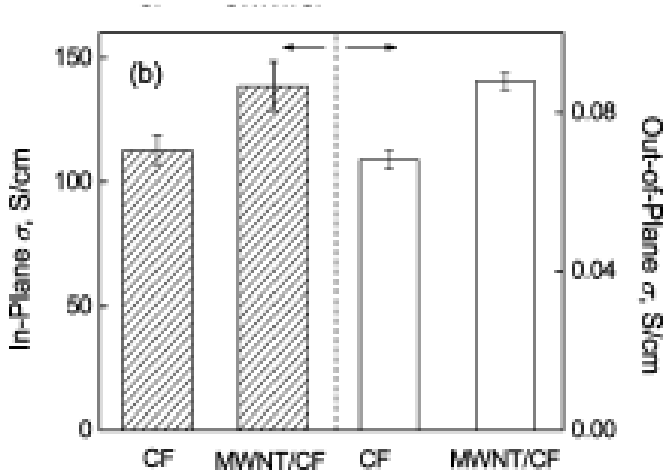


Use CNT coated fabric now to make composite

Problem - Lightning ***Property needed - Electrical conductivity***

Property

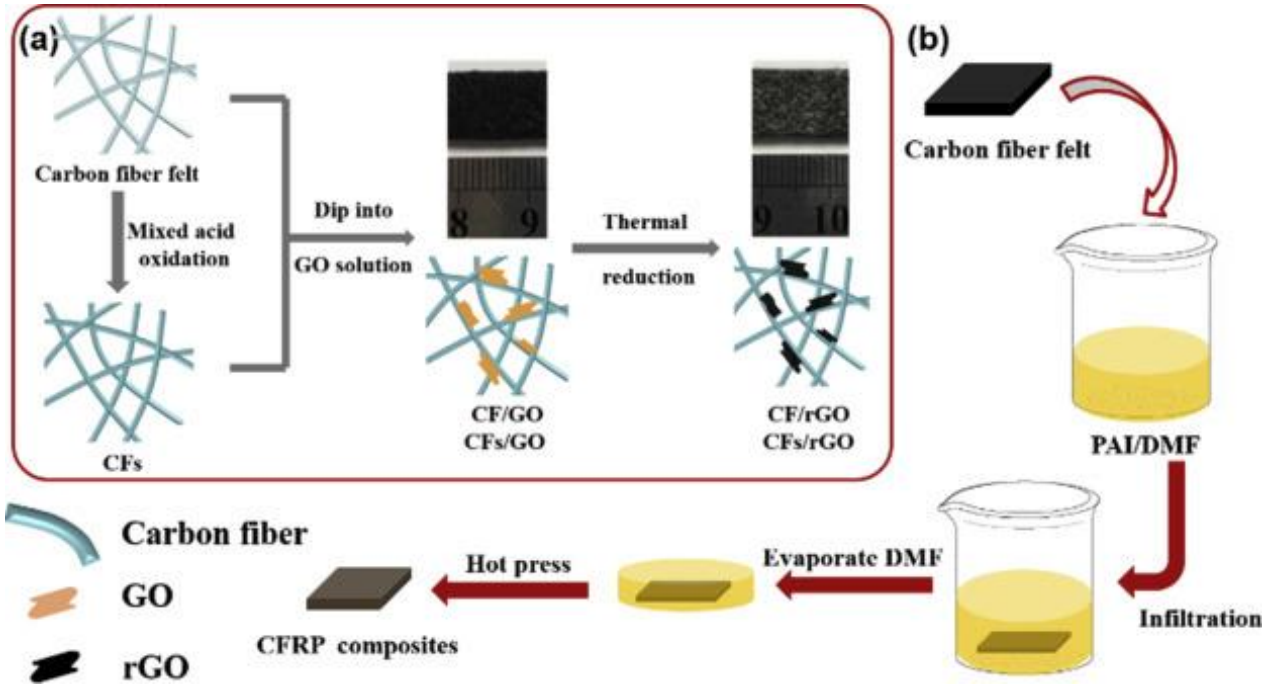
Increase in in-plane and out-of-plane electrical conductivity increased because of CNTs



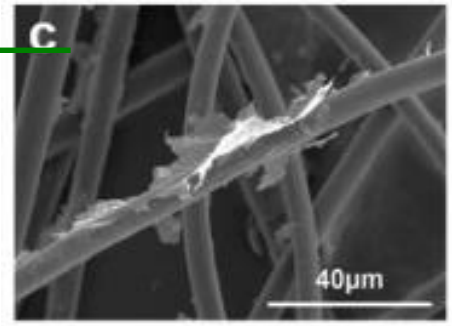
Materials Needs for Aircraft Applications (Icing)

Composition: Graphene Oxide + Polyamide + Carbon fiber composite

Processing of composites

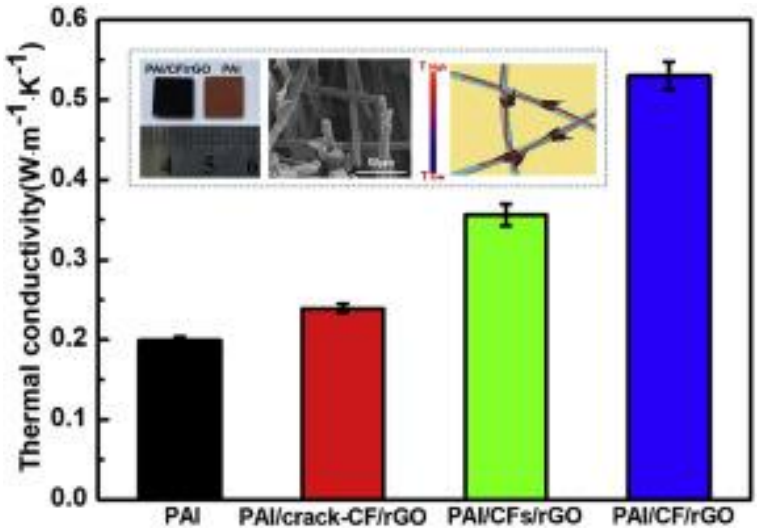


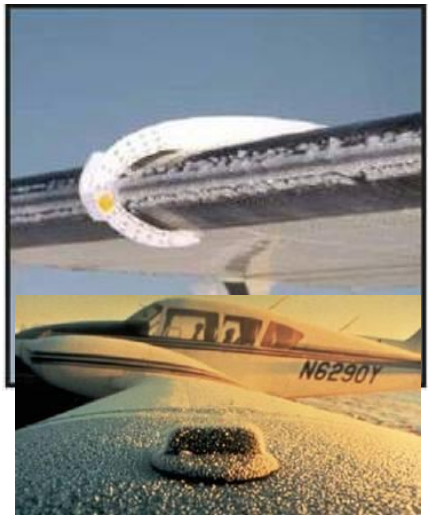
Microstructure



Property

***Significant improvements in thermal conductivity** for composites with various chemical treatments*





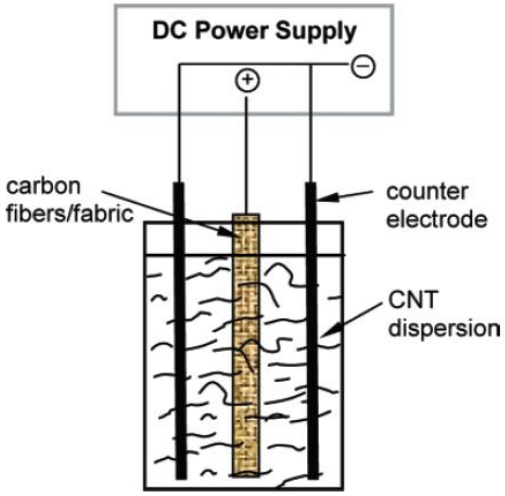
Problem - Icing

Property needed - Electrical & thermal conductivity

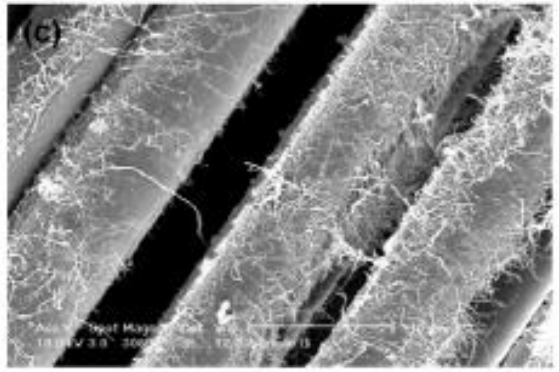
Materials Needs for Aircraft Applications (Health Monitoring)

Composition: CNT + Epoxy + Carbon fiber composite

Processing → **Microstructure** → **Processing of composites**



Electrophoretic deposition



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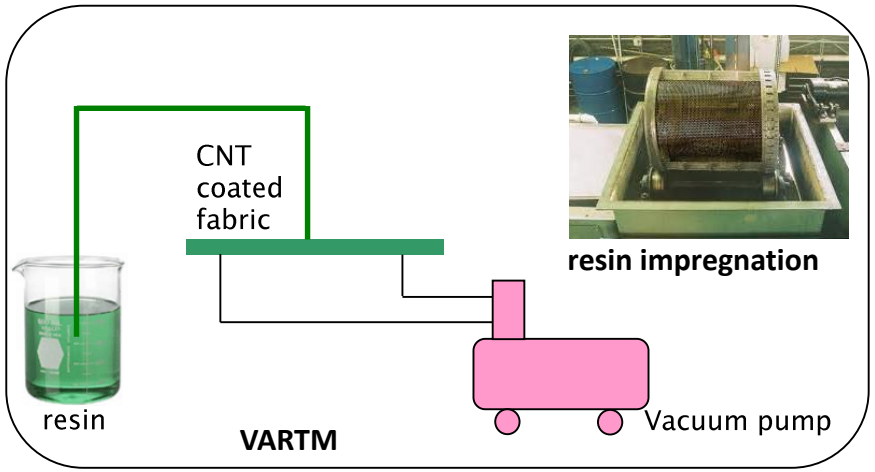
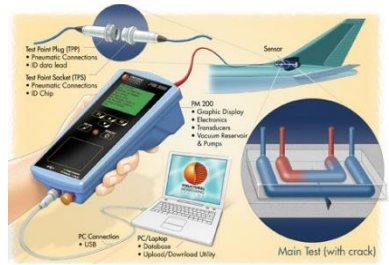


Use CNT coated fabric now to make composite

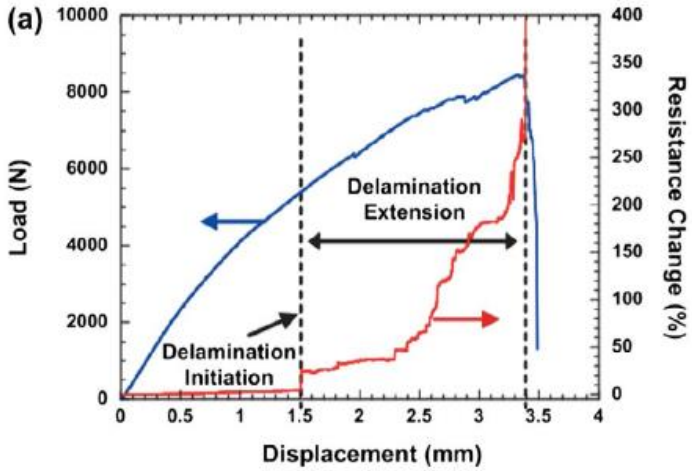


Problem - Structural health monitoring

Property needed - Electrical conductivity

Property

Internal damages which is not shown in load-displacement curves are shown as change in resistances – beneficial to detect and monitor defects in internal structures

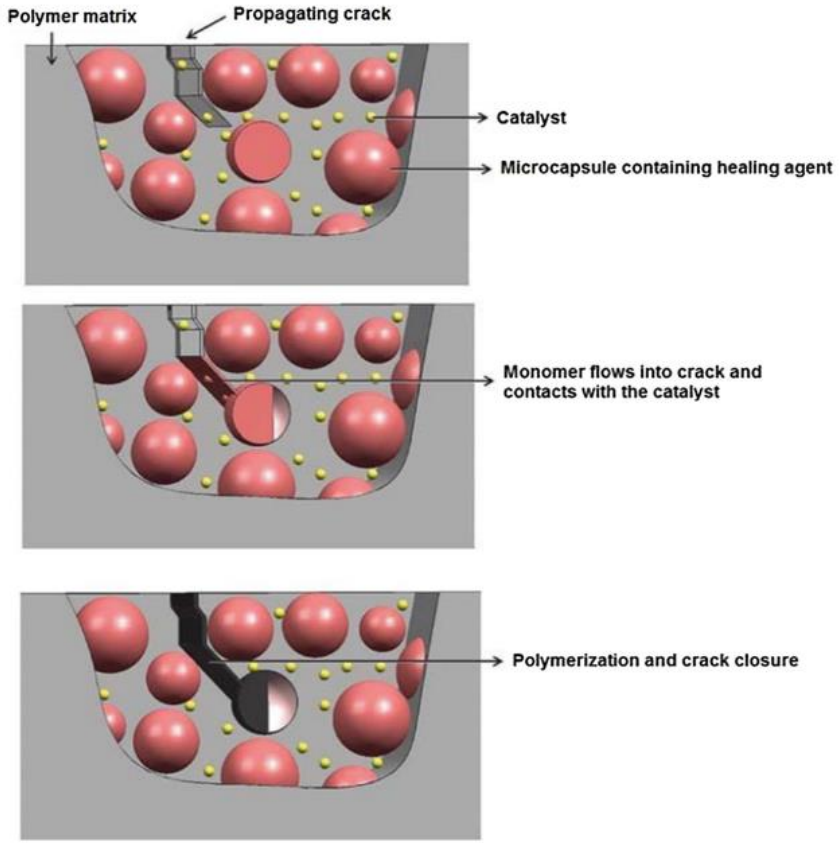


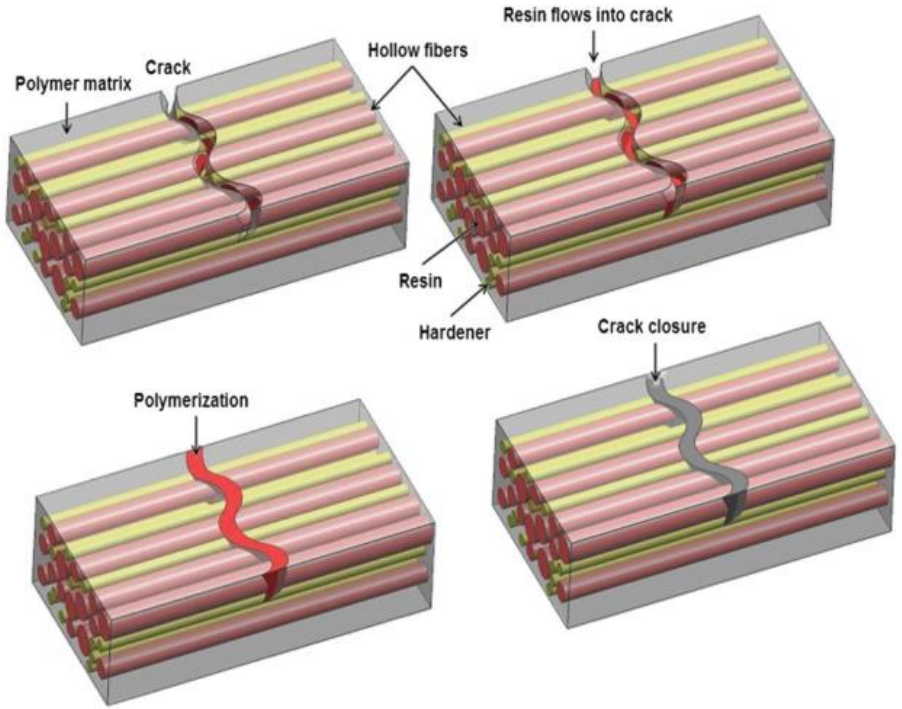
Materials Needs for Aircraft Applications (Self Healing)

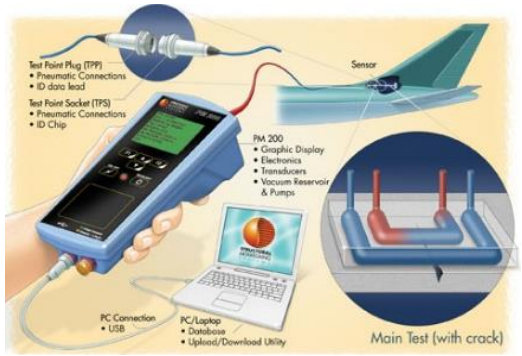
Smart materials

Composition: Polymer + Self healing capsules or fibers + Carbon fibers

Microstructures







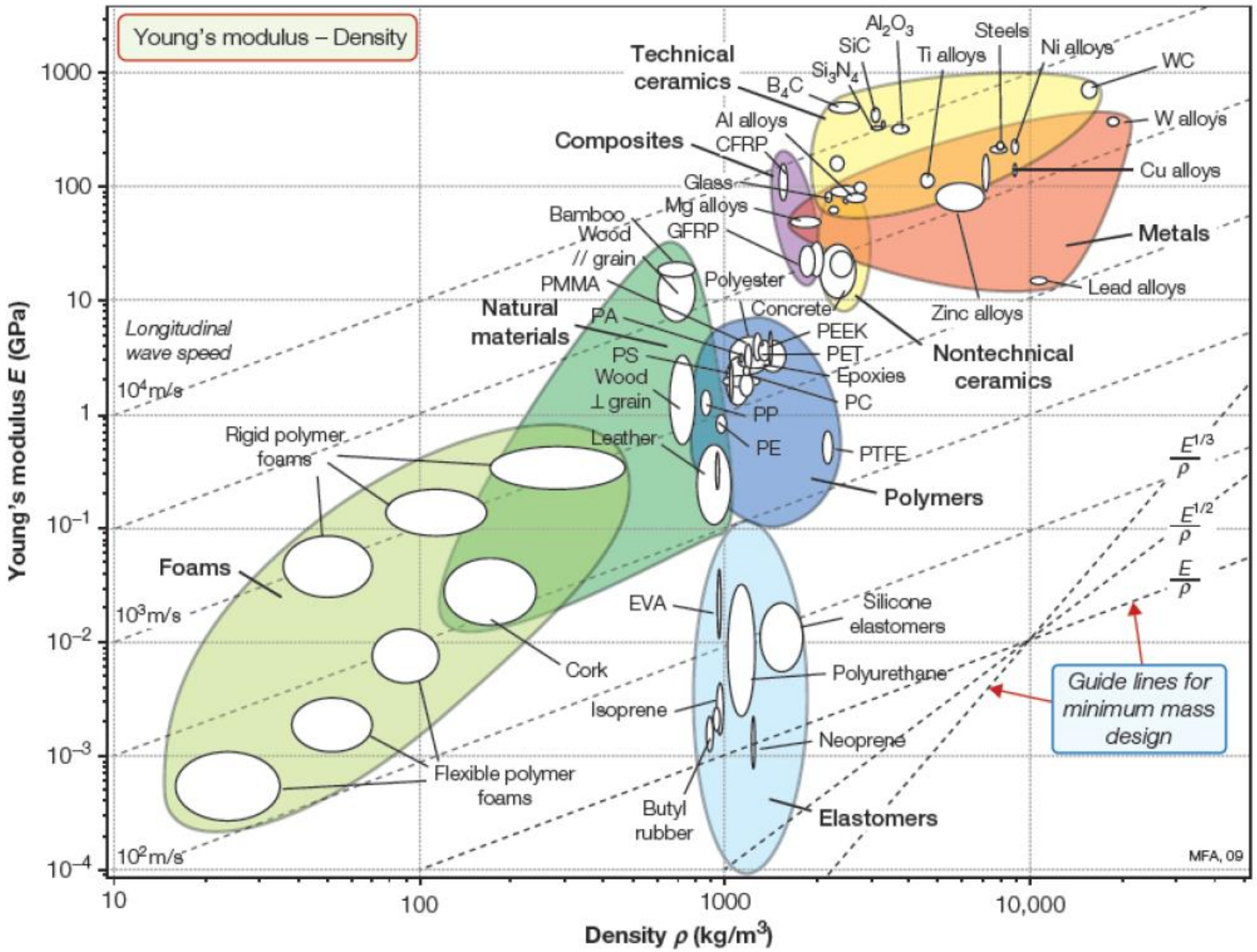
Problem - Structural health monitoring

self healing

Hollow fibers with uncured resin are used. Under cracking of fibers, resin comes out and reacts with catalyst outside. It cures and closes the cracks.

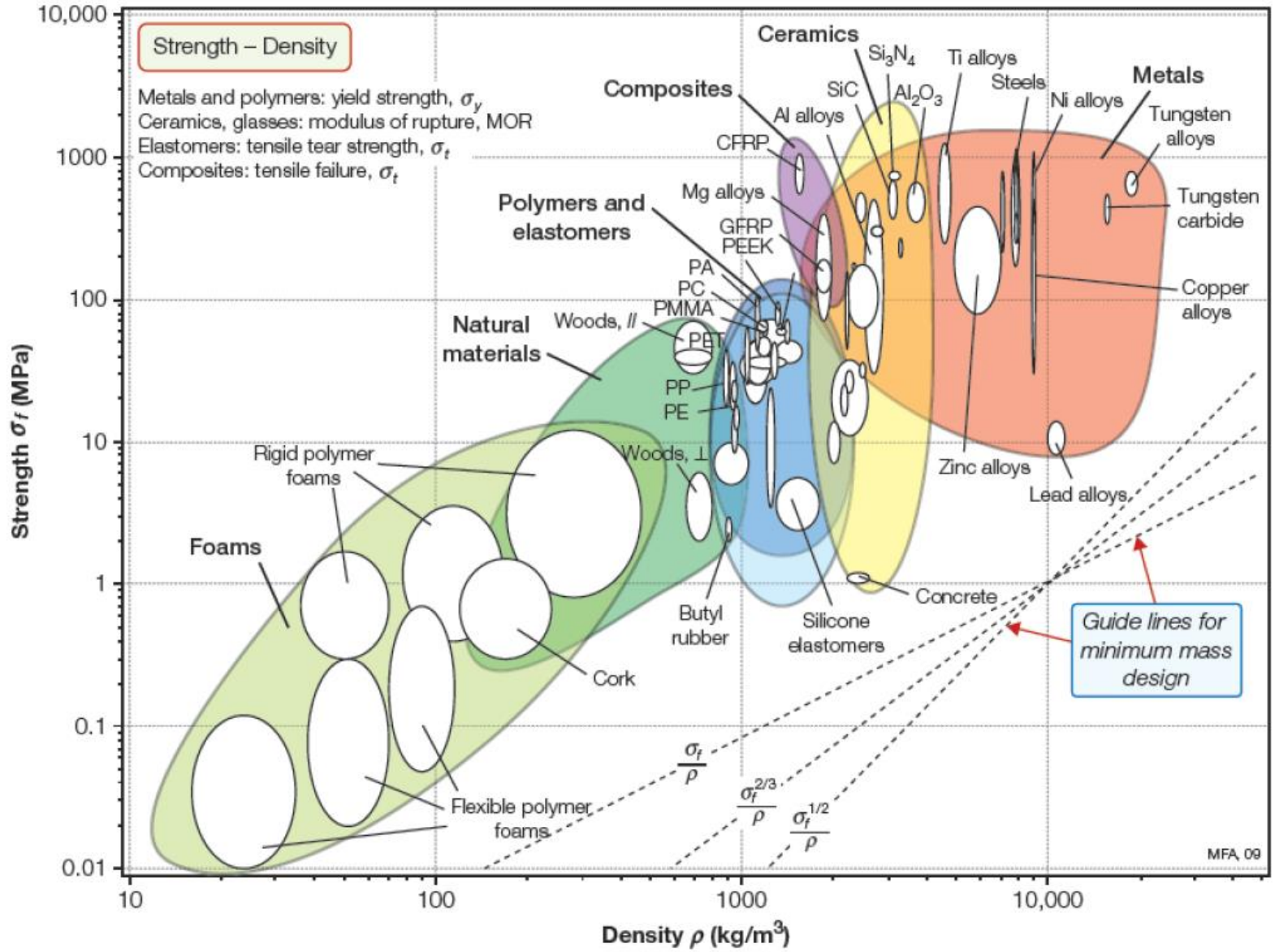
Material Property Charts

Young's Modulus vs Density



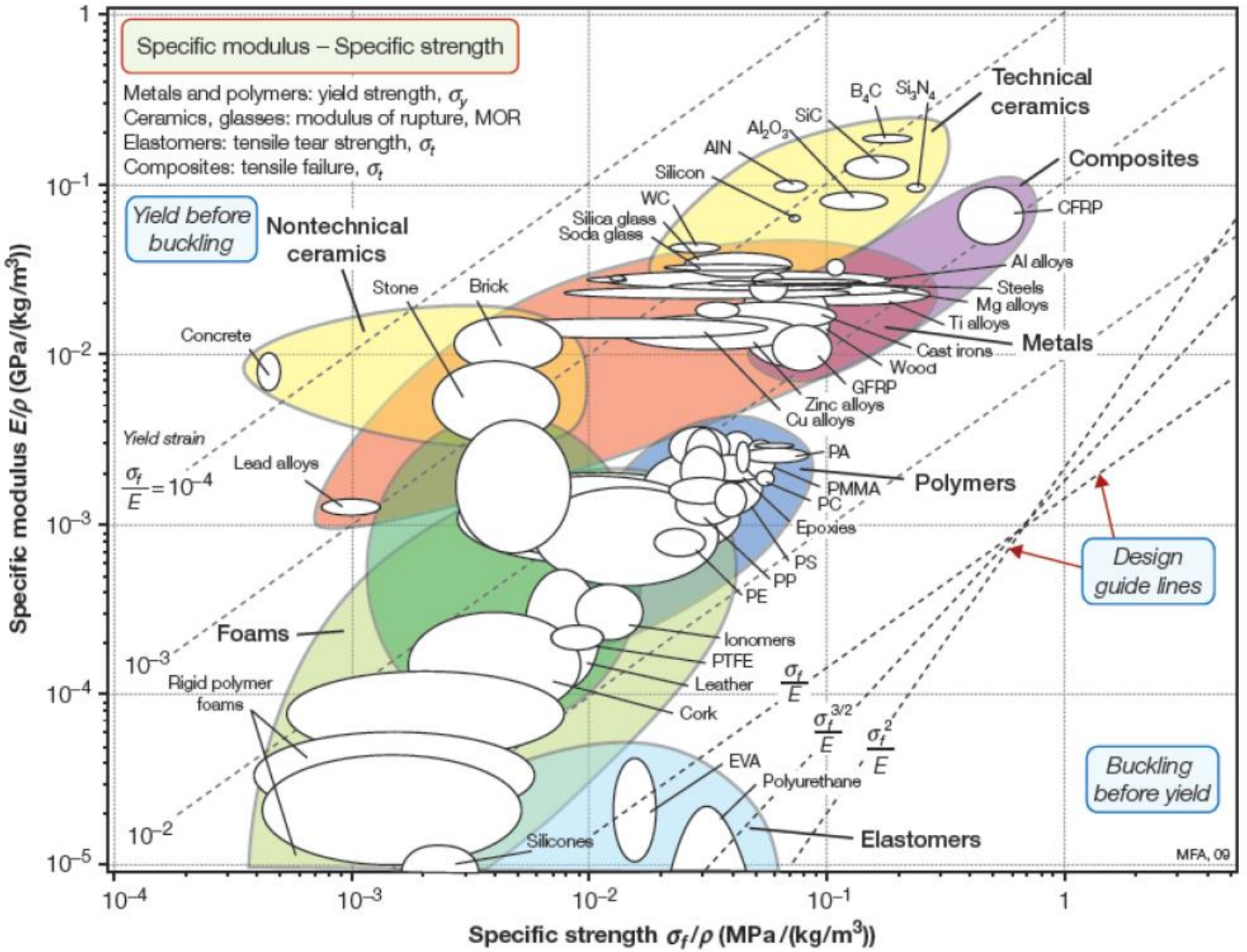
Material Property Charts

Strength vs Density



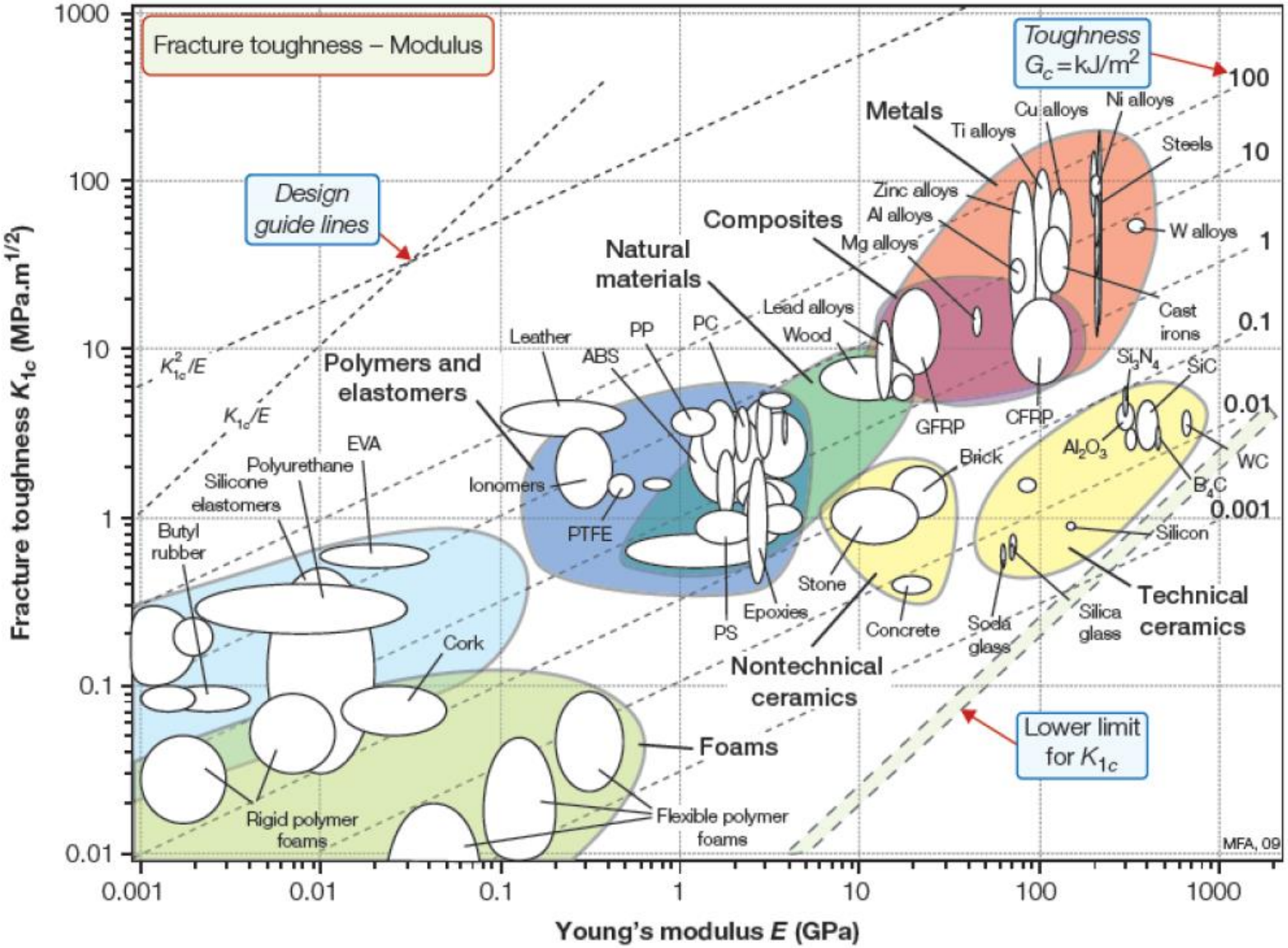
Material Property Charts

Specific modulus vs Specific strength



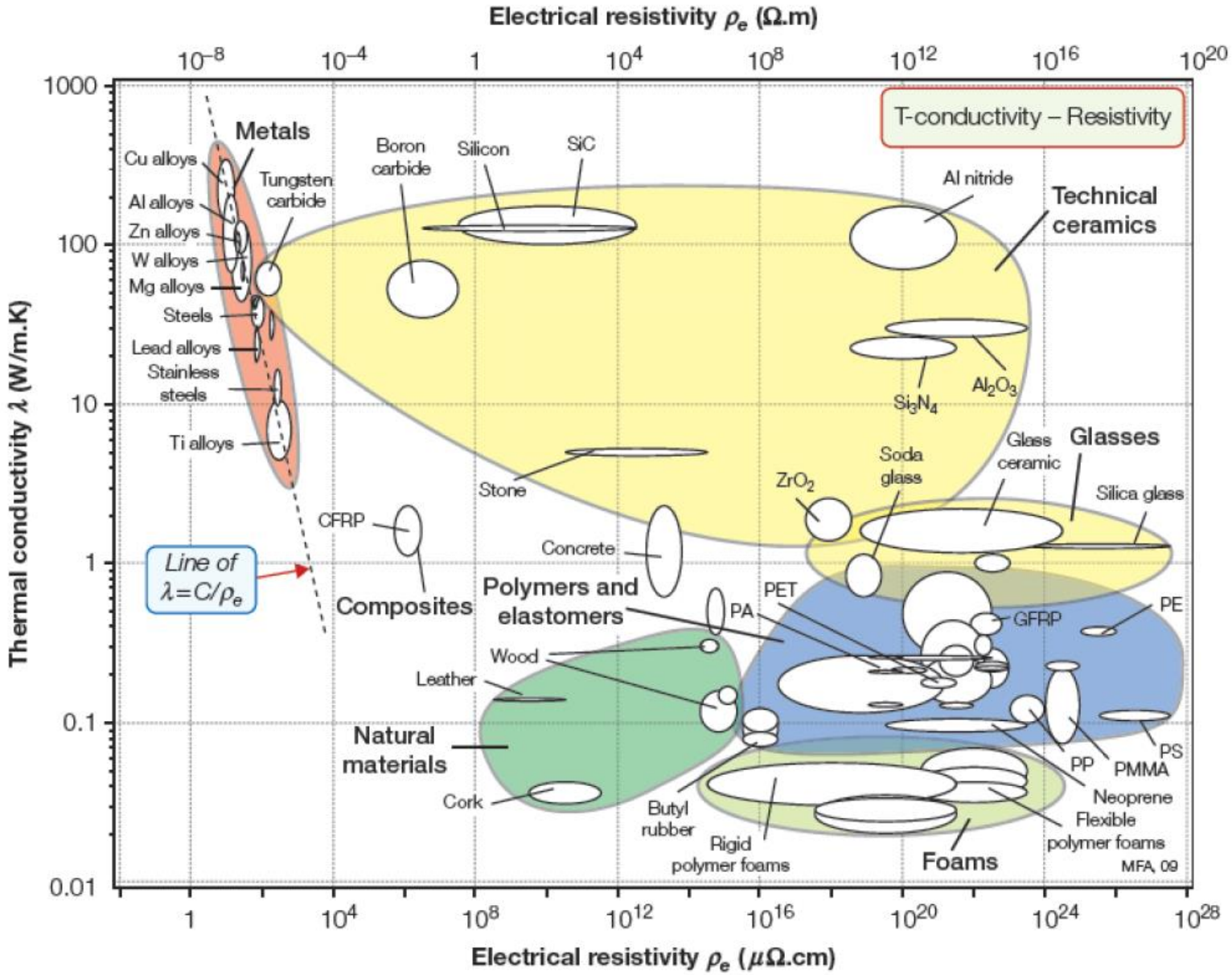
Material Property Charts

Fracture toughness vs Modulus



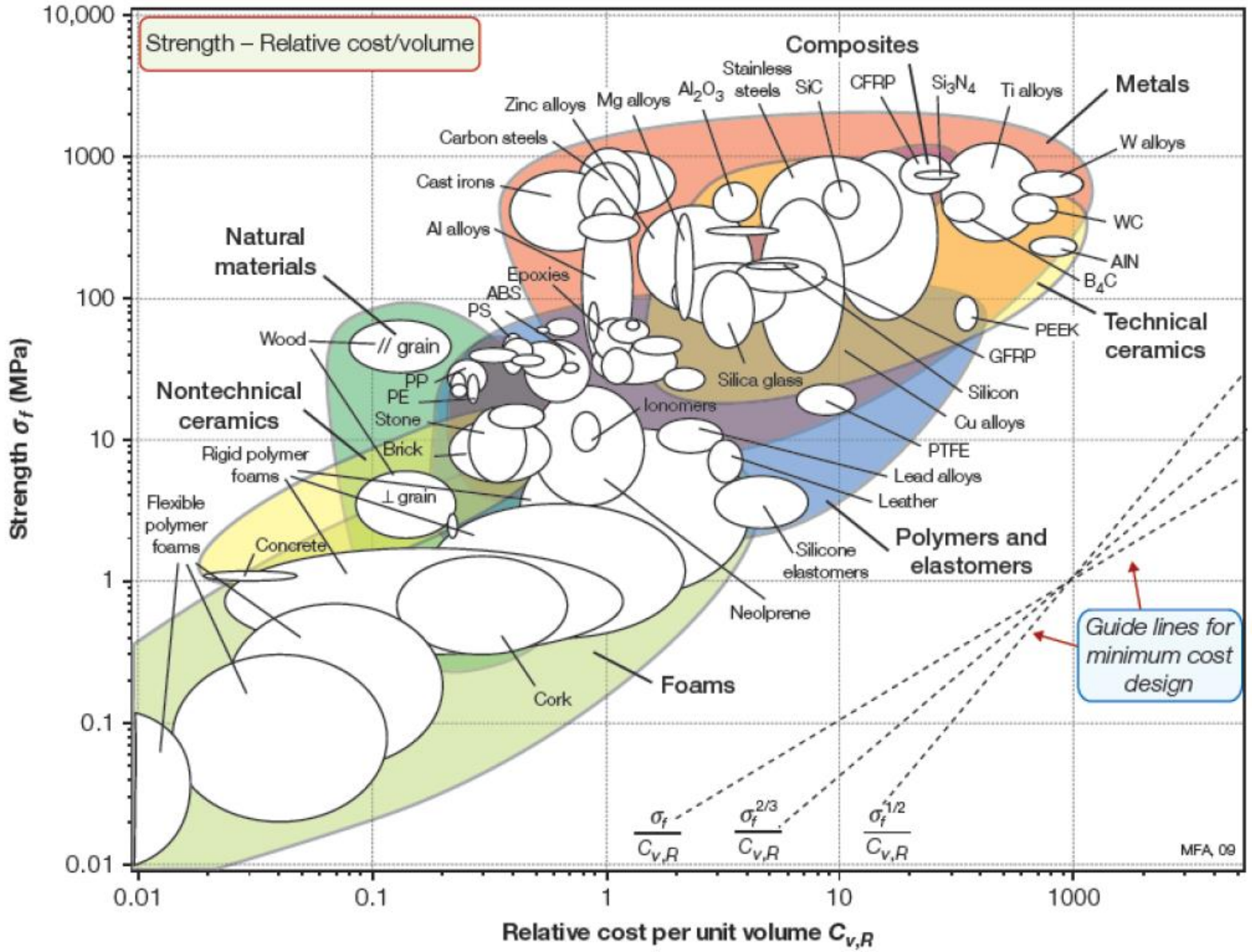
Material Property Charts

Thermal conductivity vs Electrical resistivity

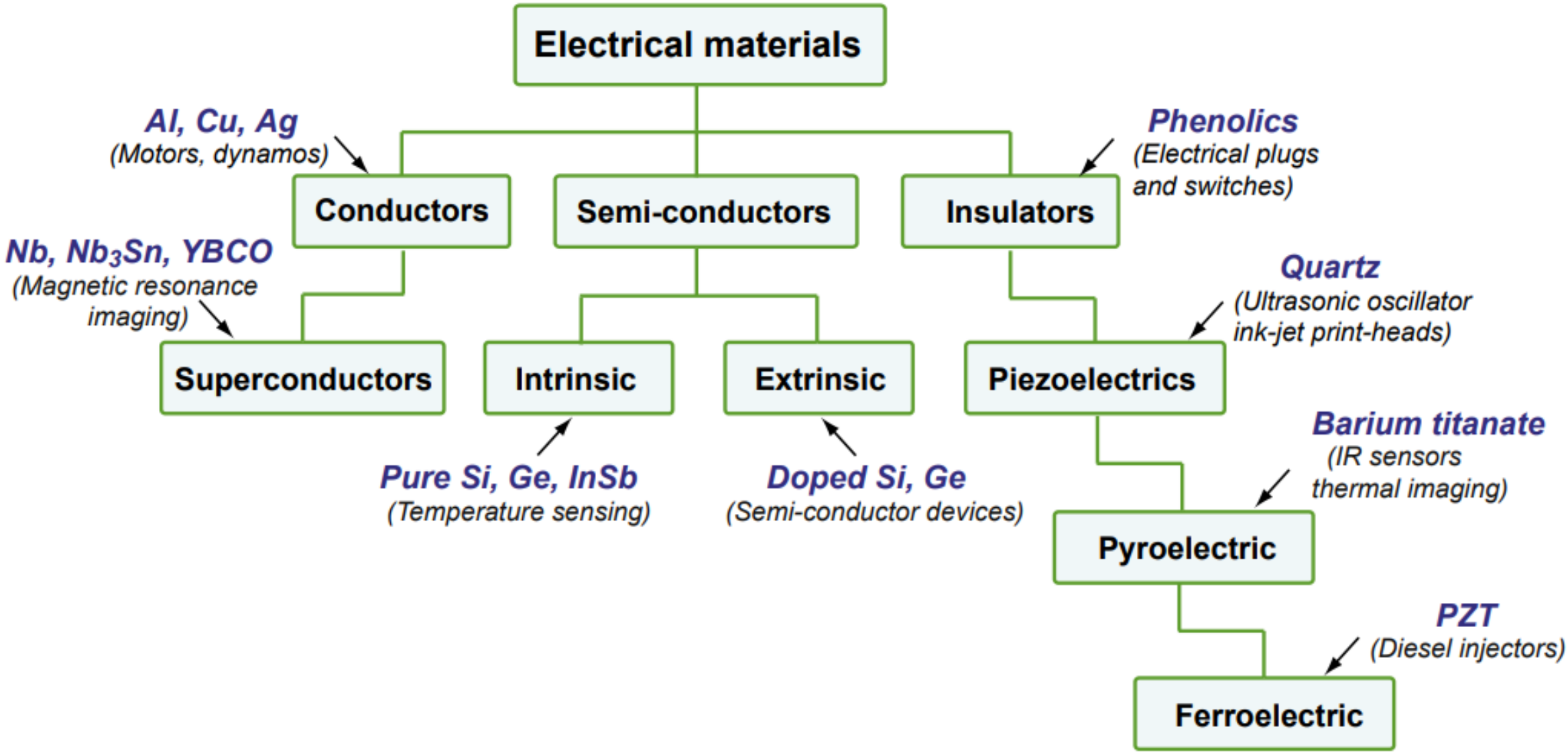


Material Property Charts

Strength vs Cost



Electrical Properties



Electrical Properties

- Ohm's Law:

$$V = I R$$

voltage drop (volts = J/C)
C = Coulomb

current (amps = C/s)

resistance (Ohms)

- Resistivity, ρ:
  -- a material property that is independent of sample size and geometry

$$\rho = \frac{RA}{l}$$

surface area of current flow

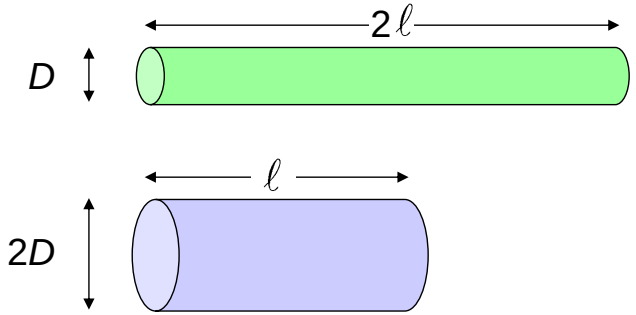
current flow path length

- Conductivity, σ

$$\sigma = \frac{1}{\rho}$$

Electrical Conduction

- Which will have the greater resistance?



$$R_1 = \frac{2\rho\ell}{\pi\left(\frac{D}{2}\right)^2} = \frac{8\rho\ell}{\pi D^2}$$

$$R_2 = \frac{\rho\ell}{\pi\left(\frac{2D}{2}\right)^2} = \frac{\rho\ell}{\pi D^2} = \frac{R_1}{8}$$

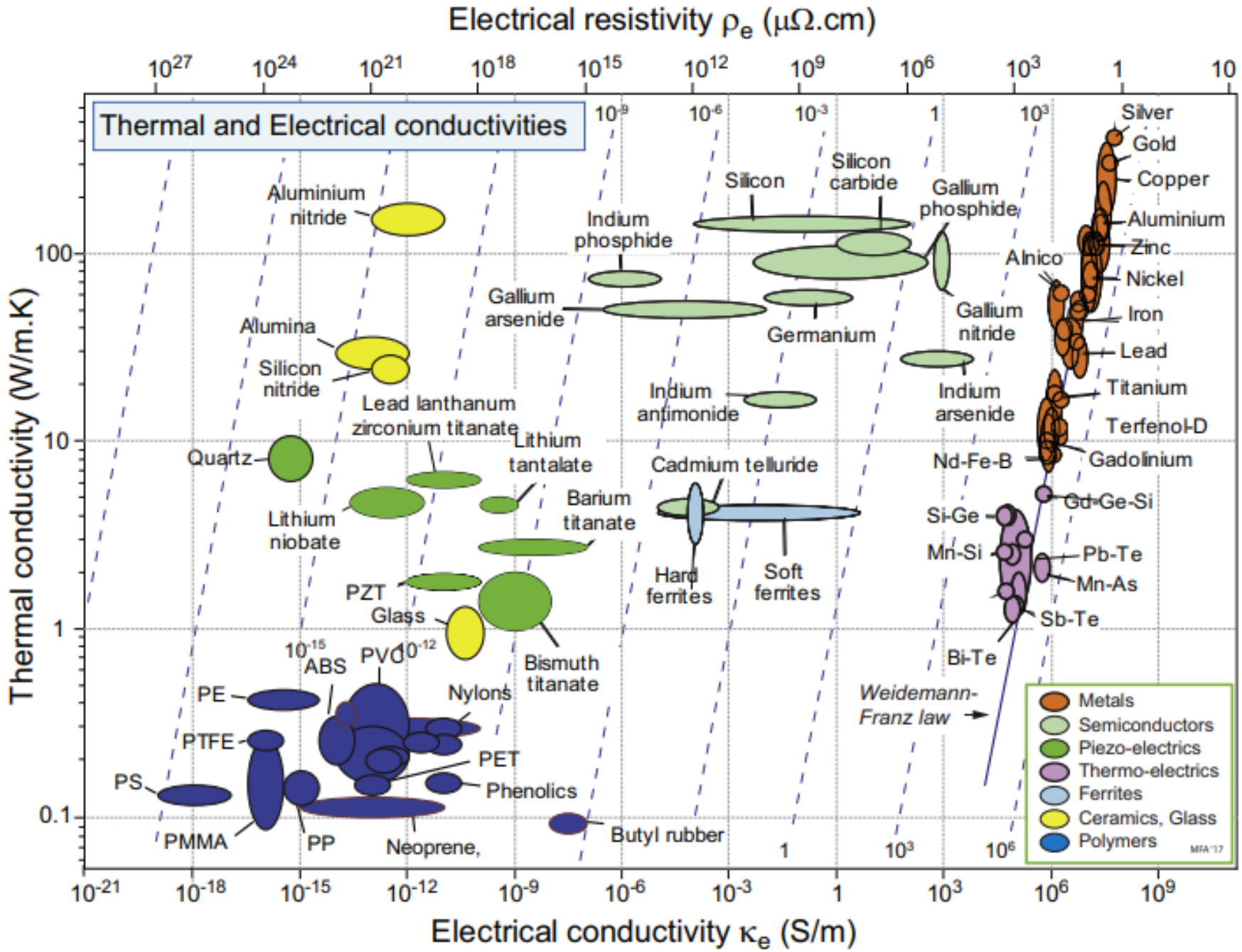
- Resistance depends on sample geometry and size.

Temperature-dependence of resistivity:

The resistivity of metals increases with temperature because thermal vibrations scatter electrons.

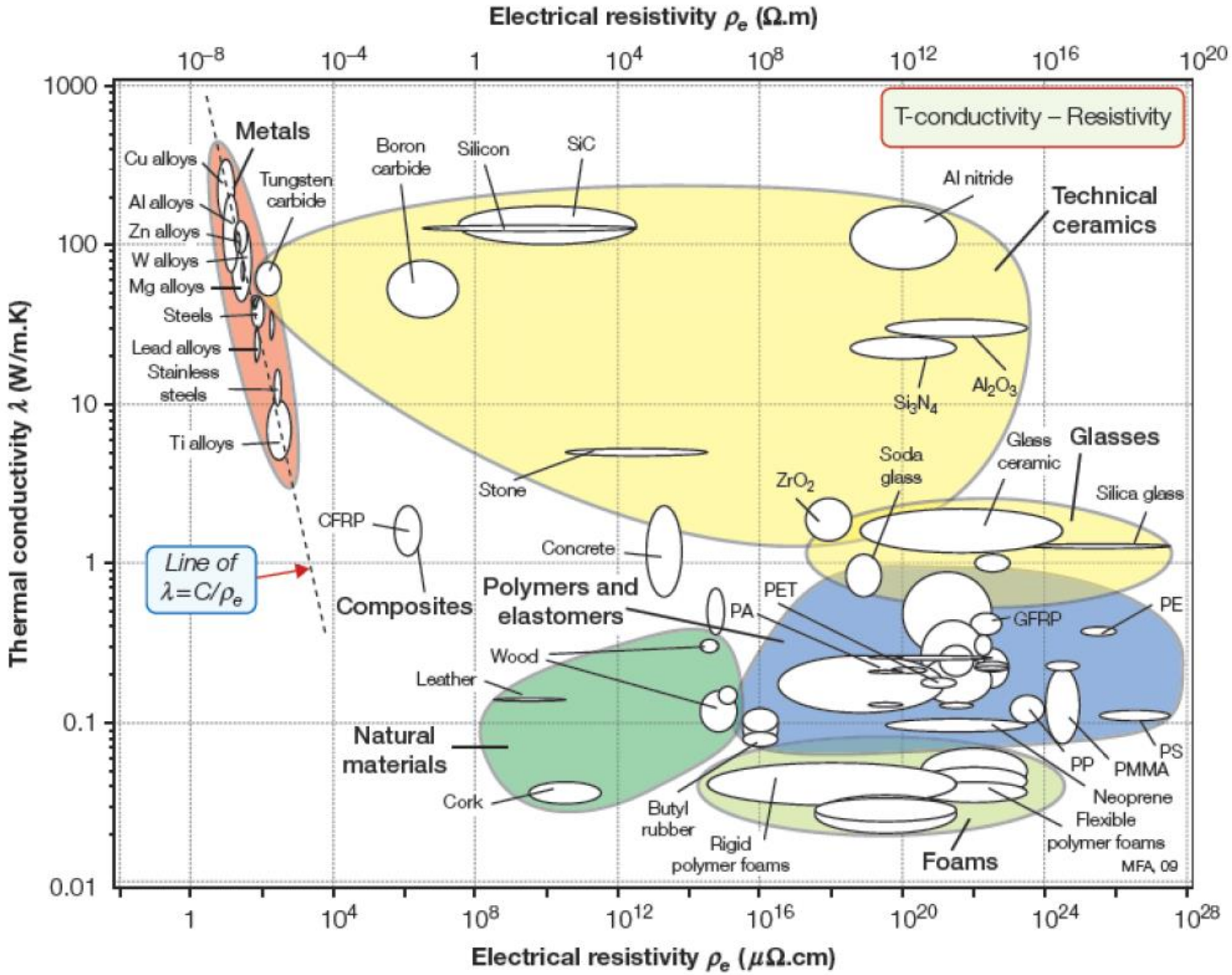
The resistivity of semiconductors, by contrast, decreases as temperature increases, because thermal energy allows more carriers to cross the band gap, entering the conduction band.

Superconducting materials suddenly lose all resistance at a critical temperature, Tc. Below Tc, the current in a superconducting material flows without any resistive loss; above Tc, superconductivity is suppressed.


Thermal and Electrical conductivities
Electrical resistivity ρ_e ($\mu\Omega$.cm)
Electrical conductivity κ_e (S/m)
Thermal conductivity (W/m.K)
Aluminium nitride
Alumina
Silicon nitride
Lead lanthanum zirconium titanate
Quartz
Lithium tantalate
Lithium niobate
Barium titanate
PZT
Glass
Bismuth titanate
PE
ABS
PVC
Nylons
PTFE
PET
PS
Phenolics
PMMA
PP
Neoprene,
Butyl rubber
Indium phosphide
Silicon
Silicon carbide
Gallium phosphide
Gallium arsenide
Germanium
Gallium nitride
Indium antimonide
Indium arsenide
Cadmium telluride
Hard ferrites
Soft ferrites
Silver
Gold
Copper
Aluminium
Zinc
Alnico
Nickel
Iron
Lead
Titanium
Terfenol-D
Nd-Fe-B
Gadolinium
Gd-Ge-Si
Si-Ge
Mn-Si
Pb-Te
Mn-As
Sb-Te
Bi-Te
Weidemann-Franz law
Metals
Semiconductors
Piezo-electrics
Thermo-electrics
Ferrites
Ceramics, Glass
Polymers
MFA '17

Material Property Charts

Thermal conductivity vs Electrical resistivity



Conductivity: Comparison

- Room temperature values $(\text{Ohm-m})^{-1} = (\Omega \text{ - m})^{-1}$

conductors

METALS	
Silver	6.8×10^{7}
Copper	6.0×10^{7}
Iron	1.0×10^{7}

semiconductors

SEMICONDUCTORS	
Silicon	4×10^{-4}
Germanium	2×10^{0}
GaAs	10^{-6}

insulators

CERAMICS	
Soda-lime glass	10^{-10} - 10^{-11}
Concrete	10^{-9}
Aluminum oxide	$<10^{-13}$

POLYMERS	
Polystyrene	$<10^{-14}$
Polyethylene	10^{-15} - 10^{-17}

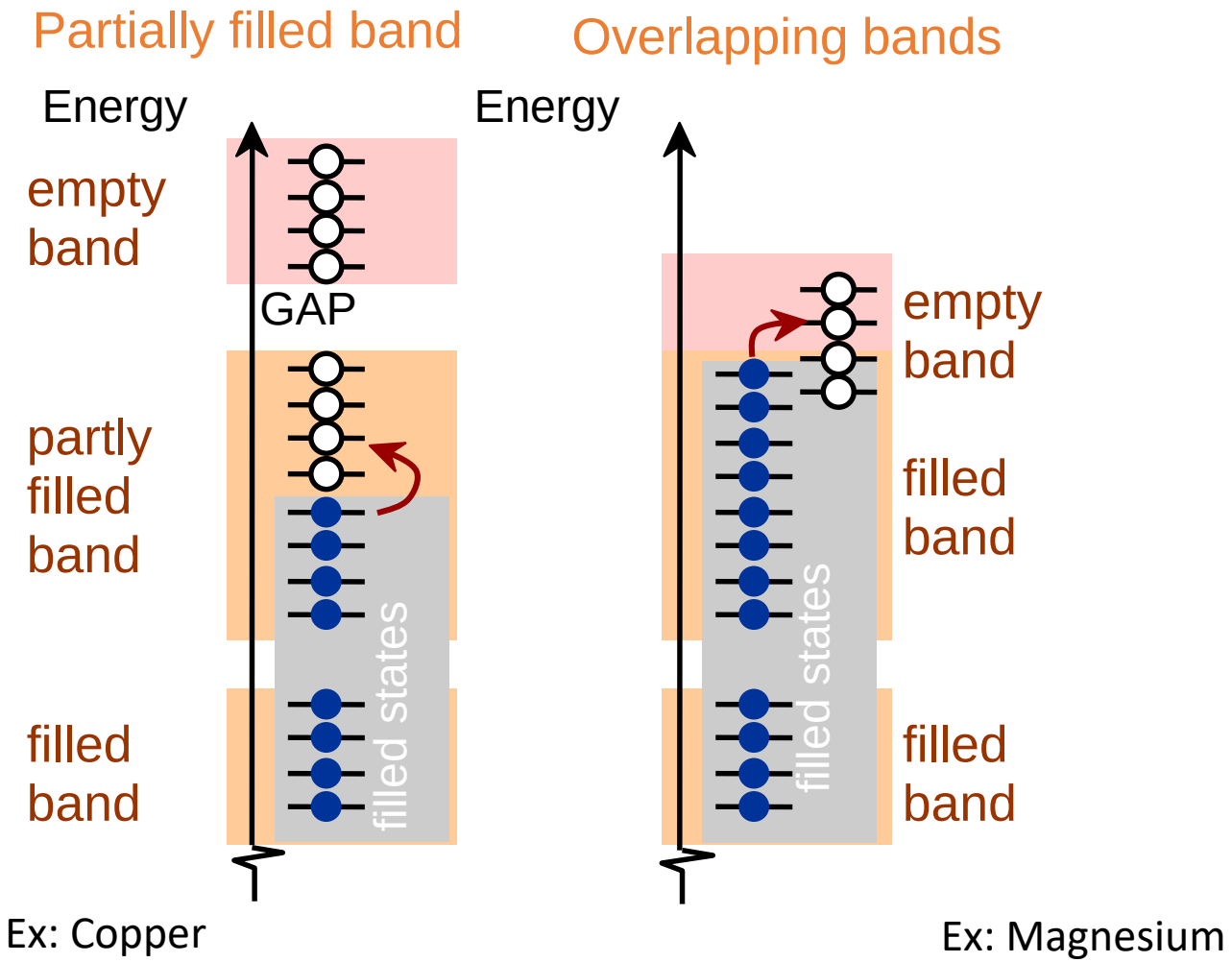
Energy Band Structures: Metals

- **Metals (Conductors):**

-- for metals empty energy states are adjacent to filled states.

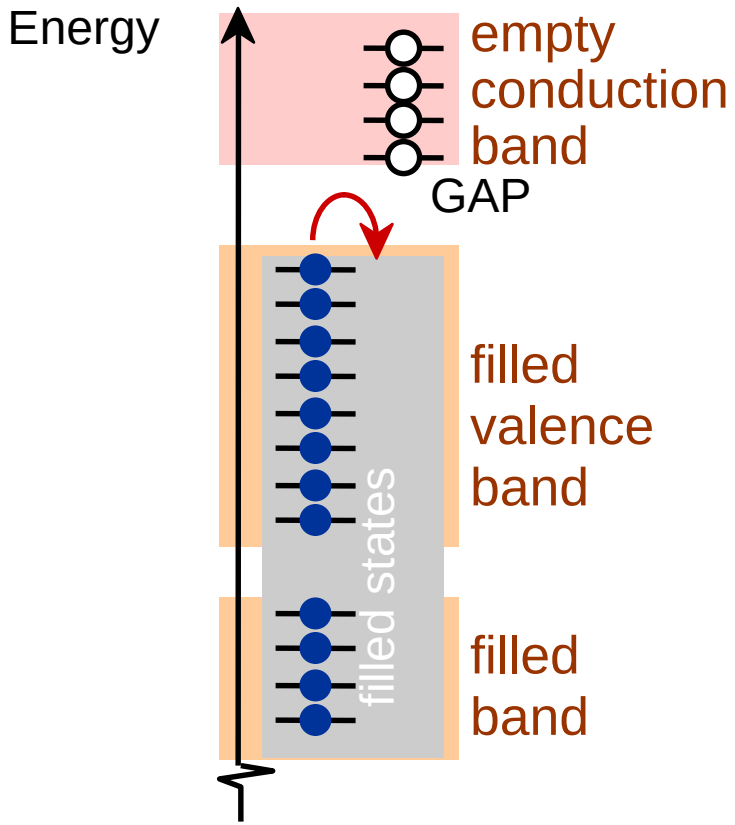
-- thermal energy excites electrons into empty higher energy states.

-- two types of band structures for metals
- partially filled band
- empty band that overlaps filled band

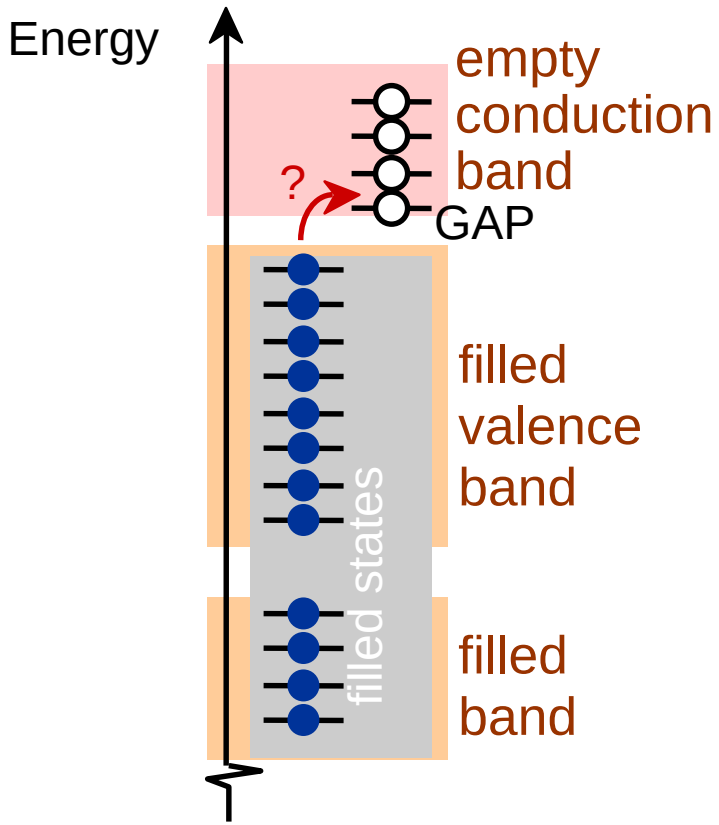


Energy Band Structures: Insulators & Semiconductors

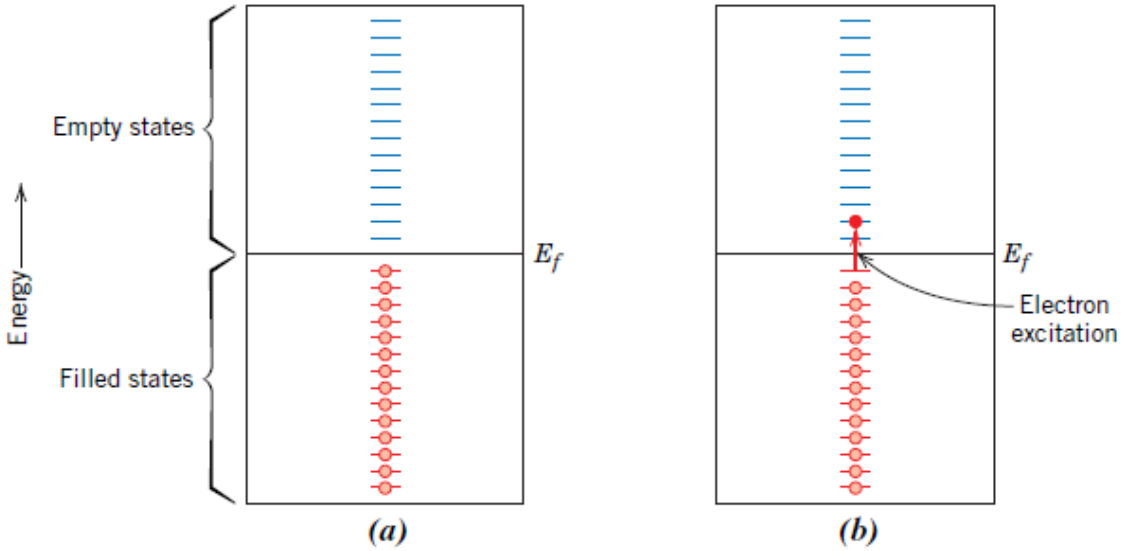
- **Insulators:**
  -- wide band gap (> 2 eV)
  -- few electrons excited across band gap



- **Semiconductors:**
  -- narrow band gap (< 2 eV)
  -- more electrons excited across band gap

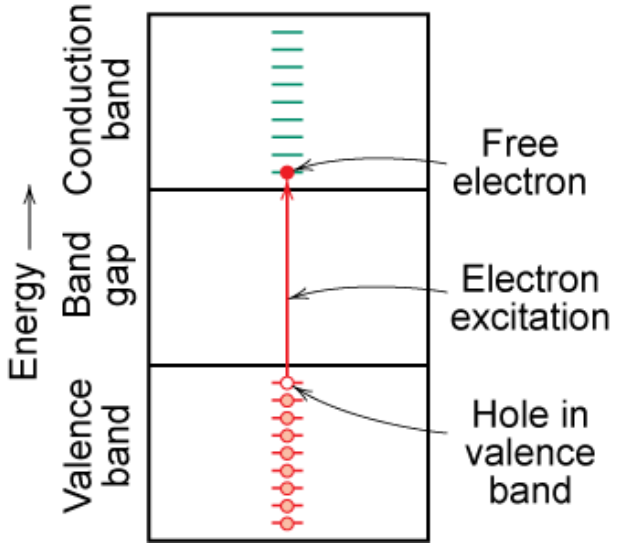


Conduction in Metals



For a metal, occupancy of electron states: (*a) before and (b) after an electron* excitation.

Charge Carriers in Insulators and Semiconductors



Two types of electronic charge carriers:

Free Electron

– negative charge

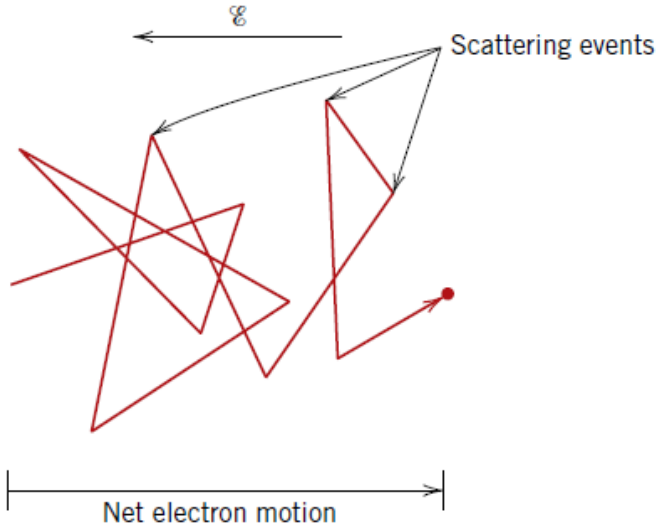
– in conduction band

Hole

– positive charge
– vacant electron state in the valence band

ELECTRON MOBILITY

- Scattering of electrons by imperfections in the crystal lattice, including impurity atoms, vacancies, interstitial atoms, dislocations, and thermal vibrations of the atoms themselves.



Drift velocity – average electron velocity

$$v_d = \mu_e \mathscr{E}$$

Constant is electron mobility – indication of frequency of scattering events

The conductivity σ of most materials may be expressed as

$$\sigma = n|e|\mu_e$$

where n is the number of free or conducting electrons per unit volume (e.g., per cubic meter) and $|e|$ is the absolute magnitude of the electrical charge on an electron (1.6×10^{-19} C). Thus, the electrical conductivity is proportional to both the number of free electrons and the electron mobility.

Resistivity in Metals

- Presence of imperfections increases resistivity
  - -- grain boundaries
  - -- dislocations
  - -- impurity atoms
  - -- vacancies

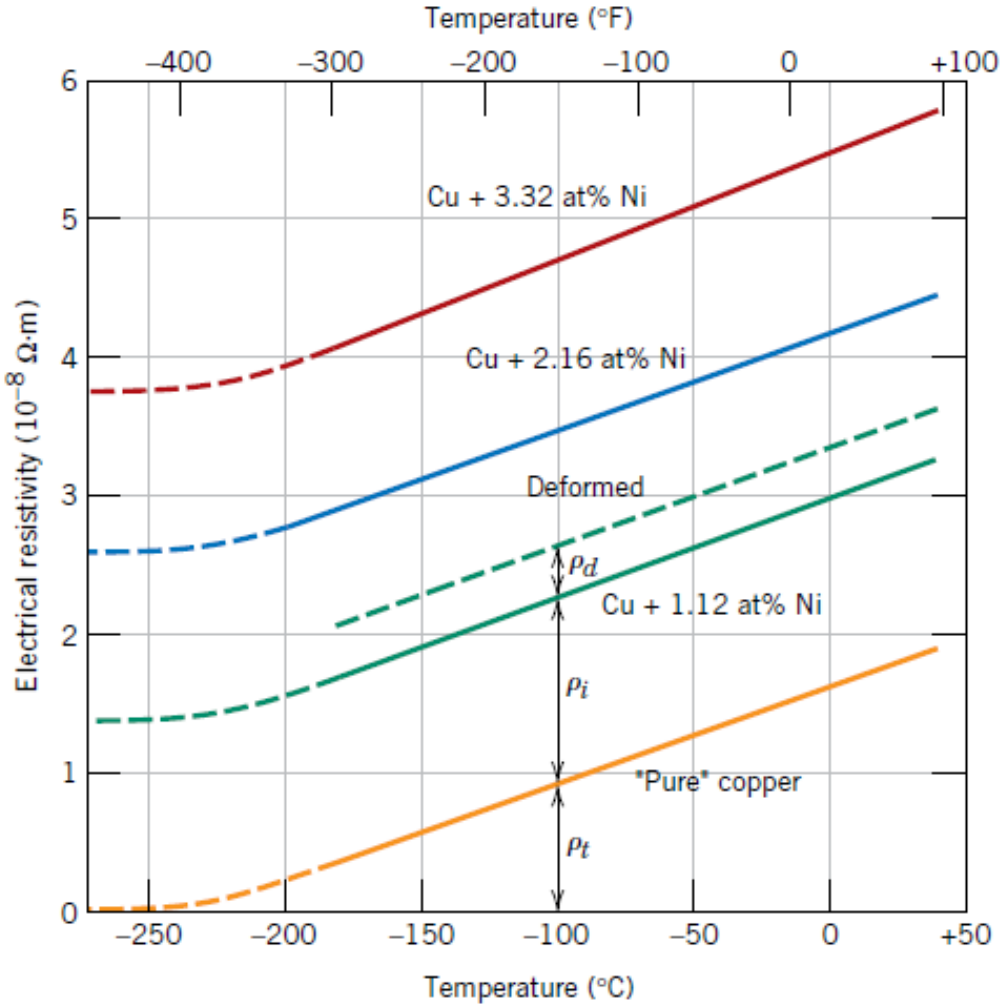
These act to scatter electrons so that they take a less direct path.

- Resistivity increases with:
  - -- temperature
  - -- wt% impurity
  - -- deformation (%CW)

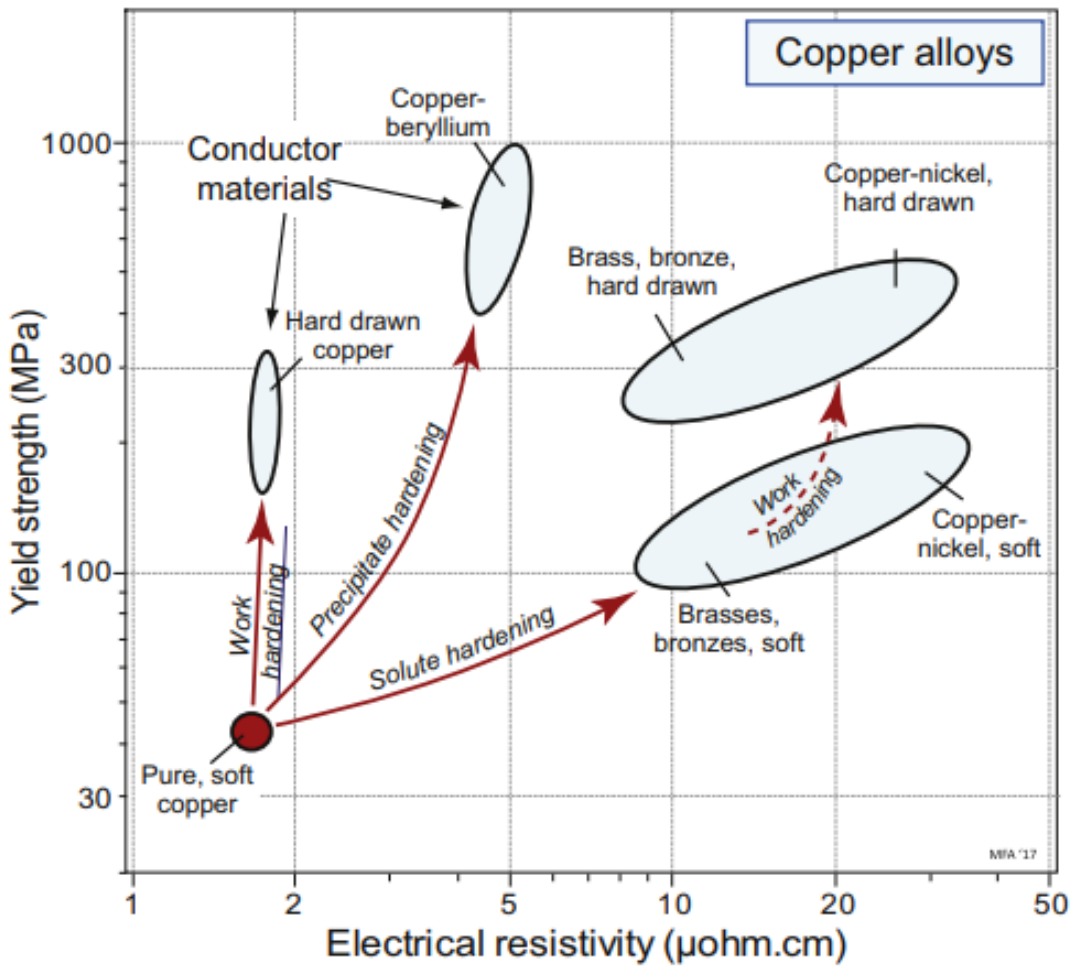
$$\rho = \rho_{thermal} + \rho_{impurity} + \rho_{deformation}$$

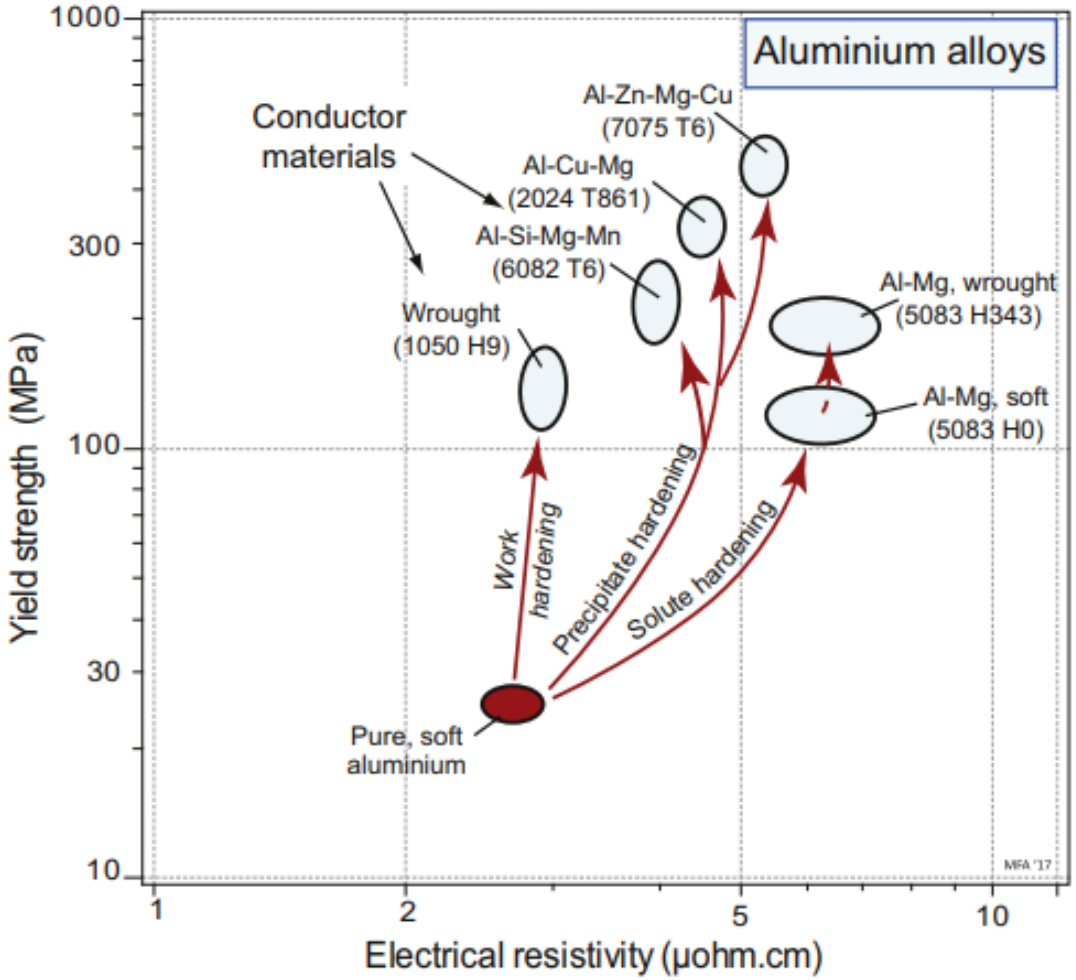
Matthiessen's rule

Metals: Influence of Temperature and Impurities on Resistivity



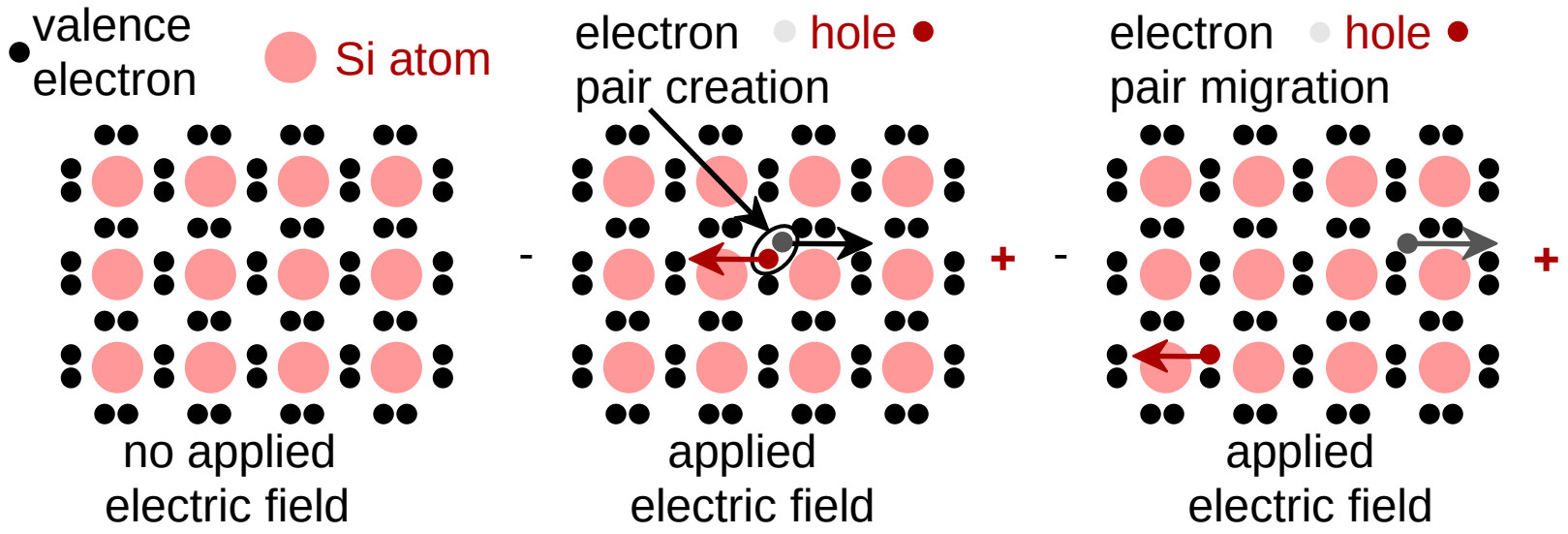
Resistivity in Metals





Semiconductors - Intrinsic Semiconduction in Terms of Electron and Hole Migration

- Concept of electrons and holes:



- Electrical Conductivity given by:

$$\sigma = n|e|\mu_e + p|e|\mu_h$$

holes/m³

hole mobility

electrons/m³

electron mobility

- for intrinsic semiconductor $n = p = n_i$

$$\therefore \quad \sigma = n_i|e|(\mu_e + \mu_h)$$

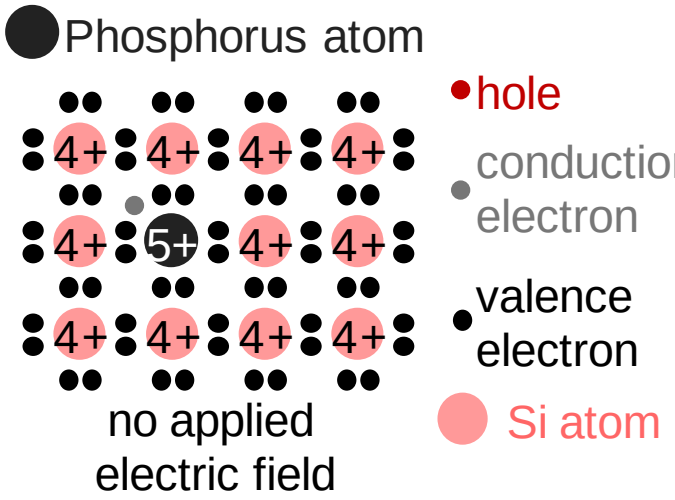
Intrinsic vs Extrinsic Conduction

- Intrinsic:
  -- case for pure Si
  -- # electrons = # holes ($n = p$)

- Extrinsic:
  -- electrical behavior is determined by presence of impurities that introduce excess electrons or holes
  -- $n \neq p$

- *n*-type Extrinsic: ($n >> p$)

$\sigma \approx n|e|\mu_e$



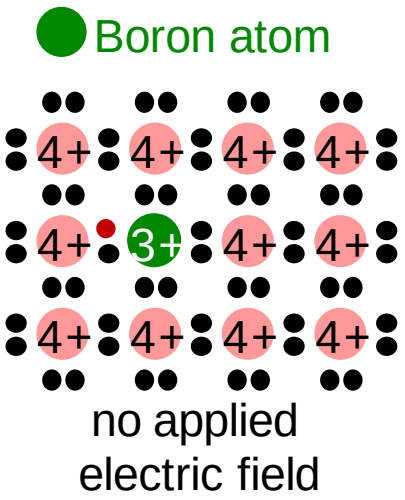
hole

conduction electron

valence electron

Si atom

- *p*-type Extrinsic: ($p >> n$)



$\sigma \approx p|e|\mu_h$

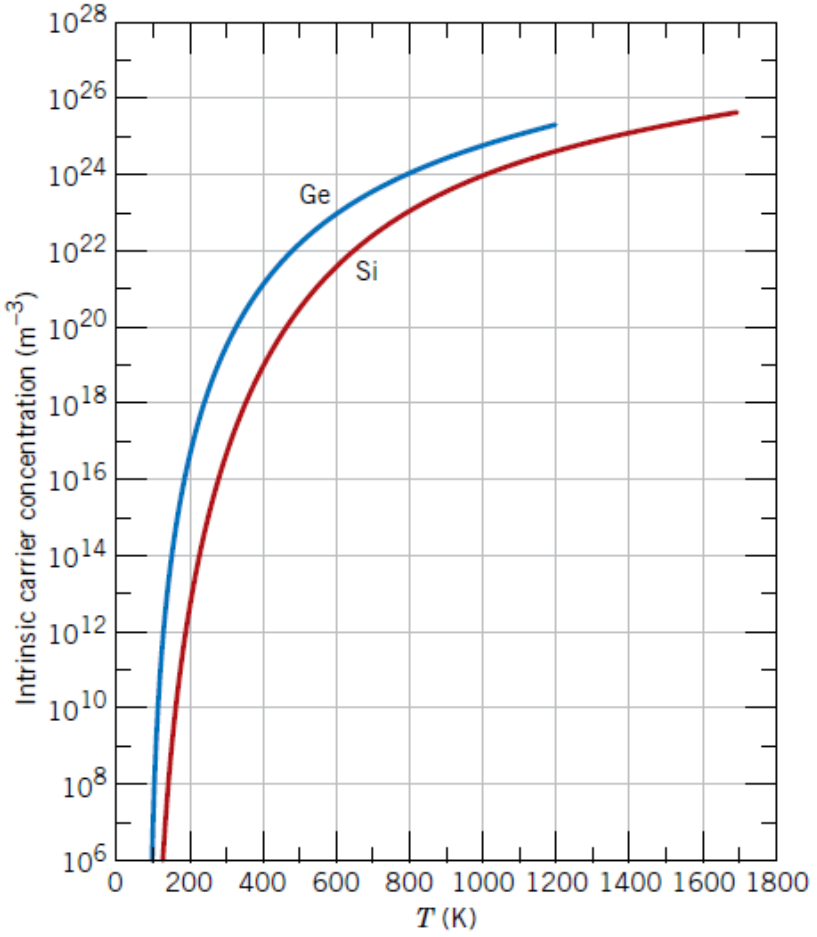
Under electric field, electron moves

Under electric field, hole moves

Carrier concentration with respect to temperature

For both intrinsic and extrinsic semiconduction, carrier concentration is a function of temperature.

Intrinsic



Intrinsic carrier concentration (logarithmic scale) as a function of temperature for germanium and silicon.

Extrinsic

the concentrations of electrons and holes increase with temperature.

at all temperatures, carrier concentration in Ge is greater than in Si. This effect is due to germanium's smaller band gap (0.67 vs. 1.11 eV)

Carrier concentration with respect to temperature

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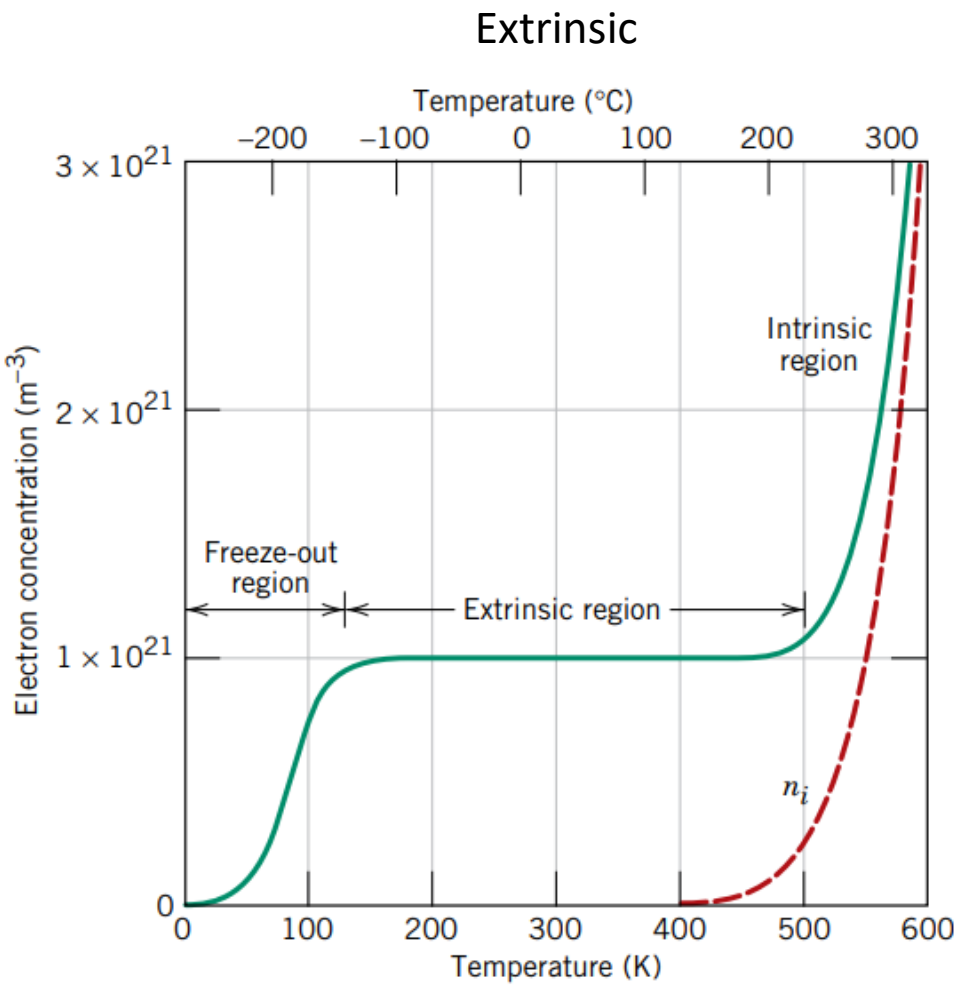


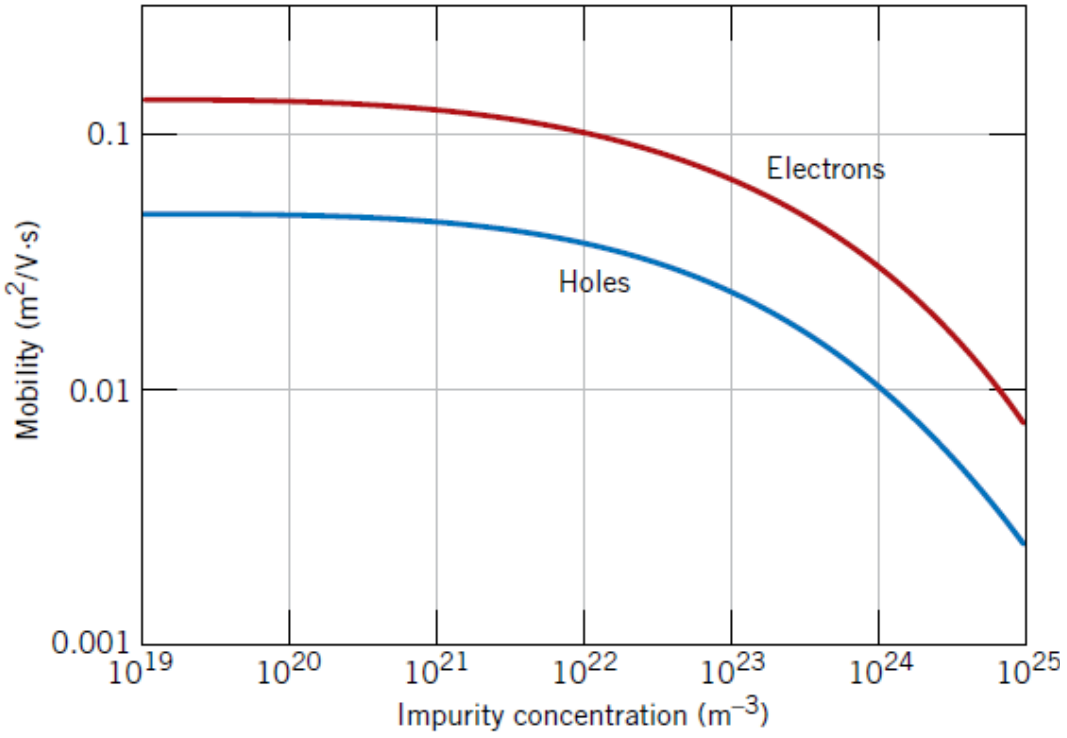
Electron concentration versus temperature for silicon (n-type) that has been doped with 10^{21} m^{-3} of a donor impurity and for intrinsic silicon (dashed line).

At intermediate temperatures (between approximately 150 K and 475 K) the material is n-type (inasmuch as P is a donor impurity), and electron concentration is constant. Also, intrinsic excitations across the band gap are insignificant in relation to these extrinsic donor excitations.

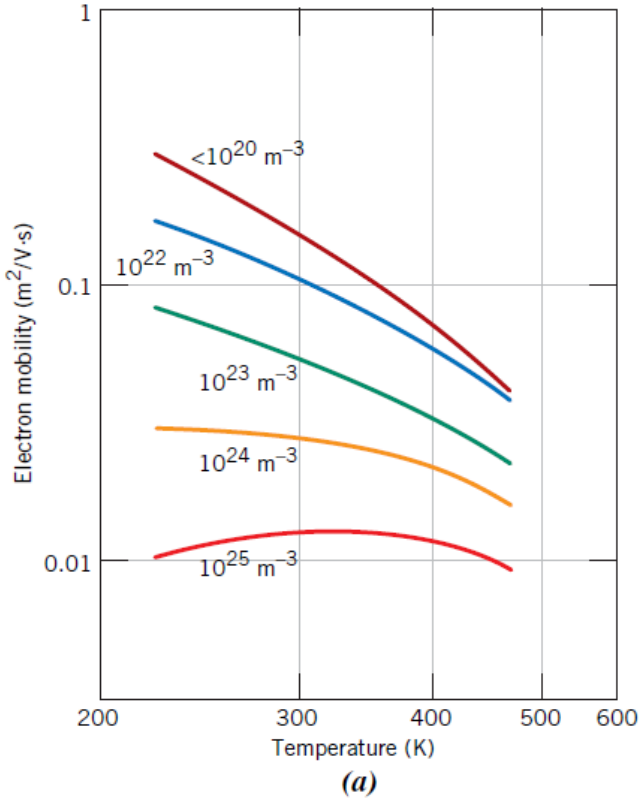
At low temperatures, the thermal energy is insufficient to excite electrons from the P donor level into the conduction band.

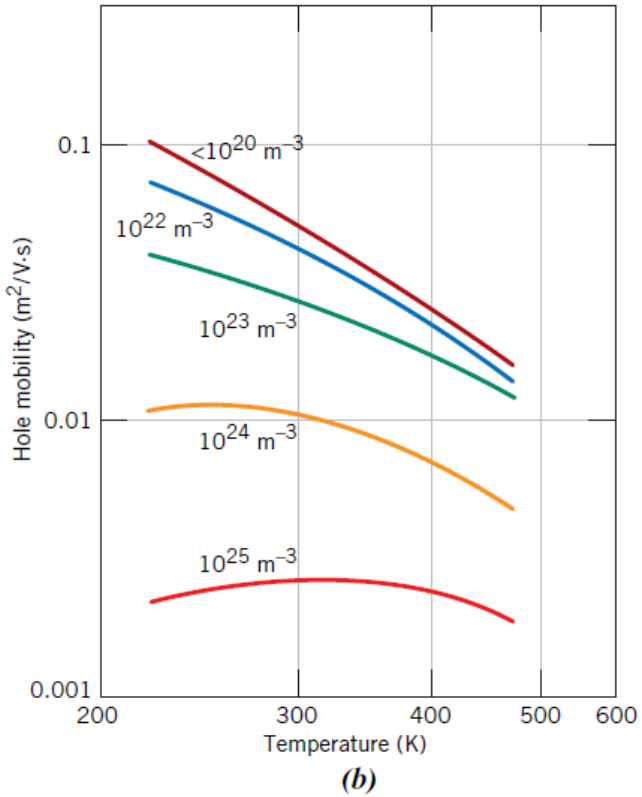
At high temperatures, charge carrier concentrations resulting from electron excitations across the band gap first become equal to and then completely overwhelm the donor carrier contribution with rising temperature.

Mobility of charge carriers with respect to dopant and temperature



For silicon, dependence of room-temperature electron and hole mobilities (logarithmic scale) on dopant concentration (logarithmic scale).





Temperature dependence of (*a*) electron and (*b*) hole mobilities for silicon that has been doped with various donor and acceptor concentrations.

Calculate the electrical conductivity of intrinsic silicon at 150°C (423 K)

Calculate the electrical conductivity of intrinsic silicon at 150°C (423 K)

$$\begin{aligned}\sigma &= n_i|e|(\mu_e + \mu_h)\\ &= (4 \times 10^{19}\ \mathrm{m}^{-3})(1.6 \times 10^{-19}\ \mathrm{C})(0.06\ \mathrm{m}^2/\mathrm{V{\cdot}s} + 0.022\ \mathrm{m}^2/\mathrm{V{\cdot}s})\\ &= 0.52\ (\Omega{\cdot}\mathrm{m})^{-1}\end{aligned}$$

Silicon (n-type) that has been doped with 10^{23} m^{-3} of a phosphorus impurity. Compute conductivity at RT and at 373K.

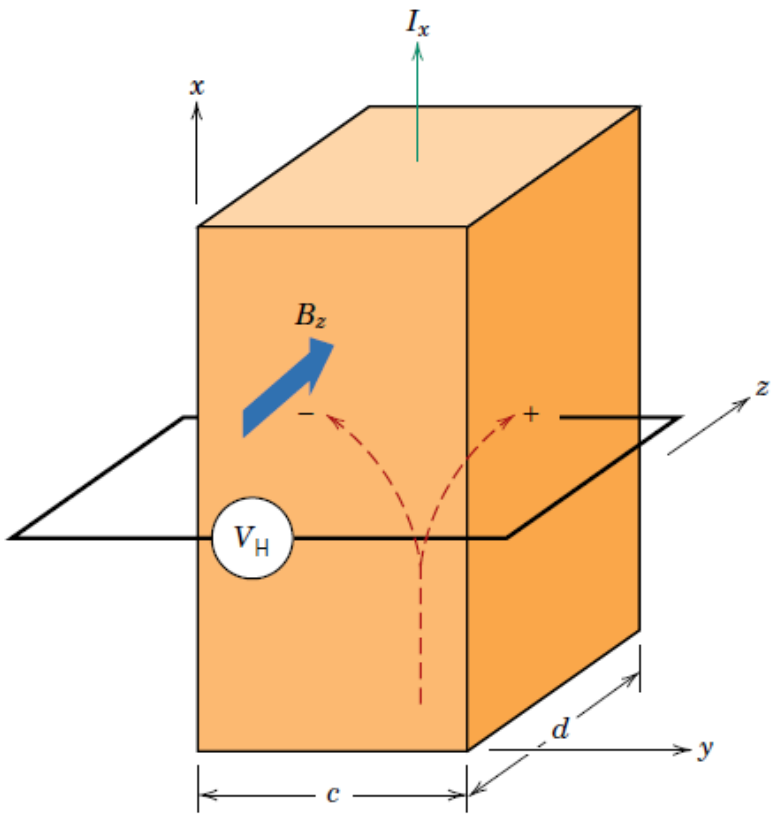
Silicon (n-type) that has been doped with 10^{23} m^{-3} of a phosphorus impurity. Compute conductivity at RT and at 373K.

$$\begin{aligned}\sigma &= n|e|\mu_e \\ &= (10^{23}\,\mathrm{m}^{-3})(1.6\times10^{-19}\,\mathrm{C})(0.07\,\mathrm{m}^2/\mathrm{V\cdot s}) \\ &= 1120\,(\Omega\cdot\mathrm{m})^{-1}\end{aligned}$$

$$\begin{aligned}\sigma &= n|e|\mu_e \\ &= (10^{23}\,\mathrm{m}^{-3})(1.6\times10^{-19}\,\mathrm{C})(0.04\,\mathrm{m}^2/\mathrm{V\cdot s}) \\ &= 640\,(\Omega\cdot\mathrm{m})^{-1}\end{aligned}$$

THE HALL EFFECT

To determine concentration of charge carrier, and mobility



A magnetic field applied perpendicular to the direction of motion of a charged particle exerts a force on the particle perpendicular to both the magnetic field and the particle motion directions.

The magnitude of V_H depends on I_x, B_z, and the specimen thickness d as

$$V_H = \frac{R_H I_x B_z}{d}$$

R_H is termed the *Hall coefficient*

$$R_H = \frac{1}{n|e|}$$

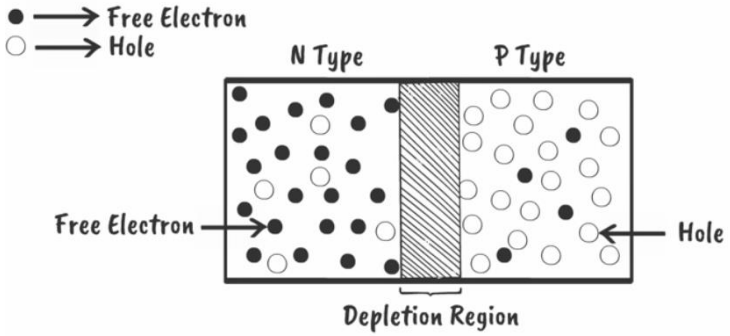
(From this, n can be determined)

Also,

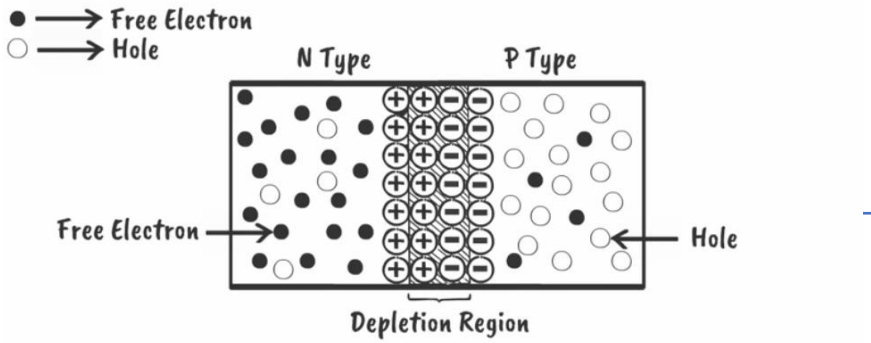
$$\mu_e = \frac{\sigma}{n|e|}$$

Semiconductor Devices

PN Diode



n type material is electrically neutral => free electrons = positive donor ions
p type material is electrically neutral => holes = negative acceptor ions



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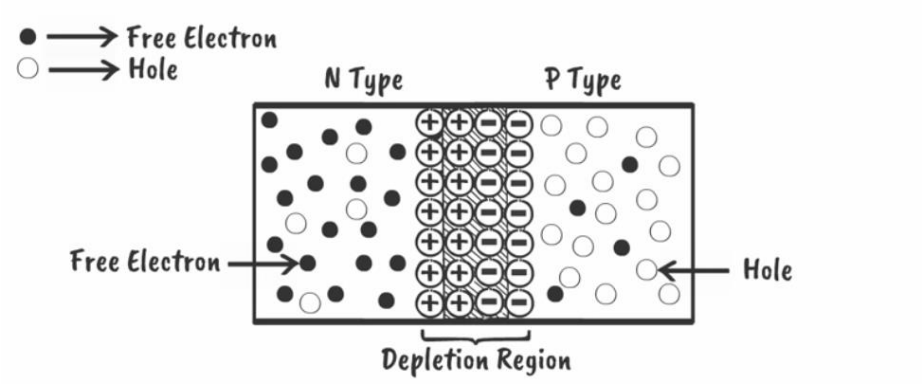
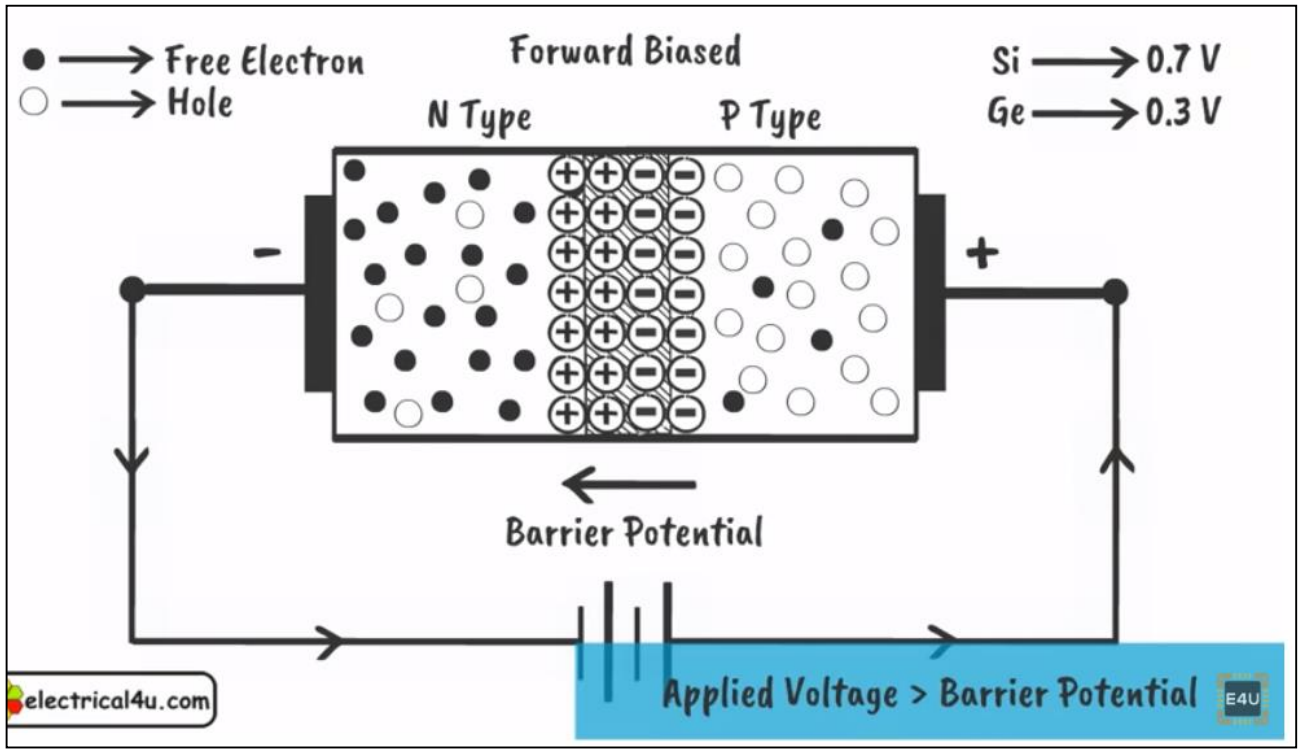


n type material is electrically neutral => free electrons = positive donor ions
p type material is electrically neutral => holes = negative acceptor ions

Depletion region generates static electric field which acts as barrier for further migration of electrons and holes across the junction

Semiconductor Devices

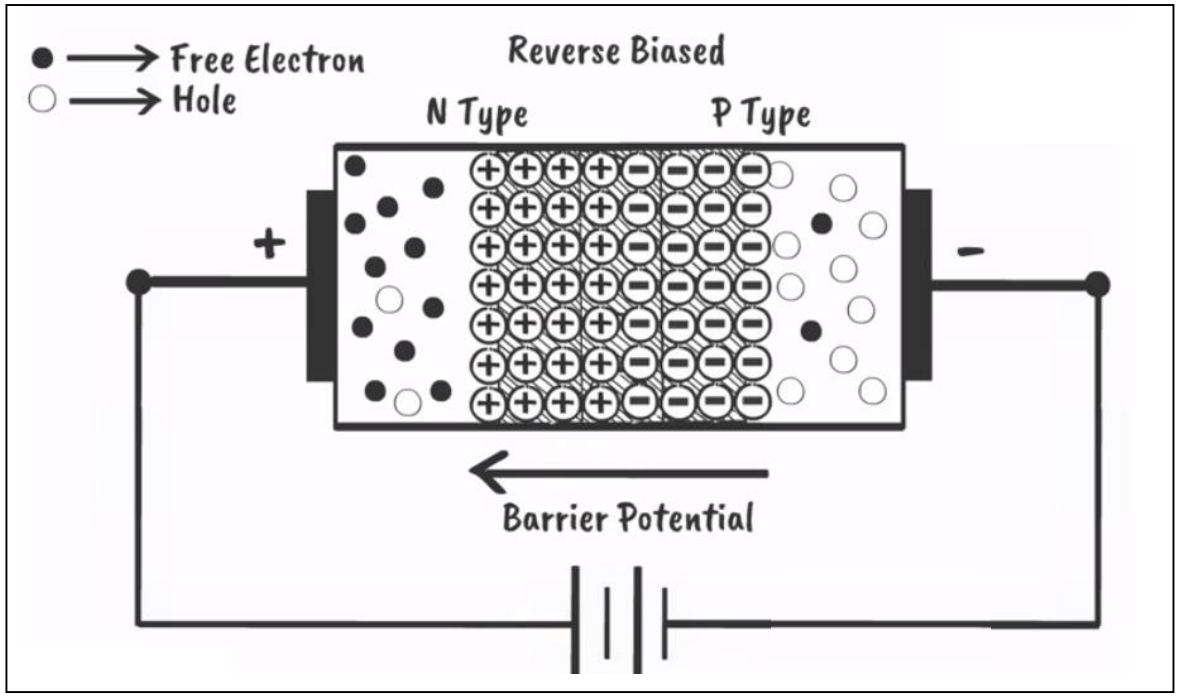
PN Diode



Only minor current flows - due to minority carrier – (electrons in p type and holes in n type) – so reverse bias blocks current

Depletion region generates static electric field which acts as barrier for further migration of electrons and holes across the junction

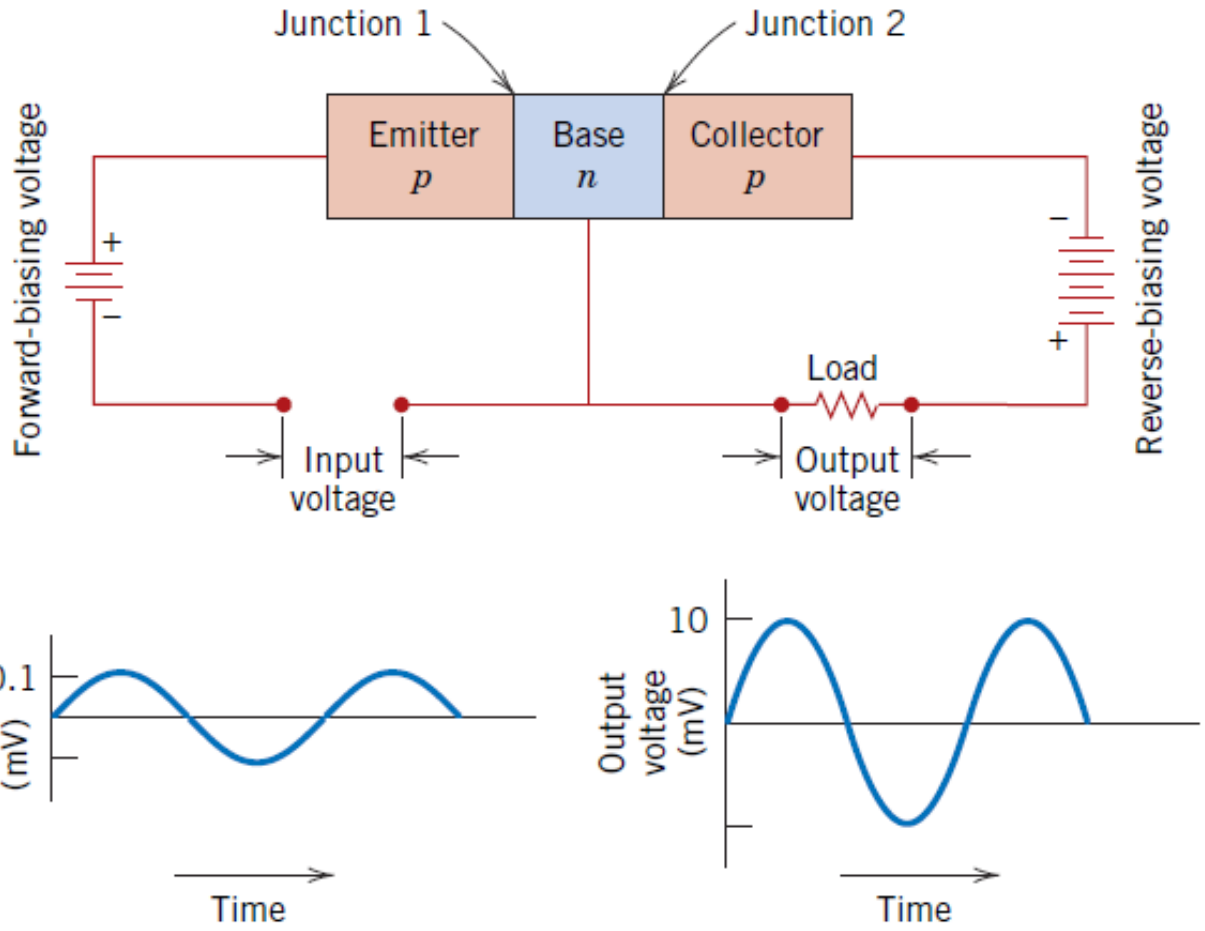
Current flows when applied voltage > barrier potential in forward bias

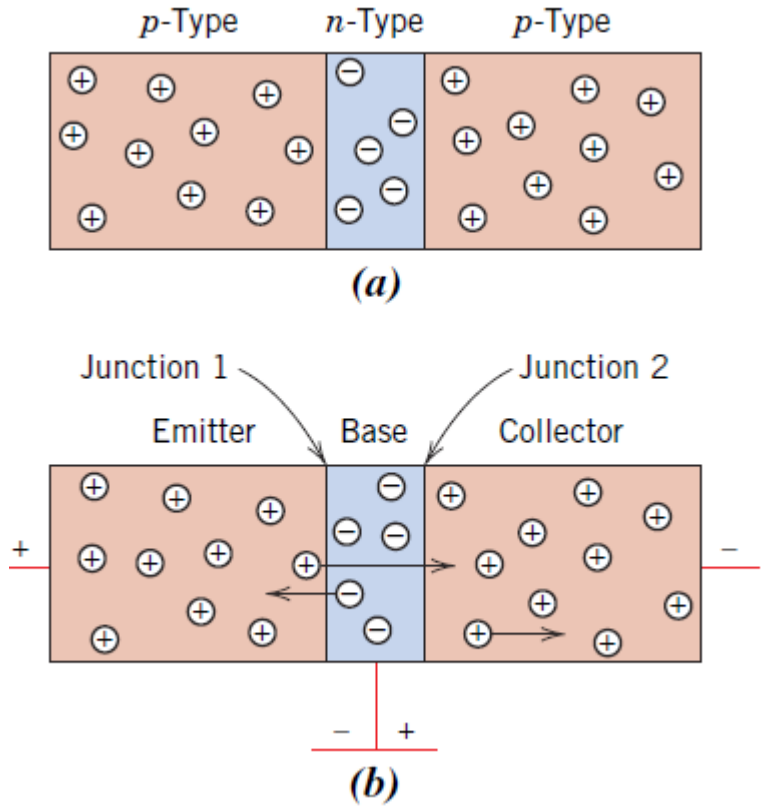


Semiconductor Devices

Transistor - For signal amplifications

PNP Junction Transistor



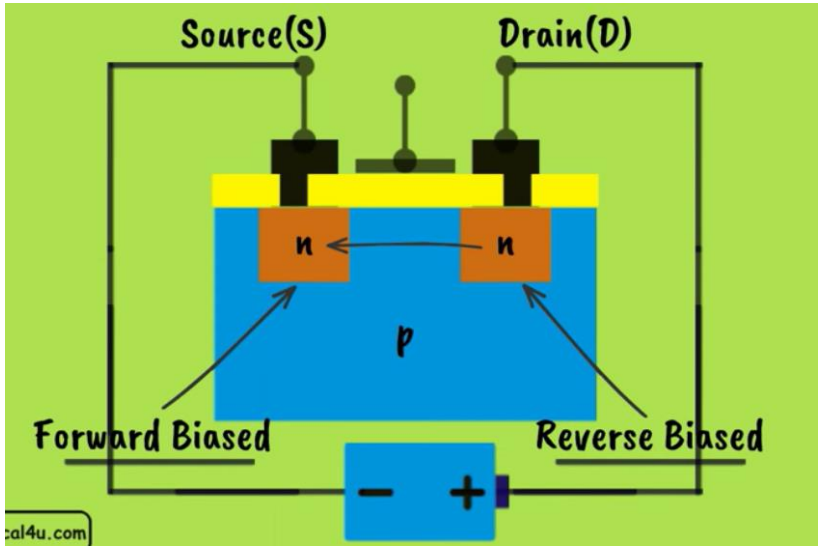


For a junction transistor (p–n–p type), the distributions and directions of electron and hole motion (a) when no potential is applied and (b) with appropriate bias for voltage amplification.

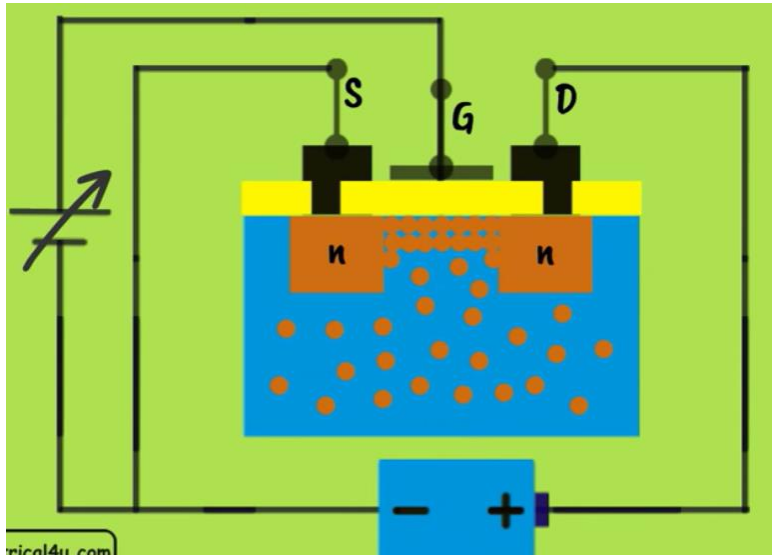
Semiconductor Devices

Transistor

MOSFET *metal-oxide-semiconductor field-effect transistor*



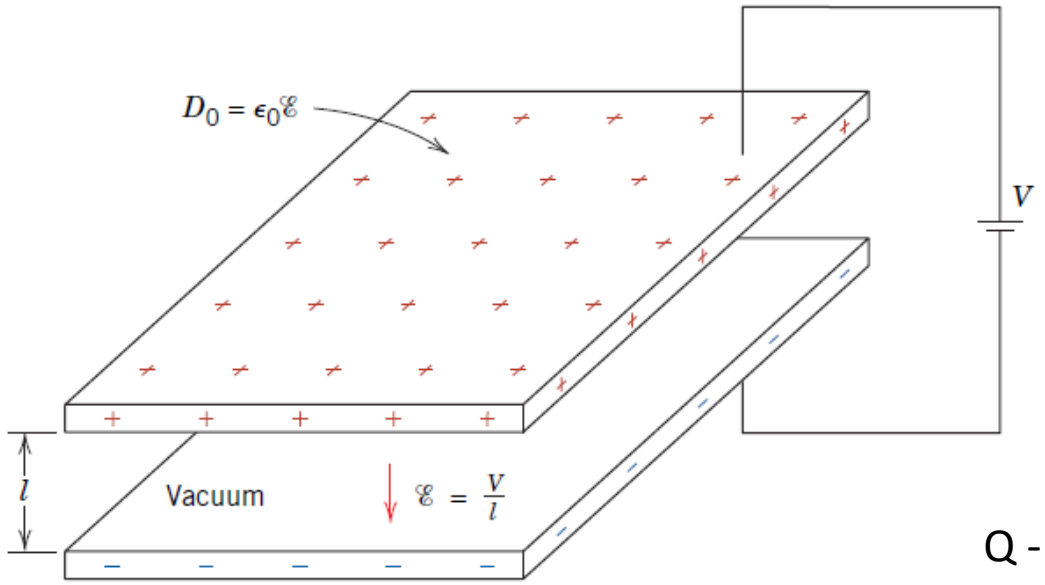
No channel between Drain and source



When gate is connected to suitable positive voltage, accumulation of attracted minority carriers (negative electrons) in P-type as shown takes place – form a channel inside p-type semiconductor – and thus channel behaves as n-type semiconductor – thus the device will conduct current as electron are attracted towards +ve terminal of the drain as the gate voltage is increased – Result is electrons will start flowing from source to drain through the channel.

Capacitor

A parallel-plate capacitor (when a vacuum is present in-between)



Capacitance *C*

$$C = \frac{Q}{V} = \epsilon_0 \frac{A}{l}$$

The units of capacitance are coulombs per volt, or farads (F).

Q - quantity of charge stored on either plate

V - the voltage applied across the capacitor

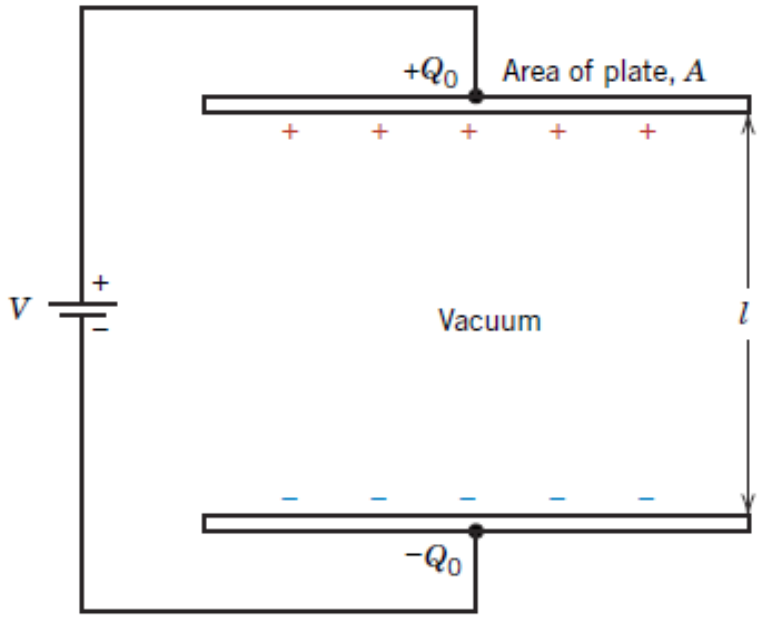
A represents the area of the plates

l is the distance between them.

Parameter ε_0, called the **permittivity** of a vacuum, is a universal constant having the value of 8.85 x10^{-12} F/m.

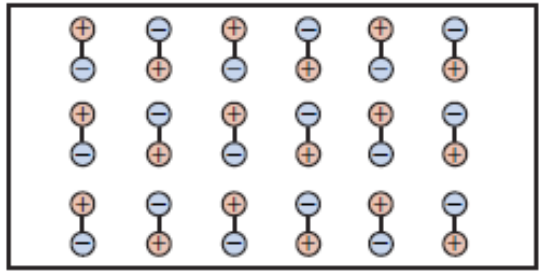
Dielectric Behaviour

Capacitor with vacuum in-between

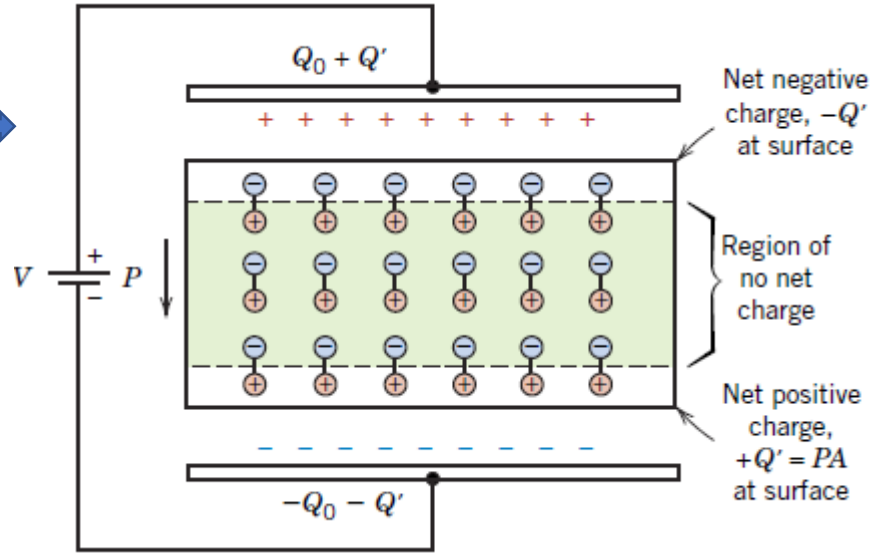


The charge stored on capacitor plates for a vacuum

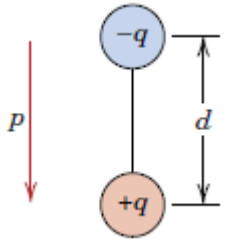
Because of dielectric, the amount of charge stored per unit volt increased



Capacitor with dielectric material in-between



The increased charge-storing capacity resulting from the **polarization of a dielectric material. P is increase in charge density because of dielectric.**

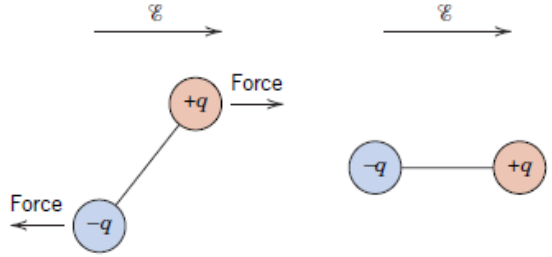


$$p = qd$$

p = dipole moment

A **dielectric** material is one that is electrically insulating (nonmetallic) and exhibits or may be made to exhibit an **electric dipole** structure—that is, there is a separation of positive and negative electrically charged entities.

Dipole under electric field



Dielectric Behaviour

If a dielectric material is inserted into the region within the plates

$$C = \epsilon \frac{A}{l}$$

ε is the **permittivity** dielectric medium

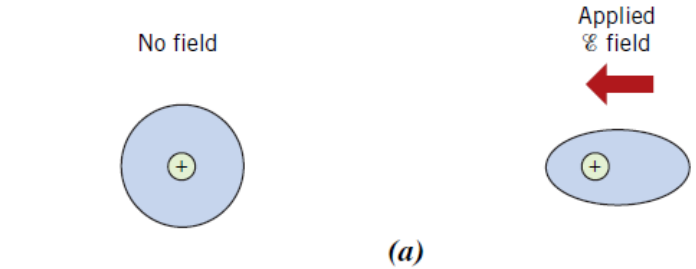
Relative permittivity or Dielectric constant, ε_r, is given by

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$

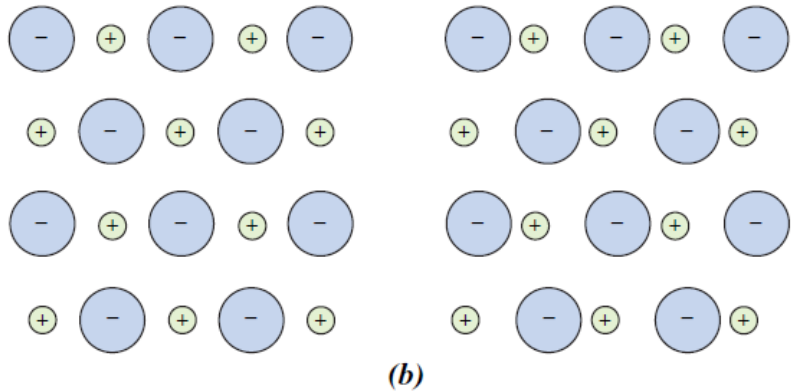
which is greater than unity and represents the **increase in charge-storing capacity upon insertion of the dielectric medium between the plates.**

The dielectric constant is one material property of prime consideration for capacitor design.

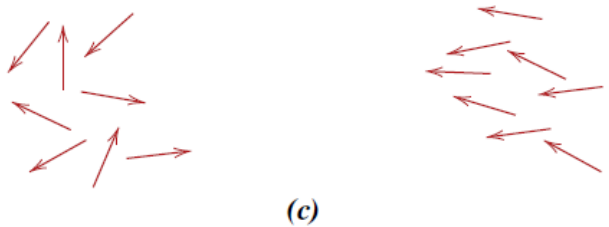
Types of Polarization



Electronic polarization that results from the distortion of an atomic electron cloud by an electric field.



Ionic polarization that results from the relative displacements of electrically charged ions in response to an electric field.

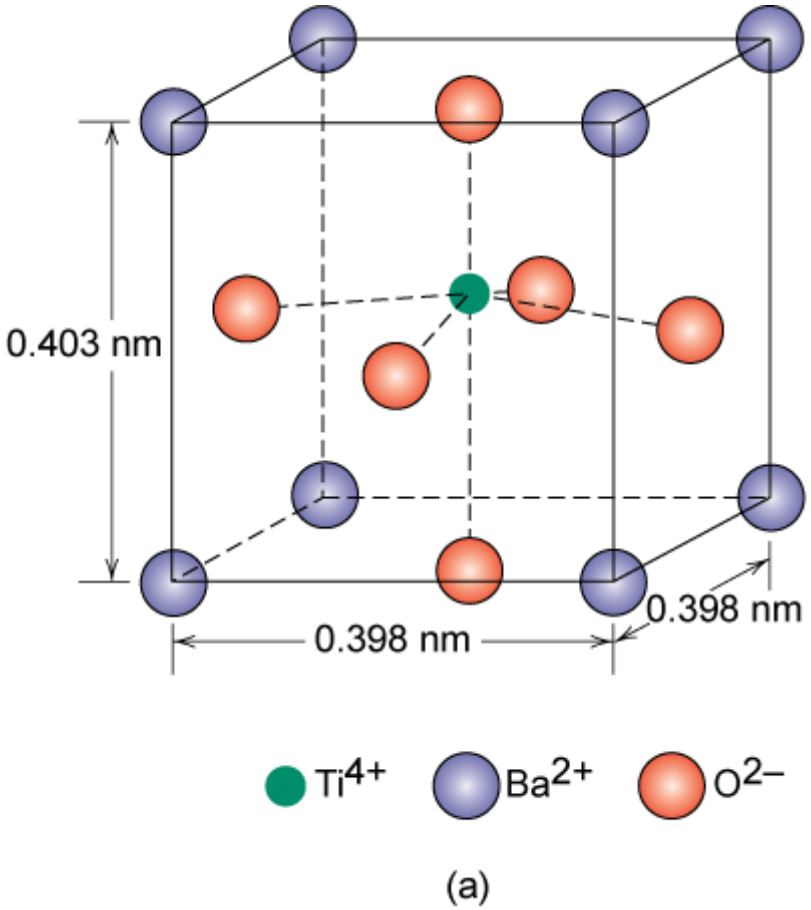


Response of permanent electric dipoles (arrows) to an applied electric field, producing **orientation polarization**.

Total polarization $P = P_e + P_i + P_o$

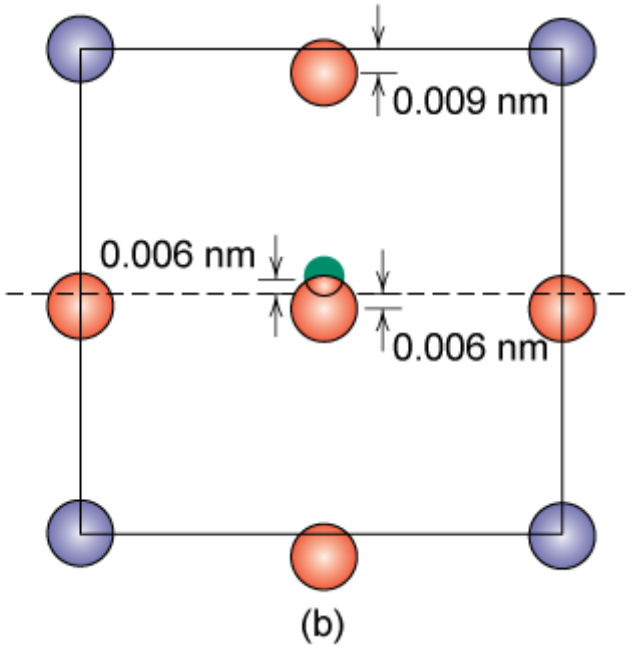
Ferroelectricity

- Experience spontaneous polarization (in the absence of electric field)
- Consists of permanent electric dipoles



Example:

$BaTiO_3$ -- ferroelectric (below its Curie temperature (120ºC))



Spontaneous polarization results because of interactions between adjacent permanent dipoles wherein they mutually align, all in the same direction.

Ferroelectrics have extremely high dielectric constants;

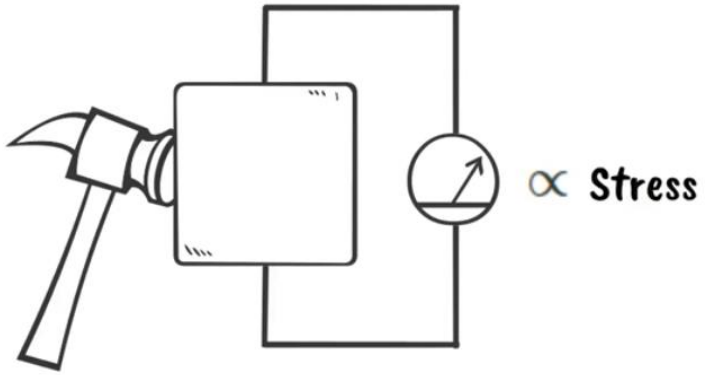
for example, at room temperature, ε_r for barium titanate may be as high as 5000.

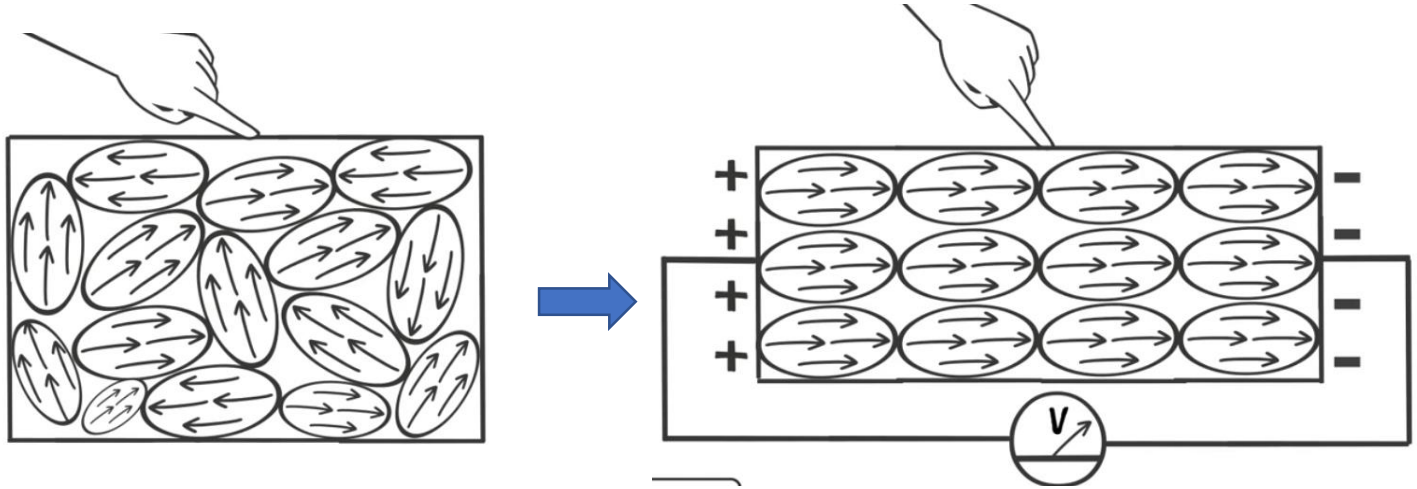
Consequently, capacitors made from these materials can be significantly smaller than capacitors made from other dielectric materials.

PIEZOELECTRICITY

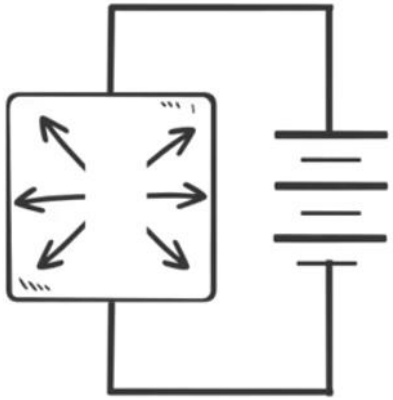
Piezoelectric ceramic materials include Quartz, titanates of barium and lead (BaTiO3 and PbTiO3), lead zirconate (PbZrO3), lead zirconate–titanate (PZT).

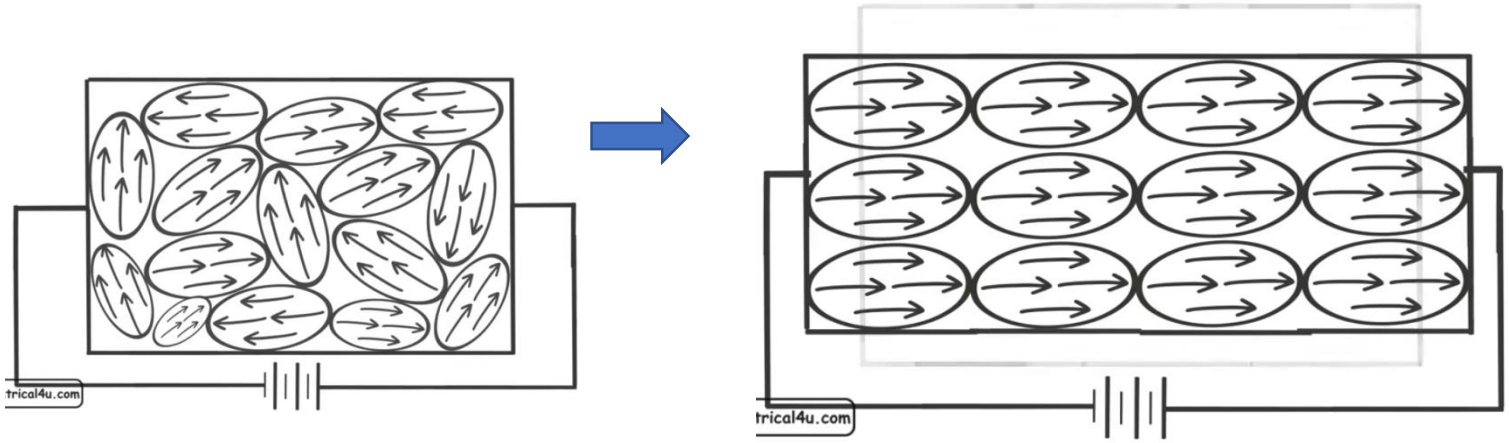
Application of stress (or strain) on a material produces electric potential – piezoelectric effect



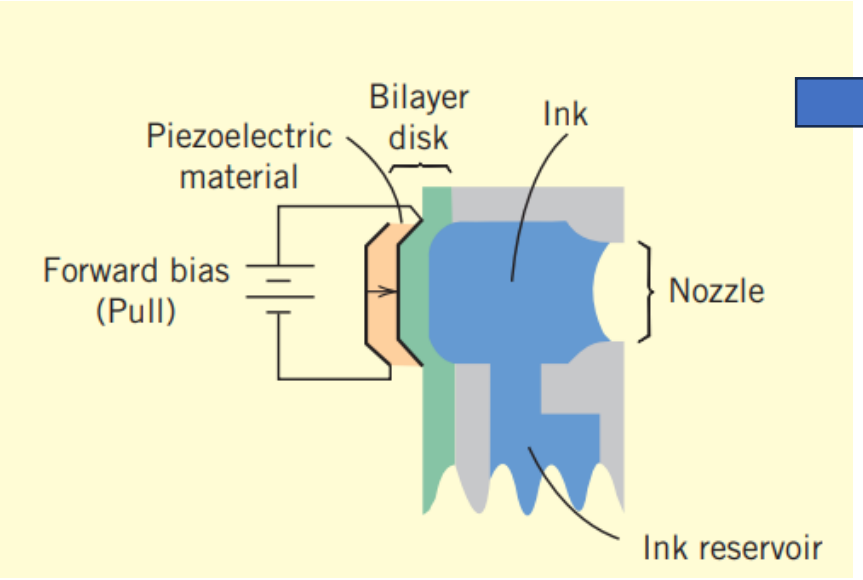


Application of electric filed on a material produces strain – inverse piezoelectric effect

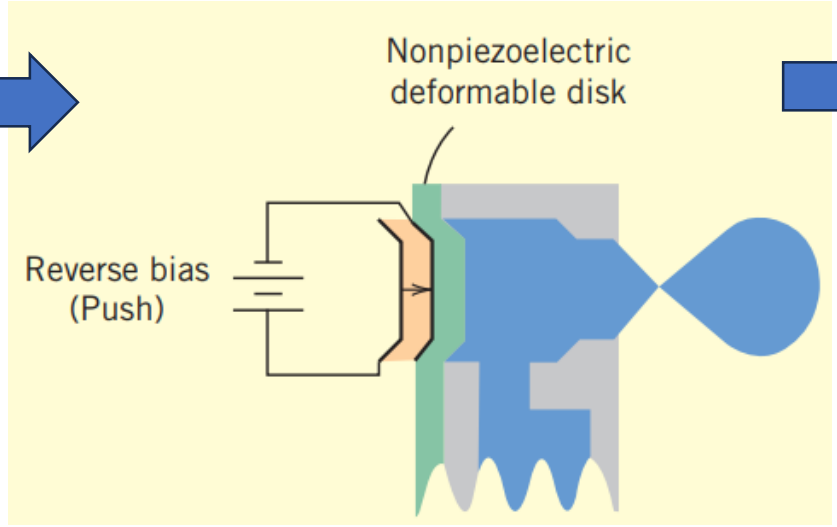




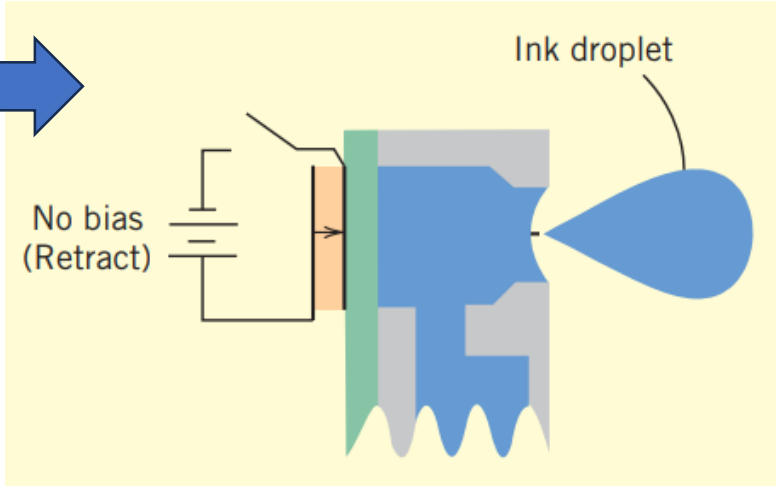
Piezoelectric Ink-Jet Printer Heads



the imposition of forward bias voltage causes the bilayer disk to flex in such a way as to pull (or draw) ink from the reservoir into the nozzle chamber

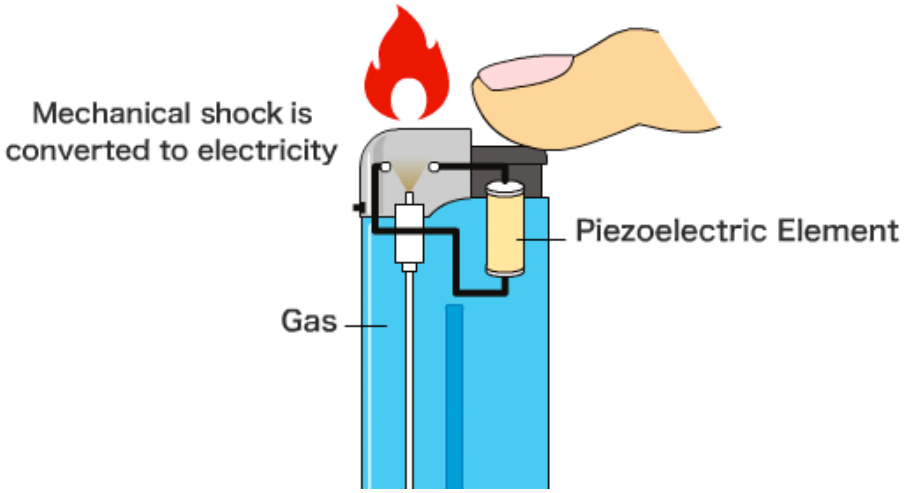


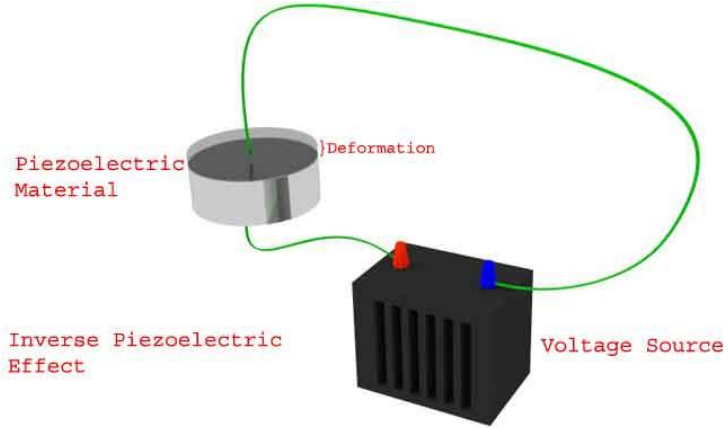
Reversing the voltage bias forces the bilayer disk to bend in the opposite direction—toward the nozzle—so as to eject a drop of ink

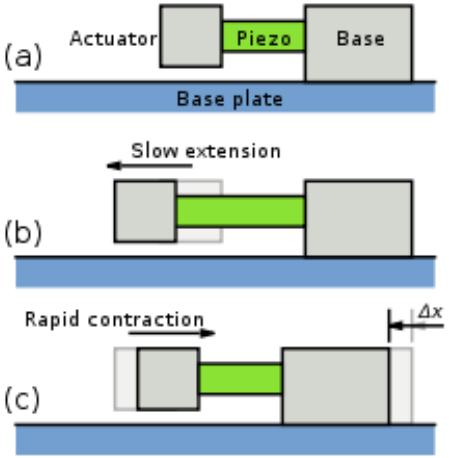


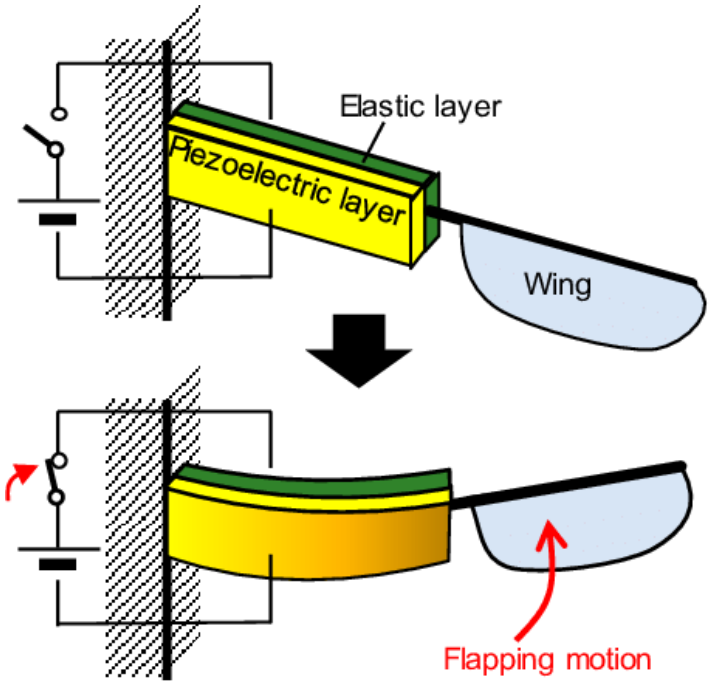
Finally, removal of the voltage causes the disk to return to its unbent configuration in preparation for another ejection sequence.

PIEZOELECTRICITY



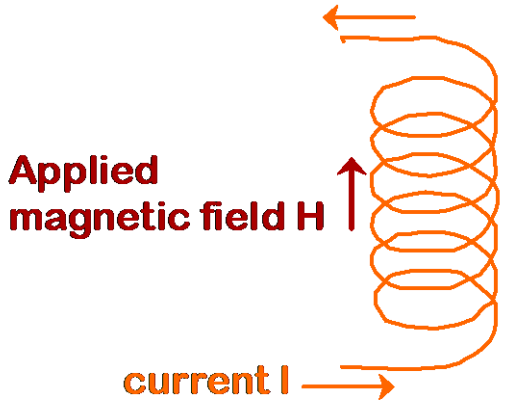






Magnetic Properties

- Created by current through a coil:



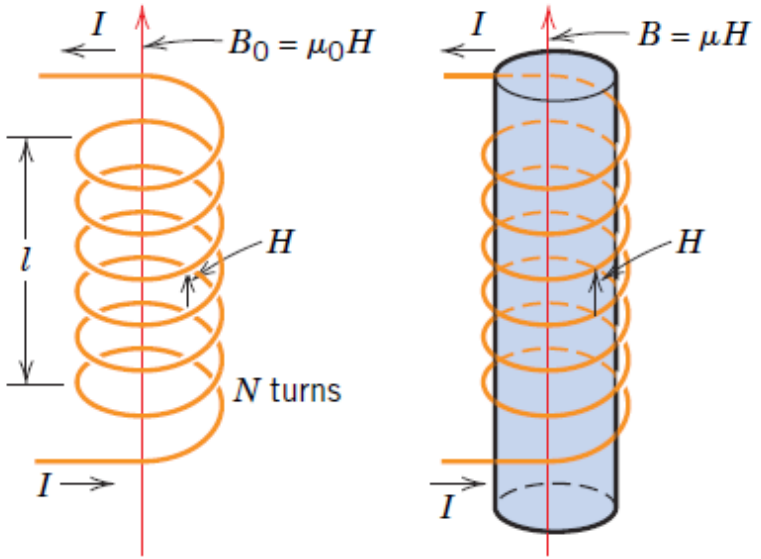
- Relation for the applied magnetic field, H:

$$H = \frac{NI}{L}$$

N = number of turns, I = current, L = length

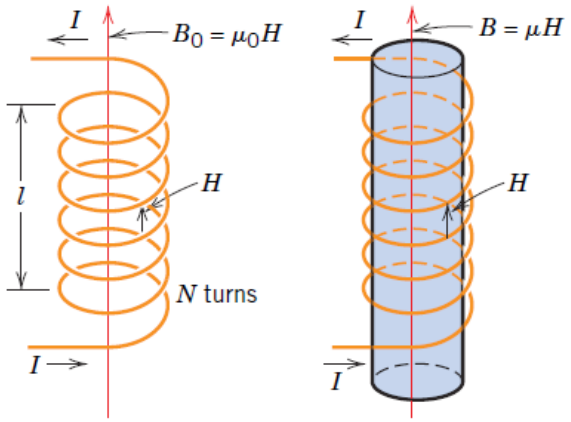
applied magnetic field (Magnetic field strength)

units = (ampere-turns/m)



Magnitude of the internal field strength within a substance that is subjected to an *H is called* **magnetic induction,** or **magnetic flux density (B)**

Magnetic induction



The magnetic field strength and flux density are related according to

$$B = \mu H$$

The parameter μ is called the **permeability,** which is a property of the specific medium. (Units – Wb/A.m or Henries/m)

In a vacuum, $B_0 = \mu_0 H$

where μ_0 is the *permeability of a vacuum,* a universal constant, which has a value of $4\pi \times 10^{-7}$ H/m

μ_r is called the *relative permeability* $\mu_r = \dfrac{\mu}{\mu_0}$

Measure of the degree to which the material can be magnetized, or the ease with which a 'B' field can be induced in the presence of an external 'H' field.

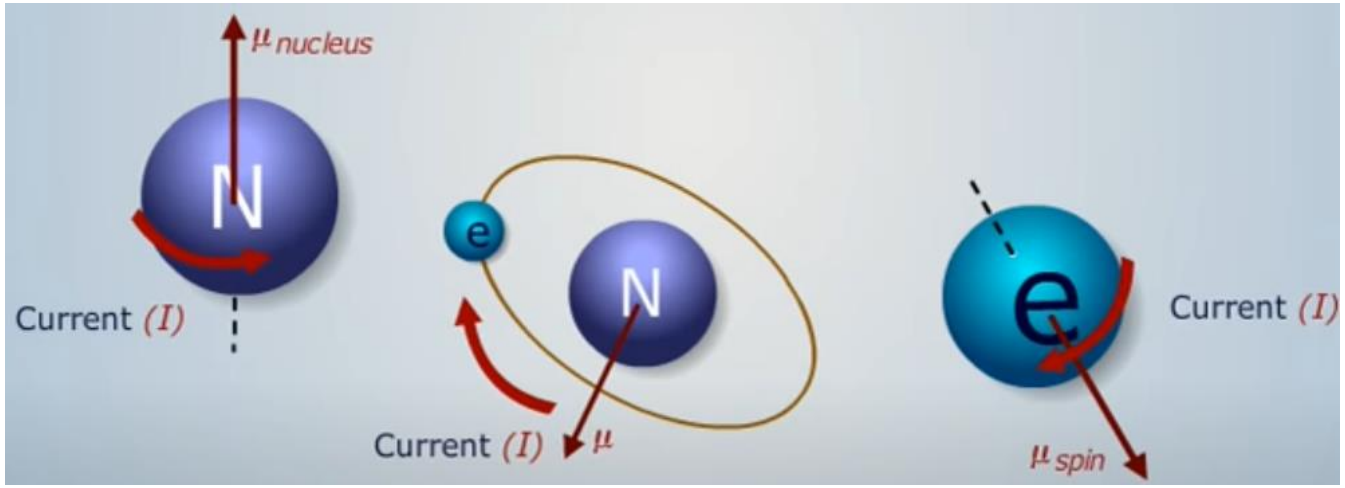
Another property, M, called the magnetization of the solid, is defined by the expression: $B = \mu_0 H + \mu_0 M$

In the presence of an 'H' field, the magnetic moments within a material tend to become aligned with the field and tend to reinforce it; the term $\mu_0 M$ in the Equation is a measure of this contribution.

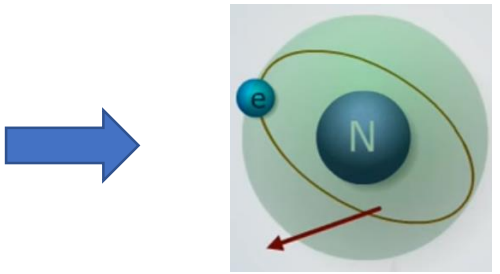
However, the magnitude of M is related to the applied field as follows: $M = \chi_m H$

χ_m is called the magnetic susceptibility and is related to relative permeability as $\chi_m = \mu_r - 1$

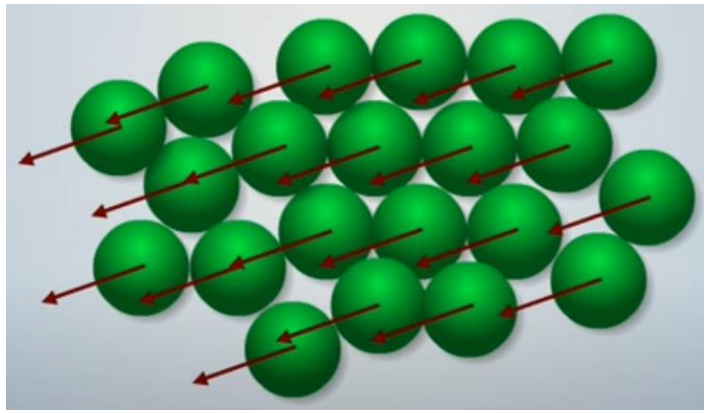
Atomic magnetic moments



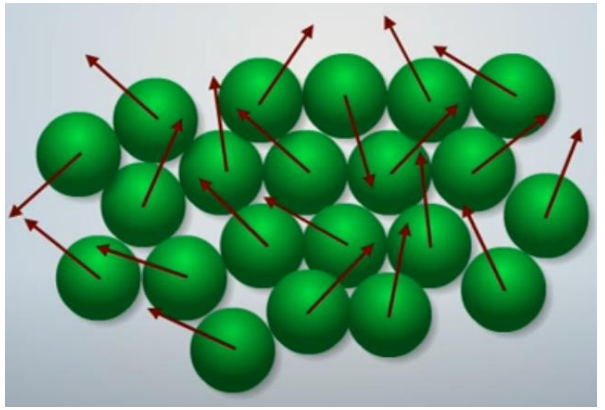
Net atomic magnetic moment



- **Electrons produce magnetic orbital, spin moments, nucleus produce spin moment**

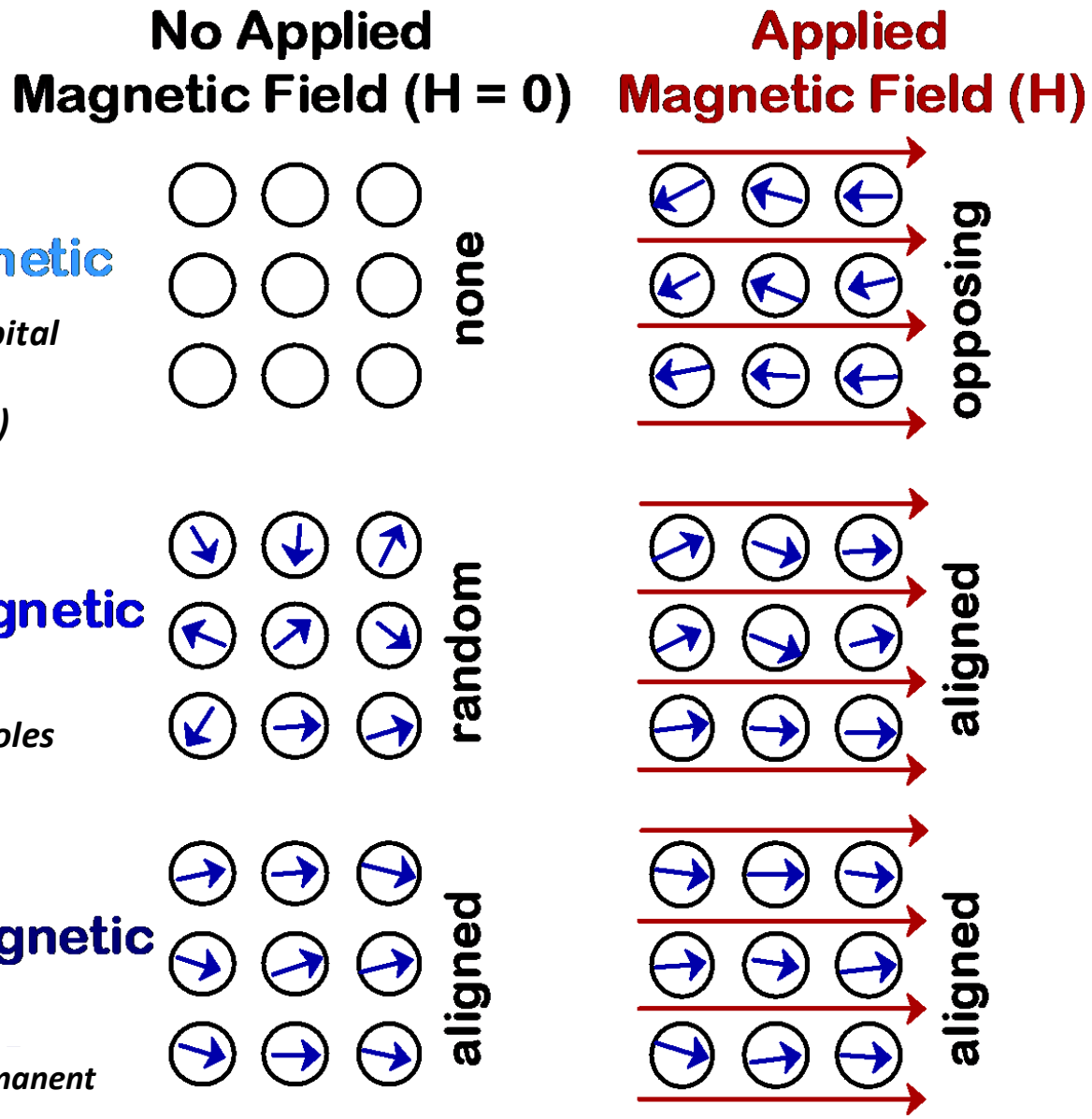


Aligned



Randomly oriented

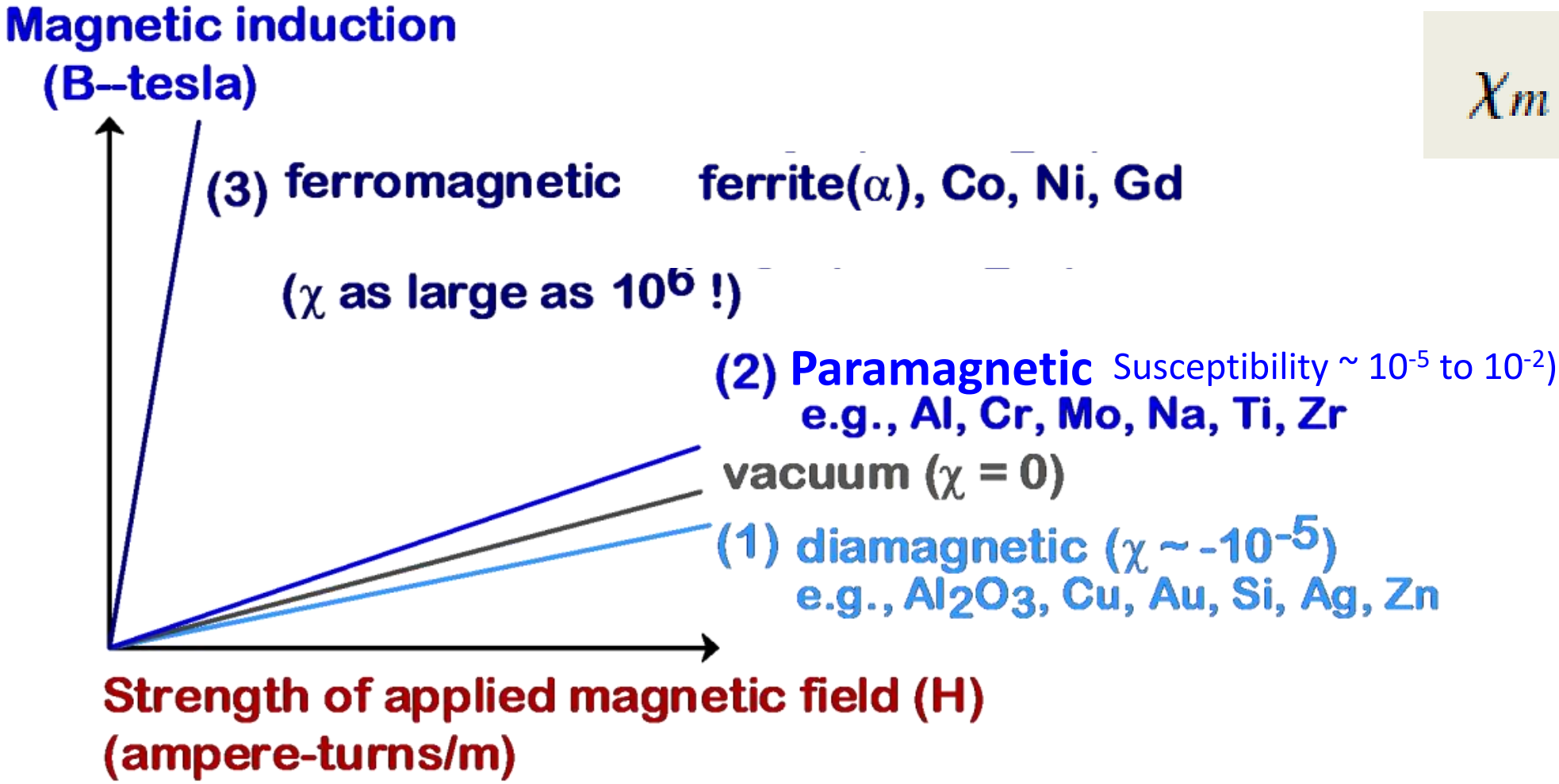
MAGNETIC MOMENTS FOR 3 TYPES



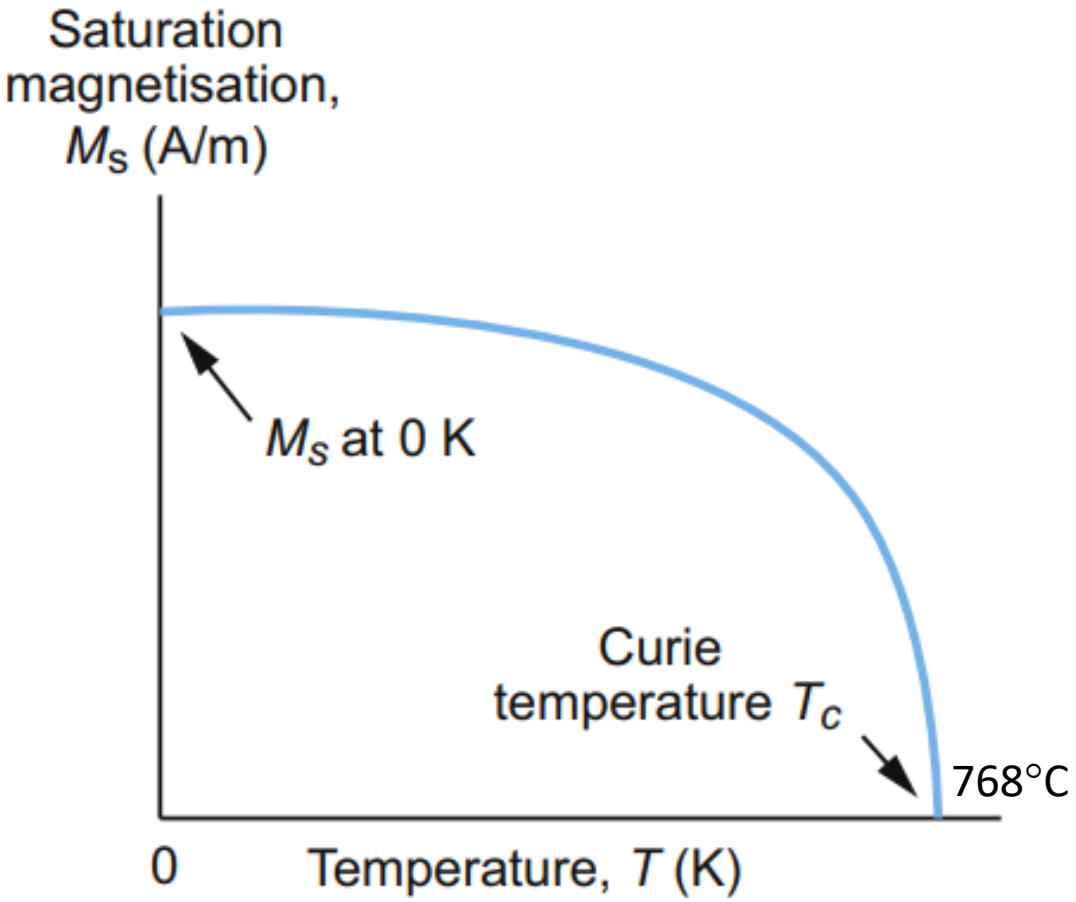
3 TYPES OF MAGNETISM

Magnetic susceptibility

$$\chi_m = \mu_r - 1$$



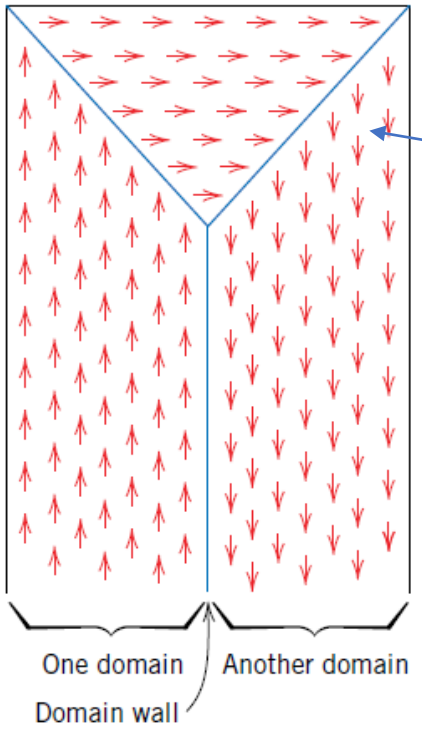
INFLUENCE OF TEMPERATURE ON MAGNETIC BEHAVIOR



With increasing temperature, the increased thermal motion of the atoms tend to randomize the direction of any moments that may be aligned.

With increasing temperature, the saturation magnetization decreases gradually and then abruptly drops to zero at what is called the **Curie temperature** T_c Above this temperature, the material is paramagnetic.

DOMAINS AND HYSTERESIS

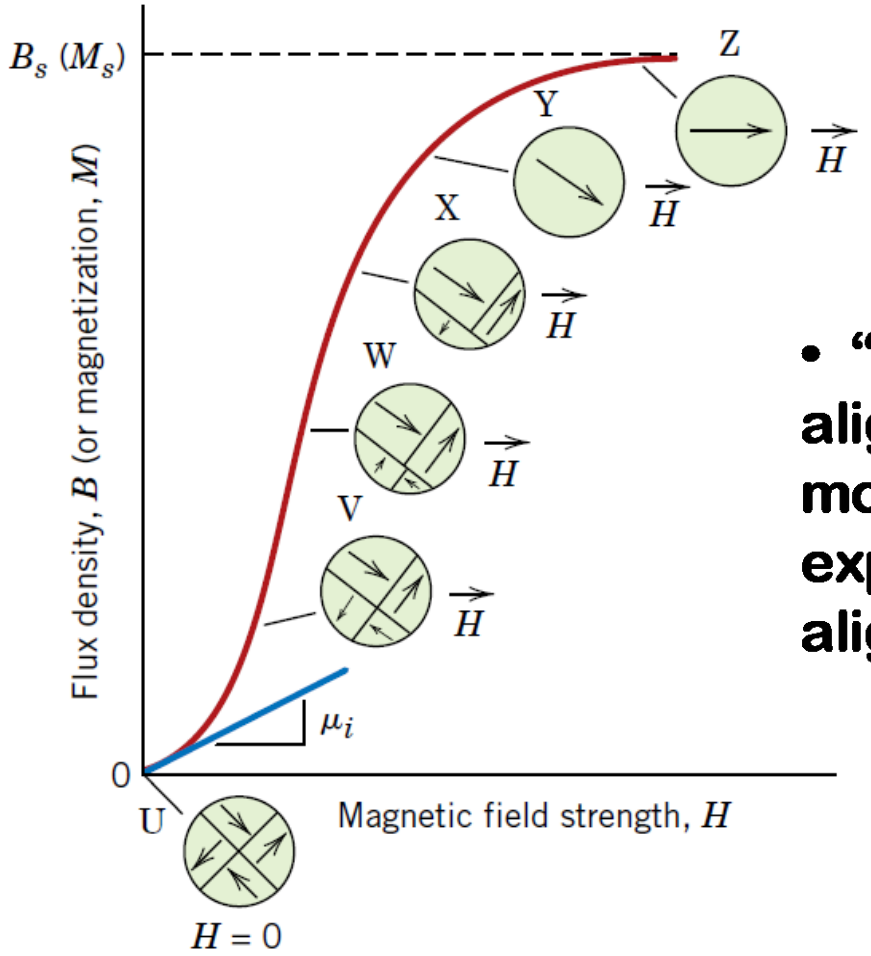


Below Tc, small-volume regions in which there is a mutual alignment in the same direction of all magnetic dipole moments

Schematic depiction of domains in a ferromagnetic material;
arrows represent atomic magnetic dipoles. Within each domain, all dipoles are aligned, whereas the direction of alignment varies from one domain to another.

FERROMAGNETIC MATERIALS

- **As the applied field (H) increases…**
  --the magnetic moment aligns with H.

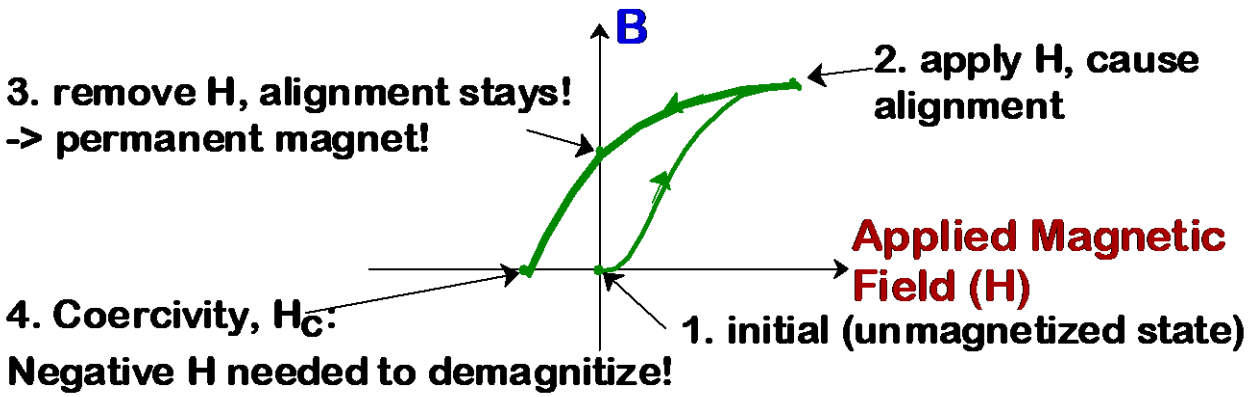


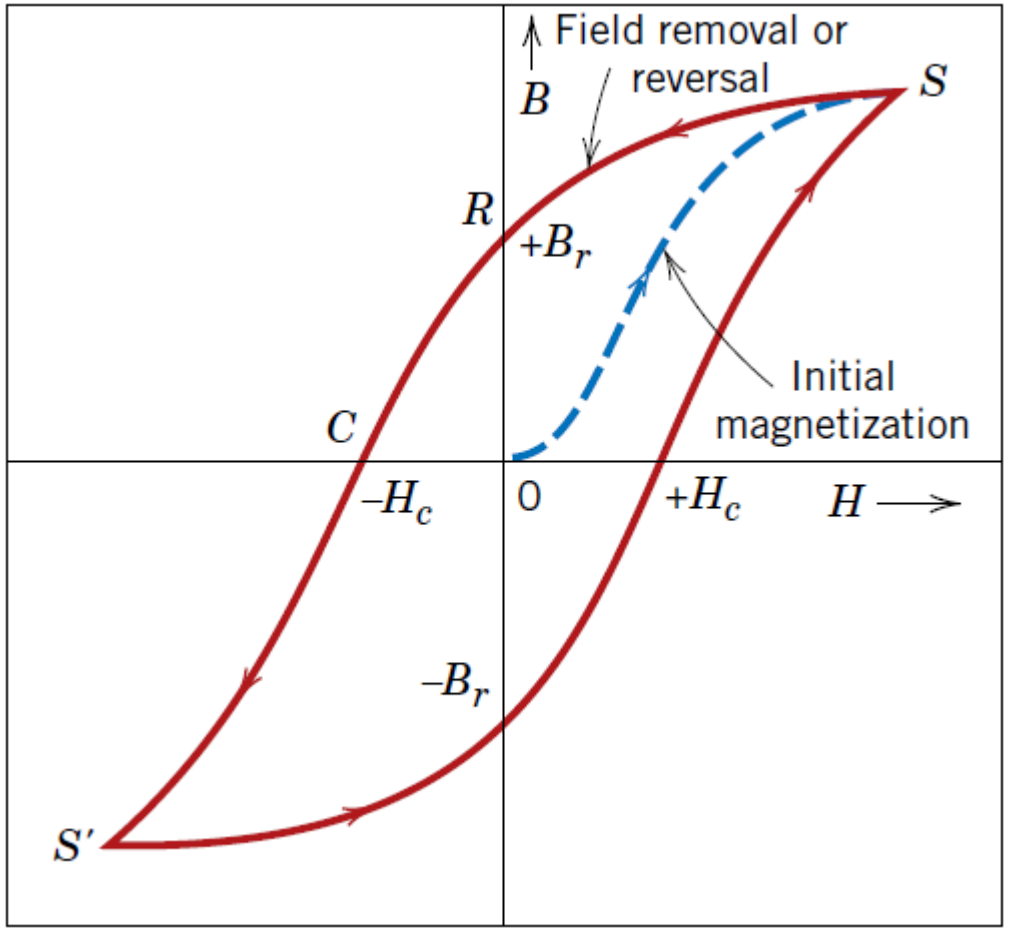
- **"Domains" with aligned magnetic moment grow at expense of poorly aligned ones!**

DOMAINS AND HYSTERESIS

FERROMAGNETIC MATERIALS

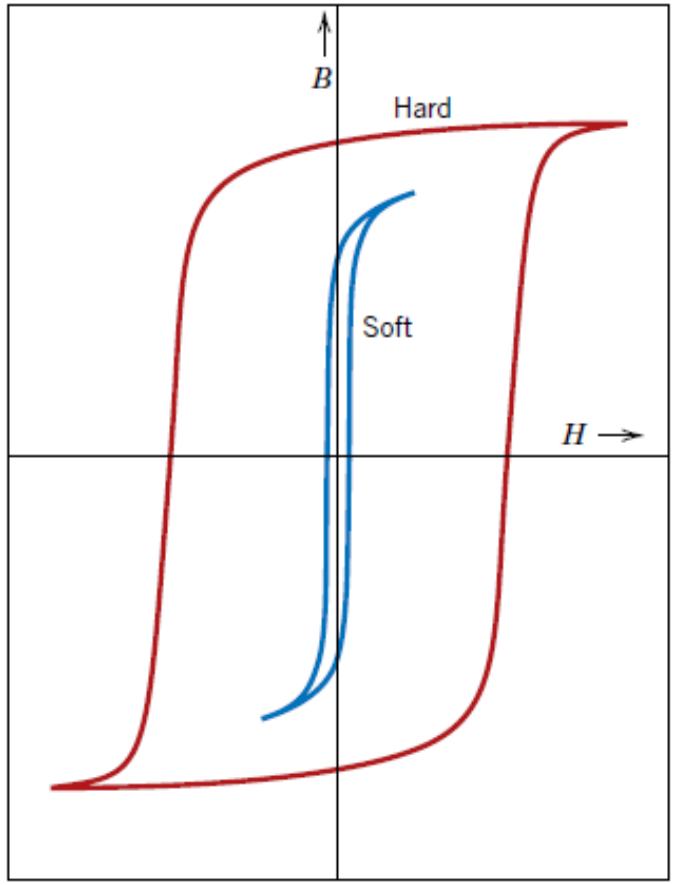
Hysteresis curve (for forward saturation)





Hysteresis curve for forward and reverse saturations

Hard vs Soft Magnets



Ferromaterials are classified as either *soft* or *hard* on the basis of their hysteresis characteristics.

Soft magnetic materials are used in devices that are subjected to alternating magnetic fields and in which energy losses must be low; one familiar example consists of transformer cores.

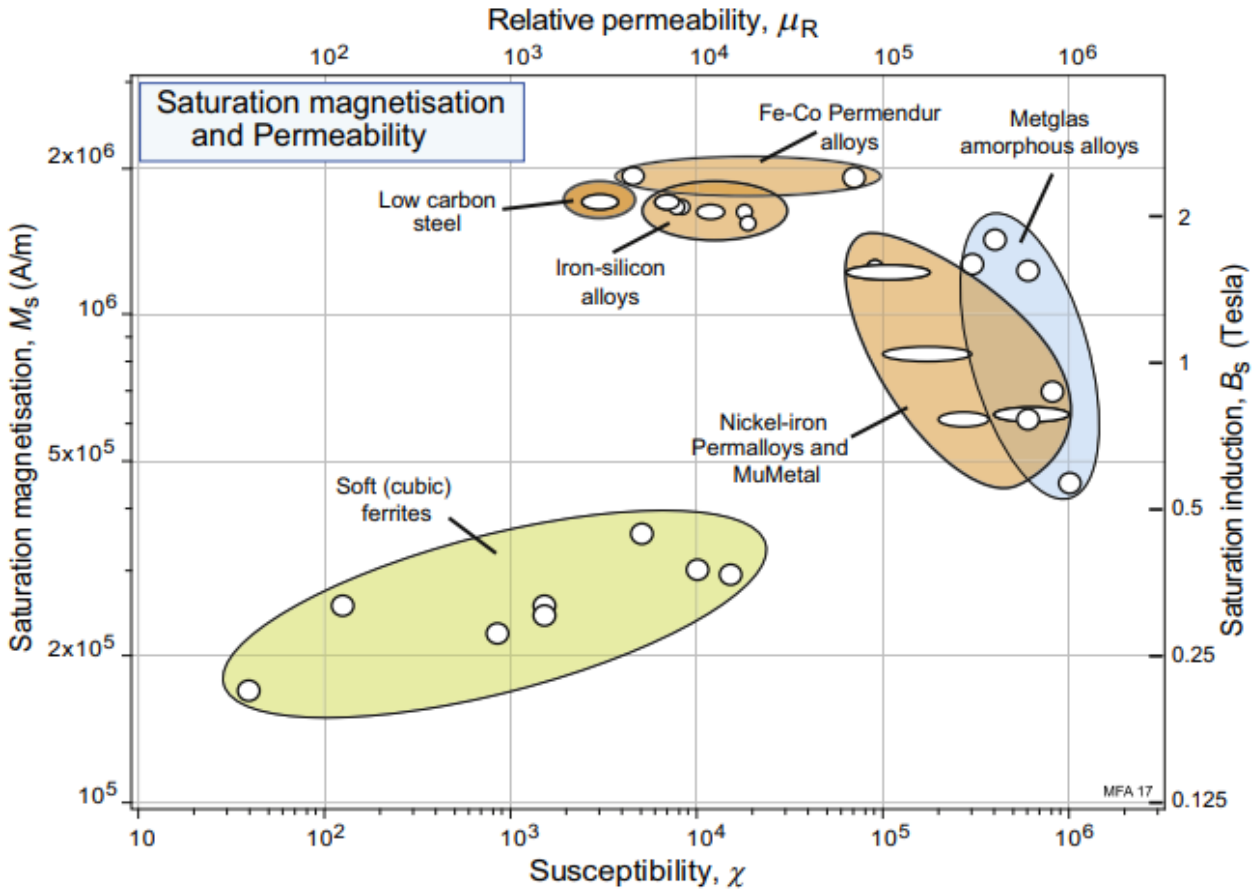
Soft magnetic materials have a low resistance to demagnetization. In terms of hysteresis behavior, a **soft magnetic material** has low coercivity, as well as high initial permeability and low hysteresis energy losses.

Hard magnetic materials are used in permanent magnets, which must have a high resistance to demagnetization.- example application in motors.
In terms of hysteresis behavior, a **hard magnetic material** has high remanence, coercivity and saturation flux density, as well as low initial permeability and high hysteresis energy losses.

Soft magnetic materials

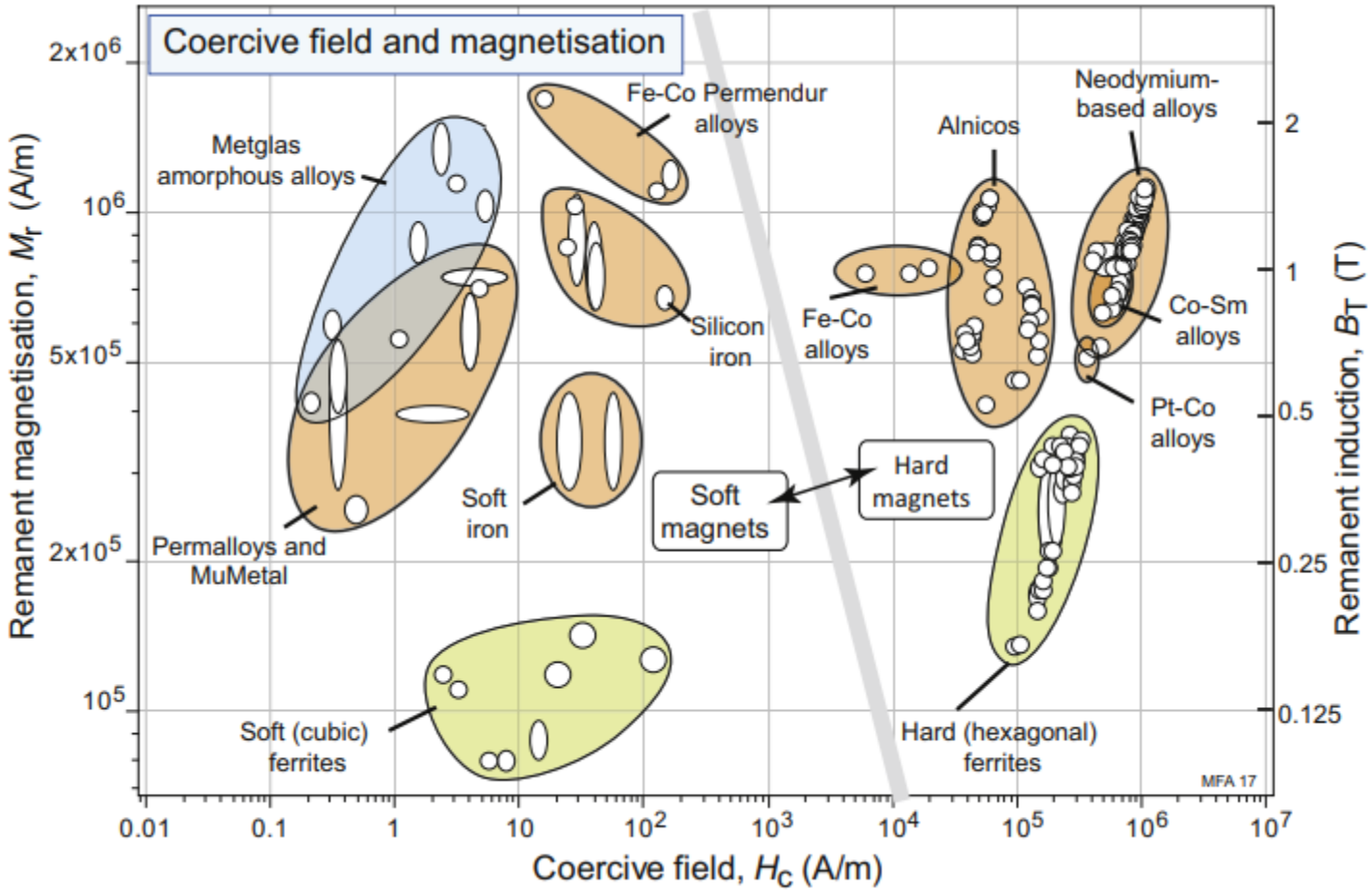
Electromagnets, transformers: - have magnetic cores that must magnetize easily when the field switches on, be capable of creating a high flux density, yet lose their magnetism when the field is switched off.

The higher the susceptibility, the greater is the magnetization.

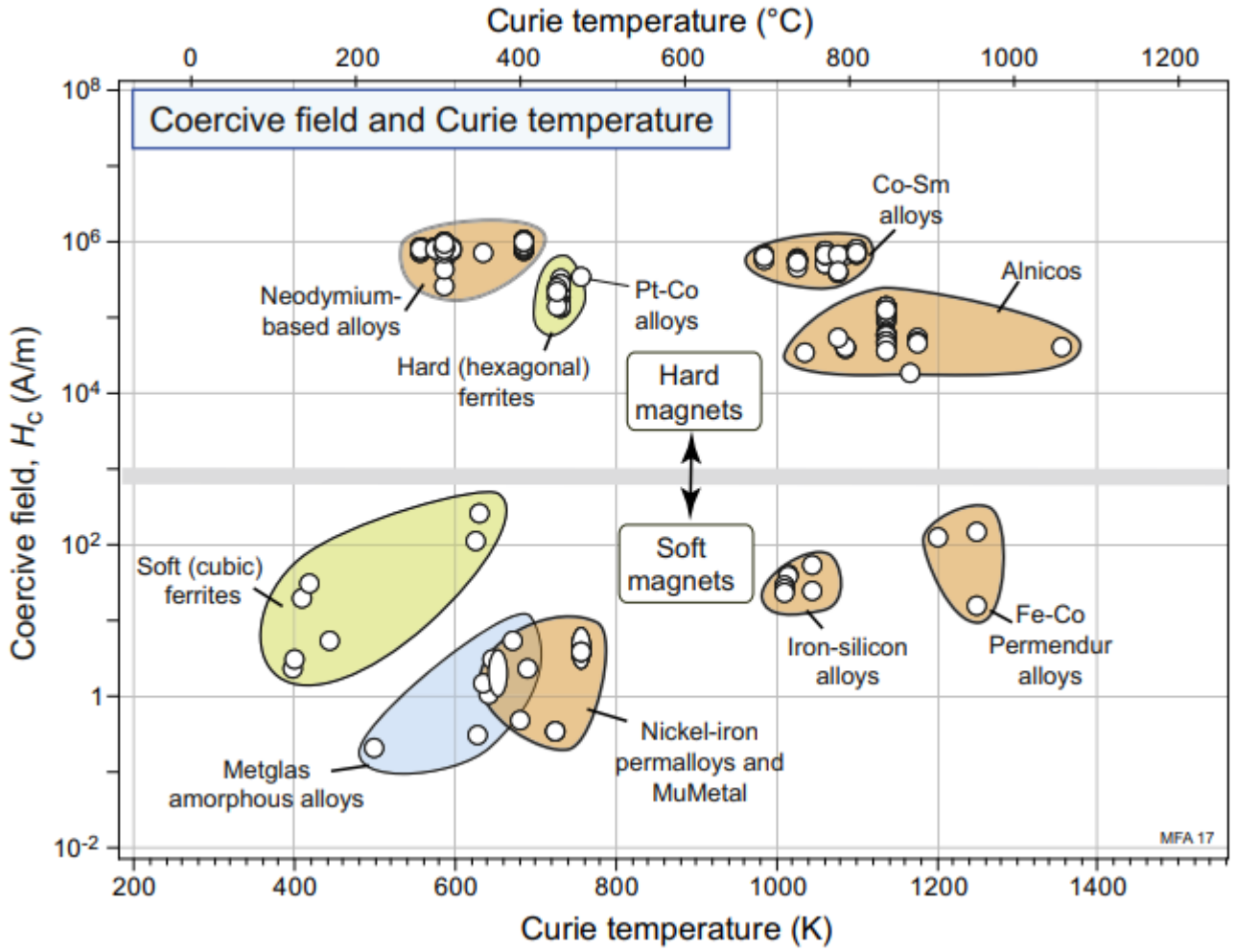


Hard magnetic materials

Motors – require high remanence and resistance to demagnetization (coercivity)



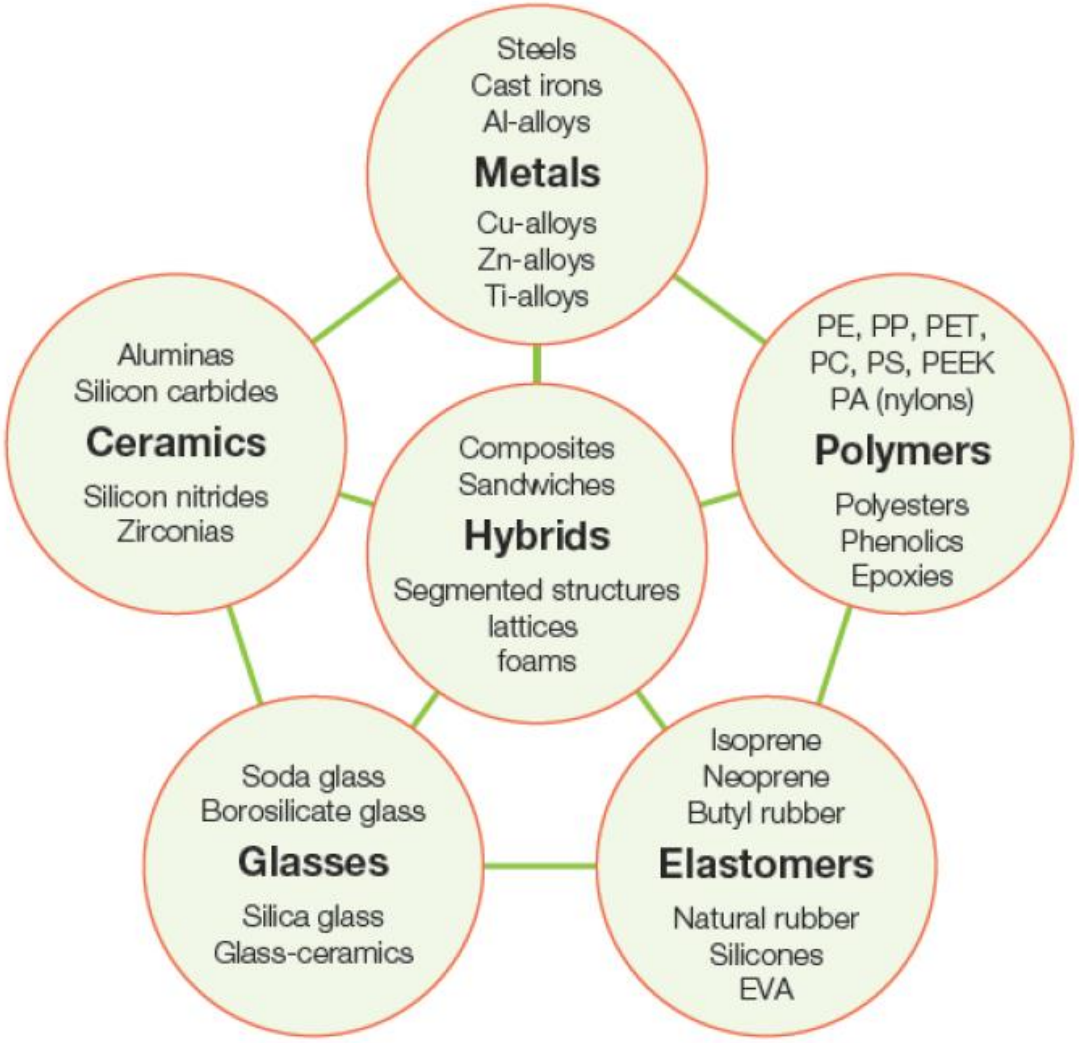
Coercive field - Curie temperature chart



A permanent magnet is needed for an aerospace application. In use the magnet may be exposed to demagnetising fields as high as 3 x 10^5 A/m and temperatures of 600 deg.C. What material would you recommend for the magnet?

Recall…

Materials Selection

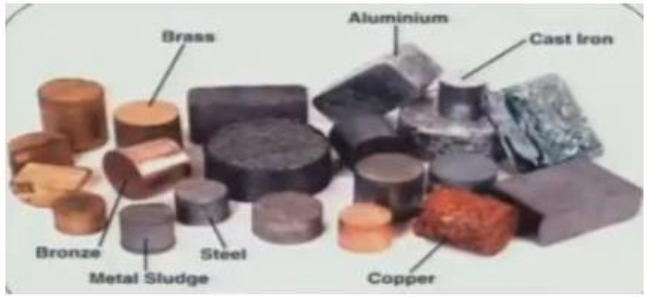


Recall…

General Properties of Metals

Steel, aluminum, magnesium, zinc, cast iron, copper, lead, etc

- High electrical conductivity
- High heat conductivity
- Ductile
- Easily deformed
- High thermal shock resistance
- Suitable for structural and load bearing applications
- Alloys utilized for development of high performance metals



Recall…

General Properties of Ceramics

Bricks, glass, refractory and abrasive ceramics

- Low electrical conductivity
- Low heat conductivity
- Brittle
- High wear resistance in high temperature applications
- Corrosion resistance
- Generally used as insulators and load bearing structures
- Advanced ceramics are utilized in integrated circuits and optical applications due to the improvements on optical and electrical properties
- Biocompatible materials

Recall…

General Properties of Polymers

Elastomers, plastics, and adhesives

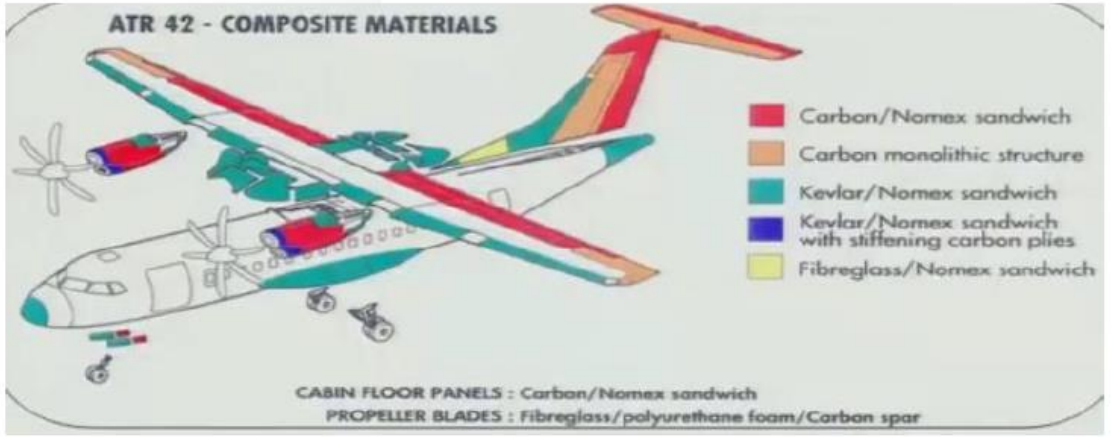
- Produced by polymerization of organic molecules into large molecular structures
- Low electrical conductivity
- Low thermal resistance
- Low strength
- Light materials

Recall…

General Properties of Composites

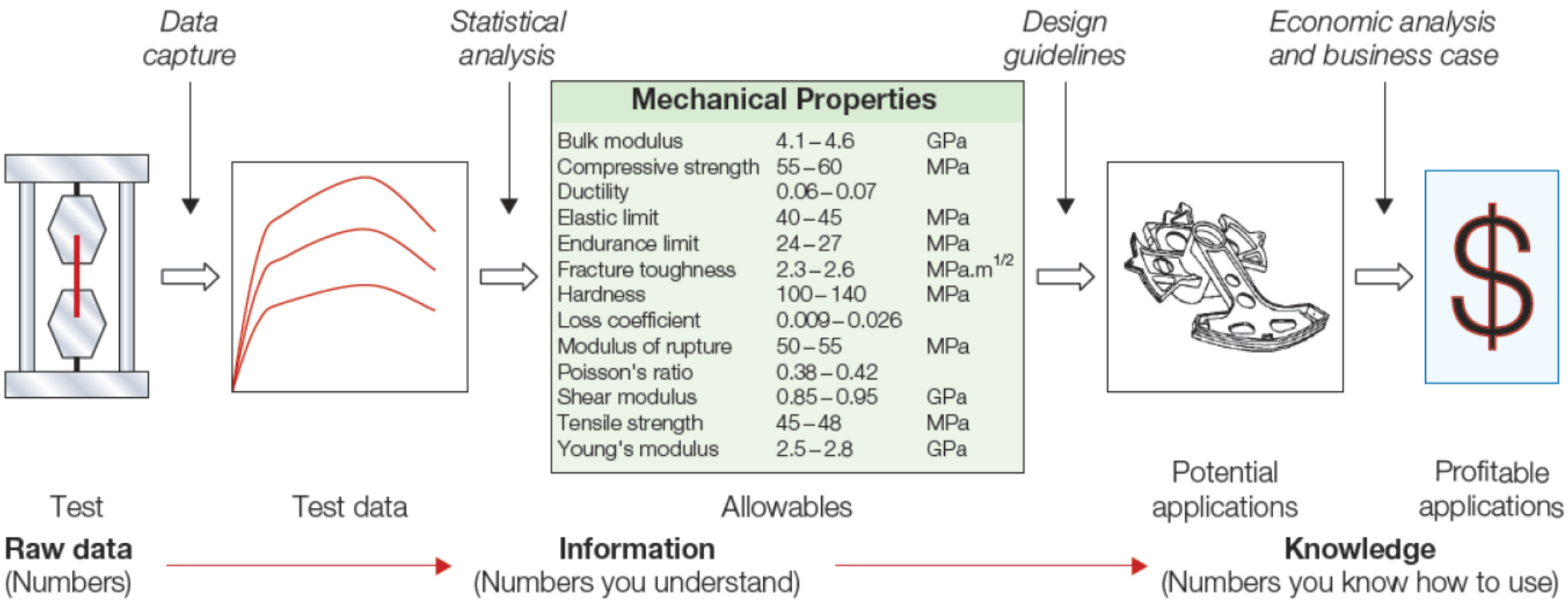
Comprised of two or more materials

- Carbon fiber reinforced polymer, polymer fiber reinforced HA
- Light
- Strong
- High fracture toughness
- High thermal shock resistance



Recall...

Types of Materials Information



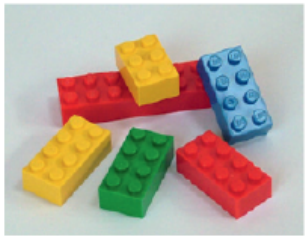
Typical Information of Materials

Recall…

Acrylonitrile butadiene styrene (ABS)

General properties				
Density	1100	-	1200	kg/m^3
Price	2.1	-	2.5	USD/kg
Mechanical properties				
Young's modulus	1.1	-	2.9	GPa
Yield strength	19	-	51	MPa
Tensile strength	28	-	55	MPa
Elongation	15	-	40	%
Hardness—Vickers	5.6	-	15	HV
Fatigue strength at 10^7 cycles	11	-	22	MPa
Fracture toughness	1.2	-	4.3	MPa.m$^{1/2}$
Thermal properties				
Glass temperature	360	-	400	K
Maximum service temperature	340	-	350	K
Minimum service temperature	150	-	200	K
Thermal conductivity	0.19	-	0.34	W/m.K
Specific heat capacity	1400	-	1900	J/kg.K
Thermal expansion coefficient	85	-	230	10^{-6}/°C
Electrical properties				
Electrical resistivity	3.3 x 10^{21}	-	3 x 10^{22}	μohm.cm
Dielectric constant	2.8	-	3.2	
Dielectric loss tangent	0.003	-	0.007	
Dielectric strength	14	-	22	MV/m
Eco-properties				
Embodied energy	91	-	1e2	MJ/kg
CO_2 footprint	3.3	-	3.6	kg/kg

The material



ABS (Acrylonitrile-butadiene-styrene) is tough, resilient, and easily molded. It is usually opaque, although some grades can now be transparent, and it can be given vivid colors. ABS-PVC alloys are tougher than standard ABS and, in self-extinguishing grades, are used for the casings of power tools. The picture shows that ABS allows detailed moldings, accepts color well, and is nontoxic and tough enough to survive the worst that children can do.

Typical uses

Safety helmets; camper tops; automotive instrument panels and other interior components; pipe fittings; home-security devices and housings for small appliances; communications equipment; business machines; plumbing hardware; automobile grilles; wheel covers; mirror housings; refrigerator liners; luggage shells; tote trays; mower shrouds; boat hulls; large components for recreational vehicles; weather seals; glass beading; refrigerator breaker strips; conduit; pipe for drain-waste-vent (DWV) systems.

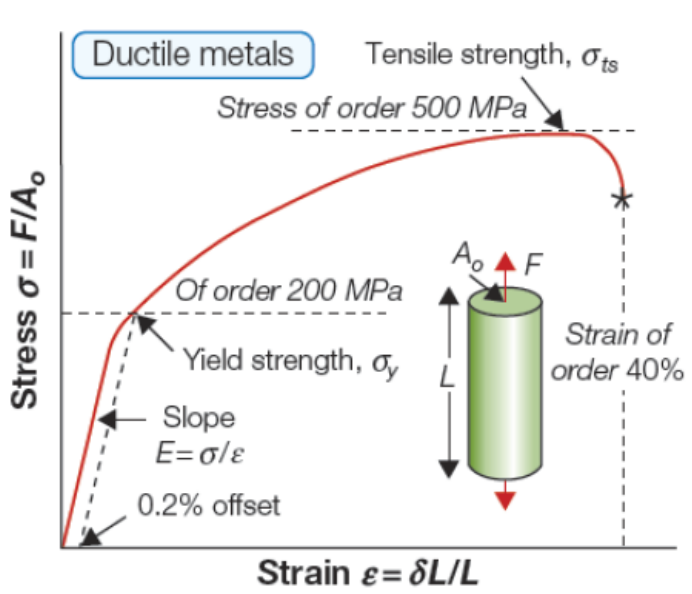
Tradenames

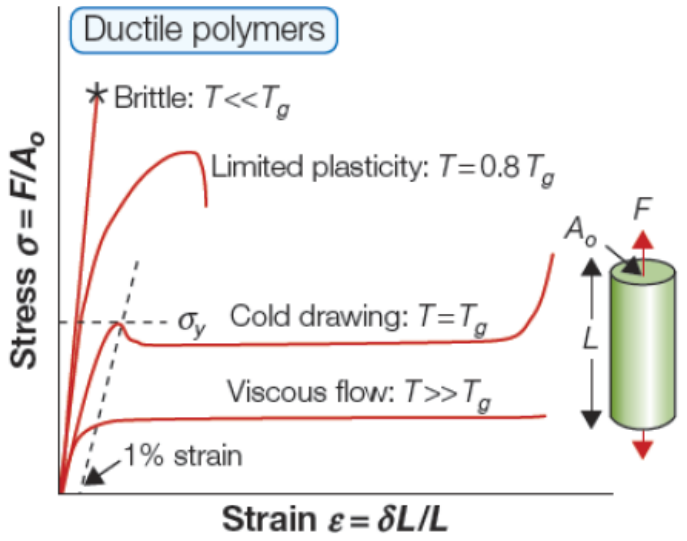
Claradex, Comalloy, Cycogel, Cycolac, Hanalac, Lastilac, Lupos, Lustran ABS, Magnum, Multibase, Novodur, Polyfabs, Polylac, Porene, Ronfalin, Sinkral, Terluran, Toyolac, Tufrex, Ultrastyr.

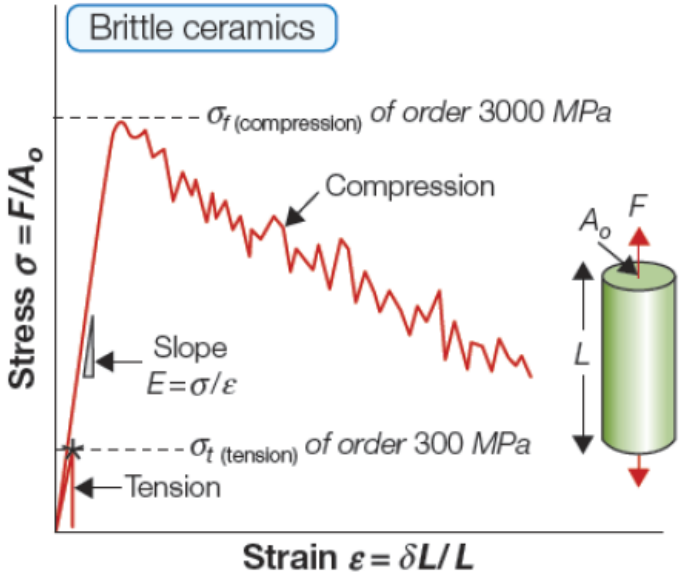
Class	Property	Symbol and Units
General	Density	ρ (kg/m^3 or Mg/m^3)
	Price	C_m ($/kg)
Mechanical	Elastic moduli (Young's, shear, bulk)	E, G, K (GPa)
	Yield strength	σ_y (MPa)
	Tensile (ultimate) strength	σ_{ts} (MPa)
	Compressive strength	σ_c (MPa)
	Failure strength	σ_f (MPa)
	Hardness	H *(Vickers)*
	Elongation	ε (–)
	Fatigue endurance limit	σ_e (MPa)
	Fracture toughness	K_{1c} (MPa.m$^{1/2}$)
	Toughness	G_{1c} (kJ/m^2)
	Loss coefficient (damping capacity)	η (–)
	Wear rate (Archard) constant	K_AMPa^{-1}
Thermal	Melting point	T_m (°C or K)
	Glass temperature	T_g (°C or K)
	Maximum service temperature	T_{max} (°C or K)
	Minimum service temperature	T_{min} (°C or K)
	Thermal conductivity	λ (W/m.K)
	Specific heat	C_p (J/kg.K)
	Thermal expansion coefficient	α (K^{-1})
	Thermal shock resistance	ΔT_s (°C or K)
Electrical	Electrical resistivity	ρ_e (Ω.m or $\mu\Omega$.cm)
	Dielectric constant	ε_r (–)
	Breakdown potential	V_b (10^6 V/m)
	Power factor	P (–)
Optical	Refractive index	n (–)
Eco-properties	Embodied energy	H_m (MJ/kg)
	Carbon footprint	CO_2 (kg/kg)

Mechanical Properties

Recall…







Stress,

Strain,

Young's modulus (elastic modulus),

Poisson's ratio

Plastic deformation,

Ductility,

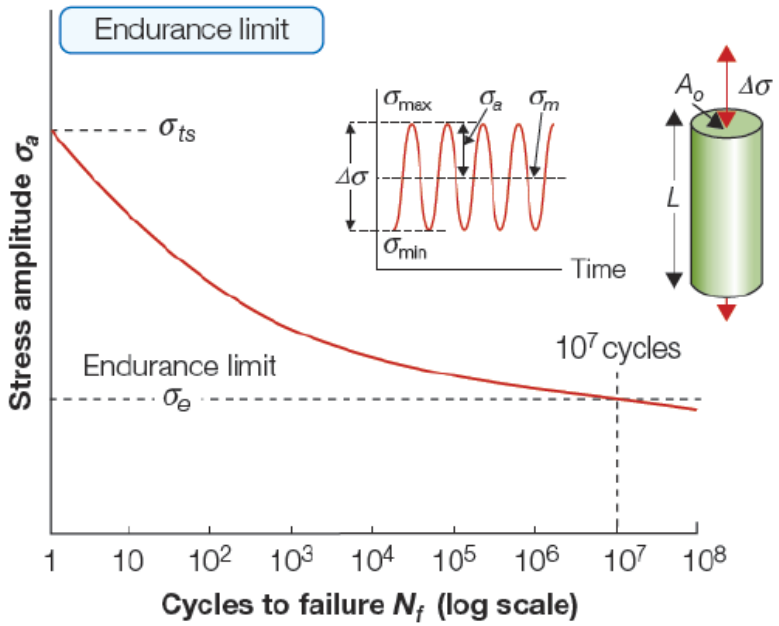
Yield strength,

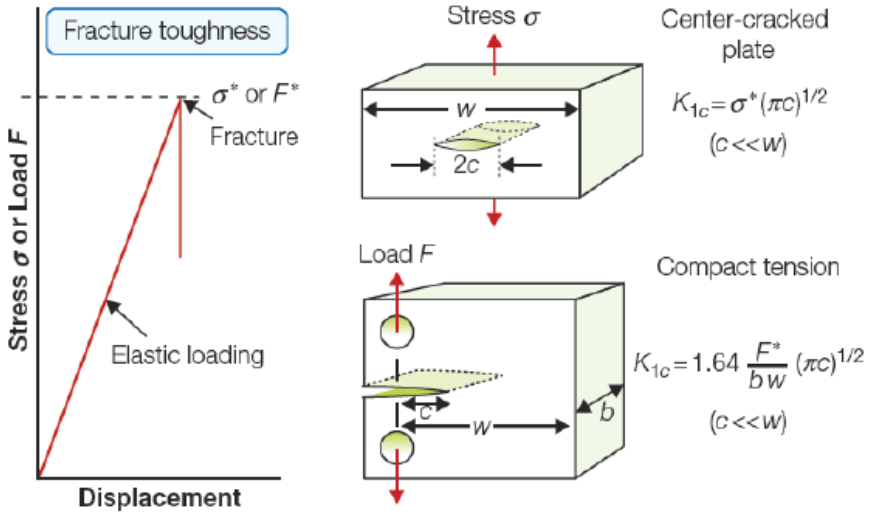
Ultimate strength,

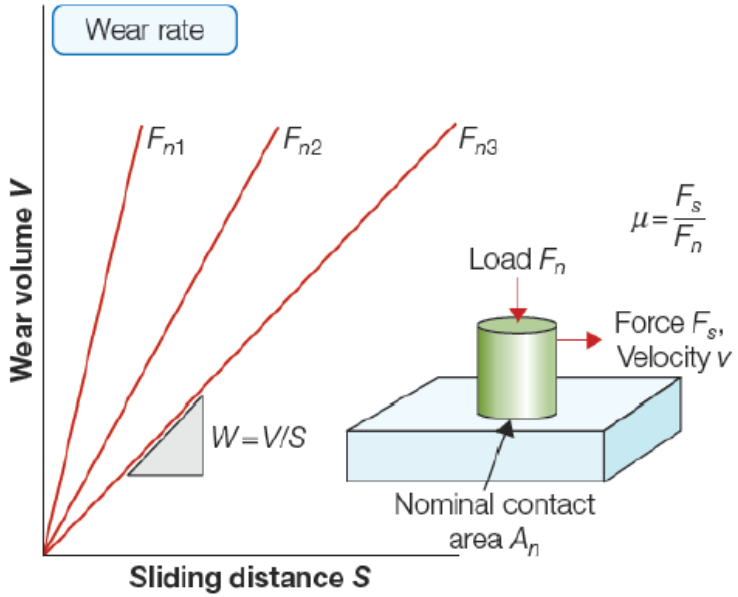
Toughness

Mechanical Properties

Recall…







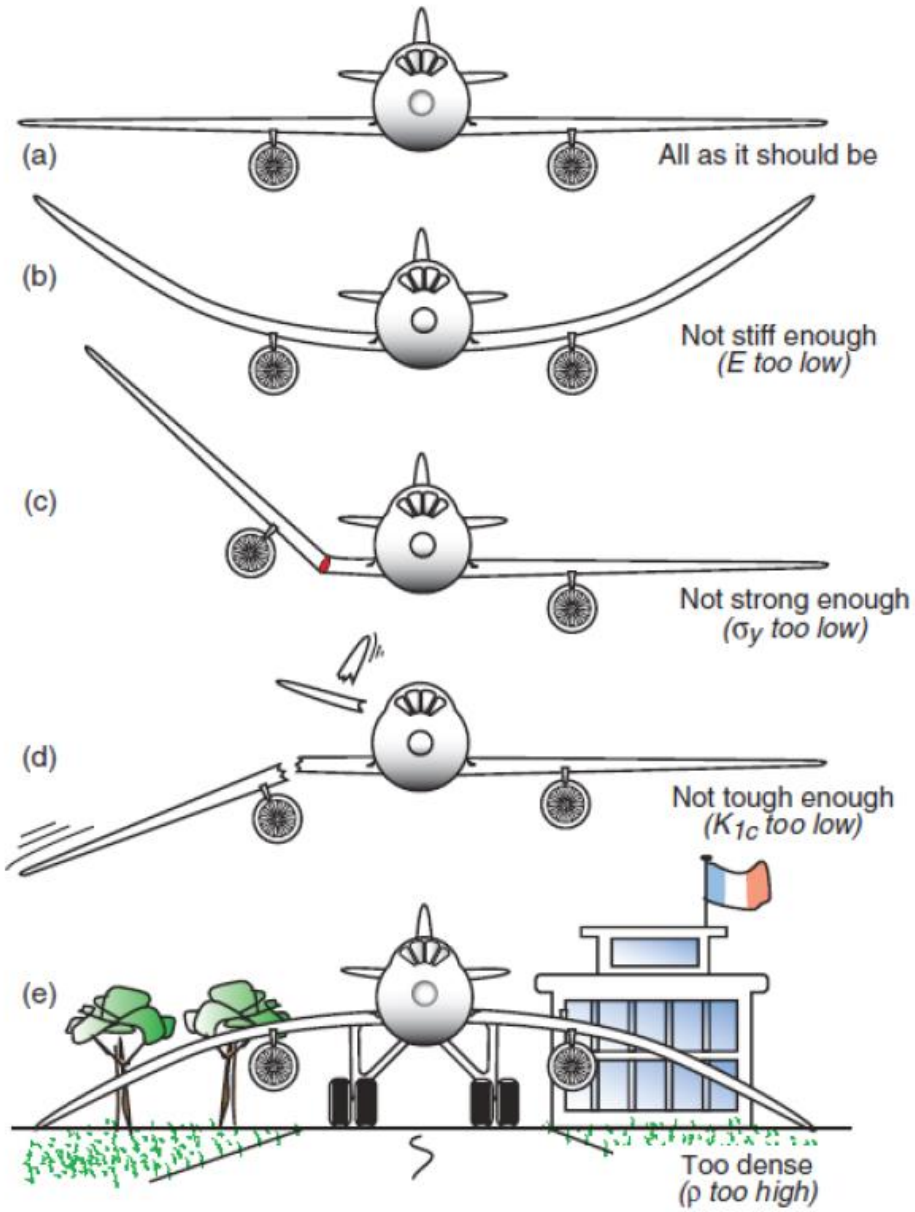
Fatigue strength,

Fracture toughness,

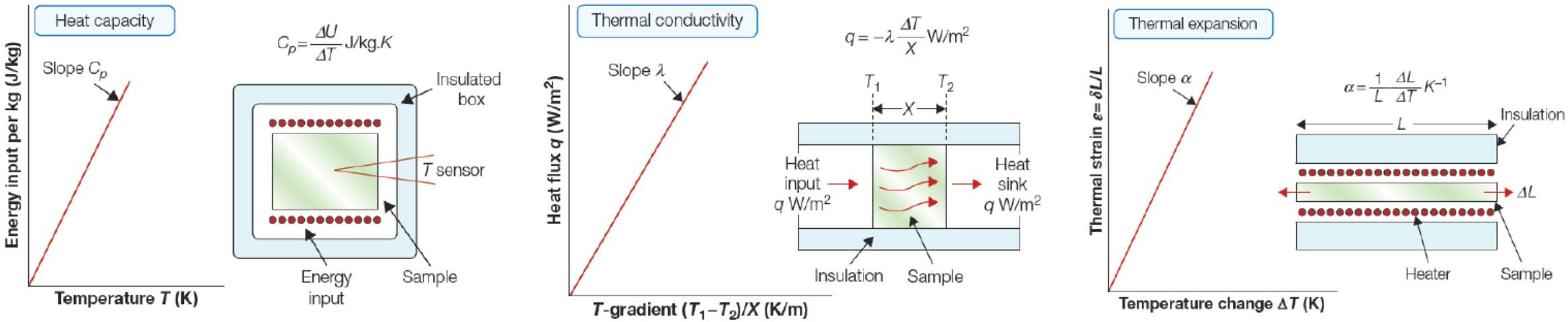
Wear resistance

Mechanical Properties

Recall…

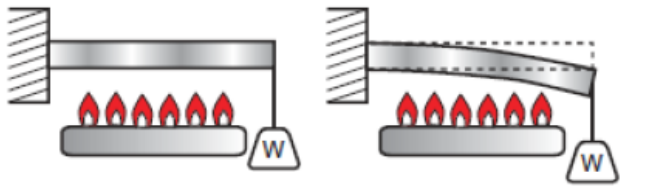


Thermal Properties

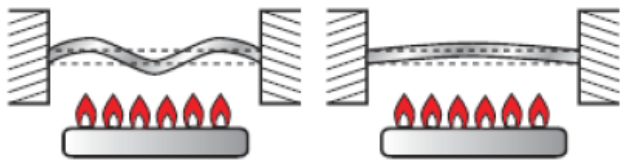


- Melting temperature
- Glass transition temperature (service temperature)
- Heat capacity
- Thermal conductivity (how much heat)
- Thermal diffusivity (rate of temperature change as the heat passes through material)
- Thermal expansion
- Thermal shock resistance

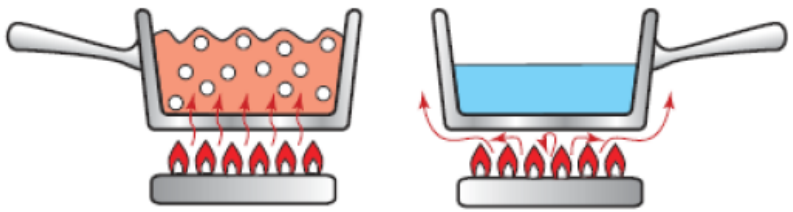
Thermal Properties



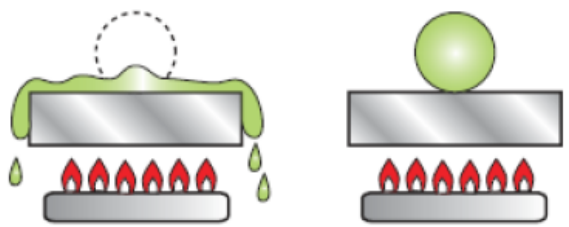
(a) High service temperature T_{max} Low service temperature T_{max}



(b) High expansion coefficient α Low expansion coefficient α



(c) High conductivity λ Low conductivity λ

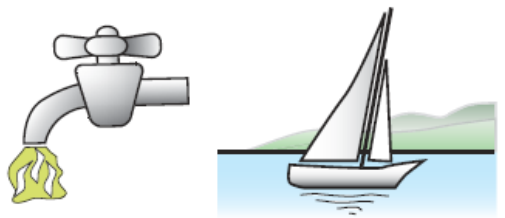


(d) High T-diffusivity a Low T-diffusivity a

Electrical Properties

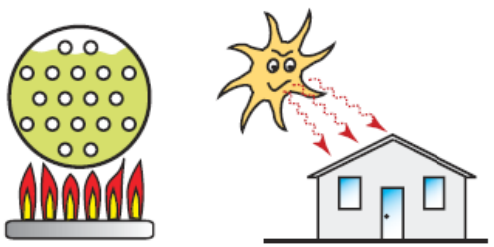
(a) Low resistivity ρ_e High resistivity ρ_e

Chemical Properties



(a) Fresh water (b) Salt water

(c) Acids and alkalis (d) Organic solvents



(e) Oxidation (f) UV radiation

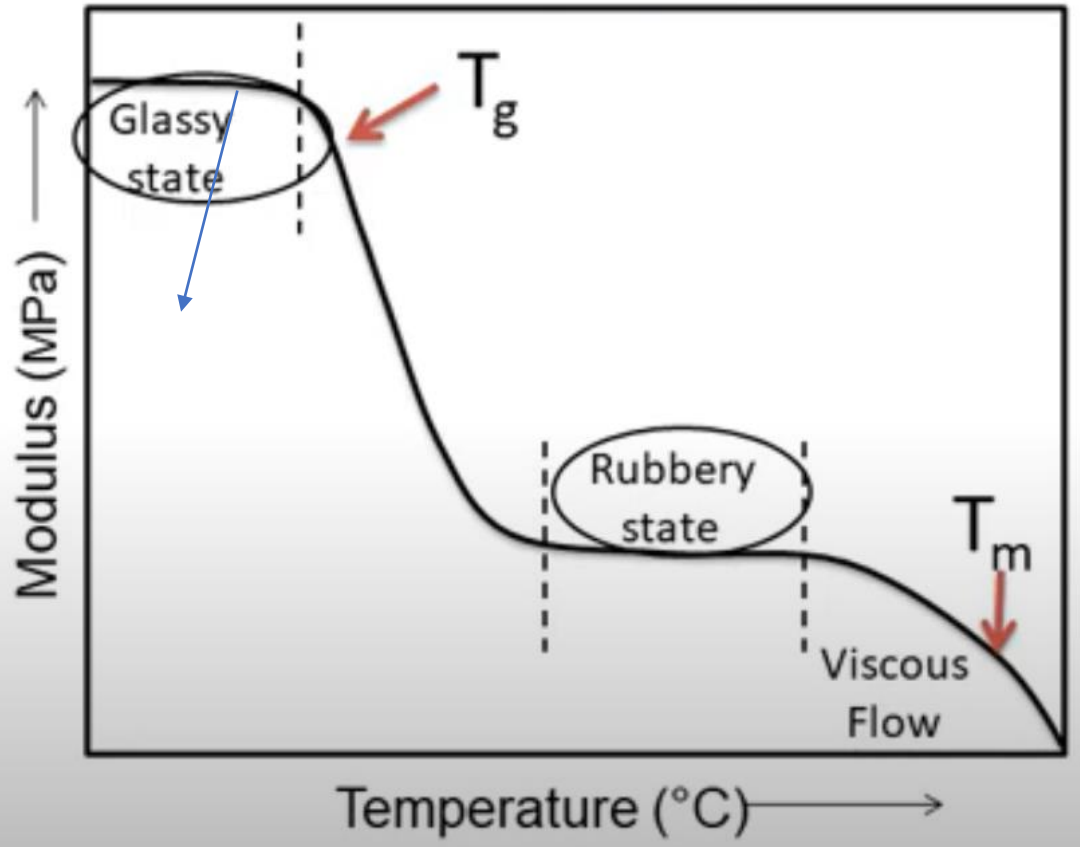
Recall…

tion Temperature

Glass Transition Temperature (T_g)

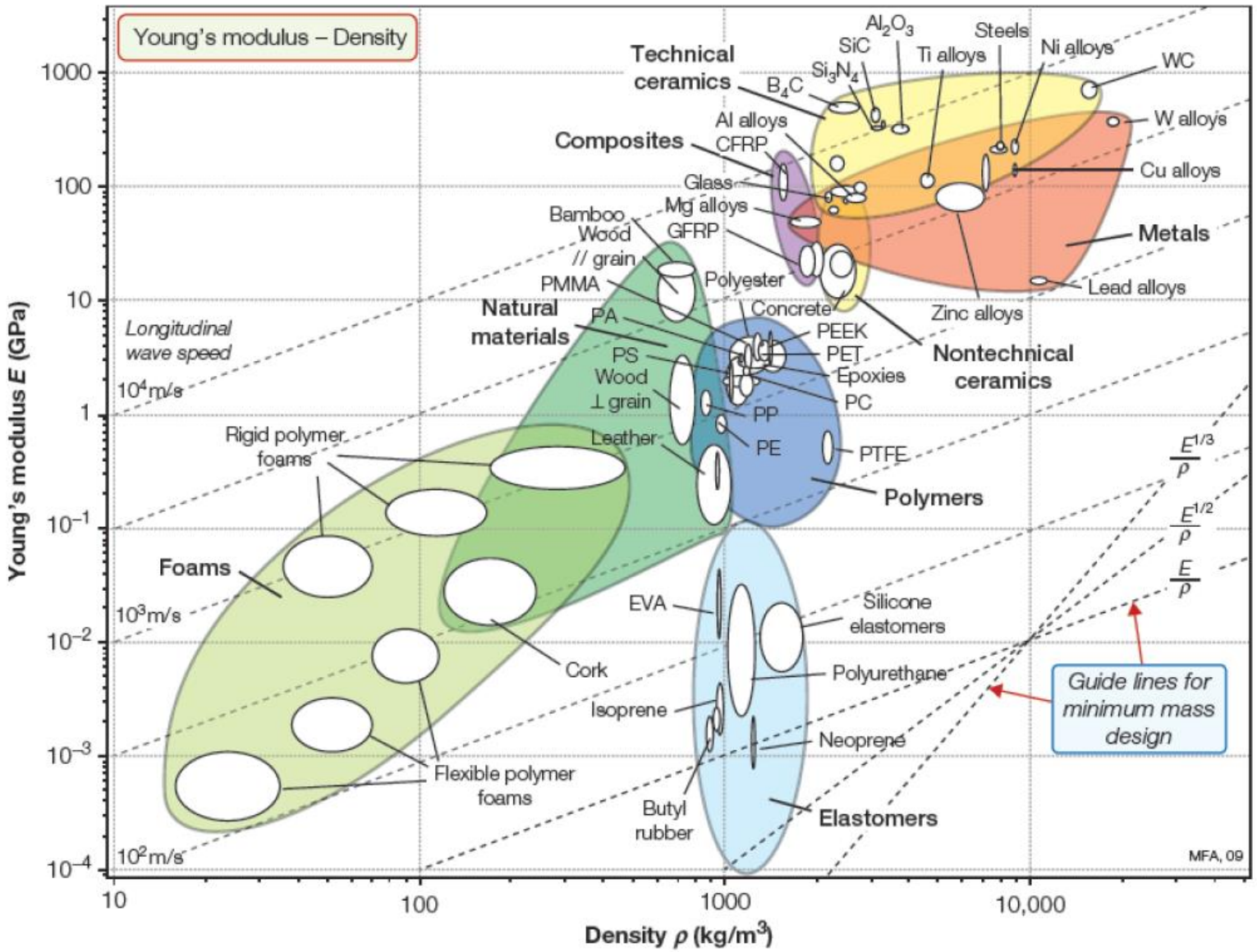
The temperature at which the polymers undergo the transition from glassy to rubbery state

Below T_g : Polymers are hard and brittle like glass, due to lack of mobility
Above T_g : Polymers are soft and flexible like rubber due to some mobility
Above T_g : Physical and mechanical properties of polymer change



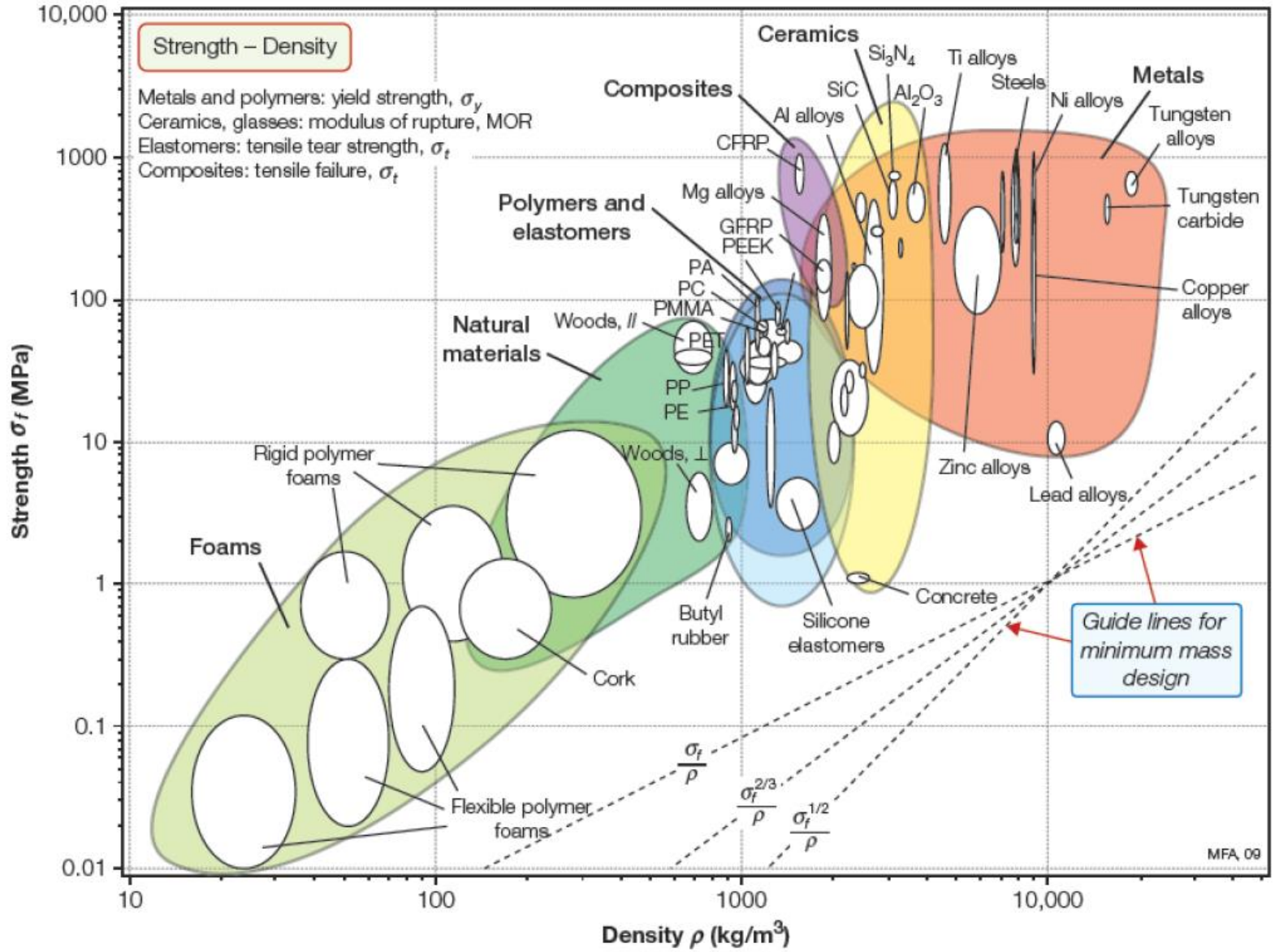
Material Property Charts

Young's Modulus vs Density



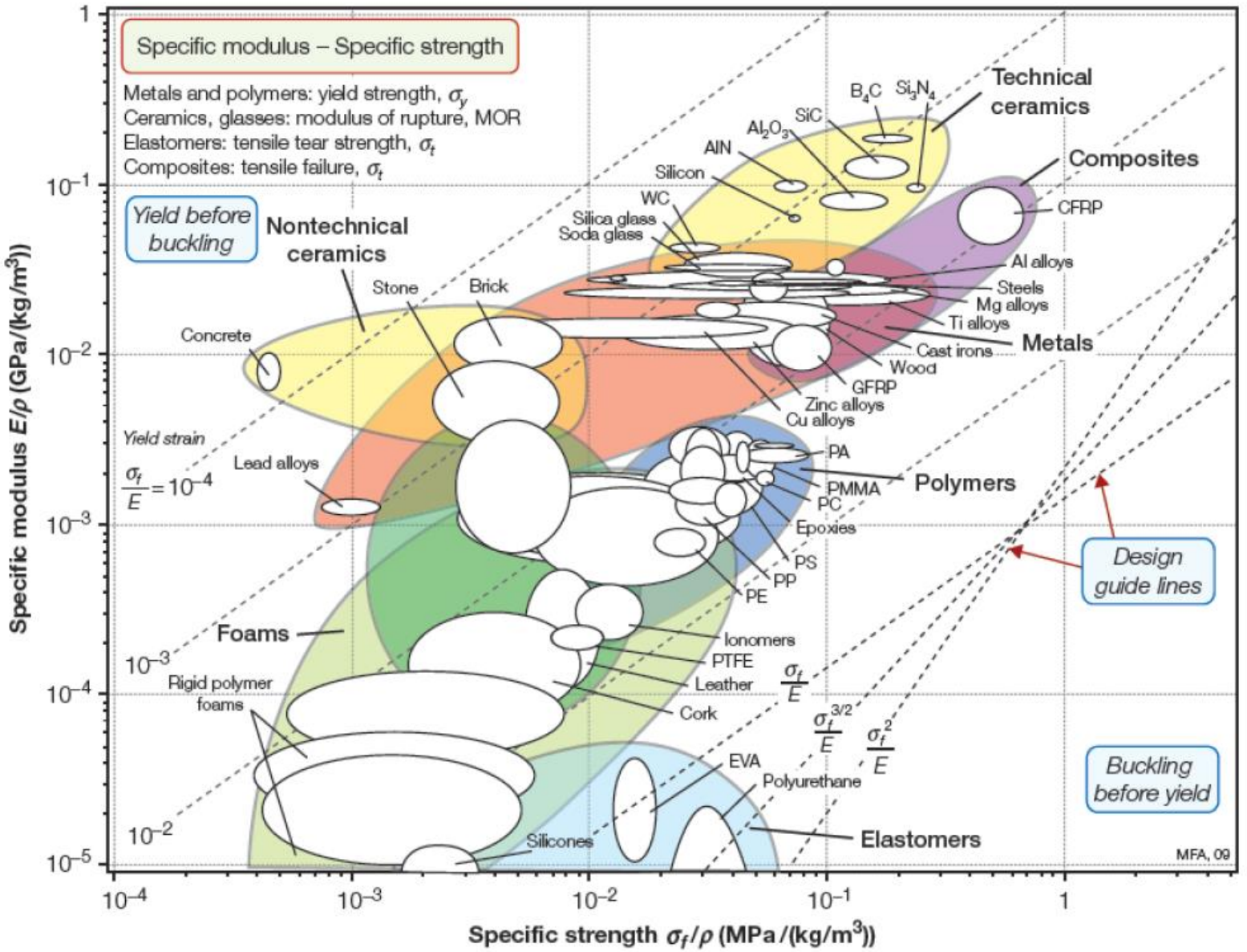
Material Property Charts

Strength vs Density



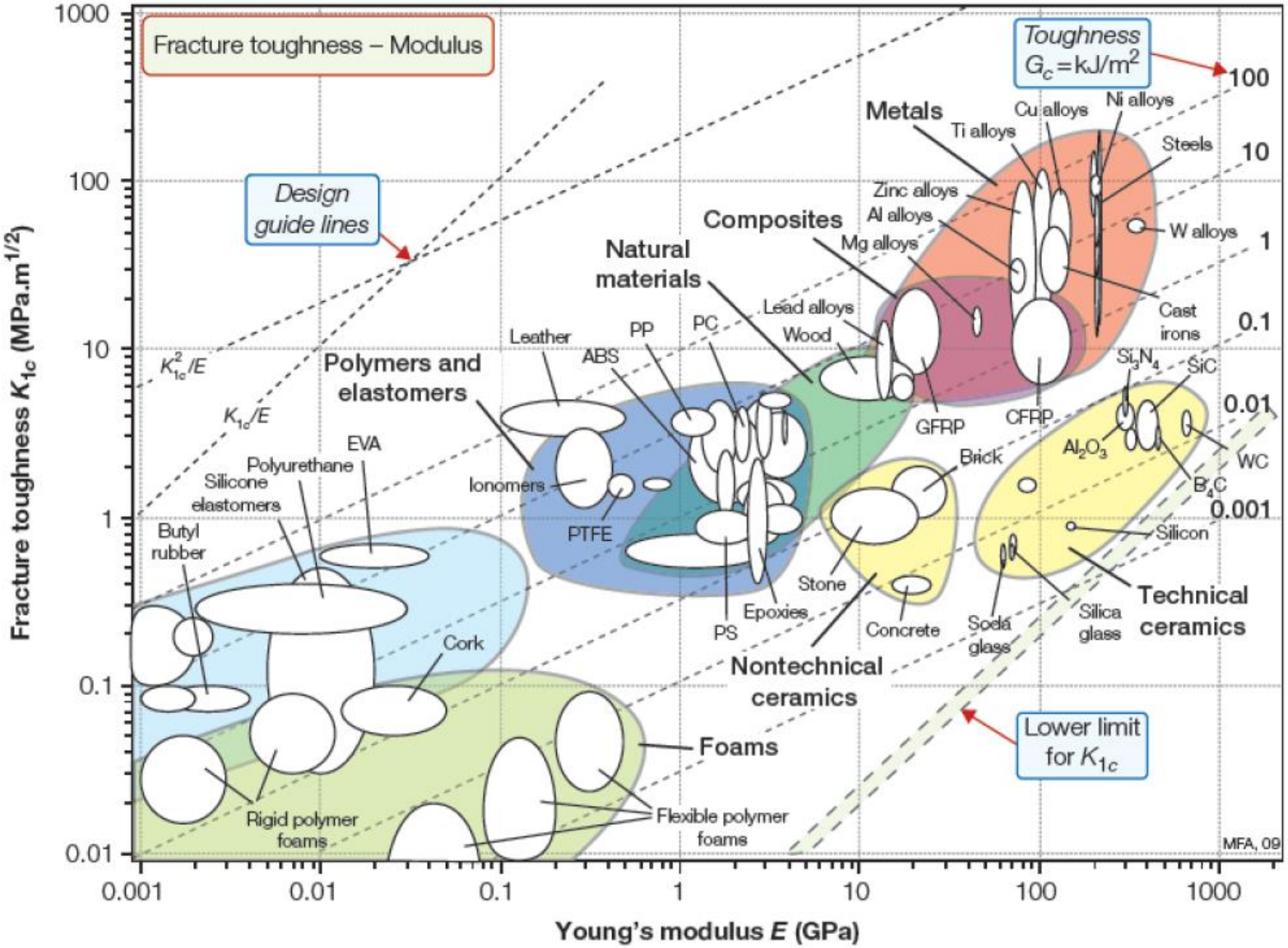
Material Property Charts

Specific modulus vs Specific strength



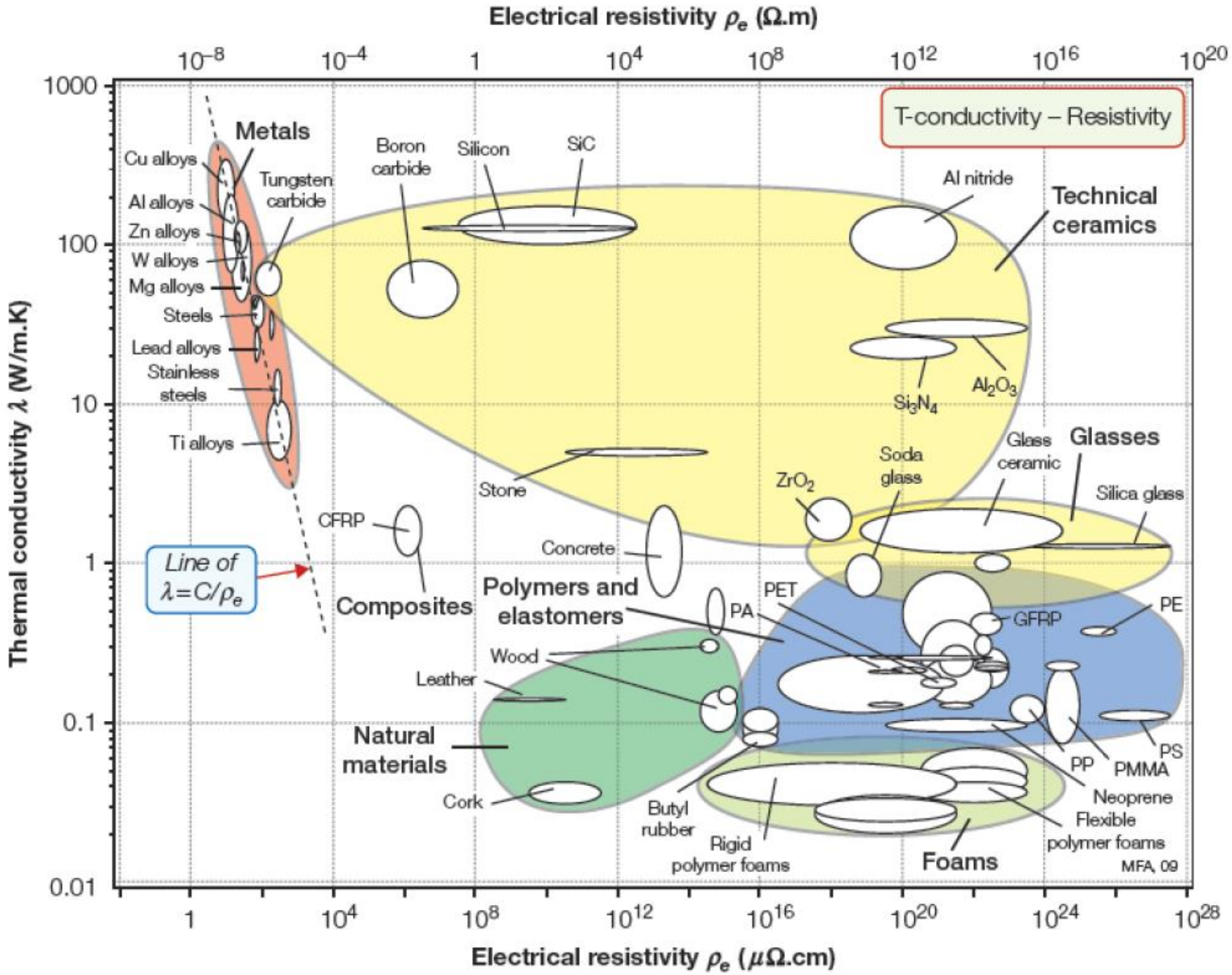
Material Property Charts

Fracture toughness vs Modulus



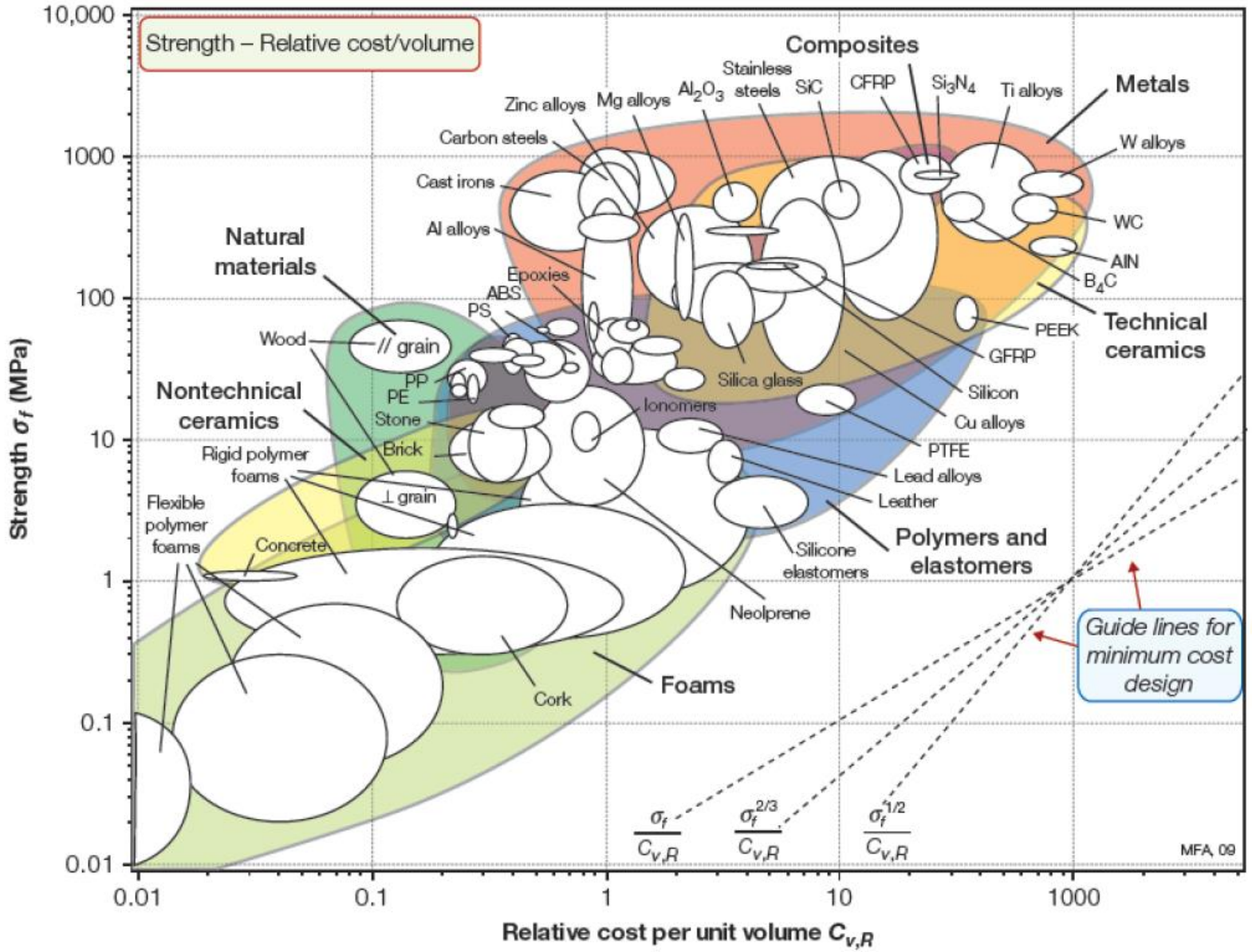
Material Property Charts

Thermal conductivity vs Electrical resistivity

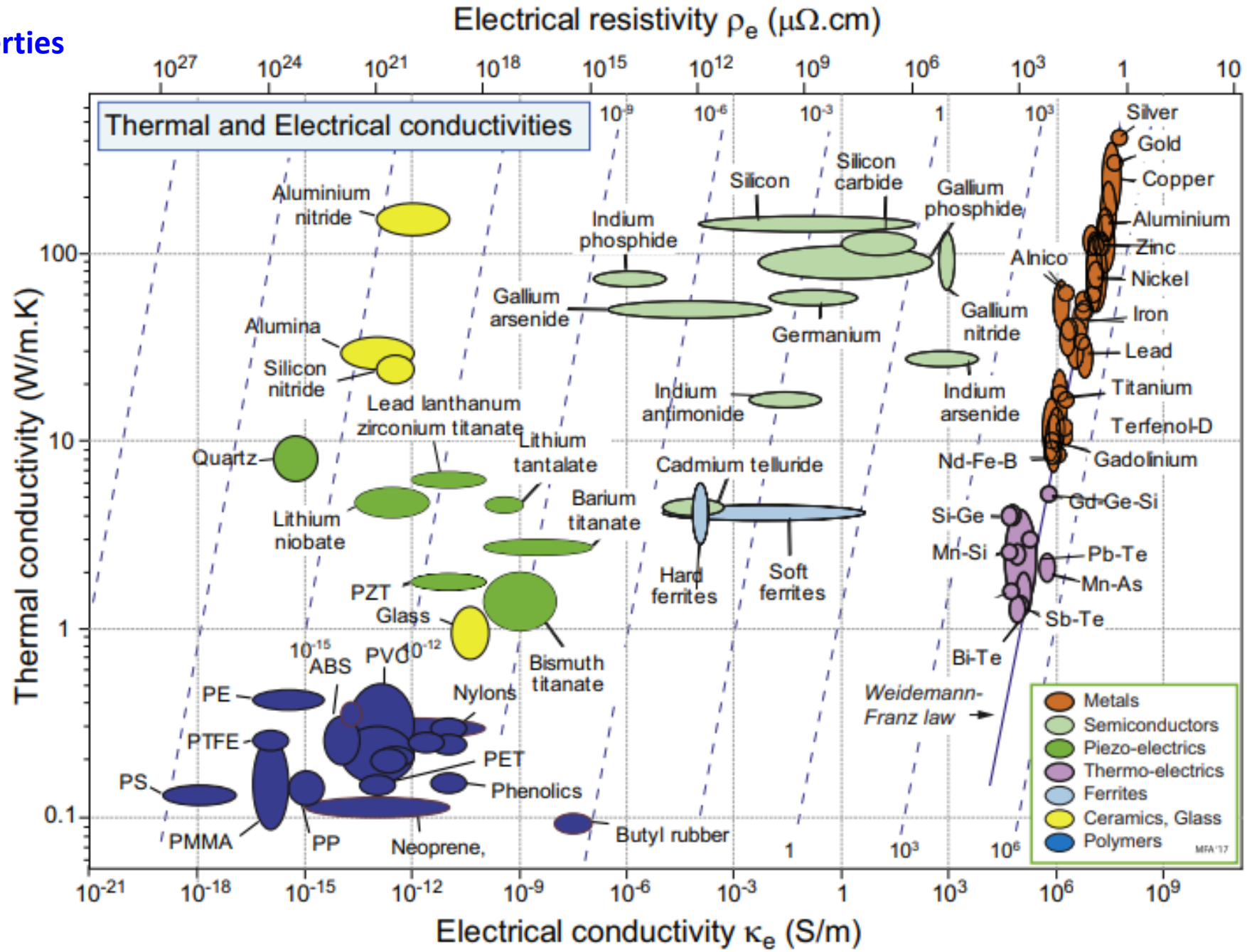


Material Property Charts

Strength vs Cost

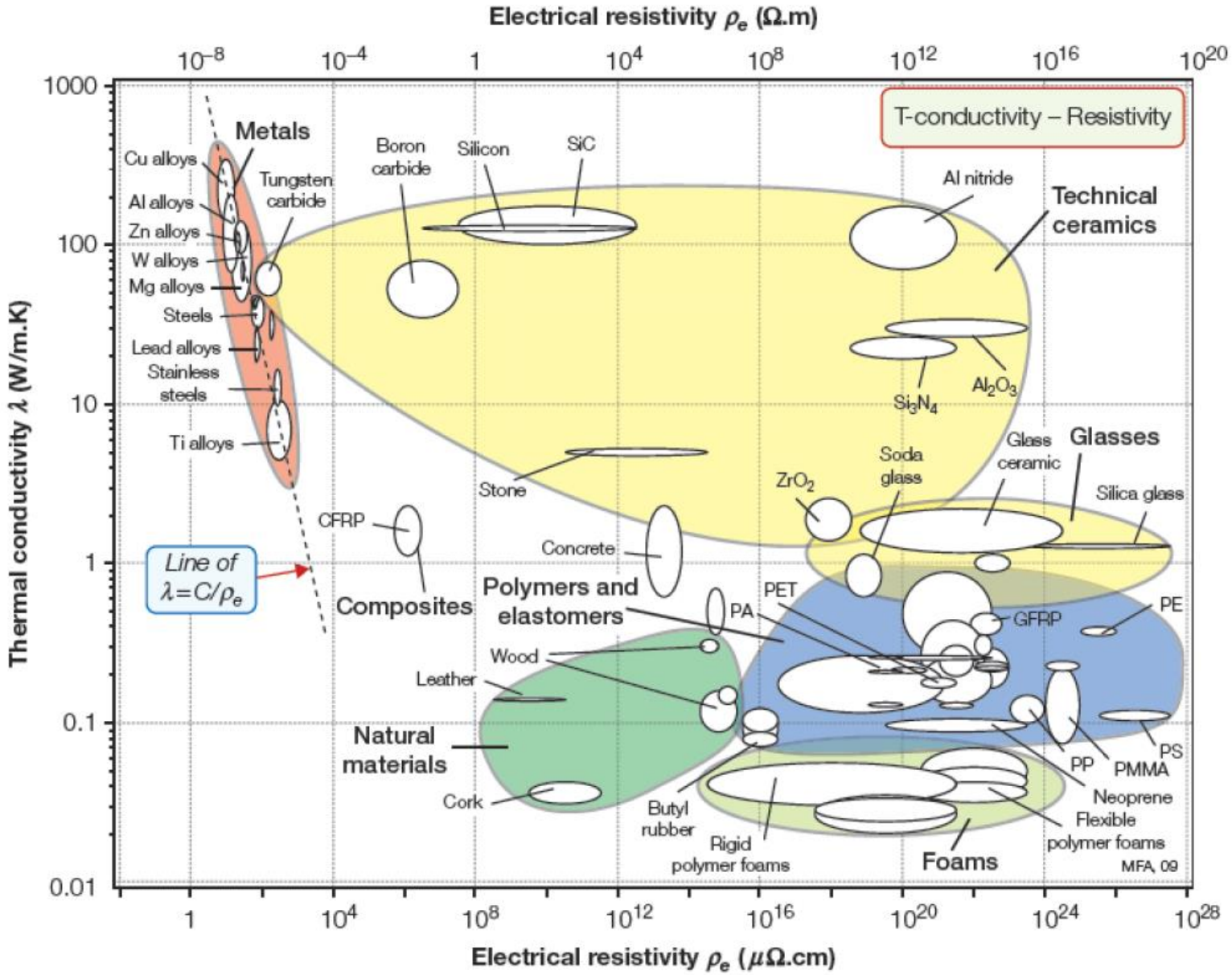


Electrical Properties

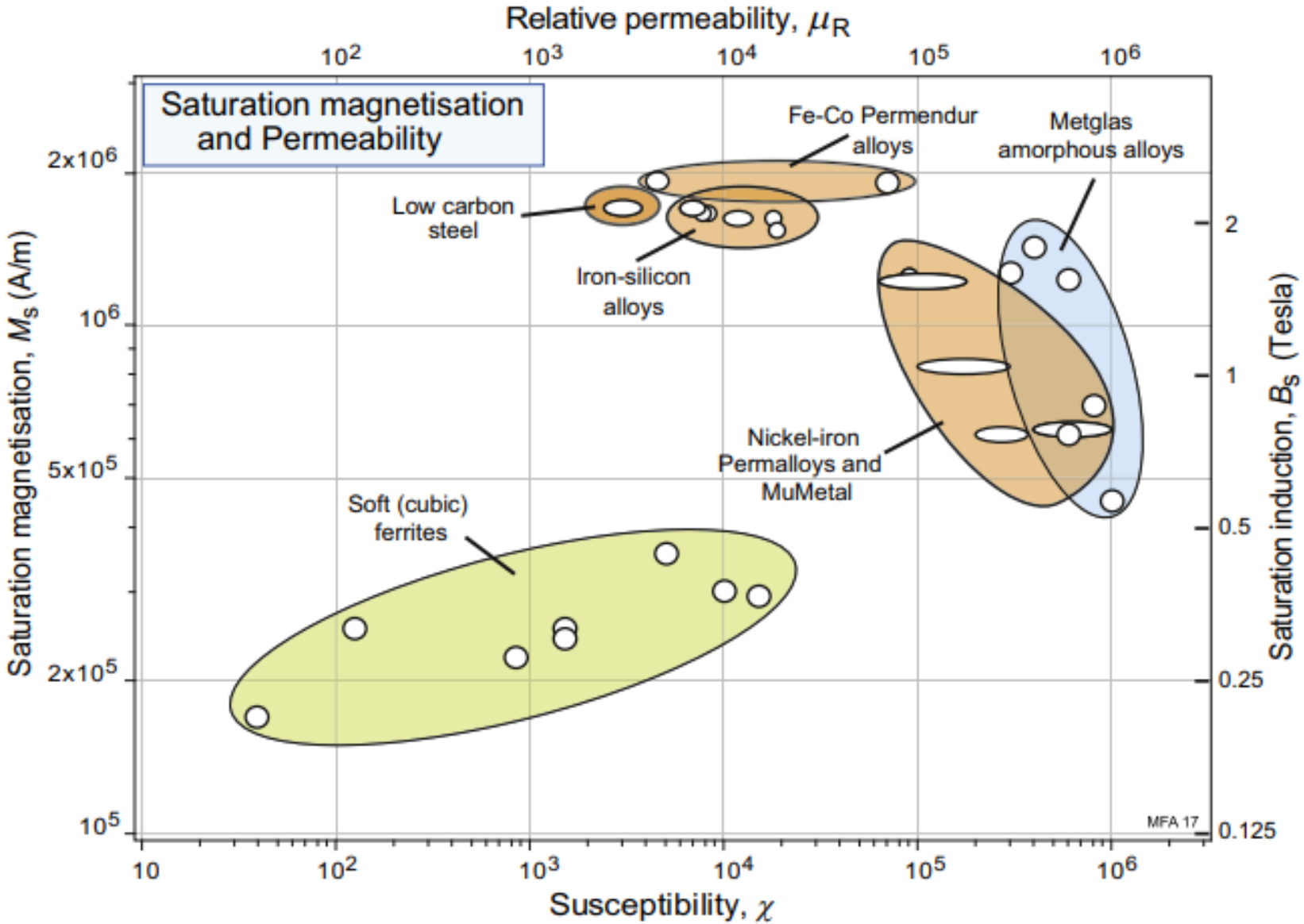


Material Property Charts

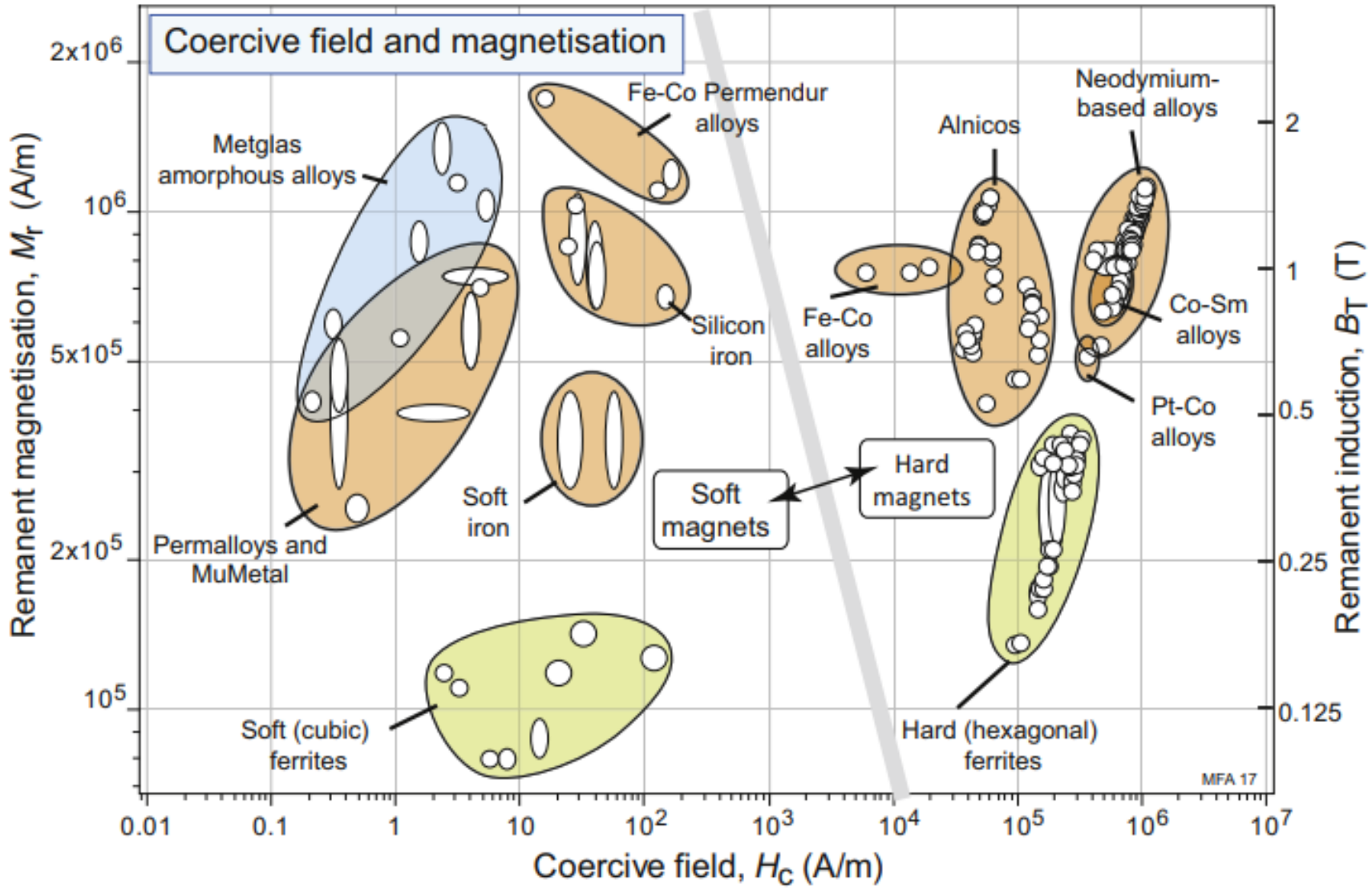
Thermal conductivity vs Electrical resistivity



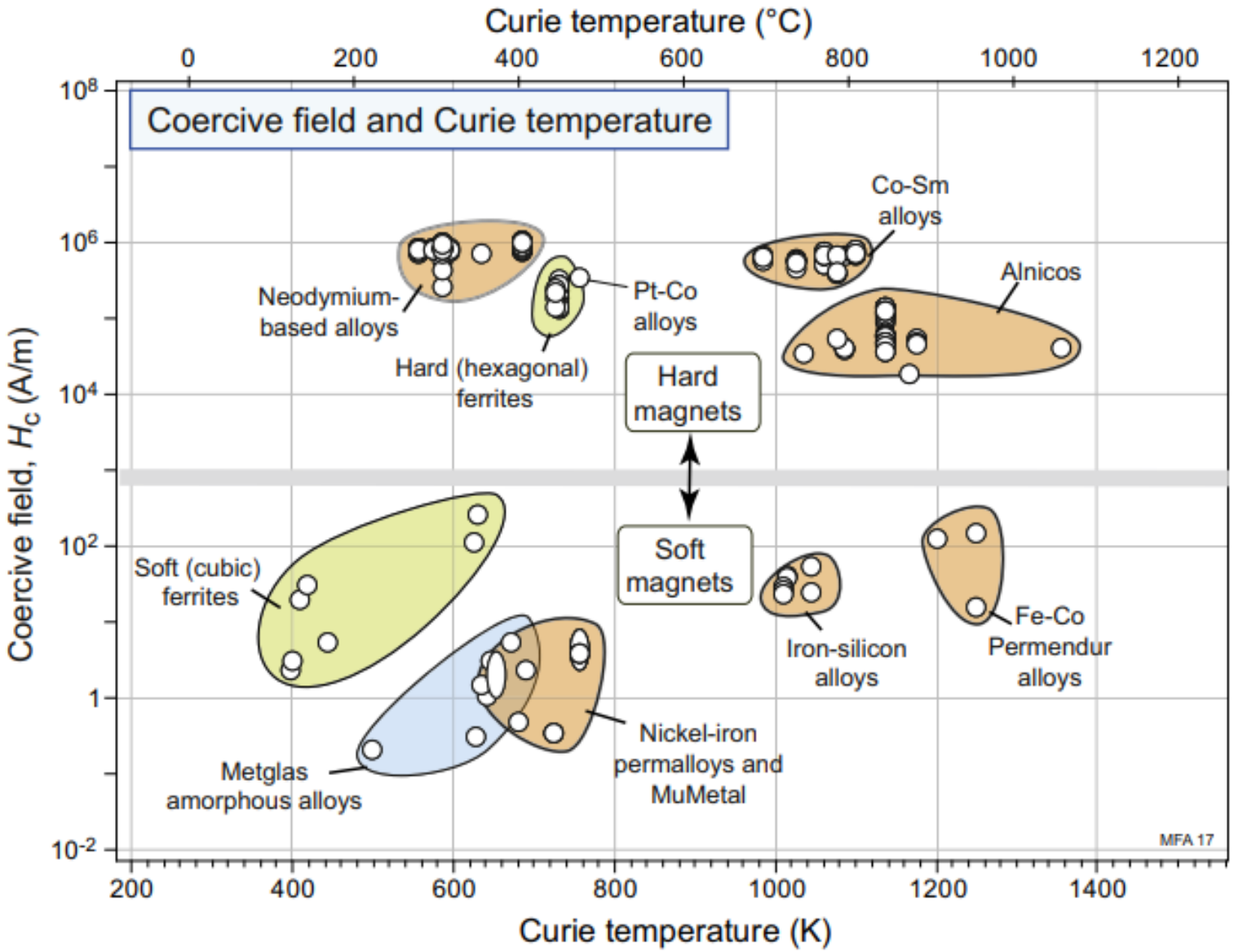
Magnetic Properties



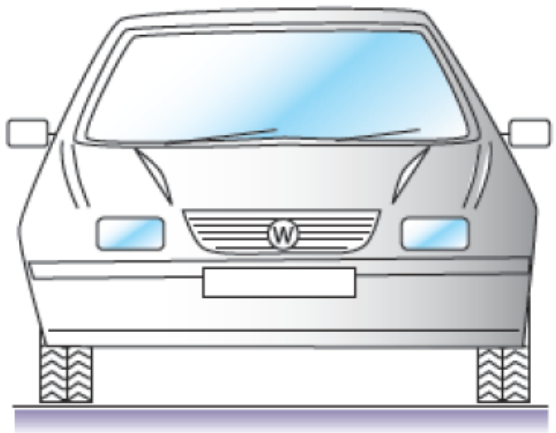
Magnetic Properties



Magnetic Properties



Selection Strategy – Choosing a car



Family car

Desired features
expressed as
- Constraints
- *Objectives*

- *Mid-sized family sedan*
- *4-door*
- *Gas fuel*
- *150+ horsepower*
- *Lowest cost of ownership*

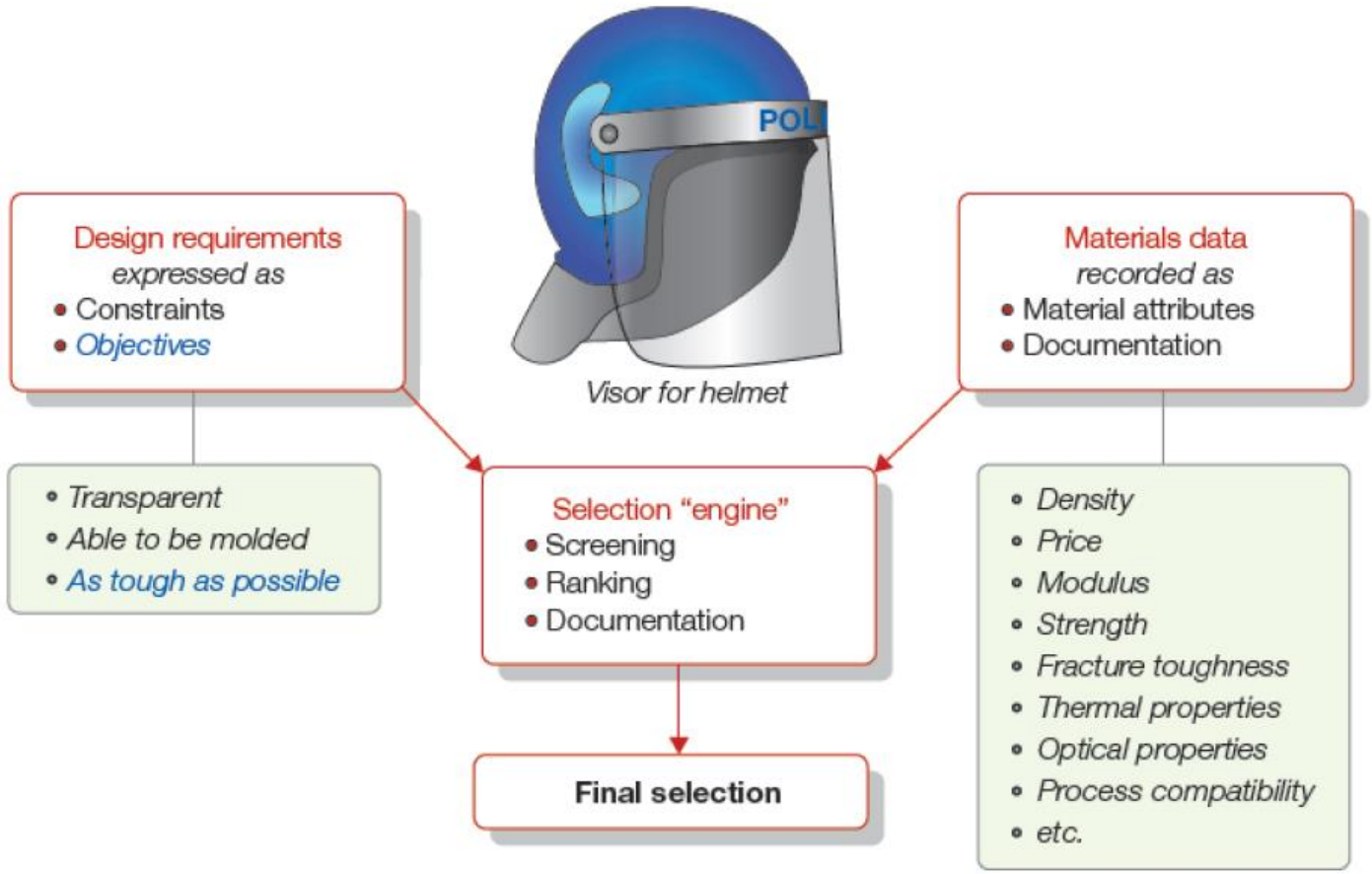
Selection "engine"
- Screening
- Ranking
- Documentation

Final selection

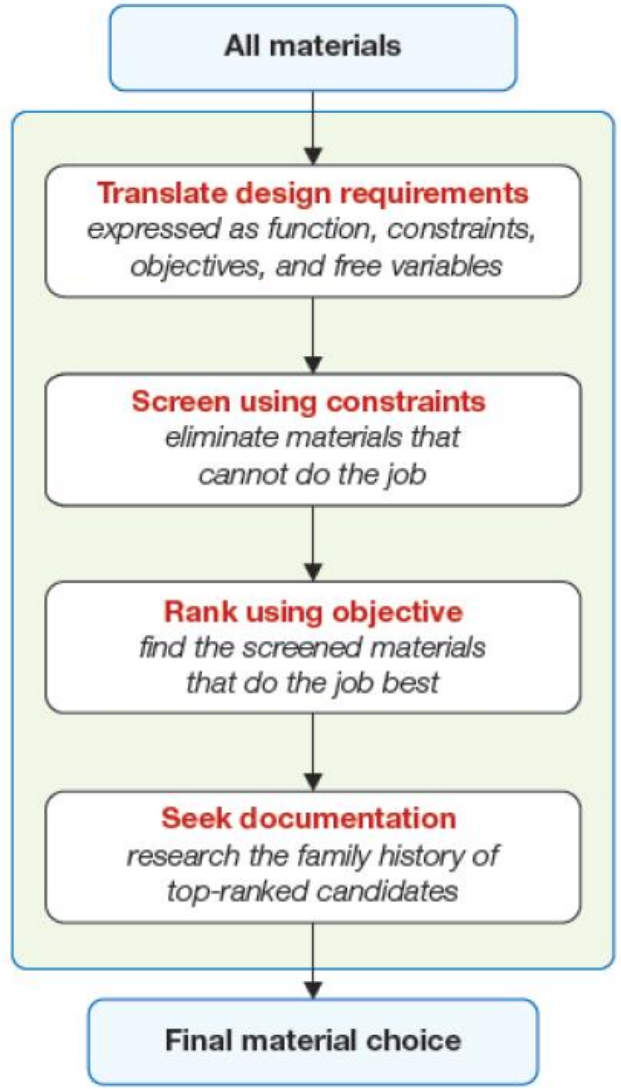
Car data
recorded as
- Features
- Economy

- *Make*
- *Model*
- *Price*
- *Dimensions*
- *Fuel type*
- *Fuel consumption*
- *CO_2 rating*
- *Cost of ownership*
- *etc.*

Selection Strategy – Choosing a visor material



Strategy for Materials Selection



Translation – how the design requirements for a component be translated into prescription for materials?

Function	What does the component do?	Support load, contain pressure, transmit heat…
Constraints*	What nonnegotiable conditions must be met? What negotiable but desirable conditions must be met?	Dimensions, loads, temperature…
Objective	What is to be maximized or minimized?	Cheap, lightweight….
Free variable	Which parameters of the problem is the designer free to change?	Dimensions, loads, temperature… *(which are not constrained)*

Design requirements for the helmet visor

A material is required for the visor of a safety helmet to provide maximum facial protection.

Translation

To allow clear vision the visor must be optically transparent. To protect the face from the front, from the sides, and from below it must be doubly curved, requiring that the material can be molded. We thus have two constraints: transparency and ability to be molded.

Fracture of the visor would expose the face to damage; maximizing safety therefore translates into maximizing resistance to fracture. The material property that measures resistance to fracture is the fracture toughness, K_{1c}. The objective is therefore to maximize K_{1c}.

Screening : Constraints impose limits on the properties or attributes (called Attribute limits) – based on this eliminate materials that cannot do the job

Ranking – Material Index - *material property or a combination of material properties to be minimized or maximized so as to enhance design performance*

Screening and ranking for the helmet visor

A search for transparent materials that can be molded delivers the following list. The first four are thermoplastics; the last two, glasses. Fracture toughness values can be found in Appendix A.

Material	Average Fracture Toughness K_{1c} MPa.m$^{1/2}$
Polycarbonate (PC)	3.4
Cellulose acetate (CA)	1.7
Polymethyl methacrylate (acrylic, PMMA)	1.2
Polystyrene (PS)	0.9
Soda-lime glass	0.6
Borosilicate glass	0.6

The constraints have reduced the number of viable materials to six candidates. When ranked by fracture toughness, the top-ranked candidates are PC, CA, and PMMA.

Documentation

It is descriptive, graphical, or pictorial: case studies of previous uses of the material, failure analyses and details of its corrosion, information, about availability and pricing, and the like. Such information is found in handbooks, suppliers' data sheets, case studies of use, and failure analyses. Documentation helps narrow the short-list to a final choice, allowing a definitive match to be made between design requirements and material attributes.

Documentation for materials for the helmet visor

At this point it helps to know how the three top-ranked candidates listed in the last examples box are used. A quick web search reveals the following.

Polycarbonate

Safety shields and goggles; lenses; light fittings; safety helmets; laminated sheet for bullet-proof glazing.

Cellulose Acetate

Spectacle frames; lenses; goggles; tool handles; covers for television screens; decorative trim, steering wheels for cars.

PMMA, Plexiglas

Lenses of all types; cockpit canopies and aircraft windows; containers; tool handles; safety spectacles; lighting, automotive taillights.

This is encouraging: All three materials have a history of use for goggles and protective screening. The one that ranked highest in our list—polycarbonate—has a history of use for protective helmets. We select this material, confident that with its high fracture toughness it is the best choice.

Disposable fork

Function	• *Disposable cutlery*
Constraints	• *Easy(and thus cheap) to mold* • *Non-toxic* • *Sufficient strength and stiffness (modulus) to withstand loads in use* • *Recyclable or renewable (e.g. wooden cutlery)* • *Ability to be colored (?)*
Objective	• *Minimize material price*
Free variables	• *Choice of material*

Bicycles – Reduce mass / reduce cost – multiple objectives possible.

Heat exchanger – Increase thermal conductivity / reduce mass – multiple conflicting objectives possible

Cooking Pan

Function	• Contain foodstuffs during cooking
Constraints	• Maximum working temperature above 300°C (boiling and frying) • High thermal conductivity, λ , to transmit and spread the heat well • Excellent corrosion resistance in salt water • Excellent corrosion resistance in mild acids (e.g. Vinegar) • Excellent corrosion resistance in alkalis (e.g. Baking soda) • Resistant to thermal shock (sudden change of temperature)
Objectives	• Minimize material cost
Free variables	• Thickness of the pan • Choice of material

Eyeglass Lens

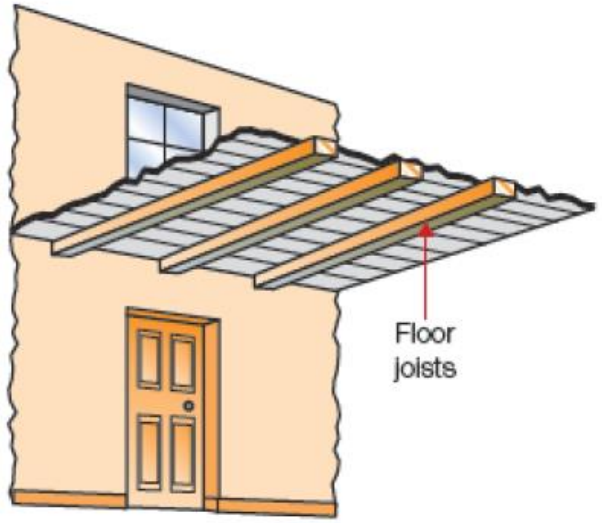
Function	• Refract light without loss of clarity
Constraints	• Optical-quality transparency • Ability to be molded or ground with precision • Excellent resistance to fresh and salt water (rain, sweat) • High hardness, or ability to carry a hard transparent coating
Objectives	• Minimize cost of material and of molding or grinding process
Free variables	• Choice of material • Choice of coating to improve scratch resistance and reflectivity

STRUCTURAL MATERIALS FOR BUILDINGS

Material	M_1 (GPa$^{1/2}$)/ (kg/m^3)	M_2 (MPa$^{2/3}$)/ (kg/m^3)	Comment
Concrete	160	14	Use in compression only
Brick	12	12	
Stone	9.3	12	
Woods	21	90	Can support bending and tension as well as compression, allowing greater freedom of shape
Cast iron	17	90	
Steel	14	45	

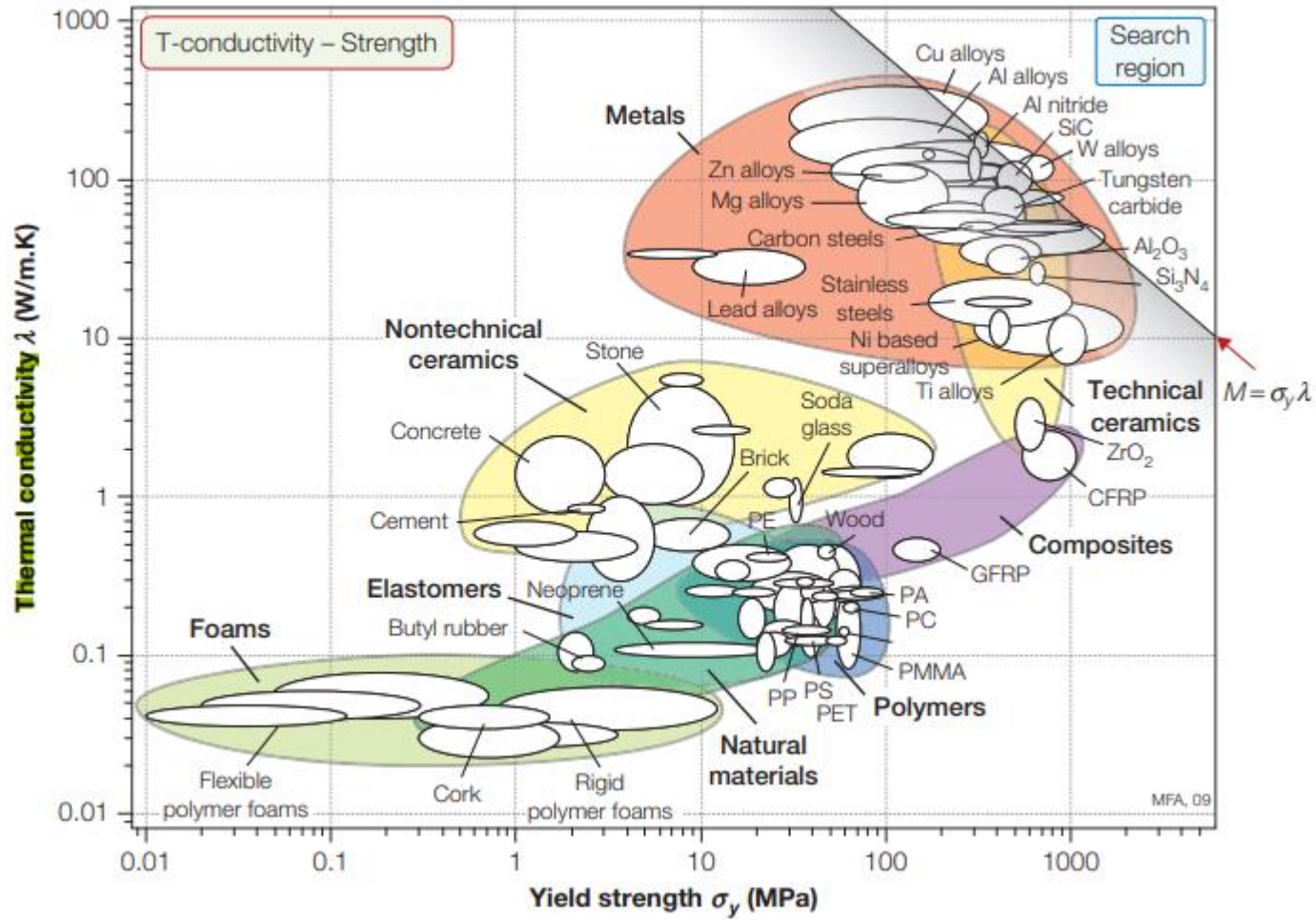
M1 = Specific modulus
M2 = Specific strength

Requirement?
For strong - specific strength should be high
For stiff – specific modulus should be high



Automobile radiator

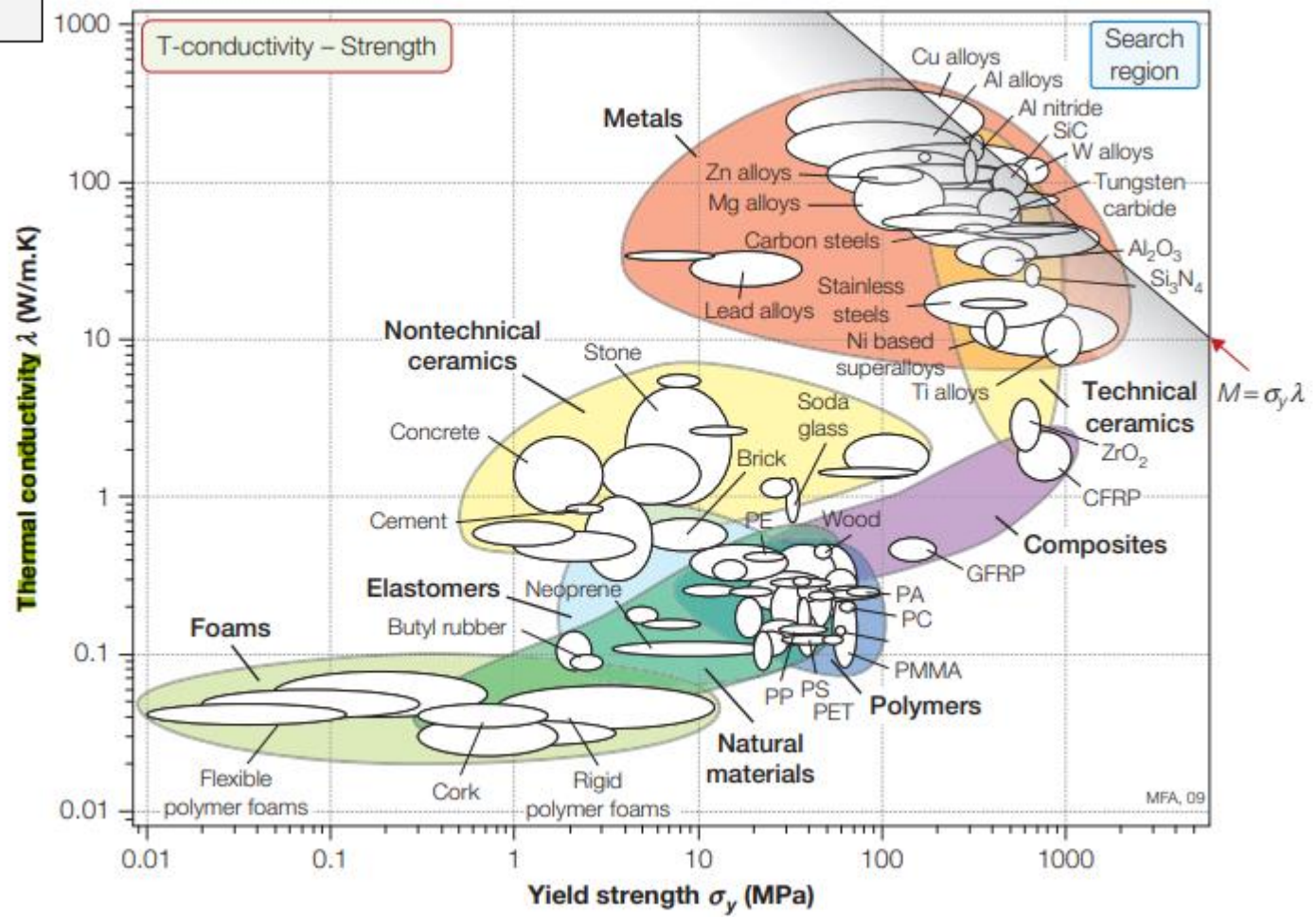
Function	• *Automobile heat exchanger*
Constraints	• *Elongation > 10%* • *Maximum use temperature > 180°C* • *Price < $5/kg* • *Durability in fresh water = very good*
Objective	• *Maximize Thermal conductivity × Yield strength, $\lambda \sigma_y$*
Free variables	• *Wall thickness* • *Choice of material*



Thermal Barrier Coatings

Function	• *Thermal barrier coating*
Constraints	• *Maximum service temperature > 1300 °C* • *Adequate strength: $\sigma_y > 400$ MPa*
Objective	• *Minimize thermal conductivity λ*
Free variable	• *Choice of material*

Alumina
Silicon carbide
Zirconia
Tungsten alloys



A low temperature furnace operating at 250 deg. C uses solid shelves of rectangular for supporting components during heat treatment.
Min required strength is 200 MPa.
Density < 6 g/cc.
What materials possible?

Use the modulus-density chart, from among the materials that appear on it:

(a)the material with the highest density

(b)the metal with the lowest modulus

(c)the polymer with the highest density

(d)the approximate ratio of the modulus of woods measured parallel to the grain and perpendicular to the grain

(e)the approximate range of modulus of elastomer

Need polymers to float in water with modulus of 0.5 GPa?

A material is required for a powerful permanent magnet that must be as small as possible and be resistant to demagnetisation by stray fields.

A material is required for a powerful permanent magnet that must be as small as possible and be resistant to demagnetisation by stray fields.

The requirement that the magnet be small and powerful requires a high remanent magnetisation, Mr. The need to resist demagnetisation implies a high coercive field, Hc. The choice, read from the chart, is the neodymium-based family of hard magnetic materials. (These are the magnets-of-choice for hybrid and electric car motors and for wind turbine generators.) These, however, are expensive, because they contain a rare earth element Nd. The Alnico family, based on aluminium, nickel, and cobalt (hence the name) are cheaper.

A soft magnetic material is needed for the core of a small high-frequency power transformer. The transformer gets hot in use; forced air cooling limits the rise in temperature to 200 deg.C. Eddy current losses are a problem if the core is electrically conducting. What material would you recommend for the core?

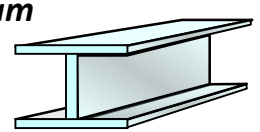
Function, constraints, objectives and free variables

It define the boundary conditions for selecting a material	
Function	What does the component do?
Constraints	What non-negotiable conditions must the material meet?
Objectives	What aspects of performance are to be maximised or minimised?
Free variables	What parameters of the problem is the designer free to change?

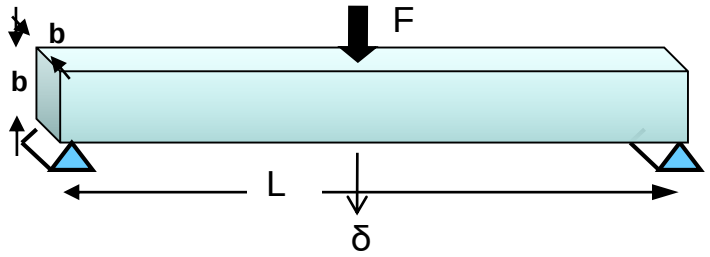
Function, constraints, objectives and free variables

Function

Beam



Stiff beam of length L and minimum mass



Square section, area $A = b^2$

Constraints

- ***Length L is specified***
- ***Must have bending stiffness > S****

Equation for constraint on A:

$$S = \frac{F}{\delta} = \frac{CEI}{L^3} = \frac{CEA^2}{12L^3}$$

m = mass
A = area
L = length
ρ = density
S = stiffness (F/δ)
This beam: $\delta = FL^3/CEI$
C = constant (here, 48)
E = Young's modulus
I = second moment of area
($I = b^4/12 = A^2/12$)

Objective

Minimize mass m:

$$m = AL\rho$$

Performance metric

$$m = \left(\frac{12L^5 S^*}{C}\right)^{1/2} \left(\frac{\rho}{E^{1/2}}\right)$$

Chose materials with largest $M = \left(\frac{E^{1/2}}{\rho}\right)$

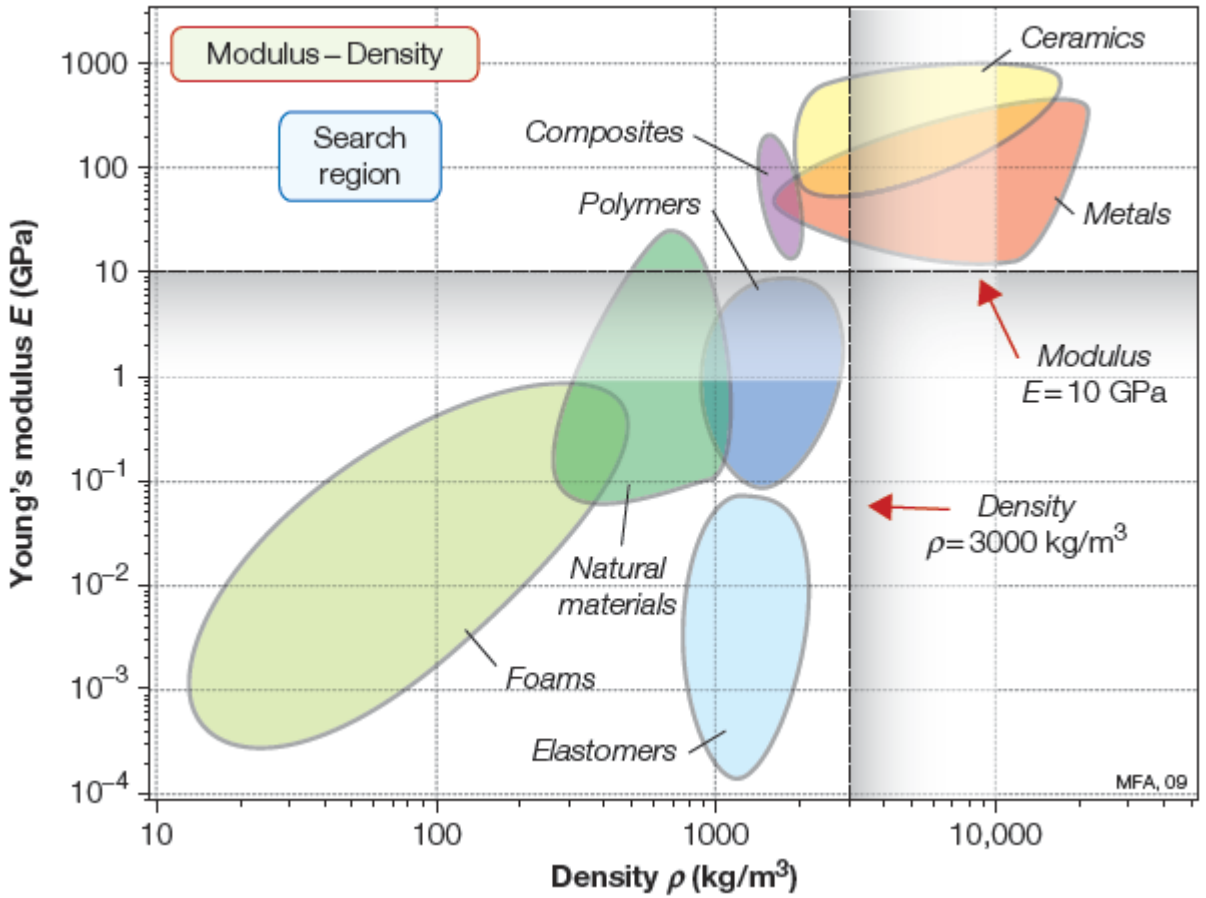
Selection Procedure

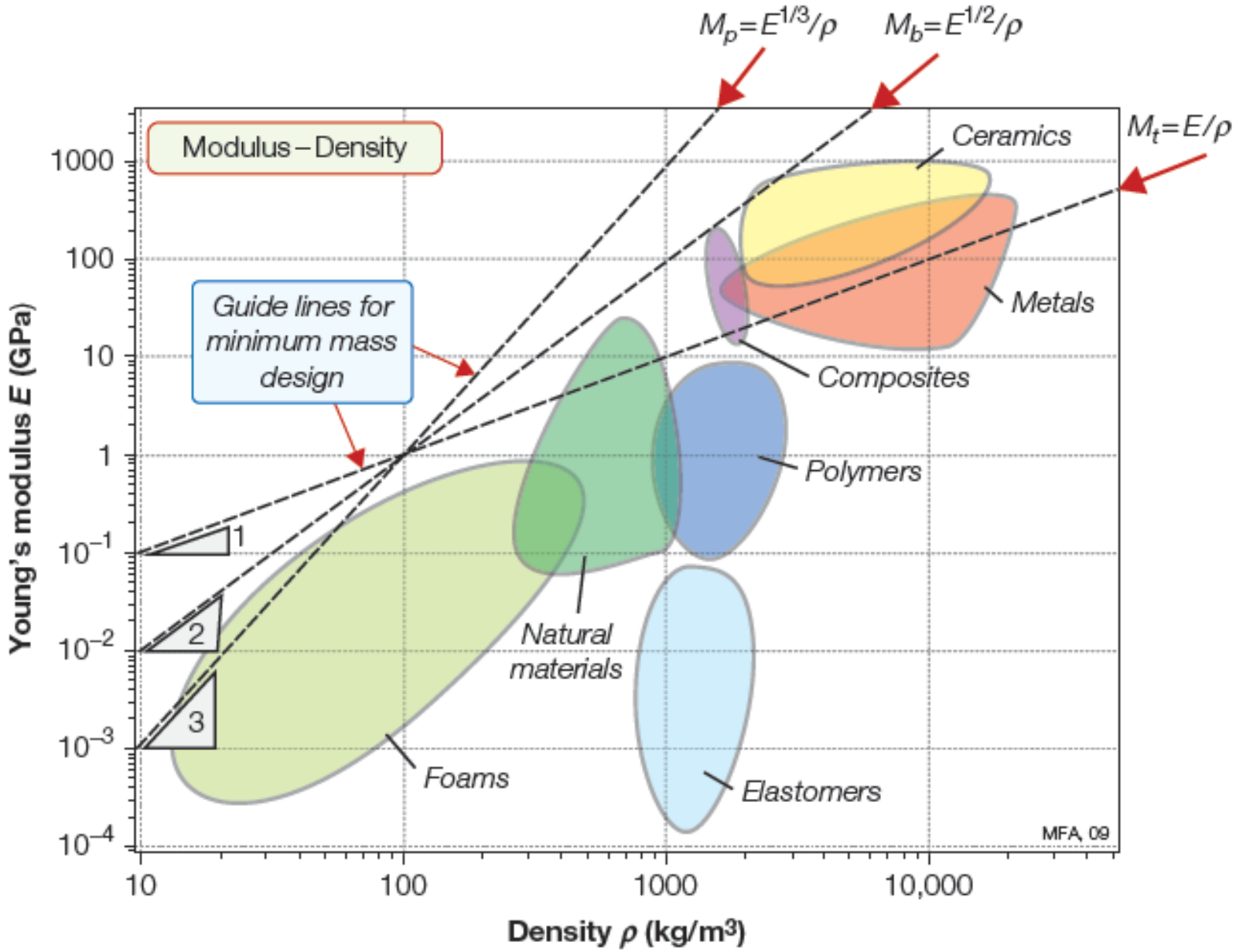
- Translation and deriving the index
- Screening: Applying attribute limits
- Ranking: Indices on charts

$$\frac{E}{\rho} = C \longrightarrow \mathrm{Log}(E) = \mathrm{Log}(\rho) + \mathrm{Log}(C)$$

$$\frac{E^{1/2}}{\rho} = C \longrightarrow \mathrm{Log}(E) = 2\,\mathrm{Log}(\rho) + 2\,\mathrm{Log}(C)$$

$$\frac{E^{1/3}}{\rho} = C \longrightarrow \mathrm{Log}(E) = 3\,\mathrm{Log}(\rho) + 3\,\mathrm{Log}(C)$$



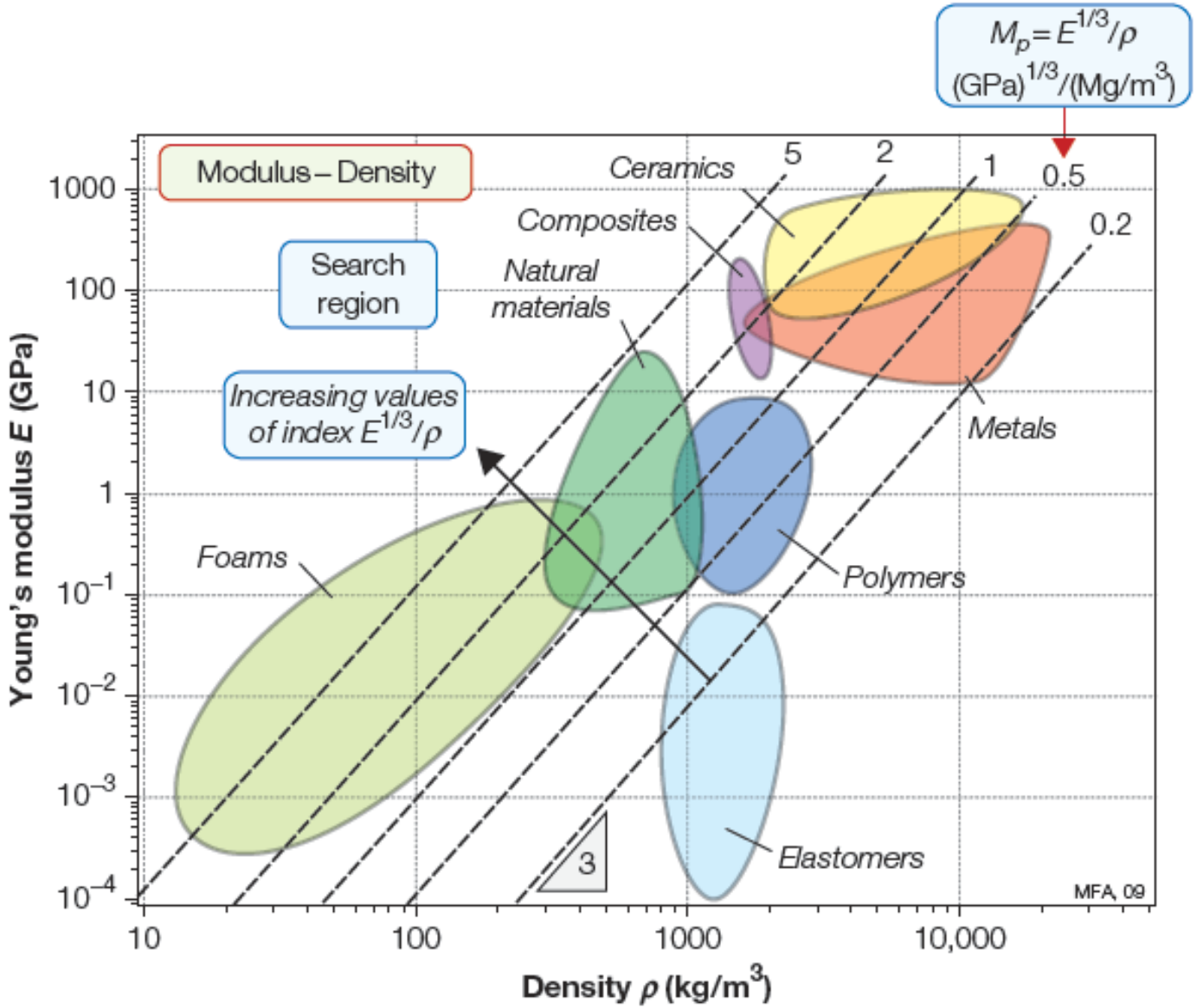

Modulus – Density
$M_p = E^{1/3}/\rho$
$M_b = E^{1/2}/\rho$
$M_t = E/\rho$
Ceramics
Metals
Composites
Polymers
Natural materials
Foams
Elastomers
Guide lines for minimum mass design
1
2
3
Young's modulus E (GPa)
1000
100
10
1
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10
100
1000
10,000
Density ρ (kg/m³)
MFA, 09

A plot of log E vs log ρ will yield a family of straight lines all having a slope of 3

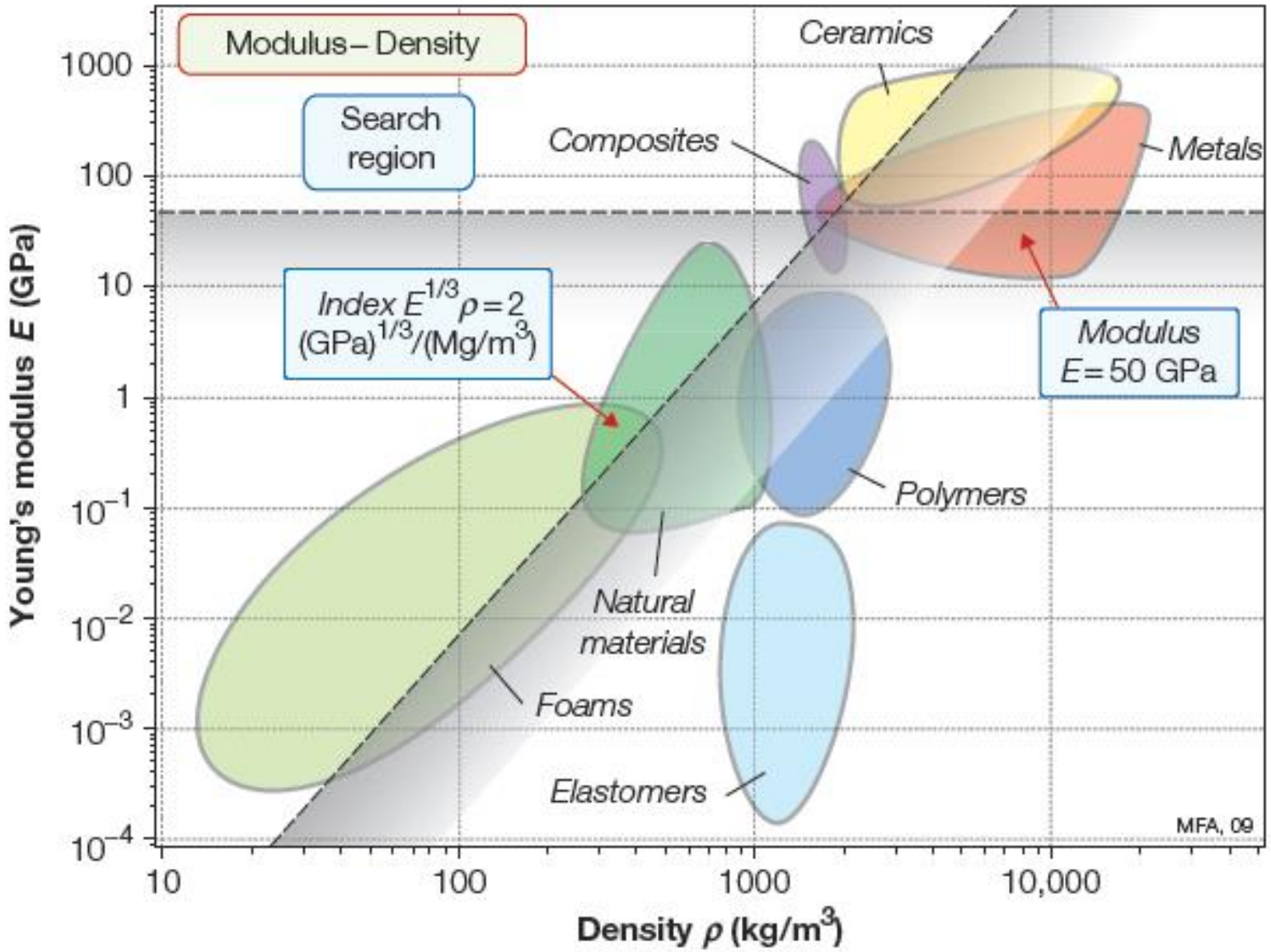
The lines are termed design guidelines

ALL materials that lie on these lines will perform equally well in terms of stiffness-per-mass

Materials lying above the line will perform better than one below → selection zone



A selection based on the index $M = E^{1/3}/\rho > 2$ $(GPa)^{1/3}$ (Mg/m^3) together with the property limit $E > 50$ GPa. The materials contained in the search region become the candidates for the next stage of the selection process.



A selection based on the index $M = E^{1/3}/\rho > 2$ $(GPa)^{1/3}$ (Mg/m^3) together with the property limit $E > 50$ GPa. The materials contained in the search region become the candidates for the next stage of the selection process.

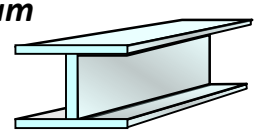
Documentation

Function, constraints, objectives and free variables

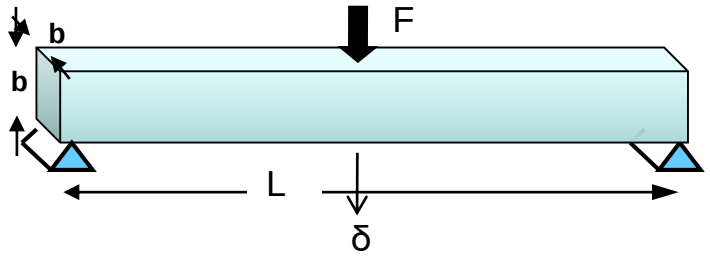
Coming back to this problem …

Function

Beam



Stiff beam of length L and minimum mass



Square section, area $A = b^2$

Constraints

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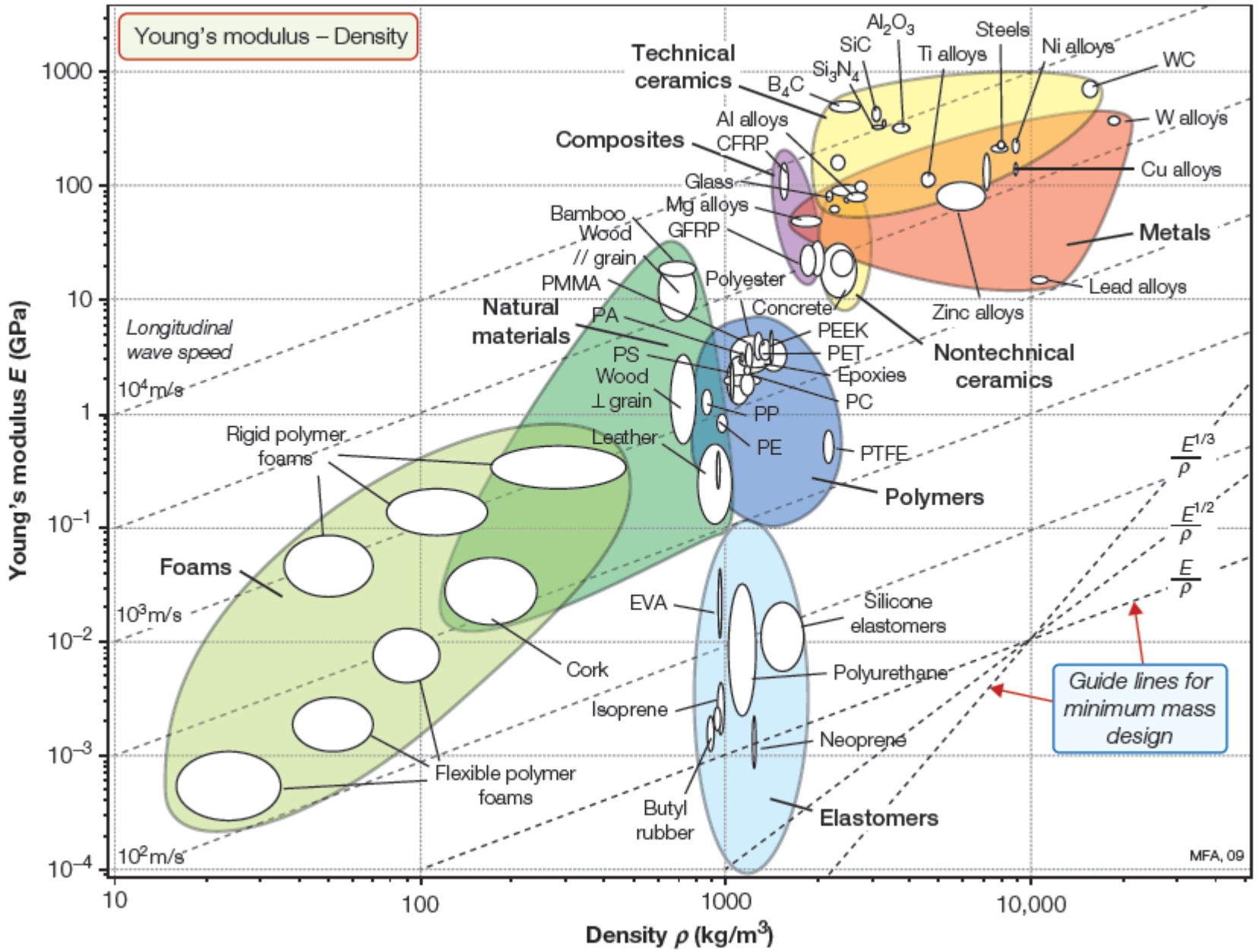
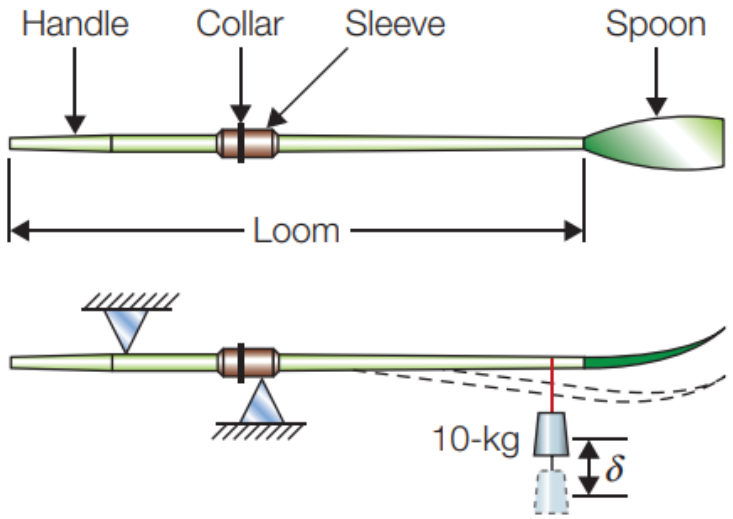
Chose materials with largest $M = \left(\frac{E^{1/2}}{\rho}\right)$

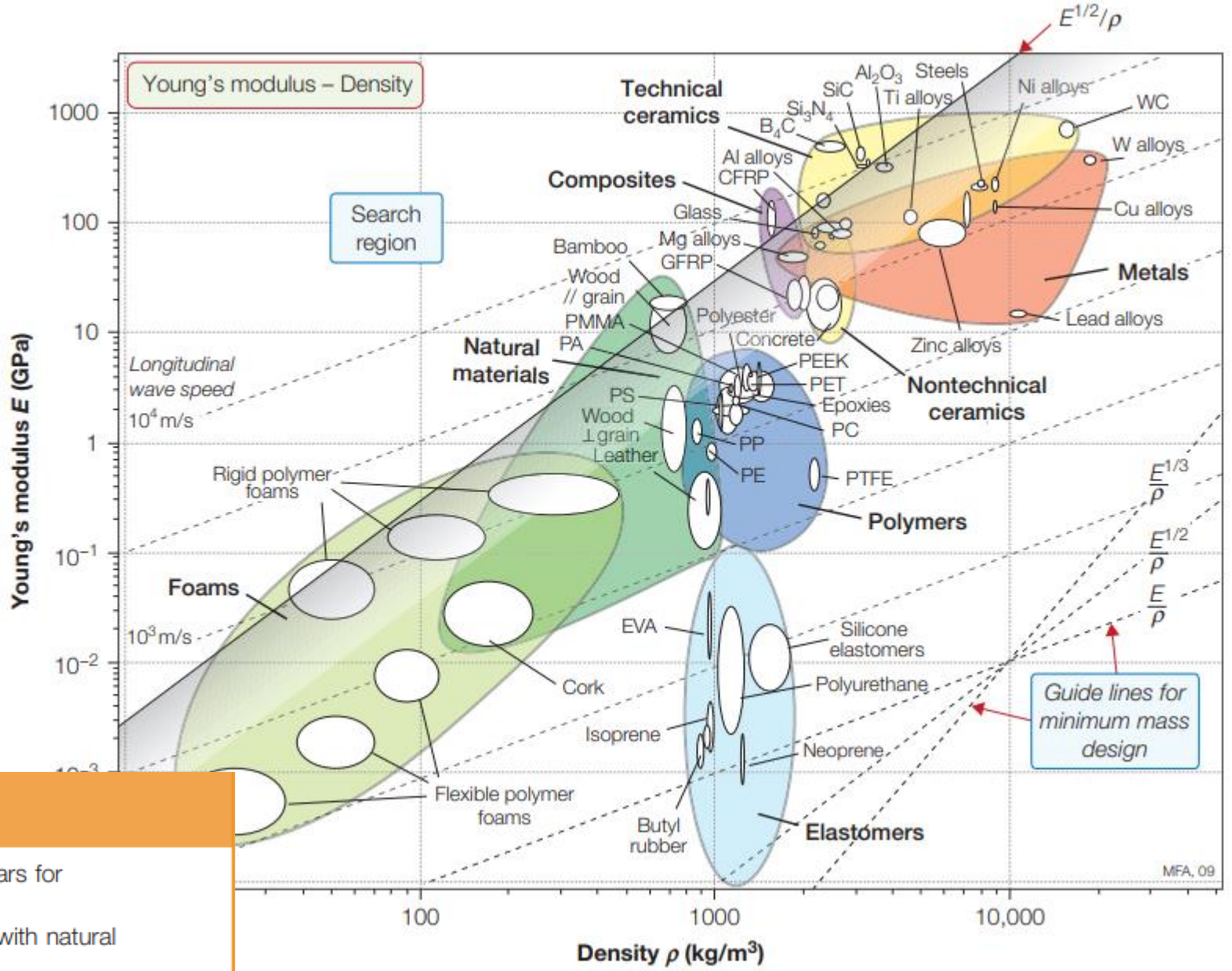
Material Property Charts

Young's Modulus vs Density



MATERIALS FOR OARS



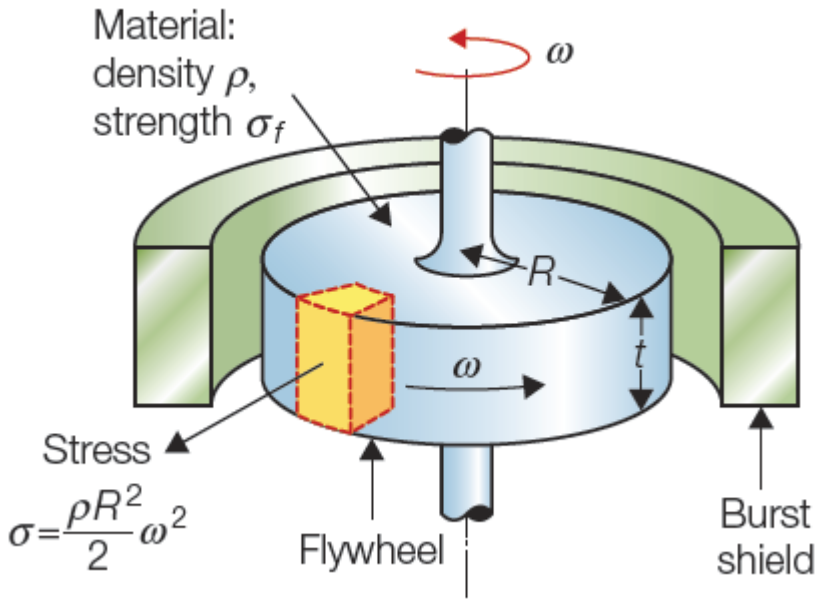


Material	Index M $(GPa)^{1/2}/(Mg/m^3)$	Comment
Bamboo	4.0–4.5	The traditional material for oars for canoes
Woods	3.4–6.3	Inexpensive, traditional, but with natural variability
CFRP	5.3–7.9	As good as wood, more control of properties
Ceramics	4–8.9	Good M but toughness low and cost high

Case Study - MATERIALS FOR FLYWHEELS

Design Requirements for a Maximum-energy Flywheel

Function	Flywheel for energy storage
Constraints	Outer radius, R, fixed Must not burst Adequate toughness to give crack tolerance
Objective	Maximize kinetic energy per unit mass
Free variable	Choice of material



A flywheel. The maximum kinetic energy it can store is limited by its strength.

Design Requirements for Fixed Velocity

Function	Flywheel for child's toy
Constraint	Outer radius, R, fixed
Objective	Maximize kinetic energy per unit volume at fixed angular velocity
Free variable	Choice of material

The objective and constraints in flywheel design thus depend on its purpose.

MATERIALS FOR FLYWHEELS

The energy U stored in the flywheel $U = \frac{1}{2} J\omega^2 \Rightarrow U = \frac{\pi}{4}\rho R^4 t\omega^2$

The mass of the disk is $m = \pi R^2 t\rho$

Kinetic energy per unit mass $\frac{U}{m} = \frac{1}{4} R^2\omega^2$

The maximum principal stress (centrifugal stress) in a spinning disk of uniform thickness $\sigma_{max} = \left(\frac{3+v}{8}\right)\rho R^2\omega^2 \approx \frac{1}{2}\rho R^2\omega^2$

$\Rightarrow \frac{U}{m} = \frac{1}{2}\left(\frac{\sigma_f}{\rho}\right)$

Upper limit

Material index for high performance flywheels, $M = \frac{\sigma_f}{\rho}$

MATERIALS FOR FLYWHEELS

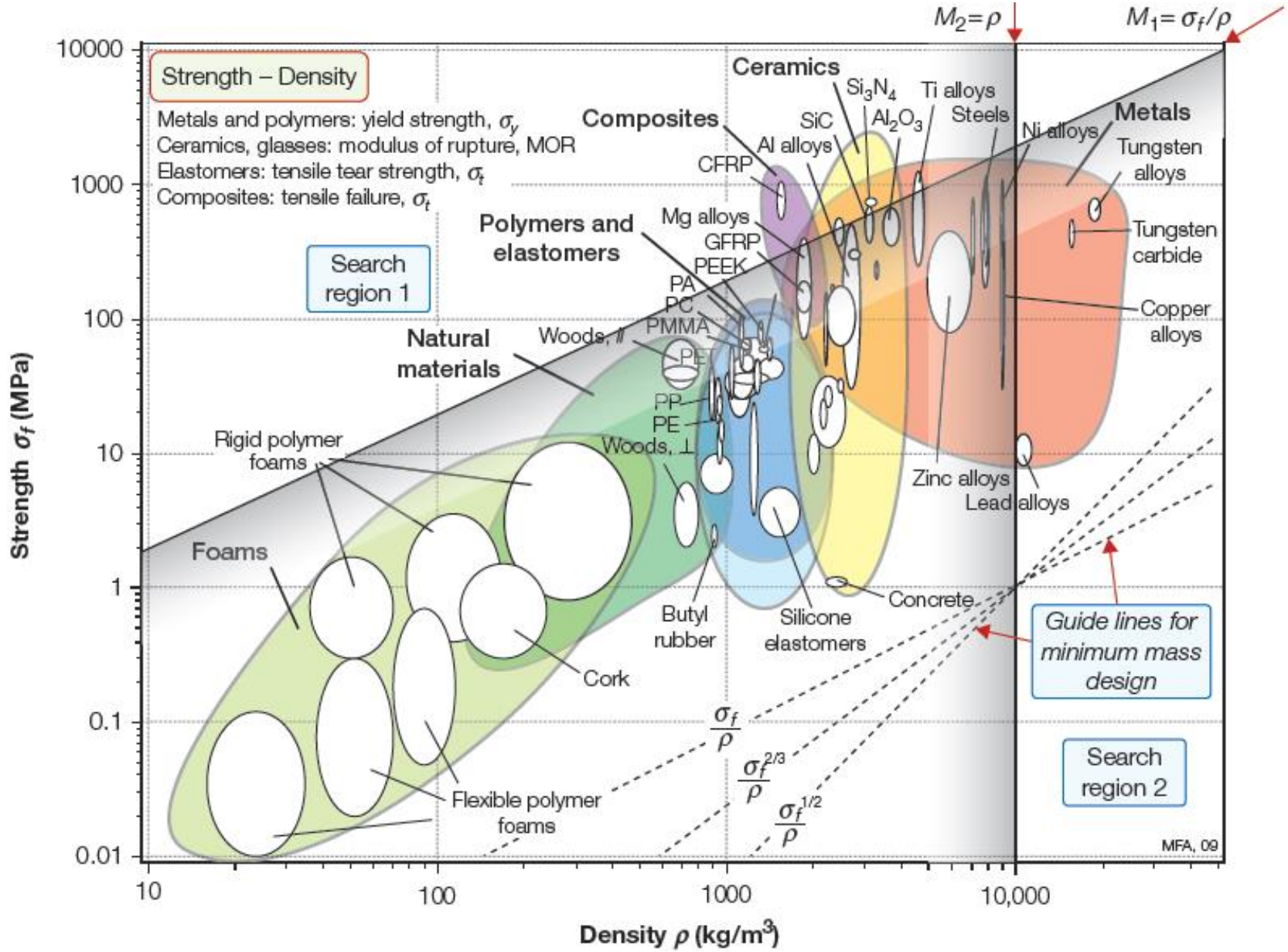
Other sort of flywheel—that of the child's toy. Here we seek the material that stores the most energy per unit volume V at constant velocity, ω.

Energy per unit volume at a given ω is $$\frac{U}{V} = \frac{1}{4}\rho R^2 \omega^2$$

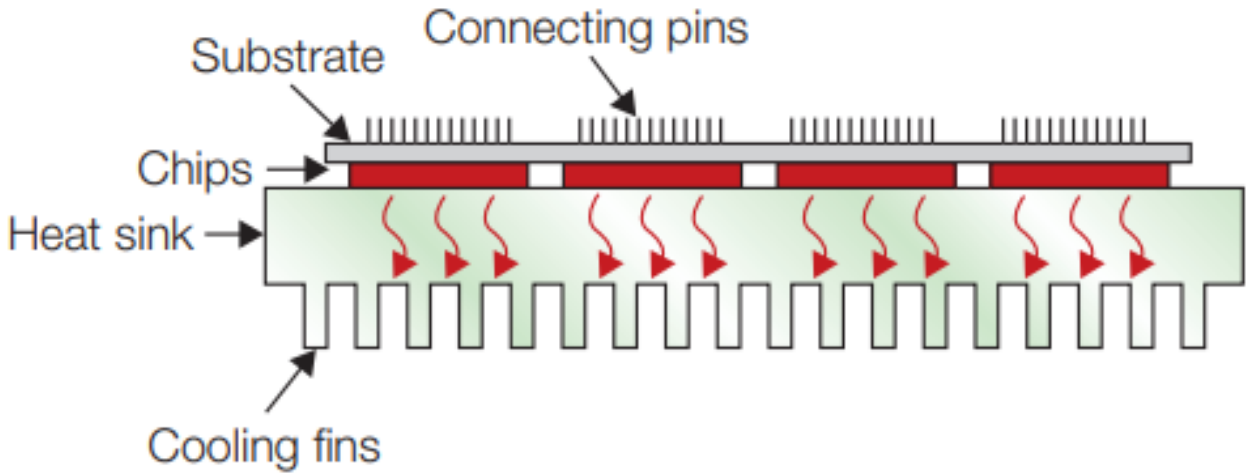
Both R and ω are fixed by the design, so the best material is now that with the greatest value of

$$M_2 = \rho$$

MATERIALS FOR FLYWHEELS

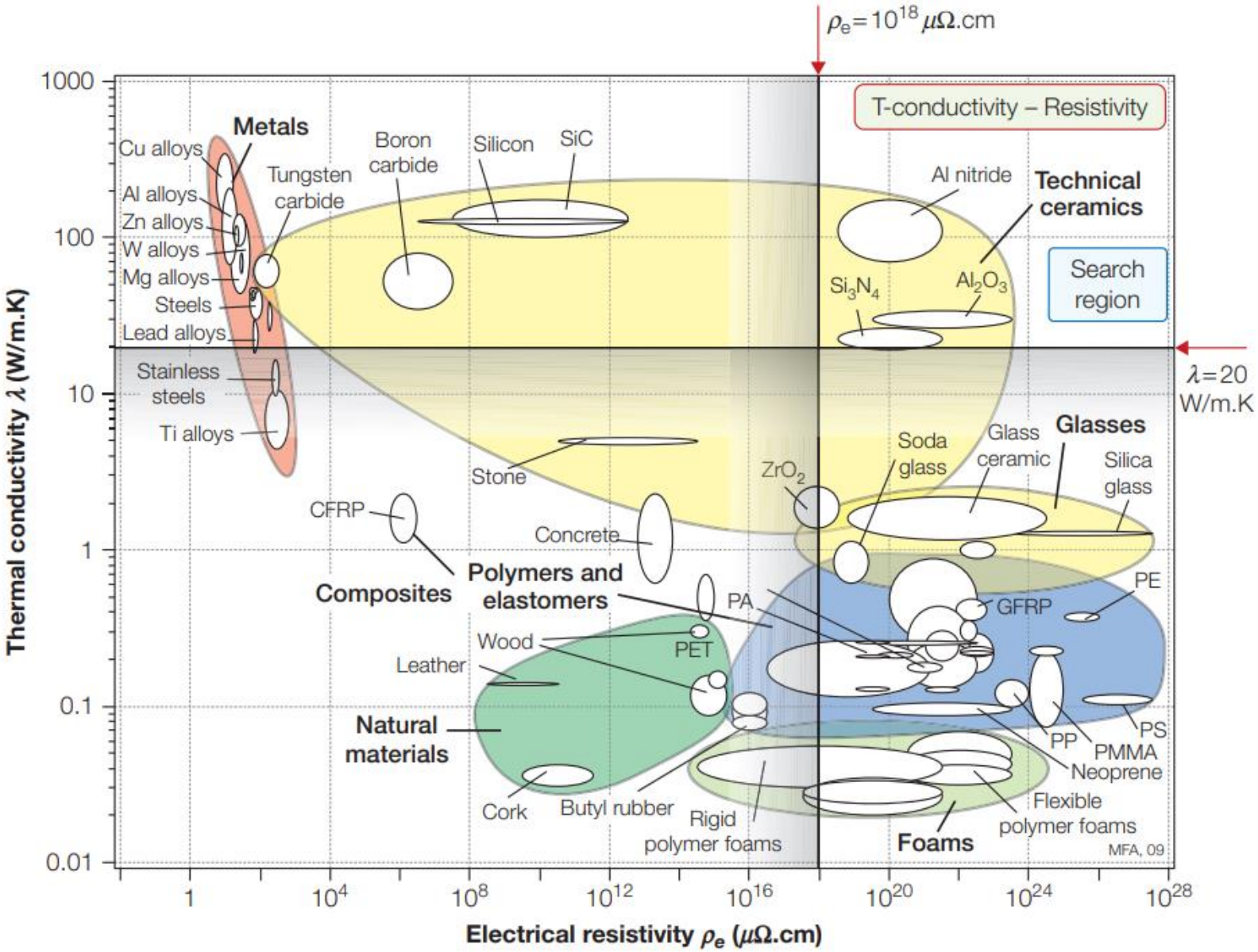


HEAT SINKS FOR HOT MICROCHIPS

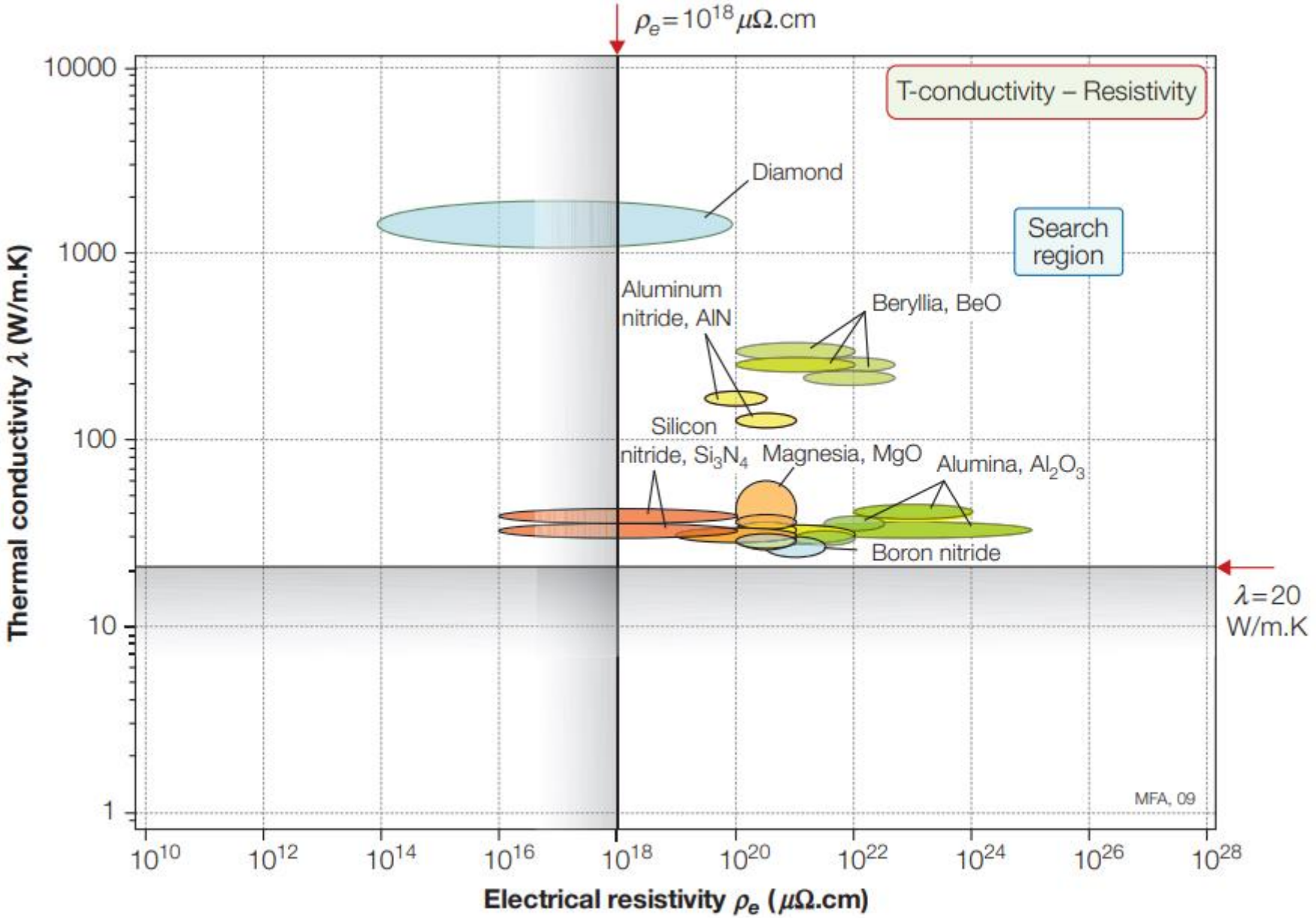


Function	Heat sink
Constraints	Material must be "good insulator," or $\rho_e > 10^{18}\ \mu\Omega$.cm Maximum service temperature > 150°C All dimensions are specified
Objective	Maximize thermal conductivity, λ
Free variable	Choice of material

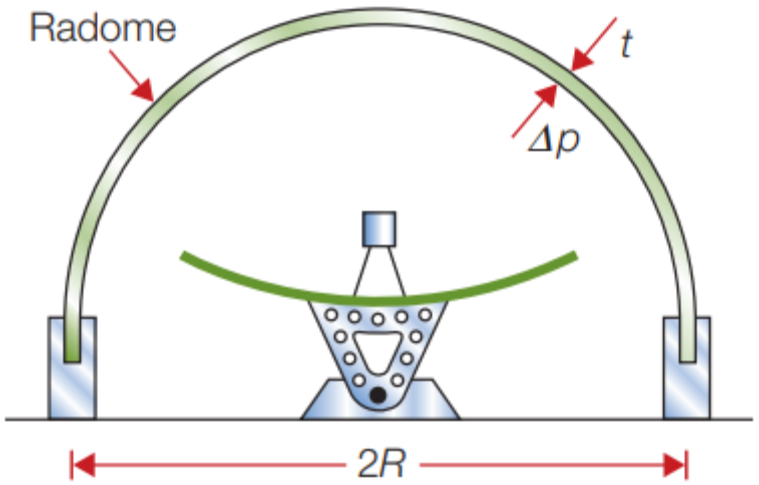
HEAT SINKS FOR HOT MICROCHIPS



HEAT SINKS FOR HOT MICROCHIPS



Materials for Radomes



The fraction of the signal that is reflected is related to the dielectric constant εr.

Dielectric loss is that due to absorption as the signal passes through the skin of the radome.

Function	Radome
Constraints	Support pressure difference Δp Tolerate temperature up to T_{max}
Objective	Minimize dielectric loss in transmission of microwaves
Free variables	Thickness of skin, t Choice of material

Materials for Radomes

Dielectric loss $\left|\frac{\Delta U}{U_o}\right| = \frac{fA^2\varepsilon_o t}{2}(\varepsilon_r \tan\delta)$ This should be minimized.

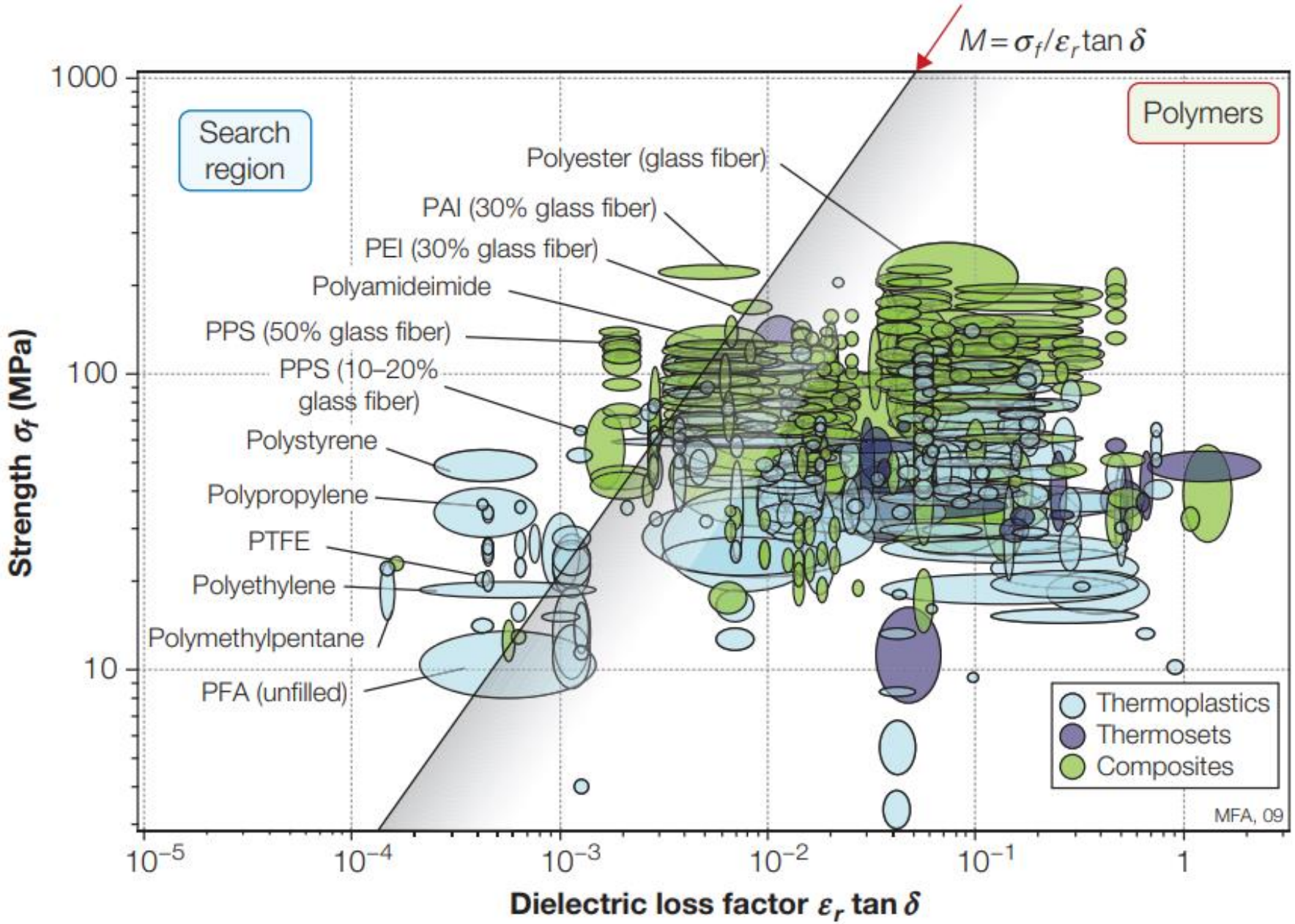
The pressure difference creates a stress $\sigma = \frac{\Delta pR}{2t}$

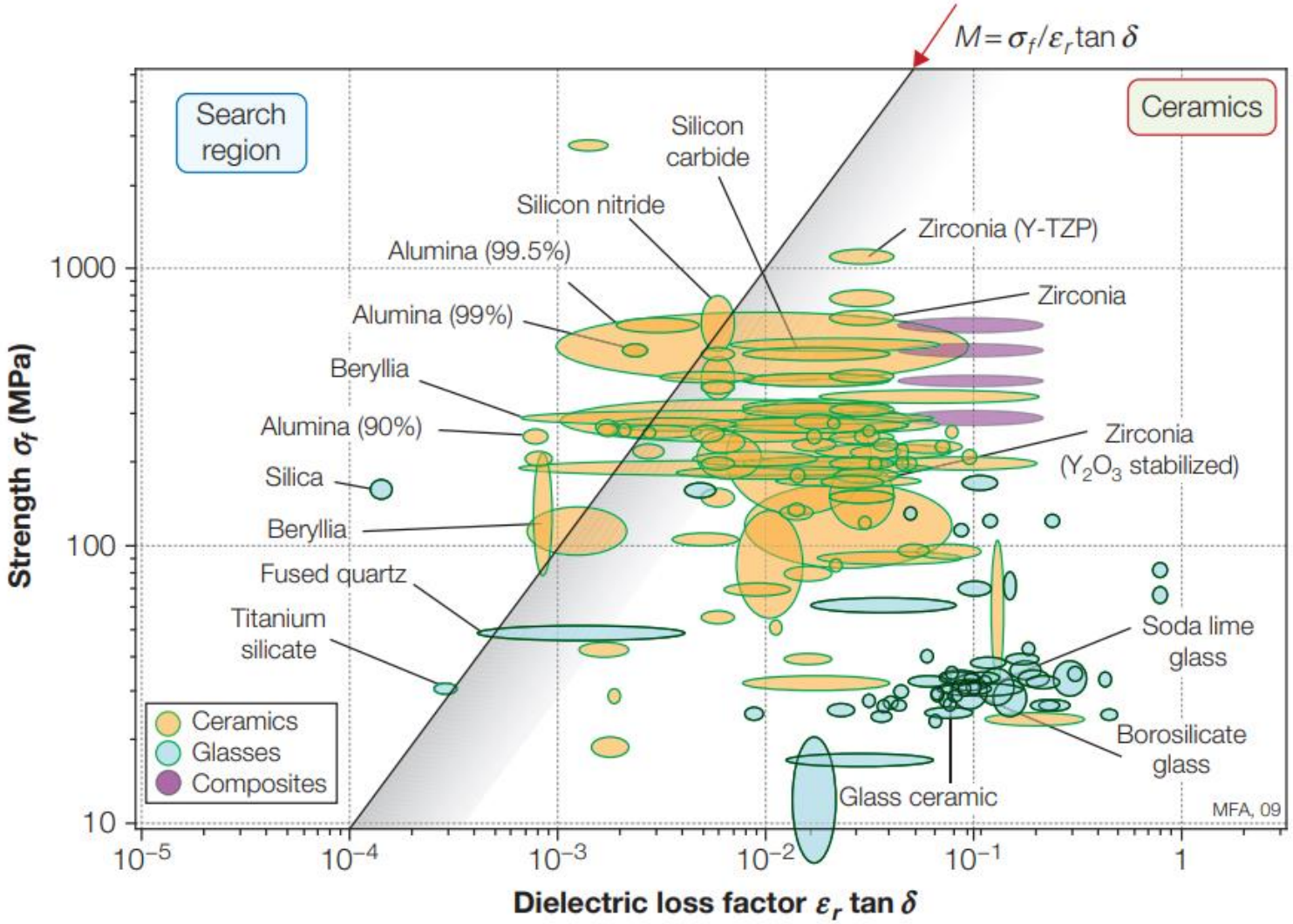
Therefore to support the pressure difference, $t \geq \frac{\Delta pR}{2\sigma_f}$

Then, $\left|\frac{\Delta U}{U}\right| = \frac{fA^2\varepsilon_o \Delta pR}{4}\left(\frac{\varepsilon_r \tan\delta}{\sigma_f}\right)$

The power loss is minimized by maximizing the index

$$M = \frac{\sigma_f}{\varepsilon_r \tan\delta}$$


$M = \sigma_f / \varepsilon_r \tan \delta$
Polymers
Search region
Polyester (glass fiber)
PAI (30% glass fiber)
PEI (30% glass fiber)
Polyamideimide
PPS (50% glass fiber)
PPS (10–20% glass fiber)
Polystyrene
Polypropylene
PTFE
Polyethylene
Polymethylpentane
PFA (unfilled)
Thermoplastics
Thermosets
Composites
MFA, 09
1000
100
10
Strength σ_f (MPa)
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
1
Dielectric loss factor $\varepsilon_r \tan \delta$



PTFE, polyethylene, polypropylene, polystyrene, and polyphenylene sulphide (PPS)	Minimum dielectric loss, but limited to near room temperature
Glass-reinforced polyester, PTFE, polyethylenes and polypropylenes, polyamideimide	Slightly greater loss, but greater strength and temperature resistance
Silica, alumina, beryllia, silicon carbide	The choice for re-entry vehicles and rockets where heating is great